

Defects to Topological Qubits

From a cracked crystal to a hard question

A dummy-to-expert workshop

One new word at a time.

Contents

I. Quantum mechanics from zero	1
1. A magnet that refuses to smear	5
2. Two coins that refuse to factor	15
II. Qubits and quantum information	27
3. A needle you can set, flip, and read	31
4. A clock that forgets	41
III. Crystals and defects	53
5. A crystal that looks the same after a slide	57
6. An empty site is not a leftover spin	69
IV. Defect-spin platforms	83
7. A diamond that glows because something is missing	87
8. A red stone that already has guests	99
9. A factory crystal and its neighbors	111
V. Interacting defect systems	125
10. Two needles that can feel each other	129
11. Three noisy bits pretending to be one	141
VI. Topology and anyons	153
12. A shoelace through a coffee cup	157
13. Two particles trade places	169

14. Two particles in a bag	183
VII. Fibonacci anyons	197
15. Two answers for one pair	201
VIII. Topological lattice models	213
16. Checks that never look at the secret	217
17. A honeycomb of compasses	229
18. Strings that only meet in allowed ways	241
19. A dance and its mirror	253
20. An actor playing a king	265
21. A recipe versus a landscape	277
IX. Engineering topology from defect clusters	289
22. The expensive states still whisper	293
23. Perturbative gadgets	305
24. Eight arrows that are not a machine	317
25. Three graphs that disagree	329
26. What diamond actually writes	345
X. Numbers, noise, and other kitchens	357
27. A number that travels with a passport	361
28. How far apart, and did the right atom land there?	377
29. A coupling wearing a nicer hat	389
30. The same shove, a different filter	401
31. Topological protection is not magic	413

32. A peak is not a braid	425
33. Programmable competitors	437
34. The coordinate that is not the atom	449
35. One microscope, three neighbors	461
36. Reading topological information	475
XI. Verdict	487
37. The strongest case, not the loudest	491
38. The strongest case against	505
39. Write the no first	519
40. Calculations with exits	531
41. The grade	545
XII. Mathematical appendices	561
A. Arrows you can add and stretch	565
B. Two lists that become four	575
C. A knob and a shoelace	585
D. Four charges and a table of crossings	595
E. Two trees that must agree	605
F. Rotate the expensive rooms away	615
G. Checks that never read the secret	623
H. Boxes and wires	631
Global evidence table	641
Glossary	659
Annotated bibliography	673

From a cracked crystal to a hard question

A diamond is almost nothing but carbon, lined up with obsessive regularity. Almost. A missing atom, a stray nitrogen, a kink in the lattice can trap an electron the way a pothole traps a marble. Laboratories already use a few of those trapped electrons as tiny compass needles: they can be pointed, flipped, and read out.

This book asks a larger question. If many such imperfections were planted on purpose, and made to talk to one another, could the crystal as a whole behave like something no single defect knows how to be? In particular: could the low-energy motion of the crowd protect a piece of quantum information the way a knot survives a tug on the rope?

The answer is not on this page. The work is to assemble enough quantum mechanics, crystal physics, and topology that the argument can fail or survive in public, with numbers attached.

You do not need a physics degree. You need algebra, a little calculus, and patience with a new word when it has earned its keep. If a later chapter needs a heavy tool, the handle will already exist.

How to read this

Read straight through once, slowly. When an equation appears, do the tiny numerical example next to it before you go on. Later chapters are the same story with more of the machinery visible, not a reward for surviving the early ones.

If you already know some of the early material, you may skip ahead — but not by much.

- Comfortable with qubits, new to solids: start at Chapter 5.
- Comfortable with color centers, new to topology: start at Chapter 12 after skimming Chapters 10 and 11.
- Comfortable with anyons, here for the hardware claim: start at Chapter 22, then go back when a word is being used as if you lived with it.

Three distinctions will be used as law:

1. A two-level atom is not yet a computer.
2. A simulation of a strange particle is not a material that contains one.
3. A long-lived spin is not automatically a protected logical bit.

Those three sentences are the plot. Everything else is us checking them.

The road

The book is one argument.

1. How a single quantum experiment is predicted.
2. How two quantum systems become one object.
3. How a laboratory turns a two-level system into a working bit.
4. How a crystal can hide an atom-like state in a defect.
5. How defects talk, cluster, and sometimes pretend to be one bigger bit.
6. How topology can protect information — and how often that word is borrowed for something weaker.
7. Whether a lattice of real defects could host the kind of anyons theorists like, including the Fibonacci kind.
8. What the energy, length, noise, and fabrication budgets actually say.
9. A verdict, with kill criteria, not a sales pitch.

The appendices are tool drawers. Open them when a chapter points at them. The glossary is a memory aid, not a prerequisite.

A promise about language

I will not ask you to swallow “Hilbert space,” “modular tensor category,” or “topologically ordered Hamiltonian” before you have a picture. When a technical word finally appears, it will name something you have already seen. If that promise breaks, the chapter is wrong, not you.

Later chapters stay honest about evidence. A measured number from a real sample is different from a calculation on an ideal lattice, which is different from a proposal. When that difference matters, it will be said in English first.

Units, when we need them

Frequencies come in hertz. Energies come in electronvolts or their tiny cousins. Temperature is kelvin. Distance is metres or nanometres. When two quantities are compared, they share a unit. You do not need the conversion table memorized; it will sit next to the first comparison that needs it.

Part I

Quantum mechanics from zero

Two chapters. One atom, then two. By the end you can predict a magnet experiment and say what is lost when you ignore half a pair.

Chapter 1. A magnet that refuses to smear

A silver atom flies through a lopsided magnet and comes out in one of two places. That is already strange. A tiny compass needle in a crooked field should spray into a smear, not pick two spots and ignore the middle.

In 1922, Otto Stern and Walther Gerlach saw the two spots. We need the smallest set of rules that predict the split, and then the second magnet, turned on its side.

No crystal yet. No computer. One atom, two answers, and the arithmetic that makes the second magnet surprising.

1.1. Two numbers that may cancel

Classically you would give each outcome a probability: a pair of nonnegative numbers that add to one. Quantum mechanics does something ruder. It first assigns each outcome a *complex number*, and turns those numbers into probabilities only at the last second.

A complex number looks like

$$z = a + ib,$$

where a and b are ordinary real numbers and $i^2 = -1$. Its partner is $z^* = a - ib$. The useful real quantity hiding inside is

$$|z|^2 = z^* z = a^2 + b^2.$$

That last thing cannot be negative. It is allowed to become a probability. The complex number itself is not: it can be negative, or point somewhere in the plane, and two of them can add to zero before anyone squares anything. That last trick is **interference**. It is why the second magnet will not behave like a polite coin.

Call the two amplitudes α and β . Stack them:

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

The funny brackets are a **ket**, pronounced “ket psi”: a column of numbers. The decoration is a name tag, so we can later tell a state from a measurement question.

Those two numbers are not free. The atom has to land somewhere, so

$$|\alpha|^2 + |\beta|^2 = 1.$$

A *state*, in this chapter, is such a column, up to an overall complex factor of magnitude one. Quantum mechanics is peculiar. It still declines overdrafts.

1.2. Up, down, and both

Give the two spots names. “Along the magnet, up” and “along the magnet, down”:

$$|z+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |z-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then the same atom is

$$|\psi\rangle = \alpha|z+\rangle + \beta|z-\rangle.$$

This is a **superposition** in the z basis. It does not say a tiny classical arrow is secretly pointing both ways. It says: if you ask the z question, these are the two complex ingredients of the answer.

To ask a question you need the opposite of a ket. Flip the column over and conjugate every entry. That row is a **bra**:

$$\langle\psi| = (\alpha^* \quad \beta^*).$$

Feed a ket to a bra and you get one complex number, the **inner product** $\langle\phi|\psi\rangle$. For our atom,

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

The space of all such columns, with this way of multiplying them, is what physicists later call a Hilbert space. You do not need that name to finish the chapter. You need the habit: add the columns, scale them by complex numbers, and compare them with an inner product.

1.3. Why a second magnet forgets the first

Idealize each Stern–Gerlach magnet as a fork with two exits. Keep one exit, block the other. That prepares the corresponding state.

A magnet turned on its side asks a different question. Call those exits $x+$ and $x-$, and define

$$|x+\rangle = \frac{|z+\rangle + |z-\rangle}{\sqrt{2}}, \quad |x-\rangle = \frac{|z+\rangle - |z-\rangle}{\sqrt{2}}.$$

The minus sign is doing real work. The two new states are orthogonal:

$$\langle x+ | x- \rangle = \frac{1}{2} (\langle z+ | + \langle z- |) (|z+\rangle - |z-\rangle) = \frac{1}{2} (1 - 1) = 0.$$

Step 1. Send a mess of atoms through a z magnet. Block $z-$. Keep $z+$. Every surviving atom is now

$$|\psi_1\rangle = |z+\rangle.$$

A second z magnet would send all of them up again. Amplitude 1, probability 1.

Step 2. Turn the next magnet on its side and keep only $x+$. The amplitude is

$$\langle x+ | z+ \rangle = \frac{1}{\sqrt{2}} (\langle z+ | + \langle z- |) |z+\rangle = \frac{1}{\sqrt{2}}.$$

So

$$P(x+ | z+) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

About half the atoms survive. Each survivor is no longer $z+$. It is

$$|\psi_2\rangle = |x+\rangle = \frac{|z+\rangle + |z-\rangle}{\sqrt{2}}.$$

Hence “chance the filter lets you through” and “state of the atom that made it” are different objects. The survivor is freshly normalized.

Step 3. Ask z again.

$$\langle z+ | x+ \rangle = \frac{1}{\sqrt{2}}, \quad \langle z- | x+ \rangle = \frac{1}{\sqrt{2}},$$

so

$$P(z+ | x+) = P(z- | x+) = \frac{1}{2}.$$

The sideways magnet erased the certainty we paid for in step 1. It did not peek at a hidden x label while leaving the z label untouched. In this ideal model, asking one question prepares an answer to that question, which is generally a mixture of answers to a different question.

For 100 atoms already in $z+$, you expect about 50 through the $x+$ door and then about 25 in each final z bucket. Small runs wobble. The conditional probabilities are still one half and one half.

1.4. Phase you can ignore, phase you cannot

If you multiply the whole ket by e^{iY} , every probability stays put:

$$|\langle a | e^{iY} \psi \rangle|^2 = |e^{iY}|^2 |\langle a | \psi \rangle|^2 = |\langle a | \psi \rangle|^2.$$

So $|\psi\rangle$ and $e^{iY}|\psi\rangle$ are the same physical state. People call this a **ray**. Remember the fact: an overall twist is invisible.

A relative twist is not. The two states

$$\frac{|z+\rangle + |z-\rangle}{\sqrt{2}} = |x+\rangle, \quad \frac{|z+\rangle - |z-\rangle}{\sqrt{2}} = |x-\rangle$$

give the same z odds and opposite x answers. Drop the distinction between overall and relative phase, and the second magnet becomes inexplicable.

1.5. From amplitude to count

Let $\{|a_j\rangle\}$ be a complete list of mutually exclusive answers: each pair is orthogonal, and there is nowhere else to go. The **Born rule** says the probability of answer j is

$$P(j) = |\langle a_j | \psi \rangle|^2.$$

Born wrote the probability reading of the wavefunction in 1926. The modern packaging uses a **projector**

$$\Pi_j = |a_j\rangle\langle a_j|.$$

Then

$$P(j) = \langle\psi|\Pi_j|\psi\rangle.$$

These projectors are idempotent, Hermitian, and they add up to the identity:

$$\sum_j \Pi_j = I.$$

That last line is why the probabilities add to one. If you actually see a unique answer j , the ideal leftover state is $|a_j\rangle$, or more generally the normalized projection

$$|\psi_j\rangle = \frac{\Pi_j|\psi\rangle}{\sqrt{\langle\psi|\Pi_j|\psi\rangle}}.$$

If you ignore which answer occurred, a single ket is no longer enough. That is the next chapter's problem.

1.6. Asking a question with a matrix

An **operator** is a linear machine that eats a ket and returns a ket:

$$A(c_1|u\rangle + c_2|v\rangle) = c_1A|u\rangle + c_2A|v\rangle.$$

In a finite list of basis states it is just a matrix. Flip it and conjugate the entries and you have the adjoint $A^\dagger$. If

$$A = A^\dagger,$$

the matrix is **Hermitian**. Ideal measurable quantities are Hermitian. Their possible readings — the **eigenvalues** — are real, and answers that belong to different readings are orthogonal. An **eigenstate** obeys

$$A|a_j\rangle = a_j|a_j\rangle.$$

For our two-state atom,

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Both are Hermitian. Direct multiplication gives

$$\sigma_z|z\pm\rangle = \pm|z\pm\rangle, \quad \sigma_x|x\pm\rangle = \pm|x\pm\rangle.$$

These Pauli matrices are dimensionless. For an ideal spin-1/2, the actual angular-momentum pieces are

$$S_z = \frac{\hbar}{2}\sigma_z, \quad S_x = \frac{\hbar}{2}\sigma_x,$$

so an S_z measurement returns $+\hbar/2$ or $-\hbar/2$, in joule-seconds. Our labels $z+$ and $z-$ are just those two eigenstates.

The two matrices do not commute:

$$[\sigma_z, \sigma_x] \equiv \sigma_z\sigma_x - \sigma_x\sigma_z \neq 0.$$

That is the compact reason they cannot share a full set of answers. The three-magnet sequence was the same fact wearing a lab coat. Drop commutativity and you may imagine a hidden pair of labels, one for z and one for x . Keep it, and that picture is already false.

The **expectation value** of A in $|\psi\rangle$ is

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle = \sum_j a_j P(j).$$

It is the average over many identically prepared runs. It need not be a possible single-shot reading. For $|x+\rangle$, a σ_z measurement returns $+1$ and -1 equally often, so the average is 0. Zero is not on the menu.

If $q = \langle \psi | A | \psi \rangle$ and $A = A^\dagger$, then $q^* = q$. Hermitian averages are real because they have nowhere complex to hide.

1.7. What happens between questions

Between measurements, an isolated system evolves in a way that never stretches or shrinks inner products. That kind of evolution is a **unitary** matrix U :

$$U^\dagger U = U U^\dagger = I.$$

If $|\psi'\rangle = U|\psi\rangle$, then

$$\langle\psi'|\psi'\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle.$$

The generator of that motion is the energy operator, the **Hamiltonian** H . Schrödinger's equation is

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle.$$

Time t is in seconds. $\hbar$ is in joule-seconds. The units match because

$$[\hbar d/dt] = (\text{J s})(1/\text{s}) = \text{J} = [H].$$

If H does not depend on time,

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar}.$$

The exponential is its power series. $Ht/\hbar$ has no units, H is Hermitian, and $U(t)$ is therefore unitary.

A concrete case:

$$H = \frac{\hbar\omega}{2}\sigma_z,$$

with ω an angular frequency in radians per second. Starting from $|x+\rangle$,

$$|\psi(t)\rangle = \frac{e^{-i\omega t/2}|z+\rangle + e^{+i\omega t/2}|z-\rangle}{\sqrt{2}}.$$

The z odds stay fifty-fifty. The relative phase crawls. Ask x and you see it:

$$P(x+; t) = |\langle x+|\psi(t)\rangle|^2 = \cos^2(\omega t/2).$$

Hence evolution can be invisible in one question and obvious in another. Amplitudes keep the phase that a single list of probabilities throws away.

1.8. What the 1922 magnet actually was

The boxes in our diagrams are not free-floating matrices. Stern and Gerlach had a source, a beam of neutral silver atoms, a magnet whose field changed from place to place, two paths, and detectors with finite patience. In a semiclassical cartoon the force is

$$\mathbf{F} \approx \nabla(\mathbf{m} \cdot \mathbf{B}),$$

with magnetic moment in joules per tesla and field in tesla. The gradient supplies inverse metres, so the right-hand side is a newton. The atom's internal answer becomes a dent in its flight path.

The 1922 paper predates the 1925 proposal of electron spin. The tidy classroom story is a later translation. Ground-state silver can be *represented* by a two-dimensional angular-momentum subspace. The real atom, magnet, and hallway occupy a much larger space. Calling the experiment “a spin-1/2 measurement” is an effective description, not a claim that the rest of the atom left the building.

The ideal analyzer also compresses several physical steps into one projection. The magnet first ties internal state to path by ordinary unitary motion of a bigger system. A detector or a blocker then makes a record. Real devices add sloppy preparation, overlapping beams, crooked fields, loss, and noisy counters. Those change the counts. They do not hold a committee meeting about the Born rule.

A two-component ket is a physical degree of freedom in an effective model. It is not yet a computer bit, a protected code, or a strange particle hiding in a crystal.

1.9. Ways to get this wrong

1. Squaring a complex amplitude instead of taking $|c|^2 = c^*c$. Drop the conjugate and the probability can become negative, or worse, complex.
2. Treating the column $(\alpha, \beta)^T$ as a basis-free object. Superposition is always relative to a question.
3. Confusing an overall phase, which does nothing, with a relative phase, which does everything.
4. Reading an expectation value as a single-shot result. Averages are not outcomes.
5. Calling a selected measurement “just more unitary evolution of the atom.” Projection throws alternatives away. The atom-plus-magnet may have been unitary *before* you conditioned on a click.
6. Assuming Hermitian means unitary. One names observables. The other preserves inner products. A Pauli matrix happens to be both. Most operators are neither.

7. Simulating these matrices on a laptop and announcing that the laptop contains a silver atom. A simulation follows a description. It does not become the thing described.
8. Finding two levels and declaring a protected qubit. Two levels are a stage. Preparation, control, readout, and protection are later auditions.

1.10. Check yourself

1. What predicts a pure-state experiment?

A normalized ket, up to an overall phase. Questions and time evolution act on it.

2. How does an amplitude become a probability?

Take $\langle a_j | \psi \rangle$ and square its magnitude: $P(j) = |\langle a_j | \psi \rangle|^2$.

3. Show that $|\alpha|^2 + |\beta|^2 = 1$ is forced by the z question.

Those two numbers *are* the probabilities of the two z answers, and the atom lands somewhere.

4. What fails if A is not Hermitian?

Eigenvalues need not be real, and answers belonging to different readings need not be orthogonal. That is why it cannot represent an ideal measurement.

5. How is unitary motion different from picking an exit?

Unitary motion keeps every inner product. Picking an exit projects and renormalizes.

6. After $z+$, then $x+$, then z again, what are the odds? What would go wrong if the second magnet merely “peeked” at x ?

One half and one half. A peek that left $z+$ intact would send every survivor back up; the data refuse that story.

You now have a state, a way to ask it a question, and a way to let it tick between questions. Next we join two atoms and discover that “the state of the pair” is not always “a state for this one and a state for that one.”

1.11. Sources

- [R037] M. Born, “Zur Quantenmechanik der Stoßvorgänge,” *Zeitschrift für Physik* **37**, 863–867 (1926). DOI: 10.1007/BF01397477.
- [R038] W. Gerlach and O. Stern, “Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld,” *Zeitschrift für Physik* **9**, 349–352 (1922). DOI: 10.1007/BF01326983.

- [R039] B. Friedrich and D. Herschbach, “Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics,” *Physics Today* **56**(12), 53–59 (2003). DOI: 10.1063/1.1650229.
- [R040] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford University Press (1958; reissued 1981). ISBN: 978-0-19-852011-5.
- [R041] E. Schrödinger, “Quantisierung als Eigenwertproblem (Vierte Mitteilung),” *Annalen der Physik* **386**, 109–139 (1926). DOI: 10.1002/andp.19263861802.
- [R004] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition*, Cambridge University Press (2010). DOI: 10.1017/CBO9780511976667.

Chapter 2. Two coins that refuse to factor

Flip two coins. Each lands heads or tails, so the pair has four possible faces: HH, HT, TH, TT. A fair pair gives each face one chance in four. A loaded pair can load those four numbers however it likes, as long as they stay nonnegative and add to one.

Replace the coins with two silver atoms of the kind that split in a magnet. Each atom still has two answers. The pair still has four labels. The surprise is not the labels. The surprise is that the pair can own four complex numbers that refuse to split into “this atom’s two” and “that atom’s two.”

No crystal yet. Two atoms, four labels, and a list that will not factor.

2.1. Four labels for a pair

Call the two atoms A and B . Give each the same two answers as in Chapter 1, written 0 and 1 instead of $z+$ and $z-$. The pair then has four joint answers:

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

The first digit belongs to A . The second belongs to B . So $|01\rangle$ means A answered 0 and B answered 1.

A *state of the pair* is four complex amplitudes, one per label, whose squared magnitudes add to one. Relative phases among those four numbers are real: they change later measurements, just as the minus sign between $z+$ and $z-$ changed the sideways magnet.

How do you write a state of A next to a state of B ? If A is $\alpha|0\rangle + \beta|1\rangle$ and B sits in $|0\rangle$, the pair is

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle = \alpha|00\rangle + \beta|10\rangle.$$

The symbol $\otimes$ is a **tensor product**: write this atom’s state next to that atom’s state, and multiply the amplitudes. We take this as the definition. The same symbol later joins operators, not just kets.

If A has d_A independent answers and B has d_B , the pair has $d_A d_B$ joint labels. Two two-answer atoms make four. A three-answer system next to a four-answer system makes twelve. Dimensions multiply; they do not add.

An operator that acts only on A becomes $M_A \otimes I_B$ on the pair, where M_A is the original operator and I_B leaves B unchanged. Do the thing here. Leave the partner alone.

Drop the product rule and you would count four labels as if they were four extra answers of a single atom. They are not. Each label is a pair of answers.

2.2. A pair you can factor

Some four-amplitude lists are just two two-amplitude lists written side by side. Take

$$|0\rangle_A \otimes |0\rangle_B = |00\rangle.$$

Atom A is definitely 0. Atom B is definitely 0. Nothing about one depends on the other.

Or take both atoms in the sideways state from Chapter 1,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Then

$$|+\rangle_A \otimes |+\rangle_B = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2}.$$

Every joint label has the same amplitude. Measure both atoms and each of the four faces appears one time in four. The two answers are independent: the probability of any pair is the product of the two single-atom probabilities.

A joint state that can be written this way — one ket for A , one ket for B , and the joining symbol between them — is a **product state**. Either two local kets multiply to the pair, or they do not.

Hence “the pair is in a product state” is a factorization claim, not a claim that the answers look uncorrelated in one particular basis. We will need that distinction in a moment.

2.3. A pair you cannot factor

Here is a different four-amplitude list:

$$|\Phi^+\rangle_{AB} = \frac{|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

People call this a Bell state. The two joint labels that appear are orthogonal, so the state is normalized:

$$\langle\Phi^+|\Phi^+\rangle = \frac{1+1}{2} = 1.$$

Measure both atoms in the $\{|0\rangle, |1\rangle\}$ basis. The only outcomes are 00 and 11, each with probability $1/2$. The answers match every time.

Could this be a product? Suppose

$$|\Phi^+\rangle = (a|0\rangle + b|1\rangle)_A \otimes (c|0\rangle + d|1\rangle)_B.$$

Expand the right side. The four amplitudes are ac , ad , bc , and bd , sitting on $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. Matching the Bell pair requires

$$ac = \frac{1}{\sqrt{2}}, \quad ad = 0, \quad bc = 0, \quad bd = \frac{1}{\sqrt{2}}.$$

The first equation makes a and c nonzero, so $ad = 0$ forces $d = 0$. Then bd cannot equal $1/\sqrt{2}$. Hence no such a, b, c, d exist.

A pure joint state that refuses this factorization is **entangled**. That is a statement about the pair, not about a secret message traveling between the atoms. The list of four amplitudes simply does not split.

There is a faster test once you trust a little more algebra. Any pure pair can be rewritten

$$|\Psi\rangle_{AB} = \sum_{k=1}^r s_k |u_k\rangle_A |v_k\rangle_B,$$

with nonnegative numbers s_k that obey $\sum_k s_k^2 = 1$, and with two orthonormal lists of local states. This rewriting is the Schmidt decomposition. The number r of nonzero s_k is the Schmidt rank. It equals 1 exactly when the state is a product. For the Bell pair, $r = 2$ and $s_1 = s_2 = 1/\sqrt{2}$.

Drop the demand that the local lists be orthonormal and you can write many states as sums of products. That does not make them product states. The Schmidt rank is the test.

2.4. When a column is no longer enough

If you know the pair is exactly $|\Phi^+\rangle$, a ket is enough. Often you do not. Maybe someone flipped a fair coin and prepared either $|00\rangle$ or $|11\rangle$. Maybe you can see only atom A. A single column cannot carry “this or that.”

The replacement is a square table of complex numbers. For a ket $|\psi\rangle$, the table is the outer product $|\psi\rangle\langle\psi|$. For the Bell pair,

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

That table is a **density operator**, also called a density matrix once a basis has been chosen. The same object covers a definite ket, a coin-flip over kets, and — in the next section — the leftover state of one atom after you ignore its partner.

Any legal density operator ρ has three jobs:

1. it equals its own adjoint, $\rho^\dagger = \rho$;
2. it never returns a negative number to a bra-ket sandwich, $\langle\chi|\rho|\chi\rangle \geq 0$ for every $|\chi\rangle$;
3. its diagonal, added up, is one: $\text{Tr } \rho = 1$.

If a preparation produces the pure state $|\psi_k\rangle$ with ordinary probability p_k , the table is

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|.$$

Ask a yes-or-no question with projector P_m . The probability of the yes labeled m is

$$\text{Pr}(m) = \text{Tr}(\rho P_m).$$

A density operator that came from a single ket satisfies $\rho^2 = \rho$, or equivalently $\text{Tr}(\rho^2) = 1$. It has rank 1. The Bell pair, looked at as a pair, is like that: $\text{Tr}(\rho_{AB}^2) = 1$.

Hence a density operator is not a more mysterious state. It is the smallest table that still predicts every probability when a ket is no longer enough. Drop Hermiticity, positivity, or unit trace, and those probabilities can become negative, complex, or fail to add to one.

2.5. Throwing one atom away

Suppose only A is on the table. You still want every prediction that a measurement on A alone can make. The right local table is obtained by summing the partner's answers out of the joint table:

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_{b=0}^1 \langle b | \rho_{AB} | b \rangle_B.$$

That sum is a **partial trace**. It is not “destroy B .” It is “write the statistics you would see if you never ask B a question.”

The useful identity, term by term, is

$$\text{Tr}_B(|ab\rangle\langle a'b'|) = \langle b'|b\rangle |a\rangle\langle a'| = \delta_{b'b} |a\rangle\langle a'|,$$

where $\delta_{b'b}$ is 1 when $b' = b$ and 0 otherwise. The Bell pair's cross terms contain $|0\rangle_B\langle 1|$ or $|1\rangle_B\langle 0|$, so they vanish. What remains is

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I_A}{2},$$

where I_A is the identity on A — the two-by-two table of ones on the diagonal and zeros off it.

The pair was pure. Either atom alone is as mixed as a two-answer system can be. Check:

$$\text{Tr}(\rho_A^2) = \text{Tr}\left(\frac{I_A}{4}\right) = \frac{1}{2}.$$

Hence globally pure does not imply locally pure. This mixedness is not “we forgot which local ket we prepared.” There is no local ket.

Operationally, every measurement you can do on A alone is predicted completely by ρ_A . An operator that acts only on A becomes $M_A \otimes I_B$ on the pair. Its average is

$$\langle M_A \rangle = \text{Tr}[\rho_{AB}(M_A \otimes I_B)].$$

The reduced table ρ_A is the unique operator that reproduces every such average:

$$\text{Tr}(\rho_A M_A) = \text{Tr}[\rho_{AB}(M_A \otimes I_B)].$$

For a product operator the partial trace is even simpler:

$$\text{Tr}_B(X_A \otimes Y_B) = X_A \text{Tr}(Y_B).$$

Linearity extends the rule to every joint table.

Drop uniqueness and you could invent many local tables that agree on one measurement and disagree on another. The partial trace is the one that agrees on all of them.

2.6. A phase only the pair can see

Compare the Bell pair with a twisted cousin

$$|\Phi_\theta\rangle = \frac{|00\rangle + e^{i\theta}|11\rangle}{\sqrt{2}},$$

where θ is a relative phase. The joint table contains the cross terms $e^{-i\theta}|00\rangle\langle 11|$ and $e^{i\theta}|11\rangle\langle 00|$. Both vanish under Tr_B , leaving $\rho_A = I/2$ for every θ .

Hence joint measurements can see the phase, and no measurement on A alone can. The information did not leave the pair. It stopped being local.

If you assumed every relative phase of a pair must show up on one of the atoms, this family would be a counterexample.

2.7. Matching answers are not enough

Take this other table:

$$\rho_{cc} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|.$$

Measure both atoms in the 0/1 basis and the answers still match every time, just as they did for $|\Phi^+\rangle$. But this table is an ordinary coin flip between two product states.

A mixed pair is **separable** if it can be written

$$\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1.$$

The superscript (k) labels a possible component, not an exponent. ρ_{cc} is separable on sight. A mixed state with no such writing is entangled.

Hence one pattern of matching bars does not prove entanglement. The Bell pair and ρ_{cc} agree on every 0/1 question and disagree as soon as you ask a sideways question.

Stricter tests exist. Bell inequalities ask whether correlations among several different questions can be explained by shared leftover randomness [R042]; [R043]; [R044]. **[Theory]** Some entangled mixed states do not violate a given Bell inequality, so “entangled” and “Bell-nonlocal” are not the same word [R044].

2.8. How mixed is mixed

The number $\text{Tr}(\rho^2)$ is the **purity**. It is dimensionless. In a system with d answers it runs from $1/d$, for the maximally mixed table I/d , up to 1, for a pure state.

For a mixed state in a finite list of answers,

$$\text{Tr}(\rho^2) < 1.$$

The decomposition of a mixed table into a story about which kets were prepared is not unique. For one two-answer atom,

$$\frac{I}{2} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -|,$$

where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. Hence the table, not a preferred bedtime story about the ensemble, contains the operational predictions.

Drop uniqueness and you may think you have learned which local states were prepared. You have not. You have learned ρ .

2.9. The off-diagonal numbers

Start with one atom in

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad \rho_+ = |+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The matrix is written in the ordered basis $(|0\rangle, |1\rangle)$. The diagonal entries are populations: chances of 0 and of 1. The off-diagonal entries are **coherences**. They carry the relative-phase information that lets two paths interfere.

A noisy partner can steal those two numbers without changing the populations at all. That is the next calculation.

2.10. A knob that only shrinks the off-diagonal

Imagine a noisy process that leaves those populations alone and only shrinks the off-diagonal numbers. Give it a knob λ between 0 and 1: keep this fraction of each coherence. One model is

$$\mathcal{E}_\lambda(\rho) := K_0 \rho K_0^\dagger + K_1 \rho K_1^\dagger,$$

with

$$K_0 = \sqrt{\frac{1+\lambda}{2}} I, \quad K_1 = \sqrt{\frac{1-\lambda}{2}} Z, \quad 0 \leq \lambda \leq 1.$$

Here I is the 2×2 identity and

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the phase-flip operator from Chapter 1. These two pieces satisfy

$$K_0^\dagger K_0 + K_1^\dagger K_1 = I,$$

so probabilities remain normalized. A map built this way — a few pieces K , each sandwiching ρ , adding up, and preserving the trace — is a quantum channel [R045]. Completely positive and trace-preserving are the adult names for “legal evolution when you do not watch everything.” You do not need those names to finish the chapter. You need the habit: sandwich, add, keep the trace.

This particular channel is **dephasing**. It attacks the phase and leaves the populations.

For a general input

$$\rho = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}, \quad a + b = 1,$$

where a and b are real populations and c is a complex coherence, direct multiplication gives

$$\mathcal{E}_\lambda(\rho) = \begin{pmatrix} a & \lambda c \\ \lambda c^* & b \end{pmatrix}.$$

Hence the populations are unchanged and the coherences shrink by λ . Applied to $|+\rangle$,

$$\rho'_+ = \frac{1}{2} \begin{pmatrix} 1 & \lambda \\ \lambda & 1 \end{pmatrix}.$$

Its purity is

$$\text{Tr}[(\rho'_+)^2] = \frac{1 + \lambda^2}{2}.$$

At $\lambda = 1$, the channel does nothing and the state is pure. At $\lambda = 0$, all coherence in this basis is gone and $\rho'_+ = I/2$. At an intermediate value, say $\lambda = 1/2$, the output is

$$\rho'_+ = \begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}, \quad \text{Tr}[(\rho'_+)^2] = \frac{5}{8}.$$

This model is pure dephasing. Energy relaxation is a different channel: it moves population, usually by exchanging energy, instead of only shrinking the off-diagonal. Drop that distinction and a long-lived excited population will impersonate a phase that is still there.

2.11. Where the missing phase went

A microscopic picture explains where the missing coherence can go. Let the environment begin in $|e\rangle$, and suppose joint unitary evolution produces

$$|0\rangle|e\rangle \mapsto |0\rangle|e_0\rangle, \quad |1\rangle|e\rangle \mapsto |1\rangle|e_1\rangle.$$

The initial $|+\rangle|e\rangle$ becomes

$$|\Psi\rangle_{SE} = \frac{|0\rangle|e_0\rangle + |1\rangle|e_1\rangle}{\sqrt{2}},$$

where S labels the system and E the environment. Tracing out E yields

$$\rho_S = \frac{1}{2} \begin{pmatrix} 1 & \langle e_1|e_0\rangle \\ \langle e_0|e_1\rangle & 1 \end{pmatrix}.$$

The leftover coherence is the overlap of the two environment states. If $|e_0\rangle$ and $|e_1\rangle$ are identical, the environment learned nothing and the coherence remains. If they are orthogonal, the environment carries a perfectly distinguishable record and local coherence vanishes.

This process — uncontrolled interaction carrying phase information into degrees of freedom you do not monitor — is **decoherence**. The name is worth keeping because it is a process, not a third kind of state next to pure and mixed. The reduced state often becomes mixed. Calling every mixed state “decohered” would be like calling every wet pavement “rained on.” Sometimes a sprinkler did it.

[Theory] Environment-induced entanglement and partial tracing are the standard open-system account of decoherence [R046]; [R047]. The total system-plus-environment state may remain pure and evolve unitarily throughout.

Hence decoherence is not a mandatory collapse postulate. In this description, system and environment can evolve unitarily while their entanglement suppresses local interference. That account explains reduced-state dynamics. It does not, by itself, settle every interpretation of quantum measurement.

2.12. What the laboratory actually has

An ideal derivation hands us ρ . A laboratory hands us detector clicks, voltages, fluorescence counts, and a preparation recipe. Reconstructing a density operator requires measurements in several bases plus statistical inference, a procedure called quantum-state tomography. The result is an estimate conditioned on calibration and a model of the measurement apparatus.

A preparation described as “the same state repeated many times” is also an experimental assertion. Slow drift can make successive trials differ. Leakage can make the assumed two-dimensional space incomplete. Selection rules and detector inefficiency can bias the apparent ensemble. Density operators remain the right language, but the chosen space and noise model must earn their keep.

[Experiment] Bell tests with separated solid-state electron spins have observed correlations incompatible with a local-realist model while closing major detection and locality loopholes under the experiment’s stated assumptions [R048]. That is much stronger evidence than seeing matching 0/1 outcomes. It still does not mean every correlated density operator is entangled.

Experimental decoherence is inferred from loss of interference or decay of off-diagonal coherence, often summarized by fitted time constants. The single parameter λ above is a channel model, not a universal law of nature. It may depend on elapsed time, pulse sequence, temperature, noise spectrum, and correlations with prior runs. Later chapters will introduce the distinct quantities T_1 , T_2 , and T_2^* rather than stuffing all unpleasantness into one Greek letter.

2.13. Ways to get this wrong

1. Treating “not a product” and “mixed” as synonyms. The Bell state is pure and entangled. The state $I/2$ is mixed but, for one subsystem, has no pair left to be entangled with.
2. Reading matching answers as entanglement. The separable state ρ_{cc} has perfect 0/1 correlation. Entanglement asks whether the full table has a separable writing, not whether one plot has matching bars.
3. Assuming a reduced mixture came from sloppy preparation. One half of a pure entangled state is mixed. Ignoring the partner is enough.
4. Calling every mixed state decohered. Intentionally flipping a fair coin to prepare $|0\rangle$ or $|1\rangle$ produces $I/2$ without any later environmental theft.
5. Treating decoherence as a mandatory collapse postulate. In the open-system description, system and environment can evolve unitarily while their entanglement suppresses local interference [R046]; [R047]. That account explains reduced-state dynamics. It does not, by itself, settle every interpretation of quantum measurement.
6. Confusing dephasing with relaxation. Pure dephasing preserves basis populations while shrinking off-diagonal terms. Relaxation transfers population, usually with energy exchange. Real channels can do both.
7. Treating an ensemble story as unique. Writing $I/2$ as a 50–50 mixture of $|0\rangle$ and $|1\rangle$ does not prove those were the actual preparations. The same operator is a 50–50 mixture of $|+\rangle$ and $|-\rangle$.
8. Simulating a Bell pair on a laptop and announcing that the laptop now contains two entangled atoms. A simulation follows a description. It does not become the thing described.

2.14. Check yourself

1. **What is the dimension of a pair whose parts have dimensions 3 and 4?**
 $3 \times 4 = 12$. Tensor-product dimensions multiply.
2. **Show that $|\Phi^+\rangle$ is not a product state.**
 Matching the four amplitudes would require $ac = 1/\sqrt{2}$, $ad = 0$, $bc = 0$, and $bd = 1/\sqrt{2}$. The first forces $a, c \neq 0$, so $d = 0$, and then bd cannot be $1/\sqrt{2}$.
3. **What fails if you read matching 0/1 answers as entanglement?**
 The separable mixture ρ_{cc} produces the same matching bars. Entanglement is the absence of a separable writing, not the presence of one correlated plot.
4. **Why does tracing out half of $|\Phi^+\rangle$ remove its cross terms?**
 Those terms contain orthogonal partner factors, and the partial trace contributes $\langle 1|0\rangle = 0$.

5. **Show that $\mathcal{E}_{1/2}$ applied to $|+\rangle$ has purity $5/8$.**

The output is $\begin{pmatrix} 1/2 & 1/4 \\ 1/4 & 1/2 \end{pmatrix}$, and $\text{Tr}[(\rho'_+)^2] = (1 + (1/2)^2)/2 = 5/8$.

6. **What fails if you call every mixed state decohered?**

A coin-flip preparation of $|0\rangle$ or $|1\rangle$ already yields $I/2$, with no later environmental record.

You now have a pair, a way to ignore half of it, and a way to watch a phase leak into something you are not watching.

2.15. Sources

- [R042] A. Einstein, B. Podolsky, and N. Rosen, “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?,” *Physical Review* **47**, 777–780 (1935). DOI: 10.1103/PhysRev.47.777.
- [R043] J. S. Bell, “On the Einstein Podolsky Rosen Paradox,” *Physics Physique Fizika* **1**, 195–200 (1964). DOI: 10.1103/PhysicsPhysiqueFizika.1.195.
- [R044] R. F. Werner, “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model,” *Physical Review A* **40**, 4277–4281 (1989). DOI: 10.1103/PhysRevA.40.4277.
- [R045] K. Kraus, “General state changes in quantum theory,” *Annals of Physics* **64**, 311–335 (1971). DOI: 10.1016/0003-4916(71)90108-4.
- [R046] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Reviews of Modern Physics* **75**, 715–775 (2003). DOI: 10.1103/RevModPhys.75.715; arXiv: quant-ph/0105127.
- [R047] M. Schlosshauer, “Decoherence, the measurement problem, and interpretations of quantum mechanics,” *Reviews of Modern Physics* **76**, 1267–1305 (2005). DOI: 10.1103/RevModPhys.76.1267; arXiv: quant-ph/0312059.
- [R048] B. Hensen *et al.*, “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres,” *Nature* **526**, 682–686 (2015). DOI: 10.1038/nature15759; arXiv: 1508.05949.

Part II

Qubits and quantum information

A two-level system is not yet a computer bit. These chapters add the lifecycle — set, flip, read — and the ways a bit forgets.

Chapter 3. A needle you can set, flip, and read

A compass needle on the desk can be tapped north, flicked south, and glanced at. Three verbs. A computer bit is the same three verbs with uglier hardware.

A two-level quantum system is not a tiny compass. It has no secret classical arrow hiding inside. Even so, we will park it, tilt it, and read it — and then check which of those verbs the laboratory actually owns.

Two rungs on an energy ladder do not give you the verbs for free. The rungs may be too close to tell apart. A tilt may knock the system onto a third rung. The detector may click for some other reason.

3.1. Prepare, twist, read

Keep the labels from Chapter 1. A spin-1/2 particle sits in a magnetic field that picks out a z -axis. The spin state with Pauli- Z eigenvalue $+1$ is $|0\rangle$. The state with eigenvalue -1 is $|1\rangle$. Another laboratory may swap the stickers. That is a convention, not physics.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We will do a phase-sensitive experiment in five steps: prepare $|0\rangle$; turn by $+\pi/2$ about the y -axis; let a phase ϕ accumulate, which is a turn about z ; turn by $-\pi/2$ about y ; measure in the $\{|0\rangle, |1\rangle\}$ basis.

The turn about axis j , where j is x , y , or z , is the matrix

$$R_j(\vartheta) = \exp\left(-\frac{i\vartheta\sigma_j}{2}\right).$$

Here $i^2 = -1$, ϑ is an angle in radians, and σ_j is the matching Pauli matrix from Chapter 1. Those matrices $R_j(\vartheta)$ are **rotations**. For the two axes we need,

$$R_y(\vartheta) = \begin{pmatrix} \cos(\vartheta/2) & -\sin(\vartheta/2) \\ \sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix}, \quad R_z(\phi) = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}.$$

The half-angles in the entries are not a typesetting accident. The first pulse gives

$$R_y(\pi/2)|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

The phase turn then gives

$$R_z(\phi)R_y(\pi/2)|0\rangle = \frac{e^{-i\phi/2}|0\rangle + e^{i\phi/2}|1\rangle}{\sqrt{2}}.$$

A Z -basis measurement right here would return each result with probability $1/2$. It would miss the relative phase entirely. The last pulse converts that phase into a population the magnet can see:

$$|\psi_f\rangle := R_y(-\pi/2)R_z(\phi)R_y(\pi/2)|0\rangle = \cos(\phi/2)|0\rangle + i\sin(\phi/2)|1\rangle.$$

The Born rule therefore predicts

$$P(0) = \cos^2(\phi/2), \quad P(1) = \sin^2(\phi/2), \quad P(0) + P(1) = 1.$$

Hence the last pulse is an analyzer: it turns a relative twist into a count. For $\phi = \pi/3$, the probabilities are $P(0) = 3/4$ and $P(1) = 1/4$. In 100 independent repetitions, about 75 zeros and 25 ones are typical. Exactly 75 is not promised. The standard deviation of the zero count is $\sqrt{100 \cdot (3/4) \cdot (1/4)} \approx 4.3$.

This little sequence is already an interferometer. The first pulse creates two amplitudes, the middle step changes their relative phase, and the final pulse recombines them. Measurement reads the resulting interference. Drop the last rotation and the phase is still there; you just cannot see it with a Z magnet.

[Experiment] Resonant control producing coherent oscillations of a single electron spin, and single-shot conversion of a spin state into a detectable charge signal, have both been demonstrated in semiconductor quantum dots [R052]; [R053]. The algebra above is ideal. The cited experiments supply concrete physical control and readout.

3.2. Two leftover angles

Every pure state in this two-dimensional space can still be written

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

with $|\alpha|^2 + |\beta|^2 = 1$. A computational-basis measurement returns 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.

A common overall phase does nothing: $|\psi\rangle$ and $e^{i\chi}|\psi\rangle$ are the same physical pure state for any real angle χ . A *relative* phase between α and β is an experiment, as the five-step run just showed.

Throw away the invisible overall phase and the leftover budget of $|\alpha|^2 + |\beta|^2 = 1$. Two real numbers remain. Call them the polar angle θ , with $0 \leq \theta \leq \pi$, and the azimuthal angle φ , with $0 \leq \varphi < 2\pi$:

$$|\psi(\theta, \varphi)\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle.$$

That is already a point on a sphere. We will now see the point as three ordinary averages.

Write the Pauli operators as

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These are the same matrices as $\sigma_x, \sigma_y, \sigma_z$ from Chapter 1. The identity I leaves every state unchanged. For a state $|\psi\rangle$, the expectation value of an operator A is $\langle A \rangle = \langle \psi | A | \psi \rangle$. Stack the three Pauli averages into a vector

$$\mathbf{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

Substitute the two-angle state and the trig collapses:

$$\mathbf{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta).$$

Its length is one, because $\sin^2 \theta + \cos^2 \theta = 1$. Hence every pure two-level state maps to the surface of a unit sphere. North is $|0\rangle$. South is $|1\rangle$. The $+x$ point is $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. The $+y$ point is $(|0\rangle + i|1\rangle)/\sqrt{2}$.

That sphere is the **Bloch sphere**. A density operator ρ for any two-level state — pure or mixed — can be written

$$\rho = \frac{1}{2} (I + \mathbf{r} \cdot \boldsymbol{\sigma}),$$

where $\boldsymbol{\sigma} = (X, Y, Z)$ and the dot means $r_x X + r_y Y + r_z Z$. Pure states have $|\mathbf{r}| = 1$ and live on the surface. Mixed states have $|\mathbf{r}| < 1$ and sit inside. The maximally mixed state $I/2$ is the center.

Laboratory noise is not, in general, a pure-state arrow that merely wobbles. It is a point that shortens. Drop the length-one condition and you have already left the surface; the leftover object is still a legal two-level state.

For any unit vector $\mathbf{n} = (n_x, n_y, n_z)$, write $\mathbf{n} \cdot \boldsymbol{\sigma} = n_x X + n_y Y + n_z Z$. The Pauli algebra gives $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$. Split the exponential into even and odd powers:

$$e^{-i\vartheta \mathbf{n} \cdot \boldsymbol{\sigma}/2} = I \cos(\vartheta/2) - i(\mathbf{n} \cdot \boldsymbol{\sigma}) \sin(\vartheta/2).$$

This unitary rotates the Bloch vector through angle ϑ about $\mathbf{n}$. The amplitudes contain $\vartheta/2$ because a two-component state is a double cover of ordinary spatial rotations: a 2π turn of the arrow is a minus sign on the ket, and only 4π brings the ket back. The wrinkle is not negotiable.

3.3. A named turn

An intended operation on the two-level system is represented, ideally, by a unitary matrix U , so that $|\psi\rangle \mapsto U|\psi\rangle$. Up to a physically irrelevant overall phase, every such unitary is a rotation of the Bloch vector:

$$U = \exp\left(-\frac{i\vartheta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right).$$

When that intended operation has a name on a circuit diagram, it is a **gate**. The symbol is a target transformation, not a recipe for the pulse that produces it.

The Pauli gates X , Y , and Z correspond, up to global phase, to π rotations about their named axes. The Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

maps $|0\rangle$ to $|+\rangle$ and converts between the Z and X measurement bases.

Hence drawing a box labeled H does not say how the laboratory produced H . Different devices synthesize the same target with microwaves, optical fields, voltages, exchange couplings, frame updates, or combinations of them. The drawn box does not know which.

3.4. A pulse with a stopwatch

A control Hamiltonian connects the target gate to time. Suppose a resonant drive produces

$$\hat{H}_d = \frac{\hbar\Omega}{2} (\cos \delta X + \sin \delta Y).$$

Here $\hat{H}_d$ is energy, $\hbar$ is Planck's reduced constant in joule-seconds, Ω is an angular frequency in radians per second, and δ is the drive phase in radians. Applying the drive for duration t , measured in seconds, gives

$$U(t) = e^{-i\hat{H}_d t/\hbar} = \exp\left[-\frac{i\Omega t}{2}(\cos \delta X + \sin \delta Y)\right].$$

The rotation angle is $\vartheta = \Omega t$. The dimensional check is

$$[\Omega t] = (\text{s}^{-1})(\text{s}) = 1,$$

as an exponent must be dimensionless. Choosing $\delta = 0$ produces an x -rotation. Choosing $\delta = \pi/2$ produces a y -rotation.

That angular speed Ω is the **Rabi frequency**. Fix Ω , pick t , and you have picked the angle.

Hence a gate is an intended map, and a pulse is a timed Hamiltonian. Confuse the two and you will treat $R_x(\pi/2)$ as if it were already a waveform.

[Experiment] Electrically or magnetically driven spin resonance implements this control logic in solid-state spins, although the microscopic coupling and calibration differ by platform [R051]; [R052]; [R054].

3.5. When two rungs become a bit

Preparing a specified start with known reliability is not “turning the apparatus on.” Waiting for relaxation toward a ground state is one method. Optical pumping, reservoir-assisted loading, measurement followed by a conditional pulse, and active reset are others.

An ideal computational-basis measurement has projectors

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|.$$

For a density operator ρ , outcome m occurs with probability

$$P(m) = \text{Tr}(M_m \rho), \quad m \in \{0, 1\}.$$

The trace Tr sums the diagonal elements of a matrix. Real instruments do not print m . They produce photons, currents, voltages, or charge-sensor traces. A classifier then maps that analog record to 0 or 1.

[Experiment] Energy-selective tunneling has been used to map a single electron's spin state to a charge transition detectable by a nearby sensor [R053]. Optically interfaced solid-state defects use spin-dependent optical dynamics to initialize and infer spin states; the relevant mechanisms and limitations vary strongly among centers [R054].

The complete operational loop is therefore:

1. Choose $|0\rangle$ and $|1\rangle$, their axis, and the conditions under which those two states stay isolated.
2. Prepare a known start and quantify whatever residual error remains.
3. Calibrate pulse amplitude, phase, frequency, and duration to implement target gates.
4. Allow the intended single-system, two-system, or sensing energy to act.
5. Rotate the desired measurement question onto the instrument's readable basis.
6. Acquire a classical record and assign an outcome.
7. Reset and repeat, while checking that calibration has not drifted.

One result is a bit. Repeated results estimate quantum probabilities. The system does not print its wavefunction, which would admittedly be convenient.

A pair of levels plus this loop is a **qubit**. The name is worth keeping because it is an operational claim, not a compliment you pay to any doublet in a spectrum. DiVincenzo organized the claim into a widely used checklist: you must be able to tell the states apart, prepare one of them, do a useful set of coherent operations, and read the result — and, for computation, keep the errors low, limit escapes into other levels, recalibrate, and couple more than one of them [R050].

Drop isolation, or drop readout, and you still have two levels. You no longer have a qubit.

3.6. A bit written on several devices

Sometimes the two answers you care about are not two levels of one device. They are two patterns spread across several devices:

$$|0_L\rangle = |\text{pattern A}\rangle, \quad |1_L\rangle = |\text{pattern B}\rangle.$$

The patterns can be entangled states of many components. Quantum error-correcting codes choose the pair of patterns so that selected physical mistakes can be spotted or undone. That condition is a property of the code and the error set, not of the typography L [R055].

That pattern-pair is an **encoded bit**. Several devices can hold one encoded bit. One device is not automatically one encoded bit.

Hence counting hardware is not counting bits. A controllable, resettable, readable doublet is a candidate qubit. A two-dimensional subspace written across several qubits is a candidate encoded bit. Programming ordinary qubits to reproduce the amplitudes of some other model is a calculation on the hardware you already have. It does not change what the hardware *is*.

3.7. What sits around the pair

A spin in a real device is not a tiny isolated Pauli matrix. The device has orbital states, nearby spins, phonons, electromagnetic modes, control wiring or optics, and a detector. The two states we have been calling $|0\rangle$ and $|1\rangle$ are a selected pair inside that larger space.

For an ideal spin-1/2 in a static field, one often writes

$$\hat{H}_0 = -\frac{\hbar\omega_0}{2}Z,$$

where ω_0 is the angular transition frequency in radians per second. A resonant field drives transitions. In an actual solid, spin-orbit coupling, hyperfine interactions, charge motion, strain, and higher-energy levels modify this cartoon. Later chapters will identify these terms for specific defects.

[Proposal] Electron spins confined in quantum dots were proposed as physical qubits with controlled exchange interactions [R051]. **[Experiment]** Subsequent experiments demonstrated ingredients including single-spin readout and coherent single-spin rotations [R052]; [R053]. A demonstration of an ingredient is evidence for that ingredient, not automatic evidence for a fault-tolerant processor.

The same caution applies to crystalline defects. **[Experiment]** Reviews of optically active solid-state spins document initialization, microwave or optical manipulation, and optical interfaces in several material platforms [R054]. Whether a particular defect qualifies as a useful qubit depends on charge state, temperature, magnetic field, collection efficiency, nearby noise, and the exact protocol. “The material contains spin” is the beginning of the device specification, not the end.

3.8. Ways to get this wrong

1. Finding two levels and declaring a bit. A transition can be too weak to drive, too broad to address, or indistinguishable from its neighbors. The system can also leak

- into a third state during control. A two-level approximation is justified only over specified energies, drives, temperatures, and times.
2. Treating the Bloch sphere as ordinary space. For a literal spin in a magnetic field, the axes can be related to physical spin components. For a superconducting circuit, a charge configuration, an orbital doublet, or a cluster encoding, the axes are abstract coordinates in a chosen basis. The drawn arrow is not necessarily pointing through the laboratory ceiling.
 3. Reading one shot as an amplitude. One computational-basis measurement returns 0 or 1. Estimating α , β , or a Bloch vector takes many identically prepared trials in several measurement bases. If preparation drifts between trials, the reconstructed state can be a fiction.
 4. Confusing relative phase with population. The states $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle - |1\rangle)/\sqrt{2}$ both produce 50–50 outcomes in the Z basis. They are nevertheless orthogonal. An analyzer rotation is needed to expose the difference. That was the whole point of the five-step run.
 5. Confusing a gate with its control waveform. The equation $R_x(\pi/2)$ names an ideal transformation. The laboratory implements a finite-duration pulse while unwanted energy terms continue to act. Pulse calibration, leakage, drift, and noise decide how closely the result matches the symbol.
 6. Calling an encoded bit the same thing as one physical doublet. Several qubits can hold one encoded bit without anyone having invented a new kind of particle. A code can catch errors actively without the hardware supplying a quiet energy gap. The nouns do not do that work for you.

3.9. Check yourself

1. **What extra claim turns two levels into a qubit?**
That the selected pair supports a usable loop: prepare, control, read, reset, and stay isolated under stated conditions.
2. **Show that a pure two-level state needs only two Bloch-sphere angles.**
Normalization removes one real parameter from α, β , and an unobservable global phase removes another, leaving θ and φ .
3. **Where is a mixed state on the Bloch sphere?**
Inside it. Its density operator is $\rho = (I + \mathbf{r} \cdot \boldsymbol{\sigma})/2$ with $|\mathbf{r}| < 1$; the center is maximally mixed.
4. **Show that the five-step run yields $P(0) = \cos^2(\phi/2)$.**
The analyzer produces $\cos(\phi/2)|0\rangle + i \sin(\phi/2)|1\rangle$. The Born rule then squares the first amplitude.
5. **What fails if several devices hold two patterns labeled 0_L and 1_L , and you count the devices as bits?**

Those two patterns are one encoded bit. “Encoded” names a chosen two-dimensional subspace, not a head-count of hardware.

6. What fails if you treat one computational-basis shot as revealing α and β ?

The shot returns 0 or 1. The amplitudes are estimated from many repeated, identically prepared runs, usually in more than one basis.

You now have a two-level system you can set, flip, and read — when those verbs actually work. Next we attach times and error rates to every verb.

3.10. Sources

- [R049] F. Bloch, “Nuclear Induction,” *Physical Review* **70**, 460–474 (1946). DOI: 10.1103/PhysRev.70.460.
- [R050] D. P. DiVincenzo, “The Physical Implementation of Quantum Computation,” *Fortschritte der Physik* **48**, 771–783 (2000). arXiv: quant-ph/0002077; DOI: 10.1002/1521-3978(200009)48:9/11<771::AID-PROP771>3.0.CO;2-E.
- [R051] D. Loss and D. P. DiVincenzo, “Quantum Computation with Quantum Dots,” *Physical Review A* **57**, 120–126 (1998). DOI: 10.1103/PhysRevA.57.120.
- [R052] F. H. L. Koppens, C. Buizert, K. J. Tielrooij, I. T. Vink, K. C. Nowack, T. Meunier, L. P. Kouwenhoven, and L. M. K. Vandersypen, “Driven Coherent Oscillations of a Single Electron Spin in a Quantum Dot,” *Nature* **442**, 766–771 (2006). DOI: 10.1038/nature05065.
- [R053] J. M. Elzerman, R. Hanson, L. H. Willems van Beveren, B. Witkamp, L. M. K. Vandersypen, and L. P. Kouwenhoven, “Single-Shot Read-Out of an Individual Electron Spin in a Quantum Dot,” *Nature* **430**, 431–435 (2004). DOI: 10.1038/nature02693.
- [R054] D. D. Awschalom, R. Hanson, J. Wrachtrup, and B. B. Zhou, “Quantum Technologies with Optically Interfaced Solid-State Spins,” *Nature Photonics* **12**, 516–527 (2018). DOI: 10.1038/s41566-018-0232-2.
- [R055] E. Knill and R. Laflamme, “Theory of Quantum Error-Correcting Codes,” *Physical Review A* **55**, 900–911 (1997). DOI: 10.1103/PhysRevA.55.900; arXiv: quant-ph/9604034.

Chapter 4. A clock that forgets

You set a clock and walk away. When you come back, two different things may have gone wrong.

The hands may have stopped. Or the hands may still be moving, but every clock on the table now shows a different time, so you can no longer read the time you set.

A two-level quantum system fails in both ways, and they are not the same failure. The extra energy that marks “excited” can drain into the surroundings. The relative twist between the two answers can smear, even while that energy is still sitting there. Those are different experiments. They produce different numbers. Calling both of them “stability” is how a flattering lifetime impersonates a useful memory.

No crystal yet. No code. Just one two-level thing, a wait, and a question that used to have a clean answer.

4.1. Still up, or already down?

Prepare the upper of the two levels. Call it $|1\rangle$. The lower level is $|0\rangle$. Wait a time t , measured in seconds. Ask: is it still up?

If you repeat the run many times, the fraction that answers “still up” is a population. It is a probability, between 0 and 1. At low enough temperature that the surroundings almost never kick the system back up, that leftover excited population drains, and it drains by the same factor in every equal interval of time.

That is an exponential. Write ρ_{11} for the excited population — the diagonal entry of the density operator from Chapter 2 that says how often you still find $|1\rangle$. The time constant of the fade is T_1 . It belongs to this one experiment: prepare excited, wait, measure population.

$$\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}.$$

At finite temperature the population does not fall all the way to zero. It approaches a thermal leftover. Then T_1 is the time constant of the approach, not the time to emptiness. Bloch-equation treatments of solid-state qubits make this “along the energy axis” reading explicit [R058].

Hence a long T_1 means the energy is still there. It does not mean you can still read a phase you wrote.

4.2. The twist that smears

You can also prepare an equal superposition, the state Chapter 1 called $|x+\rangle$ and Chapter 3 put on the equator of the Bloch sphere:

$$|+x\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

The useful number is no longer the excited population. It is the off-diagonal entry ρ_{01} , a complex number that stores the relative twist between the two answers.

In the simplest model that does not remember its own past, that entry ticks at the transition frequency and fades. Write ω for the angular transition frequency in radians per second. The $1/e$ time of the fade, under this model and this kind of wait, is T_2 . It is the decay constant of an off-diagonal entry, not a second copy of T_1 .

$$\rho_{01}(t) = \rho_{01}(0)e^{-i\omega t}e^{-t/T_2}.$$

People call a no-memory process Markovian. The word is a warning label on the model, not a property of the atom.

Why does T_1 show up inside T_2 ? Population is an amplitude squared. If the excited population fades as e^{-t/T_1} , the excited amplitude fades as $e^{-t/(2T_1)}$. The coherence ρ_{01} carries one factor of that amplitude, so energy relaxation alone already multiplies ρ_{01} by $e^{-t/(2T_1)}$.

Independent phase noise contributes a second factor, e^{-t/T_ϕ} . Here T_ϕ is just the time constant of that phase-only piece. Multiplying the factors gives

$$e^{-t/T_2} = e^{-t/(2T_1)}e^{-t/T_\phi}, \quad \boxed{\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}}.$$

Hence $T_2 \leq 2T_1$ in this model. Equality means no extra phase-only noise. It does not mean perfect pulses, perfect readout, or a good computer.

A long T_1 is not a long T_2 . The energy can sit there while the twist is already gone.

4.3. Every clock a little wrong

Chapter 3 already walked a two-pulse dance with a fixed phase. Do it again, but let the leftover frequency mismatch wander.

In a frame that rotates with the control oscillator, the two levels pick up opposite phases if their energy difference does not quite match that oscillator. Write the leftover mismatch as an angular frequency Δ , in radians per second. The leftover energy is

$$H = \frac{\hbar\Delta}{2}\sigma_z.$$

Here H is energy, $\hbar$ is Planck's reduced constant in joule-seconds, and σ_z has eigenvalue $+1$ on $|0\rangle$ and -1 on $|1\rangle$. A radian is dimensionless, so Δt is a dimensionless phase.

The pulse sequence, introduced in separated-field spectroscopy by Ramsey [R056], is

```
|0> -- pi/2 -- free evolution for t -- -pi/2 -- measure
```

Choose the first pulse so that

$$|0\rangle \longrightarrow |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Free evolution produces

$$|\psi(t)\rangle = \frac{e^{-i\Delta t/2}|0\rangle + e^{+i\Delta t/2}|1\rangle}{\sqrt{2}}.$$

The second pulse converts the relative twist into a population. With this pulse convention, the ideal probability of measuring 0 is

$$P_0^{\text{Ramsey}}(t) = \frac{1 + \cos(\Delta t)}{2}.$$

If Δ is the same on every shot, that fringe never dies. It just oscillates.

Now make the mismatch a little different on each shot. Let the extra offset be a static random variable δ with mean zero and Gaussian standard deviation σ , measured in radians per second. Averaging the phase factor over shots gives

$$\langle e^{i\delta t} \rangle = e^{-\sigma^2 t^2 / 2}.$$

The oscillating piece is now wearing a Gaussian envelope:

$$P_0^{\text{Ramsey}}(t) = \frac{1 + e^{-\sigma^2 t^2 / 2} \cos(\Delta t)}{2}.$$

The time that makes the envelope read $e^{-(t/T)^2}$ is T_2^* . The asterisk is a warning that reversible, shot-to-shot frequency junk is included. It is not an instruction to multiply T_2 by something.

$$T_2^* = \frac{\sqrt{2}}{\sigma}, \quad P_0^{\text{Ramsey}}(t) = \frac{1 + e^{-(t/T_2^*)^2} \cos(\Delta t)}{2}.$$

The dimensional check is immediate: $(\text{rad s}^{-1})^{-1} = \text{s}$, because radians carry no physical dimension. Hence a short T_2^* can sit next to a long T_1 without contradiction: the energy is still there, and the fringe is already washed out. People sometimes call the extra smear inhomogeneous broadening. The experiment is still “prepare, wait, read the fringe.”

4.4. A flip that rewinds the slow ones

A Hahn echo [R057] inserts a π pulse between two equal free-evolution intervals:

```
|0> -- pi/2 -- free tau -- pi -- free tau -- -pi/2 -- measure
```

Let the total free-evolution time be $t = 2\tau$. The π pulse reverses the sign with which a static mismatch accumulates phase. Write $y(s)$ for a sign that equals $+1$ before the pulse and -1 after it. The unwanted phase is

$$\phi_{\text{echo}} = \delta \int_0^{2\tau} y(s) ds = \delta(\tau - \tau) = 0.$$

An ideal echo therefore refocuses any mismatch that stays constant during that shot. It does not reverse energy relaxation, rapidly changing noise, or pulse error. Hahn’s pulse is a rewind button for one particular kind of phase history, not a small time machine.

For a concrete calculation, take $\sigma/(2\pi) = 100$ kHz. Then

$$\sigma = 2\pi \times 10^5 \text{ s}^{-1}, \quad T_2^* = \frac{\sqrt{2}}{2\pi \times 10^5 \text{ s}^{-1}} = 2.25 \text{ } \mu\text{s}.$$

Suppose also that this hypothetical qubit has $T_1 = 1.00$ ms and a no-memory phase-only time $T_\phi = 100 \text{ } \mu\text{s}$. The boxed relation from earlier gives $T_2 = 95.2 \text{ } \mu\text{s}$.

At $t = 10 \text{ } \mu\text{s}$, the quasistatic Ramsey contrast is

$$e^{-(10/2.25)^2} \approx 2.7 \times 10^{-9},$$

while an idealized echo limited only by the quoted exponential T_2 retains

$$e^{-10/95.2} \approx 0.900$$

of its transverse contrast. Meanwhile the excited population retains $e^{-10/1000} \approx 0.990$ of its initial excess. Hence one qubit, one time interval, three different answers. Nothing is inconsistent.

In many devices an echo time is longer than T_2^* , because echo rejects a mismatch that looks constant on one shot. That is not a theorem. A noisy π pulse can make echo worse. A nonexponential envelope may be fitted as

$$C(t) = C_0 \exp[-(t/T)^n],$$

where C_0 is dimensionless contrast, T is the reported time in seconds, and n is a dimensionless fit exponent. A useful coherence-time claim states the sequence, the observable, the envelope, the fit convention, the sample conditions, and the uncertainty.

4.5. The same noise, two sieves

Let $\delta\omega(s)$ be a time-dependent angular-frequency wobble. The accumulated phase is

$$\phi(t) = \int_0^t y(s) \delta\omega(s) ds.$$

Ramsey has $y(s) = 1$. Hahn echo has $y(s) = +1$ for the first half and -1 for the second. A zero-frequency offset survives the Ramsey integral and cancels in echo. Noise that changes appreciably between the two halves does not cancel.

Additional pulses create other sieves. They change the measurement protocol, not the identity of the material. Filter-function theory is the bookkeeping that makes this sequence dependence explicit [R059].

Hence a longer echo time is not evidence that the material became quieter. It is evidence that this particular sieve rejected more of the noise.

4.6. How close is the leftover state?

“The experiment worked” is not a number. We want a score between 0 and 1 that says how much one quantum state still looks like another.

For states ρ and σ , that score is the **fidelity**

$$F(\rho, \sigma) = \left[\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2, \quad 0 \leq F \leq 1.$$

This is the squared Uhlmann–Jozsa convention. Jozsa gives the mixed-state definition and its properties [R060]. Some authors call the unsquared square root “fidelity,” so a numerical report must say which one it is.

If the target is the pure state $|\psi\rangle$, the formula collapses to a probability:

$$F(|\psi\rangle, \rho) = \langle\psi|\rho|\psi\rangle.$$

That is the chance the leftover state still answers the ψ question. Hence a fidelity is always the fidelity *of* something. We have not yet said anything about a pulse, a readout box, or a code.

4.7. How close is the operation?

A gate is not a state. Let U be the target unitary and $\mathcal{E}$ the operation the apparatus actually applies. For a d -dimensional computational space, the **average gate fidelity** is

$$F_{\text{avg}}(\mathcal{E}, U) = \int d\psi \langle\psi|U^\dagger \mathcal{E}(|\psi\rangle\langle\psi|)U|\psi\rangle,$$

where the integral is uniform over pure input states and $d\psi$ is a normalized dimensionless measure. Nielsen derives practical formulas for this average [R061]. The score names both the object (a gate) and the average (every pure input, equally).

The complementary score $r = 1 - F_{\text{avg}}$ is the average gate infidelity. It is a probability-like dimensionless number. It is not a coherence time. It is not a worst-case error bound.

[Experiment] Randomized benchmarking estimates an average error parameter while reducing sensitivity to state-preparation and measurement errors under its modeling assumptions; interleaved benchmarking adds a target gate to estimate that gate’s contribution [R062]. It does not reveal the full noise channel. Coherent, correlated, time-dependent, or leakage errors can require additional diagnostics.

4.8. How often does the box guess right?

Readout ends in a classical decision. For binary single-shot readout, prepare $|0\rangle$ many times and prepare $|1\rangle$ many times, then average the two success rates:

$$F_{\text{RO}} = \frac{P(\hat{0} | 0) + P(\hat{1} | 1)}{2}.$$

Here $P(\hat{j} | j)$ is the conditional probability that the apparatus reports label j after j was prepared. This **balanced assignment fidelity** is one common readout-fidelity definition [R063]. Reporting the two conditional probabilities separately prevents their asymmetry from hiding inside a single average.

Hence the score includes whatever preparation errors the calibration has not independently removed. It is not Uhlmann state fidelity. It is not gate fidelity. It is a grade for a classical guess.

4.9. Population that left the room

A bit flip sends $|0\rangle$ to $|1\rangle$ and stays inside the two levels we agreed to use. Excitation to $|2\rangle$ does not. Those are different mistakes.

Let P project onto the two-level subspace spanned by $|0\rangle$ and $|1\rangle$, and let $Q = I - P$ project onto every other level. For a state ρ , the **leakage** is the population sitting outside that subspace:

$$L(\rho) = \text{Tr}(Q\rho) = 1 - \text{Tr}(P\rho).$$

It is dimensionless and lies between 0 and 1. The name is a fence, not another error inside the room.

A channel can also be assigned leakage and seepage rates: leakage moves population from P to Q , while seepage returns it [R064]. Postselecting away leaked shots may make the fidelity of the retained data look excellent while the unconditional operation is poor. The discarded fraction belongs beside the score.

Drop the fence and a high in-subspace number will impersonate a closed two-level system.

4.10. A message written on several atoms

Sometimes the thing you care about is not one physical two-level system. It is a message written across several of them, then protected by a declared protocol.

A **logical lifetime** is the decay time of a declared logical observable, or of a logical-state survival probability, under a declared noise model, control schedule, syndrome-extraction procedure, and decoder. Experimental surface-code work commonly reports the closely related logical error probability per correction cycle [R065]. The name is worth keeping because it belongs to that whole encoded experiment, not to any one physical clock.

It has units of seconds — or of cycles, if the cycle duration is supplied separately.

For a simple illustration, suppose each correction cycle lasts τ_c seconds and independently causes an absorbing logical failure with probability p_L . The survival probability after N cycles is

$$S(N) = (1 - p_L)^N = \exp\left[-\frac{N\tau_c}{T_L}\right],$$

which defines

$$T_L = -\frac{\tau_c}{\ln(1 - p_L)} \approx \frac{\tau_c}{p_L} \quad \text{when } p_L \ll 1.$$

The approximation is dimensionally sound: seconds divided by a dimensionless probability gives seconds. A logical Pauli expectation under a symmetric Pauli channel follows a different factor, so the observable and the fit must be named. “Logical lifetime” without that metadata is unfinished arithmetic.

[Experiment] Surface-code experiments report logical error per correction cycle and compare code distances. Increasing distance improved the measured logical performance in the conditions of one 2023 experiment, but the per-cycle logical error remained finite [R065]. That is evidence of active error-correction scaling. It is not evidence that the superconducting hardware acquired an emergent topological phase.

A long T_L can arise from an ordinary repetition code, a decoherence-free encoding, active correction, passive energy barriers, or topological order. The mechanism must be stated separately. No single coherence time proves a phase.

4.11. What the laboratory actually has

The ideal equations hide preparation error, measurement error, pulse distortion, drift, several decay rates, and limited sampling. A laboratory therefore has protocols and fitted data, not little tags labeled T_2 attached to atoms.

Claimed quantity	Minimal protocol	Direct observable	What it does not establish
T_1	prepare excited state; vary delay; measure population	population approach to equilibrium	phase coherence, gate accuracy, or fault tolerance
T_2^*	Ramsey/free induction	fringe contrast versus free time	irreversible dephasing alone
echo T_2	$\pi/2 - \tau - \pi - \tau$ sequence	refocused contrast versus 2τ	passive material protection or immunity to fast noise

Claimed quantity	Minimal protocol	Direct observable	What it does not establish
gate fidelity	characterize a specified gate set and metric	tomography, randomized sequences, or another declared estimator	worst-case error, readout fidelity, or zero leakage
readout fidelity	prepare labels; construct assignment matrix	conditional classical outcome frequencies	premeasurement state fidelity
leakage	resolve or infer levels outside the code space	population outside P , preferably with return dynamics	an ordinary bit/phase error rate
logical lifetime	prepare encoded states; run full protocol; decode at varied duration	logical survival or observable decay	topological order or universal fault tolerance by itself

[Experiment] Ramsey and echo data are often analyzed with more than one envelope model, because different spectra produce Gaussian, exponential, stretched-exponential, or oscillatory decay [R058]; [R059]. Choosing the model after seeing the answer can move the quoted time. Residuals, confidence intervals, and the fit window are part of the result, although brochures rarely have room for them.

Comparisons also require matched conditions: temperature, magnetic field, qubit transition, sample, control power, pulse sequence, number of refocusing pulses, and whether the value comes from a single qubit or an ensemble. A many-pulse dynamical-decoupling time should not quietly enter a table column labeled “intrinsic T_2 .” The number may be correct. The noun is doing mischief.

4.12. Ways to get this wrong

1. **Long T_1 , therefore a good memory.** False. T_1 constrains only energy relaxation. Slow frequency drift can make T_2^* short while T_1 is long. In the no-memory relation, phase-only noise can dominate $1/T_2$.
2. **Echo increased the lifetime, therefore the state became passively protected.** False. Echo is applied control. It cancels phase accumulated from sufficiently slow fluctuations. Remove the pulse and the Ramsey decay returns. Dynamical decoupling can be extremely useful without being a material energy barrier or topological protection.
3. **A 99.9% number is a 99.9% number.** False. State fidelity, average gate fidelity, postselected fidelity, readout assignment fidelity, and survival probability answer different questions. Even the same label can use squared or unsquared conventions.

Write the object, the estimator, the subspace, the conditioning, and the averaging distribution.

4. **High in-subspace gate fidelity means negligible leakage.** False. A reported gate metric may condition on remaining in the computational space, or it may have weak sensitivity to rare excursions. Report F_{avg} and leakage separately. Leakage can persist across cycles and create correlated downstream errors [R064].
5. **A logical qubit outlived one physical qubit, therefore fault tolerance was achieved.** Not necessarily. The physical comparison must be task-matched. Fault tolerance concerns controlled scaling of logical error as code resources increase below a threshold, including operations and measurements. A single favorable lifetime can be a milestone without proving scalable suppression. Conversely, an encoded qubit can be useful even when its protection is wholly non-topological.
6. **One fitted time is a property of the material.** Usually incomplete. Noise spectrum and pulse sieve jointly determine coherence decay. Sample preparation, nearby spins, charge motion, temperature, and control sequence matter. Report the protocol with the time.
7. **Physical clocks characterize an encoded message, or a long encoded lifetime characterizes a phase.** No. T_1 , T_2 , T_2^* , physical-gate fidelity, physical read-out fidelity, and physical leakage characterize a physical qubit and its controls. An encoded qubit has logical observables, logical gates, logical leakage, and a logical lifetime. A digital circuit that emulates a code may demonstrate logical behavior without an emergent many-body phase. Passive robustness of an emergent topological phase is an energy-operator property involving a gap, locality, system size, temperature, and perturbations. No single coherence time proves it.

4.13. Check yourself

1. **Show that $T_2 \leq 2T_1$ in the no-memory model.**
From $1/T_2 = 1/(2T_1) + 1/T_\phi$ and $T_\phi > 0$, the extra term is nonnegative, so $T_2 \leq 2T_1$. Equality requires $1/T_\phi = 0$.
2. **What fails if a qubit has $T_1 = 5$ ms and Ramsey $T_2^* = 2$ μ s, and you treat the long T_1 as a good memory?**
Population can relax slowly while quasistatic frequency noise rapidly washes out phase. Neither number establishes fault tolerance.
3. **Ramsey fringes decay in 3 μ s, while Hahn-echo contrast decays in 80 μ s. What distinction is directly supported?**
Much of the Ramsey decay is refocusable on the echo timescale, consistent with slow detuning noise. Echo does not prove passive protection, and the pulse sequence and envelope fits must accompany both numbers.
4. **What fails if a gate's retained shots have fidelity 0.999, but 0.02 of all shots occupy $|2\rangle$, and you report only 0.999?**

The escaped population is leakage $L = 0.02$. The high retained-shot fidelity does not erase it. Report both, plus the conditioning rule.

5. Is readout fidelity the same as the fidelity of the premeasurement quantum state?

No. Balanced readout fidelity averages two classical assignment probabilities. State fidelity compares density operators; preparation and measurement errors require separate calibration.

6. What fails if an encoded state survives for ten times the physical T_2^* and you call that topological order?

The gain could come from echo, a decoherence-free subspace, active correction, post-selection, or another encoding. Demonstrating topological order requires independent many-body evidence; logical lifetime must name its observable and protocol.

You now have several times and several scores, and none of them is allowed to impersonate another. Next we ask how a crystal makes two levels you can attach those times to.

4.14. Sources

- [R056] N. F. Ramsey, “A Molecular Beam Resonance Method with Separated Oscillating Fields,” *Physical Review* **78**, 695–699 (1950). DOI: 10.1103/PhysRev.78.695.
- [R057] E. L. Hahn, “Spin Echoes,” *Physical Review* **80**, 580–594 (1950). DOI: 10.1103/PhysRev.80.580.
- [R058] G. Ithier et al., “Decoherence in a superconducting quantum bit circuit,” *Physical Review B* **72**, 134519 (2005). DOI: 10.1103/PhysRevB.72.134519; arXiv:cond-mat/0508588.
- [R059] Ł. Cywiński, R. M. Lutchyn, C. P. Nave, and S. Das Sarma, “How to enhance dephasing time in superconducting qubits,” *Physical Review B* **77**, 174509 (2008). DOI: 10.1103/PhysRevB.77.174509; arXiv:0712.2225.
- [R060] R. Jozsa, “Fidelity for Mixed Quantum States,” *Journal of Modern Optics* **41**, 2315–2323 (1994). DOI: 10.1080/09500349414552171.
- [R061] M. A. Nielsen, “A simple formula for the average gate fidelity of a quantum dynamical operation,” *Physics Letters A* **303**, 249–252 (2002). DOI: 10.1016/S0375-9601(02)01272-0; arXiv:quant-ph/0205035.
- [R062] E. Magesan et al., “Efficient Measurement of Quantum Gate Error by Interleaved Randomized Benchmarking,” *Physical Review Letters* **109**, 080505 (2012). DOI: 10.1103/PhysRevLett.109.080505; arXiv:1203.4550.
- [R063] D. Keith et al., “Benchmarking high fidelity single-shot readout of semiconductor qubits,” arXiv:1811.03630 (2018). Stable URL: <https://arxiv.org/abs/1811.03630>.

- [R064] C. J. Wood and J. M. Gambetta, “Quantification and characterization of leakage errors,” *Physical Review A* **97**, 032306 (2018). DOI: 10.1103/PhysRevA.97.032306; arXiv:1704.03081.
- [R065] Google Quantum AI and R. Acharya et al., “Suppressing quantum errors by scaling a surface code logical qubit,” *Nature* **614**, 676–681 (2023). DOI: 10.1038/s41586-022-05434-1; arXiv:2207.06431.

Part III

Crystals and defects

A repeating crystal wants electrons to spread out. A missing tile can trap one anyway. Then a zoo of traps that are not interchangeable.

Chapter 5. A crystal that looks the same after a slide

A crystal looks the same after a slide. Start at one atom, step a fixed length to the next identical atom, and the view repeats as far as you can see. A single extra electron can sit on any of those atoms.

Now change one atom, or leave its site empty. Almost nothing has changed — except right there.

That is the puzzle. In the perfect crystal the electron prefers to spread across every site. At the altered site it can get stuck. We will write the smallest arithmetic that makes both facts true.

5.1. The same step, again and again

Pick a starting atom. Step a distance a to the next identical atom. Then take another step. Then another. The allowed landing spots are

$$\dots, -2a, -a, 0, a, 2a, \dots$$

Nothing else is allowed if you want the crystal to look unchanged after the slide. That repeating set of landing spots is a **lattice**. In one dimension it is the set $a\mathbb{Z}$ for a fixed length $a > 0$. We take this as the definition.

In one dimension the step a is the lattice constant, measured in metres. Later crystals will have two or three independent steps. The idea does not change: a lattice is the list of slides that leave the pattern looking identical.

Hence two crystals may share a lattice and still disagree about the glue. The lattice names the legal slides, not the atoms themselves. Drop that distinction and you will treat every crystal with spacing a as the same object.

5.2. A leak to the next site

Give each lattice point one local orbital for the electron, written $|n\rangle$. The label n is an integer: the electron is on site n . The orbitals do not overlap,

$$\langle m|n\rangle = \delta_{mn},$$

where the Kronecker delta is 1 when the two labels match and 0 otherwise. Chapter 1 already taught that move. A bra meets a ket and returns a number.

The electron is not nailed to its site. It can leak onto a neighbor. The energy of that leak is the **hopping** t , with $t > 0$, measured in joules or electronvolts.

Take N sites and identify site N with site 0. There is now no end of the chain to argue about. Every site has two neighbors. The only processes we allow are “leave this site and appear on the next one,” and the reverse.

The energy operator that opens only those neighboring hops is

$$H_0 = -t \sum_{n=0}^{N-1} (|n\rangle\langle n+1| + |n+1\rangle\langle n|),$$

with the indices wrapping around. The minus sign is a convention. It puts the lowest energy on the longest, gentlest wave.

Hence the electron is not an eigenstate of site number: H_0 mixes neighbors. If $t = 0$, every site *is* an eigenstate and nothing propagates. A model organized around local orbitals and nearest-neighbor leaks is often called tight binding. It is an approximation organized around that limit, not a rival law of nature [R067].

5.3. A range of allowed energies

Because every site is equivalent, try a state that advances by the same phase at every step:

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikna} |n\rangle.$$

The number k is a wavevector, measured in inverse metres, so the product ka has no units. It is a phase per step along the lattice. It is not a point on the two-level sphere from earlier chapters.

The chain closes on itself, so the phase after N steps must come back to one:

$$e^{ikNa} = 1.$$

That forces $k = 2\pi m/(Na)$ for integer m . Act with the energy operator. Each hop contributes a neighboring phase factor, and you get

$$H_0|k\rangle = -t(e^{ika} + e^{-ika})|k\rangle = -2t \cos(ka)|k\rangle.$$

So the energy of this wave is

$$E(k) = -2t \cos(ka).$$

As k runs through one complete set of distinct values, $-\pi/a < k \leq \pi/a$, the cosine runs from -1 to 1 . The allowed energies therefore fill the interval $[-2t, 2t]$. That filled interval is one **band**. Its width is $4t$, which has units of energy, as it must.

Hence translation symmetry does not pick a single energy. It picks a continuous range, labelled by k .

The wave is spread out. Each site carries probability

$$|1/\sqrt{N}|^2 = 1/N.$$

The electron is not jogging from site to site in a classical orbit. The stationary wavefunction simply has support on every site. Double the chain and you halve the probability on each site. Drop translation symmetry and this labelling by k is no longer forced.

5.4. One site that is different

Now change the on-site energy of site 0 by an amount U , also measured in energy:

$$H = H_0 + U|0\rangle\langle 0|.$$

That local change is a **defect**. We take this as the definition: a departure from the ideal lattice, here the simplest possible, one number on one site.

For an infinite chain, look for a reflection-symmetric state whose amplitudes are

$$\psi_n = \langle n|\psi\rangle = A\lambda^{|n|}, \quad |\lambda| < 1.$$

The constant A will normalize the state. The inequality $|\lambda| < 1$ is the whole demand that the amplitude fall as you walk away.

Away from the defect the eigenvalue equation only sees the two neighboring hops. For $n \geq 1$,

$$E A \lambda^n = -t(A \lambda^{n-1} + A \lambda^{n+1}),$$

so

$$E = -t(\lambda + \lambda^{-1}).$$

At site 0 the extra energy U is present, and the two neighbors are both $A\lambda$:

$$E = U - 2t\lambda.$$

Set the two expressions for E equal:

$$U = t(\lambda - \lambda^{-1}).$$

Multiply through by λ and you have a quadratic. Of the two roots, exactly one has magnitude below 1. Its magnitude and the corresponding bound-state energy are

$$|\lambda| = \frac{\sqrt{U^2 + 4t^2} - |U|}{2t}, \quad E_d = \text{sgn}(U)\sqrt{U^2 + 4t^2}.$$

The symbol E_d is the defect-state energy. The sign function $\text{sgn}(U)$ is +1 when U is positive and -1 when U is negative. Therefore $|E_d| > 2t$: the defect state sits outside the clean chain's band. For $U < 0$ it sits below the band and the amplitudes keep the same sign. For $U > 0$ it sits above the band and the signs alternate.

Hence a single altered site, on an infinite chain, binds a normalizable state whose energy is forbidden to the traveling waves of H_0 .

The falling envelope can be written $e^{-|n|a/\xi}$. That defines a localization length ξ :

$$\xi = -\frac{a}{\ln |\lambda|} = \frac{a}{\text{arsinh}(|U|/2t)}.$$

Both a and ξ have units of length. The logarithm and the inverse hyperbolic sine eat dimensionless arguments. That check catches the common habit of stuffing an energy into a logarithm without dividing by another energy.

For a concrete choice, let $U = -t$. Then

$$E_d = -\sqrt{5}t, \quad |\lambda| = \frac{\sqrt{5}-1}{2} \approx 0.618, \quad \xi \approx 2.08a.$$

Normalization gives

$$|A|^2 = \frac{1 - |\lambda|^2}{1 + |\lambda|^2} = \frac{1}{\sqrt{5}}.$$

About 0.447 of the probability sits on the defect site in this toy calculation. The rest forms exponentially decreasing tails. The total probability is still exactly 1. **[Theory]** These numbers are consequences of the chosen one-dimensional energy recipe, not predictions for diamond, sapphire, or any other real host.

If $U = 0$, then $|\lambda| = 1$ and nothing decays. If you drop $|\lambda| < 1$ on the infinite chain, the candidate is not normalizable. On a finite ring the tail wraps; the state can still pile up near site 0, but it is not the infinite-chain bound state we just solved.

5.5. Why a forbidden energy cannot travel

A clean-chain energy satisfies $E = -2t \cos(ka)$. For $|E| \leq 2t$, a real k exists, and e^{ikna} has constant magnitude: it propagates. For $|E| > 2t$, no real k works.

Write $k = i\kappa$ below the band, with inverse length $\kappa > 0$. Then

$$e^{ikna} = e^{-\kappa na}.$$

The missing real wavevector becomes an exponential tail. Hence the defect has not erected an infinitely hard wall. It has placed the state where the host offers no traveling channel.

This bound state is localized, but it is not yet an in-gap state. The chain has only one band. The extra level lives outside that band, not in a window between two bands. We will build the window in a moment. Drop the mismatch $|E| > 2t$ and the same exponential ansatz is unnecessary: a real k already exists.

5.6. A plane wave times a repeating pattern

The chain was discrete. A real crystal is a continuous medium whose potential repeats:

$$V(\mathbf{r} + \mathbf{R}) = V(\mathbf{r})$$

for every lattice translation $\mathbf{R}$. Bloch's theorem says an energy eigenfunction can then be written

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}),$$

where $\mathbf{r}$ is position, $\mathbf{k}$ is the crystal wavevector, n labels which band you are in, and the leftover function repeats in every cell: $u_{n\mathbf{k}}(\mathbf{r} + \mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r})$ [R066]. The dot product $\mathbf{k} \cdot \mathbf{r}$ is dimensionless.

That object is a **Bloch** wave. The name is worth keeping because Felix Bloch wrote the crystal-wave theorem in 1929, and because the same last name already labels the two-level sphere from earlier chapters. They are different objects. A Bloch wave is a plane-wave phase pattern multiplied by structure that repeats inside each cell.

The tight-binding wave $|k\rangle$ above is the discrete version. Other useful cartoons start from nearly free electrons instead of from local orbitals. They are approximations organized around different limits, not rival laws of nature.

Hence periodicity of V does not force a plane wave by itself. It forces a plane wave times a cell-periodic leftover. Drop the periodicity of V and Bloch's theorem does not apply.

You can add Bloch waves together to make a packet that sits near one cell. Those packets are sometimes called Wannier functions. Whether you can choose them to fall off exponentially also carries extra information about the band; our elementary chain has no such obstruction [R068]. A defect-bound state is different again. Translation symmetry is broken, and the envelope is pinned near a particular site.

5.7. A gap is a missing interval of energy

Our one-orbital chain has no internal window. It is one continuous band. The smallest lattice that does have a window uses two unequal sites, A and B , in each repeated block. Give them on-site energies $+\Delta$ and $-\Delta$, where Δ is an energy, and connect nearest neighbors with the same hopping t .

In the (A, B) listing the energy operator at wavevector k is the two-by-two matrix

$$H(k) = \begin{pmatrix} \Delta & -t(1 + e^{-ika}) \\ -t(1 + e^{ika}) & -\Delta \end{pmatrix}.$$

Solving the two-by-two eigenvalue problem gives

$$E_{\pm}(k) = \pm \sqrt{\Delta^2 + 4t^2 \cos^2(ka/2)}.$$

The upper and lower bands never meet when $\Delta \neq 0$. Their closest approach is at the edge of the k interval and has size

$$E_g = 2|\Delta|.$$

That empty window is the **gap**. E_g is an energy. It is an interval with no host traveling wave, not a hole in the crystal.

Hence two sites per repeat, with $\Delta \neq 0$, split one band into two and leave a forbidden interval between them. If $\Delta = 0$, the two bands touch and $E_g = 0$.

At zero temperature the lower band is the one electrons fill. People call it the valence band, with top edge E_v . The next available band is the conduction band, with bottom edge E_c . Then $E_g = E_c - E_v > 0$.

5.8. A state that cannot leave the altered site

Change one atom, remove it, or add a local orbital. In the atomic limit $t = 0$, a defect orbital whose on-site energy E_d sits between $-|\Delta|$ and $+|\Delta|$ is perfectly local. Turn the hops back on and the state grows tails.

If the full eigenvalue remains between the two continua, the host still has no real traveling Bloch wave at that energy, so the tails die away. This is the gapped twin of the solved chain. The decaying, normalizable wave that lives in the window is a **localized in-gap state**. Most of the probability sits near a bounded region, and the energy sits where the perfect host cannot carry a traveling wave.

An extended state is the opposite kind of object. It occupies a number of cells that grows with the sample.

Define the isolation margins

$$\delta_v = E_d - E_v, \quad \delta_c = E_c - E_d.$$

Both must be positive for a one-electron level to lie in the gap. A wider E_g permits, but does not require, larger margins. Stronger coupling to the host can broaden or shift the level. A level inside a band becomes a resonance rather than a truly bound state.

Hence a gap is room, not tenancy. A weakly bound level near a band edge may stretch over many cells. A level sitting farther from both edges often hugs the altered site more tightly. People say “shallow” and “deep” for those two habits. Deep means energy within the gap, not physical depth below a surface.

That is the attraction of a wide-gap host: it supplies a large *possible* energy window in which a defect level may sit away from both continua. **[Theory]** A wide gap does not create a suitable defect, guarantee a spin, or make a qubit. It merely makes energetic isolation more plausible. Candidate defects must still be judged by charge-state stability, wavefunctions, lattice relaxation, optical transitions, and coupling to noise [R069]; [R070]; [R071].

5.9. The nuclei also shake

The nuclei do not sit nailed at their equilibrium positions. For small displacements their collective normal modes are harmonic oscillators. A quantum of one such normal mode is a **phonon**. It is a discrete packet of lattice vibration, the sound analogue of a photon.

A phonon mode is labelled by a wavevector $\mathbf{q}$ and a branch ν . It has angular frequency $\omega_{\mathbf{q}\nu}$ in radians per second, and energy

$$E_{\text{phonon}} = \hbar\omega_{\mathbf{q}\nu}\left(n_{\mathbf{q}\nu} + \frac{1}{2}\right),$$

where $n_{\mathbf{q}\nu}$ is a nonnegative integer and $\hbar$ has units of joule-seconds.

Because electronic energies depend on nuclear positions, a defect electron couples to phonons. That coupling can create optical sidebands, shift energies, dephase a superposition, and enable nonradiative jumps by emitting several phonons. **[Theory]** Quantitative nonradiative rates need potential-energy surfaces, vibrational modes, and electron-phonon matrix elements. A band diagram alone does not supply them [R072].

Hence a static lattice was a hypothesis, not a feature of the laboratory. Drop the dependence of electronic energy on nuclear positions and the electron does not couple to phonons. Cooling reduces how many thermal phonons are already present. It does not freeze the nuclei into mathematical points.

5.10. What a real crystal adds

The tight-binding chain contains one electron, one orbital per site, static nuclei, nearest-neighbor hopping, and an adjustable scalar U . A laboratory sample contains many electrons and nuclei, surfaces, strain, impurities, isotopes, electric fields, and finite temperature. The toy model teaches a mechanism, not a materials forecast.

A real calculation usually begins from nuclear species and positions and treats the electronic structure with an approximation such as density-functional theory, often in a periodically repeated supercell. Charged defects need careful electrostatic finite-size corrections. Atomic coordinates must relax. Band-gap errors can put defect levels in the wrong place. Reviews by Van de Walle and Neugebauer and by Freysoldt and collaborators explain why defect formation energies, charge corrections, chemical potentials, and transition levels require more care than reading a dot from a band sketch [R069]; [R070].

There are also several different “energies of a defect,” and swapping them causes trouble:

1. A single-particle orbital energy labels an orbital in an approximate independent-particle calculation.

2. A thermodynamic charge-transition level comes from total-energy differences between relaxed charge states. It tells where changing the electron chemical potential changes the stable charge.
3. A vertical optical transition happens faster than nuclei rearrange, so it connects states at approximately fixed nuclear geometry.
4. A zero-phonon optical transition connects vibrational ground levels. Phonon-assisted transitions make sidebands.
5. A spin excitation can occur within one electronic configuration and may be far lower in energy than an optical transition.

These quantities need not coincide. Kohn–Sham eigenvalues are useful ingredients, but most are not automatically measured quasiparticle addition or removal energies [R069]; [R070]. Correlation can also make a many-electron defect state qualitatively richer than “one electron in one orbital.”

[Experiment] Spectroscopy can reveal absorption or emission lines, phonon sidebands, ionization thresholds, and spin resonances. Those observations constrain a defect model, but assigning a microscopic structure generally combines experiment with electronic-structure calculations. **[Theory]** Systematic searches for solid-state spin defects therefore use several criteria at once: a suitable host, a stable charge state and spin, levels separated from host bands, controllable transitions, and tolerable coupling to the environment [R071].

A wide-gap material helps in three limited ways. First, it offers more energetic room for levels that are far from both band edges. Second, visible or near-infrared transitions between defect states can occur without necessarily exciting carriers across the host gap. Third, thermal exchange with bands can be reduced when the relevant activation energy is large compared with $k_B T$, where k_B is Boltzmann’s constant and T is absolute temperature. None of that is automatic. The actual level positions, optical selection rules, and nonradiative pathways decide the outcome.

5.11. Ways to get this wrong

1. Suppose you draw a line in the gap and call it a localized state. A line on a schematic is a hypothesis. Localization has to come from a wavefunction, an inverse participation measure, spatial charge or spin density, or experimentally supported modelling. A state near an edge may be very extended. A computed state may also be an artifact of a finite supercell.
2. Suppose the host gap is wide and you conclude that every defect electron is trapped. A defect may introduce no gap state, a shallow state next to a band, or a resonance sitting inside a band. It may prefer an unwanted charge state. The gap provides room, not tenancy.

3. Suppose you treat the toy chain as the electronic structure. Our t , U , and one basis orbital omit three-dimensional bonding, multiple orbitals, spin, Coulomb repulsion, screening, spin-orbit coupling, and nuclear relaxation. Fitting a cartoon after seeing a result does not make it predictive. The value of the model is that it isolates the relation among propagation, forbidden energies, and exponential decay.
4. Suppose you find one in-gap orbital and declare a qubit. A qubit needs two controllable states, initialization, coherent gates, readout, and acceptable leakage and noise. The orbital may be empty, doubly occupied, chemically unstable, optically dark, or strongly coupled to phonons. Chapter 3's operational definition still applies.
5. Suppose "localized" is taken to mean isolated from everything. Spatial localization suppresses some overlaps, but the state still couples through electric fields, magnetic fields, strain, phonons, photons, nuclear spins, and exponentially small or dipolar interactions with other defects. Those couplings may be useful or harmful depending on what you are building.
6. Suppose a Bloch wave is a point on the Bloch sphere. A Bloch wave is a crystal wave labelled by $\mathbf{k}$. The Bloch sphere is a picture of a two-level atom. Do not lend them each other's geometry.

A localized defect orbital is a microscopic electronic state. A spin on that defect becomes a physical qubit only after control and readout are demonstrated. It is not an encoded qubit, an emergent quasiparticle, or topological order. Periodic boundary conditions in our calculation are a mathematical convenience, not a claim of protection.

5.12. Check yourself

1. **Show that $H_0|k\rangle = -2t \cos(ka)|k\rangle$.**
Each hop multiplies by a neighboring phase $e^{\pm ika}$. Adding them gives $-t(e^{ika} + e^{-ika}) = -2t \cos(ka)$.
2. **What fails if the defect ansatz drops $|\lambda| < 1$?**
The amplitudes no longer decay, so the candidate is not a normalizable bound state on the infinite chain.
3. **Show that $|E_d| > 2t$ whenever $U \neq 0$.**
 $E_d = \text{sgn}(U)\sqrt{U^2 + 4t^2}$, and $\sqrt{U^2 + 4t^2} > 2t$ as soon as $U \neq 0$.
4. **What fails if $\Delta = 0$ in the two-site cell?**
The two bands touch, $E_g = 0$, and there is no internal window for an in-gap level.
5. **What does a wide host gap guarantee?**
Only a larger possible energy window. It does not guarantee a defect level, localization, a stable charge state, a spin, an optical transition, or a qubit.

6. Why can a band diagram be insufficient even when its level positions are correct?

It omits wavefunctions, occupations, many-electron multiplets, lattice relaxation, transition matrix elements, and environmental couplings.

You now have a lattice, a band of allowed energies, a forbidden window, a state that can sit in that window, and a name for the lattice's vibrations. Next we look at the actual missing sites — substitutions, vacancies, and the mess they drag in.

5.13. Sources

- [R066] F. Bloch, “Über die Quantenmechanik der Elektronen in Kristallgittern,” *Zeitschrift für Physik* **52**, 555–600 (1929). DOI: 10.1007/BF01339455.
- [R067] J. C. Slater and G. F. Koster, “Simplified LCAO Method for the Periodic Potential Problem,” *Physical Review* **94**, 1498–1524 (1954). DOI: 10.1103/PhysRev.94.1498.
- [R068] W. Kohn, “Analytic Properties of Bloch Waves and Wannier Functions,” *Physical Review* **115**, 809–821 (1959). DOI: 10.1103/PhysRev.115.809.
- [R069] C. G. Van de Walle and J. Neugebauer, “First-principles calculations for defects and impurities: Applications to III-nitrides,” *Journal of Applied Physics* **95**, 3851–3879 (2004). DOI: 10.1063/1.1682673.
- [R070] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle, “First-principles calculations for point defects in solids,” *Reviews of Modern Physics* **86**, 253–305 (2014). DOI: 10.1103/RevModPhys.86.253.
- [R071] J. R. Weber, W. F. Koehl, J. B. Varley, A. Janotti, B. B. Buckley, C. G. Van de Walle, and D. D. Awschalom, “Quantum computing with defects,” *Proceedings of the National Academy of Sciences* **107**, 8513–8518 (2010). DOI: 10.1073/pnas.1003052107.
- [R072] A. Alkauskas, Q. Yan, and C. G. Van de Walle, “First-principles theory of non-radiative carrier capture via multiphonon emission,” *Physical Review B* **90**, 075202 (2014). DOI: 10.1103/PhysRevB.90.075202.

Chapter 6. An empty site is not a leftover spin

Replace one atom in a perfect crystal by a different element. Leave a neighboring site empty. Those two accidents can sit next to each other and become one object. An extra atom wedged off the regular sites is a different object. A whole line of the pattern that has slipped is ruder still.

None of those is automatically a leftover spin. None is automatically a qubit. We will sort the damage by geometry, then sit with one famous pair — a nitrogen atom next to a missing carbon — until charge, spin, strain, and nearby nuclei have names of their own.

6.1. Six kinds of damage

A perfect crystal repeats a block of atoms. Anything that breaks that repetition is a **structural defect**. The word names the geometry. It does not name a color, a spin, or a computer bit.

Sort the damage first by how much space it occupies.

Kind	What happened	What the electrons often notice	What not to assume
Wrong atom on a lattice site	A host atom is replaced by a different element	A different valence, size, electronegativity, and local potential	That it leaves an unpaired spin. It may just donate or accept charge.
Empty site	A lattice site has no atom	Neighboring leftover bonds may sit in the gap and the neighbors may lean in	That reconstruction has not paired every electron. Charge matters.
Extra atom stuffed in	An atom sits off the regular sites	Strong local squeeze and new bonds; the extra atom may wander	That there is only one geometry or one charge.
Two accidents bound together	Nearby defects lock into one structural unit	New symmetry and molecular-like combinations of the leftover bonds	That the pair is the sum of the two isolated level diagrams.

Kind	What happened	What the electrons often notice	What not to assume
A slipped line	A one-dimensional line defect, marked by a Burgers vector — how far the lattice slipped	A messy core plus a long-range elastic field; sometimes a band of core states	That the whole line is one localized emitter.
A mis-stacked plane	The usual stacking of planes is interrupted	A local change of stacking, a shift of the bands, and an extended squeeze	That a sheet of electronic states is a point-like two-level system.

A substitution occupies a normal lattice site in place of a host atom. A vacancy is an empty site. An interstitial occupies an extra non-lattice site. A bound pair is two of those close enough that you have to solve them together. Slipped lines and mis-stacked planes are ruder: they alter a whole line or a whole plane.

Hence “how damaged is the crystal?” is the wrong first question. The useful question is what energies and shapes the combined host-plus-damage system actually has. A pointlike accident can make a localized orbital in a wide gap. It can also make a shallow state that melts into a band, a closed shell with total spin zero, or a trap that dumps energy without shining. Extended damage can trap carriers, squeeze nearby point centers, or just add noise. The table tells you where to begin, not where the calculation ends [R070]; [R073].

A frozen squeeze of the lattice is not a seventh species of defect. Neither is a vibration. Those two will come back as characters, not as extra rows.

6.2. An empty site is not a leftover spin

Take the empty site. Three or four neighboring atoms now have a bond that used to point at the missing atom and now points at nothing. It is tempting to say “there is a dangling electron, so there is a spin.”

The neighbors often lean in and remake the bonds. Electrons can also leave or arrive until the whole object is cheaper. What remains may be a closed shell with every spin paired.

The integer that decides this is the **charge state**: how many electrons the damaged region has, relative to the neutral atom count. The name is worth keeping because the same empty site with one extra electron and the same empty site with one electron missing are different objects. They can have different shapes, different colors, and different total spins.

Hence geometry does not fix spin. Before anyone draws a little arrow on the vacancy, ask how many electrons are actually there. Drop the charge label and you have mixed several distinct objects under one drawing.

6.3. Nitrogen next to a missing carbon

We need one object we can finish on paper. Replace one carbon atom in diamond by nitrogen and remove a neighboring carbon. That pair is famous because its leftover symmetry is visible, not because every vacancy-plus-impurity pair behaves this way.

The line from the nitrogen through the empty site picks a preferred axis, usually one of diamond's [111] directions. In the ideal unsqueezed structure the three leftover carbon neighbors are equivalent. You can rotate by 120° around the axis, or reflect through any of three mirrors, and the picture looks the same.

The list of those leftover rotations and reflections is the local **point group**. For this pair it is called C_{3v} [R074]; [R075].

Hence every later coupling that hopes to be allowed has to respect the same leftover symmetries, and every leftover symmetry we break will show up as a split line. Leftover symmetry does not compute the energy scale. If the three carbons stop being equivalent, C_{3v} is already gone, and degeneracies that the point group had protected need not survive.

6.4. Combinations the three carbons allow

Let $|c_1\rangle$, $|c_2\rangle$, and $|c_3\rangle$ be the three carbon leftover-bond orbitals that face the empty site. The combinations that know about the threefold axis are

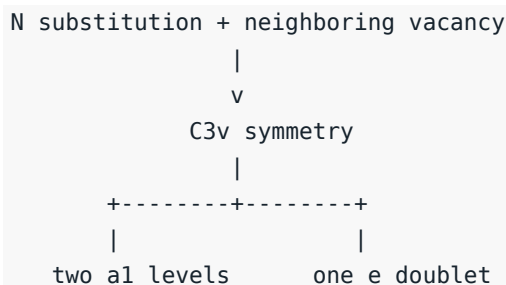
$$|a_C\rangle = \frac{|c_1\rangle + |c_2\rangle + |c_3\rangle}{\sqrt{3}},$$

$$|e_x\rangle = \frac{2|c_1\rangle - |c_2\rangle - |c_3\rangle}{\sqrt{6}}, \quad |e_y\rangle = \frac{|c_2\rangle - |c_3\rangle}{\sqrt{2}}.$$

The first combination looks the same after a 120° turn. Call it a . The other two mix into each other under that turn, so they must share an energy as long as the three carbons stay equivalent. Call that pair e .

The nitrogen leftover orbital also looks like an a . It mixes with $|a_C\rangle$ and produces two a_1 combinations at different energies. The carbon e pair has no nitrogen partner of matching shape in this small basis, so it stays doubly degenerate while ideal C_{3v} holds.

That is the first bridge from a drawing to a level diagram:



Hence leftover symmetry predicts which degeneracies are allowed and which mixings are forbidden. Real orbitals leak past four leftover bonds, and their energies need an electronic-structure calculation. Symmetry does not supply the energy scale [R070]; [R075]. Drop the threefold axis and the e pair is no longer required to share an energy.

6.5. Count the leftover electrons

Geometry is not occupancy. The negatively charged version of this pair, written NV^- , is described by six electrons in the configuration

$$a_1(1)^2 a_1(2)^2 e^2.$$

The two filled a_1 orbitals are closed shells. Two electrons remain in the e pair. Coulomb repulsion and exchange prefer those two to line up, giving a spin-triplet ground set with total electron spin $S = 1$, conventionally labeled 3A_2 .

That conclusion is not hiding in the drawing of an empty lattice site. It follows from charge state, orbital order, and the interacting many-electron problem [R074]; [R075]. **[Theory]**

The set of many-electron states that share that spin and that leftover symmetry is a **multiplet**. It names a whole family of levels, not one leftover bond and not one chosen pair of levels.

Hence spin is not “the vacancy.” Spin is what the leftover electrons do once charge and orbital order are fixed. Change the charge, or invert two orbital energies, and a different multiplet can sit on the same atoms.

6.6. A splitting with the magnet off

For the ground triplet, a useful energy rule written in frequency units is

$$\frac{H_{\text{gs}}}{h} = D \left[S_z^2 - \frac{S(S+1)}{3} \right] + E (S_x^2 - S_y^2) + \frac{\mu_B}{h} \mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S}.$$

Here H_{gs} is energy and h is Planck's constant, so each term on the right is a frequency. S_x , S_y , and S_z are dimensionless spin-1 operators. D and E are zero-field-splitting frequencies in hertz. μ_B is the Bohr magneton in joules per tesla, $\mathbf{B}$ is magnetic field in tesla, and $\mathbf{g}$ is the dimensionless electron g -tensor.

The first term separates $m_s = 0$ from $m_s = \pm 1$ even when no magnet is on. That gap is the **zero-field splitting**. It is a spin splitting, not the orbital splitting we will meet when neighbors stop being a sphere.

The second term appears when a sideways squeeze, or any other leftover-symmetry breaker, distinguishes x from y . The third term is the magnet talking to the electron spin.

For NV^- near room temperature, the axial splitting is about $D = 2.87$ GHz [R074]; [R076]. **[Experiment]** The units check: multiplying $2.87 \times 10^9 \text{ s}^{-1}$ by h gives an energy. The ideal C_{3v} model has $E = 0$. A real center commonly has a small sideways term.

Hence the magnet-off spectrum of this triplet is already split, and the size of that split is a measured frequency, not a drawing. Writing the rule does not mean “and therefore this is a qubit.” If $D = 0$ and $E = 0$, the three spin projections meet at one energy until a magnet pulls them apart.

6.7. Choosing two levels is still not a device

A magnetic field along the defect axis can pull $m_s = +1$ away from $m_s = -1$. You may then choose, for example, $|0\rangle = |m_s = 0\rangle$ and $|1\rangle = |m_s = -1\rangle$ as a two-level working space and drive it with a resonant microwave magnetic field.

NV^- also has a sharp optical line between the triplet ground and excited families near 637 nm, or about 1.945 eV [R074]. **[Experiment]** A quick check uses $hc \approx 1240 \text{ eV nm}$:

$$\frac{hc}{\lambda} \approx \frac{1240 \text{ eV nm}}{637 \text{ nm}} \approx 1.95 \text{ eV}.$$

Spin-dependent quiet pathways through intermediate singlet states let light pump the spin into a preferred level and make the fluorescence depend on which level you are on. Those pathways, plus charge-state preparation, microwave control, and photon collection, are what turn *part* of the spectrum into a laboratory qubit [R074]. **[Experiment]**

Name the layers so they cannot swap clothes. “Nitrogen next to a vacancy” is a structural pair. “NV[−]” adds a charge state. “³A₂” is a many-electron multiplet. “The $m_s = 0, -1$ pair” is a chosen two-level subspace. “A qubit” means that subspace can be initialized, steered, read out, and kept isolated enough to finish the job. Those are related sentences. They are not synonyms.

Hence two levels in a spectrum are a stage. Drop initialization, control, readout, or charge stability, and the stage is still not a device.

6.8. How expensive is the damage?

Suppose you want to know whether the crystal would rather host this accident. Compare the energy of a box of atoms that includes the damage with the energy of a perfect box. Then pay or refund the atoms you added or removed, and pay or refund the electrons. The leftover is the **formation energy**.

For defect D in charge state q ,

$$E_f(D^q) = E_{\text{tot}}(D^q) - E_{\text{tot}}(\text{bulk}) - \sum_i n_i \mu_i + q(E_F + E_{\text{VBM}}) + E_{\text{corr}}.$$

Every term is an energy, usually in electronvolts. $E_{\text{tot}}(D^q)$ and $E_{\text{tot}}(\text{bulk})$ are total energies of the damaged and perfect simulation cells. n_i counts atoms of species i added to the cell, and μ_i is that species’ chemical potential. E_F is the electron Fermi energy measured upward from the valence-band maximum; E_{VBM} places that maximum in the total-energy reference. E_{corr} patches artifacts of a finite cell. Sign conventions for n_i vary, so a paper’s convention has to be checked rather than admired from across the room [R070].

Hence formation energy predicts thermodynamic favorability under stated chemical and electronic conditions. It does not predict implantation yield, kinetic trapping, optical brightness, or coherence. Charge-transition levels mark where different q values swap which one is cheaper. A compensating impurity, a surface, a gate, or a lamp can therefore take the same atomic geometry from spin-active to spinless.

Same geometry. Different occupancy. Different story. If you omit the $q(E_F + E_{\text{VBM}})$ term, you have compared two charge states as if the surrounding electron reservoir were free.

6.9. Neighbors that are not a sphere

A free atom does not know which way is up. Neighbors pick directions. The five d -shaped clouds that shared one energy on an isolated atom split into a pair and a trio once the neigh-

bors sit at the corners of a tetrahedron or an octahedron. Which set lies lower depends on the geometry and the bonding.

That non-spherical electrostatic and covalent environment is the **crystal field**. Write H_{CF} for its contribution. Its eigenstates are orbital combinations labeled by leftover-symmetry names such as A , E , or T . A further low-symmetry dent can split an E or T family again.

Hence crystal-field splitting is an *orbital* energy splitting. It is not the spin-triplet zero-field splitting D , even though both can be present with the magnet off. One rearranges clouds. The other rearranges spin levels of an already chosen multiplet. If the neighbors were a perfect sphere, the five d clouds would still share one energy.

6.10. When orbit tugs on spin

An electron cloud that is circulating is a tiny magnet, and the electron's own spin is another tiny magnet. They notice each other. In an isolated-family shorthand the coupling is

$$H_{\text{SO}} = \lambda \mathbf{L} \cdot \mathbf{S},$$

where $\mathbf{L}$ and $\mathbf{S}$ are dimensionless orbital and spin angular-momentum operators and λ is an energy. A real defect often needs a tensor, or a version written in the leftover-symmetry language.

This is the **spin-orbit** term. It is a coupling, not a quality badge.

If the crystal field leaves an orbital degeneracy, spin-orbit can act at first order and strongly organize the levels. If the orbital ground state is a single nondegenerate cloud, orbital angular momentum is said to be quenched. Spin-orbit still sneaks in by virtually mixing excited orbitals. It can make the g -factors direction-dependent, feed the zero-field splitting, and write spin-selective optical rules [R073]; [R075]. [**Theory**]

Hence the same term can help you address the spin with light and can make the spin a better antenna for lattice motion. Useful and noisy are allowed to be the same operator. If $\lambda = 0$, those first-order and virtual mixings are gone, and you have lost both the handle and the antenna.

6.11. Two unpaired electrons talking

Two or more unpaired electrons also interact as magnets. Projected into a total-spin multiplet, their dipolar and spin-orbit-mediated effects are often summarized by

$$H_{\text{SS}} = \mathbf{S} \cdot \mathbf{D} \cdot \mathbf{S},$$

where $\mathbf{D}$ is a zero-field-splitting tensor with energy units. In axial symmetry this reduces, up to an additive constant, to $\hbar D[S_z^2 - S(S+1)/3]$. That is the same D we met on NV^- .

Hence the magnet-off split of an integer-spin multiplet can come from the electrons talking to each other, not from an applied field. A half-integer spin has a leftover doublet at zero magnetic field when time-reversal is intact. An integer spin is not guaranteed such a pair. Neither sentence is a promise of long memory. A guaranteed doublet can still jiggle.

6.12. A magnet, and then the nuclei

An applied magnetic field adds the electron Zeeman term

$$H_Z = \mu_B \mathbf{B} \cdot \mathbf{g} \cdot \mathbf{S}.$$

The leftover electron is still not alone. Some nearby nuclei have their own tiny magnets. They tug on the electron, and the electron tugs back.

If nucleus k has nuclear spin $\mathbf{I}_k$, the coupling is

$$H_{\text{hf}} = \sum_k \mathbf{S} \cdot \mathbf{A}_k \cdot \mathbf{I}_k,$$

where $\mathbf{A}_k$ is the **hyperfine** tensor for that nucleus, in energy units. It names a specific conversation: electron spin talking to nuclear spin. The tensor has an approximately isotropic contact piece, sensitive to spin density at the nucleus, and an anisotropic dipolar piece. Nuclei with spin quantum number $I_k \geq 1$ may also carry a quadrupole term $\mathbf{I}_k \cdot \mathbf{Q}_k \cdot \mathbf{I}_k$, with $\mathbf{Q}_k$ an energy tensor.

Hence the third character is the nucleus. Hyperfine coupling can resolve which nuclear configuration you have, store a state, or provide a conditional gate. Uncontrolled host nuclei instead kick the electron transition around and chew up phase. The same term is a resource when the nucleus is known and controlled, and noise when it is neither [R073]. If every nearby nucleus has $I = 0$, this conversation is silent.

6.13. A squeeze that sits still

Let $\mathbf{u}(\mathbf{r})$ be the lattice displacement at position $\mathbf{r}$, measured in metres. For small distortions the dimensionless strain tensor is

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial r_j} + \frac{\partial u_j}{\partial r_i} \right).$$

That frozen dent is the fourth character, **strain**. It is a deformation field, not a new defect species. A defect can make static strain, feel strain you apply from outside, and — in the next section — exchange energy with vibrations.

A projected defect family responds through leftover-symmetry-allowed operators O_α :

$$H_\varepsilon = \sum_\alpha g_\alpha \varepsilon_\alpha O_\alpha.$$

Here ε_α are symmetry-adapted combinations of strain components, O_α are dimensionless orbital or spin operators, and the susceptibilities g_α have energy units. Dimensionless strain times energy is energy, as it must be. Axial strain may shift a transition without lifting a doublet. Transverse strain may lower leftover symmetry, split orbitals, or mix spin projections. The allowed pattern follows from the point group. The coefficients need experiment or a microscopic calculation [R077]. **[Theory]**

Hence a static dent is already in the energy rule before anyone mentions a vibration. If $\varepsilon = 0$, this term is absent, and leftover degeneracies that strain would have split remain.

6.14. A squeeze that wiggles

Phonons, from Chapter 5, make strain time dependent. For normal mode k , write angular frequency ω_k in radians per second and let $b_k^\dagger$ and b_k create and destroy one quantum of that vibration. Then

$$H_{\text{ph}} = \sum_k \hbar \omega_k \left(b_k^\dagger b_k + \frac{1}{2} \right), \quad H_{\text{def-ph}} = \sum_{k,\alpha} \kappa_{k\alpha} (b_k + b_k^\dagger) O_\alpha,$$

where $\kappa_{k\alpha}$ is a coupling energy. A phonon can drive a transition when its energy and leftover symmetry fit. One-phonon emission or absorption, two-phonon Raman processes, and thermally activated hops through higher levels can relax the spin or scramble its phase. Which process wins depends on the defect spectrum, the phonon density of states, and the temperature [R078]. **[Theory]**

Cooling reduces how many thermal phonons are already in the lattice. It does not erase static strain, and it does not cancel every spontaneous emission.

Hence strain sits still and phonons wiggle. They use related leftover-symmetry channels and they are not the same object. Drop the operators b_k and you have kept a frozen dent while throwing away every dynamical jump the lattice can cause.

6.15. The whole energy rule

Add the pieces that this leftover family can feel:

$$H_{\text{system}} = H_{\text{CF}} + H_{\text{SO}} + H_{\text{SS}} + H_Z + H_{\text{hf}} + H_Q + H_\epsilon + H_{\text{ph}} + H_{\text{def-ph}}.$$

That sum is the working **Hamiltonian**. It names the energy rule, not a certificate that the defect is a device.

Not every term survives in every family. Leftover symmetry may forbid a matrix element. Projection may squash an orbital operator to zero. An isotope may have $I = 0$. Conversely, a term left off a neat level diagram may own the laboratory linewidth.

Hence the recipe is now visible. Start with geometry and charge. Solve the localized orbital problem. Build the many-electron states. Only then project into a spin or orbital family and keep the terms that family can feel. If you start from a spin Hamiltonian and work backwards, you have assumed the projection that the earlier steps were supposed to earn.

6.16. Why leftover symmetry can force a zero

Suppose states $|i\rangle$ and $|f\rangle$ transform under leftover-symmetry patterns Γ_i and Γ_f , while a perturbation O transforms as Γ_O . The matrix element $\langle f|O|i\rangle$ can be nonzero only if the product $\Gamma_f^* \otimes \Gamma_O \otimes \Gamma_i$ contains the totally symmetric pattern — the one that looks the same under every leftover rotation and reflection.

That selection rule can kill a transition exactly in the ideal symmetry. If the product allows the transition, leftover symmetry says only “possibly nonzero.” It does not supply the wavefunction overlap, the coupling strength, or the lifetime. Strain can mix patterns and turn a forbidden matrix element into an allowed one.

Hence symmetry predicts zeros. It does not predict numbers. Drop the leftover symmetry — by strain, by a nearby defect, by a surface — and a former zero is no longer guaranteed.

6.17. What the laboratory actually has

A fabricated sample contains a distribution, not a single perfect defect.

[Experiment] Electron-paramagnetic resonance sees spin splittings and hyperfine structure. Photoluminescence locates optical transitions. Optically detected magnetic resonance ties fluorescence to spin resonance. Stress and electric-field spectroscopy test leftover-symmetry assignments. Agreement among these measurements and many-electron calculations supports a level model more strongly than any single colored line in a spectrum [R073]; [R074]; [R075].

The local environment varies from site to site. Isotopes change hyperfine coupling. Residual impurities shift charge stability. Surfaces and interfaces bend the bands. Implantation produces vacancies, extra atoms, and extended damage on top of the intended center. Annealing moves and recombines defects, sometimes building the pair you wanted and sometimes building an excellent quiet recombination center instead. Slipped lines and mis-stacked planes add a spatially varying squeeze and may bring their own electronic states.

Ideal leftover symmetry is a reference point. Small perturbations split degenerate lines and rotate eigenstates. That is not always bad: strain can tune two emitters into resonance, and hyperfine coupling can provide a nuclear memory. Reproducible engineering still requires measuring the perturbation and leaving it in the energy rule.

The NV example is unusually well characterized. It is not a generic promise about vacancy complexes. Reviews of candidate spin defects emphasize a pile of simultaneous requirements: a stable charge state, suitable spin and optical transitions, controllable initialization and readout, long-enough coherence, and compatibility with fabrication and photonics [R073]. **[Theory/Experiment synthesis]** A material may satisfy three and fail the fourth with admirable consistency.

6.18. Ways to get this wrong

1. Suppose a vacancy leaves a dangling electron, and therefore a spin. Reconstruction and charge transfer may close the shell. Determine the charge state and the many-electron occupancy.
2. Suppose an in-gap level is a qubit. One level is not two controllable states. Even two levels need initialization, coherent control, readout, and tolerable leakage and noise.
3. Suppose a color center is a spin center. A color center absorbs or emits visible light. It can have total spin zero. Conversely, a spin-active center may have no useful optical cycle.

4. Suppose crystal-field splitting and zero-field splitting are the same. Crystal fields primarily split orbitals. Zero-field splitting separates spin sublevels and can come from electron spin–spin interaction and from spin–orbit-mediated effects.
5. Suppose static strain is a phonon. Static strain is a time-independent deformation in the model you wrote. Phonons are quantized dynamical modes. They couple through related leftover-symmetry channels and they leave different experimental fingerprints.
6. Suppose a slipped line is a large point defect. Its line core and long-range elastic field make it extended. Likewise, a mis-stacked plane is planar. Their electronic states need not be spatially isolated two-level systems.
7. Suppose density-functional orbitals are the multiplet spectrum. A one-electron orbital diagram is useful input, but exchange, correlation, excited-state relaxation, and spin–orbit coupling determine many-electron states. Quantitative optical and spin assignments need an appropriate many-body treatment and experimental checks [R070]; [R075].
8. Suppose leftover point-group symmetry is the same protection as a many-body topological phase. Local leftover symmetry can forbid a matrix element until the symmetry is broken. A many-body topological phase is a different object, with emergent excitations and global sectors. They solve different problems. A vacancy, slipped line, or mis-stacked plane may be called a topological defect in the classical theory of ordered media because it carries a winding or a Burgers vector. That vocabulary does not hand you quantum topological order, anyons, or a protected logical qubit. Arranging many structural defects on a lattice supplies microscopic degrees of freedom. Only a demonstrated many-body Hamiltonian and a phase analysis could establish an emergent topological phase.

6.19. Check yourself

1. **Show that $|a_C\rangle$ is unchanged by a cyclic permutation of the three carbon orbitals, while $|e_x\rangle$ and $|e_y\rangle$ mix.**
Cycling $c_1 \rightarrow c_2 \rightarrow c_3 \rightarrow c_1$ leaves the equal-weight sum fixed and rotates the two e combinations into linear combinations of each other.
2. **What fails if you assume an empty site leaves an unpaired spin?**
Reconstruction and charge transfer can close the shell. Total spin follows occupancy, not geometry.
3. **Show that a line at 637 nm is about 1.95 eV.**
 $hc/\lambda \approx 1240/637 \approx 1.95$ eV.

4. **What did C_{3v} leftover symmetry buy us for NV^- ?**

It organized the leftover-bond combinations into a_1 singlets and an e doublet, constrained allowed mixing, and identified how a sideways squeeze can lift degeneracy. It did not supply numerical energies.

5. **What fails if static strain is treated as a phonon?**

Static strain is a time-independent deformation. Phonons are quantized dynamical modes. They use related leftover-symmetry channels and they are not the same object.

6. **What extra facts turn a spin doublet into a qubit?**

A demonstrated route to initialize it, drive coherent operations, read it out, hold its charge state, and suppress leakage and decoherence on the operation timescale.

You can now look at a damaged crystal and ask the questions in order: what geometry remains, which orbitals are stuck, how many electrons occupy them, what total spin follows, and which of crystal field, spin-orbit, spin-spin, hyperfine, strain, and phonons split or mix the levels. You also know why the answers still do not, by themselves, make a qubit. Next we take the same checklist into diamond and ask which famous centers actually survive it.

6.20. Sources

- [R070] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle, “First-principles calculations for point defects in solids,” *Reviews of Modern Physics* **86**, 253–305 (2014). DOI: 10.1103/RevModPhys.86.253.
- [R073] G. Wolfowicz *et al.*, “Quantum guidelines for solid-state spin defects,” *Nature Reviews Materials* **6**, 906–925 (2021). DOI: 10.1038/s41578-021-00306-y.
- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001; arXiv: 1302.3288.
- [R075] Á. Gali, “Ab initio theory of the nitrogen-vacancy center in diamond,” *Reviews of Modern Physics* **91**, 015004 (2019). DOI: 10.1103/RevModPhys.91.015004; arXiv: 1906.00047.
- [R076] M. W. Doherty, N. B. Manson, P. Delaney, and L. C. L. Hollenberg, “The negatively charged nitrogen-vacancy centre in diamond: the electronic solution,” *New Journal of Physics* **13**, 025019 (2011). DOI: 10.1088/1367-2630/13/2/025019; arXiv: 1008.5224.
- [R077] P. Udvarhelyi, V. O. Shkolnikov, A. Gali, G. Burkard, and A. Pályi, “Spin-strain interaction in nitrogen-vacancy centers in diamond,” *Physical Review B* **98**, 075201 (2018). DOI: 10.1103/PhysRevB.98.075201; arXiv: 1712.02684.

- [R078] A. Norambuena, E. Muñoz, H. T. Dinani, A. Jarmola, P. Maletinsky, D. Budker, and J. R. Maze, “Spin-lattice relaxation of individual solid-state spins,” *Physical Review B* **97**, 094304 (2018). DOI: 10.1103/PhysRevB.97.094304; arXiv: 1711.10280.

Part IV

Defect-spin platforms

Diamond, sapphire, silicon carbide, and a few cousins. The question is not who owns the prettiest isolated number. It is who can field an array.

Chapter 7. A diamond that glows because something is missing

You shine a green laser into a diamond. A single point answers in red. The rest of the stone stays dark.

The carbon lattice is almost perfect. Almost. One carbon is gone. Next to the hole sits a nitrogen that does not belong. An extra electron lives on that pair, and that electron is what glows.

A jeweler would call the stone flawed. We treat the flaw as the object: first the red light, then the spin hiding inside it, then the cousins that glow at other colors.

7.1. A hole next to a nitrogen

Take one carbon out of diamond and put a nitrogen on a neighboring site. The pair is not a dye on the surface. It is a tiny object carved into the crystal: a preferred axis along the nitrogen–hole line, and no mirror that swaps the two ends.

Shine 532-nm green light on that pair. Red light comes back. Most of the red is a smear. A thin slice of it is sharp, near 637 nm. That sharp slice is the fingerprint of this particular missing-atom pair in this particular charge.

A defect that swallows one color and gives another back is a **color center**. We take this as the definition. The color is the laboratory signal. The center is one place in the crystal.

The useful member of this family is the negatively charged nitrogen-vacancy center, written NV^- . Its ground levels form a spin triplet. It will run at room temperature. Its optical line still notices local electric fields, and only a small fraction of the glow sits in that sharp 637-nm slice.

Hence a color center is a spectroscopic object, not a qubit certificate. The rest of the chapter is the extra work that turns this particular one into a controllable spin.

7.2. The thin red line

Most of the red is useless if you want two defects to look like the same atom. Vibrations in the lattice steal energy on the way out and smear the color.

The leftover slice that leaves no vibration behind is the **zero-phonon line**. The name is worth keeping because the ordinary English suggests “no photon,” and that is exactly wrong: it is the photon that took none of the lattice with it. It is the only part of the glow that can match another center, if anything can.

For NV^- that line sits at 637 nm. If the extra electron leaves, you have NV^0 instead, and the sharp line jumps to about 575 nm [R074]. Same hole, same nitrogen, different color. Already the glow is telling you the electron count.

A common room-temperature sequence looks like this:

```
532-nm light
|
v
3E excited triplet ---- red fluorescence ----> 3A2 ground triplet
|                                     ^
+-- spin-selective leak into singlets -----+
      (stronger for  $m_s = \pm 1$ )
```

NV^- has a triplet ground manifold 3A_2 , an optically excited triplet 3E , and intermediate singlet states [R074]. Green light lifts the electron into 3E . From there it can fall straight back, or it can sneak through the singlets.

That sneak path is stronger when the spin projection along the NV axis is $m_s = \pm 1$ than when it is $m_s = 0$. Repeated green flashes therefore park more and more population in $m_s = 0$. The early red flash is brighter for $m_s = 0$ than for $m_s = \pm 1$.

Hence one laser both prepares the spin and reads it, statistically. At room temperature a single ordinary flash usually does not tell you the state with certainty. You repeat. Resonant light at low temperature, or a nearby nucleus used as a helper, can do much better. Those are different experiments, with different conditions.

Drop the spin-selective leak and the same green light still makes the stone glow. It no longer parks the spin, and the brightness no longer reports m_s . The cycle is also not closed forever. Optical excitation can convert NV^- to NV^0 . Charge-repump light puts the extra electron back, probabilistically. Charge conversion and spin relaxation can both make the desired red light disappear. They are not the same failure.

7.3. How many electrons are sitting there

A structural label such as “NV” does not say how many electrons live on the defect. Change the count and you change the atom.

Chapter 6 already named that integer the charge state. The extra electron on NV^- is what gives the $S = 1$ ground state used for most sensing and qubit work. NV^0 has a different electronic and optical structure; it is not the same triplet qubit. NV^+ is generally optically dark in the usual experiments. Nearby donors and acceptors, the surface, the optical power, and any electrodes all push which count is stable [R074].

Write NV^- , not “the NV,” when you mean the qubit. The charge is part of the object.

The same warning is louder for the group-IV cousins below. SiV^- , GeV^- , SnV^- , and the less mature PbV^- are the usual spin-and-photon labels. Neutral SiV^0 is a different $S = 1$ center with a line near 946 nm. Neutral GeV and SnV have far less mature single-center qubit evidence. Stabilizing the charge is a device requirement, not a typographical detail [R082].

7.4. A balanced cousin

Picture a second object in the same hard crystal. An impurity atom M — silicon, germanium, tin, or lead — sits between two missing carbons:

```
carbon lattice ... vacancy - M - vacancy ... carbon lattice
                        ^ inversion center
```

The two holes balance. The middle of the pair is a point you can invert through and land on the same arrangement. People write that approximate symmetry D_{3d} . The NV, by contrast, has a preferred axis and no such center; that one is C_{3v} .

Because the balanced defect looks the same after inversion, a uniform electric field does not, to first order, shove its optical line around. Different copies can sit closer in color. That is the point of the geometry.

The useful negative charge MV^- has an effective electronic spin $S = 1/2$ inside orbitally degenerate ground and excited manifolds. Spin-orbit coupling splits those manifolds. Heavier impurities generally open a larger split, which quiets some phonon processes at a given cryogenic temperature — and also makes a direct microwave spin flip less convenient unless strain or field orientation mixes the states [R082].

Hence inversion symmetry is an optical-line property, not a promise that the spin is easy to turn. Drop the inversion and first-order electric-field shifts return. Keep it, and strain, higher-order Stark shifts, local charges, and fabrication damage still remain.

7.5. Microwaves that pick two of three

Now we need numbers for the NV^- ground spin, because a microwave pulse is how you turn the glow into a controlled two-level system.

Choose the NV axis as z . Let S_x , S_y , and S_z be the dimensionless spin-1 matrices, with $S_z|m_s\rangle = m_s|m_s\rangle$ and $m_s \in \{-1, 0, +1\}$. Write energy as frequency by dividing by Planck's constant h . A useful ground-state model is

$$\frac{H}{h} = DS_z^2 + \gamma_e \mathbf{B} \cdot \mathbf{S} + E(S_x^2 - S_y^2) + \mathbf{S} \cdot \mathbf{A} \cdot \mathbf{I} - \gamma_n \mathbf{B} \cdot \mathbf{I} + PI_z^2.$$

Here D is the axial splitting already present at zero field, in hertz; γ_e is the electron gyro-magnetic ratio, in hertz per tesla; $\mathbf{B}$ is the magnetic field, in tesla; E is a transverse strain or electric-field splitting, in hertz; $\mathbf{I}$ is the nitrogen nuclear-spin operator; $\mathbf{A}$ is the hyper-fine tensor, in hertz; γ_n is the nuclear gyromagnetic ratio, in hertz per tesla; and P is the quadrupole coefficient, present when the nucleus has $I \geq 1$.

That axial number D is the zero-field splitting from Chapter 6. The 2.87 GHz you will hear in every NV talk is not a magnet. It is the crystal already splitting $m_s = 0$ away from $m_s = \pm 1$ before you turn anything on.

For the common $^{14}\text{NV}^-$ center, $I = 1$. For $^{15}\text{NV}^-$, $I = 1/2$, so there is no nuclear quadrupole term. The room-temperature axial splitting is approximately $D = 2.87$ GHz, and $\gamma_e \approx 28.0$ GHz/T [R074].

Ignore strain and hyperfine structure for one minute, and align the field with z . Then

$$\frac{E_{m_s}}{h} = Dm_s^2 + \gamma_e B_z m_s.$$

At $B_z = 10$ mT = 0.010 T, the Zeeman shift has units

$$(28.0 \text{ GHz/T})(0.010 \text{ T}) = 0.280 \text{ GHz}.$$

The two microwave frequencies from $|0\rangle$ are then

$$f_{0 \rightarrow +1} = 3.150 \text{ GHz}, \quad f_{0 \rightarrow -1} = 2.590 \text{ GHz}.$$

Hence a resonant microwave pulse can select either line. Taking $|0\rangle$ and $|-1\rangle$ as the working pair leaves $|+1\rangle$ as a leakage state. This is a qubit because preparation, rotation, and measurement exist — not merely because the spectrum contains two levels.

Drop the field alignment and the two lines mix through the E term. Drop the third level from the bookkeeping and you will mis-count every pulse that leaks into $|+1\rangle$.

7.6. The nucleus next door

Most carbon in diamond is ^{12}C , nuclear spin $I = 0$. Natural diamond also contains about 1.1% ^{13}C with $I = 1/2$. Enriching the crystal in ^{12}C weakens that spin bath. A selected nearby ^{13}C can instead be used as a long-lived memory [R074]. The impurity itself may add a nucleus:

nucleus	nuclear spin I	practical consequence
^{14}N / ^{15}N	1 / 1/2	intrinsic NV hyperfine register; ^{14}N also has quadrupole structure
^{29}Si	1/2	optional intrinsic SiV nuclear memory; spin-zero Si isotopes remove it
^{73}Ge	9/2	a larger intrinsic register, but a more crowded hyperfine spectrum
^{117}Sn , ^{119}Sn	1/2	isotope-selectable SnV hyperfine degree of freedom
^{207}Pb	1/2	possible intrinsic PbV nuclear degree of freedom

The listed group-IV isotope spins and their spin-zero alternatives are summarized in the group-IV review [R082].

[Experiment] A nearby nuclear spin can extend storage and enable repetitive readout. It also creates extra spectral lines and calibration work. Calling every nucleus either “noise” or “memory” misses the choice: uncontrolled weak couplings form a bath; resolved, controllable couplings form a register.

Suppose the same ^{13}C is both unresolved and uncontrolled. Then it is part of the bath, and isotopic purification helps. Suppose it is resolved and driven. Then it is a register, and purifying it away throws the memory out with the noise.

7.7. Four cousins, four sets of conditions

Off-resonant green light, microwaves, and spin-dependent red fluorescence are the NV[−] routine. Radio-frequency fields address ^{14}N , ^{15}N , or a resolved ^{13}C . Room temperature is ordinary. Low temperature is still useful when you need the optical line itself to be narrow enough to network.

The balanced cousins do not copy that routine.

SiV⁻. Resonant light can pump and read the spin through nearly cycling transitions. Microwave or Raman control rotates it. The inversion-symmetric optical line is attractive when you want photons that look alike. At roughly 4 K, though, phonons readily swap population between ground-state orbital branches separated by only tens of gigahertz, and the spin forgets. Cooling to about 100 mK suppressed that channel enough for $T_2 = 13$ ms, $T_1 > 1$ s, and 89% single-shot spin readout in one bulk-device experiment [R083]. Those three numbers belong together with “100 mK.”

GeV⁻. The optical zero-phonon line is near 602 nm and the ground orbital splitting is larger than SiV⁻’s [R082]. Microwave control and optical readout have been demonstrated. **[Experiment]** A millikelvin experiment reported Ramsey $T_2^* \approx 1.43 \mu\text{s}$, spin-echo coherence of roughly 0.44 ms, and more than 20 ms under CPMG dynamical decoupling [R084]. “More than 20 ms” is therefore not an unconditioned free-induction time.

SnV⁻. Its zero-phonon line is near 619 nm and its ground orbital splitting is about 0.85 THz, which reduces thermal orbital excitation at helium-like temperatures relative to SiV⁻ [R082]. Strain turns on magnetic transitions that would otherwise stay dark. **[Experiment]** In a cryogenic strained device with a superconducting microwave waveguide, Hahn echo reached about 0.43 ms and CPMG-64 extended T_2 to 10(1) ms; the reported single-pulse fidelity was about 99.1% [R085]. These are one-center, pulse-sequence-specific results — not array medians.

PbV⁻. Optical emitters assigned to a lead-vacancy split-vacancy family, and large spin-orbit splittings, have been reported [R082]. Coherent spin initialization, gates, readout, and pair coupling are not yet supported at the same maturity as NV⁻ through SnV⁻. A larger splitting may improve thermal isolation and can make microwave access worse. **[Proposal]** PbV⁻ is a candidate to test, not a ready drop-in qubit.

Hence each cousin is a different bargain among optical contrast, phonon isolation, and microwave access. Quoting a T_2 without its temperature and pulse sequence is quoting a different experiment.

7.8. A table that keeps its conditions

The rows below are published benchmarks, not typical constants. Different rows use different temperatures, fields, samples, pulse sequences, and definitions. The last column is deliberately severe.

center	useful elec- tronic/optical facts	representative spin result and conditions	readout/control demonstrated	demonstrated coupling	evidence for a scalable interacting array
NV ⁻	$S = 1$; $D \approx 2.87$ GHz; 637-nm ZPL; weak ZPL fraction	$T_2 = 1.8$ ms in an isotopically engineered bulk sample at room temperature under Hahn echo [R079]	room- temperature optical prepara- tion/fluorescence and microwave control; cryogenic resonant single-shot methods	two NV electron spins at about 25 nm were entangled at room temperature with dipolar coupling 4.93(5) kHz [R080]; remote NV registers later supported a teleported CNOT [R081]	No. Pair and network-node demonstra- tions do not establish a uniform dense lattice
SiV ⁻	$S = 1/2$ Kramers branches; about 737-nm ZPL; strong ZPL emission and inversion symmetry [R082]	$T_2 = 13$ ms and $T_1 > 1$ s at about 100 mK in bulk diamond [R083]	resonant optical initializa- tion/readout, Raman and microwave control; 89% single-shot readout in that experiment [R083]	two SiV optical emitters coupled through one nanocavity showed superradi- ant/subradiant collective modes [R086]	No. The cavity experiment was a selected emitter pair, not a fabricated spin lattice
GeV ⁻	$S = 1/2$; about 602-nm ZPL; larger orbital splitting than SiV ⁻ [R082]	$T_2^* \approx 1.43 \mu\text{s}$, echo ~ 0.44 ms, CPMG > 20 ms at millikelvin temperature [R084]	optical spin readout and coherent microwave control	local hyperfine coupling is usable; no multi-GeV spin entangling result is used as evidence here	No. Long coherence of one center says nothing about placement yield or coupling uniformity

center	useful elec- tronic/optical facts	representative spin result and conditions	readout/control demonstrated	demonstrated coupling	evidence for a scalable interacting array
SnV ⁻	$S = 1/2$; about 619-nm ZPL; ground orbital splitting ~ 0.85 THz [R082]	echo ~ 0.43 ms; CPMG-64 10(1) ms in a cryogenic strained, superconducting waveguide device [R085]	optical prepa- ration/readout and strain-enabled microwave control; about 99.1% single-pulse fidelity in that device [R085]	coupling to a nearby ^{13}C was observed [R085]; photonic- interface work does not yet equal a two-SnV gate	No. Site- controlled creation exists, but a coherent interacting array has not been shown
PbV ⁻ / SiV ⁰	PbV ⁻ : less mature group-IV optical center; SiV ⁰ : distinct neutral $S = 1$ center near 946 nm	no directly comparable benchmark adopted	partial optical/spin functionality, platform dependent [R082]	no array-level coupling evidence adopted	No; exploratory. Do not average these distinct charge states into the rows above

[Experiment] The 2026 NV network result implemented an unconditional teleported controlled-NOT between remote diamond registers, with ^{13}C nuclei as data qubits and NV electrons providing local logic, readout, and photonic entanglement [R081]. That is a substantial systems result. It is also a modular, measurement-and-feed-forward interaction — not a static nearest-neighbor energy in a dense crystal array.

Likewise, the two-SiV nanocavity experiment observed coherent photon-mediated collective optical states [R086]. A shared photonic mode can couple selected emitters. That is not a deterministic two-spin gate, a uniform coupling graph, or passive many-body order. The noun “interaction” needs its object and its operating protocol.

An NV electron spin is a physical qubit. Several electron and nuclear spins controlled as a register are a processor register only if an encoding is specified. Two remote registers joined by heralded photons form a quantum network. None of those observations, by itself, is an emergent anyon, a topologically ordered medium, or a topologically protected logical qubit.

7.9. Three distances on a map

Color centers are made by putting the impurity in during chemical-vapor-deposition growth, by focused ion implantation followed by annealing, by creating vacancies near

existing dopants, or by femtosecond-laser processing. Nanopillars, waveguides, and photonic-crystal cavities then improve collection or cavity coupling. Each step can add strain, surface charge noise, paramagnetic damage, or spectral diffusion [R082].

Three distances must not be collapsed into one.

1. **Delivery resolution:** where the beam or laser deposits energy.
2. **Final defect distribution:** broadened by ion straggle, vacancy migration, and complex formation.
3. **Usable-qubit yield:** sites in the correct charge state, orientation, optical line, and coherence class after processing.

[Experiment] A 2025 SnV^- study combined focused implantation with local laser annealing and in-situ fluorescence feedback. It reported sub-50-nm site positioning, doses down to a Poisson mean of one implanted ion per site, and site-selective creation of single SnV^- emitters [R087]. The same study treated high-yield activation as an outstanding scaling problem. Sub-50-nm *site registration* is not proof that every site contains exactly one mutually resonant, coherent SnV^- at a chosen three-dimensional coordinate.

CVD growth generally gives high crystal quality but statistical location. Implantation supplies masks and coordinates but causes damage and has conversion-yield and straggle penalties. Laser feedback improves activation control and does not yet certify an interacting array. For an eventual defect Hamiltonian, fabrication must satisfy placement, orientation, charge, optical frequency, and coherence at once. Multiplying five less-than-unity yields is a quick way to make a small number. No advanced topology is required for that calculation.

7.10. What still knocks the spin around

For NV^- , important channels include the ^{13}C nuclear bath, substitutional-nitrogen electron spins, surface spins for shallow centers, magnetic-field drift, temperature and strain shifts of D , microwave amplitude error, optical spectral diffusion, and charge conversion [R074]. Isotopic purification removes much of one bath. It does not remove the surfaces or the control electronics.

For group-IV negative centers, inversion symmetry reduces first-order sensitivity of optical transitions to electric fields. It does not make the defect insensitive to strain, higher-order Stark shifts, local charges, or fabrication damage. Their defining low-temperature problem is orbital-phonon scattering. Cooling, or opening a larger spin-orbit split, suppresses thermal occupation. Strain used to enable control can also change selection rules and spectra [R082]; [R083]; [R084]; [R085].

Dense arrays add mechanisms that isolated-center optimization hides: implantation damage accumulates; optical lines crowd; microwave fields become nonuniform; dipolar neighbors become both resources and a spin bath; and surfaces sit closer in nanophotonic devices. **[Speculation]** A defect that wins in a pristine bulk sample may lose after the processing required to couple and address it. Only measurements on fabricated ensembles and arrays can settle that ordering.

7.11. Ways to get this wrong

1. Quoting the largest T_2 without its pulse sequence. T_2^* , Hahn echo, and CPMG values answer different noise questions. A many-body gate cannot automatically borrow a CPMG-protected memory time while applying incompatible interactions.
2. Treating wavelength as the whole optical interface. Zero-phonon fraction, linewidth, spectral diffusion, quantum efficiency, cavity cooperativity, collection loss, and charge stability all matter. A narrow line in bulk can broaden after nanofabrication.
3. Ignoring charge and isotope labels. $NV^0 \neq NV^-$ and $SiV^0 \neq SiV^-$. A ^{73}GeV has a different hyperfine space from a spin-zero Ge isotope.
4. Projecting one center onto N centers. Single-center coherence does not measure the distribution of coherence, transition frequencies, charge stability, or coupling strengths over an array. Pair entanglement proves a pair protocol, not an extensible lattice.
5. Equating remote coupling with a material Hamiltonian. Heralded photons, cavity-mediated optical modes, direct magnetic dipoles, and hyperfine coupling are physically different resources. A teleported gate is real quantum coupling. It is not passive local exchange.
6. Calling a controlled register topological. The NV and group-IV results in this chapter demonstrate physical qubits, memories, and network primitives. **No experiment cited here demonstrates emergent topological order in an array of diamond color centers.**

7.12. Check yourself

1. **Why must “NV” include a charge superscript in a qubit discussion?**
Because the electron count determines the spin and optical level structure; NV^- is the standard $S = 1$ qubit, while NV^0 and NV^+ are different systems.
2. **Show that at $B_z = 10$ mT the two microwave frequencies from $|0\rangle$ are 3.150 GHz and 2.590 GHz.**

The Zeeman piece is $(28.0 \text{ GHz/T})(0.010 \text{ T}) = 0.280 \text{ GHz}$, so $D \pm \gamma_e B_z$ gives $2.87 + 0.28 = 3.15 \text{ GHz}$ and $2.87 - 0.28 = 2.59 \text{ GHz}$.

3. **What fails if SiV^- is run at about 4 K rather than about 100 mK?**
Phonons swap population between the nearby orbital branches, and the spin forgets. The cited $T_2 = 13 \text{ ms}$, $T_1 > 1 \text{ s}$, and 89% single-shot readout belong with the colder run.
4. **What fails if a 10 ms CPMG result is treated as available gate time?**
That number is coherence under a specified refocusing sequence. A many-body gate that is incompatible with those pulses cannot borrow the time.
5. **What has direct two-NV dipolar entanglement established?**
That a selected pair can be coherently coupled and entangled at room temperature. It has not established controlled yield or uniformity for a large array.
6. **Show that sub-50-nm site registration is not yet a scalable interacting array.**
Registration locates a delivery site. Occupancy, charge, resonance, coherence, and designed coupling are independent, and the cited SnV study treated high-yield activation as still open.

NV^- offers mature room-temperature spin control and a comparatively difficult optical interface. Inversion-symmetric group-IV negative centers offer stronger coherent optical emission and generally demand cryogenic, phonon-aware operation. Charge state and isotope are part of the device specification. Direct dipolar, hyperfine, cavity-mediated, and remote photonic couplings have all been demonstrated in limited settings. No cited single-center, pair, cavity, network, or site-placement result establishes a scalable interacting diamond array.

7.13. Sources

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001.
- [R079] G. Balasubramanian *et al.*, “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* **8**, 383–387 (2009). DOI: 10.1038/nmat2420.
- [R080] F. Dolde *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R081] M. Iuliano *et al.*, “Unconditionally teleported quantum gates between remote solid-state qubit registers,” *Nature Communications* **17** (2026). DOI: 10.1038/s41467-026-72818-6.
- [R082] C. Bradac, W. Gao, J. Forneris, M. E. Trusheim, and I. Aharonovich, “Quantum nanophotonics with group IV defects in diamond,” *Nature Communications* **10**, 5625 (2019). DOI: 10.1038/s41467-019-13332-w.

- [R083] D. D. Sukachev *et al.*, “Silicon-vacancy spin qubit in diamond: a quantum memory exceeding 10 ms with single-shot state readout,” *Physical Review Letters* **119**, 223602 (2017). DOI: 10.1103/PhysRevLett.119.223602.
- [R084] K. Senkalla, G. Genov, M. H. Metsch, P. Siyushev, and F. Jelezko, “Germanium vacancy in diamond quantum memory exceeding 20 ms,” *Physical Review Letters* **132**, 026901 (2024). DOI: 10.1103/PhysRevLett.132.026901; arXiv: 2308.09666.
- [R085] I. Karapatzakis *et al.*, “Microwave control of the tin-vacancy spin qubit in diamond with a superconducting waveguide,” *Physical Review X* **14**, 031036 (2024). DOI: 10.1103/PhysRevX.14.031036.
- [R086] R. E. Evans *et al.*, “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* **362**, 662–665 (2018). DOI: 10.1126/science.aau4691; arXiv: 1807.04265.
- [R087] X. Cheng *et al.*, “Laser activation of single group-IV colour centres in diamond,” *Nature Communications* **16**, 5124 (2025). DOI: 10.1038/s41467-025-60373-5.

Chapter 8. A red stone that already has guests

A ruby is a hard, clear crystal that happens to be red. The red is not a dye. It is chromium sitting where aluminum should sit: a few guests in a house built of aluminum and oxygen.

Sapphire is the same house. Take the chromium out, or swap in a different guest, and the stone turns colorless, blue, yellow, or whatever the new impurity decides. Jewelers grade the color. We are going to grade something else: whether those famous guests can do the job diamond already does with one nitrogen atom next to a missing carbon.

The first laser was a ruby. Early masers used ruby too. Titanium in sapphire still powers the workhorse tunable laser in a thousand labs. So this crystal already hosts impurities the world has used for sixty years. That is the puzzle. If the guests are this famous, why is diamond still the defect people initialize, rotate, and read one at a time?

8.1. The same house, two names

Picture a crate of oxygen ions packed almost as tightly as balls. Aluminum occupies two thirds of the leftover pockets. That crystal is **corundum**. The name is worth keeping because gemstone words keep splitting one material into two reputations.

When some of those aluminum ions are replaced by chromium, people call the stone ruby. When they mean the same lattice as a window, a watch crystal, or a microwave box, they usually say sapphire. The chemistry is convenient. Chromium and aluminum both like a +3 charge, so chromium can sit on an aluminum site without dragging a compensating defect along.

Each chromium sits inside a slightly squashed cage of six oxygens. The cage is not decoration. It sets which way the chromium's three leftover electrons prefer to point, even before you turn on a magnet. Light can lift those electrons. Microwaves can shuffle them among the lowest four shelves. That is why the same small red impurity made the first laser and still shows up on an electron-spin-resonance bench.

Hence ruby and sapphire are one lattice with different guests, not two platforms. A crystal full of identical cages is not a register of labelled qubits. At the concentrations that make a ruby laser, a laser spot and a microwave resonator talk to a crowd. Thin the crowd enough that each ion is alone, and the signal disappears into the noise.

A superb microwave box is also not a many-body phase wearing a 1950s lab coat. We will keep those two compliments, and then ask the operational questions diamond already answers.

8.2. Four shelves on one ion

A chromium that has given up three electrons keeps three in its outer shell. Those three like to point the same way, so the leftover spin can take four values: $m = -3/2, -1/2, +1/2, +3/2$. Think of four shelves. A magnetic field can tilt them. Even with the field off, the squashed cage already splits them into two pairs.

That leftover split, present with no magnet, is the zero-field splitting from Chapter 6. It is an energy you can measure in a microwave lab, not a metaphor for “the crystal does something.”

Spectroscopists write the same facts in their dialect. The ion is Cr^{3+} with configuration $3d^3$. In the oxygen cage the three electrons occupy the lower-energy orbitals with parallel spins, giving an orbital-singlet ground term 4A_2 and total electron spin $S = 3/2$. The cage is slightly stretched along the crystal c -axis, and that stretch is what produces the split [R088].

Define dimensionless spin operators S_x, S_y, S_z whose S_z eigenvalues are those four m values. Let D be an energy, μ_B the Bohr magneton in joules per tesla, B_i the magnetic-field components in tesla, and $g_{\parallel}, g_{\perp}$ the dimensionless spin factors parallel and perpendicular to the c -axis. The useful ground-state Hamiltonian is

$$H = g_{\perp}\mu_B(B_x S_x + B_y S_y) + g_{\parallel}\mu_B B_z S_z + D \left[S_z^2 - \frac{S(S+1)}{3} \right].$$

At zero field, $S(S+1)/3 = 5/4$, so

$$E_{\pm 3/2} = D, \quad E_{\pm 1/2} = -D.$$

For ruby, $D < 0$. The $|\pm 3/2\rangle$ pair is therefore lower, and the separation is

$$\nu_0 = \frac{-2D}{h} \simeq 11.493 \text{ GHz},$$

where h is Planck’s constant [R088]; [R089]. Time reversal keeps $+m$ and $-m$ glued together when the field is off. That pairing is why you still have two doublets, not four lonely lines.

A quick dimensional and thermal check:

$$\frac{h\nu_0}{k_B} = \frac{(6.626 \times 10^{-34} \text{ J s})(11.493 \times 10^9 \text{ s}^{-1})}{1.381 \times 10^{-23} \text{ J K}^{-1}} \approx 0.552 \text{ K}.$$

Joule-seconds times inverse seconds gives joules. Division by joules per kelvin gives kelvin. Hence passive polarization between these doublets wants temperatures comfortably below half a kelvin. A room-temperature ruby laser proves an optical cycle exists. It does not prove the ground-state spin is sitting quietly in its lowest shelf.

For a field parallel to c , the Hamiltonian stays diagonal:

$$E_m = D \left(m^2 - \frac{5}{4} \right) + g_{\parallel} \mu_B B_z m.$$

Microwave magnetic-dipole transitions ordinarily change m by one. Tilt the field and the $|m\rangle$ states mix, the allowed strengths change, and you can find field-insensitive “clock” points. Clock transitions are promising for coherence. Promise is not a measured T_2 .

Drop time reversal and the zero-field doublets can split. Keep it, and you still have four levels, two pairs, and a leakage problem the moment you pretend the ion is a two-level atom.

8.3. Light sees volts, microwaves see microvolts

The same ion also eats red light. At cryogenic temperature two sharp lines sit near 693.6 nm and 692.2 nm, about 1.788 eV and 1.792 eV [R088]. Those photons jump from the four ground shelves to a higher pair. Spectroscopists tag the excited pair 2E . You do not need the label to remember the numbers. You need it if you open the papers.

The jump is officially spin-forbidden. Lattice vibrations sneak it through anyway. Resonant excitation can tell the ground-state branches apart, and fluorescence that rides a phonon can report their populations. That is ensemble optical initialization and ensemble detection of magnetic resonance. It is not automatically a single-shot look at one chromium spin.

Why are the two gaps so different? The oxygen cage first reorganizes electron-volt-scale orbital terms. A smaller stretch of the cage, plus the electrons’ own spin-orbit coupling, then leaves a gigahertz leftover inside the ground term. Optical photons bridge roughly 1.8 eV. Microwaves bridge tens of microelectron-volts.

Hence the optical line and the microwave line are not two pictures of one two-level atom. Treat them as one unexplained object and you hide the selection rules and the four-level leakage channels.

8.4. A box that barely leaks

Sapphire is transparent over a broad optical range, mechanically hard, chemically stable, and an exceptional microwave dielectric. Take a high-purity disk, launch a microwave that crawls around its rim, and the wave just keeps going.

How long it rings is the **quality factor** Q : stored energy divided by energy lost per radian. A huge Q is a host-device asset, not a measured impurity-spin coherence time.

[Experiment] At about 100 mK, high-purity sapphire whispering-gallery resonators have shown microwave Q in the 10^8 – 10^9 range. One 11 GHz mode with $Q = 2 \times 10^9$ corresponds to a photon energy-decay timescale $Q/(2\pi f) \approx 29$ ms [R090].

The nuclear environment is less friendly than isotopically enriched diamond. Natural aluminum is essentially all ^{27}Al , with nuclear spin $I = 5/2$, so every substitutional chromium is surrounded by a dense host nuclear-spin bath. Natural ^{52}Cr , the dominant chromium isotope, has $I = 0$, while ^{53}Cr carries $I = 3/2$ [R088]; [R090]. You can pick chromium isotopes and remove its own nucleus. You cannot turn aluminum into a spin-zero isotope.

Hence hyperfine structure and spectral diffusion are structural issues, not merely dirty-growth issues. Drop the aluminum nuclear spin from the model and the crystal you wrote down is no longer corundum.

8.5. What they measured on ruby

A laser spot on a ruby is almost never one ion. It is a crowd that happens to sit under the same beam. That crowd is an **ensemble**. A beautiful ensemble number is allowed to look like a single-spin number if you stop reading one sentence too soon.

[Experiment] Sewani and co-workers studied a 0.005% ruby crystal in a dilution refrigerator. Under 15 nW continuous resonant excitation they inferred a ruby temperature of 143 ± 7 mK, measured a maximum spin-lattice relaxation time $T_1 = 3.67 \pm 0.35$ s, and observed ODMR transitions [R088]. Their resonant confocal measurement sampled an area of roughly $1 \mu\text{m}^2$, not one ion. The paper presented coherence-time measurement as future work. It did not report Ramsey fringes, Hahn-echo T_2 , single-qubit gates, single-shot readout, or two-ion entanglement.

That distinction prevents a common numerical accident: substituting T_1 for T_2 . Seconds-long population relaxation at millikelvin temperature is encouraging. Phase memory may still die much sooner on the aluminum nuclei, chromium–chromium dipoles, field noise, and inhomogeneous strain.

[Experiment] A 2025 experiment placed a 0.05% ruby sample on a 5.5 GHz high- T_c superconducting resonator and measured ensemble spin–photon coupling from 30 mK to 16 K [R089]. At 65 mK, one transition reached collective coupling $\bar{\Omega}/2\pi \approx 21$ MHz with spin

decay $\gamma_s/2\pi \approx 44$ MHz, giving cooperativity about 15 but not the strict strong-coupling condition $\bar{\Omega} > \gamma_s$ [R089]. The sample contained an estimated 2.6×10^{15} spins. The authors inferred an average single-spin coupling of only about 0.41 Hz over all spins, or 1.25 Hz when counting the high-field-volume subset [R089].

This is valuable collective cavity physics. It is also a sharp negative result for individually selected neighbours. The impressive 21 MHz is the $\sqrt{N}$ -enhanced ensemble coupling, not one defect's coupling. In that experiment spin damping was too fast for coherent photon re-emission.

Suppose you divide 21 MHz by one spin. Then you have invented a device that the same paper's own model puts in the hertz range. The hypothesis that fails is "collective enhancement can be ignored."

8.6. Other metal guests

Sapphire accepts many transition metals because they can occupy aluminum sites in several charge states. Different valence and charge-compensation histories produce very different objects.

Species in sapphire	Established behavior	What it is for us
Cr^{3+} , $S = 3/2$	Ruby R-line fluorescence, laser/maser gain, EPR/ODMR, seconds-scale T_1 at mK, ensemble cavity coupling [R088]; [R089]	Best-characterized candidate, but the modern single-center coherent-control stack is absent
Fe^{3+} , $S = 5/2$	EPR and microwave transitions in nominally pure sapphire; useful or parasitic in frequency standards [R090]	Demonstrated ensemble spectroscopy, not optical single-spin initialization/readout
V^{2+} , $S = 3/2$, ^{51}V with $I = 7/2$	Eight-line hyperfine structure resolved by whispering-gallery ESR [R090]	Rich levels, but nuclear complexity and no demonstrated qubit lifecycle
Ti^{3+} , $3d^1$	Broad vibronic gain underlying the tunable Ti:sapphire laser, approximately 650–1100 nm [R091]	Excellent classical gain medium; broad electron–phonon optical transitions are not evidence of a narrow spin–photon qubit interface
Co, Ni, Mn and other ions	Optical absorption/EPR reported under charge-state- and growth-dependent conditions	A spectroscopic line is a candidate identifier, not a scalable center

[Experiment] Farr and co-workers detected native Fe^{3+} , Cr^{3+} , and V^{2+} impurities at parts-per-billion to parts-per-million levels in high-purity sapphire near 115 mK using 8–19 GHz whispering-gallery spectroscopy [R090]. The sensitivity is double-edged. It proves that dilute spins couple to microwave modes, and that nominally “pure” sapphire already contains uncontrolled spin species.

Hence a catalog of guests is not a catalog of qubits. A line in a spectrum identifies a candidate. It does not initialize, rotate, or read one spin.

8.7. A hole where oxygen should be

Remove an O^{2-} ion and you leave an empty oxygen site V_{O} . Charge-state language is easy to reverse, so define it with the leftover electrons, not the empty pocket.

A nominally neutral vacancy that traps two electrons is an *F* **center**. The literature uses it as a species, not as a vibe. Take one electron away and the same pocket is an F^+ center, V_{O}^+ with one leftover electron. Vacancy aggregates are labelled F_2 , F_2^+ , and so on, though historical assignments can depend on how the sample was cooked.

[Experiment] Irradiated or reduced alpha-alumina shows an *F*-center absorption band near 6.1 eV and luminescence near 3.0 eV. The F^+ center has absorption structure near 4.8 and 5.4 eV and emission near 3.8 eV [R092]; [R095]. The *F*-center emission lifetime was measured as 36 ± 4 ms, consistent with a spin-forbidden relaxed transition [R092]. **[Theory]** Modern hybrid-density-functional calculations reproduce the main absorption/emission assignments and distinguish internal vacancy transitions from transitions involving band edges [R093]. Agreement on optical energies does not establish a qubit.

The one-electron F^+ center is paramagnetic and therefore visible to EPR in suitable samples. The two electrons of the neutral *F* center can form singlet and triplet configurations involved in its luminescence. Yet, for the literature survey used here through 24 August 2026, no peer-reviewed report was located that demonstrates an isolated oxygen-vacancy center in bulk sapphire with all of the following: photon antibunching tied to that center, spin-selective optical initialization, coherent single-spin rotations, a measured single-center T_2 , and single-shot readout. The positive literature establishes ensemble color-center spectroscopy and theory [R092]; [R093]; [R095]. The missing operational stack must be recorded as missing.

Oxygen vacancies can be created by irradiation, reducing growth, or annealing, and can aggregate or change charge under illumination. Those routes offer density control. They also create displacement damage and heterogeneous local environments. Ultraviolet excitation around 5–6 eV is much less convenient for low-loss nanophotonics and biological or fiber interfaces than visible or near-infrared light. Calling the vacancy “deep in a wide gap” is necessary physics, not a device plan.

8.8. Planting chromium where you want it

Bulk ruby growth gives a uniform average concentration and random atomic locations. A focused ion beam, or a masked implant, can put chromium at a chosen depth and a chosen patch of the surface.

[Experiment] Chromium implanted at 150 keV into sapphire, followed by a 1450 °C anneal, produced substitutional ruby-like R-line luminescence in a thin implanted layer [R094]. The demonstrated fluences were ensemble-scale, 6×10^{14} to 4×10^{15} cm⁻², and the reported internal optical efficiency was about 50% under those processing conditions [R094].

This proves activation after implantation. The high anneal temperature, collision straggle, residual damage, charge-state control, and chromium diffusion all still complicate a nanometre-scale interaction graph. No sapphire result located for this chapter combined deterministic single-ion registration, verified one-Cr optical/spin behavior, and a controllable two-Cr coupling. Diamond also suffers implantation straggle and surface noise, but it has crossed the single-defect verification threshold many times [R074].

Hence implantation of chromium is a materials result, not a placement proof. An R-line from a thin layer says the guests found aluminum sites. It does not say which sites, or that two chosen guests can talk.

8.9. Scoreboard against diamond

The fairest comparison uses demonstrated capability at stated conditions, not each platform’s best hypothetical feature.

Criterion	Sapphire/corundum evidence	Diamond reference point	For a chosen-neighbour register
Single-defect identification	Ruby and vacancy work reviewed here is predominantly ensemble spectroscopy; no intrinsic sapphire antibunching-plus-spin-control demonstration was located [R088]; [R092]; [R095]	Single NV centers support room-temperature confocal fluorescence, ODMR, initialization and coherent control; mature review in [R074]	Diamond decisively ahead

Criterion	Sapphire/corundum evidence	Diamond reference point	For a chosen-neighbour register
Coherence	Cr ruby: $T_1 = 3.67 \pm 0.35$ s at mK; that study did not measure T_2 [R088]	Single NV Hahn-echo $T_2 = 1.8$ ms at room temperature in isotopically purified bulk diamond [R079]	Values are not directly comparable; sapphire's needed single-spin T_2 datum is absent
Optical interface	Narrow ruby R lines and ensemble resonant readout near 693 nm at cryogenic temperature [R088]	Single-NV spin-photon entanglement and remote-node building blocks have been demonstrated [R074]	Sapphire has spectroscopy, not a demonstrated single-spin network node
Microwave interface	Very high- Q sapphire resonators and collective coupling; 2025 ruby device had 21 MHz ensemble coupling versus 44 MHz spin decay [R089]; [R090]	Diamond spin ensembles and single NVs couple to microwave structures; single-defect control is routine [R074]	Sapphire excels as a resonator, not yet as an individually wired defect register
Placement	Random bulk doping; ensemble Cr implantation and activation after 1450 °C anneal [R094]	Implantation, delta doping, nanofabrication, and registration all demonstrated, with nonzero yield/straggle limitations [R074]	Neither is perfect; sapphire lacks the single-center end-to-end evidence
Density and selected interactions	Cr–Cr effects and collective $\sqrt{N}$ enhancement exist, but chosen pairwise coherent gates were not located [R089]	Dipolar-coupled NV pairs and multi-spin registers have been coherently controlled [R074]	Ensemble interaction is not an addressable graph
Temperature	Optical ruby operation is possible at room temperature, but the strongest quantum-spin characterization here used 65–186 mK [R088]; [R089]	NV spin initialization, control and readout operate at room temperature; high-quality optical networking usually uses cryogenics [R074]	Application-dependent, but sapphire does not show NV-like room-temperature qubit operation
Host spin bath	Unavoidable dense ^{27}Al , $I = 5/2$	^{12}C , $I = 0$, can be isotopically enriched [R074]; [R079]	Fundamental materials advantage to diamond

Claim	Status	Platform	Evidence	Reference
Optical population readout, ODMR and seconds-long T_1	Experimentally demonstrated	Cr:ruby ensemble, dilution refrigerator	Resonant PLE/ODMR and all-optical relaxation	[R088]
Coherent strong coupling of one Cr spin to one cavity photon	Not demonstrated	Cr:ruby	Inferred average single-spin coupling in hertz; collective mode failed $\tilde{\Omega} > \gamma_s$	[R089]
Trace transition-metal spins couple to sapphire microwave modes	Experimentally demonstrated	Fe, Cr, V ensembles	Avoided crossings and ESR/hyperfine spectra	[R090]
Oxygen vacancies are optically active deep defects	Experimentally demonstrated and theoretically modeled	F, F^+ in alpha-alumina	Ensemble absorption/luminescence plus hybrid-DFT assignments	[R092]; [R093]
Oxygen vacancy is an optically readable single spin qubit	No supporting demonstration located	Sapphire V_O	Required antibunching/control/readout sequence absent from located literature	[R092]; [R093]
Implanted chromium can be optically activated	Experimentally demonstrated, ensemble	Cr-implanted sapphire	R-line luminescence after high-temperature anneal	[R094]
Deterministic interacting array of sapphire defects	No supporting demonstration located	Any sapphire center	No selected pair gates, array calibration, or many-body Hamiltonian validation located	[R088]; [R089]; [R090]; [R091]; [R092]; [R093]; [R094]
Sapphire defects realize topological order	Speculation	Sapphire defect lattice	No microscopic implementation or phase diagnostic	—

A ruby ion is a physical spin. Two selected ions with a calibrated coherent interaction would be a coupled-qubit primitive. A macroscopic spin ensemble collectively coupled to one cavity mode is a different degree of freedom. None of these is automatically an

encoded qubit, an emergent anyon, or topological order. The 2025 cavity anticrossing belongs with collective hybrid systems, not with topological materials.

8.10. Ways to get this wrong

Long T_1 mistaken for long T_2 . Population can remain for seconds while phase coherence disappears rapidly. A credible array proposal needs Ramsey, echo, and driven-gate errors on the same defect, density, temperature, and geometry it proposes to use.

Collective coupling mistaken for local coupling. If N similar spins couple to one mode, the bright collective mode gains roughly $\sqrt{N}$. Dividing 21 MHz by one spin is not allowed. The measured ruby experiment’s own model put the average in the hertz range [R089]. Dark modes, inhomogeneous broadening, and the inability to address a chosen ion remain.

Color mistaken for spin readout. An absorption band says a defect has electronic transitions. Luminescence says some excitation returns radiatively. Single-photon antibunching, spin contrast, cycling stability, photon indistinguishability, and charge stability are separate tests.

A good host device mistaken for a good host defect. Sapphire’s huge microwave Q makes it an excellent resonator substrate. It also makes unwanted parts-per-billion spins measurable [R090]. Score the resonator and the qubit separately.

Random density mistaken for fabrication. Raising chromium concentration shortens average separation. It does not choose the interaction graph. It also introduces Cr pairs, energy transfer, dipolar broadening, and local strain. A designed Hamiltonian wants calibrated signs, anisotropies, and connectivities, not merely “many spins nearby.”

A theoretical vacancy spectrum mistaken for a platform. First-principles agreement with optical bands validates defect assignment [R093]. It does not supply a fabrication yield, coherence distribution, readout fidelity, or two-qubit gate.

8.11. What we can honestly claim

[**Experiment**] Sapphire supports well-characterized paramagnetic dopants, narrow chromium optical lines, long low-temperature spin relaxation, exceptionally low-loss microwave modes, optical/microwave ensemble readout, radiation-induced vacancy color centers, and implanted chromium activation [R088]; [R089]; [R090]; [R091]; [R092]; [R093]; [R094]. These are substantial ingredients for masers, sensors, frequency standards, ensemble memories, and hybrid-resonator studies.

For the specific target of a dense, controllable interacting-defect architecture, the verdict as of 24 August 2026 is **not competitive with diamond on demonstrated end-to-end qubit capability**. The decisive absences are single-center spin-selective optical readout, modern single-center T_2 and gate fidelities, deterministic verified placement, coherent gates between selected sapphire defects, and scalable nanophotonic integration around those defects. The unavoidable ^{27}Al nuclear bath is an additional materials disadvantage.

This is not proof of impossibility. **[Proposal]** A credible sapphire recovery program would first demonstrate one identified Cr^{3+} or F^+ center, measure T_2^* , Hahn-echo T_2 , optical cycling and charge stability, then create and spectrally resolve a registered pair whose coherent coupling exceeds both linewidths. Only after those milestones would an array-level Hamiltonian proposal become evidence-based. **[Speculation]** Jumping directly from ruby ensembles to Fibonacci-like order skips every hard arrow in the chain.

8.12. Check yourself

1. **Why does Cr^{3+} in ruby have four ground-state spin levels?**
Its $3d^3$ configuration gives $S = 3/2$, hence $m = \pm 3/2, \pm 1/2$. The squashed cage splits them into two pairs that stay glued by time reversal.
2. **Show that the 11.493 GHz zero-field splitting is only about half a kelvin.**
 $h\nu_0/k_B = (6.626 \times 10^{-34} \text{ J s})(11.493 \times 10^9 \text{ s}^{-1}) / (1.381 \times 10^{-23} \text{ J K}^{-1}) \approx 0.552 \text{ K}$. Strong passive polarization therefore wants sub-kelvin operation.
3. **What fails if the 3.67 s T_1 is treated as a coherence result?**
 T_1 measures population relaxation. Phase noise can make T_2 far shorter, and the cited millikelvin ruby study did not measure T_2 .
4. **Show that the 21 MHz ruby cavity result cannot be a single-spin coupling.**
About 2.6×10^{15} spins shared the mode. The paper's own inference put the average per spin at 0.41 Hz, or 1.25 Hz in the high-field-volume subset. The 21 MHz is the $\sqrt{N}$ -enhanced collective rate, and even that failed $\bar{\Omega} > \gamma_s$.
5. **What fails if an F^+ center is called a qubit because it is paramagnetic and luminescent?**
Isolation, coherent control, spin-dependent readout, stability, and measured error rates are still missing. Ensemble color is not that stack.
6. **What is sapphire's strongest clear advantage and its clearest defect-array weakness?**
Its microwave dielectric loss can be extraordinarily low. Its single-defect qubit and selected-interaction evidence is presently sparse compared with diamond.

You now have a ruby ion with four ground shelves, an 11.5 GHz split, a seconds-long T_1 that is not a T_2 , and a scoreboard that still favors diamond. Next we ask silicon carbide the same operational questions, and we will not award points for a band gap alone.

8.13. Sources

- [R088] V. K. Sewani, R. J. Stöhr, R. Kolesov, H. H. Vallabhapurapu, T. Simmet, A. Morello, and A. Laucht, “Spin thermometry and spin relaxation of optically detected Cr³⁺ ions in Al₂O₃ (ruby),” *Physical Review B* **102**, 104114 (2020). DOI: 10.1103/PhysRevB.102.104114.
- [R089] Z. Velluire-Pellat, E. Maréchal, C. Feuillet-Palma, and N. Bergeal, “Spin-photon interaction between a ruby crystal and a high-critical-temperature superconducting microwave cavity,” *Communications Physics* **8**, 236 (2025). DOI: 10.1038/s42005-025-02159-1.
- [R090] W. G. Farr, D. L. Creedon, M. Goryachev, K. Benmessai, and M. E. Tobar, “Ultrasensitive microwave spectroscopy of paramagnetic impurities of sapphire crystals at millikelvin temperatures,” *Physical Review B* **88**, 224426 (2013). DOI: 10.1103/PhysRevB.88.224426.
- [R091] P. F. Moulton, “Spectroscopic and laser characteristics of Ti:Al₂O₃,” *Journal of the Optical Society of America B* **3**, 125–133 (1986). DOI: 10.1364/JOSAB.3.000125.
- [R092] K. H. Lee and J. H. Crawford Jr., “Luminescence of the F center in sapphire,” *Physical Review B* **19**, 3217–3221 (1979). DOI: 10.1103/PhysRevB.19.3217.
- [R093] C. Wilhelmer, M. E. Turiansky, D. Waldhör, L. Cvitkovich, C. G. Van de Walle, and T. Grasser, “Optical properties of vacancies in aluminum oxide (alpha-Al₂O₃) from first principles,” *Physical Review Materials* **9**, 096202 (2025). DOI: 10.1103/5kvl-hktd.
- [R094] T. M. Hensen, M. J. A. de Dood, and A. Polman, “Luminescence quantum efficiency and local optical density of states in thin film ruby made by ion implantation,” *Journal of Applied Physics* **88**, 5142–5147 (2000). DOI: 10.1063/1.1314322.
- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001.
- [R079] G. Balasubramanian et al., “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* **8**, 383–387 (2009). DOI: 10.1038/nmat2420.
- [R095] B. D. Evans and M. Stapelbroek, “Optical properties of the F^+ center in crystalline Al₂O₃,” *Physical Review B* **18**, 7089–7098 (1978). DOI: 10.1103/PhysRevB.18.7089.

Chapter 9. A factory crystal and its neighbors

A wafer of silicon carbide is not a gem. It is already sitting in electric-car inverters: hard, wide-gap, sold by the crate. Someone can knock two neighboring atoms out of that wafer — one silicon, one carbon — and the leftover electrons still have a spin we can talk to.

That missing pair is the industrial cousin of the diamond vacancy. The crystal around it is a factory product. Wafers, ion beams, etches, electronics, and waveguides already exist for this material. We are going to see how far those tools take a spin, and then we are going to open the other drawers.

No winner. Each host is a toolbox. Silicon carbide is the drawer with the most industrial tools. Hexagonal boron nitride is a sheet. Rare-earth crystals are superb memories. Silicon is the other semiconductor. Oxides are a catalog. Diamond remains the benchmark from two chapters ago.

The question is narrower than “which material has the longest memory.” It is: which drawer lets us place, distinguish, couple, tune, and measure many spins under one compatible set of conditions?

9.1. One good spin is not a row of them

A single defect can be excellent. It can start on cue, hold still, and be heard. That is one spin.

A row of them is a different job. You need the same kind of defect in chosen sites, a way to turn one without turning the next, a way to hear each one, and a process that does not ruin the neighbors. A six-hour memory in the wrong geometry is still the wrong part. A neatly patterned grid of unidentified blinking lights is not a shortcut either.

Call the first scorecard single-defect quality: initialization, coherent control, coherence, and readout for one spin. Call the second an **array**: many spins that you can place, distinguish, couple, tune, and measure under one set of conditions. The name is worth keeping because it is a different object. Ten beautiful isolated spins are not an array.

Write the two scorecards as labels, not yet as numbers:

- Q_{single} is the quality of initialization, coherent control, coherence, and readout for one defect;

- Q_{array} is placement yield, interaction reproducibility, addressability, routing, and compatibility with fabrication.

The engineering rule is a bottleneck:

Q_{array} is limited by its weakest required operation.

A long T_2 cannot replace a missing two-spin interaction. Atomic placement cannot replace a missing readout. Multiplying record values from different samples, temperatures, and pulse sequences produces an imaginary device — usually a very photogenic one.

Drop the bottleneck and you may crown the host with the longest published memory. Keep it, and a six-hour nuclear coherence with no designed nearest neighbor is still not an array.

9.2. Two missing atoms

Silicon carbide's leading defects are a neighboring missing silicon and carbon, commonly written $V_{\text{Si}}V_{\text{C}}^0$, and a negatively charged missing silicon, V_{Si}^- .

The missing pair is a **divacancy**. The name just means two vacancies sitting next to each other. The neutral one has electronic spin $S = 1$. The silicon vacancy commonly used in 4H-SiC has $S = 3/2$.

Silicon carbide does not come in one stacking. The same atoms can pile up in more than one repeating pattern. The common wafer we will use is 4H-SiC. Different stackings and different lattice sites give different optical and spin spectra. That variety is either a tuning toolbox or a catalog-management problem, depending on the day.

Hence “a spin in SiC” is not one object. Charge, stacking, and site all change the spectrum before anyone writes a control pulse.

9.3. Dial one spin

Take a single spin-1 divacancy in 4H-SiC. Use dimensionless spin operators S_x, S_y, S_z , and write its ground-state energy as a frequency:

$$\frac{H}{h} = D \left(S_z^2 - \frac{S(S+1)}{3} \right) + E(S_x^2 - S_y^2) + \gamma_e \mathbf{B} \cdot \mathbf{S}.$$

Here H is energy in joules, h is Planck's constant in joule-seconds, D and E are axial and transverse zero-field-splitting frequencies in hertz, $\mathbf{B}$ is the magnetic field in tesla, and $\gamma_e = g\mu_B/h$ is the electron-spin gyromagnetic ratio in hertz per tesla. The electron g -factor is near 2, so $\gamma_e \approx 28 \text{ GHz T}^{-1}$.

For a field B_z along the defect axis and E small compared with the other terms, the microwave frequencies from $m_s = 0$ to $m_s = \pm 1$ are approximately

$$\nu_{\pm} = D \pm \gamma_e B_z.$$

A 2024 SiC-on-insulator experiment measured a PL6 center with $D = 1340.4 \pm 1.3 \text{ MHz}$ and a field slope of $2.82 \pm 0.02 \text{ MHz G}^{-1}$, while coherently controlling the implanted single spin at room temperature [R099]. Take $B_z = 5.0 \text{ mT} = 50 \text{ G}$. Then

$$\gamma_e B_z = (2.82 \text{ MHz G}^{-1})(50 \text{ G}) = 141 \text{ MHz},$$

so

$$\nu_- \approx 1.199 \text{ GHz}, \quad \nu_+ \approx 1.481 \text{ GHz}.$$

The units check: $\text{MHz/G} \times \text{G} = \text{MHz}$. An optical pulse initializes the spin, a microwave tone near one transition drives Rabi oscillations, and spin-dependent fluorescence or spin-to-charge conversion provides readout. **[Experiment]** Those operations have each been demonstrated for SiC divacancies, although not all record values occur in one room-temperature dense array [R096]; [R097]; [R098]; [R099].

Hence one implanted PL6 spin can be dialed. That is a Q_{single} result. It does not yet say who sits next door.

9.4. Talk to one, miss the next

Now make the same spin an addressability test. Suppose — this is a design assumption, not a reported array result — that local strain or a magnetic-field gradient separates neighboring transitions by $\Delta = 20 \text{ MHz}$. Drive one site with cyclic-frequency Rabi rate $\Omega = 2 \text{ MHz}$. A resonant π pulse lasts

$$t_{\pi} = \frac{1}{2\Omega} = 250 \text{ ns}.$$

The neighbor is not quite on resonance. The leading leftover population on the wrong site scales roughly as

$$p_{\text{xtalk}} \sim \left(\frac{\Omega}{\Delta}\right)^2 = 10^{-2}.$$

That leftover is **crosstalk**: you meant to turn one spin and you jiggled the next. The error is geometric, not moral. Larger Ω gives faster gates but more of it. Larger Δ helps selection but demands reproducible gradients or tuning, and can make nominally identical couplings unequal.

Selective control is still not an interaction. A two-spin coupling rate J must also beat the relevant linewidth and decoherence. The next chapter will put numbers on that bargain rather than hiding it under the word “scalable.”

The implanted object above is a physical defect-spin qubit. An electron plus a nearby ^{13}C nuclear spin is a small quantum register. Patterning 64 sites changes the fabrication count, not the category.

Drop the detuning and the same pulse that turns one site turns both. Keep a huge detuning and you may no longer have a uniform J_{ij} . Addressability and coupling pull in opposite directions.

9.5. Who is wired to whom

For many defects indexed by i , a minimal many-spin energy is

$$\frac{H_{\text{array}}}{h} = \sum_i \frac{H_i}{h} + \sum_{i < j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \frac{H_{\text{drive}}}{h}.$$

Here H_i is the single-defect Hamiltonian just defined, J_{ij} is a coupling frequency in hertz between sites i and j , and H_{drive} contains optical, microwave, electric, or strain control. The isotropic product $\mathbf{S}_i \cdot \mathbf{S}_j$ is only a placeholder. Magnetic dipolar coupling is anisotropic, exchange depends strongly on wavefunction overlap, and a cavity or a phonon has its own mode structure.

The list of who is coupled to whom, including the unwanted edges, is the **interaction graph**. Geometry is not the graph. Two spots on a photograph do not tell you J_{ij} .

An array proposal must therefore specify five maps:

1. **site map**: intended position $\rightarrow$ actual active defect and charge state;
2. **spectrum map**: actual defect $\rightarrow$ optical and microwave transition frequencies;
3. **control map**: control line or beam $\rightarrow$ addressed subset and crosstalk;
4. **interaction map**: geometry $\rightarrow J_{ij}$, including unwanted edges;
5. **measurement map**: physical signal $\rightarrow$ inferred local or collective observable.

Missing one map is not a small omission. It is usually where the device actually lives.

9.6. What the factory has actually shipped

[Experiment] Individual neutral divacancies in high-purity 4H-SiC have been optically isolated and coherently controlled; ensemble Hahn-echo coherence exceeded 1 ms at low temperature [R096]. In isotopically purified 4H-SiC at 5 K, a single divacancy reached $T_2 = 5.3 \pm 1.3$ s using up to 16,384 dynamical-decoupling pulses, with end-to-end spin-to-charge readout fidelity $80.8 \pm 0.6\%$ [R097]. The condition matters: “five-second coherence” is not a bare Hahn echo, not room temperature, and not an implanted dense array.

SiC also survives useful processing. **[Experiment]** Implanted V_{Si}^- centers retained nearly lifetime-limited optical emission and high spin coherence in etched nanophotonic waveguides, with control of nearby nuclear spins [R098]. **[Experiment]** In 2024, selective carbon implantation through an 8×8 mask of 100-nm-diameter holes created a designed PL6 array in SiC-on-insulator; the same study integrated an electron–nuclear register into a waveguide and reported ambient-condition entangled-state fidelities of 0.89 before and 0.88 after integration [R099]. This is unusually complete materials evidence.

[Experiment] The electronics path also advanced in 2025: room-temperature photoelectrical magnetic-resonance readout of a single V_{Si}^- spin produced a signal-to-noise ratio 1.7–2 times that of optical detection in the same study [R104]. This is evidence for integrable charge collection, not yet parallel high-fidelity readout of a coupled array.

It is not yet the desired interacting electron-spin lattice. The mask pitch and activation statistics do not imply nanometer-accurate final coordinates. One bright spot may contain zero, one, or several relevant defects. The 2024 entanglement was between one defect electron and a nearby nuclear spin, not between selected electron defects across the 8×8 pattern [R099]. SiC leads the industrial drawer because several necessary technologies coexist, not because the interaction graph has been demonstrated.

Hence Q_{single} for SiC is already serious, and pieces of Q_{array} exist. The bottleneck is still the missing electron–electron edges.

9.7. A crystal one atom thick

A monolayer or few-layer sheet of hexagonal boron nitride puts every defect close to a surface, a gate, a resonator, and a neighbor. That is attractive if you want a flat interaction graph and easy photonics. It is also attractive if you are a noisy adsorbate.

[Experiment] Single carbon-related hBN defects showed room-temperature optically detected magnetic resonance: the fluorescence changed when a microwave tone hit the spin.

In the reported material only 27 of more than 400 investigated isolated defects showed a measurable signal — about 5% — and saturated linewidths were about 35 MHz [R100]. The microscopic structures and spin multiplicities were not uniquely established.

This is a sharp example of promising single-site behavior with immature yield and identity control. A two-dimensional host shortens the route to an interface. It also shortens the route from every dirty molecule on the surface to the spin.

Drop the surface and you no longer have the sheet. Keep it, and chemical identity, uniformity, lifetime, and yield remain independent problems.

9.8. Ions that hide their electrons

Some crystals are doped with rare-earth ions such as Eu^{3+} , Er^{3+} , and Yb^{3+} . Their partly filled $4f$ shell sits inside closed outer shells, so the optical lines can be extremely narrow and the spin memories extremely long.

[Experiment] $^{151}\text{Eu}^{3+}$: Y_2SiO_5 reached 370 ± 60 minutes of hyperfine coherence at 2 K with dynamical decoupling [R101]. **[Experiment]** A single $^{171}\text{Yb}^{3+}$ ion in a YVO_4 nanophotonic cavity showed an optical linewidth below 1 MHz, spin coherence beyond 30 ms, and conditional single-shot readout above 95% [R102]. Those are superb node and memory credentials.

For a dense local energy, though, dilute dopants are commonly random in space, direct interactions are weak or inhomogeneous, and operation is generally cryogenic. Spectral multiplexing can identify many ions in one optical mode, but a frequency address book is not automatically a designed nearest-neighbor lattice.

Hence a long memory can raise Q_{single} without touching Q_{array} . The bottleneck here is the interaction map, not the clock.

9.9. The other semiconductor

Silicon is the strongest counterargument to an optical-defect-only worldview. Isotopically enriched ^{28}Si provides a quiet nuclear environment, industrial electronics are unmatched, and a scanning-tunneling-microscope hydrogen lithography method can place phosphorus donors on chosen lattice sites.

An electron in silicon can sit in more than one equivalent momentum pocket. Those pockets interfere, so the exchange energy between two nearby donors depends fiercely on the direction of the displacement, not only on the distance. That interference is a **valley** effect: the name is the shape of the energy surface in momentum space. **[Experiment/Theory]**

Atomic-scale imaging and modeling show that aligning donors along favorable crystallographic directions suppresses, but does not abolish, exchange variability [R103].

Gate-defined silicon spin qubits already support electrically controlled two-qubit logic in devices that are not color centers. Their penalties, if what you wanted was an optical defect lattice, are millikelvin operation, elaborate gate stacks and reservoirs, and the lack of a naturally bright, cycling optical transition for standard phosphorus-donor nodes. If the job is a programmable electrical spin energy rather than a spin–photon network, silicon may be the drawer you open first.

Drop the optical requirement and silicon’s Q_{array} case becomes the strongest in the chapter. Keep it, and the missing cycling transition is the bottleneck.

9.10. A drawer of oxides

“Oxide” is not one platform. The drawer includes chemically and structurally different hosts — MgO, ZnO, TiO₂, Y₂O₃, and many others. Some host transition-metal defects, oxygen vacancies, or rare-earth dopants. Rare-earth oxides overlap the previous drawer.

[Theory/Experiment] Reviews identify low nuclear-spin abundance, large band gaps, and mature thin-film processing as useful search criteria, while also emphasizing charge-state stability, optical cycling, and reproducible defect formation as independent requirements [R073]. As of the evidence surveyed here, no generic wide-gap oxide combines deterministic single-spin placement, a high-fidelity optical lifecycle, preserved coherence after nanofabrication, and a demonstrated controlled interaction graph. Oxides are a discovery space, not one backup device waiting on a shelf.

9.11. Side by side

The following matrix is deliberately about interacting arrays. “High coherence” means only what the cited experiment measured under its stated conditions. Ratings are comparative judgments, not hidden numerical data.

Host and representative spin	Native/engineered coupling route	Placement and yield	Coherence evidence	Photonics and readout	Maturity and typical temperature	Toolbox note
4H-SiC: neutral divacancy, V_{Si}^-	Nearby nuclear hyperfine demonstrated; electron dipolar/exchange at short range; cavity, strain, and electrical mediation plausible	Implantation, masks, annealing, commercial wafers; designed 8×8 PL6 pattern demonstrated, but not an atomically registered interacting electron array [R099]	> 1 ms ensemble Hahn echo at low temperature [R096]; single-spin 5.3 ± 1.3 s at 5 K with isotopic purification and 16,384-pulse decoupling [R097]	Near-infrared optical transitions, spin-to-charge readout, waveguides and SiC-on-insulator integration [R097]; [R098]; [R099]	Highest combined materials/device maturity here; room-temperature control exists, best resonant optical/readout records often cryogenic	Most complete optical-and-wafer toolbox here; activation statistics, final position, spectral disorder, and controlled electron-electron edges still open
hBN: boron vacancy and carbon-related spins	Short-range exchange/dipolar interactions are geometrically accessible in 2D; resonator coupling possible	Transfer, growth, irradiation and local writing are available; microscopic identity and reproducible spin-active yield remain weak	Room-temperature single-defect ODMR; about 5% ODMR-active yield in one >400 -defect survey and roughly 35 MHz saturated linewidth [R100]	Bright visible emitters and easy planar integration; spectral and photodynamic variability substantial [R100]	Rapidly developing, room-temperature demonstrations	High-upside planar toolbox, not yet a reproducible many-spin component library
Rare-earth crystals: Eu:YSO, Yb:YVO ₄ , Er:YSO	Weak direct ion-ion coupling; cavity-mediated and spectral-multiplexed links are natural	Dopants are usually dilute and spatially random; individual ions selected spectrally or by cavity overlap	Eu nuclear hyperfine 370 ± 60 min at 2 K with decoupling [R101]; single Yb spin >30 ms in a cavity [R102]	Exceptional narrow optical lines; single-ion cavity readout $>95\%$ conditional fidelity [R102]	Mature memories and emerging nanophotonics; usually kelvin-scale	Memory and network toolbox; less natural for a dense prescribed local interaction graph

Host and representative spin	Native/engineered coupling route	Placement and yield	Coherence evidence	Photonics and readout	Maturity and typical temperature	Toolbox note
Silicon: P donors or gate-defined electron spins	Electrically tunable exchange is strong; capacitive and resonator links available	CMOS gates plus STM donor placement can approach lattice-site precision; exchange remains sensitive to valley phase and interfaces [R103]	Long coherence in enriched ^{28}Si is established, but values depend strongly on electron versus nucleus, ensemble versus single device, and pulse sequence [R073]	Excellent electrical read-out/control; standard donor architecture lacks a bright cycling optical interface	Most mature semiconductor control stack; generally dilution-refrigerator temperatures	Strongest electrical programmable-array toolbox; less suited to an optical color-center route
Wide-gap oxides: oxygen vacancies, transition-metal or rare-earth dopants	Dipolar/exchange coupling possible in principle; oxide electronics and strain offer mediators	Thin-film processing is mature for some hosts, but quantum-active defect identity, charge state, activation and placement are host-specific [R073]	Isolated or ensemble coherence exists in selected systems; no transferable “oxide value” is meaningful [R073]	Some excellent emitters or microwave ensembles, rarely a complete single-spin lifecycle	Fragmented; temperatures range from ambient sensing to cryogenic spectroscopy	Discovery portfolio, not presently a validated interacting-array platform

Host and representative spin	Native/engineered coupling route	Placement and yield	Coherence evidence	Photonics and readout	Maturity and typical temperature	Toolbox note
Diamond baseline: NV and group-IV vacancies	Dipolar/exchange coupling, nuclear registers, and strong nanophotonic interfaces	Implantation and delta doping mature but deterministic nanometer-scale coupled arrays remain difficult	Platform-leading single-node demonstrations, with defect- and temperature-dependent trade-offs [R073]	Most mature color-center networking evidence; nanofabrication and surfaces can degrade performance	Mature, spanning room-temperature NV control to cryogenic group-IV optics	Benchmark rather than automatic winner; SiC offers stronger wafer/electronics integration, not yet stronger end-to-end coupled-array evidence

The matrix separates two claims that are often quietly merged:

- **[Experiment] single-defect quality:** SiC, rare-earth ions, silicon, diamond, and now hBN all contain individually controllable quantum systems, though with radically different evidence depth and conditions.
- **[Proposal] array suitability:** none of these hosts has demonstrated a dense, designed, coherently interacting defect lattice.

Pick the drawer for the job. For an optically addressable interacting-defect program, SiC and diamond are the two primary toolboxes, with SiC favored when monolithic wafer electronics and photonics dominate; hBN is a high-risk planar alternative; rare-earth crystals favor cavity-mediated or spectral architectures rather than short-range lattices; and selected oxides remain discovery targets. For an all-electrical exchange array, silicon moves to the front. The uncertainty is not cosmetic: a convincing two-dimensional hBN placement-and-yield result or a deterministic rare-earth implantation method could reorder the drawers.

9.12. Ways to get this wrong

1. **The record-value collage.** Combining a five-second SiC T_2 at 5 K under 16,384 pulses [R097], a room-temperature implanted array [R099], and a separate nanophotonic linewidth [R098] does not describe one device.

2. **Pattern equals placement.** A lithographic aperture locates an ion trajectory. Collision straggle, vacancy diffusion, annealing, charge conversion, and multi-defect occupancy determine the final active center. An optical spot at each mask location does not establish atomic coordinates.
3. **Addressability equals coupling.** Spectrally resolving two sites proves that controls can distinguish them. It does not measure J_{ij} , coherent exchange, or entanglement.
4. **Density equals a useful graph.** Raising defect density increases desired coupling and unwanted couplings, implantation damage, spectral crowding, and charge noise together. The crystal does not know which edges were drawn in the grant proposal.
5. **A long memory equals a fast processor.** Rare-earth nuclear coherence can be extraordinary [R101], while direct local interactions remain weak. A useful array needs a favorable ratio of gate or coupling rate to error rate, not only a large T_2 .
6. **Two-dimensional equals finished.** hBN gives geometric access, not automatically chemical identity, uniformity, lifetime, or yield. The reported single-carbon-related ODMR evidence itself documents substantial variability [R100].
7. **Semiconductor compatible equals process qualified.** SiC can host electronics and photonics, but each implantation, etch, oxide, metal, and anneal step must be tested against charge stability and spin-optical coherence. Compatibility is a research program, not a sticker.
8. **Coupled qubits equal a new phase of matter.** Even a perfect SiC or silicon spin array would initially be an ordinary interacting quantum system. A gapped many-body phase with nonlocal observables does not follow from the host matrix alone.

9.13. How fast is fast enough

Let J be a desired coupling frequency, $\Gamma_2 = 1/T_2$ an effective decoherence rate during the operation, σ_J the standard deviation of coupling disorder, Ω the control rate, and Δ_{addr} the nearest unwanted spectral detuning. A useful regime asks, schematically,

$$J \gg \Gamma_2, \quad J \gg \sigma_J, \quad \Gamma_2 \ll \Omega \ll \Delta_{\text{addr}}.$$

All symbols are frequencies in hertz, so the comparisons are dimensionally consistent. No host in the matrix has yet satisfied these inequalities across a large, designed defect array. They are the questions the next several chapters will quantify.

9.14. Check yourself

1. **What fails if the host with the longest reported T_2 is declared the winner?**
Placement, coupling, addressability, readout, yield, and compatible operating conditions remain independent bottlenecks. Q_{array} is limited by the weakest required operation.
2. **What is SiC's strongest array-level advantage?**
The coexistence of optically addressable spins with commercial wafers, implantation, electronics, mechanics, and monolithic photonics.
3. **Show that a 5.0 mT field on the cited PL6 center splits the two microwave lines to about 1.199 GHz and 1.481 GHz.**
 $\gamma_e B_z = (2.82 \text{ MHz G}^{-1})(50 \text{ G}) = 141 \text{ MHz}$, so $\nu_{\pm} = 1340.4 \pm 141 \text{ MHz}$.
4. **What fails if the 8×8 optical spots are treated as an interaction graph?**
The pattern does not give nanometer-registered electron coordinates, and the 2024 entanglement was electron–nuclear, not selected electron–electron edges.
5. **When might silicon outrank SiC?**
When the target is an all-electrical, exchange-coupled programmable spin array and a direct optical interface is not required.
6. **Show that the design-assumption crosstalk is 10^{-2} .**
With $\Omega = 2 \text{ MHz}$ and $\Delta = 20 \text{ MHz}$, $p_{\text{xtalk}} \sim (\Omega/\Delta)^2 = (1/10)^2 = 10^{-2}$. That leftover is not a measured array error; it is what the stated drive and detuning already predict.

9.15. Sources

- [R096] D. J. Christle, A. L. Falk, P. Andrich, *et al.*, “Isolated electron spins in silicon carbide with millisecond coherence times,” *Nature Materials* **14**, 160–163 (2015). DOI: 10.1038/nmat4144.
- [R097] C. P. Anderson, E. O. Glen, C. Zeledon, *et al.*, “Five-second coherence of a single spin with single-shot readout in silicon carbide,” *Science Advances* **8**, eabm5912 (2022). DOI: 10.1126/sciadv.abm5912; arXiv: 2110.01590.
- [R098] C. Babin, R. Stöhr, N. Morioka, *et al.*, “Fabrication and nanophotonic waveguide integration of silicon carbide colour centres with preserved spin-optical coherence,” *Nature Materials* **21**, 67–73 (2022). DOI: 10.1038/s41563-021-01148-3.
- [R099] H. Hu, Y. Zhou, A. Yi, *et al.*, “Room-temperature waveguide integrated quantum register in a semiconductor photonic platform,” *Nature Communications* **15**, 10256 (2024). DOI: 10.1038/s41467-024-54606-2.
- [R100] H. L. Stern, Q. Gu, J. Jarman, *et al.*, “Room-temperature optically detected magnetic resonance of single defects in hexagonal boron nitride,” *Nature Communications* **13**, 618 (2022). DOI: 10.1038/s41467-022-28169-z.

- [R101] M. Zhong, M. P. Hedges, R. L. Ahlefeldt, *et al.*, “Optically addressable nuclear spins in a solid with a six-hour coherence time,” *Nature* **517**, 177–180 (2015). DOI: 10.1038/nature14025.
- [R102] J. M. Kindem, A. Ruskuc, J. G. Bartholomew, *et al.*, “Control and single-shot readout of an ion embedded in a nanophotonic cavity,” *Nature* **580**, 201–204 (2020). DOI: 10.1038/s41586-020-2160-9; arXiv: 1907.12161.
- [R103] B. Voisin, J. Bocquel, A. Tankasala, *et al.*, “Valley interference and spin exchange at the atomic scale in silicon,” *Nature Communications* **11**, 6124 (2020). DOI: 10.1038/s41467-020-19835-1.
- [R073] G. Wolfowicz, F. J. Heremans, C. P. Anderson, *et al.*, “Quantum guidelines for solid-state spin defects,” *Nature Reviews Materials* **6**, 906–925 (2021). DOI: 10.1038/s41578-021-00306-y.
- [R104] T. Nishikawa, N. Morioka, H. Abe, *et al.*, “Coherent photoelectrical readout of single spins in silicon carbide at room temperature,” *Nature Communications* **16**, 3405 (2025). DOI: 10.1038/s41467-025-58629-1.

Part V

Interacting defect systems

Two needles that feel each other, then three noisy bits pretending to be one quieter bit.
Still local. Not yet a phase of matter.

Chapter 10. Two needles that can feel each other

Lay two tiny compass needles on a table. Neither throws a particle at the other. Each still makes a magnetic field, so the second needle can feel the first.

Twist the line that joins them. The push can grow, shrink, change sign, or vanish. Distance matters, and so does angle.

The energy of one magnetic moment sitting in the field of another is *dipole–dipole coupling*. We take that as the definition. What follows is that energy written for two electron spins in a crystal, and then a test of whether the number is large enough to use. A lonely megahertz is not a verdict.

10.1. The field of one needle

A magnetic moment is a tiny current loop. Far away, every loop with the same strength and direction looks the same. Call the first moment $\mathbf{m}_1$. Put a second moment a displacement $\mathbf{r}$ away. Write r for the length of $\mathbf{r}$, in metres, and $\hat{\mathbf{r}} = \mathbf{r}/r$ for the unit vector along the join.

The field the first moment makes at the second is

$$\mathbf{B}_1(\mathbf{r}) = \frac{\mu_0}{4\pi r^3} [3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}_1],$$

where μ_0 is the vacuum permeability, in N A^{-2} . The energy of moment $\mathbf{m}_2$ sitting in that field is $-\mathbf{m}_2 \cdot \mathbf{B}_1$. Multiply them and the two moments become an energy:

$$H_{dd} = \frac{\mu_0}{4\pi r^3} [\mathbf{m}_1 \cdot \mathbf{m}_2 - 3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})(\mathbf{m}_2 \cdot \hat{\mathbf{r}})].$$

A localized moment's field falls as $1/r^3$. Energy is moment times field, so the coupling falls the same way. Hence the power is the product of those two facts, not a slogan pasted on later.

Drop localization and the $1/r^3$ law is not a theorem. Overlapping electronic clouds are a different interaction, treated below. The far-field formula does not know about them.

10.2. Two electron spins

An electron spin is not a classical needle, but it still carries a magnetic moment. Write $\mathbf{S}$ for the angular-momentum operator, in joule-seconds, and $\hbar = h/(2\pi)$ for the reduced Planck constant. The dimensionless spin is $\mathbf{s} = \mathbf{S}/\hbar$. Its moment is $\mathbf{m} = -g\mu_B\mathbf{s}$, where g is the electron's dimensionless g -factor and μ_B is the Bohr magneton, in J T^{-1} .

Plug those moments into the energy above. The two minus signs cancel. Dividing by Planck's constant h turns joules into hertz:

$$\boxed{H_{dd} = h d(r) [\mathbf{s}_1 \cdot \mathbf{s}_2 - 3(\mathbf{s}_1 \cdot \hat{\mathbf{r}})(\mathbf{s}_2 \cdot \hat{\mathbf{r}})]}, \quad d(r) = \frac{\mu_0(g\mu_B)^2}{4\pi\hbar r^3}.$$

The quantity d is a frequency, in hertz. The derivation supplies both the angular tensor and the $1/r^3$ law. Neither is decorative.

Using $g = 2.0023$, $\mu_B = 9.2740101 \times 10^{-24} \text{ J T}^{-1}$, and $h = 6.62607015 \times 10^{-34} \text{ J s}$, the CODATA constants give [R105]

$$d(r) = \frac{52.04 \text{ MHz nm}^3}{r^3}.$$

A units check: $\mu_0\mu_B^2/r^3$ is joules. Divide by h , in joule-seconds, and you have $\text{s}^{-1} = \text{Hz}$. No leftover nanometres.

The table is that characteristic coefficient for two pointlike $g \approx 2$ electron spins. It does **not** include the angular factor or the matrix elements of a particular transition.

Separation r	$d(r)$	$1/d$	Noisy benchmark: $d/\Delta\nu^*$	Quiet benchmark: $d/\Delta\nu^*$
1 nm	52.04 MHz	19.2 ns	164	16,400
2 nm	6.505 MHz	154 ns	20.4	2,044
5 nm	416.3 kHz	2.40 μs	1.31	131
10 nm	52.04 kHz	19.2 μs	0.164	16.4
20 nm	6.505 kHz	154 μs	0.0204	2.04
50 nm	416.3 Hz	2.40 ms	0.00131	0.131

The two right columns are assumptions, not platform records. The noisy column takes free-induction coherence $T_2^* = 1 \mu\text{s}$; the quiet column takes $T_2^* = 100 \mu\text{s}$. For exponential loss of phase, the Lorentzian full width at half maximum is

$$\Delta\nu^* = \frac{1}{\pi T_2^*},$$

or 318 kHz and 3.18 kHz. A 52 kHz coupling at 10 nm sits above the quiet linewidth and inside the noisy one. Hence a bare “52 kHz” is not a gate verdict until it is compared with a linewidth measured under the same conditions.

At 1–2 nm the point-dipole arithmetic is still a useful reference, but electronic clouds may overlap. Then a different interaction, treated below, starts to matter, and the point-dipole-only model is incomplete. At 50 nm the magnetic interaction still exists. Whether you can use it depends on coherence and on how much control you can afford.

From here on, every Hamiltonian is written in frequency units: H/h is in hertz. When two defects i and j have a coherent matrix element J_{ij} , that number is whatever actually connects them. It is not automatically the same in every spin direction.

10.3. When the join is not along the field

An applied field picks a quantization axis. Call that axis z , and let θ be the angle between $\hat{r}$ and z . If the Zeeman or zero-field splitting is much larger than d , terms that rotate quickly can be dropped. Write $s^\pm = s^x \pm is^y$ for the raising and lowering operators. What remains is the energy-conserving piece:

$$\frac{H_{dd}^{\text{sec}}}{h} = d(r)(1 - 3 \cos^2 \theta) \left[s_1^z s_2^z - \frac{1}{4}(s_1^+ s_2^- + s_1^- s_2^+) \right].$$

That leftover piece is the **secular** interaction. The name is worth keeping because ordinary English suggests “worldly,” while here it means only the part a high-field experiment actually sees: the rest averages away.

The first term is a conditional frequency shift. The second swaps one spin excitation when the two transitions match. The angular factor is -2 when the defects sit along z , $+1$ when they sit in the transverse plane, and zero at the *magic angle* $\theta = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ$. Hence equal distances on a lattice can still give unequal, signed edges.

Suppose the useful two-state subspace is $H/h = j(|10\rangle\langle 01| + |01\rangle\langle 10|)$, with j in hertz. Then a complete excitation swap takes $t_{\text{swap}} = 1/(4|j|)$. A rough gate condition is $4|j|T_2 \gg 1$. Resolving the interaction in frequency generally needs $|j|$ larger than the relevant linewidth. Pulse sequences can refocus slow single-spin noise while keeping a chosen two-spin term, so T_2 , not T_2^* , may govern a carefully designed gate.

Near the magic angle the terms you dropped, and any small tilt, start to matter. Drop the hypothesis that the splitting dwarfs d , and the secular truncation is no longer justified. Pulse sequences cannot refocus relaxation, pulse error, or every fluctuating coupling.

10.4. Clouds that overlap

Press two electronic clouds close enough to share space. Electrons are indistinguishable fermions: swap them and the two-electron wavefunction changes sign. Different spin arrangements then cost different Coulomb energy.

Let $\phi_i(\mathbf{x})$ and $\phi_j(\mathbf{x})$ be the localized orbitals. The energy shift comes from the Coulomb matrix element in which their labels are crossed:

$$J_{ij}^{\text{ex}} \sim \iint d^3x d^3x' \phi_i^*(\mathbf{x}) \phi_j(\mathbf{x}) \frac{e^2}{4\pi\epsilon|\mathbf{x} - \mathbf{x}'|} \phi_j^*(\mathbf{x}') \phi_i(\mathbf{x}'),$$

where e is the elementary charge and ϵ is the host permittivity. That energy is **direct exchange**. Here the orbitals themselves overlap; later, virtual routes will not need that.

If an orbital envelope decays over a localization length a , overlap is roughly $e^{-r/a}$, so exchange commonly behaves like a polynomial times $e^{-2r/a}$. Prefactor, sign, orbital orientation, and exponent are material-specific.

[**Numerics**] Atomistic calculations for several diamond spin-center pairs found exchange larger than magnetic dipolar coupling below roughly 3 nm for NV pairs, with strong crystallographic and defect-orientation dependence [R107]. That is a calculated trend, not a demonstrated reproducible NV–NV gate. Exchange buys strength by spending placement tolerance.

Drop overlap and this term vanishes; the dipolar term remains. The two mechanisms are not interchangeable names for “spins that talk.”

10.5. A hop that is not allowed

Take two half-filled sites. An electron would like to hop from one to the other with amplitude t , in joules, but putting two electrons on one site costs U , also in joules, and U is large. Direct hopping is blocked in the low-energy subspace. A virtual doubly occupied state can still be visited and left.

Second-order perturbation theory then lowers the singlet relative to the triplet:

$$H_{\text{SE}} = J_{\text{SE}} \mathbf{s}_1 \cdot \mathbf{s}_2, \quad J_{\text{SE}} \approx \frac{4t^2}{U}.$$

This J_{SE} is an energy. Its frequency is J_{SE}/h . Hence a blocked hop still generates an effective spin–spin coupling, of a size set by how expensive the virtual visit is.

This is **superexchange**. It is not a grander name for direct exchange. Virtual charge hopping is the mechanism. If $t \propto e^{-r/a}$, the scale again falls approximately as $e^{-2r/a}$. With

an explicit bridge orbital it contains products of each defect-to-bridge hopping, so bond angles and bridge chemistry matter more than defect-to-defect distance alone.

The factor and sign can change for multiple orbitals, Hund coupling, or different filling, as Anderson's original analysis makes clear [R108]. Drop the hypothesis that U is large compared with t , and the low-energy spin-only description is no longer the right starting point.

10.6. Sharing a photon

Two optical transition dipoles talk through the electromagnetic field. In empty three-dimensional space the coherent near field scales as $1/r^3$. Retardation adds oscillatory $1/r^2$ and $1/r$ pieces once the separation is comparable to the optical wavelength. The same field also causes collective radiative decay, so a coherent level shift has to be compared with the optical homogeneous linewidth.

A resonator can replace free-space distance with a mode profile. Let a annihilate one cavity photon, σ_i^- lower emitter i , g_i be its coupling frequency, and Δ_i its detuning from the cavity. When $|\Delta_i| \gg |g_i|$, the cavity is only virtually occupied. Eliminating it leaves the scale

$$J_{12}^{\text{cav}} \approx \frac{g_1 g_2}{2} \left(\frac{1}{\Delta_1} + \frac{1}{\Delta_2} \right).$$

There is no universal $1/r^n$: g_i follows the electric-field amplitude at each site. If the cavity energy-decay rate is κ , it also contributes loss of order $(g_i/\Delta_i)^2 \kappa$. Hence the useful test is $|J_{12}^{\text{cav}}|$ larger than emitter dephasing and cavity-induced decay, not merely nonzero.

This leftover interaction is **cavity-mediated** coupling. The range is now the mode, not the Euclidean distance.

[Experiment] Two SiV centers coupled to a common diamond nanocavity exhibited spectrally resolved superradiant and subradiant optical states [R086]. That is coherent photon-mediated emitter interaction. It is not, by itself, a high-fidelity gate between long-lived ground-state spin qubits.

Drop the large-detuning hypothesis and the cavity is occupied, not a virtual mediator. Reporting $g_1 g_2 / \Delta$ without κ , temperature, and emitter linewidth is half a calculation.

10.7. Sharing a vibration

Strain is the spatial derivative of displacement. A defect whose orbital or spin energy shifts with strain acts as an elastic dipole. In an isotropic three-dimensional bulk solid,

displacement from a point force scales approximately as $1/r$. Differentiating at source and receiver gives a static elastic-dipole interaction that typically scales as $1/r^3$. **[Theory]** A 2024 lattice treatment derived this $1/r^3$ phonon-mediated law for NV centers and identified a quadrupole–quadrupole spin structure [R111]. Crystal anisotropy, surfaces, and nanobeams can change the tensor and the power law.

A quantized mechanical mode is a shared vibration both defects can lean on. Let b annihilate one quantum of that vibration, g_i^{ph} be the defect–mode coupling frequency, and Δ_i^{ph} the detuning. Dispersive elimination gives

$$J_{12}^{\text{ph}} \sim \frac{g_1^{\text{ph}} g_2^{\text{ph}}}{2} \left(\frac{1}{\Delta_1^{\text{ph}}} + \frac{1}{\Delta_2^{\text{ph}}} \right).$$

That shared quantum is a **phonon**. A waveguide Green function, rather than Euclidean distance alone, now sets the range.

An ideal single mode can link distant sites, but mechanical decay rate κ_m and thermal occupation $\bar{n} = [\exp(\hbar f_m/k_B T) - 1]^{-1}$ add noise. Here f_m is the mode frequency, k_B is Boltzmann’s constant, and T is temperature.

[Proposal] A quasi-one-dimensional diamond waveguide linking SiV centers by propagating phonons has been quantitatively analysed, including state emission and reabsorption [R109]. Treat this as a proposed network architecture, not evidence that a many-defect phonon bus already outperforms dipolar links.

Drop isotropy and the $1/r^3$ bulk law need not survive. Drop low thermal occupation and the same mode that mediates J_{12}^{ph} also injects noise of order $\kappa_m(2\bar{n} + 1)$.

10.8. Sharing a nucleus

An electron defect can feel a nearby nucleus. The coupling is a tensor $\mathbf{A}$, in hertz, that ties the electron spin $\mathbf{s}$ to the nuclear spin $\mathbf{I}$:

$$H_{\text{hf}}/\hbar = \mathbf{s} \cdot \mathbf{A} \cdot \mathbf{I}.$$

That tensor is the **hyperfine** interaction. It is how a nucleus becomes a mediator rather than just a memory.

If two electrons couple off-resonantly to the same nucleus, virtual nuclear flips generate an electron–electron scale $J_N \sim A_{1\perp} A_{2\perp} / \delta_N$, where $A_{i\perp}$ is a transverse hyperfine component and δ_N is the detuning of the virtual nuclear transition. Exact factors depend on the selected levels.

Dipolar hyperfine terms scale as $1/r_i^3$ with each electron–nucleus distance r_i , so the product can scale as $1/(r_1^3 r_2^3)$ at fixed detuning. Contact hyperfine coupling instead follows electron density at the nucleus and is short-ranged. Hence there is no single law in the electron–electron separation.

The reverse architecture — dipolar-coupled NV electrons mediating an interaction between their nuclear memories — has been developed theoretically [R110]. **[Proposal]** It illustrates a mediator hierarchy, not experimental proof of a nuclear bus between arbitrary defects. Nuclear spins offer long memory but slow coupling. Mediator initialization, spectral crowding, and electron back-action must sit in the linewidth budget.

Drop the off-resonance hypothesis and the nucleus is a real participant, not a virtual mediator. The product-of-hyperfines estimate then ceases to be the right starting point.

10.9. Electric needles

Magnetic moments are not the only pair that can feel each other at a distance. Some defect states carry a state-dependent electric moment $\mathbf{d}$, measured in coulomb-metres. In an isotropic dielectric the electrostatic energy is

$$H_{ee} = \frac{1}{4\pi\epsilon r^3} [\mathbf{d}_1 \cdot \mathbf{d}_2 - 3(\mathbf{d}_1 \cdot \hat{\mathbf{r}})(\mathbf{d}_2 \cdot \hat{\mathbf{r}})].$$

Same angular tensor. Same $1/r^3$ near field. Only the constants changed: permittivity ϵ instead of μ_0 , electric moments instead of magnetic ones.

As an illustrative upper-scale calculation, two 1-debye dipoles in a dielectric with relative permittivity $\epsilon_r = 5.7$ have coefficient $H_{ee}/h \approx 26.5 \text{ GHz nm}^3/r^3$. Here $1 \text{ D} = 3.33564 \times 10^{-30} \text{ C m}$. At 10 nm that is 26.5 MHz. If each qubit state differs by only 0.01 D, the state-dependent product is smaller by 10^{-4} , giving 2.65 kHz at 10 nm. Those dipoles are explicit assumptions, not measured NV values.

Ground-state spin levels usually couple to electric field only through spin–orbit, orbital, or strain admixture. Optical orbital transitions can couple much more strongly but decay faster. The NV electric and strain response, and its symmetry, are reviewed in [R074]. Electric coupling must beat charge-noise-induced Stark linewidth and leakage.

A large electric susceptibility is both a control knob and a larger antenna for unwanted electric fields. Drop the assumption that the state-dependent moment is a full debye, and the megahertz-scale estimate collapses by the product of the two reductions.

10.10. What a lab pair actually did

The cleanest pairwise benchmark is dipolar-coupled NV centers. **[Experiment]** A pair separated by 25 ± 2 nm had a measured effective dipolar coupling of 4.93 ± 0.05 kHz, and refocused optimal-control sequences produced electron-spin entanglement with reported fidelity above 0.82 at room temperature [R106]. Our point-dipole coefficient at 25 nm is $52.04 \text{ MHz}/25^3 = 3.33 \text{ kHz}$; an orientation or matrix-element factor of order unity makes the measured value consistent with magnetic dipolar coupling. Hence the experiment used spin echo and control engineering because free-induction noise could mask the interaction. It did not simply wait for a 4.93 kHz number to become a gate.

[Experiment] Millisecond-scale Hahn-echo coherence — up to 1.8 ms at room temperature in a specially isotopically engineered, low-spin-bath diamond sample — has been reported for single NV centers [R079]. That condition is not inherited automatically by dense implanted arrays. Damage, surfaces, unwanted paramagnetic defects, and inter-defect frequency disorder change both T_2 and yield.

Mechanism	Range or placement law	Evidence level relevant here	Coherent-use comparison
Magnetic dipole	$1/r^3$, angular tensor	Two-NV gates and entanglement demonstrated [R106]	j versus spin linewidth, $1/T_2$, pulse error
Direct exchange	roughly exponential; atomistic orientation dependence	Diamond-pair dominance below about 3 nm is numerical [R107]	J/h versus dephasing, leakage, placement disorder
Superexchange	$4t^2/U$; hopping often exponential	General mechanism established; defect implementation is material-specific [R108]	J/h versus charge/orbital noise and validity gap U/h
Free-space optical	near $1/r^3$, retarded oscillatory tail	Collective optical effects are platform-specific	coherent shift versus optical linewidth and collective decay
Cavity photon	cavity mode profile, not a simple power	Two-SiV coherent optical interaction demonstrated [R086]	J_{cav} versus emitter dephasing and induced κ loss
Static strain	often $1/r^3$ for bulk elastic dipoles	Single-defect response established [R074]; NV pair interaction derived [R111]	shift versus strain disorder and spectral diffusion
Phonon bus	mechanical Green function/mode profile	Bulk NV theory [R111]; SiV waveguide proposal [R109]	J_{ph} versus $\kappa_m(2\bar{n} + 1)$ and spin dephasing
Nuclear bus	product of hyperfine couplings	Closely related electron-mediated nuclear scheme proposed [R110]	J_N versus nuclear/electron decoherence and control time

Mechanism	Range or placement law	Evidence level relevant here	Coherent-use comparison
Electric dipole	near-field $1/(\epsilon r^3)$	Electric susceptibility established; pair gate is defect-specific [R074]	state-dependent shift versus Stark linewidth and leakage

This table refuses to rank mechanisms by coupling alone. A 10 MHz orbital interaction with a 100 MHz optical linewidth is not “stronger for quantum logic” than a 10 kHz spin interaction with a 100 Hz refocused linewidth.

Two coherently coupled defect spins are two interacting physical qubits. Several such spins projected onto a doublet may form one encoded qubit. Neither fact establishes topological order. Emergent anyons need a many-body Hamiltonian, a gapped phase, and nonlocal observables — not merely an interaction graph that resembles a drawing in a theory paper.

10.11. Ways to get this wrong

1. Quote an angular rate Ω in rad s^{-1} as if it were a cyclic frequency f in hertz. They differ by $\Omega = 2\pi f$. This chapter uses H/h and hertz throughout.
2. Treat 52.04 MHz nm^3 as every transition’s coupling. Drop the angular factor, or sit at the magic angle, and the secular term vanishes while $d(r)$ is still finite.
3. Call every indirect interaction superexchange. Virtual charge hopping is superexchange. A virtual cavity photon is photon mediation. A virtual phonon is phonon mediation. Similar second-order algebra does not make the microscopic noise identical.
4. Eliminate a cavity or mechanical mode and report only $g_1 g_2 / \Delta$. The same elimination generates inherited loss. Without κ , temperature, and emitter linewidth the coherent piece is not a budget.
5. Prefer the largest available J at any cost. Very short-range exchange can beat dipolar coupling and still be wildly nonuniform across nominally identical pairs. A target Hamiltonian may prefer a weaker, predictable edge.
6. Use the best coherence from one sample and the densest spacing from another. A valid feasibility budget must combine coupling, linewidth, yield, addressability, and temperature under the same fabrication conditions. Cross-paper arithmetic is useful for bounds, not proof of an integrated device.

10.12. Check yourself

1. **Show that magnetic dipolar coupling scales as $1/r^3$.**

A localized dipole's field scales as moment divided by r^3 . The second dipole's energy is its moment dotted into that field. The product therefore falls as $1/r^3$.

2. **Show that doubling an electron-spin separation divides d by eight.**

$d(r) \propto 1/r^3$, so $d(2r) = d(r)/8$. For $g \approx 2$, that is 52.04 MHz at 1 nm and 6.505 MHz at 2 nm.

3. **What fails if you take $d = 50$ kHz as a coherence verdict?**

The number alone cannot answer. Compare the selected matrix element with homogeneous linewidth, refocused T_2 , relaxation, detuning disorder, and control errors. A 52 kHz coupling at 10 nm sits inside a 318 kHz noisy linewidth and above a 3.18 kHz quiet one.

4. **Show that the secular angular factor vanishes at $\theta = \cos^{-1}(1/\sqrt{3})$.**

$1 - 3 \cos^2 \theta = 0$ when $\cos^2 \theta = 1/3$, so $\theta = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ$. Near that angle the terms dropped in the secular truncation, and any small tilt, start to matter.

5. **What fails if you drop the distinction between direct exchange and superexchange?**

Direct exchange needs overlapping orbitals and fermionic antisymmetry. Superexchange needs virtual hopping through energetically forbidden charge configurations, with the simplest scale $4t^2/U$. Collapse the names and you inherit the wrong range law, the wrong noise, and the wrong placement tolerance.

6. **What fails if a cavity-mediated J_{12}^{cav} is quoted without κ ?**

Position enters through mode amplitudes g_i , so there is no universal $1/r^n$. Cavity loss still contributes $(g_i/\Delta_i)^2 \kappa$. A nonzero coherent shift is not a gate if induced decay and emitter dephasing exceed it.

Those numbers and tensors are enough to write a small cluster Hamiltonian and ask whether it has an isolated low-energy doublet.

10.13. Sources

- [R105] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, "CODATA recommended values of the fundamental physical constants," NIST/CODATA database (2022 adjustment, accessed 2026), <https://physics.nist.gov/cuu/Constants/>.
- [R106] F. Dolde *et al.*, "High-fidelity spin entanglement using optimal control," *Nature Communications* 5, 3371 (2014). DOI: 10.1038/ncomms4371. arXiv:1309.4430.

- [R107] V. R. Kortan, C. Şahin, and M. E. Flatté, “Nanometer-scale exchange interactions between spin centers in diamond,” *Physical Review B* **93**, 220402(R) (2016). DOI: 10.1103/PhysRevB.93.220402. arXiv:1603.03485.
- [R108] P. W. Anderson, “Antiferromagnetism. Theory of superexchange interaction,” *Physical Review* **79**, 350–356 (1950). DOI: 10.1103/PhysRev.79.350.
- [R086] R. E. Evans *et al.*, “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* **362**, 662–665 (2018). DOI: 10.1126/science.aau4691. arXiv:1807.04265.
- [R109] M.-A. Lemonde *et al.*, “Phonon networks with silicon-vacancy centers in diamond waveguides,” *Physical Review Letters* **120**, 213603 (2018). DOI: 10.1103/PhysRevLett.120.213603. arXiv:1801.01904.
- [R110] A. Bermudez, F. Jelezko, M. B. Plenio, and A. Retzker, “Electron-mediated nuclear-spin interactions between distant nitrogen-vacancy centers,” *Physical Review Letters* **107**, 150503 (2011). DOI: 10.1103/PhysRevLett.107.150503. arXiv:1107.2617.
- [R074] M. W. Doherty *et al.*, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001. arXiv:1302.3288.
- [R079] G. Balasubramanian *et al.*, “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* **8**, 383–387 (2009). DOI: 10.1038/nmat2420.
- [R111] J. J. Nakane, K. Tahara, K. Kutsuki, and A. Yamakage, “Phonon-mediated spin-spin interaction: A general theory and application to diamond nitrogen vacancy centers,” *Physical Review B* **110**, 064428 (2024). DOI: 10.1103/PhysRevB.110.064428.

Chapter 11. Three noisy bits pretending to be one

Three compass needles sit on a table. Each jitters on its own. A truck goes by, a stray magnet swings past, the room warms a little — any of those can flip one needle, and the bit it was holding is gone.

Glue the three together with stiff springs. From across the room you no longer watch three independent jitters. You watch a single object that can move in two slow ways. Every other motion bends a spring and costs more energy than you are willing to spend.

We have not invented a quieter needle. We have written an energy for three spins and kept only a two-dimensional low-energy piece. That piece is this chapter: isolate the pair, say what a physical operator does inside it, and then be honest about what the trick buys.

11.1. Eight states, two that matter

Take three real spin-1/2 objects, numbered 1, 2, 3. Each can point up or down along a field, so the trio has eight product states. We will not treat those eight as eight independent answers. We will tie the spins so that, at low energy, only two combinations matter.

Call that tied trio a **cluster**. Later we will couple several such trios, weakly, and treat each one as a single two-level object.

Let $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ be the dimensionless spin of site i , so S_i^z has eigenvalues $\pm 1/2$. The energy inside one cluster is antiferromagnetic Heisenberg exchange plus a uniform field,

$$H_C = J \sum_{i < j} \mathbf{S}_i \cdot \mathbf{S}_j - b S_{\text{tot}}^z, \quad S_{\text{tot}}^z = \sum_{i=1}^3 S_i^z.$$

The exchange $J > 0$ and the Zeeman coefficient $b > 0$ both have units of energy, so every term in H_C does too. If a lab quotes frequencies, use J/h and b/h in hertz, where h is Planck's constant.

Three spins decompose into one total-spin $S_{\text{tot}} = 3/2$ quartet and two copies of a total-spin $S_{\text{tot}} = 1/2$ doublet. Restrict attention first to total magnetization $m = S_{\text{tot}}^z = +1/2$. In the product basis those two states can be written

$$|\tilde{0}\rangle = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}},$$

$$|\tilde{1}\rangle = \frac{|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle - 2|\uparrow\uparrow\downarrow\rangle}{\sqrt{6}}.$$

The first two arrows in $|\tilde{0}\rangle$ form a singlet. In $|\tilde{1}\rangle$ they occupy the triplet channel, coupled with spin 3 to total spin $1/2$. Direct inner products give unit norm and $\langle\tilde{0}|\tilde{1}\rangle = 0$.

Using

$$\sum_{i<j} \mathbf{s}_i \cdot \mathbf{s}_j = \frac{1}{2} \left(S_{\text{tot}}^2 - \sum_i S_i^2 \right),$$

and $S_i^2 = 3/4$, both of those states have the same energy

$$E_L = -\frac{3J}{4} - \frac{b}{2}.$$

Hence the two states span a two-dimensional eigenspace of H_C . That is not yet a useful bit: it is a degeneracy waiting for a control story.

If $J < 0$, the quartet lies lower and this pair is not the ground piece. The antiferromagnetic hypothesis is doing work.

11.2. The pair we keep

A pair of low eigenvalues is not a computer. The two states have to be controllable and readable. Unwanted states have to stay empty. Whatever we do to the three physical spins has to look, inside this pair, like the operation we meant.

When information is deliberately mapped into a chosen two-dimensional piece of a larger space, that piece is an **encoded qubit**. “Encoded” is a bookkeeping claim, not a compliment. The repetition state, this exchange-controlled pair, and an arbitrary pair of levels picked from a molecule are all encoded qubits if you put the information there on purpose.

Write the two kept states as $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$. Everything else in the eight-dimensional space of the trio is a way to leave the encoding.

Encoding is not, by itself, protection. Protection is a later calculation about a named noise.

11.3. The climb out of the pair

Any motion that is not one of the two kept states costs extra energy. That extra cost is what keeps the extra states empty — but only if the pokes trying to fill them are smaller than the cost.

Collect the two kept states with

$$P = |\tilde{0}\rangle\langle\tilde{0}| + |\tilde{1}\rangle\langle\tilde{1}|, \quad Q = I - P,$$

where I is the identity on all eight states of the three spins. The complementary piece Q is every state we have declared to be an escape.

For an exactly degenerate pair, the upward cost of an escape is

$$\Delta_{\text{leak}} = \min_{|a\rangle \in Q} (E_a - E_L),$$

provided every excluded eigenstate sits above E_L . That number is the **leakage gap**. It measures the climb *out* of the pair, not the quietness *inside* it.

If the intended pair itself has a small internal splitting, measure from the *top* of that pair to the nearest excluded level. Otherwise a large splitting inside the bit gets advertised as a large leakage gap, which it is not.

The other $S_{\text{tot}} = 1/2$ states have $m = -1/2$ and cost b . The lowest member of the $S_{\text{tot}} = 3/2$ quartet has $m = +3/2$ and costs $3J/2 - b$. Therefore

$$\boxed{\Delta_{\text{leak}} = \min\left(b, \frac{3J}{2} - b\right)}, \quad 0 < b < \frac{3J}{2}.$$

At $b = 0$, the opposite-magnetization partners sit at the same energy as the chosen pair, so the fixed- m subspace is not energetically isolated. Raising b helps until the polarized quartet is pulled down. At $b = 3J/4$, the two candidate escape costs are equal at $3J/4$.

If a perturbation has a characteristic matrix element v , also in energy units, then v/Δ_{leak} is dimensionless. The low-energy description needs that ratio small. A resonant control with angular Rabi rate Ω similarly needs $\hbar\Omega/\Delta_{\text{leak}} \ll 1$ unless a selection rule forbids the escape exactly.

A gap suppresses transitions only relative to the pokes trying to make them. It does not stop noise from acting *within* the pair. A perfectly isolated pair of levels can still dephase perfectly well.

11.4. What a physical operator looks like inside

We now need the rule that turns a real operation on the three spins into an operation on the pair. That rule is a two-by-two matrix, not a speech.

Define

$$\begin{aligned}\bar{Z} &= |\tilde{0}\rangle\langle\tilde{0}| - |\tilde{1}\rangle\langle\tilde{1}|, \\ \bar{X} &= |\tilde{0}\rangle\langle\tilde{1}| + |\tilde{1}\rangle\langle\tilde{0}|, \\ \bar{Y} &= -i|\tilde{0}\rangle\langle\tilde{1}| + i|\tilde{1}\rangle\langle\tilde{0}|.\end{aligned}$$

For any physical operator O , its first-order action on the pair is

$$O_{\text{eff}} = POP = c_0P + c_x\bar{X} + c_y\bar{Y} + c_z\bar{Z}.$$

The four coefficients are just the four matrix entries of O in the ordered basis $\{|\tilde{0}\rangle, |\tilde{1}\rangle\}$. The machine that does this is the projector P : sandwich the real operator, read the two-by-two.

For our three spins that calculation gives

$$\begin{aligned}P(\mathbf{S}_1 \cdot \mathbf{S}_2)P &= -\frac{1}{4}P - \frac{1}{2}\bar{Z}, \\ P(\mathbf{S}_2 \cdot \mathbf{S}_3)P &= -\frac{1}{4}P + \frac{1}{4}\bar{Z} - \frac{\sqrt{3}}{4}\bar{X}, \\ P(\mathbf{S}_1 \cdot \mathbf{S}_3)P &= -\frac{1}{4}P + \frac{1}{4}\bar{Z} + \frac{\sqrt{3}}{4}\bar{X}.\end{aligned}$$

Now let the three pairwise exchanges be independently tunable, J_{12}, J_{23}, J_{13} . Inside the pair,

$$PH_CP = E_{\text{off}}P + h_x\bar{X} + h_z\bar{Z},$$

where

$$E_{\text{off}} = -\frac{J_{12} + J_{23} + J_{13}}{4} - \frac{b}{2}, \quad h_x = \frac{\sqrt{3}}{4}(J_{13} - J_{23}), \quad h_z = \frac{J_{13} + J_{23} - 2J_{12}}{4}.$$

All three coefficients have energy units. Hence changing pairwise exchange rotates the encoded qubit about two nonparallel axes on its Bloch sphere. Pulse sequences generate the third axis through noncommuting rotations. That is the core exchange-only idea proposed for three-spin qubits [R114] and later demonstrated in semiconductor quantum dots [R115]. **[Theory/Experiment]** Isotropic exchange also commutes with S_{tot}^2 and S_{tot}^z , so

ideal exchange pulses do not couple this pair to the quartet. Real anisotropy, field gradients, timing errors, and coupling to higher orbital or defect levels restore leakage.

Once several clusters sit nearby, write the full energy as

$$H = \sum_C H_C + \lambda \sum_{\langle C,D \rangle} V_{CD}.$$

Here C and D label clusters, H_C is the strong energy inside cluster C , V_{CD} couples neighbors, and the dimensionless bookkeeping parameter λ says the coupling between clusters is weak. If each H_C has a well-isolated pair, the large microscopic space *may* reduce to one effective two-level system per cluster.

That word *may* is doing honest work. The states must be controllable and readable. Unwanted states must stay empty. Projected interactions must have the form you need. The fabrication process must reproduce the spectrum.

Projecting two clusters.

Let P_C project cluster C onto its pair and let $P_{\text{all}} = \prod_C P_C$. To first order in weak intercluster coupling,

$$H_{\text{eff}} = P_{\text{all}} H P_{\text{all}} = \sum_C P_C H_C P_C + \lambda \sum_{\langle C,D \rangle} (P_C P_D) V_{CD} (P_C P_D).$$

Each projected pair term can be expanded in $\bar{\sigma}_C^\mu \bar{\sigma}_D^\nu$, where $\mu, \nu \in \{0, x, y, z\}$ and $\bar{\sigma}^0 = P$. Virtual visits to Q add corrections of characteristic scale $\lambda^2 \|V\|^2 / \Delta_{\text{leak}}$. Chapter 22 will derive that statement rather than asking it to pass on charm.

Drop the isolation of each cluster and the product projector is not a low-energy description. A nonzero QVP causes leakage and virtual corrections; the small parameter is a coupling matrix element divided by Δ_{leak} , not confidence divided by typography.

11.5. A bath that cannot tell them apart

Encoding alone makes no protection claim. Protection is a statement about a specific noise and a specific calculation.

Suppose the trio couples to a bath through

$$H_{SB} = S_{\text{tot}}^z \otimes B_z,$$

where B_z is an unspecified bath operator. In the fixed- $m = +1/2$ pair,

$$PS_{\text{tot}}^z P = \frac{1}{2}P.$$

The bath therefore cannot learn whether the encoded state was $|\tilde{0}\rangle$ or $|\tilde{1}\rangle$. It adds the same phase to both. A subspace with that property, for that noise, is a **decoherence-free subspace** [R112]. **[Theory]** The name is worth keeping because ordinary English suggests that nothing decoheres, while the claim is always conditional: name the noise operators, then show that each acts as a scalar on the code.

It is not a decoherence-free subspace for independent noise $\sum_i S_i^z \otimes B_i$. Different B_i operators can distinguish the spin patterns. “The environment is probably common-mode” is not that calculation.

11.6. A bath that can spin the box

Under fully collective noise the interaction can contain all three components,

$$H_{SB} = \sum_{\alpha=x,y,z} S_{\text{tot}}^\alpha \otimes B_\alpha.$$

A collective sideways rotation changes m , so the fixed- m pair is not decoherence-free for this noise. Yet the entire total-spin-1/2 sector factorizes as

$$\mathcal{H}_{1/2} \cong \mathbb{C}_{\text{logical}}^2 \otimes \mathbb{C}_{\text{gauge}}^2.$$

Collective spin operators act on the second factor and leave the first one alone. Information stored in that first factor is a **noiseless subsystem** [R113]. **[Theory]** The second factor — a gauge spin — may wander without erasing the logical state.

This is stronger than the fixed-magnetization decoherence-free subspace against longitudinal noise, but only for genuinely collective coupling.

Three equal-exchange spins have a four-dimensional $S = 1/2$ sector. A two-state projector therefore needs a fixed magnetization, another splitting mechanism, or this noiseless-subsystem reading. Forgetting the extra two states is how the gauge degeneracy sneaks back in.

11.7. Exchange as the only knob

An **exchange-only** qubit uses controllable pair exchanges rather than separately addressed single-spin rotations [R114]; [R115]. The hardware claim is about which knobs exist, not about how quiet the bit is.

Since exchanges commute with collective rotations, they naturally stir the logical factor of the three-spin noiseless subsystem and ignore its gauge orientation.

Magnetic dipolar coupling is anisotropic. Assuming exchange is available because “spins interact” confuses the operator you have with the operator the encoding asked for. Each physical interaction must be projected from its actual form.

11.8. When a symmetry forbids an escape

A different claim is that a symmetry, not a gap slogan, is doing the forbidding.

If a symmetry generator G commutes with H_C , eigenstates can be labeled by symmetry sectors. A perturbation that also commutes with G has zero matrix elements between inequivalent sectors. That is **symmetry protection**: a vanishing matrix element, not a vibe.

In the toy model, isotropic exchange preserves total spin, while the longitudinal field preserves total magnetization. Those facts can close some leakage doors.

They do not automatically protect a logical splitting. Both logical states above share $S_{\text{tot}} = 1/2$ and $m = +1/2$, and unequal symmetry-allowed exchanges produce $\bar{X}$ and $\bar{Z}$ at first order. Hence to claim protection, identify the symmetry, the representation carried by the code, and the symmetry of the actual perturbation.

Drop the hypothesis that the perturbation commutes with G , and the vanishing matrix element is gone. Anisotropy, field gradients, and orbital admixture are the usual ways it leaves.

11.9. Local operators still see the secret

The two states we kept live on three nearby spins. You can tell them apart by looking at a local pair: $\mathbf{S}_1 \cdot \mathbf{S}_2$ is different in $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$. Unequal local exchange splits them at first order. A closed path among the three sites does not pick up a logical phase that depends on the shape of the path. Hence the logical operators are supported on three nearby spins, and the degeneracy does not depend on the topology of a macroscopic surface.

That is encoding. It is not topological order. **Topological order**, when it exists, is a property of a whole phase of matter: no small local poke can read the information or split it at first order, and the excitations that carry it can be moved around a large lattice. People later call those objects anyons. Our cluster is not one of them. The three physical spins are ordinary qubits; the pair selected by P is one encoded qubit; driving those spins so their amplitudes reproduce a desired model is digital or analog emulation. None of those facts is an emergent phase.

11.10. What the laboratory actually has

The exact Heisenberg triangle is a teaching model. Color-center hardware supplies spin-1 NV electrons, spin-1/2 or larger nuclear memories, anisotropic dipolar and hyperfine tensors, optical reset, strain, and imperfect placement. Selecting two levels of a spin-1 center can make an effective physical qubit, but it does not turn magnetic dipolar coupling into isotropic exchange.

A literature search through August 2026 is most useful when it begins with inclusion criteria. For a direct realization of the worked example, ask for: (i) at least three deliberately used spins associated with color centers; (ii) a static strong intracenter Hamiltonian; (iii) a spectroscopically isolated logical doublet; and (iv) control through projected pair interactions. Related literature sorts into four bins.

1. **Encoded center–nuclear registers exist.** [Experiment] Reiserer *et al.* encoded a diamond network memory in two nearby ^{13}C spins whose antiparallel states suppress common-mode phase shifts during repeated NV optical operations [R116]. Cramer *et al.* encoded a qubit across three ^{13}C spins and repeatedly extracted and corrected error syndromes with an NV ancilla [R117]. These are genuine encoded color-center-associated clusters, but the protection comes respectively from a decoherence-free subspace and active error correction, not from the ground doublet of a three-center exchange Hamiltonian.
2. **Spin-pair encodings have become strong experimental memories.** [Experiment] Bartling *et al.* used pairs of ^{13}C nuclei near NV nodes as antiparallel spin-pair qubits, reported intrinsic dephasing times exceeding one minute, and entangled two such encoded qubits [R118]. The “exceeding one minute” condition belongs to those measured nuclear pairs and protocols, not to arbitrary color-center clusters. This is compelling evidence for noise-selective encoding, not for passive many-body topological protection.
3. **Large local registers and coupled centers are real, but they do not satisfy all four criteria.** [Experiment] A ten-qubit diamond register combined one NV electron with its nitrogen nucleus and eight ^{13}C nuclei under gate-based control [R119]. Separately, two NV electron spins have been entangled through their direct

dipolar interaction [R106]. The former is a programmable register; the latter is an interacting pair. Neither paper reports a three-color-center low-energy encoded qubit.

4. **Multi-NV decoherence-free architectures have been proposed.** [Proposal] Yun *et al.* proposed geometric gates in a decoherence-free subspace using NV centers coupled through cavity-QED resources [R120]. This is a genuine color-center decoherence-free-subspace proposal, but its logical space and gates are engineered by drives and a mediator; it is not evidence that a fabricated, strongly coupled NV triangle naturally has the spectrum and projected exchanges derived above.

The scoped verdict is therefore precise: the literature contains genuine encoded qubits built from spins around color centers, genuine coupled color centers, and genuine multi-center decoherence-free-subspace gate proposals. In the sources identified here, no experiment or proposal simultaneously establishes the particular chain “three deliberately positioned color-center electronic spins $\rightarrow$ static exchange-dominated isolated doublet $\rightarrow$ projected exchange-only qubit.” This is a search result, not a proof that no paper exists. It marks the missing link that a defect-cluster proposal must supply rather than quietly borrowing from semiconductor quantum dots.

11.11. Ways to get this wrong

1. Call any two low levels a protected qubit. The leakage gap concerns transitions out of the code. Noise projected as $\bar{Z}$ still dephases inside it.
2. Forget the extra two states. Three equal-exchange spins have a four-dimensional $S = 1/2$ sector. A two-state projector requires fixed magnetization, another splitting mechanism, or the noiseless-subsystem reading.
3. Project and then discard Q forever. A nonzero QVP causes leakage and virtual corrections of scale $\lambda^2 \|V\|^2 / \Delta_{\text{leak}}$.
4. Assume exchange is available because spins interact. Magnetic dipolar coupling is anisotropic. Hyperfine coupling joins unlike spins. Optical and strain mediators bring driven dynamics and loss. Each must be projected from its actual operator.
5. Treat a decoherence-free subspace as generic quietness. It cancels specified correlated noise. Independent fields, coupling mismatch, relaxation, optical back-action, and control error can remain.
6. Confuse active correction with a static energy gap. Repeated syndrome extraction can extend a logical lifetime, but the protection disappears if the correction cycle stops. A leakage gap is a static spectral property. They can coexist; they are not the same mechanism.

7. Call the cluster an anyon, or call the encoding topological. Local pair correlations distinguish the two states. Local exchange splits them at first order. That is a finite encoded object, not a deconfined excitation of a many-body phase.

11.12. Check yourself

1. **Show that $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$ are orthonormal.**

Expanding in the product basis, $\langle\tilde{0}|\tilde{1}\rangle = \frac{1}{\sqrt{12}}(1 - 1 + 0) = 0$. Each coefficient list already has squared magnitudes adding to one. Equivalently, the singlet channel of the first two spins is orthogonal to the triplet channel.

2. **Show that $\Delta_{\text{leak}} = \min(b, 3J/2 - b)$ for $0 < b < 3J/2$.**

The $m = -1/2$ partners cost b . The lowest quartet member costs $3J/2 - b$. The leakage gap is the smaller of those two climbs.

3. **What fails if $b = 0$?**

The opposite-magnetization partners sit at the same energy as the chosen pair, so the fixed- m subspace is not energetically isolated. The leakage gap formula assumed $0 < b < 3J/2$.

4. **Show that $PS_{\text{tot}}^z P = \frac{1}{2}P$ on the pair.**

Both $|\tilde{0}\rangle$ and $|\tilde{1}\rangle$ have $m = +1/2$, so S_{tot}^z acts as the scalar $1/2$ on the span of P . That is why collective longitudinal noise cannot tell the encoded states apart.

5. **What fails if the noise is $\sum_i S_i^z \otimes B_i$ rather than $S_{\text{tot}}^z \otimes B_z$?**

Different B_i can distinguish the spin patterns. The pair is then not a decoherence-free subspace for that noise. “The environment is probably common-mode” is not a substitute for the calculation.

6. **Why is this encoded qubit not topological?**

Local pair operators distinguish $|\tilde{0}\rangle$ from $|\tilde{1}\rangle$, and unequal local exchange splits them at first order. The logical operators live on three nearby spins. That is a finite encoding, not a topologically ordered phase.

11.13. Sources

- [R112] P. Zanardi and M. Rasetti, “Noiseless quantum codes,” *Physical Review Letters* **79**, 3306–3309 (1997). DOI: 10.1103/PhysRevLett.79.3306.
- [R113] E. Knill, R. Laflamme, and L. Viola, “Theory of quantum error correction for general noise,” *Physical Review Letters* **84**, 2525–2528 (2000). DOI: 10.1103/PhysRevLett.84.2525. arXiv:quant-ph/9908066.

- [R114] D. P. DiVincenzo, D. Bacon, J. Kempe, G. Burkard, and K. B. Whaley, “Universal quantum computation with the exchange interaction,” *Nature* **408**, 339–342 (2000). DOI: 10.1038/35042541. arXiv:quant-ph/0005116.
- [R115] E. A. Laird, J. M. Taylor, D. P. DiVincenzo, C. M. Marcus, M. P. Hanson, and A. C. Gossard, “Coherent spin manipulation in an exchange-only qubit,” *Physical Review B* **82**, 075403 (2010). DOI: 10.1103/PhysRevB.82.075403. arXiv:1005.0273.
- [R116] A. Reiserer, N. Kalb, M. S. Blok, K. J. M. van Bemmelen, D. J. Twitchen, M. Markham, T. H. Taminiau, and R. Hanson, “Robust quantum-network memory using decoherence-protected subspaces of nuclear spins,” *Physical Review X* **6**, 021040 (2016). DOI: 10.1103/PhysRevX.6.021040. arXiv:1603.01602.
- [R117] J. Cramer, N. Kalb, M. A. Rol, B. Hensen, M. S. Blok, M. Markham, D. J. Twitchen, R. Hanson, and T. H. Taminiau, “Repeated quantum error correction on a continuously encoded qubit by real-time feedback,” *Nature Communications* **7**, 11526 (2016). DOI: 10.1038/ncomms11526. arXiv:1508.01388.
- [R118] H. P. Bartling, M. H. Abobeih, B. Pingault, M. J. Degen, S. J. H. Loenen, C. E. Bradley, J. Randall, M. Markham, D. J. Twitchen, and T. H. Taminiau, “Entanglement of spin-pair qubits with intrinsic dephasing times exceeding a minute,” *Physical Review X* **12**, 011048 (2022). DOI: 10.1103/PhysRevX.12.011048. arXiv:2103.07961.
- [R119] C. E. Bradley, J. Randall, M. H. Abobeih, R. C. Berrevoets, M. J. Degen, M. A. Bakker, M. Markham, D. J. Twitchen, and T. H. Taminiau, “A ten-qubit solid-state spin register with quantum memory up to one minute,” *Physical Review X* **9**, 031045 (2019). DOI: 10.1103/PhysRevX.9.031045. arXiv:1905.02094.
- [R106] F. Dolde *et al.*, “High-fidelity spin entanglement using optimal control,” *Nature Communications* **5**, 3371 (2014). DOI: 10.1038/ncomms4371. arXiv:1309.4430.
- [R120] M.-R. Yun, F.-Q. Guo, L.-L. Yan, E. Liang, Y. Zhang, S.-L. Su, C. X. Shan, and Y. Jia, “Parallel-path implementation of nonadiabatic geometric quantum gates in a decoherence-free subspace with nitrogen-vacancy centers,” *Physical Review A* **105**, 012611 (2022). DOI: 10.1103/PhysRevA.105.012611.

Part VI

Topology and anyons

A shoelace, a coffee cup, then two particles that trade places in a plane. Topology first.
Strange particles after.

Chapter 12. A shoelace through a coffee cup

Take a coffee cup and a shoelace. Thread the lace through the handle and tie the ends so you have a closed loop. You can slide the lace around, stretch it into a circle, squash it into a potato, even wrap it twice around the handle. You cannot pull it off the cup without cutting the lace or breaking the handle.

The lace has a shape you can change, and it has a relationship to the hole that no polite tugging can change. We will count that relationship, see why one inch of lace cannot tell you the count, and then ask why a similar stubbornness might hide a bit.

12.1. Stretching is allowed, cutting is not

Treat the handle as a nail driven through a tabletop. Lay a closed loop of string on the table. The loop may be circular, triangular, or shaped like an indecisive potato.

You may slide and stretch the string while obeying three rules:

- it must stay on the tabletop;
- it must remain a closed loop;
- it may not pass through the nail or be cut and rejoined.

If the loop does not surround the nail, you can shrink it to a tiny loop at an ordinary point. If it surrounds the nail once, shrinking gets stuck: the nail is in the way. A loop wound twice is different again. The shape is flexible. The relationship to the nail is not.

A movie of that allowed stretching is a *deformation*. We take this as the definition. The permitted moves, not the pretty shapes, are the object.

Suppose cutting the string is permitted. Then every loop can be undone, and the nail does no work. Suppose the string may pass through the nail. Then the hole stops distinguishing anything. A claim that something survives stretching is always relative to a space and a set of permitted moves.

The nail itself is not magical. We just refuse to let the string occupy one point of the table. That missing point is the hole.

Here is the whole toy world:

not wound



wound once



• = missing point

The left loop can contract without touching the missing point. The right loop cannot.

12.2. How many times around

Put the missing point at Cartesian coordinates

$$\mathbf{p} = (a, b),$$

where a and b are lengths measured along the two axes. Describe the loop by

$$\mathbf{r}(t) = (x(t), y(t)), \quad 0 \leq t \leq 1.$$

The parameter t merely tells us where we are along the loop; it has no physical units. Closing the loop means $\mathbf{r}(0) = \mathbf{r}(1)$. Missing the hole means $\mathbf{r}(t) \neq \mathbf{p}$ for every t .

From the hole, look toward the point $\mathbf{r}(t)$. Let $\theta(t)$ be the continuously unwrapped angle of that direction, measured in radians. “Unwrapped” means that after one counterclockwise turn the angle is allowed to increase from 0 to 2π , rather than jumping back to zero.

The net number of turns is

$$w = \frac{\theta(1) - \theta(0)}{2\pi}.$$

Both angles and radians are dimensionless, so w is dimensionless. Because the loop ends where it began, its final direction from the hole must equal its initial direction. The unwrapped angle can therefore differ only by a whole number of turns:

$$\theta(1) - \theta(0) = 2\pi w, \quad w \in \mathbb{Z},$$

where $\mathbb{Z}$ denotes the integers. Counterclockwise is positive by convention. Clockwise is negative.

That integer w is the *winding* of the loop about the hole. It is a count of turns, not a length or a curvature.

Take the circle of radius $R > 0$, where R has units of length:

$$x(t) = a + R \cos(2\pi t), \quad y(t) = b + R \sin(2\pi t).$$

Its direction angle is $\theta(t) = 2\pi t$. Thus

$$w = \frac{2\pi - 0}{2\pi} = 1.$$

Reverse the direction by replacing t with $-t$, interpreted modulo one, and $w = -1$. Traverse the same circle twice with $\theta(t) = 4\pi t$, and $w = 2$. Put a small circle entirely to one side of the hole, and its direction first swings one way and then swings back; the net change is zero, so $w = 0$.

For a differentiable loop, the same quantity can be calculated directly from its coordinates:

$$w = \frac{1}{2\pi} \int_0^1 \frac{[x(t) - a] \dot{y}(t) - [y(t) - b] \dot{x}(t)}{[x(t) - a]^2 + [y(t) - b]^2} dt.$$

A dot means differentiation with respect to the dimensionless parameter t . The numerator and denominator both have units of length squared, so the integrand and w are dimensionless, as required.

For our circle,

$$\dot{x}(t) = -2\pi R \sin(2\pi t), \quad \dot{y}(t) = 2\pi R \cos(2\pi t).$$

Substitution makes the fraction equal to 2π , because the numerator is $2\pi R^2$ and the denominator is R^2 . Therefore

$$w = \frac{1}{2\pi} \int_0^1 2\pi dt = 1.$$

Now squash the circle continuously into an ellipse. Its circumference and curvature change. Its winding remains one, provided the ellipse never crosses the hole. Discard the detailed shape. Keep the coarser count that is hard to change.

During a continuous deformation the loop changes continuously while remaining away from the hole, so w would have to change continuously. But w is restricted to integers. A continuous function taking integer values cannot slide from 1 to 0; it would have to pass

through nonintegers. The only escape is for the winding formula to stop being defined, which occurs when the loop hits the hole.

Hence an integer of this kind cannot drift.

Drop closedness and w need not be an integer: an open arc can change its direction by any real angle. Drop the rule that the loop miss the hole, and the formula is undefined at the instant of crossing. Either hypothesis is doing work.

12.3. One inch of string

Aim a microscope at one short segment of the right-hand loop. You can measure how bent it is. You can measure how long that scrap is. You cannot tell whether the rest of the loop closes around the hole.

A property you can decide from a bounded neighborhood is *local*. The loop's curvature at one point is local. A kink can be smoothed by changing only a short segment.

The opposite kind of property depends on the whole object, or on how it sits in the whole space. Winding is that kind. Inspecting a short segment cannot reveal whether the completed loop encloses the hole.

The contrast can be summarized without mysticism:

Question	Kind of information	Changed by
What is the loop's length?	geometric, global	ordinary stretching
What is its curvature here?	geometric, local	smoothing nearby
Does it pass through this point?	geometric / incidence	moving one segment
What is its winding about the hole?	global, survives stretching	crossing the hole, cutting, or changing closure

Almost everything about the shape can change. Selected discrete information survives a specified class of continuous changes. Topology is the study of those surviving distinctions. It is about allowed moves, not about pretty lattices.

A honeycomb, a ring, or a device with a hole drilled in it can look suggestive. Coordinates and connectivity do not, by themselves, decide which counts survive.

12.4. Why a knot can protect a bit

Suppose the bit you care about is not “is this scrap of string bent?” but “does the closed loop go through the handle?”

A local tug changes a short segment. Under the rules above, a short segment cannot change the count. To flip the bit you must cut the lace, break the handle, or walk the whole loop off the hole. Those are bigger jobs. They get harder as the relevant distance grows.

A knot in a rope is the same stubbornness in three dimensions: you cannot untie it by stretching one inch. Our tabletop loop is the flatter cousin. Either way, the information lives in how the whole thing sits, not in any one inch.

That last sentence is the useful analogy for a protected quantum bit. It is only an analogy. A classical string around a nail is not an entangled quantum state, and its winding is not yet a computer bit. After this section we will stop talking about shoelaces and start talking about labels.

The protection mechanism, if the analogy is going to earn its keep, has to look like this. A local disturbance resembles changing a short segment of string: it cannot change winding. To alter a global quantum label, local errors must form an extended chain, an excitation must travel around a route that cannot be shrunk, or the perturbation must destroy the phase that made the label meaningful. The required process becomes harder as the relevant distance grows.

But “harder” is not “impossible.” Local errors can accumulate over time. Boundaries may shorten a dangerous path. Thermal excitations can move. A finite device has residual splitting between labels. Correlated noise need not be local. Initialization and measurement can fail. A stubborn count changes the error mechanism. It does not repeal error mechanisms.

classical loop -----	quantum many-body system -----
short reshaping	local operator or local noise
winding count	nonlocal quantum label
cross hole / cut loop	extended error / gap-closing event
winding number	loop measurement or charge label

The rest of the chapter puts names on those rows. The names are not the protection. The protection, when it exists, is that a local probe cannot read or flip the label.

12.5. A movie of two loops

Suppose we have an initial loop $\mathbf{r}_0(t)$ and a final loop $\mathbf{r}_1(t)$. A continuous deformation is a movie connecting them, frame by frame, without tearing the loop or touching the hole.

Such a movie is a *homotopy*. We write it as

$$\mathbf{H}(t, s), \quad 0 \leq t \leq 1, \quad 0 \leq s \leq 1.$$

Here t locates a point along the loop and s labels the movie frame. The conditions are

$$\mathbf{H}(t, 0) = \mathbf{r}_0(t), \quad \mathbf{H}(t, 1) = \mathbf{r}_1(t),$$

$$\mathbf{H}(0, s) = \mathbf{H}(1, s),$$

and

$$\mathbf{H}(t, s) \neq \mathbf{p}$$

for every t and s . The last condition is the entire nail. The missing point $\mathbf{p}$ is often called a puncture; the space is the punctured plane.

Two loops connected by such a movie are homotopic in the punctured plane. Their winding numbers are equal. To change w , some intermediate loop must cross the puncture, cease to be closed, or become discontinuous. In the stated rules, those are singular events.

Hence homotopy classes in the punctured plane are labeled by winding.

In the punctured plane, winding number completely classifies closed loops up to homotopy when a base point is handled appropriately. In more complicated spaces, one integer may not capture every distinction. “Find one invariant” is not the same as “classify the space.” Mathematicians keep that trap polished for visitors.

12.6. A label that local tugs cannot change

All loops with $w = 1$ form one family, all loops with $w = 0$ another, and so on. Allowed local stretching never moves a loop from one family to another.

A *sector* is a collection of configurations or quantum states sharing a global label that allowed local operations cannot change. It is a bin, not a shape.

For a quantum system, let $\mathcal{H}$ denote its Hilbert space. A global label q can split that space schematically as

$$\mathcal{H} = \bigoplus_q \mathcal{H}_q,$$

where $\mathcal{H}_q$ is the set of states in sector q , and $\oplus$ means that the sectors are distinct subspaces.

In lattice models, a loop operator $W(C)$ acts along a closed path C . Its measured value can label a sector. A short, contractible loop can often be assembled from local constraints. A loop that winds around a hole, or around a periodic direction of a torus, cannot be shrunk to a point, and its value can encode genuinely global information. Kitaev's toric-code model makes this correspondence exact: noncontractible loop operators distinguish degenerate ground states [R030].

This is the quantum echo of the tabletop example, not a proof that every loop operator creates the phase we want.

A global quantum sector can also arise from an ordinary symmetry or conservation law — for example, total particle-number parity. Such a label may be useful, but sector structure alone does not imply a topologically ordered phase. One must ask what enforces the sectors, whether local measurements distinguish them, and whether they remain robust when generic local perturbations are added.

Drop the requirement that local operations cannot change the label, and the bin is just a bookkeeping convenience. Drop robustness under generic local perturbations, and a symmetry-protected split can vanish the moment you add a term that breaks the symmetry.

12.7. When many particles share the stubbornness

Imagine many particles whose lowest-energy states come in a few versions that look identical if you only inspect a small patch. No local arrow tells you which version you have. Distinguishing them takes an operator that wraps around the whole sample. Weak local noise should not mix the versions unless it first closes the energy gap that keeps them cheap to stay in.

That many-body property is *quantum topological order*. The name is worth keeping because the order lives in the phase, not in a drawing of a hole.

In the standard setting relevant here, it involves long-range entanglement and global organization that cannot be reduced to a local order parameter. Foundational work connected this order to topology-dependent ground-state degeneracy and robustness against local disorder [R122], and exactly solvable models show how nonlocal loop operators and unusual pointlike excitations realize it [R030]. Reviews [R015]; [R124] explain the broader framework and its qualifications.

No single beginner's slogan is a complete definition for every modern setting. For a two-dimensional gapped phase intended to store quantum information, however, a practical diagnostic package is:

- a local many-body Hamiltonian;
- an energy gap above a low-energy ground-state manifold;
- ground states that are locally indistinguishable in a large system;
- nonlocal operators that distinguish or transform global sectors;
- excitations with a conserved type and, in appropriate phases, unusual exchange and fusion behavior;
- stability of the phase under sufficiently weak local perturbations that do not close the gap.

Those unusual pointlike excitations, when exchanging them can do more than attach a plus or minus sign, are anyons. They are not electrons with a prettier path; they are the mobile defects of the phase.

[Theory] In an ideal gapped topologically ordered system, matrix elements of an operator O_R supported in a bounded region R take the approximate form

$$\langle \psi_a | O_R | \psi_b \rangle = c_O \delta_{ab} + \text{finite-size corrections.}$$

Here $|\psi_a\rangle$ and $|\psi_b\rangle$ are different ground states, c_O depends on the operator but not on the sector, and δ_{ab} equals one when $a = b$ and zero otherwise. In many gapped models the corrections are exponentially small in L/ξ , where L is a system length and ξ is the correlation length [R123]; [R015]. Both have units of length, so L/ξ is dimensionless, as an exponential requires.

Hence a genuinely local probe cannot learn which global ground state was chosen, nor readily convert one ground state into another. A logical bit of information can therefore be stored nonlocally.

The four questions you should now be able to ask of any “topological protection” claim are separate:

1. What is the geometry — the actual lengths, angles, coordinates, and shape?
2. What classical surviving count remains unchanged under allowed continuous deformations?
3. What are the global quantum sectors — the quantum labels that no small local operation can change?
4. Does the system possess quantum topological order, rather than merely an interesting shape, a global conserved quantity, or a programmed simulation?

A loop with winding $w = 1$ has a classical surviving count. A quantum state labeled by a nonlocal loop measurement belongs to a global quantum sector. A many-body phase with robust local indistinguishability and long-range entanglement may have quantum topological order. A logical qubit encoded in those sectors is a topological encoding. A circuit on ordinary physical qubits that prepares the model state is a digital emulation unless the hardware’s own equilibrium Hamiltonian supplies the phase.

These statements are related. They are not synonyms.

12.8. When the shoelace stops helping

The analogy worked because both stories separate local deformation from global change. It stops because a classical loop is one configuration with a visible path. A topologically ordered quantum ground state is a coherent superposition of many microscopic configurations, entangled across the system. Its order belongs to the wavefunction and Hamiltonian together, not to one attractive lattice sketch.

A literal hole can be useful in a topological code: logical operators may wind around it. Yet drilling a hole in an ordinary magnet does not manufacture topological order. Likewise, arranging defect centers around an empty site creates geometry. Only their interactions and resulting many-body state could create the required quantum order.

12.9. What the laboratory actually has

A laboratory has a finite sample or processor, imperfectly local interactions, boundaries, disorder, controls, detectors, and a temperature. It does not have an infinite plane or an exact mathematical phase label taped to the cryostat.

Evidence therefore comes in layers.

[Experiment] Superconducting-qubit processors have prepared a toric-code ground state with a circuit, measured topological entanglement entropy, and simulated anyon interferometry [R125]. This demonstrates controlled preparation and probing of a topological model state. Because the state is circuit-prepared on a programmable processor, it does not by itself show that the processor material passively enters an equilibrium topologically ordered phase.

[Experiment] Programmable Rydberg-atom arrays have implemented frustrated dynamics and measured nonlocal string observables consistent with a toric-code-type spin-liquid regime [R126]. This is analog quantum-simulation evidence for the engineered model. Whether a particular implementation supplies long-lived passive protection is a separate experimental question.

[Theory] For an ideal local Hamiltonian, stability results show that topological ground-state structure persists under an appropriate range of weak local perturbations without a phase transition [R123]. The theorem does not say that any finite device is immune to temperature, loss, drive errors, long-range couplings, or a perturbation large enough to close the gap.

For defect-engineered matter, the evidence ladder should therefore be explicit:

1. fabricate and characterize the defects;
2. establish the intended microscopic couplings;
3. derive or measure the effective many-body Hamiltonian;
4. show a gapped regime and nonlocal sector structure;
5. demonstrate local indistinguishability or equivalent diagnostics;
6. create, move, and identify the predicted excitations;
7. test robustness as size, disorder, temperature, and evolution time are varied.

A regular defect pattern completes step one and part of step two. Calling the pattern “topological” does not quietly complete steps three through seven.

12.10. Ways to get this wrong

1. Take a pretty graph for a surviving count. A honeycomb, kagome, ring, or punctured device can be geometrically suggestive. The same graph can support a trivial product state or a topologically ordered state under different Hamiltonians.
2. Take one robust number for quantum order. Winding is robust under the rules of the toy problem. It says nothing about entanglement, excitation statistics, an energy gap, or quantum coherence. A classical invariant is not quantum topological order.
3. Take a conserved global quantity for topological protection. A symmetry can divide a Hilbert space into sectors. If protection disappears when a generic local symmetry-breaking perturbation is added, the mechanism is symmetry protection, not intrinsic topological order. That may still be useful engineering. It is simply a different invoice.
4. Take an encoding for an emergent phase. Several physical spins can encode one logical qubit. A surface-code patch can use nonlocal logical operators. Neither fact alone says the hardware Hamiltonian has an emergent topologically ordered ground state. Active error correction, digital state preparation, and passive Hamiltonian protection must be reported separately.
5. Take a simulation for the simulated material. A programmable device may reproduce the amplitudes, string observables, or braiding protocol of a topological model. Those are real experimental achievements [R125]; [R126]. They establish properties of the prepared or simulated dynamics under stated controls. They do not automatically turn the underlying superconducting circuits or atoms into intrinsic anyonic matter.
6. Take “topological” for “error-free.” Topology suppresses particular local routes between sectors. It does not suppress leakage outside the modeled Hilbert space, read-out mistakes, thermal population, correlated disturbances, fabrication failures, or

control faults unless the architecture separately addresses them. Even a noncontractible logical error can be assembled from many mundane local errors. The errors have no respect for branding.

7. Merge different meanings of “defect.” An atomic vacancy in diamond is a microscopic crystal defect. A vortex can be a classical topological defect of an order parameter. An anyon is an emergent quantum excitation. A puncture or boundary in an error-correcting code is often called a defect as well. These objects can interact in an architecture, but they are not interchangeable uses of one concept.

12.11. Check yourself

1. **A circle is smoothly squashed into an ellipse without crossing the hole. What changes?**
Its geometric length and curvature generally change; its winding number does not.
2. **Show that w cannot change under a continuous deformation that misses the hole.**
Along the movie, w would have to vary continuously, but w takes only integer values. A continuous integer-valued function is constant, so the winding of every frame equals the winding of the first.
3. **What fails if the loop is allowed to pass through the hole?**
The winding formula ceases to be defined at the crossing, and afterward w may jump. The hole no longer distinguishes families of loops.
4. **Does a global quantum sector prove topological order?**
No. The sector could come from an ordinary symmetry or imposed code constraint. One must also establish the many-body phase, local indistinguishability, nonlocal operators, a suitable gap, and stability.
5. **What fails if a bounded operator O_R has matrix elements that distinguish two ground states at leading order?**
Local indistinguishability has already failed. A local probe can then read or mix the supposed global labels, and the logical bit is not stored nonlocally.
6. **What is the difference between digital emulation and emergence?**
Digital emulation uses controlled gates to prepare or evolve a model state. Emergence means the hardware’s physical many-body Hamiltonian itself produces the low-energy phase and excitations.

You now have a deformation, a winding count, a local-versus-global habit, and a reason a stubborn loop can hide a bit. Next we replace one loop around one hole with worldlines of several indistinguishable particles in two dimensions. Their winding and exchange

paths form braids, and in some systems those braids act as noncommuting operations on a degenerate quantum space.

12.12. Sources

- [R121] N. D. Mermin, “The topological theory of defects in ordered media,” *Reviews of Modern Physics* **51**, 591–648 (1979). DOI: 10.1103/RevModPhys.51.591.
- [R122] X.-G. Wen and Q. Niu, “Ground-state degeneracy of the fractional quantum Hall states in the presence of a random potential and on high-genus Riemann surfaces,” *Physical Review B* **41**, 9377–9396 (1990). DOI: 10.1103/RevModPhys.41.9377.
- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0; arXiv: quant-ph/9707021.
- [R123] M. B. Hastings and X.-G. Wen, “Quasi-adiabatic continuation of quantum states: The stability of topological ground-state degeneracy and emergent gauge invariance,” *Physical Review B* **72**, 045141 (2005). DOI: 10.1103/PhysRevB.72.045141; arXiv: cond-mat/0503554.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv: 0707.1889.
- [R124] X.-G. Wen, “Colloquium: Zoo of quantum-topological phases of matter,” *Reviews of Modern Physics* **89**, 041004 (2017). DOI: 10.1103/RevModPhys.89.041004; arXiv: 1610.03911.
- [R125] K. J. Satzinger *et al.*, “Realizing topologically ordered states on a quantum processor,” *Science* **374**, 1237–1241 (2021). DOI: 10.1126/science.abi8378; arXiv: 2104.01180.
- [R126] G. Semeghini *et al.*, “Probing topological spin liquids on a programmable quantum simulator,” *Science* **374**, 1242–1247 (2021). DOI: 10.1126/science.abi8794; arXiv: 2104.04119.

Chapter 13. Two particles trade places

Two identical beads sit on a table. Slide them until each occupies the other's old spot. Photograph the table before and after. Because the beads have no names painted on them, the two photographs are the same picture.

The walk between the photographs is not a photograph. One bead can circle the other clockwise. It can circle counterclockwise. It can go around twice. The table offers no third direction in which to lift one path over the other and iron the trip flat.

If the beads are quantum objects, and if they are forbidden to sit on the same point, that leftover memory of the walk can be a plus one, a minus one, a leftover twist, or a small matrix. We will earn those options in that order.

13.1. Time drawn upward

A photograph hides the trip. Give the trip a picture of its own. Draw the table as a horizontal line and let time run up the page. Each bead now leaves a trail.

```
time
^      \ /
|       X      exchange
|       / \
+-----> plane
```

The crossing is not a crash. One trail went around the other while both stayed on the table. If the beads may not occupy the same point, you are not allowed to pull one trail through the other.

That trail is a *worldline*. The object of study is no longer a pair of dots. It is a history.

Two histories count as the same if you can slide one drawing into the other without letting the lines meet, without moving the endpoints, and without running time backward.

Drop the collision ban and you may pull one trail through the other; the crossing becomes decoration. Keep the ban, and the drawing remembers how the beads walked.

13.2. A room would let you cheat

Stand the same two beads in a room instead of on a table. After they swap, you have an extra direction. You can lift one path over the other and continuously undo a double swap. Under the usual rules for identical point particles, the leftover memory of the walk collapses to who sits where, not how many times they wound.

On a table that extra lift does not exist. Clockwise and counterclockwise are different movies. Doing the same neighboring swap twice is a full winding of two worldlines. You cannot iron that winding flat while collisions stay forbidden.

Leinaas and Myrheim extracted this enlargement from the shape of the space of allowed arrangements. The extra possibilities are not a slogan. They are what two dimensions refuse to let you erase. **[Theory]** [R127]

Hence the same two photographs, in two spatial dimensions, may hide inequivalent walks.

13.3. The polite swap

The photographs match. The quantum state need not. After the two beads trade places counterclockwise, every amplitude can return exactly as it left. The whole state is multiplied by $+1$.

That law belongs to a *boson*. Photons are bosons. Two of them are allowed to occupy the same condition.

Suppose instead the leftover were required to be a permutation of names and nothing more. Then $+1$ would still be legal, but so would the next option, and nothing stranger. In two dimensions the menu is longer. Plus one remains on it.

13.4. The rude swap

The other tidy option is a global minus sign. After the same counterclockwise trade, every amplitude flips. The whole state is multiplied by -1 .

That law belongs to a *fermion*. Electrons are fermions. The minus sign is why two electrons refuse the same condition.

Both laws are allowed on a table. A particle that lives in two dimensions is not automatically something stranger than a boson or a fermion. Plus one and minus one still sit on the menu. In three or more spatial dimensions they *are* the menu.

Drop the two-dimensional restriction, and the extra leftover of a double swap can be ironed away. Keep it, and +1 and −1 are special cases, not a theorem that nothing else exists.

13.5. A leftover twist

On a table the state may come back multiplied by any phase

$$e^{i\theta},$$

where the real angle θ is measured in radians and counted only modulo 2π . A clockwise trade gives $e^{-i\theta}$. Plus one is $\theta = 0$. Minus one is $\theta = \pi$.

Wilczek called the general case an *anyon*, because the leftover phase can be *any* angle. **[Theory]** [R128] The name is a pun. It is also a claim: exchange writes a phase, and that phase is not required to be ± 1 .

When the leftover is only a phase, every pair of exchanges commutes. Changing the order cannot rotate one internal basis state into another. The phase becomes visible only by comparing two paths, usually by interference. **[Theory]** [R128]; [R129]

One neighboring swap writes $e^{i\theta}$. Sending one object all the way around the other is two swaps, and writes $e^{i2\theta}$. Those are different movies. Do not call a full winding “just the exchange.”

Hence Abelian anyonic statistics is a one-dimensional leftover, of which bosons and fermions are two points.

13.6. Shoelaces

Two strands are a boring lace. There is only one neighboring pair, so you cannot test whether the order of crossings matters.

Take three. Label the positions left, middle, and right. Cross the left pair counterclockwise. Cross the right pair counterclockwise. Those two movies do not look the same, and you can see the disagreement before any matrix appears.

left then right	right then left
\ / X / \ \ /	\ / X / \ \ /



A tangle of this kind, up to sliding that never lets the laces pass through each other, is a braid. Stack two tangles by placing one above the other. The collection of all such tangles on N strands, with that stacking as the product, is the *braid group*.

Write B_N for the braid group on N strands. The neighboring counterclockwise crossings are its letters. Call them $\sigma_1, \dots, \sigma_{N-1}$. The generator σ_i exchanges the objects currently at neighboring positions i and $i+1$. Its inverse σ_i^{-1} does the same exchange clockwise, and

$$\sigma_i \sigma_i^{-1} = 1.$$

Notice what is absent: $\sigma_i^2 = 1$. A braid remembers winding, not only the final permutation. Adding that missing rule by hand would collapse the shoelaces back to ordinary permutations — the three-dimensional leftover.

Well-separated crossings do not interfere:

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{when } |i - j| \geq 2.$$

Neighboring crossings obey one more sliding rule. Pull one crossing past the junction of two others and the drawing changes, but the braid does not:

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}.$$

That is the whole presentation. Far-apart letters commute. Adjacent letters satisfy the three-strand sliding rule. Squares are not the identity.

Hence two generators may fail to commute even though both are “just neighboring swaps.”

Suppose we imposed $\sigma_i^2 = 1$ after all. Then a double exchange would be trivial, clockwise would equal counterclockwise, and we would have returned to the permutation group. The relation we refused is the entire two-dimensional enlargement.

13.7. Forget the name tags

The braids are not a decoration drawn on top of physics. They *are* the collision-free histories, once you refuse to name the beads.

Start with N point-like objects in the plane. An ordered list of positions is

$$(\mathbf{r}_1, \dots, \mathbf{r}_N) \in (\mathbb{R}^2)^N,$$

where $\mathbf{r}_j = (x_j, y_j)$ has units of length. Collision arrangements form the set

$$\Delta = \{(\mathbf{r}_1, \dots, \mathbf{r}_N) : \mathbf{r}_j = \mathbf{r}_k \text{ for at least one } j \neq k\}.$$

Remove Δ , because an exchange path must keep the objects distinct. If the objects are identical, permuting their temporary labels changes no physical arrangement. The space that remains is the *configuration space*

$$\mathcal{C}_N(\mathbb{R}^2) = \frac{(\mathbb{R}^2)^N \setminus \Delta}{S_N},$$

where S_N is the group of all permutations of N labels.

A motion that starts and ends at the same physical arrangement is a loop in $\mathcal{C}_N$. Loops that deform into one another without touching Δ belong to the same class. Those classes form the fundamental group, and for particles in the plane the group is the braid group we just named from the laces:

$$\pi_1(\mathcal{C}_N(\mathbb{R}^2)) = B_N.$$

[Theory] [R127]; [R129] This equation is the hinge of the chapter. In three or more spatial dimensions, the analogous ordinary-exchange classes reduce to the permutation group.

A wavefunction need not come back to the same complex value after such a loop. Equivalently, one may lift paths to a simply connected covering space and impose a unitary transformation when a path closes downstairs. Consistency requires the transformation to depend only on the loop class and to preserve stacking. That is exactly a unitary representation of B_N . **[Theory]** [R127]; [R129]

Hence every later braid matrix is a linear action of a loop class in configuration space, not a doodle in the margin.

Drop identical-particle identification and you are studying ordered trajectories, which remember name tags the physics does not have. Drop the removal of Δ and collisions become legal; the fundamental group collapses.

13.8. Three laces, two matrices

A machine that turns each tangle into a matrix, and turns stacking tangles into multiplying matrices, is a *representation*. Write $\rho(b)$ for the matrix assigned to braid b . Composition is preserved:

$$\rho(b_1 b_2) = \rho(b_1) \rho(b_2).$$

Products of operators act on a ket from right to left. The symbol ρ names the representation. It is not a density matrix.

Overall phases can be unobservable for a closed isolated system. Relative phases between different paths can be measured by interference.

With two strands there is only one neighboring exchange, so there is no pair of generators whose order can be tested. Three strands are the smallest useful case. Consider a two-dimensional state space with orthonormal basis kets $|0\rangle$ and $|1\rangle$. Define the Pauli matrices

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and assign

$$A \equiv \rho(\sigma_1) = e^{-i\pi Z/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1-i & 0 \\ 0 & 1+i \end{pmatrix},$$

$$B \equiv \rho(\sigma_2) = e^{-i\pi X/4} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}.$$

The imaginary unit satisfies $i^2 = -1$. Both matrices are unitary: direct multiplication gives $A^\dagger A = B^\dagger B = I$, where $\dagger$ means conjugate transpose and I is the identity matrix.

Now calculate the two possible orders:

$$AB = \frac{1}{2} \begin{pmatrix} 1-i & -1-i \\ 1-i & 1+i \end{pmatrix}, \quad BA = \frac{1}{2} \begin{pmatrix} 1-i & 1-i \\ -1-i & 1+i \end{pmatrix}.$$

They are not equal. Acting on $|0\rangle$ gives, up to an overall phase,

$$AB|0\rangle \sim \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad BA|0\rangle \sim \frac{|0\rangle - i|1\rangle}{\sqrt{2}}.$$

The first state is certainly found in the $(|0\rangle + |1\rangle)/\sqrt{2}$ outcome of an X -basis measurement. The second gives that outcome with probability $1/2$. The different order is experimentally distinguishable, not merely different ink on paper.

Noncommutativity does not excuse the matrices from the sliding rule of the laces. They obey

$$ABA = BAB = -\frac{i}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

This is the three-strand braid relation. The matrices above are an Ising-type two-dimensional braid representation, up to basis and overall-phase conventions. [**Theory**] [R015]; [R134] We have used them as a clean algebraic model. Writing these matrices into a pulse sequencer would not, by itself, manufacture Ising anyons.

Hence $AB \neq BA$ is compatible with $ABA = BAB$. The first fact is noncommutativity. The second is the braid relation. Both are required.

13.9. Four leftover laws

Collect the menu. A one-dimensional representation assigns each neighboring swap a single complex number of magnitude one.

A boson still has

$$\rho(\sigma_i) = +1.$$

A fermion still has

$$\rho(\sigma_i) = -1.$$

An anyon whose leftover is only a phase has

$$\rho(\sigma_i) = e^{i\theta}.$$

Bosons and fermions are the special values $\theta = 0$ and $\theta = \pi$. All such complex numbers commute, so changing the order of exchanges cannot rotate one internal basis state into another.

The three-strand calculation was something else. The same positions supported a two-dimensional space, and the two neighboring swaps became matrices with

$$AB \neq BA.$$

When the leftover is a matrix on a space of more than one state, and at least some pair of those matrices do not commute, the statistics are *non-Abelian*. The name is worth keeping because it describes the representation, not an electric charge that forgot how to commute and not the paths literally colliding.

One exchange matrix by itself cannot reveal this. You need enough strands and a pair of operations, as in the calculation above. Arbitrary noncommuting matrices are not enough either. They must also satisfy the braid relations and unitarity. Ours did.

Hence non-Abelian statistics is a property of a representation of B_N , not a decorative phase on a single swap.

13.10. Parked particles still have leftovers

Where did that two-dimensional space come from? The beads are sitting still. Their positions are fixed. And yet $|0\rangle$ and $|1\rangle$ are two different states for the same photograph.

Imagine enclosing a pair and asking the outside world one coarse question: what kind of leftover does the pair look like from far away? Two anyons of the same kind can look like “nothing extra” or like “one leftover excitation.” Those are two answers, not a smoothie. A prepared state may be a combination of the answers. A measurement of the pair picks one.



The question mark is the leftover label. The list of allowed leftovers, for fixed positions and a fixed total leftover, is a *fusion* space. Combining is already a quantum question. The particles do not have to move to own more than one state.

This chapter needs only the fact that the space has multiple basis states. The matrices A and B acted on one such space. The next chapter will name the leftover labels, draw the trees that organize them, and show how regrouping one tree into another produces the F -matrices behind concrete braid representations.

Drop the degeneracy and there is no room for a matrix leftover: every braid can at most write a phase. Keep the degeneracy without the braid relations, and you have a degenerate subspace, not statistics.

13.11. A slow walk around

Suppose a Hamiltonian $H(\lambda)$ depends on control parameters λ that locate well-separated quasiparticles. If the system is moved slowly around a closed path while remaining in a degenerate low-energy subspace, its state can acquire a matrix-valued geometric leftover.

That leftover, after you strip off the ordinary dynamical phases that depend on how long you waited, is a *holonomy*. It is a transformation written by the shape of the walk, not by a clock. In an ideal topological phase the resulting unitary depends only on the braid class. **[Theory]** [R015]

“Slowly” and “well separated” hide physical inequalities. Motion must avoid exciting states across the many-body gap, but it must finish before decoherence and uncontrolled quasiparticle motion spoil the state. The objects must be separated by distances large compared with the phase’s correlation length so that local couplings do not substantially split the intended degeneracy. Later chapters will turn those words into energy, length, and timing budgets.

No dimensional estimate is needed here. A braid operator and a statistical angle are dimensionless. Laboratory time is measured in seconds and gaps in joules or electronvolts, but their required inequalities depend on the chosen Hamiltonian rather than braid topology alone.

Drop adiabaticity and the walk populates excited states; the leftover is no longer a holonomy of the intended subspace. Drop large separation and the degeneracy splits; the walk is then ordinary dynamical evolution of a unique ground state.

13.12. A bump that thinks it is a particle

An *emergent quasiparticle* is a localized excitation of an interacting many-body Hamiltonian, described at low energy as an effective particle even though no matching fundamental particle appears in the microscopic inventory. A phonon is an emergent quasiparticle but not an anyon. An emergent anyon is the more specific case in which the excitation’s exchanges realize a nontrivial braid representation in an effectively two-dimensional topological phase.

A defect in diamond is a microscopic imperfection with electronic and nuclear degrees of freedom. It is not an anyon merely because it is point-like or can encode a qubit. Defects might someday be the microscopic ingredients of a two-dimensional Hamiltonian whose *collective* excitations are anyons. That extra arrow — ingredients to emergent phase — is most of the engineering problem, wearing a small hat.

Fractional quantum Hall devices provide evidence for Abelian anyonic statistics. **[Experiment]** At filling factor $\nu = 1/3$, collider measurements reported correlations consistent with fractional exchange statistics, while an electronic Fabry–Pérot interferometer

reported a statistical phase associated with quasiparticle braiding. [R130]; [R131] These are experiments on collective excitations of an interacting electron system, not software labels attached to ordinary particles. The inference from an interference or correlation signal to a statistical angle remains model- and device-dependent, so “evidence consistent with” is more accurate than “a braid matrix was directly photographed.”

Hence an anyon, when it is physical, is an excitation of a phase, not a renamed microscopic defect.

13.13. A braid the hardware does not own

A programmable processor can instead prepare an encoded many-qubit state and apply gates whose logical action realizes a braid representation. Call this an *emulated braid* when the braid is implemented by a designed gate, measurement, or code-deformation sequence rather than by adiabatically transporting persistent quasiparticle eigenstates of the hardware Hamiltonian.

[Experiment] In 2023, a superconducting processor implemented a generalized stabilizer code and unitary protocol for projective Ising-type graph defects, testing fusion and non-Abelian exchange effects. [R132] The paper explicitly notes that its mobile objects were not quasiparticles that persisted as eigenstates throughout Hamiltonian evolution; gates deformed the stabilizer graph. That is a substantive emulation of braid physics, but it is not evidence that the processor material sits passively in a non-Abelian topological phase.

For a defect-engineered platform, a braid matrix programmed into microwave pulses proves control of the encoded space. It does not prove an emergent excitation spectrum. To make the stronger claim, the material or effective analog Hamiltonian must itself support the relevant phase and localized excitations, and exchanges must implement the representation while remaining inside its low-energy manifold. **[Proposal]** Engineered defect arrays could aim at this. **[Speculation]** No conclusion follows merely from arranging defects on a planar lattice or drawing crossing paths between them.

Hence a programmed braid and an emergent anyon are different claims, even when they share a matrix.

13.14. Four labels that are not synonyms

A *physical defect qubit* is a controlled microscopic degree of freedom, such as a defect spin.

An *encoded qubit* is a selected subspace spread across multiple physical degrees of freedom.

An *emulated braid* is a protocol whose logical gates reproduce a braid representation.

An *emergent quasiparticle* is an excitation of the many-body Hamiltonian. It is an anyon only if physical exchanges have the required braid statistics.

Topological order is a property of the many-body phase, not of one clever gate sequence.

These descriptions can coexist in one experiment. None follows automatically from the preceding one.

13.15. Ways to get this wrong

1. “Only bosons and fermions exist.” That conclusion uses permutation topology appropriate to three or more spatial dimensions. The planar configuration space has braid-group topology. [R127]; [R129]
2. “Every two-dimensional excitation is an anyon.” Ordinary bosons and fermions also live in two dimensions. The exchange representation decides the statistics.
3. “Any unusual phase is non-Abelian.” A scalar $e^{i\theta}$, however exotic θ looks, is Abelian. Non-Abelian statistics requires a multidimensional state space and noncommuting braid operators.
4. “ AB and BA are different matrices, so they are automatically valid braids.” Arbitrary noncommuting matrices are not enough. They must also satisfy the braid relations and unitarity. Our worked matrices passed both checks.
5. “A full winding is the same as one exchange.” One neighboring exchange is σ_i ; taking one object all the way around the other corresponds to σ_i^2 . For an Abelian anyon the phases are $e^{i\theta}$ and $e^{i2\theta}$, respectively.
6. “A quasiparticle is a tiny constituent particle.” It is a collective low-energy excitation. Its usefulness does not depend on being microscopic. Its identity does depend on the Hamiltonian and phase.
7. “Executing the right unitary proves emergence.” A conventional processor can compile A and B exactly. That proves a braid representation was emulated. Emergent anyons require the representation to arise from exchanging excitations of the system’s many-body phase.
8. “Topological means perfect.” Topological dependence is an ideal low-energy statement. Finite separation, finite temperature, disorder, nonadiabatic motion, leakage, measurement error, and stray quasiparticles can all damage a braid. [Theory] [R015]; [R030] Protection suppresses specified local errors under specified conditions. It does not negotiate with every error in the building.

13.16. Check yourself

1. **Why can two-dimensional exchange be richer than three-dimensional exchange?**

Because collision-free loops of identical-particle configurations form the braid group in two dimensions. A double exchange need not deform to the trivial path.

2. **Show that imposing $\sigma_i^2 = 1$ collapses the braid group to permutations.**

That extra relation makes a double exchange trivial, so clockwise equals counter-clockwise and winding is forgotten. What remains is the permutation of the final positions.

3. **How are a boson, a fermion, and an Abelian anyon related?**

All have one-dimensional exchange representations. Their exchange phases are respectively $+1$, -1 , and general $e^{i\theta}$; bosons and fermions are the $\theta = 0$ and $\theta = \pi$ cases.

4. **What fails if $AB \neq BA$ but $ABA \neq BAB$?**

The matrices do not represent the braid group. Noncommutativity alone is not statistics; the three-strand sliding rule is required as well.

5. **What makes an anyon non-Abelian?**

Braids act as unitary matrices on a multidimensional degenerate state space, and at least some braid operations do not commute.

6. **Does a gate sequence implementing A and B establish emergent anyons?**

No. It establishes an emulated braid representation. Emergence additionally requires exchangeable quasiparticle excitations of the many-body Hamiltonian.

You now have worldlines, a menu of exchange laws, a braid group named from shoelaces, one pair of matrices that refuse to commute, and a fusion space for those matrices to act on. The next chapter builds that space from leftover labels and trees.

13.17. Sources

- [R127] J. M. Leinaas and J. Myrheim, “On the theory of identical particles,” *Il Nuovo Cimento B* **37**, 1–23 (1977). DOI: 10.1007/BF02727953.
- [R128] F. Wilczek, “Quantum mechanics of fractional-spin particles,” *Physical Review Letters* **49**, 957–959 (1982). DOI: 10.1103/PhysRevLett.49.957.
- [R129] Y.-S. Wu, “General theory for quantum statistics in two dimensions,” *Physical Review Letters* **52**, 2103–2106 (1984). DOI: 10.1103/PhysRevLett.52.2103.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv: 0707.1889.

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0; arXiv: quant-ph/9707021.
- [R130] H. Bartolomei *et al.*, “Fractional statistics in anyon collisions,” *Science* **368**, 173–177 (2020). DOI: 10.1126/science.aaz5601; arXiv: 2006.13157.
- [R131] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, “Direct observation of anyonic braiding statistics,” *Nature Physics* **16**, 931–936 (2020). DOI: 10.1038/s41567-020-1019-1.
- [R132] M. B. Andersen *et al.* (Google Quantum AI and Collaborators), “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* **618**, 264–269 (2023). DOI: 10.1038/s41586-023-05954-4; arXiv: 2210.10255.
- [R133] A. Stern, “Anyons and the quantum Hall effect—a pedagogical review,” *Annals of Physics* **323**, 204–249 (2008). DOI: 10.1016/j.aop.2007.10.008; arXiv: 0711.4697.
- [R134] D. A. Ivanov, “Non-Abelian statistics of half-quantum vortices in p-wave superconductors,” *Physical Review Letters* **86**, 268–271 (2001). DOI: 10.1103/PhysRevLett.86.268; arXiv: cond-mat/0005069.

Chapter 14. Two particles in a bag

Put two particles in a bag and zip it. From outside you cannot see how they sit. You can only ask one coarse question: what kind of source is the whole bag?

Rattle them around inside. The answer must stay the same. Local rearrangements are invisible from outside. The bag has a label, and the label is stubborn.

Last chapter showed that exchanging two such particles can rotate a hidden state. This chapter says what that state is made of. We will write a tiny multiplication table by hand. Then we will learn two moves on diagrams: one that regroups the bag, and one that swaps two particles.

14.1. A label the bag cannot change

Imagine a loop drawn around a few quasiparticles. Ask the outside world that same coarse question: “What kind of source is inside?” Anything you do locally, inside the loop, is not allowed to change the answer.

That answer is a *charge*. Physicists often say topological charge, because a poke in one place cannot flip it. For the rest of this chapter, charge means that bag label.

Give the possible charges names such as a , b , or c . No allowed local operation can turn an isolated total charge a into a different total charge b . The empty bag — no leftover excitation — gets the special name 1. That 1 is a label, not the number one in an arithmetic drill.

Drop locality and a faraway operation may rewrite the bag from outside. Keep it, and the label is a sector in the sense of the last two chapters.

14.2. Adding two labels

Bring two bags together and treat them as one bigger bag. The combined label is not always unique. Charges a and b may be allowed to add up to several possible totals.

We write the list as a product,

$$a \times b = \sum_c N_{ab}^c c.$$

Here N_{ab}^c is a nonnegative integer. Zero means “forbidden.” One means one way. Two means two independent ways to reach the same visible total c .

The particles *fuse*. The name is worth keeping because they do not smash into debris the way billiard balls do. They are still there. We have only asked what the bigger bag looks like from outside.

The plus sign lists alternatives. It is not a recipe for a half-and-half smoothie, and it does not say that every measurement returns both answers at once. A quantum state may superpose those alternatives only when the whole experimental setting permits it.

Hence the product is a list of allowed totals, not a prepared superposition and not a probability.

14.3. The smallest interesting table

The Ising model has three charge labels: 1, ψ , and σ . Its products are

$$1 \times a = a, \quad \psi \times \psi = 1, \quad \psi \times \sigma = \sigma, \quad \sigma \times \sigma = 1 + \psi.$$

The symbol a in the first line means any of the three labels. Vacuum times anything is that thing. Two ψ charges cancel. A ψ next to a σ is still a σ . Two σ charges may total either 1 or ψ .

Write the same facts as a table. Each entry is the list of allowed totals:

$\times$	1	ψ	σ
1	1	ψ	σ
ψ	ψ	1	σ
σ	σ	σ	$1 + \psi$

Every nonempty list in this table has length one, except the last cell. Each allowed total appears once. The integer in front of a total — usually left unwritten when it equals one — is a *fusion coefficient*. It is a count of independent ways, not a probability.

All nonzero coefficients here equal one, so the model is multiplicity-free: one channel, or none.

The space of ways for a and b to total c is written

$$V_{ab}^c, \quad \dim V_{ab}^c = N_{ab}^c.$$

Its dimension is exactly that integer. For two σ charges, $V_{\sigma\sigma}^1$ and $V_{\sigma\sigma}^\psi$ are each one-dimensional, and $V_{\sigma\sigma}^\sigma$ is zero-dimensional: forbidden.

Suppose someone proposed $N_{\sigma\sigma}^\sigma = 1$ as well. Then two σ charges could total a third σ , and the table would no longer be the Ising table. The missing channel is a fact, not an omission.

14.4. A fork with three legs

A basis vector in V_{ab}^c can be drawn as a fork. Read every diagram from top to bottom: incoming charges enter at the top, and their total leaves at the bottom.



If $N_{ab}^c > 1$, put a label $\mu = 1, \dots, N_{ab}^c$ beside the vertex. Then the triple $(a, b; c, \mu)$, not merely c , specifies the channel.

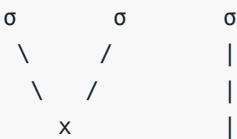
Three or more charges need a sequence of pairwise combinations. Draw that sequence as a *fusion tree*. Internal edges are intermediate charges. Choosing their labels chooses a basis. The tree is bookkeeping for a many-particle state, not a claim that one collision literally happened before another.

Drop the multiplicity labels when $N_{ab}^c > 1$, and two independent ways to reach the same total are written as one vector. The shortcut “an internal edge specifies the state” works only when every relevant coefficient is zero or one.

14.5. Three particles, two pairings

Take three Ising σ charges and fix their total to σ . First combine the left pair. Its intermediate charge x may be 1 or ψ , and either result then combines with the third σ to total σ :

Left pairing: $((\sigma \sigma)_x \sigma)_\sigma$

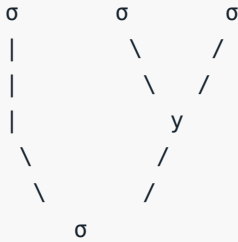




$x = 1 \text{ or } \psi$

Alternatively, combine the right pair first. Its intermediate charge y may again be 1 or ψ :

Right pairing: $(\sigma (\sigma \sigma)_y)_\sigma$



$y = 1 \text{ or } \psi$

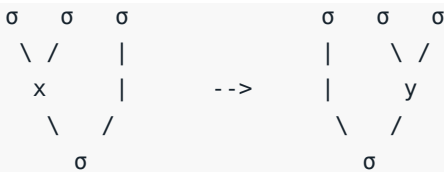
The two trees have the same three external charges and the same total. They are two bases of one space, not two different physical systems. Call that shared space the *fusion space*. It is the actual state space the last chapter's braids act on.

Check the count from the table. Left pairing: the first pair may total 1 or ψ , and both then combine with the remaining σ to give σ . Two states. Right pairing: the same two states, written in a different pairing.

Hence three σ charges of total charge σ support a two-dimensional fusion space. Changing the total-charge constraint changes the available space.

14.6. Sliding the pairing

The particles do not walk. Only the pairing slides:



That slide is an F -move. Later we will need a different move that really does exchange positions.

Denote the orthonormal left-pairing states by $|x\rangle_L$ and the right-pairing states by $|y\rangle_R$. The slide is a change of basis,

$$|x\rangle_L = \sum_{y \in \{1, \psi\}} [F_{\sigma}^{\sigma\sigma\sigma}]_{xy} |y\rangle_R.$$

In a standard gauge for Ising anyons,

$$F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

where rows are $x = (1, \psi)$ and columns are $y = (1, \psi)$ [R015]; [R023]. The explicit recoupling is therefore

$$|1\rangle_L = \frac{|1\rangle_R + |\psi\rangle_R}{\sqrt{2}}, \quad |\psi\rangle_L = \frac{|1\rangle_R - |\psi\rangle_R}{\sqrt{2}}.$$

Suppose the first two charges are definitely in channel $x = 1$. If we instead measure the last two, the amplitudes are $1/\sqrt{2}$ for $y = 1$ and $1/\sqrt{2}$ for $y = \psi$. The Born rule therefore gives probability $1/2$ for each result. That probability comes from the F -matrix. The fusion table alone does not contain it.

The normalization checks itself:

$$\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = 1.$$

Also $F^{\dagger}F = I$, where $F^{\dagger}$ is the conjugate transpose and I is the 2×2 identity. Changing pairing must preserve inner products and probabilities.

Hence an F -move is a unitary change of basis on a fixed fusion space. It does not exchange positions.

What fails if F is not unitary? Inner products, and therefore Born probabilities, would depend on which pairing you used to write the state. The two trees would no longer be two bases of one physical space.

14.7. A partner that can cancel

Each simple charge a has a partner $\bar{a}$ such that the empty bag occurs in their combination:

$$N_{a\bar{a}}^1 \geq 1.$$

That partner is the *antiparticle*. The two charges can disappear into vacuum, just as a particle and its antiparticle can. In the common simple setting, the vacuum channel occurs once. A charge can be its own antiparticle: in the Ising table, $\bar{\sigma} = \sigma$ and $\bar{\psi} = \psi$.

A physical combination experiment can still have a probabilistic outcome even when every N_{ab}^c is only zero or one. The coefficient counts basis states, not outcome probabilities. Probabilities depend on the prepared state. Conversely, $N_{ab}^c = 2$ means two orthogonal states share the same external labels a, b, c . A multiplicity label is then required even before anyone discusses amplitudes.

Drop the existence of $\bar{a}$ and there is no way for a to disappear into vacuum. The empty bag would then be unreachable from that charge, and the theory would not have duals in the sense used later.

14.8. Two ways of counting must agree

For fixed total charge d , the left pairing of three charges decomposes as

$$V_{abc}^d \cong \bigoplus_e V_{ab}^e \otimes V_{ec}^d.$$

Here e runs over allowed intermediate charges, $\oplus$ is a direct sum of alternatives, and $\otimes$ combines the two vertex spaces along a chosen path. The same space has a right-pairing decomposition

$$V_{abc}^d \cong \bigoplus_f V_{bc}^f \otimes V_{af}^d.$$

Equality of dimensions is an integer check you can run on the table before inventing any phases:

$$\sum_e N_{ab}^e N_{ec}^d = \sum_f N_{bc}^f N_{af}^d.$$

If proposed coefficients fail this equation, no choice of clever phases will rescue them.

Run it on three σ charges with total σ . Left pairing:

$$N_{\sigma\sigma}^1 N_{1\sigma}^\sigma + N_{\sigma\sigma}^\psi N_{\psi\sigma}^\sigma + N_{\sigma\sigma}^\sigma N_{\sigma\sigma}^\sigma = (1)(1) + (1)(1) + (0)(0) = 2.$$

Right pairing is the same sum with the first pair replaced by the last pair, and it also equals 2. The table is at least consistent.

For n charges, repeat the construction. External edges record the n charges, internal edges record intermediate charges, and the root records total charge. The number of admissible labelings — counting vertex multiplicities — is the dimension of the fixed-total-charge fusion space.

That is why non-Abelian charges can store information. With the total fixed, several internal labelings can remain. A local measurement around one separated particle cannot read the entire pattern. **[Theory]** In an ideal topological phase, operations supported far from one another act on this nonlocal fusion space according to the theory's data, up to corrections from finite separation and unwanted excitations [R015]; [R017]. That is a statement about the effective theory. It is not a blanket promise about a fabricated device.

Hence associativity of fusion, at the level of dimensions, is a numerical identity on the coefficients. Fail it, and there is no fusion space for three charges.

14.9. How fast the space grows

The *quantum dimension* d_a measures the growth that charge a contributes to large fusion spaces. For a unitary theory, choose the positive numbers d_a satisfying

$$d_a d_b = \sum_c N_{ab}^c d_c, \quad d_1 = 1.$$

Equivalently, form a fusion matrix $(N_a)^c_b = N_{ab}^c$. Then d_a is its largest positive eigenvalue in the relevant sector. This is a Perron–Frobenius fact, not a fit parameter [R017]; [R016].

Apply the rule to the Ising table. From $\psi \times \psi = 1$, positivity gives $d_\psi^2 = 1$, hence $d_\psi = 1$. From $\sigma \times \sigma = 1 + \psi$,

$$d_\sigma^2 = d_1 + d_\psi = 2, \quad d_\sigma = \sqrt{2}.$$

That irrational value is not the dimension of one ordinary on-site space. Finite vector spaces have integer dimensions. It encodes growth. For $2n$ Ising σ charges constrained to total charge 1, the fusion-space dimension is 2^{n-1} : two charges give one state, four give

two, six give four, and so on. The characteristic contribution per additional σ tends toward $\sqrt{2}$.

A charge with $d_a = 1$ is Abelian in a unitary fusion theory. A charge with $d_a > 1$ supports non-Abelian fusion growth. The total quantum dimension is

$$\mathcal{D} = \sqrt{\sum_a d_a^2},$$

where the sum covers all simple charges. For Ising, $\mathcal{D} = \sqrt{1 + 1 + 2} = 2$.

Hence $d_\sigma = \sqrt{2}$ is an asymptotic growth rate, not a local Hilbert-space dimension. Treating it as an on-site level count already contradicts the fact that vector-space dimensions are integers.

14.10. The same slide with more labels

Let a, b, c be three incoming charges and d their total. In the left tree, let e be the intermediate charge, μ label a basis of V_{ab}^e , and ν label a basis of V_{ec}^d . In the right tree, use f, α , and β the same way. Then

$$|((ab)_e c)_d; \mu, \nu\rangle = \sum_{f, \alpha, \beta} [F_d^{abc}]_{(e, \mu, \nu)(f, \alpha, \beta)} |(a(bc)_f)_d; \alpha, \beta\rangle.$$

This equation is the F -matrix convention of this chapter. Other sources may reverse it, conjugate it, or place labels differently. That is harmless if the convention is stated and used consistently.

An F -move still does not exchange positions. Individual entries depend on phase choices for the vertex basis states. That choice is a *gauge*. Observable probabilities and complete braid amplitudes do not depend on it.

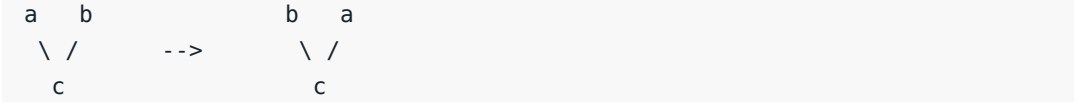
Rephase a vertex basis and the individual F -symbols change. Closed-process probabilities, consistency relations, and physically defined interference signals do not. Mistaking a gauge-dependent entry for an observable is a bookkeeping error, not a new phase of matter.

14.11. Crossing two strands

Now let the particles walk. An R -move exchanges two neighbors. For a counterclockwise exchange of a and b in total channel c , it is a unitary map

$$R_c^{ab} : V_{ab}^c \longrightarrow V_{ba}^c.$$

On a diagram it is a crossing, not a slide:



This is the move that actually braids. It is not an F -move wearing a small hat.

In a multiplicity-free theory, each allowed R_c^{ab} is a phase. In the same standard Ising gauge used above,

$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{3i\pi/8}$$

for a chosen counterclockwise convention [R015]; [R023]. Reversing the exchange uses the inverse phases.

In the left-pairing basis of three σ charges, exchanging the first two is diagonal:

$$B_{12} = \begin{pmatrix} R_1^{\sigma\sigma} & 0 \\ 0 & R_\psi^{\sigma\sigma} \end{pmatrix}.$$

To exchange the last two, first slide into the basis where those two combine, apply their R -move, and slide back:

$$B_{23} = F B_{12} F^{-1}$$

for the matrix convention of the worked example. Since F mixes the two channels and the two R phases differ, B_{12} and B_{23} do not commute. That is the small, concrete mechanism behind non-Abelian braid action.

Hence a braid of nonadjacent fusion partners generally requires both moves: F to regroup, R to exchange. Calling every basis change a braid confuses the two.

What fails if the two R phases are equal? Then B_{12} is a global phase, B_{23} is the same global phase, and they commute. Non-Abelian action on this two-dimensional space would disappear even though the fusion table was unchanged.

14.12. Why invented tables break

Regroup four charges through different sequences of F -moves. Both routes start and finish at the same trees, so they must give the same linear map. This requirement is the *pentagon equation*. Likewise, regrouping and exchanging three charges in two equivalent ways produces the hexagon equations, which tie F and R together. These coherence equations guarantee that a complicated diagram has a well-defined value independent of bookkeeping choices [R022].

Only now is it useful to name the algebraic package. A *fusion category* is a finite semisimple linear category whose simple objects are charge types, whose tensor product is fusion, whose unit object is 1, and whose associator is represented by F -matrices. Dual objects represent antiparticles. A *braided* fusion category additionally has a braiding isomorphism represented by R . A bare fusion category need not possess R -moves at all. A unitary modular tensor category adds unitarity and a nondegeneracy condition on braiding, supplying the standard algebraic package for a fully fledged two-dimensional anyon theory [R017]; [R016].

The name arrived last on purpose. The table, the slide, and the crossing are the things you compute. The category is the filing cabinet.

The same coefficients N_{ab}^c can admit inequivalent consistent F - and R -data. Fusion rules are necessary input, not a complete fingerprint [R016]. Fail the pentagon or hexagon, and a larger diagram has no well-defined value: two bookkeeping routes disagree.

14.13. What the laboratory actually has

The symbols above define an effective theory. A laboratory has a finite sample, controls, detectors, unwanted excitations, boundaries, and a microscopic Hamiltonian. It never has a meter labeled “fusion category.”

[Experiment] Evidence for fusion data must come from operationally distinguishable outcomes: create or identify excitations, combine regions, measure total charge, and show that the observed channels match a consistent rule. Interferometry is one route to charge measurement because a probe can respond to the enclosed charge’s monodromy — the effect of winding around it [R023]. A single peak or degeneracy is not enough to reconstruct N , much less F and R .

[Experiment] Evidence for an F -matrix requires comparing incompatible fusion-tree measurements while preserving phase coherence well enough to recover amplitudes, not only a list of allowed outcomes. Evidence for R requires controlled exchanges or an operationally equivalent braid and must track channel-dependent phases. Multiple sequences should satisfy braid and coherence relations within measured uncertainty.

[Theory] Even perfect agreement with a few matrices establishes only the tested effective operations. To claim emergent anyons, the excitations should arise from a many-body phase, be deconfined over the relevant scale, and carry data stable against allowed local perturbations. To claim topological protection, one must additionally bound non-topological errors, thermal excitation, finite-size splitting, and control faults. Those are physical claims beyond the algebra [R015]; [R017].

14.14. A circuit is not a crystal

A digital processor can encode the two Ising fusion-tree basis states in ordinary qubits and program the Hadamard matrix shown above. That is a digital emulation of an F -move. A cluster of defect spins can likewise encode a two-dimensional subspace. Neither fact alone creates emergent σ anyons or topological order.

Extrinsic defects can also carry nontrivial projective fusion and braiding properties, but they must be distinguished from freely propagating intrinsic quasiparticles of the host phase [R021]. The same algebra can appear in several physical settings. The hardware evidence decides which setting applies.

A collection of defect spins does not become an anyon model because its levels can be given exotic names. The proposed microscopic Hamiltonian must produce emergent excitations whose low-energy states obey consistent fusion and braiding data. Later chapters will ask whether that demanding step is physically plausible.

Hence a programmed braid and an emergent anyon remain different claims when the algebra is fusion rather than exchange.

14.15. Ways to get this wrong

1. **Reading “+” as a prepared superposition.** The fusion rule lists allowed sectors. A state and its amplitudes require preparation information.
2. **Treating N_{ab}^c as a probability.** It is an integer dimension. Probabilities come from normalized state amplitudes and a measurement protocol.
3. **Treating quantum dimension as an on-site level count.** $d_\sigma = \sqrt{2}$ cannot be the dimension of a finite local vector space. It summarizes asymptotic fusion-space growth.
4. **Calling every basis change a braid.** F regroups a fusion tree without exchanging positions. R exchanges two charges. A braid of nonadjacent fusion partners generally requires both.

5. **Ignoring total charge.** Three σ charges with total σ form the two-dimensional example above. Changing the total-charge constraint changes the available space.
6. **Dropping multiplicity labels.** The shortcut “an internal edge label specifies the state” works only when all relevant $N_{ab}^c \leq 1$. General theories need vertex labels too.
7. **Mistaking gauge-dependent entries for observables.** Rephasing a vertex basis changes individual F - and R -symbols. Closed-process probabilities, consistency relations, and physically defined interference signals are gauge invariant.
8. **Assuming fusion rules determine the phase.** The same coefficients N_{ab}^c can admit inequivalent consistent F - and R -data. Fusion rules are necessary input, not a complete fingerprint [R016].
9. **Promoting a simulator’s labels into material excitations.** A circuit may reproduce fusion-space algebra accurately while every underlying carrier remains an ordinary physical qubit. Simulation is valuable; misnaming it is not.
10. **Skipping the microscopic gap.** A beautiful set of F - and R -tables does not prove that a proposed defect Hamiltonian realizes them. One still needs a gapped many-body phase, suitable excitations, and stability under realistic perturbations.

14.16. Check yourself

1. **What does $N_{ab}^c = 2$ mean?**
There are two linearly independent fusion channels from charges a, b to total charge c . A multiplicity label is required.
2. **Show that both pairings of three Ising σ charges with total σ have dimension two.**
Left: $N_{\sigma\sigma}^1 N_{1\sigma}^\sigma + N_{\sigma\sigma}^\psi N_{\psi\sigma}^\sigma = 1 + 1 = 2$. Right: the same sum on the last pair. The two trees are two bases of one two-dimensional fusion space.
3. **What information does a fusion tree contain?**
Its external edges contain the charges, internal edges contain intermediate charges, vertices may contain multiplicity labels, and the root contains total charge.
4. **What fails if F is not unitary?**
Inner products would depend on the pairing used to write the state. Born probabilities for the same physical question would then disagree.
5. **What separates an F -move from an R -move?**
 F slides the pairing without exchanging positions. R exchanges neighboring charges in a specified fusion channel.

6. Does reproducing an F -matrix on defect-spin qubits demonstrate emergent anyons?

No. It demonstrates an encoded operation unless independent evidence establishes a many-body topological phase and its emergent excitations.

You now have a multiplication table, a way to regroup three charges, and a way to swap two of them. Next we fill the table with one lonely charge τ and watch the Fibonacci numbers appear.

14.17. Sources

- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv:0707.1889.
- [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv:cond-mat/0506438.
- [R023] P. Bonderson, K. Shtengel, and J. K. Slingerland, “Interferometry of Non-Abelian Anyons,” *Annals of Physics* **323**, 2709–2755 (2008). DOI: 10.1016/j.aop.2008.01.012; arXiv:0707.4206.
- [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” *Bulletin of the American Mathematical Society* **55**, 183–238 (2018). DOI: 10.1090/bull/1605; arXiv:1705.06206.
- [R022] G. Moore and N. Seiberg, “Classical and Quantum Conformal Field Theory,” *Communications in Mathematical Physics* **123**, 177–254 (1989). DOI: 10.1007/BF01238857.
- [R021] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, “Symmetry Fractionalization, Defects, and Gauging of Topological Phases,” *Physical Review B* **100**, 115147 (2019). DOI: 10.1103/PhysRevB.100.115147; arXiv:1410.4540.

Part VII

Fibonacci anyons

One fusion rule, a growing pile of leftover labels, and why people got excited. A golden ratio is not a device.

Chapter 15. Two answers for one pair

Here is a piece of arithmetic that should not work.

You have one leftover charge. Call it τ . Put two of them together. The pair is allowed to look like nothing at all, or like one more copy of τ :

$$\tau \times \tau = 1 + \tau.$$

We take this as the definition of the Fibonacci fusion rule. The 1 is vacuum: the two leftovers cancel. The plus sign does not add stuff to stuff. It is a fork. When you ask what the pair is, you get one of two answers, never both at once.

15.1. A leftover that cancels itself

There are only two labels in this theory. Vacuum is 1. The leftover is τ .

Vacuum behaves as an identity. For either charge a ,

$$1 \times a = a \times 1 = a.$$

A lone τ is already itself. Two of them are the interesting case, and that is the fork above.

The leftover is its own antiparticle: fuse τ with another τ and vacuum is allowed. That is why a pair can disappear. The charge is **self-dual**. Drop that, and the empty channel leaves the menu.

The plus sign still does not mean the particles break into two pieces. It means the pair has two orthogonal endings. Those endings are the two fusion channels of $\tau \times \tau$. Last chapter used that word for Ising anyons. Here the channels are 1 and τ , and nothing else.

15.2. How many histories

A line of these leftovers carries a running total, either 1 or τ . Couple on one more τ .

A previous 1 has no choice: $1 \times \tau = \tau$. A previous τ has the fork: $\tau \times \tau = 1 + \tau$.

old total	add τ	new total
1	----->	τ
τ	----->	1 or τ

A valid history of labels is one basis state of the many-particle fusion space. More leftovers mean more valid histories. We count them before we name the sequence.

Let A_n be the number of histories for n copies of τ that end with total charge 1. Let B_n count the histories that end with total charge τ . Adding one more τ gives

$$A_{n+1} = B_n, \qquad B_{n+1} = A_n + B_n.$$

The first equation is the forced move: only a previous τ can fuse with the new τ to make 1. The second adds the two ways to finish at τ . Hence the pair of counts is completely determined by the fusion rule and the starting data.

Start with one leftover. There is no way for a single τ to be empty, and one way for it to be τ . So

$$(A_1, B_1) = (0, 1).$$

Now two:

$$A_2 = B_1 = 1, \qquad B_2 = A_1 + B_1 = 1.$$

Three:

$$A_3 = B_2 = 1, \qquad B_3 = A_2 + B_2 = 2.$$

Keep going through seven:

Number n of τ anyons	total 1: A_n	total τ : B_n	sum of sectors
2	1	1	2
3	1	2	3
4	2	3	5
5	3	5	8
6	5	8	13
7	8	13	21

Number n of τ anyons	total 1: A_n	total τ : B_n	sum of sectors
-----------------------------	----------------	----------------------	----------------

The last column is bookkeeping. It is a direct sum of different total-charge sectors, not one big coherent workspace. A superselection rule forbids treating a blur between total 1 and total τ as an ordinary qubit. Computation normally fixes the total charge and stays inside one column.

Look at either column: 1, 1, 2, 3, 5, 8, ... or 1, 2, 3, 5, 8, 13, Each entry is the sum of the two before it. Those are the **Fibonacci numbers**. Define them by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$. Then

$$A_n = F_{n-1}, \quad B_n = F_n.$$

The table follows from the fusion rule alone [R135]. The theory is named for this count, not the other way around.

15.3. The same number three ways

The same recurrence can be written as a two-by-two step. In the ordered charge basis $(1, \tau)$,

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Acting with N_τ updates the pair of path counts exactly as above. Powers of N_τ therefore generate Fibonacci numbers. If $V_c^{a_1 \dots a_n}$ is the complex vector space of fusion states in which charges $a_1, \dots, a_n$ have fixed total charge c , then

$$\dim V_1^{\tau^n} = F_{n-1}, \quad \dim V_\tau^{\tau^n} = F_n.$$

Here τ^n is shorthand for n copies of τ , not an ordinary numerical power. Every fusion multiplicity in this theory is either zero or one, so each valid labeling of a fusion tree is one basis vector.

The counts grow fast. The ratio of neighbors in the table settles down. The largest eigenvalue of N_τ is the number that ratio approaches. The characteristic equation is $\lambda^2 - \lambda - 1 = 0$, and the positive root is

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

That is the **golden ratio**. It showed up as a growth rate, not as a mystic badge on the first page.

Last chapter already had a name for a growth rate of this kind. The quantum dimension d_a of a charge a is the positive number compatible with fusion,

$$d_a d_b = \sum_c N_{ab}^c d_c.$$

Set $d_1 = 1$. The rule $\tau \times \tau = 1 + \tau$ becomes

$$d_\tau^2 = 1 + d_\tau.$$

Solving $d_\tau^2 - d_\tau - 1 = 0$ gives $(1 \pm \sqrt{5})/2$. A unitary fusion theory keeps the positive Perron–Frobenius solution,

$$d_\tau = \varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

Hence φ is three things at once: the largest eigenvalue of N_τ , the limit of F_{n+1}/F_n , and the quantum dimension of τ . The state count therefore scales as a constant times φ^n .

Quantum dimension is unitless and need not be an integer. It measures asymptotic fusion-space growth, not a number of on-site energy levels [R135]; [R015]. A single τ does not have “1.618 states.”

15.4. Unequal phases after a slide

The histories are not little wires painted on the particles. They are nonlocal labels for a degenerate state space. Exchanging neighboring leftovers acts on that space.

If the exchanged pair already has a definite fusion channel, the exchange contributes a channel-dependent phase. If you exchange a different pair, first rewrite the state in a basis where that pair fuses first, apply the phases, and rewrite it back. In symbols that is the three-step recipe $F^{-1}RF$ from the last chapter.

“Change basis” and “apply unequal phases” generally fail to commute. Two neighboring exchanges therefore generate different rotations of the encoded state. Repeating them produces a large repertoire of rotations. That is why this particular fork is famous: the repertoire is large enough to approximate the gates a quantum computer wants. The density argument below turns “large” into a careful word, and then immediately takes most of the slogan back.

An engineered Hamiltonian cannot earn the name of this theory by coughing up 1.618 somewhere. It has to reproduce the fusion spaces and the braid data. If someone claims passive protection, the same data have to be the low-energy physics of a gapped two-dimensional many-body phase.

15.5. Two paths for three leftovers

Take three τ anyons and fix their total charge to τ . The table says $B_3 = 2$. Two paths, one bit.

$$\begin{aligned} |0_L\rangle &= |((\tau\tau)_1\tau)_\tau\rangle, \\ |1_L\rangle &= |((\tau\tau)_\tau\tau)_\tau\rangle. \end{aligned}$$

The subscript on parentheses is the total charge of that group. The **logical qubit** is which channel the first pair occupies, 1 or τ , while the three together stay at τ . Locally each particle still looks like τ . The bit is not stored on any one leftover.

Four leftovers of total charge 1 also give a two-dimensional space, since $A_4 = 2$. People often prefer that block so each encoded packet has vacuum total charge. Three of total τ is the smallest picture, so we compute there.

Drop the total-charge constraint and the two paths are no longer a qubit. They sit in different superselection sectors, and a blur between those sectors is not an ordinary computational space.

15.6. Swap the first pair, then the second

Adopt the convention of Ref. [R135]: fusion trees point downward, basis channels are ordered $(1, \tau)$, and the positive generator is that review's right-handed exchange with time pointing upward. Define

$$r_1 = e^{4\pi i/5}, \quad r_\tau = e^{-3\pi i/5}.$$

Exchanging anyons 1 and 2 is diagonal in our logical basis:

$$\rho(\sigma_1) = R = \begin{pmatrix} r_1 & 0 \\ 0 & r_\tau \end{pmatrix}.$$

Here σ_i is the positive exchange of anyons i and $i + 1$, and ρ is its matrix representation on the fusion space. To exchange anyons 2 and 3, first change which pair is fused, using

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

This gauge makes F real, symmetric, and $F^{-1} = F$. Therefore the three-step recipe collapses to

$$\rho(\sigma_2) = FRF = \begin{pmatrix} \varphi^{-2}r_1 + \varphi^{-1}r_\tau & \varphi^{-3/2}(r_1 - r_\tau) \\ \varphi^{-3/2}(r_1 - r_\tau) & \varphi^{-1}r_1 + \varphi^{-2}r_\tau \end{pmatrix}.$$

Apply this braid once to $|0_L\rangle$. Both entries in the first column are nonzero, so the braid creates a superposition of fusion outcomes. If the first pair is then fused and measured, this particular state yields

$$P(1) = \varphi^{-2}, \quad P(\tau) = \varphi^{-1}.$$

They sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. The probabilities are dimensionless, as probabilities must be. They belong to this prepared state and this braid. They are not universal odds for every encounter of two τ anyons.

Since R is diagonal while FRF has nonzero off-diagonal entries,

$$\rho(\sigma_1)\rho(\sigma_2) \neq \rho(\sigma_2)\rho(\sigma_1).$$

Hence neighboring exchanges fail to commute on this two-dimensional space. That calculation is the smallest visible piece of non-Abelian statistics in this theory.

Suppose instead the two channel phases were equal, $r_1 = r_\tau$. Then R would be a global phase, FRF would equal that same phase, and the two neighboring braids would commute. Unequal phases after a change of pairing are not decoration. Drop them and the non-Abelian action disappears.

15.7. A printed matrix is not yet physics

For three τ anyons with total τ , the two bases are

$$|((\tau\tau)_x\tau)_\tau\rangle, \quad |(\tau(\tau\tau)_y)_\tau\rangle,$$

where $x, y \in \{1, \tau\}$. The matrix $F_\tau^{\tau\tau\tau}$ changes between them. In the gauge above it equals the F we already used. All other allowed F -moves in this theory are scalars and can be chosen as one.

The pentagon equation says that every sequence of F -moves between two parenthesizations of four charges gives the same answer. It is the associativity audit. Even topology has paperwork.

The scalar $R_c^{\tau\tau}$ is the phase for the chosen handed exchange of two τ charges whose combined charge is c . In our convention,

$$R_1^{\tau\tau} = e^{4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{-3\pi i/5}.$$

The hexagon equations enforce compatibility between recoupling and braiding. Together, the fusion rules, unitary F -symbols, and R -symbols define the braid representation used here [R135]; [R015].

These displayed entries are not convention-free observables. Rephasing the one-dimensional fusion and splitting vertices changes individual F -symbols and sometimes R -symbol representatives, while leaving complete diagrams and probabilities invariant. That freedom is a gauge, in the sense of the last chapter: a printed matrix is meaningless until you attach its basis order, fusion-tree orientation, gauge, and braid orientation.

Reversing braid handedness replaces R by $R^{-1} = R^\dagger$. Taking the mirror theory complex-conjugates the braiding data. Other books therefore print conjugate phases without disagreeing about the physics. Compare gauge-invariant braid words, fusion probabilities, and link amplitudes — not isolated signs.

15.8. Dense, not every gate

For n anyons the braid generators still obey the two rules from Chapter 13:

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad \sigma_i \sigma_j = \sigma_j \sigma_i \quad (|i - j| \geq 2).$$

The F - and R -data turn these abstract generators into unitary matrices $\rho(\sigma_i)$ on a fixed-total-charge fusion space.

[Theory] For this theory's data, the resulting braid representations are dense, up to a physically irrelevant overall phase, in the unitary transformations on the computational fusion spaces. In particular, the two three-anyon matrices above generate a dense set of single-qubit rotations. Suitable multi-anyon braids approximate entangling operations. That is the mathematical basis of braid universality [R015]; [R136]. Numerical compiling algorithms explicitly find braid words approximating target gates [R137].

Dense means: for a target unitary U and any tolerance $\varepsilon > 0$, some finite braid word w satisfies

$$\min_{\alpha} \|\rho(w) - e^{i\alpha} U\| < \varepsilon,$$

where $\|\cdot\|$ is an operator norm and $e^{i\alpha}$ is a global phase. It does not say every gate is one exchange. It does not say the shortest braid is known.

The ordinary English word for this density is **universal**. The name is worth keeping because it is the slogan people already know, and because that slogan will mislead unless it is fenced. It does not mean every braid is a legal computational gate. It does not mean a laboratory already has a computer.

There is an important correction to the slogan that opened the “unequal phases” section. The fusion rule alone proves Fibonacci growth and fixes d_τ in a unitary theory. It does **not** by itself prove braid universality. Universality uses the particular consistent, unitary F/R data of this braided category. A fusion ring is not yet a braid representation.

15.9. Extra states you did not order

For several encoded qubits, the full fixed-charge fusion space is generally larger than the chosen tensor-product computational subspace. A careless braid may populate the extra states.

That spill is **leakage**. “Universal” does not mean “every braid is leakage-free” [R015]; [R137]. Universal compilation can suppress it arbitrarily in the ideal theory. Initialization and fusion measurement are also required for a computer.

Drop the extra states from the accounting and a compiled braid that looks perfect on the intended bits may have left the computational subspace. The gate error you quoted is then the wrong invoice.

15.10. What the laboratory actually has

The equations above describe an ideal topological theory, not a specific material.

[Theory] The $k = 3$ Read–Rezayi fractional quantum Hall state was proposed to support a non-Abelian sector related to these anyons [R139]. Experimental observation of a Hall plateau at a compatible filling is not, by itself, a measurement of the full fusion and braiding data. In the literature surveyed through August 2026, intrinsic Fibonacci exchange and fusion have not been established in a defect crystal.

[Experiment] In 2024, Xu and collaborators used 27 superconducting transmon qubits to prepare and manipulate a digital representation of a Fibonacci Levin–Wen string-net

state. They applied circuits representing creation, fusion, and braiding and measured signatures consistent with the target model [R138]. This is a substantial laboratory result. Its microscopic excitations were not passively emerging, mobile quasiparticles of a static transmon Hamiltonian. The implemented string-net target is doubled Fibonacci order, and the operations were compiled gates on ordinary physical qubits [R138].

[Experiment] A 2026 trapped-ion experiment prepared a 54-qubit state of the S_3 quantum double and demonstrated a universal gate set by combining braiding with fusion [R140]. It is relevant precisely as a control case: the anyons were digitally encoded, and the topological order was S_3 , not this theory. “Universal anyon computation” does not identify the anyon theory.

A processor that executes an F gate has simulated an algebraic move. It has not thereby acquired an equilibrium Fibonacci phase or passive topological memory.

15.11. Four things that are not the same

- **Encoded Fibonacci qubit:** information stored in a fixed-charge fusion space of τ anyons.
- **Emergent Fibonacci anyons:** quasiparticles of a gapped two-dimensional many-body phase whose adiabatic exchange realizes the theory’s braid representation.
- **Doubled Fibonacci:** the nonchiral Drinfeld-center theory produced by a Levin–Wen model with this chapter’s input data. Its full charge set and topological order are larger than the two-object chiral theory [R018].
- **Fibonacci-like digital simulation:** ordinary hardware qubits encode fusion labels while programmed gates enact F , R , or string operators. It can test the algebra without supplying intrinsic quasiparticles or passive protection [R138].

Doubled Fibonacci is not “fake Fibonacci.” It is a genuine, distinct topological order containing computationally useful Fibonacci-type sectors. Nor is a digital simulation scientifically decorative: it can verify circuits and measurement protocols. The error is to erase which physical system carries the topological order and where protection comes from.

15.12. Ways to get this wrong

1. **Inferring universality from $\tau \times \tau = 1 + \tau$ alone.** The rule fixes counting, not braid phases. Pentagon- and hexagon-consistent unitary F/R data are essential. Drop the braid data and you still have Fibonacci growth; you do not have a density theorem.

2. **Calling $d_\tau = 1.618\dots$ a local degeneracy.** A single anyon does not have “1.618 states.” Quantum dimension describes asymptotic growth of a nonlocal fusion space. Finding φ in a spectrum does not exhibit the fusion spaces.
3. **Mixing total-charge sectors.** The sum $A_n + B_n$ is not automatically one coherent Hilbert space available for encoding. Fix the boundary charge before counting logical states. What fails if you do not: you have counted a direct sum, not a qubit.
4. **Quoting bare matrices.** Changing basis phases, basis order, braid handedness, or mirror chirality changes their appearance. Compare gauge-invariant braid words, fusion probabilities, and link amplitudes — not isolated signs.
5. **Equating braid universality with a fault-tolerant device.** [Theory] Ideal adiabatic braids are insensitive to small path deformations when anyons remain separated in a gapped phase [R015]. Real systems can still suffer thermal anyon creation, quasiparticle poisoning, finite-separation splitting, diabatic transitions, control mistakes, and faulty readout. Universality says what gates can be approximated. Protection says how errors scale. Different sentence, different invoice.
6. **Ignoring leakage in multi-qubit encodings.** The physical fusion space can contain states outside the computational tensor product. A compiler must control leakage as well as gate error [R137].
7. **Replacing emergence with a circuit diagram.** A processor that executes an F gate has simulated an algebraic move. It has not thereby acquired an equilibrium Fibonacci phase or passive topological memory.
8. **Calling every non-Abelian platform Fibonacci.** Ising anyons, finite-group quantum doubles, and Fibonacci anyons have different fusion and braid data. Non-Abelian is a family description, not a model number.

15.13. Check yourself

1. **Show that $A_{n+1} = B_n$ and $B_{n+1} = A_n + B_n$ follow from the fusion rule.**
Only a previous τ can fuse with a new τ to make 1, so $A_{n+1} = B_n$. A new total τ can come from previous 1 or previous τ , so $B_{n+1} = A_n + B_n$.
2. **Show that the positive solution of $d_\tau^2 = 1 + d_\tau$ is $\varphi = (1 + \sqrt{5})/2$.**
Completing the quadratic gives $(1 \pm \sqrt{5})/2$. Unitarity keeps the positive root, which is the golden ratio. The same equation is the characteristic equation of N_τ .
3. **Where is a three-anyon logical bit stored?**
In the fusion channel 1 or τ of the first pair, with the total charge of all three fixed to τ .

4. **What fails if the two channel phases in R are equal?**

Then R is a global phase on the two-dimensional space, FRF equals that same phase, and neighboring braids commute. The non-Abelian action is gone.

5. **What fails if you infer universality from the fusion rule alone?**

You have Fibonacci growth and, in a unitary theory, $d_\tau = \varphi$. You do not have braid phases, and you do not have the density theorem. A fusion ring is not a braid representation.

6. **Did a 27-transmon experiment create intrinsic chiral Fibonacci matter?**

No. It digitally implemented a doubled-Fibonacci string-net state and associated operations on ordinary qubits [R138].

The next chapter takes a simpler phase that is *not* universal — the toric code — and shows how local Hamiltonian terms can create anyons and nonlocal logical sectors at all.

15.14. Sources

- [R135] S. Trebst, M. Troyer, Z. Wang, and A. W. W. Ludwig, “A short introduction to Fibonacci anyon models,” *Progress of Theoretical Physics Supplement* **176**, 384–407 (2008). DOI: 10.1143/PTPS.176.384. arXiv: 0902.3275.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083. arXiv: 0707.1889.
- [R136] M. H. Freedman, M. Larsen, and Z. Wang, “A modular functor which is universal for quantum computation,” *Communications in Mathematical Physics* **227**, 605–622 (2002). DOI: 10.1007/s002200200645. arXiv: quant-ph/0001108.
- [R137] L. Hormozi, G. Zikos, N. E. Bonesteel, and S. H. Simon, “Topological quantum compiling,” *Physical Review B* **75**, 165310 (2007). DOI: 10.1103/PhysRevB.75.165310. arXiv: quant-ph/0610111.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110. arXiv: cond-mat/0404617.
- [R138] S. Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6. Stable article: nature.com/articles/s41567-024-02529-6.
- [R139] N. Read and E. Rezayi, “Beyond paired quantum Hall states: Parafermions and incompressible states in the first excited Landau level,” *Physical Review B* **59**, 8084–8092 (1999). DOI: 10.1103/PhysRevB.59.8084. arXiv: cond-mat/9809384.
- [R140] C. F. B. Lo *et al.*, “Universal gates from braiding and fusing anyons on quantum hardware,” *Nature* **655**, 591–597 (2026). DOI: 10.1038/s41586-026-10709-y. Stable article: nature.com/articles/s41586-026-10709-y.

Part VIII

Topological lattice models

Checks on a grid, a honeycomb of compasses, strings on a net. Then the difference between an actor playing a king and a king.

Chapter 16. Checks that never look at the secret

A square lattice sits on a torus. Every edge carries one spin-1/2. At each vertex the four meeting edges are asked a single yes-or-no question. At each face a different four-edge question is asked.

The answers are local. They report whether that vertex or that face is violated. They do not report whether a loop of flipped edges winds through a hole of the torus.

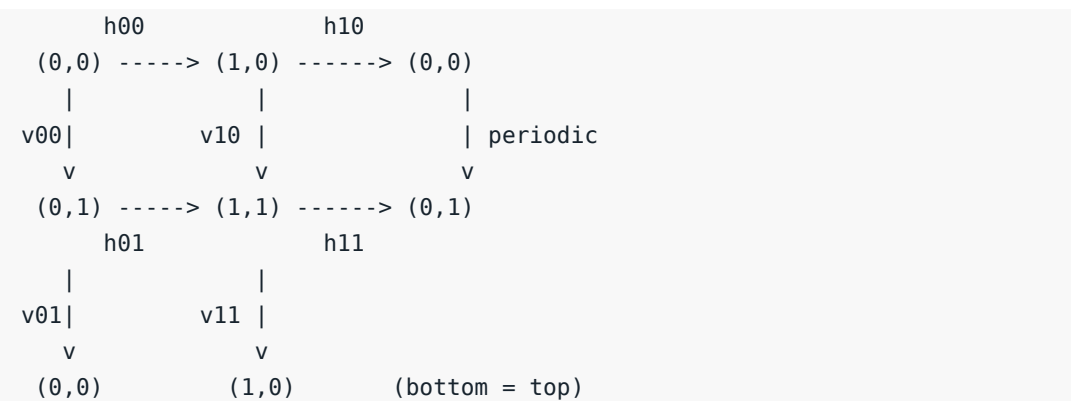
A *toric code* is this lattice model: qubits on the edges, a vertex check and a face check of the two kinds below, and either a Hamiltonian or a code built from those checks. We take that as the definition. A color-center spin is one physical qubit in a crystal. One defect per drawn edge supplies neither the checks nor a gap.

16.1. Eight qubits on a torus

Take a 2×2 square grid and identify opposite sides. Coordinates $x, y \in \{0, 1\}$ run modulo 2. Let $h_{x,y}$ be the horizontal edge from (x, y) to $(x + 1, y)$, and let $v_{x,y}$ be the vertical edge from (x, y) to $(x, y + 1)$.

There are four horizontal edges and four vertical edges. The Hilbert space has dimension $2^8 = 256$.

A flat drawing of the periodic patch is



The repeated vertices are identifications, not extra sites. A 1×1 periodic drawing is too cramped to show four distinct edges at a vertex. The 2×2 patch is the smallest that keeps the intended local pattern and still has two independent cycles.

Hence this is already a torus, not a crystal. The eight spins are labels on edges. They are not eight nitrogen-vacancy centers that inherited topology by sitting near one another.

Drop periodicity and the same local checks still make sense on a planar patch with boundary. The leftover-state count does not.

16.2. The four-edge product at a vertex

Four edges meet at a vertex: the horizontal edge going right, the horizontal edge coming in from the left, the vertical edge going up, and the vertical edge coming in from below.

Ask those four spins the X question and multiply. For the vertex at (x, y) ,

$$A_{x,y} = X_{h_{x,y}} X_{h_{x-1,y}} X_{v_{x,y}} X_{v_{x,y-1}},$$

where X_e is the Pauli X operator on edge e , and every index is modulo 2.

That four-edge product is a *star*. On a ground state the star returns +1. A -1 means that vertex is violated.

Suppose two stars share an edge. Both are products of X , so they commute. Hence every star can be given a definite value at once.

If a later face check used X on a single shared edge instead of Z on two, that star and that face would anticommute. The split of Pauli type is part of the definition, not decoration.

16.3. The four-edge product on a face

Four edges bound a face: bottom, right, top, left.

Ask those four spins the Z question and multiply. For the square whose lower-left corner is (x, y) ,

$$B_{x,y} = Z_{h_{x,y}} Z_{v_{x+1,y}} Z_{h_{x,y+1}} Z_{v_{x,y}},$$

where Z_e is Pauli Z on edge e . Indices again run modulo 2.

That four-edge product is a *plaquette*. On a ground state the plaquette returns +1. A -1 means that face is violated.

Stars are made of X . Plaquettes are made of Z . That split is the local algebra of the model.

Replace the four- Z product by four X operators on the same face and the algebra with the stars changes: a star and that face then share edges of matching Pauli type, and two canceling minus signs are no longer guaranteed.

16.4. How a pair walks around a square

Start from any state $|\psi_0\rangle$ on which every star and every plaquette already reads $+1$. Apply Z on the single edge $h_{0,0}$.

A star contains X operators. X and Z on the same edge anticommute. So this one Z anticommutes with the two stars that touch $h_{0,0}$, the stars at $(0,0)$ and $(1,0)$. Those two eigenvalues flip from $+1$ to -1 . The two violated vertices are *e anyons*.

The letter e labels a broken star. We keep it because the next construction breaks plaquettes instead.

Next apply $Z_{v_{1,0}}$. The shared endpoint $(1,0)$ is flipped a second time and returns to $+1$. The other new endpoint appears at $(1,1)$. The pair has walked. Then apply $Z_{h_{0,1}}$, moving the walker to $(0,1)$. Finally apply $Z_{v_{0,0}}$. Both remaining endpoints disappear.

The four operations multiply to one plaquette:

$$Z_{h_{0,0}}Z_{v_{1,0}}Z_{h_{0,1}}Z_{v_{0,0}} = B_{0,0}.$$

Because $B_{0,0}|\psi_0\rangle = |\psi_0\rangle$, the finished trip around one square changed nothing the checks can see. Intermediate states held a pair. The closed path is itself a check.

There is a different ending. Keep walking around a periodic direction instead of around one square. The endpoints meet and vanish, so every local check reads $+1$. The leftover loop cannot shrink. It is not a product of local plaquettes. Hence the checks have gone quiet, and a winding has moved.

Stop after three edges and one e remains. Energy is still paid, and you have not returned to the codespace.

16.5. Strings of the other kind

A path γ along the ordinary edges of the grid defines a Z string

$$W_e(\gamma) = \prod_{e \in \gamma} Z_e.$$

At an interior vertex the path uses two of the four edges, so it anticommutes with that star twice and leaves no net sign. At each endpoint it uses one edge and flips the star. Those endpoints are the e anyons just walked.

Now draw a path γ^* on the dual lattice: it runs through the centers of faces and crosses a set of ordinary edges. Its X string is

$$W_m(\gamma^*) = \prod_{e \perp \gamma^*} X_e.$$

This string flips the plaquette checks at its two endpoint faces. Those broken faces are m anyons.

Stars detect e . Plaquettes detect m . An open Z string makes e . An open X string makes m .

If both kinds of check are broken at once, the composite is $\varepsilon = e \times m$. It carries one electric violation and one magnetic violation.

A Y error on one edge is not a third independent species. Since $Y = iXZ$, it contributes to both syndromes at neighboring sites.

16.6. Why the checks commute

Each star A_v and each plaquette B_p is Hermitian and squares to the identity, so its only possible readings are $+1$ and -1 . Here v labels a vertex and p a face.

Two stars commute because they are made only of X . Two plaquettes commute because they are made only of Z .

A star and a plaquette either share no edge or share two edges. On one shared edge, $XZ = -ZX$. On two shared edges the two minus signs cancel:

$$A_v B_p = (-1)^2 B_p A_v = B_p A_v.$$

Hence every check can have a definite value at the same time. The local questions do not fight.

These checks are the *stabilizers* of the model: operators that leave the good states alone and return +1 there.

What fails if a star and a plaquette share a single edge: they anticommute, they cannot be simultaneously sharp, and a Hamiltonian built from both is no longer a sum of commuting penalties.

16.7. Energy for a broken check

If the checks are standing energy penalties, the energy of the lattice is

$$H_{\text{TC}} = -J_e \sum_v A_v - J_m \sum_p B_p,$$

where $J_e > 0$ and $J_m > 0$ are energies. Pauli products have no units, so each J must carry energy and H_{TC} does too.

A check itself is not a projector. Its eigenvalues are ± 1 . The projectors onto those two answers are

$$P_v^\pm = \frac{I \pm A_v}{2}, \quad P_p^\pm = \frac{I \pm B_p}{2}.$$

Write N_v and N_p for the numbers of vertices and faces. Then

$$H_{\text{TC}} = E_0 + 2J_e \sum_v P_v^- + 2J_m \sum_p P_p^-, \quad E_0 = -J_e N_v - J_m N_p.$$

A violated star costs $2J_e$. A violated plaquette costs $2J_m$. On a closed torus the rules force either species to appear in pairs, so the cheapest pair costs $4J_e$ or $4J_m$.

Hence the model is a commuting-projector Hamiltonian: a sum of local penalties that all commute. The exact solution, and the anyon reading of the broken checks, is Kitaev's construction [R030].

A diamond does not hand you H_{TC} . The four-body products have to be arranged, approximated, or compiled. Until they are, you have eight — or eight thousand — ordinary spins, not this energy.

16.8. Four quiet states

A ground state of H_{TC} satisfies

$$A_v|\psi\rangle = |\psi\rangle, \quad B_p|\psi\rangle = |\psi\rangle$$

for every vertex and every face. On the 2×2 torus there are eight written checks. They are not all independent.

Every edge touches two vertices and borders two faces, so

$$\prod_v A_v = I, \quad \prod_p B_p = I.$$

There are therefore three independent stars and three independent plaquettes. Six independent yes-or-no constraints on an eight-qubit space leave

$$\dim \mathcal{H}_0 = 2^{8-6} = 4.$$

Equivalently, the patch encodes $k = 8 - 6 = 2$ logical qubits. The fourfold degeneracy is exact for the ideal Hamiltonian on a torus. More generally the toric code has 4^g ground states on a closed orientable surface of genus g ; a torus is $g = 1$ [R030].

You can write one of the four states by hand. Begin with $|0\rangle^{\otimes 8}$, which already has $B_p = +1$ on every face, and average over products of stars:

$$|\psi_{00}\rangle \propto \prod_{v \neq v_0} (I + A_v) |0\rangle^{\otimes 8}.$$

One star v_0 is left out because the product of all stars is already I . The state is an equal superposition of the contractible loop configurations. Applying either of two independent wrapping operators produces the other three sectors.

Hence four ground states survive, and they are locally indistinguishable.

Count eight independent checks and you would conclude there is no leftover state. The two global products equal the identity, so that count is wrong.

16.9. Loops that wrap

On our eight-edge patch, two Z loops that wrap the torus and cannot shrink are

$$\bar{Z}_x = Z_{h_{0,0}} Z_{h_{1,0}}, \quad \bar{Z}_y = Z_{v_{0,0}} Z_{v_{0,1}}.$$

Two X loops that cross them once can be chosen as

$$\bar{X}_x = X_{h_{0,0}} X_{h_{0,1}}, \quad \bar{X}_y = X_{v_{0,0}} X_{v_{1,0}}.$$

Each closed loop commutes with every star and every plaquette. Matching pairs share one edge and anticommute: $\bar{X}_x \bar{Z}_x = -\bar{Z}_x \bar{X}_x$. Mismatched pairs share zero edges or two, and they commute. That is exactly the Pauli algebra of two encoded qubits.

These wrapping products are the *logical operators*: the smallest operations that change the ground sector without leaving a local scar.

A local operator cannot measure a wrapping eigenvalue. It also cannot turn one quiet sector into another. On an $L \times L$ lattice it takes a string of at least L single-edge Pauli operations to complete a loop that cannot shrink. That smallest wrapping weight is the code distance, $d = L$: the number of local mistakes you must stack before a winding moves and the checks still look fine.

Under weak enough local perturbations that leave a many-body gap, topological-order stability theorems say the local indistinguishability survives and the splitting among the four sectors is exponentially small in the system size, subject to their locality assumptions [Theory] [R142].

Drop the many-body gap and those theorems do not apply. Local perturbations can then mix the sectors at a scale that need not vanish with L .

16.10. A phase, not a matrix

The fusion rules of the four charges are

$$e \times e = 1, \quad m \times m = 1, \quad e \times m = \varepsilon,$$

$$e \times \varepsilon = m, \quad m \times \varepsilon = e, \quad \varepsilon \times \varepsilon = 1,$$

where 1 means vacuum, nothing left. Every pair of charges has exactly one fusion outcome. Every fixed-charge fusion space is therefore one-dimensional. There is no hidden basis for a braid to mix.

The string algebra is still not boring. Take an e around an m . Their Z string and X string cross once, so

$$W_e W_m = -W_m W_e.$$

The wavefunction picks up a phase -1 . Thus e and m have mutual semionic statistics. Taken alone, e and m have bosonic exchange statistics. Their composite ε is a fermion [R030].

These anyons are *Abelian*: every braid multiplies the state by a number, a phase. Non-Abelian anyons act with noncommuting matrices on a fusion space that has more than one dimension. The toric code has topology. It does not have universal computation by braiding.

The name is worth keeping because ordinary English hears “Abelian” as “nothing happens.” A mutual phase -1 happens. It is still a single number, so it cannot enact the matrix-valued braids of a non-Abelian fusion space.

Topological does not mean non-Abelian.

16.11. Endpoints a decoder can see

Suppose unwanted Z errors land on a set of edges E . The error operator is $Z(E) = \prod_{e \in E} Z_e$. Star checks return -1 exactly on the boundary ∂E : the vertices that touch an odd number of errored edges.

Those violated vertices are the *syndrome*: the data a decoder is allowed to see. The decoder never sees the path. It only sees the endpoints, and then has to guess a recovery.

A decoder chooses a recovery chain R with the same boundary. After recovery the net chain is the symmetric difference $E \oplus R$, because applying Z twice is the identity. Since $\partial(E \oplus R) = 0$, the leftover is closed. Two things can happen:

1. If $E \oplus R$ can shrink, it is a product of plaquettes and does nothing to the encoded sector.
2. If $E \oplus R$ winds around the torus, it is a logical $\bar{Z}$ and silently flips an encoded bit.

The same story with X errors, plaquette syndromes, and dual-lattice recovery produces logical $\bar{X}$ failures. A general Pauli $Y = iXZ$ contributes to both syndromes. Dennis and collaborators wrote this error-chain picture and tied it to statistical-mechanical decoding [R141].

Hence a vanished syndrome is not a proof of success. Both a harmless shrinkable loop and a damaging wrapping loop have no endpoints.

What fails if the decoder only cancels defects and never infers homology: it can “correct” into the wrong sector with a wrapping leftover.

16.12. A standing penalty versus asking again

The same operators appear in two physically different machines.

In *passive Hamiltonian protection*, the terms $-J_e A_v$ and $-J_m B_p$ sit in the device Hamiltonian all the time. Leaving the ground space costs energy. At low temperature a bath must supply that energy to create a pair. A gap also supports stability against weak enough local perturbations [**Theory**] [R142]. This is genuine Hamiltonian protection only if the physical system actually realizes those interactions and lives in their gapped phase.

It is not a self-correcting quantum memory in two dimensions. Once a pair exists, either pointlike anyon can walk across the lattice without increasing the number of excitations. The highest energy along a logical-string path therefore stays of order J_e or J_m . It does not grow with L . At nonzero temperature, diffusion can finish a wrapping loop, so making the lattice bigger without bound does not make the passive lifetime bigger without bound [**Theory**] [R143]; [R144]. The gap makes pairs expensive to create. It does not pin the pairs that already exist.

In *active syndrome extraction*, hardware measures the stars and plaquettes again and again, usually with extra ancilla qubits and a gate circuit. Classical software compares rounds, guesses space-time error chains, and updates a correction or a Pauli frame. The checks need not be energy terms in the data-qubit Hamiltonian at all. Measurement errors make time another decoding direction, which is why one round of tidy-looking syndrome data is not enough [R141]; [R145].

A static H_{TC} whose spectrum is thermally occupied is an analog many-body model. A circuit can prepare a toric-code wavefunction without making H_{TC} the hardware’s equilibrium Hamiltonian. Repeated check measurement is active quantum error correction, even if no four-body energy penalty exists. A material with a robust emergent $\mathbb{Z}_2$ topological phase is stronger evidence than a prepared state or a short digital braid. Eight defect spins on the eight edges of a drawing are still eight defect spins. They become this model only after the stars, the plaquettes, and a many-body gap are shown to exist.

The two strategies can be stacked, but their error models differ. Passive penalties change transition energies. Active correction buys information and spends control, measurement, and decoding. Calling both “the toric code” is fine only after you say which machine you mean.

16.13. What the laboratory actually has

The ideal square-lattice Hamiltonian is built from weight-four products. Most elementary spin couplings in candidate hardware are one-body or two-body, so the exact commuting-projector Hamiltonian needs direct multiqubit interactions, perturbative stand-ins, extra constraints, or digital gates. Each route brings extra energy scales or control errors.

For a defect array, assigning one defect spin to each edge is only the first line of the recipe. One must still demonstrate the star and plaquette constraints and a many-body gap **[Proposal]**. A diamond device with tidy single-spin coherence is a different object than this lattice model.

[Experiment] Satzinger and collaborators prepared and characterized a toric-code ground state using 31 superconducting qubits, measured a topological entanglement signature, and performed anyon operations [R125]. That is strong evidence a programmable processor can synthesize and probe the state. It is not evidence that the processor's native equilibrium material realizes a passive toric-code Hamiltonian.

[Experiment] Repeated stabilizer extraction has been demonstrated in planar surface-code devices, which use boundaries instead of a periodic torus but share the same local check algebra. Krinner and collaborators ran a distance-three, 17-qubit surface code through repeated correction cycles [R145]. That is evidence for active extraction and decoding, not for thermally passive storage.

[Experiment] A trapped-ion processor has also prepared toric-code topological order with measurement and feed-forward under periodic connectivity [R146]. Again the preparation protocol is the resource that did the work. Turning the controls off does not leave behind an autonomous topological material.

For defect-engineered crystals the toric code is therefore a benchmark, not a free inheritance. A convincing claim would need spectroscopy or dynamics that establish the intended effective terms, a separated low-energy manifold, nonlocal ground sectors, and controlled string excitations. Long coherence of individual defects, a few entangled neighbors, or a software-defined stabilizer graph answers a different question.

16.14. Ways to get this wrong

1. Calling every stabilizer a projector. A_v and B_p have eigenvalues ± 1 . The projectors are $(I + A_v)/2$ and $(I + B_p)/2$. The distinction matters when you compare Hamiltonian coefficients.
2. Forgetting the global relations. Counting eight checks on eight qubits would predict no leftover state. Two products of checks equal the identity, leaving six independent constraints and two logical qubits.

3. Treating a vanished syndrome as proof of success. Both a harmless shrinkable loop and a damaging wrapping loop have no endpoints. Decoding has to infer the homology class, not merely cancel defects.
4. Equating an energy gap with a self-correcting memory. The two-dimensional toric code has a finite pair-creation cost and no energy barrier that grows with system size [R143]; [R144]. Passive suppression is useful. Indefinite thermal protection does not follow.
5. Equating Abelian braiding with triviality. The -1 mutual phase is topological and measurable. It is still a single number, so it cannot enact the matrix-valued braids of a non-Abelian fusion space.
6. Equating digital preparation with emergence. A gate sequence may produce the exact wavefunction and the anyon algebra. Emergent topological order also requires that a local physical Hamiltonian make that sector its robust low-energy physics.
7. Assuming a torus demonstration is a device layout. A torus is mathematically clean. Laboratories usually use planar patches with boundaries. Boundaries change the degeneracy and the shape of the logical operators. They keep the local star and plaquette lesson.
8. Mistaking a diamond array for this model. One defect per edge is a hardware sketch. Without the four-body checks and a many-body gap, the crystal is still a pile of ordinary qubits.

16.15. Check yourself

1. **Show that a star and a plaquette commute.** Same-type checks use the same Pauli type. A star and a plaquette overlap on zero edges or two, so their XZ minus signs occur an even number of times.
2. **Show that the eight-edge torus encodes two logical qubits.** Eight physical qubits and six independent stabilizers: three stars plus three plaquettes, because $\prod_v A_v = I$ and $\prod_p B_p = I$. So $k = n - r = 8 - 6 = 2$, and there are four ground states.
3. **What creates an e pair, and what creates an m pair?** An open direct-lattice Z string flips star checks at its endpoints and creates e anyons. An open dual-lattice X string flips endpoint plaquettes and creates m anyons.
4. **What fails if a decoder treats a vanished syndrome as success?** Both a shrinkable leftover and a wrapping leftover have empty boundary. The wrapping leftover is a logical operator.
5. **Why are the anyons Abelian if the braiding phase is not $+1$?** Fusion outcomes are unique, so each fusion space is one-dimensional. Braiding multiplies the state by a phase such as -1 rather than mixing a vector of fusion amplitudes.

6. **What fails if you take the many-body gap for a self-correcting memory?**
 Once a pair exists, either anyon can walk without raising the excitation number. The barrier along a wrapping path stays of order J_e or J_m and does not grow with L .

Local commuting checks can open a gapped subspace whose leftover labels are nonlocal. Open strings make endpoint anyons. Closed shrinkable strings are stabilizers. Closed wrapping strings are logical operators. Crossing strings supply the braiding phase. A many-body gap is not by itself a scalable finite-temperature memory. Active extraction is not passive Hamiltonian protection. One-dimensional fusion spaces make these anyons Abelian. The next lattice we meet will try to get related order from two-body honeycomb bonds instead of four-spin checks.

16.16. Sources

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0. arXiv: quant-ph/9707021.
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* **43**, 4452–4505 (2002). DOI: 10.1063/1.1499754. arXiv: quant-ph/0110143.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195. arXiv: 1001.0344.
- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* **42**, 065303 (2009). DOI: 10.1088/1751-8113/42/6/065303. arXiv: 0810.4584.
- [R144] B. M. Terhal, “Quantum error correction for quantum memories,” *Reviews of Modern Physics* **87**, 307–346 (2015). DOI: 10.1103/RevModPhys.87.307. arXiv: 1302.3428.
- [R145] S. Krinner *et al.*, “Realizing repeated quantum error correction in a distance-three surface code,” *Nature* **605**, 669–674 (2022). DOI: 10.1038/s41586-022-04566-8. arXiv: 2112.03708.
- [R125] K. J. Satzinger *et al.*, “Realizing topologically ordered states on a quantum processor,” *Science* **374**, 1237–1241 (2021). DOI: 10.1126/science.abi8378. arXiv: 2104.01180.
- [R146] M. Iqbal *et al.*, “Topological order from measurements and feed-forward on a trapped ion quantum computer,” *Communications Physics* **7**, 205 (2024). DOI: 10.1038/s42005-024-01698-3. arXiv: 2302.01917.

Chapter 17. A honeycomb of compasses

Draw a honeycomb. At every vertex sit a spin-1/2. Three bonds leave each site, and those three bonds point along three different lattice directions. Color the directions x , y , and z .

The neighbor rule is not “point the same way.” Along an x -colored bond the two spins couple only through their x components. Along a y bond, only y . Along a z bond, only z . Each spin therefore hears three different instructions at once.

A classical arrow cannot obey all three. The x , y , and z pieces of a quantum spin do not commute, so the same site cannot settle them independently.

This locked pairing of bond direction with spin direction is a *compass* interaction: on each nearest-neighbor bond of type $\alpha \in \{x, y, z\}$, the energy contains $\sigma_i^\alpha \sigma_j^\alpha$ and no other spin components. We take that as the definition. It is not the Heisenberg magnet, where every component talks to every component.

17.1. One hexagon that will not flip

The frustration is organized, not random. Walk around a single hexagon and pick one spin component at each corner. The product of those six operators is a loop the energy cannot change.

Label the six sites clockwise by 1, 2, ..., 6. Let the bond types around the perimeter be

```

      1
    y /   \ z
      6     2
    x |     | x
      5     3
    z \   / y
      4
  
```

perimeter sequence: (1,2)= z , (2,3)= x , (3,4)= y ,
(4,5)= z , (5,6)= x , (6,1)= y

Every vertex also has one bond leaving the hexagon. Its type is the missing member of $\{x, y, z\}$. So the outward types at sites 1 through 6 are x, y, z, x, y, z .

Put one spin-1/2 at each vertex. Write σ_i^α for the dimensionless Pauli operator of component $\alpha \in \{x, y, z\}$ at site i . The loop we want is

$$W_p = \sigma_1^x \sigma_2^y \sigma_3^z \sigma_4^x \sigma_5^y \sigma_6^z.$$

Operators on different sites commute, and each Pauli operator squares to the identity. Therefore

$$W_p^2 = 1,$$

so a measurement of W_p can return only $w_p = +1$ or $w_p = -1$. That two-valued loop eigenvalue is the *flux* through the hexagon. No ordinary magnetic field has to thread the ring.

Now check that the energy cannot change it. Take the z -bond term on sites 1 and 2,

$$K_{12} = \sigma_1^z \sigma_2^z.$$

At site 1, σ^z anticommutes with the σ^x sitting in W_p . At site 2, it anticommutes with σ^y . Passing K_{12} through W_p therefore produces two minus signs:

$$K_{12}W_p = (-1)^2 W_p K_{12} = W_p K_{12}.$$

The same two-minus-sign argument works for every perimeter bond. An outward bond uses the same component as W_p at its hexagon endpoint, so it commutes there and acts as the identity on the other five plaquette sites. Every bond term commutes with W_p , and so does their sum:

$$[H, W_p] = 0.$$

Hence every plaquette flux is conserved in the ideal model [R017]. The coloring is structural, not decoration. If a fabricator quietly replaces an x coupling by an isotropic one, the two-minus-sign argument may no longer apply.

A single spin operator generally flips two neighboring fluxes. For example, σ_i^α anticommutes with the two plaquette operators adjacent to the α bond from site i , and commutes with the others. Acting on a simultaneous flux eigenstate changes those two w_p signs. On a closed periodic lattice, fluxes are therefore created in pairs. This is the honeycomb cousin of creating two toric-code anyons at the ends of a short string.

17.2. Three colored energies

Let $\langle ij \rangle_\alpha$ mean a nearest-neighbor bond of type α . The Kitaev honeycomb energy is

$$H_K = -J_x \sum_{\langle ij \rangle_x} \sigma_i^x \sigma_j^x - J_y \sum_{\langle ij \rangle_y} \sigma_i^y \sigma_j^y - J_z \sum_{\langle ij \rangle_z} \sigma_i^z \sigma_j^z.$$

The real numbers J_x, J_y, J_z have units of energy. Pauli products are dimensionless, so every term and H_K have units of energy. When we draw the standard phase diagram we take $J_\alpha \geq 0$. Many sign choices can be related by spin rotations or gauge choices. Boundaries and finite geometries deserve separate care.

This is the compass interaction written as a Hamiltonian. It is not the isotropic Heisenberg interaction $J \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j$. It also contains no explicit six-spin W_p term. The loop conservation law comes out of the algebra of the two-body terms.

Because all W_p commute with H_K and with one another, an energy eigenstate can be labeled by a set $\{w_p\}$. Each complete assignment is a *flux sector*.

On a torus there are also two global Wilson-loop labels, and not every apparent sign assignment is independent. Those global constraints matter for exact state counting. The local lesson is that $\{w_p\}$ is conserved.

For the translationally invariant model on the honeycomb lattice, the ground state lies in the flux-free sector, conventionally $w_p = +1$ on every plaquette [R017]; [R148]. **[Theory]** “Flux-free” does not mean “feature-free.” The leftover spectrum can still be gapless or topological.

Chapter 16 obtained topology from explicit four-spin stabilizers. A proposed defect array is more likely to supply pairwise exchange, dipolar, or mediated couplings. The honeycomb model is the existence proof that strictly local two-body spin interactions can generate flux sectors, fractionalized quasiparticles, and topological order collectively. **[Theory]** It is an exact result for a special Hamiltonian, not evidence that an arbitrary honeycomb-shaped array becomes topological [R017]; [R148].

17.3. Four halves for two answers

Inside any one sector the interacting spins can be rewritten as free fermions hopping through a frozen pattern of signs. The bookkeeping objects are Hermitian and square to one. Each is half of an ordinary complex fermion in the algebraic sense.

At each site introduce four of them,

$$b_i^x, \quad b_i^y, \quad b_i^z, \quad c_i,$$

which anticommute when distinct and square to one. Represent the spin operators as

$$\sigma_i^\alpha = ib_i^\alpha c_i.$$

A *Majorana* operator γ is exactly that kind of object: $\gamma^\dagger = \gamma$ and $\gamma^2 = 1$. The name is worth keeping because it marks an emergent variable used to solve a many-spin problem, not a microscopic electron literally sawn in half.

Four Majoranas span more states than one spin-1/2. The physical subspace is selected by the local constraint

$$D_i = b_i^x b_i^y b_i^z c_i = +1.$$

Changing all four Majoranas at a site by a minus sign changes no physical spin operator. That redundancy is a local $\mathbb{Z}_2$ gauge freedom, not an extra laboratory knob.

Choose an orientation for every α bond from site i to site j , and define

$$u_{ij} = ib_i^\alpha b_j^\alpha, \quad u_{ij} = \pm 1.$$

Substitution into one bond term gives

$$-J_\alpha \sigma_i^\alpha \sigma_j^\alpha = i J_\alpha u_{ij} c_i c_j,$$

with the displayed sign tied to the chosen bond orientation. Thus

$$H_K = i \sum_{\langle ij \rangle_\alpha} J_\alpha u_{ij} c_i c_j.$$

Every u_{ij} commutes with this ideal Hamiltonian. Once their signs are fixed, H_K is quadratic in the c Majoranas and can be diagonalized as a free-fermion hopping problem. The gauge-invariant product of u_{ij} around a plaquette equals W_p up to the fixed orientation convention. Individual u_{ij} signs depend on the gauge. The loop product is physical.

The original sites carry physical spins. The b and c Majoranas live in an enlarged space subject to $D_i = +1$. A fixed- u solution before projection contains unphysical states. A flux is an emergent many-body label. None of those facts demonstrates a localized, independently wired Majorana zero mode in a material, and none creates an encoded qubit by itself.

The reorganization splits two kinds of excitation:

- fluxes, which change the loop eigenvalues; and
- matter Majoranas, whose allowed energies depend on the flux background.

Their combination is the useful physics. Neither ingredient, treated as an unconstrained bookkeeping variable, is by itself a physical spin excitation.

17.4. When three lengths make a triangle

Use a two-site unit cell with sublattices A and B . Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be the translations connecting a chosen z bond to neighboring x and y bonds. In the flux-free gauge, the positive matter-Majorana excitation energy has the form

$$E(\mathbf{k}) = 2|f(\mathbf{k})|, \quad f(\mathbf{k}) = J_z + J_x e^{i\mathbf{k} \cdot \mathbf{a}_1} + J_y e^{i\mathbf{k} \cdot \mathbf{a}_2},$$

where $\mathbf{k}$ is crystal momentum. The factor of 2 depends on the Hamiltonian normalization. The condition for a zero does not.

The three complex terms in f can sum to zero exactly when the lengths J_x, J_y, J_z can form a triangle. Therefore the *gapless B phase* obeys

$$J_x \leq J_y + J_z, \quad J_y \leq J_z + J_x, \quad J_z \leq J_x + J_y.$$

The name is Kitaev's label for the interior of that triangle. The isotropic point $J_x = J_y = J_z$ sits inside it and has two Majorana Dirac cones.

Outside the triangle are three gapped regions:

$$\begin{aligned} A_x : J_x &> J_y + J_z, \\ A_y : J_y &> J_z + J_x, \\ A_z : J_z &> J_x + J_y. \end{aligned}$$

[Theory] At zero field the A phases have Abelian topological order. In a strongly anisotropic limit their low-energy theory becomes toric-code-like [R017]; [R148]. The unperturbed B phase is gapless, so it is not yet the gapped non-Abelian phase often advertised in one sentence too few.

Drop any one triangle inequality and $f(\mathbf{k})$ cannot vanish. The leftover spectrum then has a finite gap, and the isotropic cones are gone.

17.5. How pairs manufacture a loop

Take $J_z \gg J_x, J_y > 0$. A z bond has Hamiltonian $-J_z \sigma_i^z \sigma_j^z$. Its aligned states have energy $-J_z$ and its anti-aligned states have energy $+J_z$, leaving a low-energy doublet separated by $2J_z$. Treat each aligned z dimer as one effective spin.

A single x or y bond leaves this subspace. The first process that returns every disturbed dimer while distinguishing a closed loop occurs at fourth order. Degenerate perturbation theory gives, in the standard convention,

$$H_{\text{eff}} = \text{constant} - J_{\text{eff}} \sum_p \tilde{W}_p + \dots, \quad J_{\text{eff}} = \frac{J_x^2 J_y^2}{16 J_z^3},$$

where $\tilde{W}_p$ is a four-effective-spin plaquette operator on the contracted lattice [R017]. The dimensional check is

$$[J_{\text{eff}}] = \frac{(\text{energy})^4}{(\text{energy})^3} = \text{energy}.$$

Hence nothing inserted a four-body microscopic force. Repeated virtual excursions generated it. The price is equally visible: if $J_x, J_y \ll J_z$, the emergent topological scale is much smaller than the microscopic coupling.

What fails if the anisotropy is only moderate: higher-order terms and residual dimer-breaking processes are no longer small, and the effective toric-code description is not a controlled expansion.

17.6. A field, a gap, and Ising anyons

Break time-reversal symmetry with a weak Zeeman perturbation

$$H_h = - \sum_i (h_x \sigma_i^x + h_y \sigma_i^y + h_z \sigma_i^z),$$

where each h_α is an energy. For a physical magnetic field it contains the appropriate g factor and Bohr magneton. In third-order perturbation theory, nonzero $h_x h_y h_z$ generates an effective three-spin interaction with scale

$$\kappa \sim \frac{h_x h_y h_z}{J^2},$$

where J denotes a representative Kitaev exchange. Again $[\kappa] = (\text{energy})^3/(\text{energy})^2 = \text{energy}$. This term opens a mass gap at the Dirac cones.

[Theory] In the weak-field effective model, the resulting phase has Majorana-band Chern number $\nu = +1$ or -1 and vortices bind Majorana zero modes, producing non-Abelian Ising topological order [R017]; [R148]. A generic microscopic magnetic field also destroys exact flux conservation, so the all-field problem is not simply the same free-Majorana solution with a larger gap.

The word *Ising* now needs three separate boxes, because it has been asked to do three different jobs.

1. Conventional Ising magnetic order breaks a $\mathbb{Z}_2$ symmetry and is diagnosed by a local order parameter such as magnetization. That is not the phase under discussion.
2. Ising topological order has topological charges 1 (vacuum), σ (vortex), and ψ (fermion), with

$$\sigma \times \sigma = 1 + \psi, \quad d_\sigma = \sqrt{2}.$$

Its degeneracy and braiding live in a nonlocal fusion space. No local magnetization is required.

3. Fibonacci topological order instead has 1 and τ , with

$$\tau \times \tau = 1 + \tau, \quad d_\tau = \varphi = \frac{1 + \sqrt{5}}{2}.$$

These fusion rules are not relabelings. Braiding Ising anyons alone does not provide a dense set of arbitrary qubit unitaries. In standard encodings it supplies Clifford operations and needs an additional non-topological resource for universality. Fibonacci braiding is computationally universal in the sense developed in Chapter 15.

So the field-gapped honeycomb model is a rigorous route from two-body interactions to Ising-type non-Abelian anyons. It is not a realization, approximation, or synonym of Fibonacci order. Do not call the honeycomb Fibonacci. The two phases disagree about the only data that define them.

17.7. What the laboratory actually has

The microscopic route most often discussed uses spin-orbit-entangled magnetic ions in edge-sharing octahedra. Interference between exchange paths can suppress ordinary Heisenberg exchange and favor bond-directional Kitaev coupling. **[Proposal]** Jackeli and Khaliullin derived this mechanism for certain Mott insulators [R147]. It is a materials design principle, not a proof that every edge-sharing honeycomb implements H_K .

Candidate compounds — including honeycomb iridates and α -RuCl₃ — also contain Heisenberg exchange, symmetric off-diagonal Γ terms, longer-range couplings, lattice distortions, phonons, stacking faults, and interlayer coupling [R149]; [R150]. α -RuCl₃ develops zigzag magnetic order near 7 K at zero field, depending on sample and stacking history [R151]. **[Experiment]** That local symmetry-breaking order is direct evidence that the clean zero-field sample is not the ideal Kitaev spin liquid.

Neutron and Raman measurements have nevertheless found broad continua and field-dependent spectra compatible with substantial Kitaev interactions and fractionalized descriptions [R149]; [R150]; [R151]. **[Experiment]** Compatibility is weaker than unique identification: magnons, disorder, and multiparticle continua can overlap similar frequency ranges.

A sharper proposed signature is the thermal Hall conductance of one chiral Majorana edge mode. The ideal two-dimensional theory predicts

$$\frac{\kappa_{xy}}{T} = \frac{1}{2} \frac{\pi^2 k_B^2}{3h_P} = \frac{\pi k_B^2}{12\hbar},$$

where κ_{xy} is transverse thermal conductance, T is temperature, k_B is Boltzmann's constant, h_P is Planck's constant, and $\hbar = h_P/(2\pi)$. The units are W K⁻².

Kasahara and collaborators reported a half-quantized plateau-like signal in field-driven α -RuCl₃ and interpreted it as a Majorana edge contribution [R152]. **[Experiment]** Later measurements found strong sample dependence and substantial phonon or bosonic contributions [R150]; [R153]. A 2026 acoustic study directly inferred phonon Hall viscosity and attributed a substantial intrinsic thermal Hall response to phonons [R154]. **[Experiment]** As of 2026, the field-induced non-Abelian Kitaev phase in α -RuCl₃ remains actively disputed rather than experimentally settled [R150]; [R154].

No cited experiment has taken engineered color-center defects, arranged them into this exact compass Hamiltonian, and established intrinsic honeycomb-model Ising order. Still less has it obtained Fibonacci order. For defect engineering, this chapter supplies a Hamiltonian target and a warning label: geometry, pairwise coupling, and a suggestive continuum are not sufficient. One must establish the interaction tensor, many-body gap, flux or gauge structure, and topological response.

17.8. Ways to get this wrong

1. Calling a honeycomb a Kitaev material because the graph has hexagons. A graph specifies who is connected. The Kitaev Hamiltonian also specifies which spin component couples on every edge and how large competing terms are. An isotropic Heisenberg magnet drawn on hexagons is a different model.

2. Seeing Majoranas in the algebra and announcing that the sample contains Majorana zero modes. The c_i operators form a delocalized matter band and live in an enlarged, gauge-redundant representation. A localized zero mode appears at a vortex only in the gapped topological regime and after the physical-state constraint is enforced. A numerical parton spectrum without gauge projection is not yet a physical excitation spectrum.
3. Assuming flux conservation survives realistic control fields. It is exact for H_K and for selected integrable extensions. Generic Zeeman, Heisenberg, Γ , disorder, and drive terms can mix flux sectors. Small perturbations need not immediately destroy a gapped topological phase, but they do destroy the shortcut of labeling every eigenstate by exact w_p values.
4. Hearing “non-Abelian” and writing “Fibonacci.” Ising and Fibonacci anyons both have multidimensional fusion spaces, but their fusion rules, quantum dimensions, braid representations, and computational power differ. The honeycomb model’s canonical non-Abelian phase is Ising-type. Calling it Fibonacci discards the very topological data that define the phase.
5. Treating a finite programmed simulation as material emergence. A gate array can encode the Majorana Hamiltonian, prepare a flux sector, and measure the expected spectrum. That is a digital emulation by ordinary physical qubits unless the simulator’s own equilibrium Hamiltonian has the phase and protection. Simulation is valuable evidence about the model. It is not evidence that intrinsic anyons inhabited the processor between gate pulses.
6. Taking an exactly solvable point for a complete architecture. Exact solvability identifies the phase cleanly. An architecture must additionally initialize a sector, cool below the many-body gap, create and move excitations, read out fusion information, tolerate disorder, and scale boundaries and defects. The theorem does not route the wires.

17.9. Check yourself

1. **Show that every plaquette flux commutes with the bond Hamiltonian.** A perimeter bond anticommutes with two factors in W_p , giving two canceling minus signs; an outward bond uses the same component as the plaquette factor at their shared site.
2. **What is physical: u_{ij} or W_p ?** An individual u_{ij} depends on gauge choice. The loop product, equivalently the flux W_p , is gauge-invariant.
3. **Show that the zero-field matter spectrum is gapless precisely when J_x, J_y, J_z form a triangle.** $E(\mathbf{k}) = 2|f(\mathbf{k})|$ vanishes if and only if the three complex terms in f can sum to zero, which is the triangle inequalities. The isotropic point lies in this B region.

4. **What fails if you take the ungapped B phase for protected non-Abelian anyons?** The bulk is gapless. A weak time-reversal-breaking perturbation can gap its Dirac Majoranas and produce Ising topological order; the ungapped bulk is not the desired protected phase.
5. **Why is the non-Abelian honeycomb phase not Fibonacci?** Its vortices obey $\sigma \times \sigma = 1 + \psi$ with $d_\sigma = \sqrt{2}$, not $\tau \times \tau = 1 + \tau$ with $d_\tau = \varphi$. Their braid computational powers differ as well.
6. **What fails if $J_x, J_y \ll J_z$ is treated as a free topological scale?** Virtual processes from two-body bonds can generate a loop interaction, but its scale $J_x^2 J_y^2 / (16 J_z^3)$ may be far smaller than the microscopic exchange.

You now have a two-body compass model that conserves flux, a free-fermion solution inside each sector, and a field-gapped phase whose anyons are Ising, not Fibonacci.

17.10. Sources

- [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005. arXiv: cond-mat/0506438.
- [R147] G. Jackeli and G. Khaliullin, “Mott Insulators in the Strong Spin-Orbit Coupling Limit: From Heisenberg to a Quantum Compass and Kitaev Models,” *Physical Review Letters* **102**, 017205 (2009). DOI: 10.1103/PhysRevLett.102.017205. arXiv: 0809.4658.
- [R148] M. Hermanns, I. Kimchi, and J. Knolle, “Physics of the Kitaev Model: Fractionalization, Dynamic Correlations, and Material Connections,” *Annual Review of Condensed Matter Physics* **9**, 17–33 (2018). DOI: 10.1146/annurev-conmatphys-033117-053934. arXiv: 1705.01740.
- [R149] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, “Concept and realization of Kitaev quantum spin liquids,” *Nature Reviews Physics* **1**, 264–280 (2019). DOI: 10.1038/s42254-019-0038-2. arXiv: 1903.08081.
- [R150] Y. Matsuda, T. Shibauchi, and H.-Y. Kee, “Kitaev quantum spin liquids,” *Reviews of Modern Physics* **97**, 045003 (2025). DOI: 10.1103/3m4m-3v59. Stable URL: APS.
- [R151] A. Banerjee *et al.*, “Neutron scattering in the proximate quantum spin liquid α -RuCl₃,” *Science* **356**, 1055–1059 (2017). DOI: 10.1126/science.aah6015. arXiv: 1702.01688.
- [R152] Y. Kasahara *et al.*, “Majorana quantization and half-integer thermal quantum Hall effect in a Kitaev spin liquid,” *Nature* **559**, 227–231 (2018). DOI: 10.1038/s41586-018-0274-0. arXiv: 1805.05022.
- [R153] É. Lefrançois *et al.*, “Evidence of a Phonon Hall Effect in the Kitaev Spin Liquid Candidate α -RuCl₃,” *Physical Review X* **12**, 021025 (2022). DOI: 10.1103/PhysRevX.12.021025. arXiv: 2111.05493.

- [R154] A. Shragai, E. Horsley, S. Kim, Y.-J. Kim, and B. J. Ramshaw, “Phonon Hall viscosity and the intrinsic thermal Hall effect of α -RuCl₃,” *Nature* **652**, 1166–1172 (2026). DOI: 10.1038/s41586-026-10420-y. Stable URL: Nature.

Chapter 18. Strings that only meet in allowed ways

Draw colored strings on the edges of a honeycomb. A blank edge is empty. A colored edge carries one type of string. At each three-way meeting, not every trio is allowed.

Two of one color and one of another can be fine. Three of the same color can be fine. A single colored string arriving at two blanks is not.

Do not treat any one drawing as the state. The object we care about is a superposition of many legal drawings, with amplitudes. A local move can create a small closed loop, slide it across a meeting, and swallow it into the surrounding net.

A *Levin–Wen model* is this lattice Hamiltonian: a finite set of labels on the edges of a trivalent lattice, a local rule saying which three labels may meet, and a second local rule that inserts and recouples contractible loops. We take that as the definition. Enforcing the meeting rule alone is not a topological phase.

18.1. Which three labels may meet

The smallest useful color set is the one from Fibonacci fusion. Call the blank edge vacuum and write it 1. Call the nontrivial color τ . Both are their own reverse. Their fusion rules are

$$1 \times a = a, \quad \tau \times \tau = 1 + \tau,$$

where a may be 1 or τ . The plus sign still means two possible meeting channels, not two particles sitting in a pile.

On a lattice where three edges meet, the allowed unordered triples are

$$(1, 1, 1), \quad (1, \tau, \tau), \quad (\tau, \tau, \tau),$$

including permutations. A vertex with exactly one τ is forbidden. The triple (τ, τ, τ) is the interesting one: two τ strings may meet and continue as τ .

That list is the *branching rule*. It is a local constraint on three incident labels. Enforcing it does not by itself make a topological phase.

Drop the triple (τ, τ, τ) and the remaining rule is Abelian: two τ strings may meet only as vacuum. The later loop algebra then cannot reproduce Fibonacci fusion.

18.2. Superpositions of legal nets

The branching rule rejects illegal meetings. It does not decide which legal drawing is the ground state.

Stop treating any one legal drawing as the answer. The ground state we want is a coherent superposition of many of them. A local move may create a small closed loop, slide it across a vertex, and absorb it into the surrounding net. Closed, branching drawings then appear at every length scale.

That fluctuating legal picture is a *string-net*. Levin and Wen showed that this picture can be written as the exact ground state of commuting local projectors [R018]. **[Theory]**

The toric code was the quieter cousin of this picture: local projectors, very simple string rules, and labels that commute past each other. Here a meeting can branch, and the rewrite amplitudes are not ± 1 bit flips.

An edge label is a microscopic basis state of the lattice. A closed labeled net is a component of a many-body wavefunction. An endpoint or ribbon that survives all local equivalences can be an emergent quasiparticle. None of those is, merely by being written on paper, a defect-center spin or a digitally emulated anyon.

18.3. One hexagon, two drawings

Take a single hexagonal plaquette whose six outward legs all carry 1. The branching rule then leaves only two boundary configurations:

```
|0> : six boundary edges are 1
|R> : six boundary edges are  $\tau$ , forming a ring
```

Insert a τ loop into $|0\rangle$ and push it outward until it lies on the boundary. Because $\tau \times 1 = \tau$, this gives $|R\rangle$. Call this loop-insertion operator B^τ . In this two-state sector, fusing a second τ loop uses $\tau \times \tau = 1 + \tau$, so

$$B^\tau|0\rangle = |R\rangle, \quad B^\tau|R\rangle = |0\rangle + |R\rangle.$$

Therefore, in the ordered basis $(|0\rangle, |R\rangle)$,

$$B^\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Let

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

be the golden ratio. The quantum dimensions are $d_1 = 1$ and $d_\tau = \varphi$, and the squared total quantum dimension of the input labels is

$$\mathcal{D}^2 = d_1^2 + d_\tau^2 = 1 + \varphi^2 = \varphi + 2.$$

Hence a loop did not get flipped like six independent bits. It was inserted and fused. The empty drawing and the ring are no longer separate classical pictures. They mix.

If the six outward legs are not all vacuum, this two-by-two matrix is not the whole story. Intermediate fusion channels appear, and the isolated-hexagon calculation cannot leave its sector without an F -move.

18.4. Weighting the loops

The empty net is not the ground state. Average over loop types, weighted by their quantum dimensions:

$$B_p = \frac{1}{\mathcal{D}^2} \sum_{s \in \{1, \tau\}} d_s B_p^s = \frac{I + \varphi B_p^\tau}{\varphi + 2},$$

where $B_p^1 = I$. This weighted average is the *plaquette projector*. It is not a six-bit flip. It is a filter that accepts the particular superposition the input data demand.

In our two-state sector,

$$B_p = \frac{1}{\varphi + 2} \begin{pmatrix} 1 & \varphi \\ \varphi & \varphi^2 \end{pmatrix} = |g\rangle\langle g|,$$

with

$$|g\rangle = \frac{|0\rangle + \varphi|R\rangle}{\sqrt{\varphi + 2}}.$$

This calculation does three useful things. It shows that $B_p^2 = B_p$, that a ground-state component is not just the empty net, and that the relative amplitude φ is fixed by the input labels. A loop was inserted, fused, and coherently recoupled.

This tiny sector hides the interesting bookkeeping. If outward legs carry τ , several intermediate fusion channels are possible. Moving the inserted loop through a vertex then changes the parenthesization of three fusions. That is an F -move, and we will need it as soon as the hexagon is no longer isolated.

Equal weights in place of d_s would generally destroy idempotence. The empty net and the ring would then not sit in a rank-one projector.

18.5. Labels on every edge

Zoom out from one hexagon. Choose an oriented trivalent lattice, often a honeycomb. Each oriented edge e carries a label i_e from a finite set $I = \{0, 1, \dots\}$. The label 0, also written 1, is vacuum. Reversing an edge replaces i by its dual i^* . Fibonacci labels are self-dual, so orientation is visually quiet there. It matters for a general set of labels.

A basis vector is a complete assignment

$$|\{i_e\}\rangle = \bigotimes_e |i_e\rangle.$$

The fusion multiplicity N_{ij}^k is the number of independent ways labels i and j can fuse to k . For the multiplicity-free models used here, N_{ij}^k is either zero or one. A trivalent vertex with appropriately oriented labels i, j, k is allowed when $N_{ij}^{k^*} > 0$. Input data with multiplicities need an extra vertex-channel index. Suppressing that index is a simplification, not a theorem that nature dislikes indices.

The quantum dimension $d_i > 0$ is the positive solution of

$$d_i d_j = \sum_k N_{ij}^k d_k.$$

For Fibonacci fusion this gives $d_\tau^2 = 1 + d_\tau$, whose positive root is φ . Those are the same numbers that weighted the hexagon.

Hence the honeycomb labels are the microscopic basis of the many-body Hilbert space, not a drawing laid over some other degrees of freedom.

18.6. A projector at each vertex

Define Q_v to be diagonal in the edge-label basis:

$$Q_v|i, j, k\rangle = \begin{cases} |i, j, k\rangle, & N_{ij}^{k*} > 0, \\ 0, & N_{ij}^{k*} = 0. \end{cases}$$

Thus $Q_v^2 = Q_v$. A state with $Q_v = 0$ violates the local fusion rule. This operator is the *vertex projector*: the branching rule written as an energy term. It accepts allowed meetings and kills forbidden ones.

In the Hamiltonian, changing a satisfied vertex from eigenvalue one to zero raises the energy by J_v , before accounting for any linked plaquette violations.

One could instead define a penalty $J_v(1 - Q_v)$. It differs from $-J_v Q_v$ only by an additive constant. The physics is less impressed by constants than accountants are.

What fails if Q_v is omitted: illegal triples such as a single τ at a vertex remain in the low-energy space, and the later loop operators no longer act inside a fusion-consistent subspace.

18.7. Changing the order of fusion

Fusion can be performed in different orders:

$$((a \times b) \rightarrow x) \times c \rightarrow d \quad \longleftrightarrow \quad a \times ((b \times c) \rightarrow y) \rightarrow d.$$

The unitary change of basis is

$$|((ab)xc)d\rangle = \sum_y [F_d^{abc}]_{xy} |a(bc)y; d\rangle.$$

Here a, b, c, d are labels, while x, y are allowed intermediate fusion channels. The matrices F_d^{abc} obey the pentagon equation. That consistency condition says that two sequences of recouplings around a five-leg tree give the same answer. It is precisely what makes neighboring plaquette operations mutually compatible [R018]; [R155]. **[Theory]**

In a common unitary gauge, the only nontrivial two-dimensional Fibonacci matrix is

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix},$$

where rows and columns are ordered by intermediate channel $(1, \tau)$. The matrix is real, symmetric, and unitary. Its signs and phases depend on gauge, but consistent gauge changes do not alter measurable topological data.

When a τ loop is pushed through a vertex whose outward legs already carry τ , this matrix is the amplitude book. Without it, the hexagon calculation does not know how to leave the two-state sector.

Drop the pentagon equation and neighboring B_p need not commute. The model then ceases to be a commuting-projector Hamiltonian.

18.8. How a loop joins the net

For each possible string type s , define B_p^s by a graphical algorithm:

1. insert a closed loop of type s just inside plaquette p ;
2. fuse that loop into the boundary edges;
3. apply an F -move at each boundary vertex to recouple it through the existing net;
4. sum coherently over every allowed intermediate channel.

For a hexagon this is a circuit of six local recouplings, not a statement that the final operator is a simple product of six Pauli matrices. Its matrix elements also depend on labels on the legs leaving the plaquette.

The loop operators reproduce the fusion algebra,

$$B_p^r B_p^s = \sum_t N_{rs}^t B_p^t.$$

The special weighted sum

$$B_p = \frac{1}{\mathcal{D}^2} \sum_s d_s B_p^s, \quad \mathcal{D} = \sqrt{\sum_s d_s^2},$$

is a projector. Quantum-dimension identities establish $B_p^2 = B_p$; the pentagon equation establishes $[B_p, B_{p'}] = 0$, including for adjacent plaquettes. The branching and recoupling data also give $[Q_v, B_p] = 0$ [R018]; [R155].

There is a short reason the weights work. Let $\Omega = \sum_s d_s s$ denote the formal weighted loop. Fusion with a label r gives

$$r \times \Omega = \sum_{s,t} d_s N_{rs}^t t = d_r \Omega.$$

Fusing two weighted loops therefore gives $\Omega \times \Omega = \mathcal{D}^2 \Omega$. Dividing loop insertion by $\mathcal{D}^2$ makes it idempotent. The plaquette projector is a normalized vacuum-flux filter.

A six-edge bit flip with convenient coefficients is a different Hamiltonian. It need not reproduce this fusion algebra.

18.9. Energy for both rules

Call each vertex v and each plaquette p . Let Q_v be the dimensionless projector that accepts allowed labels at vertex v , and let B_p be the dimensionless projector that applies the plaquette rule at p . Let the positive energies J_v and J_p set the cost of violating those rules. The many-body Hamiltonian is

$$H = -J_v \sum_v Q_v - J_p \sum_p B_p.$$

Every term has units of energy. Authors often set $J_v = J_p = 1$; that chooses an energy unit rather than discovering free electricity.

The vertex terms reject illegal meetings. The plaquette terms declare locally equivalent legal nets to be parts of one quantum state.

The jobs line up like this:

microscopic edge degree of freedom	
↓	
allowed fusion at every vertex	$Q_v = 1$
↓	
coherent loop fluctuations	$B_p = 1$
↓	
nonlocal ground sectors and emergent anyons	

That is an exact theoretical construction, not a material recipe. The difference between those two sentences is much of the remaining book.

18.10. When every projector agrees

Because all terms commute, the model is frustration free. A ground state $|\Psi_0\rangle$ satisfies

$$Q_v |\Psi_0\rangle = |\Psi_0\rangle, \quad B_p |\Psi_0\rangle = |\Psi_0\rangle$$

for every v and p . **[Theory]** These equations say, respectively, that the wavefunction has support only on legal nets and is invariant under the weighted insertion and recoupling of a contractible loop [R018]. Tensor-network constructions make this coherent superposition explicit [R159].

“Condensed” does not mean every edge is likely occupied. It means closed nets can fluctuate over arbitrarily large scales in the ideal ground state, with local reconnections related by the input data. That proliferation of closed, branching nets at all length scales is *string-net condensation* [R018]. The name is worth keeping because it names a phase of the wavefunction, not a box of threads, and because it is not the same job as taking an already-emergent bosonic particle and identifying it with the vacuum.

On a surface with noncontractible cycles, a local move cannot always erase the global winding information. Distinct global sectors form the topological ground space.

Enforcing only $Q_v = 1$ produces a constrained Hilbert space of legal drawings. Without coherent B_p dynamics it can remain a classical collection of nets.

18.11. Leftover ends

A local operator can violate one or more projectors. A failed vertex constraint resembles an electric charge. A failed plaquette constraint resembles flux. In a non-Abelian string net, however, the fully mobile quasiparticles are generally charge–flux composites. Ribbon operators create them in pairs, move them along paths, and reveal fusion and braiding through the algebra of intersecting ribbons.

The input label τ on an edge is not itself one of those quasiparticles. It is a microscopic basis state. An emergent anyon is an excitation of the collective Hamiltonian that survives all local equivalences.

The systematic home of those bulk superselection sectors is the *Drinfeld center* $Z(\mathcal{C})$ of the input fusion category $\mathcal{C}$ [R156]; [R157]. **[Theory]** The center equips an object with consistent information about how it passes every string type. That extra passing rule is the emergent braiding data. One does not have to put a microscopic braid operator on every edge.

What fails if the edge label is treated as the anyon: you have counted input data, not excitations of H .

18.12. The mirror that comes for free

For modular Fibonacci input,

$$Z(\text{Fib}) \simeq \text{Fib} \boxtimes \overline{\text{Fib}},$$

where the overline denotes the mirror, with opposite braiding chirality. The four bulk charges may be labeled

$$(1, \bar{1}), \quad (\tau, \bar{1}), \quad (1, \bar{\tau}), \quad (\tau, \bar{\tau}),$$

with quantum dimensions $1, \varphi, \varphi, \varphi^2$, respectively. On a torus the ideal doubled-Fibonacci model consequently has four topological ground sectors [R156]; [R158]. **[Theory]**

This is *doubled-Fibonacci* order: a Fibonacci sector paired with its time-reversed partner. Its net chirality cancels. The input label τ is not itself an emergent anyon, and the bulk is not a single chiral Fibonacci theory. Chapter 19 will examine what that does — and does not — mean for computation.

Drop the mirror sector and you have described a different phase. Ordinary two-dimensional Levin–Wen with Fibonacci input does not produce a lone chiral Fibonacci liquid.

18.13. What a crystal does not give you

The Levin–Wen Hamiltonian is an exact existence proof and a precise benchmark. It is not a list of naturally available couplings in diamond, sapphire, or silicon carbide.

A proposed defect implementation would first need an isolated local Hilbert space for every effective edge, with $|1\rangle$ and $|\tau\rangle$ encoded in two reliably controlled states. It would then need a large energy penalty implementing the three-edge branching projector Q_v . Harder still, it would need the plaquette matrix elements dictated by a product of F -moves and the exact ratio $d_1 : d_\tau = 1 : \varphi$.

Microscopic defect interactions are commonly pairwise at leading order, whereas B_p is a correlated, label-dependent operation around an entire plaquette and its attached legs. It is not merely a uniform ring exchange. **[Proposal]** Ancilla-mediated perturbative gadgets, periodic drives, or digital circuits could synthesize such an operator, but each route must separately establish the effective coefficients, unwanted terms, many-body gap, and regime of validity. The code and Turaev–Viro literature establishes mathematical and computational constructions [R155]; [R158]; it does not thereby establish a passive defect-material phase.

[Speculation] A defect-cluster architecture might encode one edge label per cluster and use mediators to generate Q_v and B_p . The microscopic challenge is not “make spins on a honeycomb.” It is to reproduce the category-consistent multi-edge recoupling amplitudes strongly enough that residual two-body terms, disorder, leakage, and temperature do not

close the topological gap. No result in the sources used for this chapter demonstrates that hierarchy in a crystalline defect array.

Exact commutation is not required throughout an entire topological phase. Suitable commuting-projector topological order is stable to sufficiently weak local perturbations while its gap remains open [R142]. **[Theory]** But “sufficiently weak” is a condition to derive or measure for the proposed effective Hamiltonian, not permission to ignore a term because it makes the diagram untidy.

18.14. Ways to get this wrong

1. Calling the edge label an anyon. The microscopic τ label is input data. Emergent anyons are excitations of the collective Hamiltonian and belong to $Z(\text{Fib})$.
2. Equating branching with topological order. Enforcing $Q_v = 1$ produces a constrained Hilbert space. Without coherent B_p dynamics, it can remain a classical collection of legal nets.
3. Replacing recoupling with a bit flip. Fibonacci B_p depends on fusion channels and F -symbols. A six-edge flip with convenient coefficients is generally a different Hamiltonian.
4. Calling every superposition a condensate. String-net condensation includes scale-spanning closed-net fluctuations and topological equivalences. A prepared superposition of two drawings does not prove a phase.
5. Confusing string-net condensation with anyon condensation. The former constructs a ground-state liquid of fluctuating nets. The latter changes topological order by condensing an emergent bosonic sector.
6. Dropping the mirror sector. A Fibonacci input to the ordinary two-dimensional Levin–Wen construction gives doubled Fibonacci, not a lone chiral Fibonacci phase.
7. Treating digital enforcement as passive emergence. Measuring projectors and applying corrections can emulate or stabilize a code space. Passive topological order requires the material Hamiltonian itself to energetically select the phase.
8. Assuming exact solvability is the protection mechanism. Commuting projectors make analysis clean. Protection instead depends on locality, a persistent many-body gap, system size, temperature, and control of perturbations. Exact algebra is the map; it is not the weather.

18.15. Check yourself

1. **Show that Q_v is a projector.** It is diagonal in the edge-label basis with eigenvalues 1 on allowed triples and 0 on forbidden ones, so $Q_v^2 = Q_v$.

2. **What does B_p^s do?** It inserts an s -type loop into plaquette p , fuses it into the boundary, and recouples the resulting fusion trees with F -moves.
3. **Show that the weights $d_s/\mathcal{D}^2$ make B_p a projector.** The formal loop $\Omega = \sum_s d_s s$ obeys $\Omega \times \Omega = \mathcal{D}^2 \Omega$. Dividing by $\mathcal{D}^2$ makes insertion idempotent, so $B_p^2 = B_p$.
4. **What fails if you enforce $Q_v = 1$ without coherent B_p ?** You obtain a constrained space of legal nets. Without loop recoupling they need not form a topological liquid.
5. **What topological order follows from Fibonacci input?** The bulk is $Z(\text{Fib})$, equivalent to Fibonacci times mirror Fibonacci: doubled-Fibonacci order.
6. **What fails if the edge label τ is treated as an emergent anyon?** τ is microscopic input. The bulk charges live in $Z(\text{Fib})$ and include the mirror sector.

18.16. Sources

- [R018] Michael A. Levin and Xiao-Gang Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv: cond-mat/0404617.
- [R155] Lukasz Fidkowski, Michael Freedman, Chetan Nayak, Kevin Walker, and Zhenghan Wang, “From string nets to nonabelions,” *Communications in Mathematical Physics* **287**, 805–827 (2009). DOI: 10.1007/s00220-009-0757-9; arXiv: cond-mat/0610583.
- [R156] Alexei Kitaev and Liang Kong, “Models for gapped boundaries and domain walls,” *Communications in Mathematical Physics* **313**, 351–373 (2012). DOI: 10.1007/s00220-012-1500-5; arXiv: 1104.5047.
- [R157] Alexander Kirillov Jr., “String-net model of Turaev–Viro invariants,” arXiv:1106.6033 (2011). DOI: 10.48550/arXiv.1106.6033; stable URL: arXiv:1106.6033.
- [R158] Robert König, Greg Kuperberg, and Ben W. Reichardt, “Quantum computation with Turaev–Viro codes,” *Annals of Physics* **325**, 2707–2749 (2010). DOI: 10.1016/j.aop.2010.08.001; arXiv: 1002.2816.
- [R159] Oliver Buerschaper, Miguel Aguado, and Guifré Vidal, “Explicit tensor network representation for the ground states of string-net models,” *Physical Review B* **79**, 085119 (2009). DOI: 10.1103/PhysRevB.79.085119; arXiv: 0809.2393.
- [R142] Sergey Bravyi, Matthew B. Hastings, and Spyridon Michalakis, “Topological quantum order: Stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195; arXiv: 1001.0344.

Chapter 19. A dance and its mirror

Two dancers circle clockwise. Film them. Play the film: the same steps, the same handedness. Stand a mirror next to the floor. In the glass, clockwise has become counterclockwise.

One film has a handedness. The pair of films has both. The pair is not a third, handedness-free dance. Two opposite corkscrews in the same drawer do not become spoons.

Chapter 15 gave one fusion rule. Chapter 18 built a lattice Hamiltonian from that rule. Neither sentence said whether you bought the dance, the mirror, or both.

19.1. A braid that notices a mirror

A braid of two non-Abelian quasiparticles is a dance with a direction. Play it normally and the fusion state is transformed by a unitary matrix U . Reflect the laboratory. Clockwise has become counterclockwise, and — after consistent choices of basis — the matrix is the complex conjugate U^* .

A medium that supplies only one orientation of that movie is *chiral*. Reverse space and the braiding data change. That is a physical claim, not a nickname.

The chiral Fibonacci theory, written $\mathcal{F}$, is that one-orientation medium. Its nontrivial charge is τ . We will not use *Fibonacci* by itself as the name of a phase or a quasiparticle. That word is the family name. It does not say which member you are holding.

Drop the orientation and you have not specified a theory. The fusion rule $\tau \times \tau = 1 + \tau$ is shared by the dance and the glass. The braid matrices are not.

19.2. The reverse braid

Stand the laboratory in a mirror and you get a second theory. Call it the *mirror* chiral Fibonacci theory, written $\overline{\mathcal{F}}$. Its corresponding charge is $\bar{\tau}$.

The bar means reversed braiding, not antiparticle. Both τ and $\bar{\tau}$ are self-dual: each is its own partner under fusion.

Hence $\mathcal{F}$ and $\overline{\mathcal{F}}$ are not the same phase. One is a left-handed dance. The other is the dance you see in the glass. A Hamiltonian that realizes one need not realize the other.

What fails if you read the bar as “antiparticle”? You will look for a charge that annihilates τ and miss that τ already annihilates itself.

19.3. A pair of labels

Put the two theories side by side as independent bookkeeping layers. An excitation must then carry a pair of labels:

$$(\text{charge in } \mathcal{F}, \text{ charge in } \overline{\mathcal{F}}).$$

That pair is the useful picture of doubling. It is not yet the definition, and it need not mean that a device literally contains two physical films. The formal construction packages a theory together with all consistent ways its strings can pass around other strings. For a modular input, that package happens to factor as theory times mirror [R160].

The result is the *doubled-Fibonacci* theory, written DFib:

$$\text{DFib} = \mathcal{F} \boxtimes \overline{\mathcal{F}}.$$

The name is worth keeping because ordinary English hears a copy, or a cancellation into something dull. The construction packages a theory with its reverse. You still have two corkscrews. You do not have spoons.

[Theory] A Levin–Wen model whose input is the Fibonacci fusion category has bulk anyons described by the Drinfeld center of that category. Because the chiral Fibonacci category is modular, its center is equivalent to a product of the chiral theory and its reverse-braided mirror [R018]; [R160]. The standard commuting-projector construction therefore targets DFib, not $\mathcal{F}$ alone.

That distinction is not decoration. It changes the bulk charge list, the torus degeneracy, the edge, the thermal Hall response, and the operations you must control if you want a computer. A defect lattice aimed at the wrong member of the family may be beautifully engineered and still answer a different question.

19.4. Four charges, not two

The chiral theory $\mathcal{F}$ has 1 and τ . Its mirror $\overline{\mathcal{F}}$ has $\bar{1}$ and $\bar{\tau}$. Therefore DFib has four simple bulk charges. Give them names that keep their sector:

Doubled-Fibonacci charge	Pair label	Quantum dimension	Topological spin
vacuum 1	$(1, \bar{1})$	1	1
left-handed charge x	$(\tau, \bar{1})$	φ	$e^{4\pi i/5}$
right-handed charge y	$(1, \bar{\tau})$	φ	$e^{-4\pi i/5}$
diagonal charge b	$(\tau, \bar{\tau})$	φ^2	1

Here $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. The displayed spin convention chooses $\theta_\tau = e^{4\pi i/5}$; reversing every braid conjugates it. The quantum dimension of a pair is the product of the component dimensions [R160]; [R015].

Hence the doubled-Fibonacci theory is not the two-object theory $\{1, \tau\}$ with a longer name. It contains two simple charges, x and y , each carrying one chiral Fibonacci fusion subtheory, plus the diagonal charge b .

The symbol b is a label, not a claim that this non-Abelian charge is an ordinary pointlike boson. It has trivial topological spin but quantum dimension $\varphi^2 > 1$. Drop that distinction and you will treat b as a spare copy of vacuum.

19.5. Fuse one side at a time

Pair labels make the arithmetic almost clerical. Fuse each factor on its own:

$$(a, \bar{c}) \times (a', \bar{c}') = (a \times a', \bar{c} \times \bar{c}').$$

For example,

$$x \times y = (\tau, \bar{1}) \times (1, \bar{\tau}) = (\tau, \bar{\tau}) = b.$$

There is only one outcome. Now fuse that result with itself:

$$\begin{aligned} b \times b &= (\tau \times \tau, \bar{\tau} \times \bar{\tau}) \\ &= (1 + \tau, \bar{1} + \bar{\tau}) \\ &= \mathbf{1} + x + y + b. \end{aligned}$$

The complete nontrivial fusion rules are therefore

$$\begin{aligned}
x \times x &= \mathbf{1} + x, & y \times y &= \mathbf{1} + y, \\
x \times y &= b, & x \times b &= y + b, \\
y \times b &= x + b, & b \times b &= \mathbf{1} + x + y + b.
\end{aligned}$$

Hence x fuses with itself exactly as τ did in Chapter 15, and y does the same job in the mirror. The diagonal charge b is the new object. It is not a spare copy of τ .

What fails if you fuse only the first components and ignore the bars? You recover the chiral table and lose y and b as distinct bulk charges.

19.6. Add the squared sizes

For any anyon theory, add up the squared sizes of the simple charges. The *total quantum dimension* $\mathcal{D}$ is defined by

$$\mathcal{D}^2 = \sum_a d_a^2,$$

where the sum runs over simple charges and d_a is the quantum dimension of charge a . In this example,

$$\begin{aligned}
\mathcal{D}_{\text{DFib}}^2 &= 1 + \varphi^2 + \varphi^2 + \varphi^4 \\
&= (1 + \varphi^2)^2, \\
\mathcal{D}_{\text{DFib}} &= 1 + \varphi^2 = \varphi + 2.
\end{aligned}$$

The last equality uses $\varphi^2 = \varphi + 1$. Hence $\mathcal{D}_{\text{DFib}}$ is the square, in categorical dimension, of one chiral sector — as a double should be [R160].

On a torus, the ideal doubled-Fibonacci theory has four ground-state sectors, one per simple bulk charge. The chiral Fibonacci theory has two. If someone hands you a ground-space count of two and calls it doubled, they have miscounted.

19.7. Opposite braids do not cancel every rotation

If only x charges are braided, the mirror component is vacuum. Their fusion spaces, F -moves, and braid matrices are exactly those of $\mathcal{F}$. If only y charges are braided, the matrices are their complex conjugates.

For diagonal charges, both components move. A braid word represented by U in $\mathcal{F}$ is represented schematically by

$$U \otimes U^*$$

on the factorized doubled fusion space.

This does **not** make every braid the identity. Opposite chiralities cancel net handedness. They do not cancel every rotation. Suppose the two factors were forced to move together as an inseparable pair. Then the computational action would be that of $U \otimes U^*$, not of U , and the usual chiral-sector argument would no longer apply.

You also cannot point to the fusion rule of b and silently reuse that argument. Which charge is mobile and measurable is part of the computer specification.

19.8. A leftover you can heat

A topological phase is chiral when its low-energy topological data have a net handedness. Reversing spatial orientation replaces its braiding by the reverse braiding.

You can give that leftover a number. Call it the *chiral central charge* c_- , defined modulo the appropriate invertible phase. At an interface with the vacuum it controls the universal low-temperature thermal Hall coefficient [R015]

$$\frac{\kappa_{xy}}{T} = c_- \frac{\pi^2 k_B^2}{3h},$$

where κ_{xy} is thermal Hall conductance in watts per kelvin, T is temperature in kelvin, k_B is Boltzmann's constant, and h is Planck's constant. The dimensions work: k_B^2/h has units W K^{-2} , the same as κ_{xy}/T .

The chiral Fibonacci theory has nonzero c_- . The mirror chiral Fibonacci theory has the opposite value. In the doubled-Fibonacci theory the sum vanishes [R160]; [R015]. Equivalently, the topological spins of x and y are complex conjugates, while the diagonal charge has spin one.

Hence zero c_- does not imply trivial topological order, and it does not prove that a microscopic Hamiltonian respects time reversal. It says that the doubled topological data have no net chiral anomaly.

19.9. How a string slides past every partner

A Levin–Wen edge may carry only the input labels 1 or τ . The bulk theory has four charges. The extra objects are not a counting error. They are input labels that have been told how to slide past every other string.

That extra rule is a *half-braiding*: a consistent recipe for sliding one string past every possible partner. The consistency condition is what promotes recoupling data into genuine two-dimensional bulk braiding data.

For an input fusion category $\mathcal{C}$, package every object together with every consistent half-braiding. The resulting theory is the Drinfeld center $Z(\mathcal{C})$.

[Theory] Levin–Wen edge labels are input labels, while deconfined bulk quasiparticles are simple objects of $Z(\mathcal{C})$ [R018]; [R161]. For $\mathcal{C} = \mathcal{F}$, modularity gives

$$Z(\mathcal{F}) \simeq \mathcal{F} \boxtimes \overline{\mathcal{F}} = \text{DFib}.$$

Hence microscopic edge labels are not a census of emergent quasiparticles. Drop the half-braiding and you are left with a grammar of meetings, not a list of bulk charges.

19.10. Commuting projectors pick both

In the ideal string-net Hamiltonian,

$$H = - \sum_v Q_v - \sum_p B_p,$$

all local projectors commute. The ground state is assembled by local recouplings without selecting a net direction of propagation. The center supplies every consistent overcrossing together with the reverse-braided information needed for a nonchiral bulk. This is why the construction naturally yields doubled orders [R018]; [R161].

This is a statement about this construction, not a ban on chiral phases in lattice systems. Noncommuting local Hamiltonians can realize chiral topological order. “Lattice” does not mean “doubled.” “Standard Levin–Wen commuting projector” does.

19.11. An edge that cannot sit still

At a boundary between a standalone chiral Fibonacci phase and the trivial vacuum, the nonzero net chirality obstructs a completely gapped edge [R156]; [R015]. Low-energy modes must carry the anomaly. Edge disorder may complicate them, but it cannot simply erase the net chiral energy flow while the bulk phase remains unchanged.

The doubled-Fibonacci theory is different. Opposite edge chiralities can couple and gap one another. **[Theory]** It admits gapped boundaries described by suitable boundary data; in one categorical description the condensable diagonal combination is built from $\mathbf{1} \oplus b$ [R156].

“Admits” is the important verb. A particular termination may still be gapless because of its couplings or symmetries. A sketch with a circle around it is not a boundary condition.

A gapped doubled-Fibonacci boundary can also rearrange which charges are distinguishable or confined at that boundary. A proposal that uses holes, punctures, or boundary quasiparticles must specify its boundary type.

19.12. Which charge you can move

Consider four $x = (\tau, \bar{1})$ charges constrained to total charge $\mathbf{1}$. There are two fusion paths. If a is the intermediate charge of the first pair, define

$$|0_L\rangle : a = \mathbf{1}, \quad |1_L\rangle : a = x.$$

This two-dimensional fusion space is a logical qubit. Braiding the x charges acts with the same dense braid-group representation as braiding charges in the chiral Fibonacci theory. **[Theory]** Those representations can approximate arbitrary logical unitaries, up to overall phase and with standard encoding and compilation qualifications [R136]; [R015]. The y sector offers the complex-conjugate construction.

Hence an ideal doubled-Fibonacci phase can support universal topological quantum computation, provided the architecture can create, move, braid, fuse, and measure a selected chiral simple-object sector — x or y — while controlling leakage and unwanted sector mixing. The presence of the mirror sector does not cancel those gates.

Three limitations belong with that verdict.

1. The fusion rule alone is insufficient; universality uses the F - and R -data and access to the correct charge sector.
2. If hardware exposes only the diagonal charge b , or forces mirror charges to move as inseparable pairs, universality does not follow from the ordinary chiral-sector argument.

3. Universal braid representations do not supply initialization, charge measurement, low thermal anyon density, or an adequate many-body gap. Those are physical requirements, not categorical corollaries.

[Theory] Turaev–Viro codes make the computational statement constructive: under suitable access to encoded surfaces and topological operations, doubled theories derived from universal modular data can implement quantum computation [R158]. This is support for the ideal-model verdict, not evidence that a proposed defect Hamiltonian realizes those operations.

Keep the four layers separate. A physical spin or transmon is a microscopic qubit. A fusion space of several x charges can encode a logical qubit. A circuit that stores the amplitudes of that fusion space digitally emulates the doubled-Fibonacci model. An emergent x charge requires a many-body phase whose low-energy bulk superselection sectors are those of DFib. Only the last case makes braiding passive geometric motion inside an intrinsic topological medium.

19.13. Twenty-seven qubits, no crystal

The mathematical distinction is settled. Its material realization is not.

[Experiment] In 2024, Xu and collaborators used 27 superconducting transmon qubits to digitally prepare a three-plaquette state of the Levin–Wen model built from Fibonacci fusion-category data, then implemented open-string, recoupling, and braid circuits [R138]. They reported average vertex- and plaquette-projector expectation values of 0.94 and 0.58 for their optimized prepared state, under the device and measurement conditions given in that paper. The target string-net is a doubled-type topological model even though the paper uses a shorter anyon label.

The same experiment explicitly states that its braids involved no Hamiltonian dynamics of quasiparticle excitations and therefore did not inherit protection from a many-body energy gap [R138]. The transmons executed gates that emulated the desired fusion-space transformations. That is an important experimental result. It is not evidence of an intrinsic chiral Fibonacci material phase, nor of an equilibrium doubled-Fibonacci phase passively stabilized in defect matter.

[Proposal] A static defect-spin architecture based on a Levin–Wen target should therefore write DFib in its requirements. It must derive a low-energy Hamiltonian with the correct center, gap, ribbon operators, and controllable sector-resolved excitations. No result cited in this chapter establishes that diamond, sapphire, silicon carbide, or another defect host currently meets those conditions.

A chiral Fibonacci target would be a different engineering program. It would need a mechanism for broken time-reversal symmetry and nonzero net chirality. It would not be obtained merely by deleting the word “doubled” from a commuting-projector string-net proposal.

19.14. Ways to get this wrong

Call every object τ . In the doubled-Fibonacci bulk, x , y , and b are inequivalent. Writing τ for all three discards topological spin and fusion information.

Confuse input strings with output anyons. The two Levin–Wen edge labels $1, \tau$ define local Hilbert-space states. The four charges $1, x, y, b$ classify emergent bulk excitations. They live at different descriptive levels.

Think “doubled” means trivial. Net thermal Hall response cancels, but non-Abelian fusion spaces remain. The doubled-Fibonacci theory has four bulk sectors, total quantum dimension $\varphi + 2$, and nontrivial braiding.

Think opposite chirality cancels every gate. It cancels net chiral central charge. A braid confined to the x sector is not accompanied by a y -sector braid, so its computational action remains that of the chiral Fibonacci theory.

Assume a gappable edge is already gapped. Existence of a consistent gapped boundary does not specify the boundary Hamiltonian or its gap.

Equate a prepared wavefunction with an equilibrium phase. A circuit may prepare a small state with ideal-model correlations and may enact an ideal braid matrix. An analog topological phase additionally requires a local Hamiltonian, a robust many-body gap, scalable bulk, and adiabatic quasiparticle motion. Digital simulation tests the model; it does not cause the processor’s microscopic excitations to become doubled-Fibonacci anyons.

19.15. Check yourself

1. **What topological order does a standard Levin–Wen model with the Fibonacci fusion category as input produce?**

The doubled-Fibonacci order $Z(\mathcal{F}) \simeq \mathcal{F} \boxtimes \overline{\mathcal{F}}$, not a single chiral Fibonacci order.

2. **Show that $x \times y = b$ and $b \times b = 1 + x + y + b$.**

Fuse the pair labels independently: $(\tau, \bar{1}) \times (1, \bar{\tau}) = (\tau, \bar{\tau}) = b$, and $(\tau \times \tau, \bar{\tau} \times \bar{\tau}) = (1 + \tau, \bar{1} + \bar{\tau}) = 1 + x + y + b$.

3. **Show that $\mathcal{D}_{\text{DFib}} = \varphi + 2$.**

$\mathcal{D}^2 = 1 + \varphi^2 + \varphi^2 + \varphi^4 = (1 + \varphi^2)^2$, so $\mathcal{D} = 1 + \varphi^2$. Then $\varphi^2 = \varphi + 1$ gives $1 + \varphi^2 = \varphi + 2$.

4. **What fails if hardware exposes only the diagonal charge b ?**

The ordinary chiral-sector universality argument does not apply. The fusion rule of b is not the fusion rule of τ , and a braid of inseparable pairs acts as $U \otimes U^*$, not as U .

5. **What fails if a digital braid is offered as evidence of an intrinsic doubled-Fibonacci phase?**

The braid demonstrates controlled emulation of the model's state and operations. It does not establish a protecting many-body Hamiltonian or emergent quasiparticles.

6. **Why is the doubled-Fibonacci theory nonchiral but nontrivial?**

The two sectors have opposite chiral central charges, so the net value vanishes, while their non-Abelian fusion and braiding data remain.

A Levin–Wen model with the Fibonacci fusion category as input targets DFib. Its four bulk charges are $1, x, y, b$. The charges x and y carry opposite chiral Fibonacci braid data. The doubled edge can be gapped even though a single chiral edge cannot. Ideal universal computation is available by controlling one chiral charge sector, not merely by observing the fusion rule. Chapter 20 will use those distinctions to audit what programmable hardware has actually prepared, fused, braided, and protected.

19.16. Sources

- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv: cond-mat/0404617.
- [R160] M. Müger, “From subfactors to categories and topology II: The quantum double of tensor categories and subfactors,” *Journal of Pure and Applied Algebra* **180**, 159–219 (2003). DOI: 10.1016/S0022-4049(02)00248-7; arXiv: math/0111205.
- [R161] C.-H. Lin, M. Levin, and F. J. Burnell, “Generalized string-net models: A thorough exposition,” *Physical Review B* **103**, 195155 (2021). DOI: 10.1103/PhysRevB.103.195155; arXiv: 2012.14424.
- [R156] A. Kitaev and L. Kong, “Models for gapped boundaries and domain walls,” *Communications in Mathematical Physics* **313**, 351–373 (2012). DOI: 10.1007/s00220-012-1500-5; arXiv: 1104.5047.
- [R136] M. H. Freedman, M. J. Larsen, and Z. Wang, “A modular functor which is universal for quantum computation,” *Communications in Mathematical Physics* **227**, 605–622 (2002). DOI: 10.1007/s002200200645; arXiv: quant-ph/0001108.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv: 0707.1889.
- [R158] R. König, G. Kuperberg, and B. W. Reichardt, “Quantum computation with Turaev–Viro codes,” *Annals of Physics* **325**, 2707–2749 (2010). DOI: 10.1016/j.aop.2010.08.001; arXiv: 1002.2816.

- [R138] S. Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6; stable article: [nature.com/articles/s41567-024-02529-6](https://www.nature.com/articles/s41567-024-02529-6).

Chapter 20. An actor playing a king

An actor walks onstage in a crown. He speaks the king's lines. He makes the king's choices. The audience follows the plot. After the curtain he takes the crown off. He is still an actor. The castle is still plywood.

A programmable chip can do the same job with anyons. It can hold the labels vacuum and τ . It can apply the braid matrix. It can return the fusion histogram the theory wrote down. That is a real experiment. It is not a crystal whose own energy landscape is those anyons.

20.1. Four things that look alike

When a paper says “we braided Fibonacci anyons,” four different objects may be hiding under that one sentence.

The first object is the actual two-level thing in the fridge: a superconducting circuit bit, a trapped ion, a nuclear spin in a drop of chloroform. Nobody braided that object around another object. It sat on a chip, or in a trap, and waited for a pulse.

The second object is a pattern of several of those bits that we *choose* to read as a charge. The charge lives in the bookkeeping. The fridge still contains transmons, or ions, or nuclei.

The third object is a snapshot of the whole register. At one instant the entanglement can be real, the loops can have the right answers, and the fusion table can match the model. The snapshot was written by a circuit. It starts rotting the moment the circuit stops.

The fourth object is an excitation the material itself would produce if you just cooled it. No circuit standing over the wavefunction with a script. The energy landscape does the selecting.

Those four pictures need four names, or the next paper will glue them together. The hardware bit is a *physical qubit*. A pattern of several bits, read through a target model, is an encoded anyonic degree of freedom. A snapshot with the right model observables is a digitally prepared topological wavefunction. An excitation of the laboratory Hamiltonian is an intrinsic anyon.

The name *encoded* is worth keeping because it marks the gap between “the theory is on the chip” and “the chip is the theory.”

[Experiment] Superconducting and trapped-ion processors have prepared wavefunctions with Fibonacci, finite-group quantum-double, and twist-defect structure. They have also run circuit operations that stand for fusion and braiding [R162]; [R132]; [R163]; [R164]; [R138]; [R165]; [R140]; [R166]. None of that shows that the processor's own Hamiltonian has a Fibonacci phase, a protecting many-body gap, or thermally stable quasiparticles.

A future defect array can still be useful as a digital processor even if it never becomes Fibonacci matter. Digital success is a real capability. It is not the same capability as a material that hosts the anyons by itself.

20.2. Braid two, then ask how they fuse

Let $\mathbf{1}$ denote vacuum and τ a Fibonacci charge, with

$$\tau \times \tau = \mathbf{1} + \tau.$$

Four τ anyons whose total charge is vacuum encode a two-dimensional logical space. Define $|\bar{0}\rangle$ as the state in which the first pair and the last pair each fuse to $\mathbf{1}$. Define $|\bar{1}\rangle$ as the state in which those pairs each fuse to τ , while all four still have total charge $\mathbf{1}$.

Create the pairs from vacuum, so the initial state is $|\bar{0}\rangle$. Exchange the middle two anyons once. In a common phase convention, the braid generator σ_2 gives

$$\sigma_2|\bar{0}\rangle = \phi^{-1}e^{4\pi i/5}|\bar{0}\rangle + \phi^{-1/2}e^{-3\pi i/5}|\bar{1}\rangle,$$

where

$$\phi = \frac{1 + \sqrt{5}}{2}$$

is the golden ratio. The phases depend on chirality and convention. The probabilities in this experiment do not. Fuse the original pairs and measure their channels:

$$P(\mathbf{1}) = |\phi^{-1}|^2 = \phi^{-2} \approx 0.382, \quad P(\tau) = |\phi^{-1/2}|^2 = \phi^{-1} \approx 0.618.$$

The normalization works because $\phi^2 = \phi + 1$, so $\phi^{-2} + \phi^{-1} = 1$. The ratio is especially convenient:

$$\frac{P(\tau)}{P(\mathbf{1})} = \phi.$$

Hence a single braid of the middle pair, followed by fusion of the original pairs, must return the golden ratio. That is the logic behind the fusion signature used in recent superconducting experiments [R138]; [R165]. It is more informative than merely applying a precomputed 2×2 braid matrix, because creation, motion in an encoded string-net, and fusion readout are all represented on a spatial many-qubit register.

What fails if you keep only the final 2×2 matrix and skip the spatial register? You can still match the histogram. You have not shown that intermediate states represented spatially separated quasiparticles.

The digital implementation still has a mundane circuit underneath:

```

initialize physical qubits
      |
prepare target vacuum wavefunction  U_prep
      |
create two encoded anyon pairs      U_create
      |
move/exchange middle pair           U_braid
      |
change to a fusion basis             U_fuse
      |
measure physical qubits -----> fusion histogram

```

The anyon worldlines describe the logical action of these gates. No transmon literally walks around another transmon.

20.3. A circuit is not a cooling step

Let N be the number of physical qubits. A digital experiment typically begins with the product state $|0\rangle^{\otimes N}$ and applies a compiled circuit

$$|\psi_{\text{prep}}\rangle = U_{\text{prep}}|0\rangle^{\otimes N}.$$

The target may be the ground state $|G\rangle$ of an ideal Hamiltonian

$$H_{\text{target}} = - \sum_v Q_v - \sum_p B_p,$$

where Q_v enforces an allowed fusion or branching rule at vertex v , and B_p enforces the desired flux around plaquette p . For a Fibonacci Levin–Wen model, Q_v can involve three edge qubits and B_p can involve many qubits around a plaquette [R138].

The circuit can be exact, variationally synthesized, measurement-assisted, or adaptive. The experiment then measures quantities such as $\langle Q_v \rangle$, $\langle B_p \rangle$, Wilson loops, or entanglement combinations. Agreement says that the *prepared state* resembles the target. It does not say that the laboratory Hamiltonian equals H_{target} .

In equilibrium preparation, by contrast, one would engineer $H_{\text{physical}} \approx H_{\text{target}}$, cool below its gap, and let the Hamiltonian select and stabilize the phase. That stronger test belongs to Chapter 21.

Suppose the circuit is exact, so $|\psi_{\text{prep}}\rangle = |G\rangle$. The snapshot can still be perfect. The idle Hamiltonian of the chip can still be something else. A prepared ground-state wavefunction is not a phase.

20.4. Moving a label with gates

An open string operator $W_a(\gamma)$ of charge type a acts along a path γ . Ideally it creates a pair of charges at the endpoints while leaving the intervening region locally in the code space:

$$|a, \bar{a}; \gamma\rangle = W_a(\gamma)|G\rangle.$$

For digital string nets, a local rewrite of the graph can change how fusion is parenthesized, or slide a string, without changing the topological content. That rewrite is an *F-move*. Every compiled braid on a string-net is mostly a pile of these local rewrites, not a transmon taking a walk.

An *R-move* supplies the exchange phase when strings cross. Both moves become multi-qubit gates and are finally decomposed into native one- and two-qubit operations.

A compiled braid is therefore a product

$$U_{\text{braid}} = U_m U_{m-1} \cdots U_2 U_1,$$

where each U_j is a local graph deformation, string extension, or crossing operation. The order matters. Showing that two such products give different final observables is the operational signature of non-Abelian behavior.

What fails if two compiled words that differ by a local deformation give different histograms? Then the hardware is not implementing the topological move you named. It is implementing some other circuit that happens to share a label.

20.5. Fusion is a measurement

The symbol $a \times b = \sum_c N_{ab}^c c$ lists allowed total charges c . A fusion experiment must resolve which channel occurred. Let $\Pi_c^{(ab)}$ be the projector onto channel c for the chosen pair. For density matrix ρ ,

$$P(c) = \text{Tr}(\rho \Pi_c^{(ab)}).$$

Hardware rarely measures $\Pi_c^{(ab)}$ directly. Instead, a circuit maps the fusion basis to computational-basis bit strings, after which physical-qubit readout estimates $P(c)$. Other experiments measure stabilizers or a nonlocal string whose eigenvalue identifies the charge. Error mitigation can improve the estimate. It must be reported separately from raw protection.

Hence a fusion histogram is a statement about a measurement circuit and a readout channel, not automatically a statement about an anyon the material produced.

20.6. Four kinds of “protected”

Four mechanisms get sold under the same adjective.

First: the material’s own energy gap slaps your hand if you wander off the code space. Local perturbations split logical states only weakly as the system grows. Nobody has to keep typing corrections. That is *passive* protection, and it is the strongest mode in this chapter. The Hamiltonian is doing the work, not the control software.

Second: you keep measuring syndromes, decoding, and correcting. The protection is real. It is also a job. That is active error correction.

Third: you run the experiment several noisy ways and guess what a quieter machine would have said. Dynamical decoupling, readout correction, zero-noise extrapolation, twirling, and postselection all live here. They improve an estimate. They do not create a protected logical memory.

Fourth: inside the ideal model, some selected pretend-errors act as a global phase. That demonstrates the model’s algebra. It does not make the hardware immune to its own uncontrolled noise.

The 2024 Fibonacci experiment states the point unusually clearly: its braids used no Hamiltonian dynamics of quasiparticle excitations and therefore lacked the topological protection supplied by an energy gap [R138]. The 2025 IBM experiment likewise required deep circuits and composite error mitigation [R165].

A correct final braid is still not enough by itself. Suppose calibration produces a direct unitary V equal to the desired logical braid on the encoded two-dimensional subspace. Measuring the expected fusion probabilities proves that V was implemented there. It does not prove that intermediate states represented spatially separated quasiparticles, nor that local path deformations leave the operation unchanged.

Stronger evidence combines several tests: model constraints before and after the operation, localized endpoint excitations, path or order dependence, fusion-channel readout, noncommuting braid sequences, and negative controls. Even that package establishes a digital realization unless a native Hamiltonian and protection test are also present.

20.7. What the laboratory actually has

The table is current through 24 August 2026. It includes direct Fibonacci demonstrations and the closest non-Abelian digital comparators, because “non-Abelian,” “universal,” and “Fibonacci” are not interchangeable. Qubit counts distinguish the active experimental register from the total chip where the paper makes that distinction.

Date and primary source	Hardware and qubits	Target model and preparation	Fusion measurement	Braid operation and evidence	Protection mode	Intrinsic-versus-emulated verdict
May 2023, Xu <i>et al.</i> [R162]	Programmable square-lattice superconducting processor; up to 68 transmons	Toric/surface-code states with twist defects; finite-depth gate circuits prepare stabilizer ground states	Stabilizers and logical/string observables distinguish $\sigma \times \sigma = \mathbf{1} + \varepsilon$ channels	Pauli-string operations and graph deformations implement Ising-type logical gates and noncommuting/projective braids	No passive gap; circuit preparation and readout mitigation	Emulated projective Ising anyons , not Fibonacci and not native transmon excitations

Date and primary source	Hardware and qubits	Target model and preparation	Fusion measurement	Braid operation and evidence	Protection mode	Intrinsic-versus-emulated verdict
May 2023, Andersen <i>et al.</i> [R132]	Google superconducting processor; 25 transmons in a 5×5 register	Generalized surface-code stabilizer state; four CZ layers prepare the ground state	Destructive stabilizer and nonlocal Pauli-string measurements reveal vacuum or fermion after fusing twist vertices	Local two-qubit Clifford graph-deformation gates move and braid degree-3 vertices; fusion after crossing reveals changed fermion parity	No repeated error correction and no passive gap; paper identifies future error correction as necessary	Encoded twist defects with projective Ising statistics; spatial braid, but not intrinsic or Fibonacci
July 2023, Fan <i>et al.</i> [R163]	Liquid-state NMR, ^{13}C -labelled chloroform; 2 nuclear-spin qubits at 295 K	Minimal disk-code subspace representing three boundary Fibonacci anyons; shaped RF pulses prepare logical states	No pair-fusion experiment; logical-state tomography/readout in the two-spin basis	Fifteen compiled braid blocks approximate a logical Hadamard; randomized-benchmarking fidelity reported as 97.18% [R163]	Selected simulated local disturbances act as a global phase in the ideal code; physical dephasing remains, so no passive protection	Direct Hilbert-space emulation of three Fibonacci anyons; no spatial quasiparticles or Fibonacci material
February 2024, Iqbal <i>et al.</i> [R164]	Quantinuum H2 trapped-ion processor; 27 qubits on an encoded kagome lattice	Adaptive measurement-and-feed-forward circuit prepares the D_4 quantum-double ground-state wavefunction	Anyon pairs are created and fused; charge/interference measurements diagnose outcomes	Ribbon operations move charges through a Borromean ring spacetime braid; interferometry distinguishes the intrinsically non-Abelian process within the model	Adaptive preparation and verification, but no passive D_4 Hamiltonian gap during storage/operations	Digitally emergent D_4 anyons in a prepared wavefunction; non-Abelian but not Fibonacci, and not intrinsic ion excitations

Date and primary source	Hardware and qubits	Target model and preparation	Fusion measurement	Braid operation and evidence	Protection mode	Intrinsic-versus-emulated verdict
July 2024, Xu <i>et al.</i> [R138]	Flip-chip, tunable-transmon square lattice; 27 neighboring qubits mapped to a three-plaquette honeycomb string net	Doubled-Fibonacci Levin–Wen state; product-state loops, U_S , CNOT and variationally synthesized F -move circuits; final preparation depth 53	Two anyon pairs are recombined; tail/fusion qubits are measured in the computational basis to recover channel probabilities and monodromy	Open-string creation plus five-qubit F - and three-qubit R -moves braid the middle pair; measured $d_\tau \approx 1.60$ [R138]	Echo and readout mitigation only; authors explicitly say there is no Hamiltonian topological protection	Most direct 2024 digital Fibonacci-string-net demonstration , but ordinary transmons emulate doubled-Fibonacci anyons
July 2025, Mineev <i>et al.</i> [R165]	IBM superconducting hardware: 7-qubit charge register and 9-qubit braid register selected on the 133-qubit Heron processor <i>ibm_torino</i> ; a 3-qubit vacuum also ran on <i>ibm_peekskill</i>	Doubled-Fibonacci string-net condensate via dynamical string-net preparation: modular- S , exact F -moves, tails, and graph sewing	Graph deformation maps charge/fusion information to single-qubit/root-edge readout; reported charge-measurement accuracy 94% [R165]	Four five-qubit F -moves exchange the middle $\tau 1$ anyons; an R - then F -move fuses a pair; measured probability ratio 1.65 ± 0.14 , consistent with ϕ [R165]	Dynamical decoupling, twirling, zero-noise extrapolation, readout mitigation; no passive gap or demonstrated logical lifetime	Emulated doubled-Fibonacci anyons in a digitally prepared string net; exact local moves improve faithfulness, not intrinsicness

Date and primary source	Hardware and qubits	Target model and preparation	Fusion measurement	Braid operation and evidence	Protection mode	Intrinsic-versus-emulated verdict
July 2026, Lo <i>et al.</i> [R140]	Quantinuum H2 trapped ions; 54 qubits	Quantum double $D(S_3)$; adaptive circuit gauges charge-conjugation symmetry and prepares the non-Abelian ground-state wavefunction	Charge-pair braiding and fusion implement logical $\mathcal{X}$ - and $\mathcal{Z}$ -basis measurements on fusion-space qutrits	Ribbon/pull-through braids entangle topological qutrits; braiding plus fusion prepares a magic state and supplies a universal gate set	Measurement/forward in preparation and fusion-based operations; no native passive S_3 Hamiltonian protection demonstrated	Emulated S_3 quantum-double anyons ; universal through braid <i>plus fusion</i> , not Fibonacci braid universality
August 2026, Hayata, Hidaka & Kikuchi [R166]	Quantinuum H2 trapped ions; 18 model qubits plus 2 ancillas for compiled multi-control gates	$(2 + 1)$ -D q -deformed $SU(2)_3$ Yang–Mills model restricted to its integer-spin/Fibonacci sector; Trotterized real-time evolution	No anyon-pair fusion readout; computational basis/Wilson-loop observables test fusion-constrained gauge dynamics	No exchange braid; circuits explicitly execute up to 47 sequential F-moves while evolving the gauge model [R166]	Dynamical decoupling and parallelization mitigate dominant idling error; no passive topological protection	Fibonacci fusion algebra is digitally encoded , but the experiment is gauge dynamics, not a Fibonacci braid demonstration

Two trends matter. First, the hardware has progressed from direct matrix emulation on two nuclear spins to spatial many-qubit string nets and adaptive non-Abelian wavefunctions. Second, the interpretation has not crossed the digital–intrinsic boundary. Larger registers and better topological observables do not, by themselves, change what Hamiltonian the laboratory built.

The two superconducting Fibonacci experiments are complementary. Xu *et al.* used 27 transmons to represent a three-plaquette fixed-point string-net state and measured entanglement and braid/fusion data [R138]. Mineev *et al.* reduced the local graph resources, used exact graph moves, and made fusion readout economical, but relied on very deep circuits

and extensive mitigation [R165]. Both are significant experiments. Neither is a passively protected topological computer.

The 2026 ion results broaden the lesson. Lo *et al.* obtained universality from S_3 anyons by combining braid operations with fusion measurements [R140]. That is not the Fibonacci claim that braiding alone densely generates logical gates. Hayata, Hidaka, and Kikuchi encoded the Fibonacci fusion algebra in a non-Abelian lattice-gauge simulation and pushed long sequences of F -moves, but did not braid or fuse localized anyon pairs [R166]. “Contains Fibonacci mathematics” is therefore a wider category than “demonstrates Fibonacci fusion and braiding.”

20.8. Ways to get this wrong

Call every non-Abelian object Fibonacci. Ising twist defects obey $\sigma \times \sigma = \mathbf{1} + \varepsilon$, not $\tau \times \tau = \mathbf{1} + \tau$. D_4 and S_3 quantum doubles have still richer charge sets. They are valuable comparators, but their braid representations and computational power differ. A universal protocol using S_3 braiding *and fusion* is not evidence of Fibonacci braid universality.

Treat a snapshot as a stable phase. A finite-depth or adaptive circuit can prepare a state with target stabilizers and long-range entanglement. Once the circuit stops, the hardware evolves under its own control and noise Hamiltonian. Without a sustained target Hamiltonian or repeated error correction, the state is not passively stabilized.

Confuse a better estimate with a better memory. Zero-noise extrapolation estimates a hypothetical low-noise observable from several noisy runs. Readout mitigation inverts a calibrated measurement channel. Postselection discards detected bad shots. None produces a logical qubit that remains correct in an individual run. They improve scientific inference, not stored quantum information.

Ask “protected against what?” The NMR experiment showed robustness against a programmed class of local disturbances inside its encoded model [R163]. That is a legitimate test of a theoretical invariant. It does not mean the liquid-state nuclear spins were protected against RF miscalibration, relaxation, or dephasing; the measured gate error was in fact dominated by ordinary hardware decoherence [R163]. Always ask: protected against which physical errors, for how long, and relative to what unencoded control?

Infer a material from a compiled answer. A direct logical unitary can reproduce the same final fusion histogram as a spatial braid. Better demonstrations localize endpoints, move them with local operations, compare braid orders, and perform fusion or interferometric readout. Still, digital locality is programmed locality. Intrinsic emergence additionally requires the physical Hamiltonian to generate and stabilize the quasiparticle sector.

Ignore how small the chip is. Three plaquettes can display exact algebraic relations, but cannot establish asymptotic degeneracy splitting, a correlation length well below system size, a stable phase under generic perturbations, or a logical-error rate that improves with

code distance. Small systems are excellent microscopes for topological data and poor rulers for thermodynamic protection.

20.9. Check yourself

1. **What is the clean operational sequence for a digital braid experiment?**

Prepare a target vacuum wavefunction, create encoded anyon pairs, move/exchange them with compiled local gates, map to a fusion basis, and measure the fusion channel.

2. **Show that $P(\tau)/P(1) = \phi$ after braiding the middle pair of four Fibonacci anyons.**

The braid amplitudes have squared magnitudes ϕ^{-2} and ϕ^{-1} . Their ratio is ϕ , and $\phi^2 = \phi + 1$ makes the two probabilities add to one.

3. **Did the superconducting experiments contain physically intrinsic Fibonacci anyons?**

No. Ordinary transmon qubits digitally represented doubled-Fibonacci string-net states and operations. The experiments did not realize a native Fibonacci Hamiltonian with passive protection.

4. **What fails if adaptive measurement and feed-forward is called passive?**

Adaptive circuits can prepare or actively maintain a state. Passive protection comes from the physical Hamiltonian and gap without continual corrective intervention. The label then names the wrong agent.

5. **What fails if every non-Abelian digital result is filed under Fibonacci?**

Ising, D_4 , and S_3 have different fusion rules, braid representations, and computational power. A universal S_3 protocol that uses braiding *and* fusion is not evidence of Fibonacci braid universality.

6. **What can a defect-spin array borrow from these results without becoming topological matter?**

State-preparation circuits, encoded fusion spaces, compiled F/R moves, stabilizer or Wilson-loop measurements, adaptive feedback, and error-mitigation methods.

You can now read a braid paper and ask which of the four objects it actually built. Next we replace the compiled preparation $U_{\text{prep}}|0\rangle^{\otimes N}$ with the stronger demand that the laboratory Hamiltonian itself look like the target.

20.10. Sources

- [R162] Shibo Xu *et al.*, “Digital Simulation of Projective Non-Abelian Anyons with 68 Superconducting Qubits,” *Chinese Physics Letters* **40**, 060301 (2023). DOI: 10.1088/0256-307X/40/6/060301.
- [R132] Trond I. Andersen *et al.*, “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* **618**, 264–269 (2023). DOI: 10.1038/s41586-023-05954-4.
- [R163] Yu-ang Fan *et al.*, “Experimental quantum simulation of a topologically protected Hadamard gate via braiding Fibonacci anyons,” *The Innovation* **4**, 100480 (2023). DOI: 10.1016/j.xinn.2023.100480; arXiv:2210.12145.
- [R164] Mohsin Iqbal *et al.*, “Non-Abelian topological order and anyons on a trapped-ion processor,” *Nature* **626**, 505–511 (2024). DOI: 10.1038/s41586-023-06934-4; arXiv:2305.03766.
- [R138] Shibo Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6; arXiv:2404.00091.
- [R165] Zlatko K. Mineev *et al.*, “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* **16**, 6225 (2025). DOI: 10.1038/s41467-025-61493-8.
- [R140] Chiu Fan Bowen Lo *et al.*, “Universal gates from braiding and fusing anyons on quantum hardware,” *Nature* **655**, 591–597 (2026). DOI: 10.1038/s41586-026-10709-y; arXiv:2601.20956.
- [R166] Tomoya Hayata, Yoshimasa Hidaka & Yuta Kikuchi, “Digital quantum simulation of q -deformed SU(2) Yang–Mills theory on a trapped-ion quantum computer,” *Physical Review Research* **8**, 033137 (2026). DOI: 10.1103/vlpv-n8dy; arXiv:2601.13530.

Chapter 21. A recipe versus a landscape

A recipe is a list of moves. Cream the butter. Add the eggs. Bake for forty minutes. When you follow it, a cake appears. The cake is real. If you leave the kitchen, the recipe does not keep the cake from going stale.

A landscape is not a list. Water sits at the bottom of a valley because every nearby step is uphill. You can walk away. The water stays. The ground is doing the work, not a cook standing over the pot.

A circuit that prepares a toric-code state is a recipe. A crystal whose own energy prefers that state is a valley. Both can put the same wavefunction on the table. Only one of them is still doing the job after you stop typing gates.

21.1. Two jobs with the same snapshot

The last chapter was the recipe. A processor applies timed gates, writes a snapshot, and measures it. That is *digital* simulation. The hardware is following instructions about a model, not becoming the model.

The stronger claim is that the material itself is the valley. Local interactions stay on. The desired long-range-entangled sector sits lower than everything nearby. Excitations cost a fixed energy. Fabrication errors do not flatten the valley. The environment does not hand out enough anyons, and move them far enough, to enact a logical operator. That is an *analog* realization. The hardware is the landscape, not a script about the landscape.

A defect spin can still be a physical qubit. Several defects can still encode a logical bit. A processor can still write a Fibonacci-like snapshot. None of those facts says an undriven array sits in a topologically ordered phase.

The energy landscape of a closed quantum system is a Hermitian operator H , the Hamiltonian. Every later inequality in this chapter is a statement about H , not about one pretty wavefunction.

The analog target is a family of materials

$$H_{\text{physical}} = H_{\text{topological}} + V,$$

where the unwanted term V is local enough and weak enough that it does not change the phase. An encoded logical qubit may live in either a digital snapshot or an analog Hamiltonian. Emergent anyons and equilibrium topological order require the Hamiltonian.

Only then can one discuss *passive* protection supplied by energy and locality, rather than by a controller that keeps measuring syndromes.

21.2. Four edges of one square

Put one qubit on each edge of a square lattice. Let X_e and Z_e denote Pauli operators on edge e . For every vertex v and plaquette p , define

$$A_v = \prod_{e \ni v} X_e, \quad B_p = \prod_{e \in \partial p} Z_e.$$

Each product contains four edges on the square lattice. A star and a plaquette share either zero or two edges, so all A_v and B_p commute. The static toric-code Hamiltonian is

$$H_{\text{TC}} = -J_s \sum_v A_v - J_p \sum_p B_p, \quad (21.1)$$

where $J_s > 0$ and $J_p > 0$ are energies, measured in joules (J) or electronvolts (eV). A ground state obeys $A_v = B_p = +1$ for every term. On a torus, the ideal model has four ground states distinguished by noncontractible loop operators [R030]. **[Theory]**

Now isolate one plaquette, label its edges 1, 2, 3, 4, and define $B = Z_1 Z_2 Z_3 Z_4$. During a digital time step of duration δt , the target evolution from the plaquette term $-J_p B$ is

$$U_p(\delta t) = \exp\left(+i \frac{J_p \delta t}{\hbar} B\right), \quad (21.2)$$

where $\hbar$ has units J·s. The exponent is dimensionless because $J_p \delta t / \hbar$ has units (J)(s)/(J s) = 1.

A gate processor can synthesize this four-body evolution using parity computation. Let

$$W = \text{CNOT}_{1 \rightarrow 4} \text{CNOT}_{2 \rightarrow 4} \text{CNOT}_{3 \rightarrow 4}.$$

Conjugation gives $W^\dagger Z_4 W = Z_1 Z_2 Z_3 Z_4 = B$. With $R_z(\phi) = \exp(-i\phi Z/2)$, choose $\phi = -2J_p \delta t / \hbar$. Then

$$W^\dagger R_z(\phi)_4 W = \exp\left(+i \frac{J_p \delta t}{\hbar} B\right). \quad (21.3)$$

Repeating corresponding gate blocks for all stars and plaquettes implements evolution under Eq. (21.1). At the exactly commuting point the ideal factors commute, so this particular decomposition has no Trotter error. Real gates still have control errors, decoherence, and an idle Hamiltonian that is not Eq. (21.1).

Compare that circuit with a material containing a permanent term $-J_p B$. If $b = \pm 1$ is the measured eigenvalue of B , its energy contribution is $-J_p b$. Changing b from $+1$ to -1 costs $2J_p$. A local X_e anticommutes with the two adjacent plaquettes, so in a periodic bulk it creates two magnetic anyons and costs $4J_p$. Similarly, a local Z_e creates two electric anyons and costs $4J_s$. Thus the periodic model's lowest bulk pair-creation gap is

$$\Delta_{\text{pair}} = 4 \min(J_s, J_p), \quad (21.4)$$

while the energy assigned to one well-separated anyon is $\epsilon_a = 2J_s$ or $2J_p$. Boundaries can permit a single excitation, so one must state which gap convention and geometry are being quoted [R030]; [R141]. **[Theory]**

Hence the circuit and the static Hamiltonian can produce the same unitary or the same state. Only the static Hamiltonian supplies the energy penalty continuously. If the circuit prepares $|\psi_{\text{TC}}\rangle$ and then the hardware idles under a trivial H_{idle} , the target state was prepared, not made into an equilibrium phase.

21.3. The hill a local shovel must climb

Order the eigenenergies of the full interacting Hamiltonian as $E_0 \leq E_1 \leq \dots$. If the topological ground manifold occupies the lowest q levels, define the bulk many-body gap

$$\Delta(L) = E_q(L) - E_{q-1}(L), \quad \Delta_{\text{topo}} = \liminf_{L \rightarrow \infty} \Delta(L), \quad (21.5)$$

where L is the sample's linear size in metres and Δ is an energy. Material topological order requires $\Delta_{\text{topo}} > 0$, not merely a nonzero level spacing in a small cluster.

That number Δ_{topo} is the *many-body gap*. The name is worth keeping because it is a property of the whole interacting spectrum in the large-sample limit, not of one defect or one cluster. A defect's optical gap, a cluster's leakage gap, and the collective topological gap are three different numbers.

The gap sets a microscopic response time $\tau_\Delta = \hbar/\Delta_{\text{topo}}$. Slow, weak changes compared with this scale can avoid exciting the bulk. They still need not preserve the logical state if they split the ground manifold or move real anyons.

What fails if you quote only a two-site splitting and call it Δ_{topo} ? You have measured a cluster, not the thermodynamic gap above the topological manifold.

21.4. A shove that stays nearby

Write a lattice Hamiltonian as

$$H = \sum_X h_X,$$

where X is a bounded set of nearby sites and h_X acts only on those sites. “Local” normally means finite range or sufficiently fast decay with the diameter of X . It does not mean that the global operator norm $\|H\|$ is small. That norm generally grows with the number of sites.

For an unwanted perturbation $V = \sum_X V_X$, a useful local strength is

$$g_{\text{loc}} = \max_i \sum_{X \ni i} \|V_X\|, \quad (21.6)$$

which has units of energy. The engineering ratio is $g_{\text{loc}}/\Delta_{\text{topo}}$, not $\|V\|/\Delta_{\text{topo}}$.

That per-site number is the *local strength*. An extensive perturbation can look huge in the global norm while remaining weak at every site — and one strong local defect can be dangerous despite a small average.

Stability theorems prove that broad classes of local commuting-projector Hamiltonians retain separated spectral bands and exponentially narrow ground bands below a nonzero local perturbation threshold, subject to topological-order conditions [R142]. **[Theory]** The theorem does not supply one universal threshold such as “ten percent” for every material.

Defect arrays make locality subtle. Dipolar interactions fall as $1/r^3$, rather than ending at a chosen neighbor. In spatial dimension D , the shell sum scales as $\int dr r^{D-1}/r^\alpha$, so it converges at long distance only for $\alpha > D$. A $1/r^3$ tail is summable in two dimensions, but that fact alone does not put it under every finite-range stability theorem. The real tail, crosstalk, and disorder must be retained or bounded, not erased with a hopeful nearest-neighbor sketch.

21.5. How far a local change is felt

For local observables O_i and O_j separated by distance r , define their connected ground-state correlation

$$C(r) = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle.$$

A gapped short-range system generally has

$$|C(r)| \leq C_0 e^{-r/\xi}, \quad (21.7)$$

where C_0 carries the observable's units and ξ is the *correlation length* in metres [R167]. **[Theory]** It is the reach of ordinary local questions, not the length of the nonlocal entanglement that topological order is allowed to keep.

A dimensional estimate is $\xi \sim \hbar v_{\text{LR}}/\Delta$, where v_{LR} is a characteristic information-propagation velocity in m/s: (J s)(m/s)/J = m.

Finite ξ does not contradict topological order. Ordinary local correlators can decay rapidly while the ground state contains nonlocal entanglement and loop information. At the exact toric-code fixed point, many connected local correlations vanish beyond zero range. Generic perturbations make ξ finite rather than destroying the phase.

The practical requirement is

$$L \gg \xi, \quad (21.8)$$

and, for localized non-Abelian anyons, their separation $R \gg \xi$. Otherwise their topological sectors overlap appreciably and “nonlocal encoding” is mostly a label on a small device.

21.6. Two ground states that look the same up close

Topological ground states are locally indistinguishable in the thermodynamic limit: a bounded local operator cannot reliably tell which logical sector the system occupies. Weak local perturbations then dress the states rather than immediately selecting one sector [R142]. This is zero-temperature phase stability, not a lifetime theorem.

On a finite torus, virtual anyons can wind around a noncontractible cycle and annihilate. The process couples logical sectors. For a gapped local phase, its characteristic splitting has the form

$$\delta E_0(L) \sim C \Delta_{\text{topo}} e^{-L/\xi}, \quad (21.9)$$

up to model-dependent powers, paths, and coefficients. Fusion-state splitting for anyons separated by R is similarly exponentially small in R/ξ [R015]. **[Theory]** Here C is dimensionless, so δE_0 has energy units.

During storage time t_{store} , this splitting produces a relative phase of order $\delta E_0 t_{\text{store}}/\hbar$. A necessary condition is

$$\frac{\delta E_0 t_{\text{store}}}{\hbar} \ll 1. \quad (21.10)$$

For controlled adiabatic motion, a familiar ideal window is $\hbar/\Delta_{\text{topo}} \ll t_{\text{op}} \ll \hbar/\delta E_0$: slow relative to bulk excitation, fast relative to resolving the unwanted splitting [R015]. **[Theory]** Real control adds diabatic errors, noise, and timing constraints.

What fails if you treat $\delta E_0 = 0$ at finite L ? The relative phase still accumulates. Over a long enough t_{store} the sectors are resolved, and “exact degeneracy” was a finite-size slogan.

21.7. Heat keeps handing out anyons

At nonzero temperature, the environment keeps creating excitations.

At temperature T in kelvin, a weakly coupled equilibrium system has density matrix

$$\rho_T = \frac{e^{-H/(k_B T)}}{\text{Tr}(e^{-H/(k_B T)})},$$

where Boltzmann’s constant k_B has units J/K. A dilute anyon species of energy ϵ_a has an equilibrium occupation proportional to

$$n_a \sim \frac{g_a}{a^2} e^{-\epsilon_a/(k_B T)} \quad (21.11)$$

in two dimensions, where a is the lattice spacing and g_a is a dimensionless degeneracy factor. This expression is only the dilute, noninteracting estimate. The expected *number* also grows with area. Thus $k_B T \ll \epsilon_a$ is necessary for suppression but does not guarantee an anyon-free macroscopic sample [R141]; [R169]; [R143]. **[Theory]**

As a scale exercise, suppose — not claim — that $J_s = J_p = \hbar \times 1 \text{ GHz}$. Since $\hbar \times 1 \text{ GHz}/k_B \approx 48 \text{ mK}$, one anyon costs $2J \approx k_B \times 96 \text{ mK}$, and a bulk pair costs $4J \approx k_B \times 192 \text{ mK}$. At $T = 20 \text{ mK}$, the pair Boltzmann factor is $e^{-192/20} \approx 6.8 \times 10^{-5}$, but the single-anyon density factor is $e^{-96/20} \approx 8.2 \times 10^{-3}$. The arithmetic is favorable and still not magic: area, bath spectral density, and motion determine the logical error rate.

In two-dimensional toric-code-like memories, the environment can create an anyon pair for a fixed energy and then move one anyon across the sample without climbing a taller hill. A large sample is therefore not automatically a better passive memory.

21.8. Creation is only half a fault

Once an anyon exists, local hopping can extend its error string. If motion is diffusive in two dimensions with diffusion constant D in m^2/s , its root-mean-square distance after time t is

$$\ell_{\text{diff}}(t) = \sqrt{4Dt}, \quad (21.12)$$

which has units $\sqrt{(\text{m}^2/\text{s})(\text{s})} = \text{m}$. A code of distance d and spacing a needs, roughly, $\ell_{\text{move}}(t_{\text{store}}) \ll da$, together with a sufficiently low creation rate.

Static disorder can Anderson-localize ideal toric-code defects, giving wavefunctions that decay over a localization length ξ_{loc} [R170]. **[Theory]** **[Numerics]** This can slow coherent propagation. It does not raise the creation gap, and a thermal bath can supply inelastic hops between localized states. At finite density, anyon interactions also invalidate a one-particle localization argument.

Hence localization is a kinetic aid to be demonstrated under the actual noise model, not a synonym for topological protection. Deliberate pinning also conflicts with operations that require controlled anyon motion.

What fails if you treat localization as a gap? Creation remains cheap. A bath can still hop the anyon from well to well. The energy barrier never grew.

21.9. What is actually in the fridge

A fabricated defect array has more than the target Hamiltonian:

$$H_{\text{lab}} = H_{\text{target}} + V_{\text{static}} + H_{\text{drive}}(t) + H_{\text{bath}} + H_{\text{system-bath}}. \quad (21.13)$$

The terms represent fabrication disorder and unwanted couplings, time-dependent control, environmental modes, and their coupling to the spins. An analog claim must survive the first correction and quantify the last two. Turning on drives that Floquet-engineer an effective Hamiltonian can be valuable, but then heating, micromotion, calibration, and drive failure belong in the protection claim.

A persuasive material demonstration would not stop at preparing a target state. It would establish a bulk gap and its size scaling; show that local observables cannot distinguish candidate ground sectors beyond finite-size corrections; measure nonlocal loop or braiding data; and demonstrate that weak local perturbations do not close the gap or change the topological data. Thermal measurements must separately estimate excitation creation and transport. No single finite-device signature proves the thermodynamic phase.

For defect-cluster architectures, $H_{\text{physical}} \approx H_{\text{string-net}}$ remains a **[Proposal]** until the derived microscopic terms, corrections, gap, and diagnostics are supplied. Agreement between a programmed circuit and target fusion data is evidence for control and simulation, not evidence that the undriven material has emergent anyons.

21.10. Immortality is a taller claim

A *self-correcting quantum memory* has a storage lifetime that grows without bound with system size at fixed nonzero temperature, without active syndrome measurement and feedback [R169]. That is stronger than a long lifetime in one finite sample.

The ordinary two-dimensional toric code has a constant energy barrier. Create an anyon pair for a fixed cost, move one endpoint across the lattice with no additional growing energy cost, then annihilate the pair. The resulting string can be a logical operator. Under the assumptions of geometrically local two-dimensional stabilizer Hamiltonians, the Bravyi–Terhal no-go theorem proves an $O(1)$ energy barrier, ruling out conventional self-correction in that class [R168]. **[Theory]** This theorem is powerful but not a license to make claims about every non-Abelian, long-range, driven, or nonequilibrium model.

Accordingly, a gapped 2D topological phase can suppress local matrix elements, exponentially reduce finite-size splitting, and lower thermal anyon density, yet still need active error correction for arbitrarily long storage. Analyses of the toric code coupled to thermal reservoirs likewise distinguish stability against static perturbations from thermal stability [R143]. **[Theory]** “Passive protection” should therefore be reported as a measured improvement over specified time, temperature, size, and bath — not as an unqualified promise of immortality.

A useful necessary-condition sheet is

$$\begin{array}{l}
 \Delta_{\text{topo}} > 0, \quad L/\xi \gg 1, \\
 g_{\text{loc}}/\Delta_{\text{topo}} < c_{\text{stab}}, \\
 k_B T/\epsilon_a \ll 1, \\
 \delta E_0 t_{\text{store}}/\hbar \ll 1, \\
 \ell_{\text{move}}(t_{\text{store}}) \ll da.
 \end{array} \tag{21.14}$$

Here c_{stab} is a model-dependent dimensionless stability threshold. These inequalities are necessary, not jointly sufficient. One must also count creation sites, include leakage out of encoded clusters, establish the target topological data, and model the bath.

21.11. Ways to get this wrong

A prepared ground-state wavefunction is not a phase. A phase is a stable property of a Hamiltonian family and its low-energy sector, not one state produced by a pulse sequence.

A single-particle or cluster gap is not the topological gap. The required Δ_{topo} is the many-body gap above the topological manifold after all effective corrections are included.

Global perturbation norm is the wrong criterion. An extensive perturbation can have large $\|V\|$ while remaining weak per site. Conversely, one strong local defect can be dangerous despite a small average.

Finite correlation length is not short-range entanglement. Topological phases have short-ranged ordinary connected correlations but long-range entanglement.

Exponential splitting is not exact degeneracy. At finite L , δE_0 is generally nonzero and matters over long enough time.

Localization is not a gap. Localization slows motion; it does not prevent thermal creation and may fail under bath-assisted or interacting dynamics.

Perturbative stability is not thermal self-correction. The former concerns a weak static change of Hamiltonian near zero temperature. The latter concerns open-system dynamics for growing size at nonzero temperature.

Active digital correction is not passive material protection. Useful error correction remains useful. The category label is what changes.

21.12. Check yourself

1. **A circuit exactly implements $e^{-iH_{\text{TC}}t/\hbar}$. Has it realized an analog toric-code phase?**

No. The idle physical Hamiltonian must itself lie in the phase; a timed gate sequence implements dynamics but need not supply an equilibrium energy landscape.

2. **Show that a local X_e on the periodic toric code costs $4J_p$.**

X_e anticommutes with the two adjacent plaquettes. Each flipped plaquette changes its energy by $2J_p$, so the pair costs $4J_p$.

3. **What fails if you use $\|V\|$ instead of g_{loc} ?**

$\|V\|$ usually grows with array size even when every site remains weakly perturbed. The engineering ratio then looks fatal while the local theory is still in the phase — or, conversely, one strong defect is hidden by a small average.

4. **Show that if $J_s = J_p = h \times 1 \text{ GHz}$, the pair Boltzmann factor at $T = 20 \text{ mK}$ is about 6.8×10^{-5} .**

$h \times 1 \text{ GHz}/k_B \approx 48 \text{ mK}$, so a pair costs $4J \approx k_B \times 192 \text{ mK}$ and $e^{-192/20} \approx 6.8 \times 10^{-5}$.

5. **What fails if $k_B T \ll \Delta_{\text{pair}}$ is taken as passive fault tolerance?**

The number of creation sites, the bath, anyon mobility, code geometry, and the constant energy barrier can still produce logical errors. Diluteness is not emptiness, and a large sample is not automatically safer.

6. **What does the 2D stabilizer no-go theorem actually rule out?**

A growing energy barrier — and hence conventional self-correction — within its assumptions of geometrically local two-dimensional stabilizer Hamiltonians. It does not classify every conceivable topological material.

You can now tell a compiled circuit from a Hamiltonian in the phase, and you have a short list of inequalities that phase still has to satisfy. Next we ask whether a microscopic defect Hamiltonian actually produces the required low-energy terms — and how large the leftover corrections become.

21.13. Sources

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0; arXiv:quant-ph/9707021.
- [R167] M. B. Hastings and T. Koma, “Spectral gap and exponential decay of correlations,” *Communications in Mathematical Physics* **265**, 781–804 (2006). DOI: 10.1007/s00220-006-0030-4; arXiv:math-ph/0507008.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195; arXiv:1001.0344.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv:0707.1889.
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* **43**, 4452–4505 (2002). DOI: 10.1063/1.1499754; arXiv:quant-ph/0110143.

- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* **11**, 043029 (2009). DOI: 10.1088/1367-2630/11/4/043029; arXiv:0810.1983.
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* **88**, 045005 (2016). DOI: 10.1103/RevModPhys.88.045005; arXiv:1411.6643.
- [R170] C. Stark, L. Pollet, A. Imamoglu, and R. Renner, “Localization of toric code defects,” *Physical Review Letters* **107**, 030504 (2011). DOI: 10.1103/PhysRevLett.107.030504; arXiv:1101.6028.
- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* **42**, 065303 (2009). DOI: 10.1088/1751-8113/42/6/065303; arXiv:0810.4584.

Part IX

Engineering topology from defect clusters

The expensive states still whisper. Two-body crystals try to fake four-body rules. Pretty arrows get audited.

Chapter 22. The expensive states still whisper

A cluster has more states than you will keep. The coupling that joins two clusters does not stay inside the pair you came for: it mixes that pair with expensive levels and returns. The return trip is a new operator on the cheap pair.

Suppose the cheap states sit at energy zero, the expensive ones sit at least Δ above them, and the mixing has scale g , with $|g|/\Delta$ small. Then any leftover operator that starts cheap and ends cheap must use the mixing at least twice. The first such term has scale g^2/Δ .

That leftover is the working Hamiltonian of this chapter. Delete the expensive states before recording the return trips and you lose it. Keep every state and the page is too large to read.

22.1. A trip that starts cheap and ends cheap

Picture two subspaces. The cheap one holds the states you will keep. The expensive one costs at least an extra energy Δ . A weak coupling of strength g joins them. Both g and Δ are energies, and we take $|g|/\Delta$ small.

One use of the coupling cannot start cheap and finish cheap: it lands in the expensive subspace. Two uses can:

$$\text{cheap } a \xrightarrow{g} \text{expensive } m \xrightarrow{g} \text{cheap } b.$$

The leftover amplitude has the scale

$$\frac{g \ g}{E_{\text{low}} - E_m} \approx -\frac{g^2}{\Delta}.$$

Three facts sit in that expression. Two couplings produce g^2 . A large expensive gap suppresses the process as $1/\Delta$. If the intermediate state lies above the states you kept, the denominator is negative.

The expensive state appears as an intermediate amplitude, not as a final state you would find if you measured energy. That kind of contribution is **virtual**.

Hence a second-order leftover on the cheap subspace is an energy of order g^2/Δ , not of order g .

If the coupling becomes strong enough to populate the expensive state for real, calling the contribution virtual is wishful accounting. The visit is virtual only while $|g|/\Delta$ is small.

The final sign of a particular spin coupling also depends on matrix-element phases and on how you write the leftover operator. “Second order is negative” is therefore too crude. The energy correction from coupling only to higher states is negative-semidefinite. An exchange coefficient extracted after you subtract a constant can still be positive. The two-site example later will do exactly that.

The same scale, drawn as a chain:

```
cheap states
  |
  | weak coupling v
  v
expensive subspace, cost Delta
  |
  | back to the cheap subspace
  v
leftover coupling ~ v^2 / Delta
```

22.2. Three states, one expensive

Take three orthonormal states: two cheap ones $|L\rangle$ and $|R\rangle$, and one expensive one $|e\rangle$. Let the unperturbed Hamiltonian be

$$H_0 = \Delta|e\rangle\langle e|,$$

where $\Delta > 0$ has units of energy. The cheap states sit at energy zero. Let a coupling of strength g , also an energy, join each cheap state to the expensive one:

$$V = g(|e\rangle\langle L| + |e\rangle\langle R| + \text{h.c.}).$$

Here “h.c.” means the Hermitian conjugate, which supplies the return paths. We assume $|g|/\Delta \ll 1$.

The machine that keeps the cheap pair is

$$P = |L\rangle\langle L| + |R\rangle\langle R|.$$

Everything you threw away is

$$Q = |e\rangle\langle e|.$$

They add to the whole space, $P + Q = I$. Applied twice, each is itself. They kill each other:

$$P^2 = P, \quad Q^2 = Q, \quad PQ = QP = 0.$$

P is a **projector**. You have already used projectors as measurement questions. Here the same machine just means “keep this subspace.” Q is the complementary projector: the expensive states you threw off the working page.

Because $PVP = 0$, there is no first-order cheap-to-cheap door. The second-order leftover is

$$H_{\text{eff}}^{(2)} = -PVQ \frac{1}{\Delta} QVP = -\frac{g^2}{\Delta} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}_{\{|L\rangle, |R\rangle\}}.$$

A microscopic coupling that never directly joins $|L\rangle$ to $|R\rangle$ still generates

$$\langle L | H_{\text{eff}} | R \rangle = -\frac{g^2}{\Delta}.$$

Hence a virtual path through a single expensive state writes an off-diagonal cheap coupling that the original V did not contain.

The antisymmetric combination $(|L\rangle - |R\rangle)/\sqrt{2}$ is dark: it stays at zero energy. The symmetric combination couples to $|e\rangle$ and slides down by $-2g^2/\Delta$.

This toy can be solved exactly. Its lower bright energy is

$$E_- = \frac{\Delta - \sqrt{\Delta^2 + 8g^2}}{2} = -\frac{2g^2}{\Delta} + \frac{4g^4}{\Delta^3} + O\left(\frac{g^6}{\Delta^5}\right).$$

The expansion displays control rather than merely asserting it. The leading retained shift is of order g^2/Δ . Here symmetry kills the cubic term, so the first omitted energy correction is fourth order. Drop that symmetry and a second-order truncation leaves third-order leftovers.

22.3. Split the space before you expand

Write the full Hamiltonian as

$$H = H_0 + \lambda V.$$

The operator H_0 is the part you can solve. V has units of energy and holds the perturbation. The number λ is a dimensionless bookmark: set it to 1 after you have decided that the physical ratio is small.

The projector P selects the retained subspace. Its partner $Q = I - P$ selects every eliminated state. Choose P from eigenstates of H_0 , so $PH_0Q = 0$. In blocks,

$$H = \begin{pmatrix} PHP & PHQ \\ QHP & QHQ \end{pmatrix}.$$

The diagonal blocks stay inside one subspace. The off-diagonal blocks join cheap to expensive. A first-order projection, PHP , simply deletes those blocks. We are going to record the return trips instead.

Let E_a be an unperturbed energy in P and E_m one in Q . The separation is

$$\Delta = \min_{a \in P, m \in Q} |E_m - E_a|.$$

For a genuinely cheap P , the relevant differences $E_m - E_a$ are positive. The useful small parameter is not “ V is small” by itself. It is

$$\epsilon = \frac{|\lambda|v}{\Delta} \ll 1,$$

where v is the off-diagonal coupling scale — for example a suitable operator norm of $PVQ + QVP$. Both v and Δ are energies, so ϵ is a pure number.

Hence the expansion is controlled by a ratio of two energies, not by the mere existence of a splitting of H into $H_0 + V$.

If a level in Q approaches a level in P , then Δ shrinks, ϵ grows, and the cheap model you chose has to grow with it. The divergence is a warning label, not free coupling strength.

22.4. The exact leftover after you throw them out

Write an exact eigenstate as $|\psi\rangle = |p\rangle + |q\rangle$, with $|p\rangle = P|\psi\rangle$ and $|q\rangle = Q|\psi\rangle$. Projecting $H|\psi\rangle = E|\psi\rangle$ gives two equations:

$$PHP|p\rangle + PHQ|q\rangle = E|p\rangle,$$

$$QHP|p\rangle + QHQ|q\rangle = E|q\rangle.$$

If $E - QHQ$ can be inverted, the second line solves for the expensive piece:

$$|q\rangle = (E - QHQ)^{-1}QHP|p\rangle.$$

Feed that back into the first line and you get an exact, energy-dependent cheap operator

$$H_{\text{eff}}(E) = PHP + PHQ(E - QHQ)^{-1}QHP.$$

That is the projection logic associated with Feshbach [R171]. The inverse is the denominator you already met. Replacing E and QHQ by their unperturbed values starts the series.

For a degenerate cheap manifold of energy E_0 , the result through second order is

$$H_{\text{eff}} = E_0 P + \lambda PVP + \lambda^2 PVQ \frac{1}{E_0 - QH_0Q} QVP + O\left(\frac{|\lambda|^3 v^3}{\Delta^2}\right)$$

on a bounded finite problem whose relevant perturbation scales are represented by v . Equivalently, when every eliminated state lies above E_0 ,

$$H_{\text{eff}}^{(2)} = -PVQ \frac{1}{QH_0Q - E_0} QVP.$$

The two forms are the same. The second makes the minus sign hard to drop, because $QH_0Q - E_0$ is positive.

Hence the cheap leftover through second order is the first-order block PHP , plus a sum of virtual paths through Q , plus a remainder of order v^3/Δ^2 .

In a basis $|a\rangle, |b\rangle \in P$ and $|m\rangle \in Q$,

$$\langle a|H_{\text{eff}}^{(2)}|b\rangle = \sum_{m \in Q} \frac{\langle a|V|m\rangle \langle m|V|b\rangle}{E_0 - E_m}.$$

Every allowed virtual path contributes. Paths may add or cancel. A small denominator fattens a term and, at the same time, threatens the expansion.

22.5. A small twist that almost splits the blocks

There is another way to write the same folding. Choose an anti-Hermitian generator $S^\dagger = -S$, taken to be off-diagonal, and transform

$$\tilde{H} = e^S H e^{-S}.$$

Because e^S is unitary, H and $\tilde{H}$ have the same exact spectrum. Expand $S = \lambda S_1 + \lambda^2 S_2 + \dots$ and use

$$e^S H e^{-S} = H + [S, H] + \frac{1}{2}[S, [S, H]] + \dots.$$

Split $V = V_d + V_{od}$ into the part that stays inside a subspace and the part that joins cheap to expensive. Cancelling the first-order off-diagonal block requires

$$[S_1, H_0] = -V_{od}.$$

Therefore

$$\langle a|S_1|m\rangle = \frac{\langle a|V|m\rangle}{E_a - E_m}.$$

Project the twisted Hamiltonian back into P and you recover

$$H_{\text{eff}} = PH_0P + \lambda PV P + \frac{\lambda^2}{2}P[S_1, V_{od}]P + O(\lambda^3).$$

If the retained states are not exactly degenerate, the second-order matrix element uses the Hermitian average of the two denominators:

$$\frac{1}{2} \sum_m V_{am} V_{mb} \left(\frac{1}{E_a - E_m} + \frac{1}{E_b - E_m} \right).$$

Hence the same second-order leftover can be written as a change of basis that removes the cheap-to-expensive block order by order.

Different honest choices of cheap basis can make two leftover Hamiltonians look different while leaving their cheap predictions unitarily equivalent [R025]. State the convention before you compare coefficients.

This small twist is the **Schrieffer–Wolff** transformation. The name is worth keeping because it is the standard label for this particular folding, written by Schrieffer and Wolff for a different cheap problem and now used wherever a cluster or a many-body system has a cheap block that still hears the expensive states [R024]; [R025]. It is degenerate perturbation theory dressed as a change of basis.

22.6. Two sites that hop and then pay

The Hubbard model describes fermions that hop between sites and pay an on-site energy for double occupancy [R172]. Take two sites, labeled 1 and 2, occupied by exactly two electrons. Let $c_{i\sigma}^\dagger$ create an electron of spin $\sigma \in \{\uparrow, \downarrow\}$ on site i , and write $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. The pieces are

$$H_0 = U \sum_{i=1}^2 n_{i\uparrow} n_{i\downarrow},$$

$$V = -t \sum_{\sigma} \left(c_{1\sigma}^\dagger c_{2\sigma} + c_{2\sigma}^\dagger c_{1\sigma} \right).$$

Here $U > 0$ is the double-occupancy bill and t is the hopping matrix element; both are energies. We set $\lambda = 1$ and assume $|t|/U \ll 1$.

The retained sector P has one electron per site:

$$|\uparrow, \uparrow\rangle, \quad |\uparrow, \downarrow\rangle, \quad |\downarrow, \uparrow\rangle, \quad |\downarrow, \downarrow\rangle.$$

All four have unperturbed energy $E_0 = 0$. The thrown-away sector Q holds the two doublon-hole states

$$|D_1\rangle = |\uparrow\downarrow, 0\rangle, \quad |D_2\rangle = |0, \uparrow\downarrow\rangle,$$

both at energy U . In this two-site problem the gap is $\Delta = U$, and $PVP = 0$: one hop always creates a doublon and a hole.

Use total-spin states

$$|S\rangle = \frac{|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle}{\sqrt{2}},$$

$$|T_0\rangle = \frac{|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle}{\sqrt{2}}, \quad |T_+\rangle = |\uparrow, \uparrow\rangle, \quad |T_-\rangle = |\downarrow, \downarrow\rangle.$$

A doubly occupied single orbital must be a spin singlet. The polarized triplets are Pauli blocked. The two hopping paths from $|T_0\rangle$ cancel. With a consistent choice of fermionic phases,

$$V|S\rangle = -\sqrt{2}t(|D_1\rangle + |D_2\rangle), \quad V|T_\mu\rangle = 0.$$

The total squared matrix element from the singlet into Q is $4t^2$. The denominator is $0 - U = -U$, so

$$\delta E_S^{(2)} = -\frac{4t^2}{U}, \quad \delta E_T^{(2)} = 0.$$

The singlet drops relative to the triplets. That is antiferromagnetic exchange: positive J favors antiparallel spins. In operator form inside the singly occupied sector,

$$H_{\text{eff}}^{(2)} = J \left(\mathbf{S}_1 \cdot \mathbf{S}_2 - \frac{1}{4} n_1 n_2 \right), \quad J = \frac{4t^2}{U} > 0,$$

where $\mathbf{S}_i$ is the spin-1/2 operator at site i , and $n_i = n_{i\uparrow} + n_{i\downarrow}$. At half filling inside P , $n_1 n_2 = 1$. Since $\mathbf{S}_1 \cdot \mathbf{S}_2 = -3/4$ in the singlet and $+1/4$ in a triplet, the formula gives $-J$ and 0, respectively.

Hence the leftover of two hops through a doubly occupied intermediate is an antiferromagnetic Heisenberg coupling whose coefficient is $4t^2/U$.

People often drop the constant $-J/4$ and write $J\mathbf{S}_1 \cdot \mathbf{S}_2$. That shifts every retained energy by $+J/4$ but keeps the singlet-triplet splitting J . Confusing the two conventions produces apparent disagreements of exactly one constant.

Dimensional check. Because $[t] = [U] = \text{energy}$,

$$[J] = \frac{[t]^2}{[U]} = \text{energy}.$$

For a deliberately hypothetical example, take $t = 1$ meV and $U = 10$ meV. Then $t/U = 0.1$ and

$$J^{(2)} = 4 \frac{(1 \text{ meV})^2}{10 \text{ meV}} = 0.4 \text{ meV}.$$

These are teaching numbers, not a claim about a particular defect.

The two-site problem also shows the error. The exact singlet-triplet splitting is

$$J_{\text{exact}} = \frac{\sqrt{U^2 + 16t^2} - U}{2} = \frac{4t^2}{U} - \frac{16t^4}{U^3} + O\left(\frac{t^6}{U^5}\right).$$

At the illustrative ratio $t/U = 0.1$, $J_{\text{exact}} \approx 0.385$ meV, about 3.9% below the second-order value. Odd orders vanish in this specific two-site, half-filled model. That is a symmetry and path-counting bonus, not a general Schrieffer–Wolff rule. Drop the two-site restriction, or leave half filling, and those odd orders need not vanish. Systematic t/U expansions for the Hubbard model generate further-neighbor and multi-spin terms at higher order

[R026]. The superexchange mechanism and its antiferromagnetic sign have a broader history beyond this minimal model [R173].

22.7. When “small” is actually small

For a finite cluster with characteristic off-diagonal matrix element v , the hierarchy is

$$\begin{array}{ccc} |\lambda|v & \ll & \Delta \\ |\{z\}| & & |\{z\}| \\ \text{mixing} & & \text{leakage cost} \end{array} .$$

Then the thrown-away amplitude inside a dressed cheap eigenstate is generally of order $\epsilon = |\lambda|v/\Delta$. A generated second-order energy is of order

$$\epsilon^2 \Delta = \frac{|\lambda|^2 v^2}{\Delta},$$

and a generic third-order remainder is of order

$$\epsilon^3 \Delta = \frac{|\lambda|^3 v^3}{\Delta^2}.$$

These are scaling estimates, not universal coefficient-one error bars. Multiple bonds, near-degeneracies, and many virtual paths supply combinatorial factors. In an array of clusters with coordination number z , a conservative local check is often $zv/\Delta \ll 1$, followed by an explicit bound or a convergence study for the actual local Hamiltonian. A global operator norm grows with system size and can be an unnecessarily blunt criterion. Local Schrieffer–Wolff and linked-cluster formulations organize extensive systems more carefully [R025].

Hence a useful derivation reports at least four items: the retained space P , the smallest relevant gap Δ , the dimensionless expansion parameter, and the leading omitted operators with their scale.

Reporting only the attractive term is not perturbation theory. It is advertising.

22.8. What a laboratory still has to measure

The Schrieffer–Wolff transformation is [**Theory**]. A laboratory has energy levels, transition matrix elements, drives, disorder, dissipation, and finite temperature. To claim that a measured defect cluster realizes a leftover Hamiltonian derived this way, you have to connect those observables to H_0 , V , P , and Q .

For a defect cluster, P might be an isolated spin doublet. The gap Δ might be a many-spin excitation gap, an orbital splitting, or a charge-transfer energy. Those are physically different denominators and they can have different noise sensitivity. The coupling V might be magnetic dipole coupling, direct exchange, strain coupling, or a driven interaction. A magnetic dipole term that already acts inside the doublet contributes at first order through PVP ; do not relabel it “superexchange.” A charge-transfer or mediator-assisted path that leaves P and returns can contribute at second or higher order.

For real superexchange through a ligand or an extra defect orbital, several intermediate energies and hopping amplitudes replace the single U . Their contributions can compete. Orbital occupancy plus Hund coupling can even flip the leftover sign. The two-site formula $4t^2/U$ is a derivation template, not a universal materials formula [R173].

A credible check is spectroscopic: measure or calculate the cheap manifold and the leakage levels, estimate the matrix elements that join them, diagonalize at least the smallest microscopic cluster, and compare its cheap spectrum and projected observables with the truncated leftover model. **[Numerics]** Exact diagonalization is especially useful here, because it tests coefficients and truncation errors without assuming the series is already accurate. Agreement at one tuned point is weaker than agreement across a range of v/Δ .

[Proposal] Later cluster constructions may use this expansion to ask whether the two-body defect couplings you actually have generate the encoded interactions you want. The transformation itself does not guarantee that the desired term is large, that unwanted terms cancel, or that the leftover phase is topological.

Eliminating expensive states does not create a new physical device. The crystal still contains every microscopic defect and mediator state. H_{eff} describes an encoded cheap sector of that same system. A generated spin interaction does not establish emergent anyons or topological order; those are many-body properties that need the right phase, the right gap, and the right nonlocal observables. A digitally programmed gate sequence that *looks* second-order is yet another category: it may emulate the same operator without owning the static microscopic hierarchy derived here.

22.9. Ways to get this wrong

1. **Wrong denominator sign.** For a retained energy E_0 and a higher intermediate energy E_m , use $E_0 - E_m < 0$. Equivalently, write an explicit minus sign and the positive cost $E_m - E_0$. Do not do both.
2. **Throwing the expensive states away before the return trips.** Replacing H by PHP loses $PVQ(E_0 - QH_0Q)^{-1}QVP$. First-order projection and second-order elimination answer different questions.

3. **No spectral separation.** If a Q level approaches P , its denominator becomes small, mixing becomes large, and the cheap model you chose has to grow. The divergence is a warning label, not free coupling strength.
4. **Comparing gauges term by term.** Leftover Hamiltonians related by a unitary rotation inside P can share out coefficients differently while predicting the same spectrum. Compare invariant predictions, or transform operators consistently [R025].
5. **Forgetting dressed observables.** Physical cheap states contain a Q -sector admixture of order v/Δ . An observable O should be transformed as $Pe^SOe^{-S}P$, not merely replaced by POP , when that correction matters.
6. **Hiding lower-order unwanted terms.** A desired interaction generated at fourth order may be much smaller than an allowed first- or second-order nuisance. Symmetry must forbid, cancel, or let you tune those terms. Notation cannot.
7. **Calling a small coefficient accurate because it exists.** If virtual paths nearly cancel, the desired coefficient can be smaller than the nominal remainder. Control of the full Hamiltonian does not guarantee a small *relative* error in a specially suppressed term.
8. **Using a closed-system Hamiltonian to claim coherence.** Block diagonalization does not remove phonons, fluctuating fields, thermal activation, or control noise. Coupling to a bath can cause real leakage, and leftover noise operators have to be projected too.
9. **Equating generated interactions with a topological phase.** A two-body microscopic model may generate multi-body operators, but correct operator content, coefficient hierarchy, system size, temperature, and perturbative stability still have to be established.

22.10. Check yourself

1. **What do P and Q mean?** P projects onto the cheap states retained in the leftover model; $Q = I - P$ projects onto the expensive states you threw away.
2. **Show that a virtual process that leaves P and returns at second order has scale v^2/Δ .** It uses the coupling twice to leave and return, and the intermediate resolvent supplies one inverse excitation energy, so the leftover is $v \cdot (1/\Delta) \cdot v$.
3. **What fails if an intermediate energy E_m approaches the cheap energy E_0 ?** The denominator $E_0 - E_m$ becomes small, $\epsilon = v/\Delta$ is no longer small, and the cheap model you chose has to grow. The leftover formula is not a source of free coupling.

4. **Show that Hubbard exchange is antiferromagnetic even though the singlet correction is negative.** Only the singlet gains the energy $-4t^2/U$; the triplets remain at zero. Writing this splitting as $J(\mathbf{S}_1 \cdot \mathbf{S}_2 - 1/4)$ gives $J = 4t^2/U > 0$.
5. **What must accompany a second-order leftover Hamiltonian?** The definition of P , the leakage gap Δ , a dimensionless small parameter such as v/Δ , and the scale and operator form of the leading omitted corrections.
6. **What fails if a Schrieffer–Wolff-generated plaquette term is taken as proof of topological order?** The term is one leftover operator. The many-body phase and its gap, excitations, robustness, and finite-size behavior still need separate evidence.

22.11. Sources

- [R024] J. R. Schrieffer and P. A. Wolff, “Relation between the Anderson and Kondo Hamiltonians,” *Physical Review* **149**, 491–492 (1966). DOI: 10.1103/PhysRev.149.491.
- [R025] S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* **326**, 2793–2826 (2011). DOI: 10.1016/j.aop.2011.06.004; arXiv: 1105.0675.
- [R171] H. Feshbach, “Unified theory of nuclear reactions,” *Annals of Physics* **5**, 357–390 (1958). DOI: 10.1016/0003-4916(58)90007-1.
- [R172] J. Hubbard, “Electron correlations in narrow energy bands,” *Proceedings of the Royal Society A* **276**, 238–257 (1963). DOI: 10.1098/rspa.1963.0204.
- [R026] A. H. MacDonald, S. M. Girvin, and D. Yoshioka, “ t/U expansion for the Hubbard model,” *Physical Review B* **37**, 9753–9756 (1988). DOI: 10.1103/PhysRevB.37.9753.
- [R173] P. W. Anderson, “New Approach to the Theory of Superexchange Interactions,” *Physical Review* **115**, 2–13 (1959). DOI: 10.1103/PhysRev.115.2.

Chapter 23. Perturbative gadgets

Four spins sit around a square. Topology wants them to pay an energy only when their four answers multiply to the wrong sign. The crystal, left to itself, will not write that rule. It knows how to couple two neighbors. It does not know how to couple four at once.

So we add a fifth spin and make it expensive to flip. Each of the four can knock that helper, weakly, two at a time. A process that knocks all four and then puts the helper back where it started leaves a four-body term on the cheap subspace.

That term is the whole trick. It is also a bill. The helper is extra hardware. The four-body energy is tiny compared with the two-body knocks that produced it. And the first leftovers are not four-body at all — they are ordinary pair terms, larger, and they have to be cancelled by hand.

23.1. Four neighbors and one expensive helper

Picture the square with the helper in the middle:

```
      target 1
        |
target 4 -- ancilla -- target 2
        |
      target 3

strong cost to excite ancilla:  $\Delta$ 
weak two-body links:           $g_i$ 
```

The helper has a large excitation energy Δ . One weak knock can send it into the expensive subspace. A process that stays cheap must eventually bring it home. If all four targets take part before that return, the leftover amplitude can remember the product of four spin operators.

The expensive helper is a **mediator**. It never stays excited in the process we care about. It only carries the visits.

Those visits are virtual: they live in an amplitude, not as a stable population that an energy measurement would find. Each visit contributes a weak coupling in the numerator and an energy denominator. Four visits therefore suggest

$$K \sim \frac{g^4}{\Delta^3},$$

where g , Δ , and the effective coupling K are energies. The units check is $E^4/E^3 = E$, as a Hamiltonian coefficient must be.

Hence a four-body leftover generated by four two-body knocks has scale g^4/Δ^3 , not scale g .

The rule of thumb is useful. It hides combinatorial factors and lower-order debris. We now expose both.

23.2. The first leftover is a pair, not a square

Let the four targets be spin-1/2 particles numbered $i = 1, 2, 3, 4$. The Pauli operator Z_i has eigenvalues $z_i = \pm 1$. Let the mediator have ground state $|0\rangle_a$, excited state $|1\rangle_a$, and excitation cost $\Delta > 0$. Define

$$H_0 = \Delta|1\rangle_a\langle 1|, \quad V = X_a S, \quad S = \sum_{i=1}^4 g_i Z_i.$$

Here X_a flips the mediator, and g_i is the energy of a two-body $X_a Z_i$ coupling. Every term in V is two-local. The projector onto the low-energy mediator state is

$$P = |0\rangle_a\langle 0| \otimes I_{1\dots 4}.$$

This toy is unusually friendly because all Z_i commute. For fixed target eigenvalues, the mediator Hamiltonian is the matrix

$$\begin{pmatrix} 0 & S \\ S & \Delta \end{pmatrix}.$$

Its lower eigenvalue, now read as an operator on the targets, is

$$E_-(S) = \frac{\Delta - \sqrt{\Delta^2 + 4S^2}}{2} = -\frac{S^2}{\Delta} + \frac{S^4}{\Delta^3} - \frac{2S^6}{\Delta^5} + O\left(\frac{S^8}{\Delta^7}\right).$$

The first nonconstant term appears at second order, not fourth. Expanding it gives

$$-\frac{S^2}{\Delta} = -\frac{\sum_i g_i^2}{\Delta} I - \sum_{i < j} \frac{2g_i g_j}{\Delta} Z_i Z_j.$$

The identity merely shifts every energy. The six pair couplings are unwanted and parametrically larger than the term we want. At fourth order, however,

$$\frac{S^4}{\Delta^3} \supset \frac{24g_1g_2g_3g_4}{\Delta^3} Z_1Z_2Z_3Z_4.$$

The factor $24 = 4!$ counts the orders in which the four links can act. Choosing the sign of one g_i sets the sign of the four-body term. Direct two-body counterterms can cancel the pair terms through fourth order. For completeness, the fourth-order coefficient multiplying a particular $Z_i Z_j$ is

$$\frac{4g_i g_j (g_i^2 + g_j^2) + 12g_i g_j \sum_{k \neq i, j} g_k^2}{\Delta^3}.$$

A calibrated counterterm must negate this coefficient together with the second-order coefficient. After that cancellation,

$$H_{\text{eff}} = K Z_1 Z_2 Z_3 Z_4 + O(g^6/\Delta^5), \quad K = \frac{24g_1g_2g_3g_4}{\Delta^3}.$$

An adapter of this kind is a **gadget**. The name is worth keeping because it is an engineering part, not a new law of nature: two-body couplings in, a hoped-for many-body term out, and a loss of scale in between.

Hence the four-body term is real, but it is not the first leftover. The first leftover is a pair, larger by two powers of g/Δ , and it must be cancelled before the four-body term is the leading cheap operator.

This construction is explicit. One mediator. Four mediator–target links. Six nominal target–target counterterm links. It is a derivation, not a promise that a particular crystal supplies those ten links.

The mediator and the four targets are physical or cluster-encoded spins. The operator $Z_1 Z_2 Z_3 Z_4$ is an effective interaction. Neither fact alone creates an emergent anyon. Only an extended, gapped lattice Hamiltonian in the correct phase has toric-code quasiparticles. Digitally applying a four-qubit gate would simulate the same operator in time. It would not provide a passive static Hamiltonian.

23.3. How small the four-body piece is

For equal magnitudes $|g_i| = g$, define the dimensionless expansion parameter $\eta = g/\Delta$. Then

$$\frac{|K|}{\Delta} = 24\eta^4.$$

The sixth-order contribution to the same four-body operator is $-960g^6/\Delta^5$, so its magnitude relative to the leading K is $40\eta^2$. At $\eta = 0.05$, the leading four-body scale is only $1.5 \times 10^{-4}\Delta$, the sixth-order correction is 10% of it, and the leading unwanted pair scale $2g^2/\Delta = 5 \times 10^{-3}\Delta$ is about 33 times larger. Holding the residual pair term below $0.1K$ consequently requires roughly 0.3% cancellation in this illustrative cell. At $\eta = 0.025$, the sixth-order fraction falls to 2.5%, but K/Δ falls to 9.375×10^{-6} .

Hence perturbation theory sends an invoice either way: make η smaller and the series improves while K collapses. **[Theory]** These numbers follow directly from the expansion above; they are not measured device performance.

As a units example, suppose — not claim — that $\Delta/h = 1$ GHz, where h is Planck's constant. At $\eta = 0.05$, $K/h = 150$ kHz, and the dynamical time $\hbar/K = 1/[2\pi(K/h)] \approx 1.1 \mu\text{s}$. The arithmetic shows why a large microscopic gigahertz number need not imply a large topological gap.

To make a toric-code plaquette, use Z_i on its four edges. To make a star, replace each target Z_i by X_i . On a periodic square lattice with E target edges, there are $V + P = E$ stabilizers, so this naive design uses one mediator per stabilizer: E mediators for E targets, four mediator-link incidences and six counterterm incidences per stabilizer. Each target must participate in four gadgets. This count excludes routing devices and whatever encoding turns a defect cluster into one target spin.

23.4. From one cell to a lattice

Write a gadget Hamiltonian as

$$H = H_0 + V,$$

where H_0 has a low-energy subspace P , its complement is $Q = I - P$, and the smallest P -to- Q excitation energy is Δ . A sequence contributing at order n has the schematic form

$$PVQ \frac{1}{E_0 - QH_0Q} QVQ \cdots QVP,$$

with n copies of V and $n - 1$ resolvents. If every weak matrix element has scale g , its scale is g^n/Δ^{n-1} . Jordan and Farhi give systematic gadgets at arbitrary perturbative order, while Bravyi, DiVincenzo, Loss, and Terhal show how bounded-strength local target Hamiltonians can be simulated with an extensive error $O(\epsilon N)$ without making simulator couplings grow with the number N of target spins [R174]; [R175]. **[Theory]** Those are Hamiltonian-simulation results. They do not remove device connectivity or coherence requirements.

Three separate gaps must not be conflated.

1. **Penalty gap** Δ : the cost of leaving a gadget's encoded subspace.
2. **Effective stabilizer coefficient** K : typically suppressed as g^n/Δ^{n-1} .
3. **Topological many-body gap** Δ_{top} : the lowest allowed excitation energy of the extended effective model.

For the ideal toric-code Hamiltonian

$$H_{\text{TC}} = -K_A \sum_v A_v - K_B \sum_p B_p,$$

an individual violated stabilizer costs $2K_A$ or $2K_B$. On a closed surface, local operations create violations in pairs, so the smallest such pair costs $4 \min(K_A, K_B)$. Thus the simulator's useful low-energy gap is of order K , not Δ . A basic hardware budget is

$$\max(k_B T, \hbar/T_2, |\delta J|, |\delta K|) \ll \Delta_{\text{top}} \sim K \ll g \ll \Delta.$$

Here $k_B T$ is thermal energy, T_2 is the relevant coherence time, δJ is a residual unwanted two-body energy, and δK denotes stabilizer disorder. Every entry has energy units.

Hence the useful protection is set by the suppressed coefficient K , not by the mediator penalty that produced it.

The inequality is a design criterion, not a universal phase boundary. One gadget cell is not an extensive proof. Shared targets permit mixed virtual paths involving neighboring mediators. The number of paths grows with coordination, and operator-norm bounds based on the whole V can grow with system size. Parallel-gadget constructions, locality-based Schrieffer–Wolff analyses, or exact subsystem symmetries are needed to keep local errors controlled [R175]; [R177]. Finally, proving that an approximate effective Hamiltonian has the right ground space is weaker than proving a stable thermodynamic gap and the correct excited anyons.

23.5. What the papers actually built

Target	Microscopic construction	First target order / resource cost	What is established	Important limit
Toric-code phase from the honeycomb model	Nearest-neighbor XX , YY , or ZZ bonds; strong J_z dimers encode effective spins	Fourth order; two physical spins per effective dimer	In the anisotropic limit, $K = J_x^2 J_y^2 / (16 J_z ^3)$ multiplies an effective four-dimer plaquette operator [R017]; [R176]. [Theory]	Natural and exactly solvable, but its non-Abelian phase is Ising type, not Fibonacci.
Toric code by code gadgets	Each target edge qubit is encoded into four physical qubits; inter-gadget couplings are two-body	Fourth order on a square lattice; four physical qubits per target qubit	Encoded star and plaquette terms arise; odd orders vanish and second order is a constant because the gadget detects single-qubit errors [R177]. [Theory]	The low-energy model is approximate, frustrated, and has a suppressed gap.
Finite-group quantum doubles $D(G)$	Two-body code gadgets using $ G $ -level qudits	Four physical qudits per target edge in the square-lattice construction [R177]	Includes non-Abelian anyonic models when G is non-Abelian. [Theory]	A non-Abelian quantum double is not automatically Fibonacci; local dimension and operator control grow with group structure.
Simplified quantum doubles	A hopping/clock gadget generates an ordered operator product	Order follows the clock path; König reduces quantum-double interactions to simpler local terms [R178]. [Theory]	Ordered noncommuting products demand more structure than the commuting parity toy above.	

Target	Microscopic construction	First target order / resource cost	What is established	Important limit
String-net PEPS	Strong two-body terms define an encoded PEPS subspace; weak bonds generate a parent Hamiltonian	Finite order set by the local interaction region; additional virtual degrees of freedom per PEPS tensor	A two-body parent construction is proved for a broad PEPS class, with an explicit double-semion string-net example [R018]; [R179]. [Theory]	Double semion is Abelian. The paper is not an experimental doubled-Fibonacci defect design.
Exact two-body alternative	Enlarged “composite particles” and two-body interactions	Extra local Hilbert space rather than a small perturbation parameter	Ocko and Yoshida construct nonperturbative two-body models for toric-code and quantum-double topological codes [R180]. [Theory]	Avoids perturbative gap suppression but retains substantial local-state and coupling complexity.

The honeycomb result is a particularly clean reality check. In the strong- J_z limit, each z -bond has a two-dimensional low-energy doublet. A closed sequence of two J_x and two J_y bond operations returns all dimers to that subspace. Lower orders only shift energies; fourth order produces the plaquette term with coefficient $J_x^2 J_y^2 / (16|J_z|^3)$ [R017]; [R176]. This is the same numerator-and-denominator logic as the toy mediator, but lattice symmetry and exact solvability organize the unwanted terms for us.

The distinction between “non-Abelian” and “Fibonacci” is load-bearing. König’s and Brell *et al.*’s group-based constructions reach quantum doubles of finite groups, some with non-Abelian excitations [R177]; [R178]. Levin–Wen models are broader categorical string nets [R018]. The PEPS gadget framework explicitly treats double semion and argues for a broad class of suitable PEPS [R179], but that is not the same as a worked, defect-mapped two-body Hamiltonian for doubled Fibonacci. **[Proposal]** Extending such machinery to a particular doubled-Fibonacci tensor and then to a fabricable defect interaction graph remains an architecture problem, not a solved consequence of citing “string nets.”

23.6. What a defect has to answer

A defect device does not begin with abstract Pauli symbols. It begins with electronic and nuclear levels, selection rules, dipolar tensors, exchange paths, strain, optical excited states, and disorder. Mapping a gadget requires answers to six concrete questions.

Where does Δ come from? It may be a crystal-field splitting, a driven dressed-state gap, or an intra-cluster penalty. Whichever it is, the same control fields and strain that set Δ must leave a well-isolated low-energy doublet. Leakage levels turn the simple two-level denominator into several denominators, possibly with opposite signs.

Can each required operator be implemented? The parity cell needs $Z_i X_a$; a complete toric code also needs the basis-rotated star analogue. A fixed magnetic dipole interaction generally supplies a geometry-dependent tensor, not independently switchable clean Pauli products. Echo or dressing can average terms, but then the Hamiltonian becomes Floquet-engineered rather than simply static.

Can one mediator reach four targets without reaching unwanted neighbors? Dipolar couplings decay rather than stop. Exchange is shorter-ranged but highly sensitive to atomic placement. The toy's six counterterms include diagonal target pairs that may be harder to realize than the four radial links. Published code-gadget constructions solve locality in an abstract lattice, not defect implantation and routing [R177]; [R179].

Are signs and magnitudes calibratable? At $\eta = 0.05$, our toy needed sub-percent cancellation merely to make residual pair terms one tenth of K . Spatial disorder changes both g_i and denominators. A small fractional error relative to g can therefore be a large error relative to K .

Can the system be initialized and cooled into P ? The penalty suppresses real mediator excitation only when thermal energy, drive-induced transitions, and linewidths are small compared with Δ . Topological defects are controlled by the much smaller Δ_{top} , so satisfying $k_B T \ll \Delta$ is not enough.

Does coherence outlast the effective dynamics? The relevant time is $\hbar/K$, not $\hbar/g$. Mediator dephasing can also be inherited by the effective term because the virtual admixture is nonzero. A useful calculation must project both the Hamiltonian and the noise operators.

[Experiment] None of the gadget papers reviewed in this chapter reports a crystal-defect array that realizes its full two-body simulator Hamiltonian and diagnoses the resulting topological phase [R175]; [R177]; [R178]; [R179]; [R180]. Their results are theory constructions. Defect hardware may eventually supply ingredients, but combining the required connectivity, anisotropy, scale hierarchy, cooling, and calibration is presently a proposal-level mapping exercise.

23.7. Ways to get this wrong

1. **Quoting the penalty gap as protection.** The mediator may cost Δ , while anyons cost only order K . Thermal and coherent protection must be budgeted against the latter.
2. **Ignoring lower orders.** Our desired term first appeared at fourth order, but larger pair terms appeared at second order. “First desired order” does not mean “first nonzero order.”
3. **Cancelling on paper, not in hardware.** A counterterm is another physical interaction with its own noise, range, and calibration error.
4. **Extrapolating one cell to a lattice.** Shared-spin virtual paths can create cross-gadget operators. Extensivity and a thermodynamic gap need separate analysis.
5. **Calling an encoded target spin topological.** The four-qudit code gadget of Brell *et al.* is a local error-detecting encoding; topology belongs to the extended low-energy phase [R177].
6. **Calling every non-Abelian model Fibonacci.** Non-Abelian quantum doubles and the honeycomb Ising phase have different fusion and braid data from Fibonacci anyons.
7. **Confusing an effective ground Hamiltonian with its excitations.** Correct ground-space order does not automatically guarantee the desired quasiparticle dispersion, localization, or braiding under higher-order corrections.
8. **Taking g/Δ vanishingly small.** This improves the series while collapsing K , slowing operations and tightening temperature and coherence requirements.

23.8. Check yourself

1. **Show that an order- n virtual process usually scales as g^n/Δ^{n-1} .** It contains n weak matrix elements and $n - 1$ intermediate energy denominators.
2. **What did the one-mediator example cost?** One ancilla, four mediator–target links, six nominal pair counterterms, and fourth-order gap suppression for one four-spin term.
3. **What fails if Δ is quoted as the toric-code gap?** Δ penalizes leaving the gadget subspace; stabilizer violations live inside the low-energy theory and cost order $K \ll \Delta$. Thermal and coherent protection must be budgeted against K .

4. **Show that reducing g/Δ improves the relative sixth-order error while collapsing K faster.** For equal $|g_i|$, the relative sixth-order piece is $40\eta^2$ while $|K|/\Delta = 24\eta^4$. Halving η quarters the relative error and multiplies K by $1/16$.
5. **Do existing non-Abelian quantum-double gadgets establish doubled Fibonacci defect hardware?** No. They establish Hamiltonian reductions for a different family; the category, local Hilbert space, physical couplings, and phase diagnostics still have to be mapped.
6. **What must be shown after deriving one plaquette term?** Controlled parallel composition, small local errors compared with K , a stable thermodynamic gap, and the correct extended ground and excitation structure.

A two-body gadget can generate a four-body term. Its coefficient is generally high order in g/Δ . Mediators, counterterms, and shared-cell paths are real resources. The useful topological gap is order K , not the penalty Δ . Toric-code and finite-group quantum-double reductions are established theoretically. A defect-specific doubled-Fibonacci construction is not.

23.9. Sources

- [R174] S. P. Jordan and E. Farhi, “Perturbative gadgets at arbitrary orders,” *Physical Review A* **77**, 062329 (2008). DOI: 10.1103/PhysRevA.77.062329; arXiv:0802.1874.
- [R175] S. Bravyi, D. P. DiVincenzo, D. Loss, and B. M. Terhal, “Quantum simulation of many-body Hamiltonians using perturbation theory with bounded-strength interactions,” *Physical Review Letters* **101**, 070503 (2008). DOI: 10.1103/PhysRevLett.101.070503; arXiv:0803.2686.
- [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv:cond-mat/0506438.
- [R176] J. Vidal, K. P. Schmidt, and S. Dusuel, “Perturbative approach to an exactly solved problem: Kitaev honeycomb model,” *Physical Review B* **78**, 245121 (2008). DOI: 10.1103/PhysRevB.78.245121; arXiv:0809.1553.
- [R177] C. G. Brell, S. T. Flammia, S. D. Bartlett, and A. C. Doherty, “Toric codes and quantum doubles from two-body Hamiltonians,” *New Journal of Physics* **13**, 053039 (2011). DOI: 10.1088/1367-2630/13/5/053039; arXiv:1011.1942.
- [R178] R. König, “Simplifying quantum double Hamiltonians using perturbative gadgets,” *Quantum Information and Computation* **10**, 292–324 (2010). DOI: 10.26421/QIC10.3-4-9; arXiv:0901.1333.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv:cond-mat/0404617.

- [R179] C. G. Brell, S. D. Bartlett, and A. C. Doherty, “Perturbative 2-body parent Hamiltonians for projected entangled pair states,” *New Journal of Physics* **16**, 123056 (2014). DOI: 10.1088/1367-2630/16/12/123056; arXiv:1407.4829.
- [R180] S. A. Ocko and B. Yoshida, “Nonperturbative gadget for topological quantum codes,” *Physical Review Letters* **107**, 250502 (2011). DOI: 10.1103/PhysRevLett.107.250502; arXiv:1107.2697.

Chapter 24. Eight arrows that are not a machine

Here is the nicest picture in the book.

```
host crystal
-> addressable defect spins
-> strongly coupled cluster
-> isolated two-state pseudospin
-> edge label {1, tau}
-> vertex and plaquette operators
-> doubled-Fibonacci phase
-> emergent anyons and a logical qubit
```

Each arrow looks like a finished step. It is not. The words are hypotheses. We will take them one at a time, name the object sitting on each side, and write down a measurement that would make the arrow fail.

The first honest calculation is three spins and one coupling. Everything after that is a bill.

A defect electron or nucleus is a physical spin. A cluster doublet is an encoded leftover: two slow states built out of several real spins. Calling those two states 1 and τ is a dictionary, not an anyon. A lattice whose Hamiltonian really is the Levin–Wen one can have emergent quasiparticles. A processor that prepares the same amplitudes is a digital copy. Only the many-body phase supplies intrinsic topological order. The diagram above mixes all four of those objects as if they were the same noun.

24.1. Three defects, not twenty

Someone will ask for five to twenty defects. That interval is not a result. The evidence below does not show a designed, mutually coupled array of five to twenty optically addressable centers that forms a protected low-energy edge. So we start with three effective spin-1/2 objects: the smallest open antiferromagnetic chain with a spinful ground doublet. That is modesty demanded by the algebra, not a secret optimum.

Let $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$ be the dimensionless spin-1/2 operator at site i , with $\mathbf{S}_i^2 = 3/4$. Energies live in the coupling J , measured here in joules. A spectroscopist may quote $J/\hbar$ in hertz, where $\hbar$ is Planck's constant. Consider

$$H_C = J(\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3), \quad J > 0.$$

This is an ideal isotropic exchange model. **[Theory]** It is a worked target, not a claim that three nearby NV, SiV, or SiC centers naturally realize equal antiferromagnetic Heisenberg exchange. Exchange-only encodings based on three spins are established theoretically, and controlled three-electron versions have been pursued in quantum dots [R114]. That does not hand the same Hamiltonian to color centers.

Define the end-spin sum $\mathbf{S}_{13} = \mathbf{S}_1 + \mathbf{S}_3$ and the total spin $\mathbf{S} = \mathbf{S}_{13} + \mathbf{S}_2$. Then

$$H_C = \frac{J}{2} (\mathbf{S}^2 - \mathbf{S}_{13}^2 - \mathbf{S}_2^2).$$

The spectrum follows without diagonalizing an 8×8 matrix:

End spin S_{13}	Total spin S	Energy	Degeneracy
1	1/2	$-J$	2
0	1/2	0	2
1	3/2	$J/2$	4

The ground manifold is a doublet. The smallest climb out of it is

$$\Delta_C = 0 - (-J) = J.$$

A convenient basis, in site order 1, 2, 3, is

$$|\tilde{\uparrow}\rangle = \sqrt{\frac{2}{3}} |\uparrow\downarrow\uparrow\rangle - \sqrt{\frac{1}{6}} (|\uparrow\uparrow\downarrow\rangle + |\downarrow\uparrow\uparrow\rangle),$$

with $|\tilde{\downarrow}\rangle$ obtained by flipping every spin. Let

$$P_C = |\tilde{\uparrow}\rangle\langle\tilde{\uparrow}| + |\tilde{\downarrow}\rangle\langle\tilde{\downarrow}|$$

keep only this pair, and let $\tilde{\mathbf{S}} = P_C \mathbf{S} P_C$ be the leftover spin-1/2 operator. That leftover pair is a **pseudospin**. The name is worth keeping because it is two slow states made of three real spins, not a fourth particle that appeared in the crystal.

Angular-momentum addition gives

$$P_C \mathbf{S}_1 P_C = P_C \mathbf{S}_3 P_C = \frac{2}{3} \tilde{\mathbf{S}}, \quad P_C \mathbf{S}_2 P_C = -\frac{1}{3} \tilde{\mathbf{S}}.$$

Hence the three-spin chain, if it really is this Hamiltonian, has a unique ground doublet whose local magnetic answers stand in the ratio $2 : -1 : 2$.

The signs are an experiment. If you do not see the predicted spectrum and those matrix elements, this cluster model is already dead. There is no reason to draw a lattice around it.

24.2. One link between two leftover pairs

Take two clusters, A and B , with an endpoint interaction

$$V_{AB} = j \mathbf{S}_{3,A} \cdot \mathbf{S}_{1,B},$$

where j is an energy. With $P = P_A P_B$, first-order projection writes

$$P V_{AB} P = \frac{4j}{9} \tilde{\mathbf{S}}_A \cdot \tilde{\mathbf{S}}_B.$$

Virtual leakage adds corrections of order j^2/J , so the controlled regime is $|j|/J \ll 1$. The units check: j^2/J is still an energy. **[Theory]** This derivation establishes a two-body leftover exchange and nothing more. It does not generate a string-net vertex projector, a plaquette recoupling operator, or Fibonacci statistics.

Hence the first honest intercluster leftover is a Heisenberg coupling of two pseudospins, smaller than the microscopic endpoint coupling by $4/9$, plus a leakage correction of order j^2/J .

Each reduction swallows internal motion and leaves a slower pair. A later gadget swallows mediator motion and leaves a still slower vertex or plaquette term. The output is useful only if every reduction has a clean input and less leftover junk than the thing it is supposed to produce. Here the junk is leakage, disorder, leftover lower-order interactions, and decoherence. Each reduction also shrinks the energy. That is the whole tension: a hierarchy can simplify the space while making the desired physics too faint to see.

Nor is the leftover pair automatically quiet. Uniform magnetic noise couples to $\tilde{\mathbf{S}}$ at first order because

$$P_C(\mathbf{S}_1 + \mathbf{S}_2 + \mathbf{S}_3)P_C = \tilde{\mathbf{S}}.$$

The cluster suppresses leakage when noise lacks energy near J . It does not suppress ordinary logical dephasing. Encoded and decoherence-free are not synonyms.

24.3. Two names on two states

A Fibonacci string-net edge admits two labels: vacuum 1 and the nontrivial label τ , with the fusion rule

$$\tau \times \tau = 1 + \tau.$$

A cluster doublet has the right local dimension, so we may define

$$|1\rangle_e \equiv |\tilde{\uparrow}\rangle_e, \quad |\tau\rangle_e \equiv |\tilde{\downarrow}\rangle_e$$

for edge e . **[Proposal]** This is a dictionary between two bases. The physical spin operator does not acquire Fibonacci fusion data by being given a Greek name.

On a trivalent graph, the allowed unordered vertex triples are

$$(1, 1, 1), \quad (1, \tau, \tau) \text{ and permutations,} \quad (\tau, \tau, \tau).$$

Let Q_v project the three edges that meet at vertex v onto those allowed states. That is a three-edge constraint.

Hence dimension two does not write the vertex rule for you. Drop the constraint and the two names are only a relabeling of a spin.

24.4. A plaquette that is not a product of Paulis

Let $\varphi = (1 + \sqrt{5})/2$ be the golden ratio, the quantum dimension of τ . The total squared quantum dimension is

$$\mathcal{D}^2 = 1 + \varphi^2.$$

For plaquette p , let B_p^s mean “insert a loop carrying label s and recouple it into the surrounding edges using the Fibonacci F -symbols.” The doubled-Fibonacci plaquette projector is [R018]

$$B_p = \frac{B_p^1 + \varphi B_p^\tau}{1 + \varphi^2}.$$

The ideal Levin–Wen Hamiltonian is

$$H_{\text{LW}} = -K_v \sum_v Q_v - K_p \sum_p B_p, \quad K_v, K_p > 0,$$

where K_v and K_p are energies. **[Theory]** The projectors commute in the ideal construction and realize doubled, not single chiral, Fibonacci order [R018]. The plaquette operator changes superpositions of edge labels with precise irrational relative amplitudes. An arbitrary six-spin interaction is not “close enough” merely because it touches a hexagon.

Generic perturbative constructions can reduce locality for broad classes of encoded PEPS parent Hamiltonians, and an explicit double-semion string-net example exists [R179]. **[Theory]** That result is an existence argument in an abstract two-body model, not a defect-specific doubled-Fibonacci blueprint. For a gadget whose desired term first appears at order n , dimensional analysis gives a scale of the form

$$K_{\text{eff}} \sim c j \left(\frac{j}{\Delta_m} \right)^{n-1},$$

where Δ_m is a mediator excitation energy and c is a dimensionless, construction-dependent coefficient. Every extra perturbative order reduces the output when $|j|/\Delta_m \ll 1$. Lower-order unwanted terms must be canceled or shown harmless.

Hence producing one desired term in perturbation theory is not producing H_{LW} . Nature is allowed to notice the orders you erased from the blackboard.

24.5. Two inequalities that have to hold

Define δ_C as the energy norm of intra-cluster disorder, Γ as a logical decoherence rate in s^{-1} , σ_{eff} as the energy scale of spatial variation in effective couplings, and H_{err} as all unwanted effective terms. Here k_B is Boltzmann’s constant, T is temperature in kelvin, $\hbar = h/(2\pi)$, and $\|\cdot\|$ is the operator norm. Let Δ_{topo} be the actual many-body gap above the intended ground sector, not simply the coefficient printed before one projector. A minimally viable passive hierarchy is

$$J \gg \max(|j|, \delta_C, k_B T),$$

and

$$\Delta_{\text{topo}} \gg \max(k_B T, \hbar \Gamma, \sigma_{\text{eff}}, \|H_{\text{err}}\|).$$

Both sides of each inequality have energy units. These are necessary, not sufficient. You still need locality, a system larger than the correlation length, a way to initialize, and a way

to identify topological sectors. Stability theorems protect suitable topologically ordered commuting-projector models against sufficiently weak local perturbations [R142]. They do not say that a badly approximated microscopic Hamiltonian lies in that phase.

Hence a large cluster coupling J does not, by itself, protect a logical qubit. The many-body gap after every projection and gadget suppression is the scale that has to beat temperature, decoherence, disorder, and leftover junk.

24.6. Arrow by arrow

Link	Status	What is actually supported	Falsifiable next test
Crystal → addressable defect spin	[Experiment], host-dependent	Individual solid-state defect spins can be initialized, controlled, and read out; this is mature for selected centers, not for arbitrary defects.	Demonstrate the chosen charge state, coherence, and readout under the density and temperature required by the array.
Separate defects → coherent cluster	[Experiment] for small diamond systems; [Speculation] at 5–20 designed centers	Coupled diamond color centers separated by $98 \pm 3 \text{ \AA}$ were coherently controlled [R181], and two single defect spins were entangled at room temperature [R080]. A sensed three-electron-spin cluster has shown coherent dynamics [R182]. None is the proposed large regular cluster.	Fabricate repeated three-center units and reproduce one coupling graph and spectrum across the chip.
Three spins → isolated pseudospin	[Theory] for the model above; [Proposal] for defects	The exact projection works if nearly isotropic antiferromagnetic J is realized. Exchange-only logic has a broader theoretical basis [R114].	Resolve the $-J, 0, J/2$ multiplets and the projected $2 : -1 : 2$ local response.

Link	Status	What is actually supported	Falsifiable next test
Pseudospin $\rightarrow$ edge label $\{1, \tau\}$	[Proposal]	A two-state Hilbert space has the correct dimension. No fusion constraint follows from dimension alone.	Tomographically implement one Q_v and verify all allowed and forbidden triples.
Two-body defect coupling $\rightarrow Q_v, B_p$	[Theory] in generic gadget frameworks; [Speculation] in defect hardware	Locality reduction is possible in abstract models [R179]. No cited defect experiment implements the Fibonacci F -symbol amplitudes as a static effective plaquette operator.	Measure the full effective operator, including phases and all lower-order residual terms, on one vertex and one plaquette.
Approximate projectors $\rightarrow$ doubled-Fibonacci phase	[Theory] for the ideal model; [Numerics] for selected perturbations [R183]; [Speculation] for this architecture	The ideal phase and some phase boundaries are known. The basin of attraction cannot be inferred from term names alone.	On increasing patches, show a stable gap, correlation-length saturation, and the expected ground-sector/topological data under measured disorder.
Phase $\rightarrow$ emergent Fibonacci-type excitations	[Theory] conditional on the phase	Excitations of the doubled theory carry the corresponding non-Abelian data. A finite circuit demonstrating fusion does not prove a material phase.	Create separated excitations, perform path-deformation tests, and recover fusion/braid matrices insensitive to local path details.
Emergent excitations $\rightarrow$ logical qubit	[Proposal] for defect hardware	Topological encodings and braiding are known theoretically. Initialization, motion, fusion measurement, and error budgets are unspecified here.	Demonstrate a logical operation whose error decreases with separation or code distance under fixed local noise.

This table is the architecture. The arrow diagram was only its table of contents.

24.7. What the laboratory actually has

[Experiment] Small interacting diamond systems exist. Neumann and colleagues coherently controlled a pair of color centers at a measured separation of $98 \pm 3 \text{ \AA}$ [R181]. Dolde and colleagues generated entanglement between individual defect electron spins [R080]. Rosenfeld and colleagues observed coherent dynamics involving an NV center and two dark electronic spins [R182]. These results establish that “defects can interact coherently” is not fantasy. They do not establish equal antiferromagnetic Heisenberg bonds, repeatable cluster spectra, or a two-dimensional array.

[Experiment] Cluster-sized registers also exist in a different sense. Bradley and colleagues controlled a ten-qubit diamond register consisting of one NV electron spin and nine nuclear spins, at 3.7 K, and entangled as many as seven spins [R119]. That is impressive evidence for multispin control. It is not ten engineered defect centers, not a passive exchange cluster, and not an analog string-net Hamiltonian. The electron-mediated, pulse-controlled register belongs in the digital/control column.

[Experiment] Programmable processors have now dynamically prepared Fibonacci string-net states and exercised fusion and braiding protocols [R165]. This demonstrates control of the encoded mathematics, not spontaneous emergence from a static defect Hamiltonian. It is valuable as a measurement recipe and a source of target data. It does not validate the material hierarchy.

For sapphire, corundum, SiC, or diamond, the selection must be made from measured interaction tensors, fabrication distributions, charge stability, and readout — not from the beauty of the host crystal. **[Speculation]** No source cited here supports treating a five-to-twenty-defect cluster in any of these hosts as an available component. The count is a resource hypothesis to be optimized after a microscopic Hamiltonian exists.

24.8. Eight tests that can still kill it

The proposal survives only if all of the following can be tested and retained.

1. **A1 — Repeatable local object [Proposal].** The selected defect and charge state behave as the same effective spin across a dense array. Spectroscopy must bound site-to-site Hamiltonian variation.
2. **A2 — Cluster window [Proposal].** A reproducible cluster has one doublet separated by Δ_C , with Δ_C exceeding $k_B T$, disorder, the energy scale of leakage-driving controls, and intercluster coupling by stated margins.
3. **A3 — Useful projection [Theory $\rightarrow$ experiment].** Measured microscopic operators project to the predicted leftover tensors. For the worked model, endpoint exchange must approach $4j/9$ and corrections must scale as j^2/J .

4. **A4 — No cluster-count assumption [Speculation].** Three, five, or twenty spins are acceptable only if they improve a measured objective such as gap, addressability, noise susceptibility, or gadget order. More spins are otherwise more fabrication opportunities for the universe to express an opinion.
5. **A5 — Operator completeness [Proposal].** Available interactions plus mediators generate both Q_v and B_p with the Fibonacci matrix elements. Hamiltonian tomography must include unwanted terms; fitting only the desired coefficients fails this assumption.
6. **A6 — Surviving energy scale [Proposal].** The measured Δ_{topo} , obtained after every projection and gadget suppression, exceeds $k_B T$, $\hbar \Gamma$, coupling disorder, and residual interactions.
7. **A7 — Phase evidence [Numerics $\rightarrow$ experiment].** Finite-size calculations using measured parameters show convergence toward doubled-Fibonacci topological data, and success persists across the measured disorder distribution. One carefully tuned small patch is insufficient.
8. **A8 — Operational topology [Proposal].** Excitations can be created, moved, and fused without closing the gap, and a nonlocal observable is less sensitive to local perturbations as system size grows.

A sensible experimental ladder is therefore: one cluster spectrum; two-cluster projection; one three-edge vertex; one plaquette operator; then several connected plaquettes. Stop at the first failed inequality. A tiny patch can falsify a microscopic mapping. It cannot by itself prove a thermodynamic phase.

End-to-end verdict — [Assessment]. The hierarchy is **highly speculative but physically coherent**. No link violates a known principle, and the first projection is explicit. Experimental support presently reaches only small coherent spin systems and digitally controlled registers. The defect-specific cluster Hamiltonian, Fibonacci vertex and plaquette operators, surviving topological gap, and protected logical operation are all unestablished. Neither a five-to-twenty-defect component nor defect-based Fibonacci feasibility may be used as an input assumption.

24.9. Ways to get the chain wrong

The local-dimension fallacy. Two levels can encode 1 and τ , a coin toss, or whether lunch happened. Fusion rules live in constraints and recoupling amplitudes, not in the number two.

The five-to-twenty fallacy. No scaling law in this chapter selects that interval. Increasing cluster size may create a cleaner code space, but it also increases spectral crowding, placement burden, control cross-talk, and the number of disorder parameters. Until a calculation and experiment identify an optimum, the interval has status **[Speculation]**.

The gadget-equals-phase fallacy. Producing one desired term in perturbation theory is not producing the target Hamiltonian. Lower-order terms can dominate; higher-order terms can be smaller than linewidths; the perturbative parameter can be too large for the expansion and too small for observation. Annoyingly, nature is permitted to notice all orders.

The digital-equals-emergent fallacy. Pulse sequences can prepare a string-net wavefunction and compile braids [R165]. That is [**Experiment: digital emulation**]. Passive order requires the material Hamiltonian, its gap, and its perturbative stability to do the work between pulses.

The gap-label fallacy. J is a cluster leakage gap, not a topological gap. K_p is a Hamiltonian coefficient, not automatically the many-body gap. Only the spectrum of the full disordered effective model determines Δ_{topo} .

The protection fallacy. The worked cluster is spinful, so magnetic noise acts within its logical doublet. Even a genuine two-dimensional topological phase at nonzero temperature contains thermally activated anyons. Topological order does not repeal thermodynamics.

The chirality fallacy. The Levin–Wen target here is doubled Fibonacci. Calling it simply “Fibonacci” hides the time-reversed sector and can lead to incorrect edge and control claims.

24.10. Check yourself

1. **Why use three defects rather than assume five to twenty?** Three is the smallest exactly tractable cluster with a spinful ground doublet, while the larger count has no demonstrated defect-cluster basis in the cited evidence.
2. **Show that endpoint exchange between two such clusters projects to $(4j/9)\tilde{\mathbf{S}}_A \cdot \tilde{\mathbf{S}}_B$.** Each endpoint spin projects as $P_C \mathbf{S}_{\text{end}} P_C = (2/3)\tilde{\mathbf{S}}$, so the product of the two left-over factors is $4/9$. Corrections of order j^2/J sit outside this first-order statement.
3. **What fails if you map $|\tilde{\uparrow}\rangle$ to $|1\rangle$ and stop?** You have named a local basis. Vertex constraints, plaquette recoupling, a gapped phase, and emergent excitations are still required. Dimension two does not write fusion.
4. **What is the decisive energy scale?** The full model’s measured or reliably calculated Δ_{topo} , after projection and gadget suppression — not the bare cluster coupling J .
5. **What fails if the first cluster does not show the predicted doublet and the $2 : -1 : 2$ local response?** The cluster model is already dead. There is no reason to fabricate a lattice around it.

24.11. Sources

- [R114] D. P. DiVincenzo, D. Bacon, J. Kempe, G. Burkard, and K. B. Whaley, “Universal quantum computation with the exchange interaction,” *Nature* **408**, 339–342 (2000). DOI: 10.1038/35042541.
- [R181] P. Neumann, R. Kolesov, B. Naydenov, *et al.*, “Quantum register based on coupled electron spins in a room-temperature solid,” *Nature Physics* **6**, 249–253 (2010). DOI: 10.1038/nphys1536.
- [R080] F. Dolde, I. Jakobi, B. Naydenov, *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R182] E. L. Rosenfeld, L. M. Pham, M. D. Lukin, and R. L. Walsworth, “Sensing coherent dynamics of electronic spin clusters in solids,” *Physical Review Letters* **120**, 243604 (2018). DOI: 10.1103/PhysRevLett.120.243604.
- [R119] C. E. Bradley, J. Randall, M. H. Abobeih, *et al.*, “A ten-qubit solid-state spin register with quantum memory up to one minute,” *Physical Review X* **9**, 031045 (2019). DOI: 10.1103/PhysRevX.9.031045.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv:cond-mat/0404617.
- [R179] C. G. Brell, S. D. Bartlett, and A. C. Doherty, “Perturbative 2-body parent Hamiltonians for projected entangled pair states,” *New Journal of Physics* **16**, 123056 (2014). DOI: 10.1088/1367-2630/16/12/123056; arXiv:1407.4829.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195; arXiv:1001.0344.
- [R183] M. D. Schulz, S. Dusuel, K. P. Schmidt, and J. Vidal, “Topological phase transitions in the golden string-net model,” *Physical Review Letters* **110**, 147203 (2013). DOI: 10.1103/PhysRevLett.110.147203; arXiv:1212.4109.
- [R165] Z. K. Mineev, K. Najafi, S. Majumder, *et al.*, “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* **16**, 6225 (2025). DOI: 10.1038/s41467-025-61493-8; arXiv:2406.12820.

Chapter 25. Three graphs that disagree

Draw a honeycomb and put a label on every edge. A string-net vertex term checks the three labels that meet at a corner. A plaquette term recouples the six labels around a hexagon. That is one drawing.

Now move each label to the midpoint of its edge, and join two midpoints whenever the original edges share a vertex. The new drawing is not a honeycomb. It is a lattice of triangles and hexagons.

Now let every pair of midpoints couple as the laboratory Hamiltonian couples them, including the slow $1/r^3$ tail and any helper parked at a vertex. The energy sees a third drawing still.

A pretty tiling is not a Hamiltonian. The energy written in the crystal must know the same meetings as the string net, or the operators copied from Levin and Wen act on the wrong graph.

25.1. The map the operators were written for

A string-net Hamiltonian asks two adjacency questions: which labels meet at a vertex, and which labels surround a plaquette. That asked-for adjacency is the *target graph*.

Call it $G = (V, E)$. Here V is the set of vertices and E is the set of edges. In the primary drawing, every vertex of G is trivalent: three edges meet there. Those three edges are the three labels that a vertex term will later check.

Hence the operators were written for G , not for a photograph of the chip.

The host crystal does not automatically supply G . It supplies defects at whatever positions fabrication can hold, and couplings that depend on distance, orientation, crystal axis, and leftover damage. Between G and those couplings sit two more drawings, and they are allowed to disagree.

The chain is

topological data $\longrightarrow$ target graph $\longrightarrow$ cluster positions
 $\longrightarrow$ physical couplings $\longrightarrow$ effective operators.

Drop any middle arrow and the remaining ones do not compose. That is how a lattice proposal quietly acquires magic wires.

25.2. Clusters at the midpoints

Chapter 24 put several physical defects into one low-energy cluster, and asked that cluster to carry a string-net edge label. So put one encoded cluster C_e at the midpoint of each target edge e . The cluster stores the label — say 1 or τ for the Fibonacci input theory — not a microscopic defect spin directly.

Those midpoints, with an edge between two clusters whenever the corresponding target edges share a vertex, form the *placement graph*.

The distinction is easiest to see one graph at a time. The first panel below is the logical honeycomb: labels live on its edges. The second moves one encoded cluster to every edge midpoint; joining clusters whose honeycomb edges share a vertex produces kagome triangles and hexagons. The third adds the couplings and helpers that the laboratory Hamiltonian may actually contain.

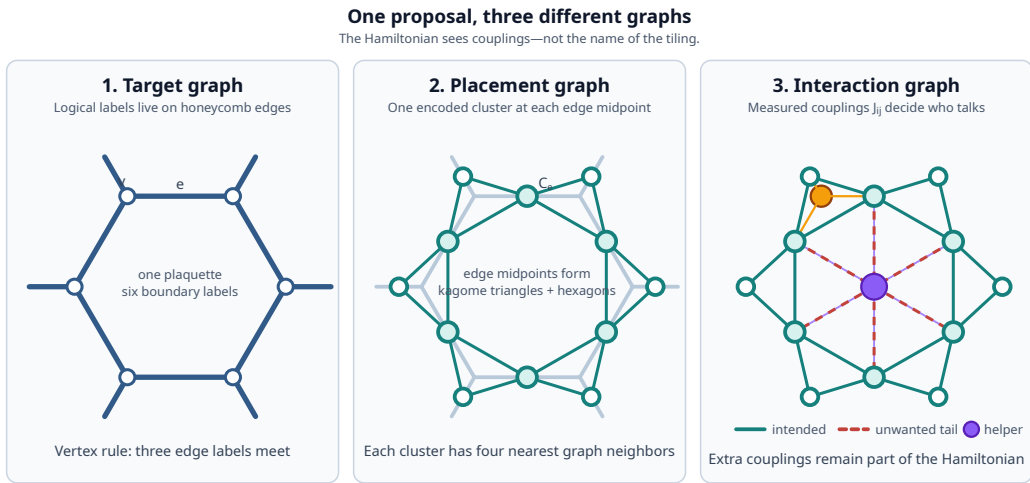


Figure 25.1.: The target, placement, and interaction graphs are different objects. Honeycomb edge labels become kagome-positioned encoded clusters; mediators and unwanted long-range couplings then enlarge the physical interaction graph.

The six central cluster sites alone are incomplete. Each C_i also belongs to a triangle of three edge centers near each end of its original honeycomb edge. Repeat the construction and the clusters occupy a kagome lattice.

Two clusters are neighbors in the placement graph precisely when their target edges share an endpoint. That construction is the *line graph* $L(G)$ of G : the vertices of $L(G)$ are the edges of G . The name is worth keeping because a line graph is not a plot of a line; it is this mechanical swap of vertices for edges.

An edge of a degree-three graph has two endpoints and meets two other edges at each endpoint, so every cluster has

$$z_{L(G)} = 2(3 - 1) = 4$$

nearest constraint-neighbors. Hence “honeycomb target” and “kagome device” can both describe the same architecture without contradiction. They are two of the three drawings, not two competing crystals.

If you drop the shared-vertex rule and join every pair of nearby clusters, you have already left $L(G)$. That larger drawing is the next object, not this one.

25.3. Who actually couples

If each nearby pair of clusters coupled only along placement-graph edges, the placement graph would also be the pair graph of the Hamiltonian. Dipolar tails, leftover exchange, cavities, drives, and perturbative gadgets do not stop at the pencil line. Who actually couples is the *interaction graph*.

The ideal Levin–Wen energy has a vertex term Q_v and a plaquette term B_p ,

$$H_{\text{SN}} = - \sum_{v \in V} J_v Q_v - \sum_p J_p B_p,$$

where J_v and J_p are energies. Q_v checks the allowed fusion of the three labels incident on vertex v . B_p inserts and recouples a loop around plaquette p . On a honeycomb plaquette, B_p acts on six boundary edge labels and also depends on nearby labels through the recoupling data. The exact commuting-projector construction is **[Theory]**, not an interaction already known to occur among color centers [R018].

Hence drawing G does not write H_{SN} . The Hamiltonian can see only the interaction graph.

The defects inside C_e are physical spins. The doublet of C_e is an encoded local degree of freedom. The labels on G are string-net variables. Excitations are emergent anyons only if the many-body Hamiltonian actually enters the topological phase. Programming a circuit, or measuring a few fusion-like outcomes, does not by itself make intrinsic anyons.

25.4. One length, then a factor of 0.19

Give each logical honeycomb edge length a , measured in metres. Two incident edge mid-points are separated by

$$d_1 = \frac{\sqrt{3}}{2}a.$$

The numerical factor is dimensionless, so $[d_1] = [a] = \text{m}$. All four nearest placement-graph bonds of a cluster have this same length in the regular embedding. That equality is the main geometric advantage of the primary choice.

Suppose, only to expose the tail, that pair coupling magnitudes scale as $J(r) = C/r^3$, where C has units of energy times cubic metres. The next kagome distance is $d_2 = \sqrt{3} d_1$. Then

$$\frac{J(d_2)}{J(d_1)} = \left(\frac{d_1}{d_2}\right)^3 = \frac{1}{3\sqrt{3}} \approx 0.19.$$

Hence a fifth of the nearest-neighbor scale already sits on the next kagome shell. Chapter 26 must retain such non-target terms or provide a demonstrated refocusing mechanism.

This is a derived geometric estimate, not a universal defect coupling. Magnetic dipoles also have angular factors. Exchange need not follow a power law. Driven interactions can have another range entirely. Still, 0.19 is too large to erase with a hopeful adjective. Drop the r^{-3} hypothesis and the number changes; do not drop the extra edges.

A vertex helper M_v may sit near each original honeycomb vertex and couple to the three surrounding C_e . A plaquette helper M_p may sit at each hexagon center and couple radially to six boundary clusters. The spokes meet only at their helper, so a planar layout needs no geometric crossing. A six-neighbor helper and an effective six-edge operator are not the same thing. The latter still requires a controlled gadget or projection.

The geometry has five jobs:

1. keep the logical vertex degree compatible with the branching rule;
2. provide usable and sufficiently uniform separations;
3. avoid unwanted interaction edges or make them correctable;
4. leave physical room for vertex and plaquette helpers without wire crossings;
5. tolerate placement error, missing defects, annealing, surfaces, and crystal-axis constraints.

Fail any one of those and the remaining four do not rescue the Hamiltonian.

25.5. Why three edges at a vertex

Let z denote vertex degree: the number of target edges incident on a target vertex. The standard Levin–Wen construction is naturally expressed on a trivalent graph, $z = 3$, where a local branching constraint tests three labels [R018]. Higher-degree vertices can be resolved into trees of trivalent vertices using additional short edges and choices of fusion basis. That resolution adds degrees of freedom and operators. It is not free simplification.

If edge clusters are used on a regular degree- z graph, their line graph has degree

$$z_{\text{edge clusters}} = 2(z - 1).$$

A triangular target ($z = 6$) therefore gives degree 10 among edge clusters. A square or kagome target ($z = 4$) gives degree 6. A trivalent target gives degree 4. Hence lower degree usually means fewer intended pair channels, fewer spectral collisions, and fewer ways for one bad site to be everyone’s problem.

Drop the trivalent hypothesis and you have not merely redrawn the wallpaper. You have changed the local algebra the vertex term was written to test.

25.6. A five-percent wobble

For a distance-dependent coupling, fabrication error becomes coupling disorder. If $J(r) = Cr^{-3}$ and a bond length changes by a small amount δr , differentiation gives

$$\frac{\delta J}{J} \approx -3 \frac{\delta r}{r}.$$

The quantities δr and r must use the same length unit, so the ratio is dimensionless. A 5% radial error therefore means roughly 15% coupling error before angular and defect-to-defect variation are counted. This is geometry, not a forecast for a particular wafer.

Equal drawing lengths also do not guarantee equal spin couplings. For magnetic dipoles the bond direction relative to quantization axes matters. Defects in crystallographically inequivalent orientations may therefore turn a one-length lattice into a several-coupling Hamiltonian. Pulse engineering might average that anisotropy. Until derived for the selected defects, that is **[Proposal]**.

Drop the power-law hypothesis and the factor of three changes. Keep it, and a millimetre-scale drawing error is not a rounding error in J .

25.7. Drawn edges versus real couplings

Write a target adjacency matrix A_{ij} , equal to one when clusters i and j should couple at the leading pairwise level and zero otherwise. Let the realized pair coupling be J_{ij} . A useful geometry makes large $|J_{ij}|$ coincide with $A_{ij} = 1$, while keeping

$$\epsilon_{\text{graph}} = \frac{\max_{A_{ij}=0} |J_{ij}|}{\min_{A_{ij}=1} |J_{ij}|}$$

small. This dimensionless ratio does not prove topological stability. It reveals graph mismatch before a heroic many-body calculation begins.

Long-range dipolar coupling means $A_{ij} = 0$ rarely implies $J_{ij} = 0$. Hence ϵ_{graph} is a diagnostic, not a license to delete the tail. If you count only drawn edges, you have already answered a different question from the one the Hamiltonian asks.

25.8. A helper in the middle

Every named tiling below is planar, so nearest-neighbor target edges can be drawn without crossings. The harder question is whether the operator-incidence graph is still planar after helpers are added. One helper per vertex is easy. One helper per face is geometrically natural but may demand coordination equal to face size: six on honeycomb, three and twelve on decorated honeycomb, and four or eight on square–octagon.

A central helper can share a site, cavity, or mode among several bonds. Pairwise microscopic couplings do not automatically become the required product operator. **[Theory]** Perturbative constructions can generate multi-body terms only with energy denominators, unwanted corrections, and a reduced effective scale. Chapter 23 supplied that warning. A lattice cannot repeal perturbation theory.

Hence a six-leg star of pair couplings is not B_p . Drop the gadget derivation and you are left with six ordinary bonds, not a plaquette recoupling.

25.9. A drawn dot is not a defect

A drawn coordinate is not a guaranteed active defect. The fabrication stack includes pattern registration, aperture or beam size, ion straggle, vacancy diffusion during annealing, conversion yield, depth distribution, and damage-induced noise.

[Experiment] Toyli and colleagues used electron-beam-defined apertures plus nitrogen implantation to fabricate patterned single-spin and spin-array sites in diamond; for 20 keV

^{15}N , their modeling gave about 9 nm lateral straggle, under those implantation conditions [R189]. **[Experiment]** In a different 2017 experiment, focused Si implantation produced SiV centers with approximately 32 nm lateral standard deviation and 48 ± 21 nm accuracy relative to nanocavities; the measured conversion yield reached about 2.5% at 100 keV before an added electron-irradiation step [R190]. Those are not interchangeable “placement precision” numbers. They concern different ions, energies, devices, and reported metrics.

A geometry should therefore permit repeated masks or direct-write coordinates, post-fabrication characterization, and spare sites or repair strategies. None of the cited experiments demonstrates a dense, uniformly coupled lattice of encoded clusters realizing a string-net Hamiltonian.

Drop the distinction between beam width and annealed endpoint spread, and you have already understated the disorder that enters J_{ij} .

25.10. Which graphs are even on the table

The table separates the target graph from the edge-cluster graph. “Mediator burden” means local vertex and face operators, not merely nearest-neighbor drawing lines.

Candidate target	Target degree	Edge-cluster degree	Spacing/graph issue	Crossing or mediator issue	Fabrication verdict
Honeycomb	3	4; positions are kagome	One intended midpoint spacing in regular embedding; dipolar tails remain	No planar crossings; 3-leg vertex and 6-edge face	Primary: simplest trivalent incidence and a single repeated coordinate motif
Triangular	6	10	Dense, strongly overconnected for edge encoding	No geometric crossings, but a six-label vertex must be decomposed	Reject as direct string-net graph; useful only for a different derived model

Candidate target	Target degree	Edge-cluster degree	Spacing/graph issue	Crossing or mediator issue	Fabrication verdict
Kagome	4	6 if its edges carry labels	Triangles and hexagons create two face scales; degree exceeds trivalent target	4-label vertices require resolution; 3- and 6-face mediators	Keep as the <i>placement graph</i> of honeycomb edge clusters, not the primary logical graph
Decorated honeycomb / star (3, 12 ²)	3	4	Intra-triangle and inter-triangle bonds naturally form at least two classes	3-edge and 12-edge faces; large face operator or mediator tree	Interesting for Kitaev-type physics, but too many scales for the first defect prototype
Ruby	4	6	Crowded local triangles and colored bond classes	Higher local degree and mediator congestion	Strong theoretical precedent for two-body color-code physics, not a direct Fibonacci edge-label map
Square–octagon (4.8.8)	3	4	Incident midpoint distances are unequal in regular embedding	3-leg vertices; separate 4- and 8-edge face machinery	Fallback: orthogonal registration is attractive, but two bond/face scales must be calibrated
Square	4	6	One site spacing but wrong degree for direct trivalent branching	4-label vertex resolution and 4-edge faces	Natural for toric-code stabilizers, not the cleanest doubled-Fibonacci map

Candidate target	Target degree	Edge-cluster degree	Spacing/graph issue	Crossing or mediator issue	Fabrication verdict
Brick-wall honeycomb	3	4	Graph-equivalent to honeycomb but geometrically anisotropic	Same incidence as honeycomb	A mask-friendly deformation of the primary, not a distinct topological candidate

The verdict column is a choice of graph, not a report of a material. Each later section earns one of those rows.

25.11. Honeycomb

The honeycomb wins by being boring in the correct places. It is trivalent, bipartite, planar, and has one face size. Levin and Wen’s construction directly uses trivalent string-net branching [R018]. Kitaev’s exactly solved honeycomb model further shows that a simple trivalent interaction graph can support emergent gauge structure and non-Abelian Ising-type excitations for suitable bond-dependent interactions, although that phase is not Fibonacci order [R017]. **[Theory]** This precedent validates the importance of the graph, not the proposed defect Hamiltonian.

Hence the primary target is honeycomb because the local algebra matches, not because hexagons are fashionable.

Its disadvantage is already sitting in the worked example. Edge encoding turns the cluster positions into degree-four kagome. Plaquette terms still span six clusters, and long-range tails add unwanted edges. Borrow Kitaev’s anyons and you have borrowed the wrong phase.

25.12. Six neighbors

A triangular lattice places six nearest neighbors around every vertex. That is useful for frustrated magnets and quantum dimer models; a gapped resonating-valence-bond phase was established theoretically in the triangular-lattice quantum dimer model [R187]. **[Theory]** It does not follow that triangular geometry implements the Fibonacci string net needed here.

As a direct target it replaces each trivalent branching check with a six-way junction. Resolving that junction needs internal fusion-tree edges. With edge clusters, the local pair graph has degree ten. Dense coordination may strengthen connectivity, but it also multiplies unintended dipolar paths and addressability demands.

We reject it for the first architecture. Keep the triangular lattice for a different derived model; do not rename its dimers τ .

25.13. Kagome is not the target

Kagome has four edges meeting at each vertex and contains triangular and hexagonal faces. It is famous for frustration; theory has found fractionalized phases in specific kagome Hamiltonians [R188]. Again, “supports a spin liquid in some model” is not “supports this string net with available couplings.”

Kagome nevertheless has a privileged role here: it arises automatically as the physical mid-point lattice of honeycomb edge variables. In that role, each kagome triangle corresponds to one logical honeycomb vertex and each kagome hexagon corresponds to one logical honeycomb plaquette.

Hence the exact incidence match is more valuable than using kagome as the target graph itself. Take kagome as the target and you have already given up trivalent branching and accepted two face scales.

25.14. Decorated honeycomb

Replace every honeycomb vertex by a triangle and one obtains the decorated honeycomb, or star, lattice with face sequence (3, 12, 12). It remains trivalent. **[Theory]** Yao and Kivelson solved a Kitaev-type model on this lattice and found a chiral spin liquid with non-Abelian anyons in part of its phase diagram [R184]. Those excitations are associated with Kitaev/Majorana physics, not doubled Fibonacci string-net order.

For fabrication, decoration provides internal triangles that could coincide with strongly coupled mini-clusters. It also creates two geometric bond classes — within and between triangles — and faces of size three and twelve. That may be useful if the microscopic Hamiltonian naturally has two scales. It is a liability if uniformity is already the scarce resource. A twelve-boundary plaquette operation also needs a helper tree or several perturbative stages.

We retain it as a research variant, not the primary layout. Drop the two-scale hypothesis and the decoration is a cost with no corresponding microscopic gift.

25.15. Ruby

The ruby lattice is four-valent and uses colored bond types in the best-known topological construction. **[Theory]** Kargarian, Bombin, and Martin-Delgado gave a two-body spin Hamiltonian on the ruby lattice whose low-energy sector realizes topological color-code order [R185]. This is genuine and relevant evidence that microscopic two-body interactions can encode richer low-energy stabilizers.

It is not evidence for a ruby-lattice Fibonacci phase in defect clusters. Ruby's degree-four junctions, small triangles, and several bond classes raise control and placement burdens. Its main value to this architecture is methodological: study whether a two-body parent Hamiltonian can generate the required low-energy constraints.

Do not borrow the color-code result and relabel its anyons τ . The anyons belong to the Hamiltonian that produced them.

25.16. Squares mixed with octagons

The Archimedean square–octagon tiling has face sequence (4, 8, 8) and degree three, so it preserves trivalent branching. Kitaev-type models on it support a rich set of Abelian and non-Abelian phases in theory [R186], though those results again do not establish Fibonacci order or defect feasibility.

Its straight, orthogonal square motif is friendly to lithographic registration. The price appears even before materials physics. At a vertex, the angles between incident edges are 90° , 135° , 135° . For equal target-edge length a , adjacent midpoint distances are

$$d_{90} = a \sin 45^\circ = \frac{a}{\sqrt{2}}, \quad d_{135} = a \sin 67.5^\circ \approx 0.924a.$$

A pure r^{-3} interaction would differ by

$$\left(\frac{d_{135}}{d_{90}} \right)^3 \approx 2.23$$

between those two intended bond lengths. The square and octagonal plaquettes also require four-edge and eight-edge operations. Calibration, geometric distortion, or driven equalization is therefore mandatory.

Hence the fallback is viable because its logical degree and planarity are right — not because the regular embedding is uniform. Drop the equal-edge hypothesis and you might flatten the 2.23; you then owe a derivation of the distortion that did it.

25.17. A square, and a brick-wall honeycomb

The square lattice hosts many toric-code drawings, but its degree-four vertices do not directly match a trivalent Fibonacci fusion tree.

A brick-wall embedding is graph-isomorphic to honeycomb: bend the edges until rows and columns look rectangular without changing which vertices connect. Topological lattice models tolerate such graph-preserving deformation at the ideal level. Physical coupling strengths do not. Brick-wall coordinates may simplify a write pattern, but unequal bond lengths or orientations must be carried into the Hamiltonian.

We regard it as a fabrication variant of the primary honeycomb, not the fallback phase graph. Isomorphism of G is not isomorphism of the interaction graph.

25.18. What the laboratory can put down

[Experiment] Patterned arrays and individual nanostructure targets have been implanted [R189]; [R190]. These capabilities support coordinate lists at tens-to-hundreds-of-nanometres scales in particular experiments, not arbitrary interaction matrices.

The proposed string-net prototype needs substantially more:

- several defects must form each reproducible cluster doublet;
- intra-cluster couplings must dominate inter-cluster couplings;
- every intended edge must have the correct sign, tensor structure, and strength;
- unwanted long-range couplings must be included or suppressed;
- missing or wrong-charge defects must be found and repaired;
- vertex and plaquette terms must emerge with a gap larger than disorder and decoherence.

No cited experiment meets that list. The geometry decision is therefore **[Proposal]**: use a shallow planar patterned layer in a host whose chosen defect species can be created and read out, first testing isolated triangles and hexagons. Diamond has direct patterned-implantation evidence [R189]; [R190]. Silicon carbide may offer wafer-scale fabrication advantages, but this chapter does not assume equal placement yield, coherence, or coupling without host- and defect-specific evidence.

A graph-first sequence should test one logical vertex, then two vertices sharing an edge cluster, then one hexagonal plaquette with neighboring legs. Measure each projected constraint before tiling the coordinate cell.

25.19. Ways to get this wrong

Confusing target and placement lattices. Call the device “kagome” without saying that kagome sites encode honeycomb edges, and the branching map is already lost. Call it “honeycomb” and you can conceal the degree-four physical neighbor graph.

Counting drawn edges, not couplings. Dipolar interactions do not stop at the nearest-neighbor pencil line. Evaluate J_{ij} , angular factors, and ϵ_{graph} . What fails if you set every $A_{ij} = 0$ entry to $J_{ij} = 0$ is the tail that Chapter 26 still has to write.

Equating equal distance with equal interaction. Crystal orientation, strain, charge state, and quantization axis remain. Geometry controls only part of J_{ij} .

Treating a helper as a many-body term. A six-leg star of pair couplings is not B_p . The effective operator and its correction terms must be derived.

Borrowing an anyon from another Hamiltonian. Decorated-honeycomb and square-octagon Kitaev models provide non-Abelian precedents [R184]; [R186]; ruby models provide color-code order [R185]. None thereby realizes Fibonacci anyons in the proposed defects.

Using nominal beam size as final defect uncertainty. Straggle, annealing, registration, and stochastic conversion survive the beam specification [R189]; [R190].

Optimizing only nearest-neighbor spacing. Making a small strengthens desired and undesired interactions, crowds clusters and helpers, and can merge damage volumes. Making a large improves addressability while weakening local interactions. Chapter 28 must optimize the full error and coupling budget.

Assuming planarity removes crossings in Hilbert space. A planar drawing can still require nonplanar control connectivity or a high-degree cavity mode. “No crossed ink” is a necessary convenience, not a Hamiltonian theorem.

25.20. Which graph we keep

Primary — honeycomb logical graph, kagome cluster positions. It matches trivalent branching without auxiliary fusion-tree vertices; all nearest edge-midpoint separations are equal in the regular embedding; its line graph has degree four; vertex and hexagonal plaquette helpers fit without planar crossings; and the pattern repeats with one coordinate motif. Fabrication remains challenging because clusters, yields, long-range tails, and six-edge effective terms are unproved.

Fallback — square-octagon logical graph. It preserves degree three and planarity while offering an orthogonal square motif that may suit registration, routing, or device boundaries. It loses on uniform spacing: its regular edge-midpoint layout has two intended distances and two face sizes, so it demands calibration or engineered couplings.

Choose it only if fabrication tests show that orthogonal patterning and access outweigh the factor-of-about-2.23 geometric spread in a nominal r^{-3} bond scale.

Both choices are proposals, not reports of an observed topological material. The selected target graph is trivalent honeycomb. One encoded cluster occupies each edge midpoint. The physical nearest-constraint graph is degree-four kagome. A vertex term addresses three clusters and a hexagonal plaquette term at least six boundary clusters. The regular nearest midpoint spacing is $\sqrt{3}a/2$. Long-range pair terms stay in the Hamiltonian. Square–octagon is the fallback, with two intended midpoint distances and four- and eight-edge faces.

The next job is no longer to choose a tiling. It is to write the zero-field, Zeeman, dipolar, exchange, strain, phonon, and drive terms that can actually produce that graph and its required operators.

25.21. Check yourself

1. **Show that honeycomb edge clusters occupy a kagome lattice of degree four.**
Two cluster sites are adjacent when their honeycomb edges share a vertex. That line-graph construction is kagome, and $z_{L(G)} = 2(3 - 1) = 4$.
2. **What are the three graphs that must not be conflated?**
The string-net target graph, the physical placement graph, and the realized interaction graph including unwanted and mediated couplings.
3. **Show that a pure r^{-3} tail on the next kagome shell is about 0.19.**
The next distance is $d_2 = \sqrt{3}d_1$, so $J(d_2)/J(d_1) = (d_1/d_2)^3 = 1/(3\sqrt{3}) \approx 0.19$.
4. **What fails if you take planarity as a proof that mediators are easy?**
Geometric edge crossings disappear, but high-degree mediators and effective multi-body operators still require a physical derivation. A six-leg star is not B_p .
5. **What is the primary/fallback decision?**
Primary: honeycomb target with kagome edge-cluster positions. Fallback: square–octagon target, accepted only with compensation for its two spacing and face classes, including the factor-of-about-2.23 spread in a nominal r^{-3} bond scale.
6. **What fails if beam width is used as “placement accuracy”?**
Beam width, lateral straggle, final defect distribution, and registration to a device are different quantities. None of the cited numbers may be used without its definition and conditions.

A pretty tiling is not a Hamiltonian. Honeycomb is the target graph because it is trivalent and has one face size. Kagome is where the clusters sit. Who couples is a third drawing, and the 0.19 dipolar tail is already enough to keep the extra edges.

25.22. Sources

- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv:cond-mat/0404617.
- [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv:cond-mat/0506438.
- [R184] H. Yao and S. A. Kivelson, “An exact chiral spin liquid with non-Abelian anyons,” *Physical Review Letters* **99**, 247203 (2007). DOI: 10.1103/PhysRevLett.99.247203; arXiv:0708.0040.
- [R185] M. Kargarian, H. Bombin, and M. A. Martin-Delgado, “Topological color codes and two-body quantum lattice Hamiltonians,” *New Journal of Physics* **12**, 025018 (2010). DOI: 10.1088/1367-2630/12/2/025018; arXiv:0906.4127.
- [R186] G. Kells, J. Kailasvuori, J. K. Slingerland, and J. Vala, “Kaleidoscope of topological phases with multiple Majorana species,” *New Journal of Physics* **13**, 095014 (2011). DOI: 10.1088/1367-2630/13/9/095014; arXiv:1012.5276.
- [R187] R. Moessner and S. L. Sondhi, “Resonating valence bond phase in the triangular lattice quantum dimer model,” *Physical Review Letters* **86**, 1881–1884 (2001). DOI: 10.1103/PhysRevLett.86.1881; arXiv:cond-mat/0007378.
- [R188] L. Balents, M. P. A. Fisher, and S. M. Girvin, “Fractionalization in an easy-axis kagome antiferromagnet,” *Physical Review B* **65**, 224412 (2002). DOI: 10.1103/PhysRevB.65.224412; arXiv:cond-mat/0110005.
- [R189] D. M. Toyli, C. D. Weis, G. D. Fuchs, T. Schenkel, and D. D. Awschalom, “Chip-scale nanofabrication of single spins and spin arrays in diamond,” *Nano Letters* **10**, 3168–3172 (2010). DOI: 10.1021/nl102066q.
- [R190] T. Schröder *et al.*, “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* **8**, 15376 (2017). DOI: 10.1038/ncomms15376; stable full text.

Chapter 26. What diamond actually writes

Three red points sit a few nanometres apart inside a diamond. Each is an NV^- : one carbon gone, a nitrogen next door, an extra electron that glows. You already know the 2.87 GHz split and the microwave that parks that electron in $m_s = 0$ or $m_s = -1$. The leftover $m_s = +1$ is leakage, not a third answer.

Those three electrons can feel one another. A magnetic dipole falls as $1/r^3$. At five nanometres that is hundreds of kilohertz, not gigahertz. Diamond also gives each defect a crystal axis, a nitrogen nucleus, strain, phonons, and whatever microwave you shine in.

A host-supported microscopic Hamiltonian is the operator whose every term is either a thing diamond supplies or a thing we write down in order to set it to zero. We take that as the definition. Then we throw unused states away twice: first each spin-1 down to a driven qubit, then each triangle down to two cluster states. The shopping list becomes an operator only after we have a picture we can actually solve.

26.1. Three vacancies in a triangle

Take three NV^- centers with the same crystal axis, and call that axis local z . Put them, ideally, at the vertices of an equilateral triangle in the plane perpendicular to z . A diamond (111) plane has that symmetry. Fabricating many identical nanometre-scale triangles is a proposal, not a demonstrated device.

Each ground-state NV electron has spin $S = 1$. Write $|0\rangle$ for $m_s = 0$ and $|1\rangle$ for $m_s = -1$. Define Pauli operators τ^α , for $\alpha = x, y, z$, by

$$\tau^z|0\rangle = |0\rangle, \quad \tau^z|1\rangle = -|1\rangle.$$

The omitted $m_s = +1$ state is leakage, not a third qubit value.

Use a static field B_0 along z , and apply a microwave near the $|0\rangle \leftrightarrow |1\rangle$ line. Sit in a frame that rotates with that microwave. Fast leftover oscillations can be dropped if the leftover frequencies are huge compared with the couplings. That drop is the *rotating-wave approximation*. What remains of the dipole is called **secular**: the part that does not shake at gigahertz.

For two identical, resonant NV qubits the secular pair energy is

$$\frac{H_{ij}^{\text{sec}}}{h} = \frac{c_{ij}}{4} (1 - 3 \cos^2 \theta_{ij}) (\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y) + \text{one-body shifts}.$$

Here h is Planck's constant, $c_{ij} > 0$ is a frequency defined below, and θ_{ij} is the angle between the pair displacement and z . For the planar triangle, $\theta_{ij} = \pi/2$. If all three sides are equal, write $c_{ij} = c$. The cluster Hamiltonian is then

$$\frac{H_C}{h} = \frac{c}{4} \sum_{i < j \in C} (\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y). \quad (26.1)$$

This toy is small enough to solve. In the one-excitation basis $\{|100\rangle, |010\rangle, |001\rangle\}$, every diagonal entry is $-c/4$, and every off-diagonal hop is $-c/2$. Therefore the symmetric state

$$|W_1\rangle = \frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}}$$

has quasienergy $-5hc/4$, while the two states with phases $1, e^{\pm 2\pi i/3}, e^{\mp 2\pi i/3}$ have quasienergy $+hc/4$. Those rotating-frame numbers are **quasienergies**: they are defined only while the drive and its clock are on, and they are defined only up to the drive frequency. The name is worth keeping because a quasienergy is not a laboratory energy; turn the drive off and the number is no longer the splitting you will measure.

Flip every qubit and the two-excitation state

$$|W_2\rangle = \frac{|011\rangle + |101\rangle + |110\rangle}{\sqrt{3}}$$

also has quasienergy $-5hc/4$. The fully polarized states have quasienergy $+3hc/4$. Thus, in the ideal resonant rotating frame, $|W_1\rangle$ and $|W_2\rangle$ form a doublet separated from the nearest cluster states by

$$\Delta_C/h = \frac{3c}{2}. \quad (26.2)$$

Hence the cluster gap of the ideal triangle is $3c/2$ in frequency units, between the symmetric W -state doublet and the chiral states.

Define cluster Pauli operators X_C, Y_C, Z_C with $Z_C|W_1\rangle = |W_1\rangle$ and $Z_C|W_2\rangle = -|W_2\rangle$. If P_C projects onto this doublet, direct evaluation gives, for any site i in the symmetric triangle,

$$P_C \tau_i^z P_C = \frac{1}{3} Z_C, \quad P_C \tau_i^x P_C = \frac{2}{3} X_C, \quad P_C \tau_i^y P_C = \frac{2}{3} Y_C. \quad (26.3)$$

Equation (26.3) is the microscopic-to-encoded dictionary. It is also a warning. A physical pair interaction remains an encoded pair interaction at first order. Projection shrinks and rotates coefficients. It does not spontaneously manufacture a plaquette operator.

Drop the equal-side, equal-axis hypotheses and the doublet of (26.2) splits. The dictionary (26.3) is then a different matrix, not a small footnote.

26.2. Every term diamond actually writes

We write H/h , so every coefficient is in hertz. Electron spin operators S_i^α are dimensionless and have eigenvalues $m_s = 0, \pm 1$. Each NV has a local orthonormal frame (x_i, y_i, z_i) , with z_i along the nitrogen–vacancy axis. The selected array uses only one of diamond’s four NV orientations. Mixing orientations would produce different splittings and interaction tensors.

Now the shopping list becomes an operator. A host-supported microscopic Hamiltonian is

$$\begin{aligned} \frac{H}{h} = & \sum_i \left[D_i((S_i^{z_i})^2 - \frac{2}{3}) + E_{x,i}((S_i^{x_i})^2 - (S_i^{y_i})^2) + E_{y,i}\{S_i^{x_i}, S_i^{y_i}\} \right] \\ & + \sum_i \gamma_e \mathbf{B}_i \cdot \mathbf{S}_i + \sum_i (\mathbf{S}_i \cdot \mathbf{A}_i \cdot \mathbf{I}_i + P_i(I_i^{z_i})^2 - \gamma_n \mathbf{B}_i \cdot \mathbf{I}_i) \\ & + \sum_{i < j} c_{ij} [\mathbf{S}_i \cdot \mathbf{S}_j - 3(\mathbf{S}_i \cdot \hat{\mathbf{r}}_{ij})(\mathbf{S}_j \cdot \hat{\mathbf{r}}_{ij})] \\ & + \sum_{i < j} J_{ij}^{\text{ex}} \mathbf{S}_i \cdot \mathbf{S}_j + \frac{H_{\text{strain}}(t) + H_{\text{ph}} + H_{\text{drive}}(t)}{h}. \end{aligned} \quad (26.4)$$

The anticommutator is $\{A, B\} = AB + BA$. Every line is a thing diamond actually supplies, or a thing we write down in order to set it to zero.

Zero-field and static strain. D_i is the axial zero-field splitting. For the NV^- ground state near room temperature, $D \approx 2.87$ GHz [R074]. Subtracting $2/3$ only removes an irrelevant trace. The frequencies $E_{x,i}$ and $E_{y,i}$ describe transverse symmetry breaking from static strain and electric fields. The NV ground-state spin Hamiltonian and its electric response are experimentally established [**Experiment**] [R074]; [R191].

It is useful to separate controllable or fluctuating strain:

$$\frac{H_{\text{strain}}(t)}{h} = \sum_i \left[\delta D_i(t)(S_i^{z_i})^2 + \epsilon_{x,i}(t)((S_i^{x_i})^2 - (S_i^{y_i})^2) + \epsilon_{y,i}(t)\{S_i^{x_i}, S_i^{y_i}\} \right]. \quad (26.5)$$

The coefficients already include the appropriate spin–strain susceptibility and therefore have units of hertz. Mechanical control of NV spin transitions has been demonstrated [**Experiment**] [R192]; [R193]. Equation (26.5) does not assume a strain-mediated interaction between distant defects.

Zeeman term. $\gamma_e = g_e \mu_B / h \approx 28 \text{ GHz/T}$ is the electron gyromagnetic ratio in frequency units, $\mathbf{B}_i$ is the magnetic field in tesla, and $\mathbf{S}_i$ is dimensionless. Thus $\gamma_e B$ has units s^{-1} , as required.

Hyperfine term. $\mathbf{I}_i$ is the dimensionless ^{14}N or ^{15}N nuclear spin; $\mathbf{A}_i$ is its hyperfine tensor in hertz. P_i is the ^{14}N quadrupole coefficient, and $\gamma_{n,i}$ is the nuclear gyromagnetic ratio. Nearby ^{13}C nuclei add terms of the same form. Hyperfine coupling is not optional bookkeeping: its megahertz-scale structure can exceed the dipolar coupling in the example below [R074]. The proposal therefore requires isotopic ^{12}C enrichment and preparation of every nitrogen nucleus in a specified state. Otherwise “identical NVs” is false.

Magnetic dipole–dipole term. $\mathbf{r}_{ij}$ points from defect i to j , $r_{ij} = |\mathbf{r}_{ij}|$, and $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$. Its coefficient is

$$c_{ij} = \frac{\mu_0 (g_e \mu_B)^2}{4\pi h r_{ij}^3}. \quad (26.6)$$

The units check is $(\text{T m/A})(\text{J/T})^2/(\text{J s m}^3) = \text{s}^{-1}$. Numerically, $c_{ij} \approx 52 \text{ MHz}(1 \text{ nm}/r_{ij})^3$. Coherent coupling and entanglement between separate NV electron spins have been demonstrated [**Experiment**] [R080]. This supports the term, not the proposed array yield.

Exchange. J_{ij}^{ex} is an isotropic overlap-mediated exchange frequency. Diamond does not provide a verified, lithographically tunable long-range NV–NV exchange bus. Because electronic wave-function overlap decays rapidly with separation, the 5–10 nm toy array below sets $J_{ij}^{\text{ex}} \approx 0$. We retain the symbol in (26.4) to state what was discarded, not to borrow a convenient Heisenberg interaction. In particular, the antiferromagnetic open-chain Hamiltonian sometimes used for an abstract three-spin encoded qubit does not follow from this NV geometry.

Phonons. Let b_q destroy a phonon in normal mode q , with frequency ν_q . A compact supported form is

$$\frac{H_{\text{ph}}}{h} = \sum_q \nu_q b_q^\dagger b_q + \sum_{i,q} (b_q + b_q^\dagger) \left[g_{iq}^\parallel (S_i^{z_i})^2 + g_{iq}^x Q_i^x + g_{iq}^y Q_i^y \right], \quad (26.7)$$

where $Q_i^x = (S_i^{x_i})^2 - (S_i^{y_i})^2$, $Q_i^y = \{S_i^{x_i}, S_i^{y_i}\}$, and each g is a frequency. Phonons can coherently drive a selected mechanical mode [R192]; [R193], but an uncontrolled bath produces relaxation and dephasing. Measured NV longitudinal relaxation depends strongly on temperature [**Experiment**] [R194]. We do not replace that bath with a noiseless interaction.

Drive. A microwave magnetic field $\mathbf{B}_{1,i}(t)$ gives

$$\frac{H_{\text{drive}}(t)}{h} = \sum_i \gamma_e \mathbf{B}_{1,i}(t) \cdot \mathbf{S}_i. \quad (26.8)$$

Optical fields can initialize and read the NV through excited electronic states [R074], but those states are outside (26.4). Adiabatic elimination would add light shifts, Raman couplings, scattering, and optical decay together. We do not keep the useful term and misplace the decay under the rug.

Hence every symbol in (26.4) is either measured, estimated, or set to zero on purpose. Drop the “set to zero” column and you have changed the host.

26.3. From spin 1 to a driven qubit

Let $P_i = |0\rangle\langle 0| + |1\rangle\langle 1|$. In this subspace,

$$P_i S_i^z P_i = \frac{\tau_i^z - \mathbb{1}}{2}, \quad P_i S_i^x P_i = \frac{\tau_i^x}{\sqrt{2}}, \quad P_i S_i^y P_i = \frac{\tau_i^y}{\sqrt{2}}. \quad (26.9)$$

For a field along the NV axis, the two transition frequencies from $m_s = 0$ are approximately $\omega_- = D - \gamma_e B_0$ and $\omega_+ = D + \gamma_e B_0$. A drive near ω_- , with ω_- , ω_+ , and their separation all much larger than couplings and Rabi frequencies, permits the rotating-wave and leakage expansions. Projecting (26.6) and discarding terms oscillating near gigahertz frequencies gives the secular pair Hamiltonian used in the triangle.

This derivation is **[Theory]**. It describes a Floquet quasienergy Hamiltonian sustained by a phase reference. Turning off the resonant drive restores the large laboratory energy difference between $|W_1\rangle$ and $|W_2\rangle$. The encoded doublet is therefore not a passive ground-space doublet.

Hence the first projection is a driven two-level truncation, not a spontaneous encoding. The three microscopic NV spins are still physical spins. The two states retained at each NV are still physical qubit levels. Nothing in (26.9) has invented an anyon.

What fails if the leftover frequencies are not huge compared with the couplings is the rotating-wave step itself: gigahertz terms that were dropped come back, and (26.1) is no longer the pair energy.

26.4. From a triangle to two cluster states

Let $P = \prod_C P_C$ project every triangle onto the doublet of (26.2), and let V contain inter-cluster dipolar bonds, detuning disorder, and weak drives. Schrieffer–Wolff perturbation theory – the downstairs-and-upstairs expansion of Chapter 22 – gives

$$\frac{H_{\text{eff}}}{h} = P \frac{H_C + V}{h} P - P \frac{V}{h} Q \left(Q \frac{H_C - E_0}{h} Q \right)^{-1} Q \frac{V}{h} P + O\left(\frac{v^3}{(\Delta_C/h)^2}\right), \quad (26.10)$$

where $Q = 1 - P$, E_0 is the ideal doublet energy, and v bounds the frequencies in V . Hidden states still matter: a traveler can briefly leave the retained space, take a shortcut, and return. Those virtual excursions are the second-order terms.

For a physical bond between site i of cluster C and site j of cluster D , (26.3) yields

$$P \tau_i^{\alpha} T_{ij}^{\alpha\beta} \tau_j^{\beta} P = M_{\alpha} M_{\beta} \Sigma_C^{\alpha} T_{ij}^{\alpha\beta} \Sigma_D^{\beta}, \quad (M_x, M_y, M_z) = \left(\frac{2}{3}, \frac{2}{3}, \frac{1}{3}\right), \quad (26.11)$$

with repeated α, β summed, T_{ij} the physical dipolar tensor in hertz, and $\Sigma_C = (X_C, Y_C, Z_C)$. A uniform resonant transverse drive projects as $P_C \Gamma \sum_i \tau_i^x P_C = 2\Gamma X_C$. A detuning $\sum_i \delta_i \tau_i^z / 2$ gives $(\sum_i \delta_i) Z_C / 6$ at first order. Unequal detunings also mix the doublet with the chiral states and generate corrections of order $\sigma_{\delta}^2 / (\Delta_C/h)$.

Consequently the toy model is

$$\frac{H_{\text{eff}}}{h} = \sum_C (2\Gamma_C X_C + \bar{\delta}_C Z_C) + \sum_{\langle C,D \rangle} \sum_{\alpha, \beta} \tilde{J}_{CD}^{\alpha\beta} \Sigma_C^{\alpha} \Sigma_D^{\beta} + \delta H^{(2)}/h, \quad (26.12)$$

not a string-net Hamiltonian. Here $\tilde{J} = MTM$, and $\delta H^{(2)}/h$ contains shifts and additional two-cluster or, when two bonds share a virtual cluster excitation, small multicluster terms of scale at most roughly $u^2 / (\Delta_C/h)$. Their coefficients are fixed by geometry and denominators. They are not automatically Fibonacci F -symbols.

Hence a first-order intercluster bond remains a bilinear bond between cluster Pauli operators, with projection factors $(2/3, 2/3, 1/3)$. A cluster doublet is an encoded pseudospin. A periodically driven array of those pseudospins is an analog emulator. Neither label, alone or together, establishes emergent anyons or equilibrium topological order.

Choose $B_0 = 20$ mT. Using $D = 2.87$ GHz and $\gamma_e = 28$ GHz/T gives $\omega_- \approx 2.31$ GHz and $\omega_+ \approx 3.43$ GHz. For an illustrative triangle side $r = 5$ nm, (26.6) gives $c \approx 52/5^3$ MHz = 0.416 MHz, so $\Delta_C/h \approx 0.624$ MHz. At a representative 10 nm intercluster separation, the unprojected dipolar scale is $u \leq 52$ kHz before angular and projection factors. A weak drive $\Gamma \sim 20$ kHz would satisfy $u, \Gamma \ll \Delta_C/h \ll \omega_{\pm}$.

The dimensional hierarchy is therefore plausible on paper. The complete regime is stricter:

$$\omega_{\pm} \gg c \gg u, \Gamma, \sigma_{\delta}, \Gamma_{\text{decoh}}, \quad k_B T/h \ll \Delta_{\text{desired}}/h, \quad (26.13)$$

while nuclear states must be fixed and the rotating-frame phases synchronized. The last thermal inequality refers to a desired many-body gap, which (26.12) has not been shown to possess. Nanoscale aperture masks with sub-10 nm features have confined implanted NV clusters, including clusters of up to three centers [**Experiment**] [R195]; ion straggle and conversion statistics mean this is not deterministic placement of the specified 5 nm equilateral triangle, much less a honeycomb-edge array. The numbers are a falsifiable design point [**Proposal**], not an experimental device claim.

The leading controlled corrections are now explicit:

$$\|\delta H^{(2)}/h\| = O\left(\frac{u^2 + \Gamma^2 + \sigma_{\delta}^2}{\Delta_C/h} + \frac{c^2 + \Omega^2}{\min(\omega_-, \omega_+)} + \frac{c^2 + \Omega^2 + E_{\perp}^2}{\Delta_{\text{leak}}/h}\right), \quad (26.14)$$

where Ω is a physical microwave Rabi frequency, $E_{\perp}^2 = E_x^2 + E_y^2$, and Δ_{leak} is the relevant separation to $m_s = +1$. The first fraction is cluster leakage, the second is the nonsecular or Bloch–Siegert correction, and the third is spin-1 leakage. Equation (26.14) is scaling, not an equality. A device calculation must diagonalize its actual tensor geometry.

What fails if u is not small compared with Δ_C/h is the Schrieffer–Wolff truncation: the traveler does not return, and the retained doublet is no longer the slow space.

26.5. What the operator still is not

The hoped-for chain is NV electron spins, then coupled three-NV clusters, then cluster pseudospins, then a honeycomb-edge model, then topological order. The last step is a question mark. A low-energy projector only translates the microscopic interactions that are present. It does not award extra interactions for architectural enthusiasm.

The doubled-Fibonacci Levin–Wen target has a local branching constraint Q_v at every trivalent vertex and a plaquette operator B_p that inserts a loop and recouples edge labels with the Fibonacci F -symbols [R018]. For the honeycomb-edge architecture, (26.12) is missing:

1. a three-edge vertex projector with the correct allowed Fibonacci fusion channels;

2. the conditional, many-edge off-diagonal plaquette action with the correct relative amplitudes and phases;
3. a mechanism canceling larger one-body and two-body terms without also closing the intended gap;
4. evidence that residual terms in (26.14) lie inside the doubled-Fibonacci phase's stability region;
5. a passive laboratory-frame ground manifold. The present doublet is resonantly dressed.

Higher-order gadgets could be proposed to address items 1–3, but they require mediators and produce amplitudes suppressed by products of u/Δ_C . Those mediators are not in the selected microscopic Hamiltonian. Therefore **[Theory]** the derived model is an anisotropic, driven pseudospin model; **[Speculation]** it might become an ingredient of a separately engineered gadget construction; it is not presently a derivation of Fibonacci order.

Three questions, answered in order. What roads does diamond provide? Known. Which roads remain inside the chosen subspace? Calculable. Do the remaining roads have the pattern required by the target topological model? No. That is a successful derivation, not a failed sales pitch.

26.6. What the laboratory actually has

[Experiment] Single-NV zero-field, Zeeman, hyperfine, electric/strain, microwave, optical, and phonon-relaxation physics are well characterized [R074]; [R191]; [R192]; [R193]; [R194]. **[Experiment]** Dipolar coupling has been used to entangle two individually addressed NV electron spins [R080]. These facts justify the terms in (26.4).

What is not established is equally specific. References [R074]; [R191]; [R192]; [R193]; [R080]; [R194]; [R018]; [R195] do not report a mesoscopic array of identical 5 nm three-NV triangles with fixed orientation, initialized nuclear states, uniform detunings, and individually phased drives. They do not measure the cluster gap in (26.2), nor a many-body topological gap, topological degeneracy, string operator, or anyonic excitation in this material architecture. The microscopic ingredients and the proposed architecture therefore remain separated by a fabrication-and-control gap.

A measured pair coupling is a physical bond. A projected cluster doublet is an encoded leftover. Neither object is a vertex projector, a plaquette recoupling, or a topological gap.

26.7. Ways to get this wrong

1. **Calling a rotating frame a passive phase.** Quasienergy degeneracy depends on a drive and a clock. It is not thermal ground-state protection. What fails if the drive is turned off is the doublet of (26.2) itself: the laboratory splitting between $|W_1\rangle$ and $|W_2\rangle$ returns.
2. **Replacing dipole coupling with friendly exchange.** Isotropic antiferromagnetic exchange makes elegant encoded-spin algebra, but it is not a supported 5–10 nm NV coupling here. Keep J_{ij}^{ex} in (26.4) and set it to zero on purpose; silently restoring it changes the host.
3. **Ignoring the nitrogen nucleus.** An unresolved hyperfine manifold is not a small footnote compared with a sub-megahertz cluster gap.
4. **Treating phonons only as a bus.** The same coupling permits relaxation and dephasing. A coherent-mode proposal must include loss and thermal occupation.
5. **Confusing first-order projection with a gadget.** Equations (26.3) and (26.11) preserve operator locality. Multibody terms require virtual processes and arrive with denominators and competitors.
6. **Inferring a phase from a Hamiltonian’s vocabulary.** “Honeycomb,” “cluster,” and “plaquette” are nouns. The spectrum, gap, ground-state degeneracy, and quasi-particle data carry the physics.
7. **Using the toy numbers as tolerances.** Five and ten nanometres are assumed separations, not achieved placement distributions or yields.

26.8. Check yourself

1. **Why is exchange written and then set to zero?**
To make the microscopic audit explicit. Long-range tunable NV–NV exchange is not a supported resource for the chosen spacing; silently using it would change the host.
2. **Show that the ideal triangle has cluster leakage gap $\Delta_C/h = 3c/2$.**
In the one-excitation basis the symmetric W -state sits at $-5c/4$ and the chiral states at $+c/4$, so the gap is $3c/2$. The two-excitation partner $|W_2\rangle$ matches.
3. **What fails if you replace the dipole with isotropic antiferromagnetic exchange?**
The secular pair energy is no longer the XXZ form of (26.1), and the abstract three-spin Heisenberg encoding of Chapter 24 does not follow from this NV geometry.

4. **Show that an intercluster dipolar bond remains bilinear at first order.**

Equation (26.3) sends each physical Pauli to a cluster Pauli times $(2/3, 2/3, 1/3)$, so (26.11) is still a two-cluster coupling $\tilde{J} = MTM$.

5. **Which target terms are absent?**

The Fibonacci branching projector and the conditional F -symbol plaquette recoupling operator, along with a demonstrated passive topological gap.

6. **What fails if the resonant drive is turned off?**

Quasienergy degeneracy of $|W_1\rangle$ and $|W_2\rangle$ disappears. The laboratory-frame splitting returns, and the encoded doublet is no longer a passive ground-space pair.

A same-orientation NV array is described by (26.4). Spin-1 projection gives (26.9) and a secular dipolar tensor. An ideal resonantly driven three-NV triangle has the doublet (26.2). Local operators project according to (26.3). Intercluster bonds produce (26.11) with corrections scaling as (26.14). None of that is a static encoded doublet, isotropic exchange, a vertex projector, a plaquette recoupling, or a topological gap.

26.9. Sources

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, *et al.*, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001.
- [R191] F. Dolde, H. Fedder, M. W. Doherty, *et al.*, “Electric-field sensing using single diamond spins,” *Nature Physics* **7**, 459–463 (2011). DOI: 10.1038/nphys1969.
- [R192] A. Barfuss, J. Teissier, E. Neu, A. Nunnenkamp, and P. Maletinsky, “Strong mechanical driving of a single electron spin,” *Nature Physics* **11**, 820–824 (2015). DOI: 10.1038/nphys3411.
- [R193] E. R. MacQuarrie, T. A. Gosavi, A. M. Moehle, N. R. Jungwirth, S. A. Bhave, and G. D. Fuchs, “Coherent control of a nitrogen-vacancy center spin ensemble with a diamond mechanical resonator,” *Optica* **2**, 233–238 (2015). DOI: 10.1364/OP-TICA.2.000233.
- [R080] F. Dolde, I. Jakobi, B. Naydenov, *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R194] A. Jarmola, V. Acosta, K. Jensen, S. Chemerisov, and D. Budker, “Temperature- and magnetic-field-dependent longitudinal spin relaxation in nitrogen-vacancy ensembles in diamond,” *Physical Review Letters* **108**, 197601 (2012). DOI: 10.1103/PhysRevLett.108.197601.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110.

- [R195] T.-Y. Hwang, J. Lee, S.-W. Jeon, *et al.*, “Sub-10 nm precision engineering of solid-state defects via nanoscale aperture array mask,” *Nano Letters* **22**, 1672–1679 (2022). DOI: 10.1021/acs.nanolett.1c04699.

Part X

Numbers, noise, and other kitchens

Energy, distance, gaps, noise, protection, and two other labs trying the same dream. A number travels with a passport.

Chapter 27. A number that travels with a passport

Someone writes 13 ms on the whiteboard and sits down. The room now has a coherence time for silicon vacancies. It does not.

The 13 ms is real, and it is narrower than that sentence. It was recorded on one SiV^- electron spin, near 100 mK, after a CPMG-32 pulse train, in diamond with $10^{-3}\%$ ^{13}C [R083]. Warm the stone, skip the pulses, or put the carbon-13 back, and you are measuring a different object.

A recorded number together with those conditions is a *passport*. We take this as the definition. The rest of the chapter treats a frequency, a time, or a fidelity without a passport as unusable in a Hamiltonian. The number can still tell a story. It should not cross an engineering border.

27.1. Three objects, one unit

A slide can show 2.87 GHz, 4.93 kHz, and 12.7 kHz in the same column. Hertz is a crowded room. Those three numbers are not three sizes of one thing.

The first is a static level splitting: it picks a local basis. The NV^- ground-state zero-field splitting is one of those. It tells $m_s = 0$ from $m_s = \pm 1$ before a magnet is turned on.

The second is a coupling: it moves information or makes entanglement. The fitted dipolar matrix element of a 25 nm NV pair is one of those.

The third is a decay or a linewidth: it erases phase. A Ramsey width, an echo width, and a phonon-broadened optical line are three of those, and they are not interchangeable with each other either.

All three can be quoted in megahertz. The shared unit does not make them the same object. Hence a large on-site splitting does not rescue a tiny intercluster coupling, and a millisecond echo does not promise a millisecond of undriven many-body evolution. “Works at room temperature” does not mean thermal equilibrium prepares the spin you wanted.

Two comparisons will keep coming back:

useful coupling > uncontrolled broadening and drift,
effective gap > thermal and disorder scales.

This chapter fills the left- and right-hand sides with numbers that still have their passports. Filling those sides does not produce a topological gap. That object is a spectrum, and it is not on this page.

Treat each entry as a record with mandatory fields:

quantity		platform and charge state		sample		temperature
		field/strain/drive protocol		measurement definition		source

Drop any field and the number is free to wander into the wrong inequality.

27.2. Hertz become kelvin once you name the map

Laboratories speak frequency. Thermometers speak kelvin. A Hamiltonian, strictly, speaks joules. Those are three names for the same interval once the conversion is named.

Let ν be ordinary cyclic frequency, in hertz. Let $\omega = 2\pi\nu$ be angular frequency, in radians per second. Let E be energy, in joules or electronvolts. Define

$$E = h\nu = \hbar\omega, \quad T_E = \frac{E}{k_B} = \frac{h\nu}{k_B},$$

where h is Planck's constant, $\hbar = h/(2\pi)$, and k_B is Boltzmann's constant. The number T_E is the **equivalent temperature** of that energy. The name is worth keeping because it is not the temperature of the cryostat. It is the temperature at which $k_B T$ would match E .

The constants $h = 6.62607015 \times 10^{-34}$ J s, $k_B = 1.380649 \times 10^{-23}$ J K⁻¹, and the elementary charge are exact in the SI [R196]. Therefore

$$\begin{aligned} 1 \text{ GHz} &\leftrightarrow 4.135667696 \text{ } \mu\text{eV} \leftrightarrow 47.9924307 \text{ mK}, \\ 1 \text{ MHz} &\leftrightarrow 0.004135667696 \text{ } \mu\text{eV} \leftrightarrow 47.9924307 \text{ } \mu\text{K}, \\ 1 \text{ kHz} &\leftrightarrow 4.135667696 \times 10^{-6} \text{ } \mu\text{eV} \leftrightarrow 47.9924307 \text{ nK}, \\ 1 \text{ K} &\leftrightarrow 86.1733326 \text{ } \mu\text{eV} \leftrightarrow 20.8366191 \text{ GHz}. \end{aligned}$$

The arrows are conversions. They are not a license to call a gigahertz an energy. Gigahertz is cyclic frequency. Microelectronvolts are energy. Millikelvin here is equivalent temperature. Write $E = h\nu$ or $E/h = \nu$ when you mean the Hamiltonian coefficient.

The dimensional check is visible: (J s)(s⁻¹) = J, and J/(J K⁻¹) = K.

We will quote Hamiltonian coefficients as cyclic frequencies. If a term has energy J , the table gives J/h in hertz. If a paper writes $H = \hbar g(\dots)$, then g is normally angular frequency

and the cyclic frequency is $g/(2\pi)$. Lose that factor of 2π and every later comparison is off by six.

Drop the distinction between T_E and the cryostat set point, and a 100 mK refrigerator looks “cold” for a 5 kHz splitting. It is not. The splitting’s equivalent temperature is a fraction of a microkelvin.

27.3. Three measured numbers through that map

NV zero-field splitting. For the NV^- ground-state triplet near room temperature, $D/h \approx 2.87$ GHz [R197]. The energy and equivalent temperature are

$$E_D = h(2.87 \text{ GHz}) = 11.87 \text{ } \mu\text{eV}, \quad T_D = 0.1377 \text{ K}.$$

At 300 K, $k_B T/h \approx 6.251$ THz, far above D/h . Hence room-temperature optical polarization works because the driven optical cycle is out of thermal equilibrium, not because the $m_s = 0$ state is thermally frozen out.

A measured NV–NV coupling. In an implanted pair separated by 25 ± 2 nm at room temperature, the fitted secular dipolar coupling was 4.93 ± 0.05 kHz [R080]. Its energy and equivalent temperature are

$$E_{dd} = 2.039 \times 10^{-5} \text{ } \mu\text{eV}, \quad T_{dd} = 2.366 \times 10^{-7} \text{ K}.$$

That tiny equivalent temperature does not forbid a driven gate at 300 K. It does say that thermal equilibrium cannot select states separated only by this bare pair coupling.

A cryostat set point. At $T = 100$ mK,

$$k_B T = 8.617 \text{ } \mu\text{eV}, \quad \frac{k_B T}{h} = 2.084 \text{ GHz}.$$

An SiV orbital splitting whose cyclic frequency is 46 GHz has equivalent temperature 2.21 K, so its upper branch is strongly suppressed at 100 mK. A 5 kHz dipolar splitting is not. One thermometer reading can therefore be cold for one degree of freedom and effectively infinite-temperature for another.

Suppose you quote only the refrigerator temperature. Then you have said nothing about which splittings equilibrium can see.

27.4. A decay time is not a gap

Let T_2 be a coherence time. For a purely exponential coherence envelope $\exp(-t/T_2)$, the corresponding Lorentzian full width at half maximum is

$$\Delta\nu_{\text{FWHM}} = \frac{1}{\pi T_2}.$$

This is a model-dependent conversion, not a definition of every linewidth. Gaussian Ramsey decay, spectral diffusion, stretched exponentials, and pulse-filtered noise do not obey the same formula. Keep both the protocol and the envelope whenever you can.

A gate duration t_g gives an operation rate $1/t_g$. It is not automatically a Hamiltonian gap. For example, $t_g = 20$ ns gives $1/t_g = 50$ MHz; the resonant Rabi frequency for a π rotation may differ by factors set by the pulse convention. Write “inverse duration” rather than relabel it as an energy.

What fails if you drop the exponential hypothesis? The same T_2 number can be assigned to a quasi-static, calibratable detuning and to an irreversible Markovian linewidth. Those two noises last the same time on a Ramsey plot and disagree about whether a later pulse can undo them.

27.5. Who sits in the upper level

For two nondegenerate levels separated by energy Δ , thermal equilibrium gives

$$\frac{p_1}{p_0} = \exp\left(-\frac{\Delta}{k_B T}\right),$$

where p_0 and p_1 are the lower- and upper-level probabilities. The dimensionless ratio

$$R_T = \frac{\Delta}{k_B T}$$

is the **thermal-resolution ratio**. It answers one question only: does equilibrium even notice the split?

When $R_T \ll 1$, the two levels are nearly equally populated unless a drive, a measurement, or a reservoir engineers a nonequilibrium state. When $R_T \gg 1$, thermal excitation is exponentially suppressed. Degeneracy multiplies the Boltzmann factor, so the equation must be amended when several states share an energy.

For an architecture we also need

$$R_C = \frac{|J|/h}{\Delta\nu_{\text{noise}}}, \quad N_{\text{coh}} = \frac{|J|T_{2,\text{relevant}}}{h}.$$

Here J is the coupling energy, $\Delta\nu_{\text{noise}}$ is the relevant cyclic-frequency broadening, and N_{coh} is roughly the number of coupling cycles available inside the applicable coherence time. “Applicable” means free-induction or Hamiltonian-compatible protection, not whichever decoupling record is largest.

Hence R_T is not R_C , and neither is a many-body gap. A system can be thermally blind to a splitting and still gate it, or thermally resolved and still too noisy to use.

27.6. What the laboratory actually wrote down

The table below is not a leaderboard. It contains conditional brackets from identifiable experiments or, where marked, a transparent conversion anchored by a measurement. A range may represent tensor components, inequivalent defect configurations, or protocols on the same sample; the “conditions” column says which. It is not silently promoted to a population-wide distribution.

The rows favor samples relevant to interacting defects: bulk or nanostructured hosts, implanted pairs where coupling was actually measured, and protocols whose temperature is stated. Records based on aggressive dynamical decoupling are retained but quarantined from undriven analog evolution. Absence of a comparable single-defect metric is recorded as absence, not replaced with an ensemble number from a different task.

All frequency-valued Hamiltonian entries are E/h . “Equivalent T_E ” means E/k_B , not the operating temperature. Uncertainties and ranges keep their source-specific meaning.

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Ground-state zero-field splitting	NV^- , $D/h \approx 2.87$ GHz	$11.87 \mu\text{eV}$; $T_E = 0.1377$ K	Single NVs in bulk type-IIa diamond; room-temperature optical/MW work. The last digits shift with temperature, strain, and sample, so $D/h = 2.87$ GHz is a working value, not a universal exact constant.	[Experiment] [R197]; [R198]
Inequivalent zero-field splittings	Neutral divacancies in 4H-SiC, about 1.30–1.34 GHz	$5.38\text{--}5.54 \mu\text{eV}$; $T_E = 62.4\text{--}64.3$ mK	Single c-axis divacancy configurations in high-purity semi-insulating 4H-SiC at 20 K. The range brackets crystallographically inequivalent configurations, not temperature drift [R096].	[Experiment] [R096]
Ruby zero-field transition	$\text{Cr}^{3+} : \text{Al}_2\text{O}_3$, 11.493 ± 0.004 GHz	$47.53 \pm 0.02 \mu\text{eV}$; $T_E = 551.6 \pm 0.2$ mK	Dilute Cr^{3+} ensemble in ruby measured in a dilution refrigerator; the optically inferred local temperature bottomed at 143 ± 7 mK although the cryostat reached 20 mK. This is an ensemble transition, not a demonstrated single-defect qubit array [R088].	[Experiment] [R088]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Hyperfine tensor	$NV^- - ^{14}N$, $ A /h \approx 2.14\text{--}2.70\text{ MHz}$	$0.00885\text{--}0.01117\text{ }\mu\text{eV}$; $T_E = 0.103\text{--}0.130\text{ mK}$	Room-temperature EPR/ENDOR on NV ensembles. The bracket is the magnitude of axial versus transverse tensor components; it is not center-to-center scatter. Nearby ^{13}C sites can have very different hyperfine couplings [R197].	[Experiment] [R197]
Direct electron-spin coupling	NV^- pair, $4.93 \pm 0.05\text{ kHz}$	$(2.039 \pm 0.021) \times 10^{-5}\text{ }\mu\text{eV}$; $T_E = 0.2366 \pm 0.0024\text{ }\mu\text{K}$	Two ^{15}N -implanted NVs, separation $25 \pm 2\text{ nm}$, room temperature; value is the fitted secular dipolar interaction for that orientation. The same experiment measured double-quantum phase accumulation four times faster, but that is not a fourfold larger microscopic dipole coupling [R080].	[Experiment] [R080]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Dipolar engineering envelope	Electron spins at 10–50 nm: roughly 0.4–100 kHz before angular cancellation	$1.7 \times 10^{-6} - 4.1 \times 10^{-4} \mu\text{eV}$; $T_E \approx 0.02 - 4.8 \mu\text{K}$	[Theory] Derived from electron dipolar r^{-3} scaling and order-unity angular factors, anchored by the measured 25 nm pair above. It is a geometry-conditioned design envelope, not a survey of fabricated pairs. Exchange at atomic separations is excluded.	[Theory] anchored by [R080]
Static strain response	SiV ⁻ orbital transitions, order 0.1–1 PHz per unit strain; demonstrated optical tuning 150 GHz	The observed tuning is 620 μeV , or 7.20 K equivalent	Individual SiVs in an actuated diamond cantilever at 4 K. The susceptibility bracket distinguishes spin-sensitive and orbital/symmetry channels; 150 GHz is the observed tuning span of one optical line, not a coherent spin–spin coupling [R200].	[Experiment] [R200]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Phonon-induced broadening	SiV ⁻ optical D line about 0.10–0.48 GHz over 4–20 K	0.41–1.99 μeV ; $T_E = 4.8\text{--}23\text{ mK}$	Low-strain HPHT/CVD bulk samples; low-temperature linewidth followed approximately $\Gamma = (-1.05 + 24.26T/\text{K})\text{ MHz}$ before crossover, saturating near the lifetime limit around 4 K. This is a measured phonon-sensitive linewidth, not a single-mode coupling constant [R201].	[Experiment] [R201]
Optical linewidth	Same low-strain SiV ⁻ D transition: near 100 MHz at 4 K, increasing with temperature	0.414 μeV ; $T_E = 4.80\text{ mK}$ at 100 MHz	Single-site photoluminescence excitation linewidth in bulk material. Nanostructure strain, spectral diffusion, power broadening, and line choice can widen it; do not assign this bulk value to an implanted nanodevice [R201].	[Experiment] [R201]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
Coherence without/with refocusing	Coupled NV pair: $T_2^* = 22.6\text{--}27.8\ \mu\text{s}$; double-quantum echo $T_2 = 150\text{--}514\ \mu\text{s}$	Exponential-equivalent widths: 11.5–14.1 kHz for T_2^* , and 0.62–2.12 kHz for echo	The two values in each range are the two implanted centers in the same 25 nm pair at room temperature. Echo and Ramsey are separate protocols and must not be merged into one “coherence” number [R080].	[Experiment] [R080]
Coherence under strong protection	SiV [−] : $T_2^* \approx 1.5\text{--}13\ \mu\text{s}$ in $10^{-3}\%$ ^{13}C material; $T_2 = 13 \pm 1.7\ \text{ms}$ with CPMG-32	For 13 ms, exponential-equivalent width $\approx 24\ \text{Hz}$	Single SiV near 100 mK. Natural-abundance diamond in the same study gave $T_2^* \sim 0.3\ \mu\text{s}$. The 13 ms value is pulse-protected memory, not evidence for 13 ms of untouched analog dynamics [R083].	[Experiment] [R083]
Coherence in another host	4H-SiC divacancy, Hahn-echo $T_2 \approx 1.2\text{--}1.25\ \text{ms}$	Exponential-equivalent width $\approx 255\text{--}265\ \text{Hz}$	Isolated neutral divacancies in high-purity semi-insulating 4H-SiC at 20 K. The bracket covers reported single centers/configurations under Hahn echo, not shallow implanted arrays [R096].	[Experiment] [R096]

Quantity	Platform and defensible value	Explicit conversion	Conditions and what the range means	Status/source
MW gate-duration and optical-initialization examples	NV ⁻ : a 20 ns near-inversion MW pulse in the measured Rabi sequence; a 1.3 μ s, 532 nm polarization pulse	Inverse durations: 50 MHz and 0.77 MHz; these are rates, not gaps	Single NV in type-IIa HPHT diamond at 300 K. No average gate fidelity was attached to the 20 ns pulse in this study, so it is a duration example rather than a platform-wide gate benchmark [R198].	[Experiment] [R198]
Readout and initialization fidelity	NV ⁻ : electron initialization $\geq 99.7 \pm 0.1\%$; single-shot electron readout $93.2 \pm 0.5\%$ using a 40 μ s window	Readout inverse duration 25 kHz, not an energy splitting	Single NV in bulk diamond below 10 K using resonant, spin-selective optical excitation. Room-temperature off-resonant fluorescence normally requires repeated averaging and must not inherit this fidelity [R199].	[Experiment] [R199]
Slow cryogenic optical operations	SiV ⁻ : optical pumping timescale about 30 ms; single-shot readout about 89% with 20 ms integration	Inverse durations about 33 Hz and 50 Hz	Single SiV near 100 mK with field aligned to the defect axis; the long coherence experiment used still longer preparation/readout pulses. These durations concern optical cycling and collection, not microwave gate speed [R083].	[Experiment] [R083]

The table deliberately contains awkward comparisons. That is the point. NV centers can be controlled quickly at room temperature, yet a demonstrated 25 nm dipolar coupling sits in the kilohertz range. SiV centers offer narrow, stable optical transitions, yet their orbital phonons force millikelvin operation for the longest spin memory. SiC can show millisecond Hahn echoes at 20 K, but that datum comes from isolated defects in high-quality material, not a densely implanted interacting lattice. Ruby supplies a robust ensemble transition at $E/h = 11.493$ GHz, while modern evidence for individually initialized, coupled Cr^{3+} defect arrays with the complete control stack remains missing [R088].

27.7. One pair, three ledgers

Use the measured room-temperature NV pair as a deliberately honest example. Let the coupling be $J/h = 4.93$ kHz. For the two centers, take $T_2^* = 22.6\text{--}27.8$ μs and echo times 150–514 μs , all from the same sample and protocol family [R080]. Then

$$N_{\text{coh}}^* = \frac{J}{h} T_2^* = 0.11\text{--}0.14,$$

whereas

$$N_{\text{coh}}^{\text{echo}} = 0.74\text{--}2.53.$$

Hence the free-induction result is below one coupling cycle. Echo improves the ledger, and the actual entangling sequence used tailored double-quantum phase accumulation plus refocusing [R080]. A static many-body Hamiltonian cannot automatically borrow that pulse sequence. The pulses may average away the interaction one hoped to retain.

For thermal equilibrium at 300 K,

$$R_T = \frac{h(4.93 \text{ kHz})}{k_B(300 \text{ K})} \approx 7.9 \times 10^{-10}.$$

The same experiment can demonstrate driven entanglement while being nowhere near passive thermal polarization in the interaction eigenbasis. There is no contradiction. Driven quantum control and equilibrium many-body protection are different products.

If an exponential $T_2^* = 25$ μs were the correct model, then $\Delta\nu = 1/(\pi T_2^*) = 12.7$ kHz. Comparing a 4.93 kHz coupling gives $R_C \approx 0.39$. This estimate is conservative only for that line shape and protocol. A quasi-static, calibratable detuning and an irreversible Markovian linewidth can yield the same Ramsey timescale but have very different

consequences for control. The cure is a measured noise spectrum, not a more decorative decimal place.

27.8. Ways to get this wrong

1. **Compare D to the noise and call the coupling safe.** Suppose $D/h = 2.87$ GHz and the linewidth is a kilohertz. Then the on-site splitting is safe, and the interaction that would generate a topological stabilizer has not been mentioned. The relevant comparison is the projected intercluster term against projected noise.
2. **Borrow an optical width for a spin width.** Suppose the optical D line is 100 MHz. That number can matter for photon indistinguishability and resonant readout. Drop the distinction of operators, and you have assigned the optical noise spectrum to the spin. Optical and microwave transitions sample different operators.
3. **Call an inverse time an energy.** Suppose a pulse lasts 20 ns. Then $1/(20 \text{ ns}) = 50$ MHz is useful bookkeeping. What fails if you call it a 50 MHz gap? You have skipped the Hamiltonian. The resonant Rabi frequency for a π rotation may differ by the pulse convention.
4. **Confuse the cryostat with the sample.** Suppose the refrigerator reaches 20 mK. In the ruby experiment that did not imply a 20 mK optically illuminated spin ensemble; the inferred local temperature was 143 ± 7 mK at low optical power [R088]. Laser heating, imperfect thermal anchoring, and microwave dissipation belong in the passport.
5. **Import protection that changes the Hamiltonian.** Dynamical decoupling extends memory by reversing selected couplings. What fails if the same toggling-frame average cancels J ? The longer T_2 is numerically impressive and architecturally weightless. High-fidelity resonant NV readout below 10 K does not imply passive low-temperature order [R199]. Conversely, weak thermal polarization does not prevent active optical initialization.
6. **Call a driven pair a topological qubit.** A pulse sequence creating entanglement between two NVs is an experiment on driven physical qubits. It is not an encoded topological qubit, an emergent anyon, or evidence for topological order. A future cluster projection would produce new effective couplings and new effective noise; those scales must be derived rather than copied from the physical-spin table.
7. **Cherry-pick records.** A record coherence time often combines an unusually pure host, a deep defect, low defect density, low temperature, and many refocusing pulses. A dense implanted array tends to move several of those conditions in the wrong direction. What fails if you combine the best coherence from one sample, the best

optical linewidth from another, and the closest pair from a third? You have described a crystal that was never grown. Keep T_2^* , echo T_2 , and dynamically decoupled T_2 as separate columns, and keep local temperature under illumination, not merely refrigerator temperature.

[Proposal] Until a candidate cluster is fabricated, use scenario ranges rather than a single “best” input: a pessimistic case tied to implanted or shallow data, a base case tied to reproducible bulk or device data, and an optimistic case that requires every enabling condition to be stated. Do not combine the best coherence from one sample, the best optical linewidth from another, and the closest pair from a third as though they were one crystal. That imaginary sample is always excellent.

[Experiment] Across demonstrated defect platforms, local cyclic frequencies E/h span roughly 1–12 GHz in the specific NV, 4H-SiC divacancy, and ruby cases tabulated here, while directly measured tens-of-nanometres electron dipolar interactions can sit at kilohertz scales [R197]; [R080]; [R096]; [R088]. Optical strain susceptibilities can be enormous, but static optical tuning is not itself a coherent interdefect interaction [R200]. Phonon processes can change linewidths strongly between 4 and 20 K for SiV centers [R201].

[Theory] These facts establish a hierarchy problem, not impossibility. Projection and perturbative gadgets usually reduce useful intercluster scales further, while disorder and leakage remain. Any viable proposal must calculate the effective J_{eff} , the effective linewidth, and the many-body gap from the same conditional parameter set.

[Speculation] It is possible that strain-enhanced, phonon-mediated, exchange-assisted, or cavity-mediated designs move beyond bare dipolar kilohertz scales. This chapter supplies no experimental basis for assigning such a design a particular topological gap. A proposed coupling belongs in a separate row until measured.

27.9. Check yourself

1. **Show that if $H = \hbar g S_x$ with $g = 2\pi \times 5$ MHz, the cyclic frequency in this ledger is 5 MHz.**
 $g/(2\pi) = 5$ MHz; the energy is $\hbar g = h(5 \text{ MHz})$.
2. **What are the energy and equivalent temperature of 1 MHz?**
 $0.0041357 \mu\text{eV}$ and $47.992 \mu\text{K}$.
3. **Show that an NV splitting with $D/h = 2.87$ GHz does not thermally polarize the spin at 300 K.**
 Its equivalent temperature is only 0.1377 K. Optical pumping produces the useful nonequilibrium polarization.

4. What fails if you insert the 13 ms SiV coherence into a static analog simulation?

It used CPMG-32 near 100 mK in isotopically purified diamond. The pulse sequence and isotope conditions must be compatible with the intended Hamiltonian; the protection can average away the interaction one hoped to keep.

5. What must accompany a readout fidelity?

Platform, sample, temperature, optical protocol, integration time, state definition, and whether the result is single-shot.

6. What fails if you compare the zero-field splitting to the noise and declare the coupling safe?

D is an on-site splitting. The relevant comparison is the projected useful coupling, and ultimately the many-body gap, against projected noise.

A number without a passport is a rumor with units. Hamiltonian coefficients are cyclic frequencies E/h . Equivalent temperature is E/k_B , not the cryostat. The laboratory's local cyclic frequencies span roughly 1–12 GHz in the tabulated NV, 4H-SiC divacancy, and ruby cases, while a measured 25 nm electron dipolar coupling sits at 4.93 kHz. Driven entanglement at 300 K and thermal resolution of that same coupling are different products. The next chapter asks how far apart the defects actually are.

27.10. Sources

- [R196] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th ed., version 3.01, 2024. Stable URL: <https://www.bipm.org/en/publications/si-brochure>.
- [R197] S. Felton *et al.*, “Hyperfine interaction in the ground state of the negatively charged nitrogen vacancy center in diamond,” *Physical Review B* **79**, 075203 (2009). DOI: 10.1103/PhysRevB.79.075203.
- [R080] F. Dolde *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545; arXiv: 1212.2804.
- [R198] L. Robledo *et al.*, “Spin dynamics in the optical cycle of single nitrogen-vacancy centres in diamond,” *New Journal of Physics* **13**, 025013 (2011). DOI: 10.1088/1367-2630/13/2/025013; arXiv: 1010.1192.
- [R199] L. Robledo *et al.*, “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* **477**, 574–578 (2011). DOI: 10.1038/nature10401; arXiv: 1301.0392.
- [R083] D. D. Sukachev *et al.*, “Silicon-vacancy spin qubit in diamond: a quantum memory exceeding 10 ms with single-shot state readout,” *Physical Review Letters* **119**, 223602 (2017). DOI: 10.1103/PhysRevLett.119.223602; arXiv: 1708.08852.

- [R200] S. Meesala *et al.*, “Strain engineering of the silicon-vacancy center in diamond,” *Physical Review B* **97**, 205444 (2018). DOI: 10.1103/PhysRevB.97.205444; arXiv: 1801.09833.
- [R201] K. D. Jahnke *et al.*, “Electron–phonon processes of the silicon-vacancy centre in diamond,” *New Journal of Physics* **17**, 043011 (2015). DOI: 10.1088/1367-2630/17/4/043011; arXiv: 1411.2871.
- [R096] D. J. Christle *et al.*, “Isolated electron spins in silicon carbide with millisecond coherence times,” *Nature Materials* **14**, 160–163 (2015). DOI: 10.1038/nmat4144; arXiv: 1406.7325.
- [R088] V. K. Sewani *et al.*, “Spin thermometry and spin relaxation of optically detected Cr^{3+} ions in Al_2O_3 (ruby),” *Physical Review B* **102**, 104114 (2020). DOI: 10.1103/PhysRevB.102.104114; arXiv: 2007.07493.

Chapter 28. How far apart, and did the right atom land there?

Two NV centers, ten nanometres apart, would have a point-dipole coefficient of 52 kHz if they sat exactly on the dots you drew. Draw the dots, fire a nitrogen through a hole, anneal, and look for the glow. You have specified a target. You have not specified a pair.

The energy is set by the actual displacement and by whether the intended charge-stable spin formed at each end. If the two centers land 10 nm apart on average but wander by several nanometres, the couplings smear. If only a few sites hold the intended spin, the picture is mostly holes.

A commanded spacing is not a device. We take this as the problem: the Hamiltonian cares about the measured positions of centers that pass optical, spin, charge, and coherence tests. Everything else is a proxy.

28.1. The beam is not the stop

An ion is told a coordinate. The crystal answers with at least five lengths, and they are not five names for resolution.

1. Where the incoming ion crosses the surface. That is the beam spot, or the hole in the mask.
2. Where the ion actually stops, after rattling through the solid.
3. How far a mobile vacancy wanders during the anneal before it is captured, recombines, clusters, or dies at a surface.
4. The chance that the delivered ion or created vacancy becomes the named structural defect, in the named charge.
5. The measured position *after* you keep only the centers that also pass optical, spin, charge, and coherence tests.

The sideways and downward spray in (2) is **implantation straggle**. The name is worth keeping because the brochure quotes the beam, and the crystal answers with this extra scatter.

A sub-10-nm beam can feed a 20-nm straggle distribution. A sharply stopped implanted nitrogen can capture a vacancy that has wandered. A center can sit in the right place and

still be optically dark. The instrument brochure normally quotes the first number, because that is the number the instrument controls. The Hamiltonian cares about the fifth.

A useful schematic is

```
commanded site
|
+-- surface entry: beam/aperture distribution
|
+-- stopping point: entry convolved with ion straggle
|
+-- annealed complex: vacancy migration and capture
|
+-- created center: structural/charge conversion succeeds
|
`-- usable center: position measured; spin passes acceptance tests
```

“Convolved” means that independent random displacements add as random variables, not as marketing adjectives. For independent Gaussian one-dimensional errors with standard deviations $\sigma_1, \sigma_2, \dots$, the total standard deviation is

$$\sigma_{\text{total}} = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots}.$$

Hence independent errors combine under a square root. Systematic registration offsets must instead be added as biases and calibrated. They do not disappear under that square root.

What fails if you quote only the beam? You have specified entry, not stopping, not capture, and not the conditional distribution of centers that later pass a spin test.

28.2. Two spins and a cube of distance

Take two localized electron spins separated by a displacement vector $\mathbf{r}$. Its length is r , and $\hat{\mathbf{r}} = \mathbf{r}/r$ is its direction. Let $\mathbf{S}_1$ and $\mathbf{S}_2$ be dimensionless spin operators, so their eigenvalues are pure numbers. For electron Landé factors g_1 and g_2 , the point-dipole energy is

$$H_{dd} = h\nu_d(r) [\mathbf{S}_1 \cdot \mathbf{S}_2 - 3(\mathbf{S}_1 \cdot \hat{\mathbf{r}})(\mathbf{S}_2 \cdot \hat{\mathbf{r}})],$$

where h is Planck’s constant and

$$\nu_d(r) = \frac{\mu_0}{4\pi} \frac{g_1 g_2 \mu_B^2}{hr^3}.$$

Here μ_0 is the vacuum permeability and μ_B is the Bohr magneton. The frequency ν_d is the **dipolar coefficient**. It is not automatically the observed transition splitting or gate rate. Those additionally contain the angular factor, spin-state matrix elements, detuning, and pulse-sequence conventions.

For two electron-like spins with $g_1 = g_2 = 2.0023$, using rounded 2022 CODATA values [R208],

$$\frac{\mu_0}{4\pi} \approx 10^{-7} \text{ N A}^{-2}, \quad \mu_B = 9.274010 \times 10^{-24} \text{ J T}^{-1},$$

$$h = 6.62607015 \times 10^{-34} \text{ J s},$$

we obtain

$$\nu_d(r) \approx \frac{52.04 \text{ MHz nm}^3}{r^3}.$$

A tesla is $\text{T} = \text{N (A m)}^{-1}$. Therefore

$$(\text{N A}^{-2}) \frac{(\text{J T}^{-1})^2}{(\text{J s}) \text{m}^3} = (\text{N A}^{-2}) \frac{(\text{A m}^2)^2}{(\text{N m s}) \text{m}^3} = \text{s}^{-1}.$$

So ν_d is a frequency. Substituting $r = 1 \text{ nm} = 10^{-9} \text{ m}$ gives $5.204 \times 10^7 \text{ s}^{-1} = 52.04 \text{ MHz}$. The cubic scaling then supplies every other row.

Separation r	r^3 (nm ³)	Dipolar coefficient ν_d	Square-lattice density proxy $1/r^2$ (sites/ μm^2)
1 nm	1	52.0 MHz	1,000,000
2 nm	8	6.50 MHz	250,000
5 nm	125	416 kHz	40,000
10 nm	1,000	52.0 kHz	10,000
20 nm	8,000	6.50 kHz	2,500
50 nm	125,000	416 Hz	400

The density column is only the geometric value for one site per square of side r . Real honeycomb, kagome, cluster, routing, and optical-access layouts have different areas per site. The dipolar values agree in scale with **[Experiment]** coherent coupling of two NV electron spins whose inferred separation was $9.8 \pm 0.3 \text{ nm}$; the observed coupling was in the tens-of-kilohertz range after geometry was included [R181]. That experiment established a coupled pair, not a deterministic 10,000-site-per-square-micrometre array.

The angular operator matters. With both spin moments quantized along a unit vector $\hat{\mathbf{z}}$, a common secular coefficient contains

$$A(\theta) = 1 - 3 \cos^2 \theta,$$

where θ is the angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{z}}$. It ranges from -2 to $+1$ and vanishes at the magic angle $\theta = \cos^{-1}(1/\sqrt{3}) \approx 54.7^\circ$. Hence “10 nm gives 52 kHz” means a coefficient scale. It does not promise a 52 kHz usable edge.

Drop the angular factor, and a pair sitting at the magic angle looks like a missing coupling rather than a geometry. Keep it, and 52 kHz is an envelope, not a delivered gate.

28.3. When two nanometres is not a dipole

At 1–2 nm, treating each defect as a point magnetic dipole is no longer a complete microscopic model. Electronic wavefunctions can overlap. Charge configuration, relative crystallographic orientation, strain, and the exact atomic realization matter.

The leftover short-range piece is **exchange**. It is not a polite correction to the table: it often changes roughly exponentially rather than as a universal power law. **[Numerics]** Atomistic calculations for diamond spin centers found strongly orientation-dependent exchange that can exceed dipolar coupling for some NV-pair geometries below roughly 3 nm [R107].

This does **not** mean that every 2 nm pair receives a convenient, strong, identical exchange link. It means the 6.50 MHz dipolar entry is only the long-range term in a site-specific Hamiltonian. At these distances one must calculate or measure the actual pair, check that two recognizable defects and their desired charge states remain stable, and include exchange, tunnelling, altered orbital levels, and implantation damage. Adding a guessed “exchange bonus” to the table would be fiction with units.

What fails if you keep only the r^{-3} term below 2 nm? You have kept the piece that is universal and dropped the piece that can dominate.

28.4. How a wobble becomes a noisy coupling

Let the intended separation be r_0 , and let δr be a small error in the separation along the bond direction. Because $\nu_d = Cr^{-3}$, where $C = 52.04 \text{ MHz nm}^3$ for the electron-like case,

$$\frac{dv_d}{dr} = -3Cr^{-4} = -\frac{3v_d}{r}.$$

To first order,

$$\frac{\delta v_d}{v_d} \approx -3 \frac{\delta r}{r_0}.$$

Suppose each endpoint has an independent one-axis placement standard deviation σ_{site} along the bond. The relative displacement then has

$$\sigma_r = \sqrt{2} \sigma_{\text{site}}, \quad \frac{\sigma_v}{v_d} \approx \frac{3\sqrt{2} \sigma_{\text{site}}}{r_0}.$$

This is an optimistic radial-only estimate. It omits angular-factor disorder, non-Gaussian tails, missing sites, and exchange variability. Requiring radial dipolar disorder below 10% gives

$$\sigma_{\text{site}} < \frac{0.10}{3\sqrt{2}} r_0 \approx 0.0236 r_0.$$

Target spacing	Maximum independent per-site σ_{site} for 10% radial coupling scatter
1 nm	0.024 nm
2 nm	0.047 nm
5 nm	0.118 nm
10 nm	0.236 nm
20 nm	0.471 nm
50 nm	1.18 nm

These are not claimed fabrication capabilities. They are derived tolerances. Even a 2 nm per-site standard deviation would produce an estimated $3\sqrt{2}(2/10) \approx 0.85$, or 85%, radial coupling scatter around a 10 nm design. Moving to 50 nm relaxes the relative tolerance, but the bare coefficient has fallen to 416 Hz. Hence spatial design has no free lunch. It does provide an itemized bill.

The linear expression fails when σ_r/r_0 is not small. It can also assign probability to negative scalar distances if used carelessly. A proper disorder model samples three-dimensional endpoint positions, computes $\mathbf{r}$, evaluates the full dipolar tensor and any short-range terms, and rejects atomically impossible or merged-defect configurations. Report quantiles of J , not just J at the mean distance: because r^{-3} is convex, rare close pairs create a long strong-coupling tail.

What fails if you quote only J at the mean distance? Convexity of r^{-3} puts extra weight on the close tail, so the typical coupling and the mean coupling need not agree.

28.5. Usable only if every yes holds

A site is usable only if several things happen in a row. Define the following events for one target site: A , the intended ion or vacancy is delivered; C , the desired structural complex forms; Q , it has the usable charge state; O , it can be initialized and read out; S , its spin and coherence pass specification; and G , its final geometry lies inside tolerance. The usable-site probability is exactly

$$p_{\text{use}} = P(A)P(C | A)P(Q | A, C)P(O | A, C, Q)P(S | A, C, Q, O)P(G | A, C, Q, O, S).$$

That product is a **yield**. It is a chain of conditional probabilities, not a single success.

The factors need not be independent. “Single-ion detected” addresses A , not the rest. “20% conversion” addresses a version of C under stated processing conditions, not p_{use} . “33 nm placement” is meaningful only after specifying whether it is lateral, depth, one standard deviation, a mean radial offset, or a confidence interval.

If N sites were independent and every one were required, the defect-free array yield would be p_{use}^N . At $p_{\text{use}} = 0.96$, even 100 sites give $0.96^{100} \approx 1.69\%$. At 1,000 sites the value is about 1.9×10^{-18} . Repair, repeated implantation, movable qubits, redundant layouts, or tolerance of missing sites can change this conclusion. Silently replacing site yield by array yield cannot.

A related trap appears when the number of created centers per target is Poisson distributed with mean λ . The probability of exactly one center is

$$P(1) = \lambda e^{-\lambda},$$

which is largest at $\lambda = 1$, where $P(1) = e^{-1} \approx 36.8\%$. Dose tuning alone cannot beat that ceiling under a Poisson model. Feedback or postselection changes the process rather than winning an argument with the exponential.

Hence a cheerful percentage is not a device yield until the denominator, the conditioning, and the array size are named.

28.6. What the laboratory actually has

The following rows deliberately preserve different definitions. They should not be compressed into a leaderboard called “resolution.”

Host and process	Surface-entry control	Straggle/stopping distribution	Vacancy motion	Reported creation yield	Final measured placement or usability statement
Diamond, 20 keV N component of implanted CN ⁻ through a PMMA mask	80 nm diameter aperture	[Numerics] SRIM stopping depth 30 ± 10 nm and about 9 nm straggle	800 °C, 2 h anneal; diffusion contribution not separately measured	[Experiment] mean 3.5 NV centers from about 50 N atoms per aperture, approximately 7% N-to-NV conversion, over 49 sites [R202]	Emission sites followed the mask; the paper did not establish a sub-10-nm final usable-position distribution [R202]
Diamond, focused Si implantation for SiV	Typical beam FWHM below 40 nm	[Numerics] 19 nm lateral straggle; combining a 40 nm FWHM beam ($\sigma \approx 17$ nm) and straggle predicted $\sigma \approx 26$ nm	High-temperature anneal; vacancy capture controls conversion	[Experiment] up to about 2.5% at 100 keV; electron irradiation plus re-anneal raised a tested bulk-sample value to about 20% [R190]	[Experiment] created single SiVs had 32 nm one-axis standard deviations; radial offset 40 ± 20 nm. Cavity targeting gave 48 ± 21 nm [R190]
Diamond, femtosecond vacancy writing plus local annealing and optical feedback	Diffraction-scale optical focus, not an ion beam	No implanted-ion straggle	Vacancy creation and local annealing are the mechanism; feedback stops once emission is detected	[Experiment] approximately 96% single-NV creation yield under the reported feedback protocol [R203]	[Experiment] about 33 nm in-plane positioning deviation; this is not equivalent to 33 nm three-dimensional nearest-neighbour control [R203]

Host and process	Surface-entry control	Straggle/stopping distribution	Vacancy motion	Reported creation yield	Final measured placement or usability statement
Diamond, nitrogen delta-doping during growth plus irradiation/anneal	No deterministic lateral coordinate	[Experiment] a roughly 1–2 nm doped layer; cap thickness selected nominal NV depths from about 5 to 100 nm	Vacancies supplied after growth and annealed into the N layer	Not reported as a deterministic per-site useful yield in this study	[Experiment] tight depth engineering, but no lateral array placement; shallow nominal 5 nm layers retained $T_2 > 100 \mu s$ under reported conditions [R204]
Diamond, delta-doped N plus localized electron irradiation	200 keV electron beam with a 20 nm spot	Electrons create vacancies along a narrow path; the N layer, rather than ion stopping, sets depth	850 °C, 11 min anneal; [Numerics] diffusion-capture modeling accompanied the measurement	[Experiment] NV number was dose-tunable from about 0 to 10; the reported device metric was a threefold increase in expected single-electron-spin magnetometer yield, not an absolute ion-to-NV conversion [R207]	[Experiment] about 4 nm depth confinement and lateral standard deviation 46(1) nm in 280 nm pillars; mean Hahn-echo $T_2 = 98(37) \mu s$ for characterized NVs [R207]

Host and process	Surface-entry control	Straggle/stopping distribution	Vacancy motion	Reported creation yield	Final measured placement or usability statement
4H-SiC, 30 keV carbon implantation through a lithographic mask	Mask-defined target arrays	[Numerics] implantation selected shallow silicon vacancies at roughly 40 nm depth	Host- and defect-specific activation; diamond diffusion numbers do not transfer	[Experiment] $19 \pm 4\%$ silicon-vacancy generation efficiency and $34 \pm 4\%$ probability of a single emitter at optimized dose [R205]	Predetermined arrays were demonstrated, but those yield figures do not by themselves establish sub-10-nm final pair-spacing distributions [R205]

The SiV row is the bookkeeping done out loud. Suppose the beam FWHM is 40 nm. Converting to a Gaussian standard deviation gives $40/2.355 \approx 17$ nm. Combining that with 19 nm straggle gives $\sqrt{17^2 + 19^2} \approx 25.5$ nm, consistent with the paper's predicted 26 nm. The measured created-center distribution was broader, at 32 nm per axis [R190]. Hence a “below 40 nm” beam was not a below-40-nm final location. The discrepancy is why the endpoint has to be measured.

The CN-mask row supplies another propagation, not a measurement. If entry points were uniform across a circular aperture of radius $a = 40$ nm, one Cartesian coordinate would have $\sigma_x = a/2 = 20$ nm. Combining that idealized aperture distribution with 9 nm straggle gives $\sqrt{20^2 + 9^2} \approx 22$ nm before adding alignment, diffusion, channeling tails, or optical-localization uncertainty. This is an illustrative use of the reported inputs, **not** a measured final NV standard deviation [R202]. A narrow pitch between apertures does not make centers within an aperture narrowly spaced.

Vacancy diffusion can create the center or erase the intended geometry. **[Experiment]** In nitrogen-rich type-Ib diamond implanted with focused Ar ions, Racke and colleagues measured 0.04–0.79 created NV centers per implanted Ar ion over 12–132 keV [R206]. **[Numerics]** Their model of vacancy loss to the surface placed an upper bound of about 300 nm on the single-vacancy diffusion length for an 800 °C anneal [R206]. That upper bound is not a claim that every NV moves 300 nm, nor can it be inserted as a Gaussian placement error. The implanted nitrogen is comparatively immobile; a mobile vacancy must reach a neighboring lattice site to form the complex. The result is a reaction-and-capture distribution conditioned on nitrogen density, depth, surfaces, damage, charge state, temperature, and time.

The methods do different jobs. Delta doping can make depth thin while leaving lateral position random. A focused beam can target lateral coordinates while ion scattering broadens depth and lateral position. Laser feedback can raise center-creation yield without providing atomic-scale three-dimensional registration. **[Experiment]** By 2025, delta doping combined with localized electron-created vacancies had demonstrated about 4 nm depth confinement and 46(1) nm lateral confinement in 280 nm diamond pillars, but not the sub-nanometre relative placement required for a uniform 10 nm dipolar graph [R207]. Using such hybrid methods for a dense interacting lattice remains **[Proposal]** until the final pair distribution and usable-array yield are measured.

A regularly spaced fluorescence image is an array of physical emitters. It is not yet an array of coherently coupled qubits. A few coupled defects are not an encoded cluster. A fabricated interaction graph is not evidence of a topological phase. Spatial success supplies one necessary input to the microscopic Hamiltonian. It does not certify emergence, a many-body gap, or a logical qubit.

28.7. Ways to get this wrong

1. **Quote the beam and forget the crystal.** Suppose the ion-optical spot is 10 nm. That specifies entry, not stopping. The reported SiV experiment had a nominal beam smaller than 40 nm, 19 nm calculated lateral straggle, and a 32 nm measured one-axis created-center spread [R190]. Drop any of the three and the remaining number is a different object.
2. **Treat a transport simulation as an endpoint measurement.** Transport simulations are valuable **[Numerics]**. They model ion trajectories and damage under assumptions about an initially ideal target. What fails if you stop there? Charge-state stability, annealed-complex location, channeling tails in a particular crystal, and usable-spin yield have not been measured. Validate simulated distributions with microscopy, spectroscopy, or coupling-based localization.
3. **Call conversion yield deterministic placement.** A detected single ion can still fail to form the requested center. Conversely, a 96% laser-feedback creation result does not mean that 96% of sites land inside a 1 nm tolerance or pass a coherence threshold [R203]. Drop the denominator, the conditioning, or the acceptance tests, and the percentage is free to wander.
4. **Ignore tails and nearest unwanted neighbours.** Mean and standard deviation are insufficient when rare close pairs dominate r^{-3} coupling. Damage-related paramagnetic defects may also sit nearer than the intended qubit and couple more strongly. What fails if you report only the mean distance? Convexity of r^{-3} puts extra weight on the close tail. Report a three-dimensional point-process distribution, tail quantiles, and identities of dark spins where possible.

5. **Import diamond statistics into another host.** Stopping distributions depend on ion species, energy, crystal orientation, density, channeling, and target composition. Vacancy migration and complex formation depend on the host's defect chemistry. The 300 nm diamond upper bound [R206], 7% CN-to-NV result [R202], and SiC yield [R205] are separate conditioned experiments, not interchangeable constants. For sapphire in particular, a proposed array has no defensible spatial budget until the chosen defect, charge state, implantation species, anneal, and final spin acceptance measurement are specified.
6. **Assume stronger is automatically better.** At 1–2 nm, exchange, tunnelling, altered defect identity, and damage may defeat the intended isolated-spin model [R107]. At 50 nm, placement is easier but 416 Hz bare dipolar coupling may lose to linewidth and decoherence. The viable window is where a *measured distribution* of useful couplings outruns the Chapter 27 noise rates without leaving the Hamiltonian model's domain.

28.8. Check yourself

1. **What fails if a focused beam has a 10 nm spot and you take that as the final defect uncertainty?**
Ion straggle, registration bias, annealing/capture physics, and the conditional distribution of successfully created usable centers remain.
2. **Show that the electron-spin dipolar coefficient at 20 nm is 6.50 kHz.**
 $52.04 \text{ MHz nm}^3 / 20^3 = 6.50 \text{ kHz}$, before angular and matrix-element factors.
3. **What fails if you treat the 1 nm table entry as a complete coupling prediction?**
Wavefunction overlap and site-specific exchange, tunnelling, charge, and structural changes can rival or dominate the point-dipole term.
4. **Show that independent 1 nm bond-direction placement errors at a 10 nm pitch give about 42% radial dipolar scatter.**
 $3\sqrt{2}(1/10) \approx 0.42$ in the small-error approximation; angular disorder is additional.
5. **Does a 20% conversion yield imply a 20% usable-site yield?**
No. Charge state, readout, coherence, and geometric-tolerance probabilities remain, with potentially correlated conditioning.
6. **What must a fabrication paper report for this architecture?**
Three-dimensional final-position distributions, non-Gaussian tails and biases, exact creation-yield denominator, charge/spin acceptance, missing and extra defects, pair-coupling distribution, and array-level yield under stated processing conditions.

A 10 nm drawing is a wish. For electron-like spins the point-dipole coefficient is $52.04 \text{ MHz nm}^3/r^3$, giving 52.0 MHz, 6.50 MHz, 416 kHz, 52.0 kHz, 6.50 kHz, and 416 Hz at 1, 2, 5, 10, 20, and 50 nm. Angular factors and short-range exchange stay separate. Small radial errors give $\sigma_J/J \approx 3\sqrt{2}\sigma_{\text{site}}/r$, so 10% coupling uniformity wants per-site bond-direction error below about $0.0236r$. The laboratory has useful pieces — coupled 10 nm-class pairs, nanometre-scale stopping calculations, tens-of-nanometre measured final placement, and process-dependent creation yields — but not a dense, defect-free, sub-nanometre-tolerance interaction graph. Spacing is not a device.

28.9. Sources

- [R181] P. Neumann *et al.*, “Quantum register based on coupled electron spins in a room-temperature solid,” *Nature Physics* **6**, 249–253 (2010). DOI: 10.1038/nphys1536; arXiv:1004.5090.
- [R107] V. R. Kortan, C. Şahin, and M. E. Flatté, “Nanometer-scale exchange interactions between spin centers in diamond,” *Physical Review B* **93**, 220402(R) (2016). DOI: 10.1103/PhysRevB.93.220402; arXiv:1603.03485.
- [R202] P. Spinicelli *et al.*, “Engineered arrays of NV color centers in diamond based on implantation of CN^- molecules through nanoapertures,” *New Journal of Physics* **13**, 025014 (2011). DOI: 10.1088/1367-2630/13/2/025014; arXiv:1008.1483.
- [R190] T. Schröder *et al.*, “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* **8**, 15376 (2017). DOI: 10.1038/ncomms15376; stable full text: PMC5458551.
- [R203] Y.-C. Chen *et al.*, “Laser writing of individual nitrogen-vacancy defects in diamond with near-unity yield,” *Optica* **6**, 662–667 (2019). DOI: 10.1364/OP-TICA.6.000662.
- [R204] K. Ohno *et al.*, “Engineering shallow spins in diamond with nitrogen delta-doping,” *Applied Physics Letters* **101**, 082413 (2012). DOI: 10.1063/1.4748280; arXiv:1207.2784.
- [R205] J. Wang *et al.*, “Efficient generation of an array of single silicon-vacancy defects in silicon carbide,” *Physical Review Applied* **7**, 064021 (2017). DOI: 10.1103/PhysRevApplied.7.064021.
- [R206] P. Räcke, L. Pietzonka, J. Meijer, D. Spemann, and R. Wunderlich, “Vacancy diffusion and nitrogen-vacancy center formation near the diamond surface,” *Applied Physics Letters* **118**, 204003 (2021). DOI: 10.1063/5.0046031.
- [R207] S. Kim *et al.*, “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* **16**, 9803 (2025). DOI: 10.1038/s41467-025-64758-4.
- [R208] P. J. Mohr *et al.*, “CODATA recommended values of the fundamental physical constants: 2022,” *Reviews of Modern Physics* **97**, 025002 (2025). DOI: 10.1103/RevModPhys.97.025002; stable constants database: NIST Reference on Constants, Units, and Uncertainty.

Chapter 29. A coupling wearing a nicer hat

Write J next to two defects. Write Δ_{topo} on the same line. The second symbol has not been earned.

J is a coupling between neighbors — maybe fifty megahertz if you write it as a frequency. A many-body gap is the energy of the first excitation that leaves the ground-state family of the whole lattice. Those are different objects. Projection, a fourth-order gadget, temperature, disorder, and decoherence can only reduce what you may spend. They cannot mint the gap from J by renaming.

We will write every cut in the same units. There is no measured material Δ_{topo} to insert at the end.

29.1. Same units, or the comparison is empty

Several scales compete in one spectrum. A ledger forces every scale into the same units and stops you from counting the same joule twice. Energies are not literally spent. They still have to be ordered.

We will usually quote an energy E as a frequency E/h , where h is Planck's constant. A table entry of 1 MHz means the energy $h \times 10^6 \text{ s}^{-1}$, not an angular frequency. Temperature converts by

$$\frac{k_B T}{h} = 20.8366 \frac{\text{GHz}}{\text{K}} T = 20.8366 \frac{\text{MHz}}{\text{mK}} T.$$

Here k_B is Boltzmann's constant and T is thermodynamic temperature in kelvin. The numerical conversion follows from the exact SI values of k_B and h [R196]. Writing energies as ordinary frequencies keeps stray factors of 2π from breeding.

Hence 1 mK is 20.8366 MHz in these units, and a kilohertz coupling is 4.8×10^{-5} mK. Mix cyclic frequency with angular frequency, or mix the cryostat with the sample, and the ledger is already lying.

29.2. One tile is not a phase

Start with an ideal effective term

$$H_p = -KB_p,$$

where $K > 0$ is an energy and B_p is a dimensionless operator with eigenvalues $+1$ and -1 . Calling p a plaquette is suggestive. One tile is still not a topological phase.

Its two energies are $-K$ and $+K$, so reversing its eigenvalue costs

$$\Delta_p = (+K) - (-K) = 2K.$$

That subtraction, not the symbol K by itself, is the excitation energy. In a closed lattice, constraints can require excitations to appear in pairs, in which case the lowest allowed bulk process may cost $4K$. Boundaries can change that factor. A genuine value of Δ_{topo} has to come from the spectrum and boundary conditions of the stated model.

Now suppose K is not sitting on the table. In the strongly anisotropic limit of Kitaev's honeycomb model, fourth-order perturbation theory generates a plaquette coefficient of the form [R017]

$$K = \frac{J_x^2 J_y^2}{16\Lambda^3}.$$

Here J_x and J_y are weak bond energies and Λ denotes the strong bond scale (called J_z in the original model). Setting $J_x = J_y = J$ gives

$$K = \frac{J^4}{16\Lambda^3} = \frac{J}{16} \left(\frac{J}{\Lambda} \right)^3.$$

The dimensions check: $[J^4/\Lambda^3] = \text{energy}$. If $J/\Lambda = 0.1$, the coefficient is $J/16,000$, before disorder or other corrections enter. **[Theory]** This fourth-order result is established for the specified honeycomb limit [R017]. It is not a derived coefficient for a diamond, SiC, or sapphire defect array, and the honeycomb phase in this limit is not doubled Fibonacci order.

For a conservative toy ledger below, we define

$$\Delta_{\text{topo}} = 2K.$$

[Assumption] That factor of two represents one violated effective term. It is deliberately less favorable than a periodic pair threshold of $4K$. It remains an illustrative assumption, not a material gap claim.

What fails if you quote K as the gap? You have named the coefficient of a local term, not the first allowed excitation of the lattice. Pair creation and boundaries can change the factor, and a crystal does not inherit K from a paper that chose it freely.

29.3. The first state that is not family

Let $E_0 \leq E_1 \leq E_2 \leq \dots$ be the eigenenergies of the finite effective Hamiltonian. A topologically ordered system may have several nearly degenerate ground states. Let $\mathcal{G}$ denote that low-energy family. Define

$$\Delta_{\text{topo}} = E_{\text{first outside } \mathcal{G}} - E_{\text{top of } \mathcal{G}}.$$

That difference is the **topological gap**. The name is worth keeping because it is a many-body spectral object, not a new hat for J .

The finite-size width inside $\mathcal{G}$ is a different quantity,

$$\delta_{\text{fs}} = E_{\text{top of } \mathcal{G}} - E_{\text{bottom of } \mathcal{G}}.$$

Both have units of joules. We will quote Δ_{topo}/h and δ_{fs}/h in hertz.

A cluster leakage gap Λ , an optical transition energy, and a bare nearest-neighbor coupling J are not Δ_{topo} . They constrain the derivation of the effective Hamiltonian. Only diagonalization, a controlled analytic solution, or defensible many-body numerics can supply its gap. Exactly solvable commuting-projector string-net models demonstrate how a local Hamiltonian can possess topological order [R018], but their overall energy scale is freely chosen in the abstract model. A crystal does not inherit that normalization.

A defect spin is a physical degree of freedom. A cluster doublet is an encoded local degree of freedom. A circuit that prepares a string-net wavefunction is a digital emulation unless the undriven hardware Hamiltonian already has the phase. Δ_{topo} is a property of the many-body Hamiltonian and its spectrum. A logical qubit additionally requires controlled initialization, operations, and readout inside that family.

Measuring a two-spin avoided crossing establishes none of the last three items by itself.

Drop the distinction between δ_{fs} and Δ_{topo} , and a small splitting inside the family looks like a small gap. Small δ_{fs} is good. Small Δ_{topo} is bad.

29.4. Each virtual hop takes a cut

Write the microscopic Hamiltonian as

$$H = H_0 + V,$$

where H_0 separates a retained cluster subspace from leakage states by energy Λ , and V contains weaker intercluster couplings of characteristic local strength J . The dimensionless ratio

$$\epsilon = \frac{J}{\Lambda}$$

is the **expansion parameter**. Every later power of it is a cut you can see.

For an interaction first generated at perturbative order n , dimensional analysis gives

$$K_n = c_n \frac{J^n}{\Lambda^{n-1}} = c_n J \epsilon^{n-1},$$

where c_n is a dimensionless coefficient fixed by the actual graph, matrix elements, denominators, and interference among virtual paths. Perturbative gadgets can generate higher-body interactions from two-body couplings, but their target terms and errors must be bounded together [R209]. **[Theory]** The scaling law is a bookkeeping form; neither n nor c_n may be guessed after looking at a desired answer.

Suppose the first omitted contribution is approximately

$$R_{n+1} = c_{n+1} J \epsilon^n.$$

Then

$$\frac{|R_{n+1}|}{|K_n|} \approx \left| \frac{c_{n+1}}{c_n} \right| \epsilon.$$

A small ϵ improves control but shrinks K_n . A large ϵ increases the nominal term while undermining the expansion. This is a real optimization problem, not permission to set $\epsilon = 0.9$ and write “perturbative” in the caption.

A crooked bond becomes a crooked K . For $K = c J^n \Lambda^{1-n}$, logarithmic differentiation gives

$$\frac{\delta K}{K} = n \frac{\delta J}{J} - (n-1) \frac{\delta \Lambda}{\Lambda} + \frac{\delta c}{c}.$$

A spatially common drift in J is amplified by n . If n independent bond factors each have the same small fractional standard deviation s , their product instead has fractional standard deviation approximately $\sqrt{n} s$ to leading order. Correlation therefore matters. Quoting one implantation-position standard deviation is not yet a distribution of K .

Hence fourth order is not a free improvement. It is a cut of order ϵ^3 on J , paid for in validity, and then amplified again if the bonds wobble.

29.5. Temperature, disorder, forgetfulness, and size

Temperature. Thermal excitation is governed by $k_B T$. In a dilute, noninteracting estimate, the occupation of an excitation of energy Δ_{topo} carries a Boltzmann factor

$$p_1 \sim e^{-\Delta_{\text{topo}}/k_B T}.$$

[**Theory**] This captures activation but not diffusion, entropy, boundaries, or interaction among anyons. Reviews of finite-temperature quantum memories emphasize that a nonzero gap alone does not make a two-dimensional memory self-correcting [R169]. For N approximately independent places where an excitation can occur, the crude expected count is $N p_1$. Requiring it below a target $p_\star$ gives

$$\frac{\Delta_{\text{topo}}}{k_B T} \gtrsim \ln\left(\frac{N}{p_\star}\right).$$

This system-size condition is stricter than merely $\Delta_{\text{topo}} > k_B T$.

Disorder. Define σ_J here as the root-mean-square energy variation of the *relevant projected local terms* after static calibration. This definition matters. Raw microscopic coupling variation can partly renormalize K , generate random effective fields at lower perturbative order, or mix the retained and leakage subspaces. The dangerous comparison is local:

$$\sigma_J \ll \Delta_{\text{topo}}.$$

A total operator norm that grows with the number of sites is not the right local yardstick. Nor is an RMS value sufficient when distributions have long tails or spatial correlations. [**Theory**] Stability theorems prove persistence of spectral bands for certain topologically ordered commuting-projector Hamiltonians under sufficiently weak, bounded, short-range local perturbations [R142]. They do not provide a universal percentage tolerance for a proposed defect device.

Decoherence broadening. Define the effective decoherence rate by

$$\Gamma = \frac{1}{T_{2,\text{eff}}},$$

where $T_{2,\text{eff}}$ is the decay time, in seconds, of the projected degree of freedom under the stated control sequence. Then the associated energy scale is $\hbar\Gamma$, with $\hbar = h/(2\pi)$. In ordinary frequency units,

$$\frac{\hbar\Gamma}{h} = \frac{1}{2\pi T_{2,\text{eff}}}.$$

This convention must not be confused with a spectroscopic full width at half maximum, which can carry a different factor. Also, one scalar T_2 cannot describe leakage, non-Markovian noise, or correlated errors. The comparison $\hbar\Gamma \ll \Delta_{\text{topo}}$ is necessary for resolving coherent many-body dynamics, not sufficient for fault tolerance.

Finite size. For a gapped local topological phase, virtual processes that wind around the sample can split the nominal ground-state family. A common asymptotic form is

$$\delta_{\text{fs}} \sim A\Delta_{\text{topo}}e^{-L/\xi},$$

where L is the shortest noncontractible linear size, ξ is a correlation length in the same units, and A is dimensionless. **[Theory]** Exponentially narrow low-energy bands under appropriate weak local perturbations are supported by stability results [R142], but the prefactor and even the useful asymptotic regime are model-dependent. Exactly solvable commuting-projector points can have zero splitting; generic perturbations restore a nonzero one.

What fails if you keep only $\Delta_{\text{topo}} > k_B T$? You have ignored entropy, system size, local disorder, linewidth, and the splitting inside the family. Any one of those can eat the leftover.

29.6. The leftover has to beat all four

The compact screening criterion is

$$\Delta_{\text{topo}} \gg k_B T, \quad \sigma_J, \quad \hbar\Gamma, \quad \delta_{\text{fs}}$$

with two additions:

1. unwanted projected terms and perturbative remainders must also be locally much smaller than Δ_{topo} ; and

2. thermal performance should be tested against $\Delta_{\text{topo}}/k_B T \gtrsim \ln(N/p_\star)$, not only against one.

For a numerical screen, define

$$\mathcal{M} = \frac{\Delta_{\text{topo}}}{\max(k_B T, \sigma_J, \hbar\Gamma, \delta_{\text{fs}})}.$$

$\mathcal{M} < 1$ fails even the scale-ordering test. $\mathcal{M} > 1$ only passes that test. The symbol $\gg$ demands a margin tied to a target error probability and system size; it is not a decorative greater-than sign.

Hence a ratio of 4.86 is not a comfortable gap. It is a number that still has to face $\ln(N/p_\star)$.

29.7. Three toy ledgers

The following is a sensitivity calculation, not a prediction. Every uncited number in the input block is deliberately labeled as an assumption.

[Proposal; Assumptions] We use the fourth-order toy coefficient $K = J^4/(16\Lambda^3)$ motivated by the specified honeycomb limit [R017], set $\Delta_{\text{topo}} = 2K$, set the finite-size prefactor $A = 1$, and assume the quantities in this table after projection:

Input	Optimistic	Base	Pessimistic
cluster separation Λ/h	1000 MHz	250 MHz	50 MHz
weak coupling J/h	300 MHz	50 MHz	5 MHz
expansion ratio $\epsilon = J/\Lambda$	0.30	0.20	0.10
physical temperature T	0.010 mK	1.0 mK	100 mK
residual effective disorder σ_J/h	0.100 MHz	0.025 MHz	0.001 MHz
projected $T_{2,\text{eff}}$	1 ms	0.10 ms	0.010 ms
linear size L/ξ	10	5	2

The “optimistic” temperature is 10 μK . That is an aggressive assumed spin temperature, not a cited capability of a dense defect array. The other temperatures, couplings, disorder values, and projected coherence times are likewise scenario inputs. Experiments have separately demonstrated coherent coupling between individual diamond defect spins [R080]

and long electronic-spin coherence under specialized material and decoupling conditions [R211]; those results do not validate simultaneous achievement of this table.

Using ordinary frequency units avoids repeated factors of h :

$$\frac{K}{h} = \frac{1}{16} \frac{(J/h)^4}{(\Lambda/h)^3}, \quad \frac{\Delta_{\text{topo}}}{h} = 2 \frac{K}{h}.$$

For the base row,

$$\frac{K}{h} = \frac{1}{16} \frac{(50 \text{ MHz})^4}{(250 \text{ MHz})^3} = 0.025 \text{ MHz},$$

so $\Delta_{\text{topo}}/h = 0.050 \text{ MHz} = 50 \text{ kHz}$. The units reduce as $\text{MHz}^4/\text{MHz}^3 = \text{MHz}$.

Applying the same arithmetic to all rows gives:

Derived quantity	Optimistic	Base	Pessimistic
K/h	0.50625 MHz	0.025 MHz	0.0003125 MHz
Δ_{topo}/h	1.0125 MHz	0.050 MHz	0.000625 MHz
$k_B T/h$	0.20837 MHz	20.8366 MHz	2083.66 MHz
σ_J/h	0.100 MHz	0.025 MHz	0.001 MHz
$\hbar\Gamma/h$	0.000159 MHz	0.001592 MHz	0.015915 MHz
assumed δ_{fs}/h	0.0000460 MHz	0.000337 MHz	0.0000846 MHz
scale margin $\mathcal{M}$	4.86	2.40×10^{-3}	3.00×10^{-7}

All displayed digits are arithmetic outputs, not measurement precision. The temperature conversion uses the SI constants [R196]. Decoherence uses $1/(2\pi T_{2,\text{eff}})$. Finite-size splitting uses the explicit assumption $A = 1$ and the listed L/ξ .

Verdict. [Proposal] The base and pessimistic scenarios fail because $k_B T$ exceeds the toy gap; the pessimistic scenario also has decoherence broadening and effective disorder above the gap. The optimistic scenario passes only the weakest ordering test, with temperature setting $\mathcal{M} = 4.86$. For a toy array with $N = 1000$ excitation locations and an assumed target $p_\star = 0.01$, the dilute estimate demands

$$\frac{\Delta_{\text{topo}}}{k_B T} \gtrsim \ln(1000/0.01) = 11.51,$$

whereas the optimistic ratio is 4.86. It therefore fails that stated thermal target too. These conclusions trace only to the displayed toy formula, the exact unit conversion, and the labeled assumptions. They are not claims that a real defect architecture has any of the three gaps.

The exercise also reveals a design trap. Reducing J/Λ from 0.30 to 0.10 makes fourth-order perturbation theory more comfortable, but—together with the assumed absolute scales—collapses the toy gap from megahertz to hundreds of hertz. A feasibility proposal must optimize validity and gap simultaneously, then include the leading unwanted terms.

29.8. What the laboratory actually has

[Experiment] Coupled individual defect spins and long coherence have each been demonstrated in diamond [R080]; [R211]. They are important ingredients. They are not a measurement of a bulk topological gap, a correlation length, or a topological ground-state family.

For the defect-cluster architecture assessed here, the gap budget currently lacks four experimentally connected objects:

1. a fabricated lattice with the required interaction graph;
2. spectroscopy validating the projected many-body Hamiltonian and its unwanted terms;
3. finite-size scaling that separates a ground-family splitting from the excitation gap; and
4. evidence that the fitted phase has the claimed topological data rather than merely a similar low-energy spectrum.

Consequently, there is no supported material value of Δ_{topo} to insert. **[Proposal]** A defensible program would first measure small-cluster spectra and distributions, fit a microscopic Hamiltonian without discarding inconvenient terms, compute its phase diagram and gap with uncertainty propagation, and then compare increasingly large patches. A gap should be reported with host, defect species, charge state, geometry, field, strain, temperature, boundary conditions, and model-fitting procedure attached.

Temperature also needs care. Refrigerator temperature, phonon-bath temperature, and the effective temperature of driven spins need not coincide. Similarly, a single-defect echo T_2 measured under a pulse train is not automatically the $T_{2,\text{eff}}$ of a continuously interacting cluster lattice. Combining the best coupling from one device, the best coherence from another, and the lowest cryostat temperature from a third produces a brochure, not a budget.

29.9. Ways to get this wrong

1. **Call the largest energy the gap.** Suppose the zero-field splitting is a gigahertz and the fourth-order topological term is a kilohertz. Then the largest energy is safe, and the gap of the phase has not been mentioned. The gap is spectral and phase-specific.
2. **Use the target Hamiltonian's normalized gap.** Writing a string-net Hamiltonian with unit coefficients sets a convention. It does not show that a defect implementation supplies one joule, one kelvin, or one megahertz [R018]. Drop the conversion and the crystal inherits a paper's choice of one.
3. **Ignore lower-order unwanted terms.** A desired fourth-order plaquette term loses if a symmetry-breaking field survives at first or second order. Compare local projected remainder norms with Δ_{topo} .
4. **Treat RMS disorder as a complete model.** Rare bad bonds can nucleate low-energy excitations. Correlated drift and heavy tails are different from independent Gaussian variation. What fails if you quote only an RMS? You have described a typical bond, not the worst one the lattice actually has.
5. **Equate a nonzero gap with passive memory.** Thermally created anyons can diffuse and enact a logical operator without paying an energy proportional to their travel distance. Two-dimensional topological order can be stable as a phase while being a poor self-correcting memory at nonzero temperature [R169].
6. **Confuse ground-state splitting with the bulk gap.** Small δ_{fs} is good; small Δ_{topo} is bad. A finite patch may show both. What fails if spectroscopy swaps their labels? A narrow family looks like a small gap, and the architecture is declared dead for the wrong reason — or alive for the wrong reason.
7. **Use T_2 as an energy gap.** T_2 gives a linewidth scale only after a convention and noise model are specified. It says nothing by itself about topological order.
8. **Promote digital evidence to an analog gap.** A programmable device can prepare and manipulate states with anyonic fusion rules while its native Hamiltonian remains an ordinary qubit Hamiltonian. The relevant analog claim is that the material's low-energy spectrum and phase realize the order without a gate sequence continually synthesizing it.
9. **Extrapolate a perturbation series outside control.** Increasing J/Λ raises the target coefficient and every omitted order. The remainder must be calculated or bounded, not wished smaller.
10. **Report a single favorable point.** A phase needs a finite region of parameter space. Fabrication produces a distribution, so uncertainty propagation and finite-size scaling belong in the claim.

29.10. Check yourself

1. **What defines Δ_{topo} ?**

The energy from the top of the finite-size ground-state family to the first state outside it, for a stated Hamiltonian and boundary condition.

2. **Show that if $K = J^4/(16\Lambda^3)$ and $J/\Lambda = 0.1$, then $K = J/16,000$.**

Substitute: $K = (J/16)(J/\Lambda)^3 = J/16 \times 10^{-3} = J/16,000$. Projection and perturbative order can reduce the effective coefficient to $c_n J(J/\Lambda)^{n-1}$, and the spectral gap adds a model-dependent factor. A bare coupling J is not the gap.

3. **Show that 1 mK is 20.8366 MHz in these units.**

$k_B T/h = 20.8366$ MHz, using exact SI constants [R196].

4. **Show that if $T_{2,\text{eff}} = 100 \mu\text{s}$ and $\Gamma = 1/T_{2,\text{eff}}$, then $\hbar\Gamma/h = 1.59$ kHz.**

$1/(2\pi T_{2,\text{eff}}) = 1.59$ kHz.

5. **What fails if you only require $\Delta_{\text{topo}} > k_B T$?**

Entropy and system size matter; a dilute estimate requires roughly $\Delta_{\text{topo}}/k_B T \gtrsim \ln(N/p_*)$ for the stated target. Disorder, decoherence, finite-size splitting, and left-over projected terms remain.

6. **What evidence would turn a budgeted gap into a gap claim?**

A derived or fitted effective Hamiltonian with quantified errors, followed by controlled analytic or numerical spectroscopy and finite-size scaling; experimental spectroscopy is stronger still.

A coupling is a handshake. A gap is a spectrum. The screening inequality is $\Delta_{\text{topo}} \gg k_B T, \sigma_J, \hbar\Gamma, \delta_{\text{fs}}$, plus small projected remainders and a logarithmic thermal penalty. The three ledgers in this chapter are scenarios only. There is no measured material Δ_{topo} to put in the last cell.

29.11. Sources

- [R017] Alexei Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv: cond-mat/0506438.
- [R018] Michael A. Levin and Xiao-Gang Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv: cond-mat/0404617.
- [R209] Roberto Oliveira and Barbara M. Terhal, “The complexity of quantum spin systems on a two-dimensional square lattice,” *Quantum Information & Computation* **8**, 900–924 (2008). arXiv: quant-ph/0504050.

- [R142] Sergey Bravyi, Matthew B. Hastings, and Spyridon Michalakis, “Topological quantum order: Stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195; arXiv: 1001.0344.
- [R169] Benjamin J. Brown, Daniel Loss, Jiannis K. Pachos, Chris N. Self, and James R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* **88**, 045005 (2016). DOI: 10.1103/RevModPhys.88.045005; arXiv: 1411.6643.
- [R196] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th ed., updated 2026. DOI: 10.59161/AUEZ1291; stable publication page.
- [R080] F. Dolde *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R211] N. Bar-Gill, L. M. Pham, A. Jarmola, D. Budker, and R. L. Walsworth, “Solid-state electronic spin coherence time approaching one second,” *Nature Communications* **4**, 1743 (2013). DOI: 10.1038/ncomms2771.

Chapter 30. The same shove, a different filter

A truck rolls past the building. Three nearby defects all feel the same magnetic shove. Those three spins were taught to impersonate one quiet pair: two levels we keep, and a crowd of others we promised not to visit.

The shove does not wait outside the encoding. It is an operator on the three physical spins. Sandwich it with the projector onto the quiet pair and you get a two-by-two matrix acting *inside* that pair. Sandwich the leftover piece and you get a ticket out.

So a cluster is not a wall. It is a *filter*. More physical spins do not automatically mean less noise. They mean different signs and different gains on baths that were already in the crystal.

30.1. Three places a shove can go

Imagine three microphones wired to report one number. Average their outputs and independent hiss partly cancels. Shake all three together and averaging does nothing. Reverse one sign and the common shove may cancel — but now a gradient becomes the signal.

A cluster encoding is that wiring. After this paragraph the microphones are gone: the microscopic defects are the sites, and the matrix elements inside the chosen pair are the signs and the gains.

There are three places a perturbation can go.

1. It can shift the two kept levels differently. The relative twist smears. That is *dephasing*, the same failure Chapter 4 measured as a fading off-diagonal, now written on the encoded pair.
2. It can mix the two kept levels. Population moves inside the pair. That is relaxation or excitation of the encoded qubit.
3. It can leave the pair. That is leakage, the climb into Q from Chapter 11.

Static fabrication errors add a fourth practical problem: different clusters implement different Hamiltonians. Slow drift makes that disorder time dependent.

A cluster can reject a uniform magnetic field and still be painfully sensitive to a field gradient, a fluctuating exchange bond, or a pulse that addresses several constituent defects at once. Drop the hypothesis that “more spins average the bath,” and a perfectly common shove survives the filter unchanged. The universe has read the wiring too.

30.2. Three spins and a local field

Take three spin-1/2 defects with strong, approximately isotropic exchange. Let $|\uparrow\rangle$ and $|\downarrow\rangle$ be eigenstates of the single-spin operator S^z , with eigenvalues $+1/2$ and $-1/2$. We use dimensionless spin operators. An energy-valued field $\xi_i(t)$, measured in joules (J), couples as

$$\delta H_z(t) = \sum_{i=1}^3 \xi_i(t) S_i^z.$$

Within the total-magnetization sector $S_{\text{tot}}^z = +1/2$, choose the logical states

$$|0_L\rangle = \frac{|\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle}{\sqrt{2}},$$

$$|1_L\rangle = \sqrt{\frac{2}{3}}|\uparrow\uparrow\downarrow\rangle - \frac{|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle}{\sqrt{6}}.$$

These are the same kind of three-spin pair Chapter 11 kept. Let $P = |0_L\rangle\langle 0_L| + |1_L\rangle\langle 1_L|$ project onto this doublet. Define logical Pauli operators

$$\tau_z = |0_L\rangle\langle 0_L| - |1_L\rangle\langle 1_L|, \quad \tau_x = |0_L\rangle\langle 1_L| + |1_L\rangle\langle 0_L|.$$

Direct multiplication gives

$$PS_1^zP = \frac{1}{6}I - \frac{1}{6}\tau_z - \frac{\sqrt{3}}{6}\tau_x,$$

$$PS_2^zP = \frac{1}{6}I - \frac{1}{6}\tau_z + \frac{\sqrt{3}}{6}\tau_x,$$

$$PS_3^zP = \frac{1}{6}I + \frac{1}{3}\tau_z.$$

Therefore

$$P\delta H_zP = h_0I + h_x\tau_x + h_z\tau_z,$$

where

$$h_0 = \frac{\xi_1 + \xi_2 + \xi_3}{6}, \quad h_x = \frac{\sqrt{3}}{6}(-\xi_1 + \xi_2),$$

$$h_z = \frac{-\xi_1 - \xi_2 + 2\xi_3}{6}.$$

The identity term $h_0 I$ supplies a common phase and does not damage the logical state. Low-frequency h_z noise dephases it. Noise in h_x near the logical angular frequency ω_L , measured in radians per second, drives logical transitions.

Now set $\xi_1 = \xi_2 = \xi_3 = \xi_c$. Both h_x and h_z vanish because

$$P(S_1^z + S_2^z + S_3^z)P = \frac{1}{2}I.$$

Hence a perfectly common longitudinal fluctuation is rejected to first order. **[Theory]** That is decoherence-free behavior for one specified operator, not immunity to magnetic noise in general. Drop uniformity — let ξ_3 differ from ξ_1 and ξ_2 — and h_z returns at once.

For equal, mutually uncorrelated site noise, define the two-sided energy-noise spectrum

$$S_\xi(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} \langle \xi_i(t) \xi_i(0) \rangle.$$

Its units are J^2 s: the integrand has J^2 , and dt contributes seconds. The projected spectra are

$$S_{h_x}(\omega) = S_{h_z}(\omega) = \frac{1}{6} S_\xi(\omega).$$

That factor of six is useful. It is not a blanket sixfold lifetime improvement. Dephasing depends on the low-frequency filter function. Relaxation samples noise near ω_L . Leakage samples higher frequencies near the cluster gap.

The state

$$|Q\rangle = \frac{|\uparrow\downarrow\uparrow\rangle + |\downarrow\uparrow\uparrow\rangle + |\uparrow\uparrow\downarrow\rangle}{\sqrt{3}}$$

belongs to the spin-3/2 manifold rather than the encoded doublet. Local field gradients have nonzero matrix elements $\langle Q | S_i^z | 0_L \rangle$ and $\langle Q | S_i^z | 1_L \rangle$, so noise at the cluster excitation energy can cause leakage. A uniform longitudinal field, proportional to total S^z , cannot make that transition. Clustering has suppressed one correlation channel while exposing gradient and leakage channels.

30.3. Every bath, the same sandwich

Let the ideal cluster Hamiltonian be H_C , and let its logical doublet be separated from all other cluster states by a minimum energy Δ_C , measured in J. Define $Q = I - P$ as the projector onto those unwanted states. Write the microscopic disturbance as

$$\delta H(t) = \sum_a \xi_a(t) O_a.$$

The label a includes site, Cartesian component, and physical mechanism. For example, O_a might be a spin component, an orbital or strain operator, a local charge projector, or a bond operator $\mathbf{S}_i \cdot \mathbf{S}_j$. The coefficient $\xi_a(t)$ carries whatever units make $\xi_a O_a$ an energy.

Every operator, once trapped inside the pair, has the form

$$P O_a P = c_{a0} I + \frac{1}{2} \sum_{\mu=x,y,z} g_{a\mu} \tau_\mu.$$

Here c_{a0} is an irrelevant common shift. The numbers $g_{a\mu}$ are the cluster's **noise susceptibilities**. The name is worth keeping because they are the wiring gains: how much this physical operator, after projection, looks like each logical Pauli.

The effective logical field is

$$b_\mu(t) = \sum_a g_{a\mu} \xi_a(t), \quad P \delta H P = \text{common shift} + \frac{1}{2} \sum_\mu b_\mu \tau_\mu.$$

If the logical splitting points along z , slow b_z produces phase wander. Components b_x, b_y at ω_L cause transitions. Hence which physical disturbance is “longitudinal” is decided *after projection*, not by its name in the laboratory.

How two baths covary at a frequency is written as a *cross-spectrum*,

$$S_{ab}(\omega) = \int dt e^{i\omega t} \langle \delta \xi_a(t) \delta \xi_b(0) \rangle.$$

The diagonal is each bath's power at frequency ω . The off-diagonal is whether two baths shake together. Then

$$S_{\mu\nu}^{(L)}(\omega) = \sum_{a,b} g_{a\mu} g_{b\nu} S_{ab}(\omega).$$

This equation is the noise model. The diagonal entries describe logical noise power. The off-diagonal entries retain correlations between logical axes. A Markovian weak-noise calculation would make the logical relaxation rate proportional to $[S_{xx}^{(L)}(\omega_L) + S_{yy}^{(L)}(\omega_L)]/\hbar^2$.

The dimensions check: $J^2 s / (J^2 s^2) = s^{-1}$. Exact prefactors depend on one-sided versus two-sided spectral conventions, so a simulator must state its convention rather than smuggling in a factor of two.

The output is not a single coherence time. It is this frequency-dependent matrix, plus whatever still lives in Q . Drop the off-diagonal blocks of S_{ab} and you can reverse the predicted benefit of the encoding.

30.4. When hiss cancels and when it does not

Suppose the same operator acts at N sites with logical coefficients g_i . If site noises are independent and have the same spectrum S_0 , then

$$S_L = S_0 \sum_i g_i^2.$$

If they are perfectly correlated, then

$$S_L = S_0 \left(\sum_i g_i \right)^2.$$

For a delocalized average, $g_i = 1/N$: independent noise falls as $1/N$, while common noise stays unchanged. For a decoherence-free sign pattern with $\sum_i g_i = 0$, common noise cancels. Conversely, if every g_i has the same order-one sign, common noise power grows as N^2 .

Hence correlation is neither automatically good nor automatically bad. It must be contracted with the projected susceptibility vector. **[Theory]** What fails if you quote only $\sum_i g_i^2$ is the common-mode case: the encoding that looked like averaging becomes an antenna.

30.5. The states we promised not to visit

Projection onto P is only half the story. The discarded block QO_aP is the ticket out of the pair.

For an unwanted eigenstate $|m\rangle$ with energy difference $E_m - E_L = \hbar\omega_{mL}$, the weak-noise transition rate contains

$$\Gamma_{L \rightarrow m} \propto \frac{1}{\hbar^2} \sum_{a,b} \langle L|O_a|m\rangle \langle m|O_b|L\rangle S_{ab}(\omega_{mL}).$$

Those matrix elements $\langle m|O_a|L\rangle$ are exactly the contents of QO_aP . A large Δ_C helps only when the environment has little spectral weight at $\Delta_C/\hbar$ and controls are slow or spectrally narrow compared with that frequency.

Strong pulses, resonant phonons, and charge switching do not promise to be polite. Every microscopic noise operator therefore needs both pieces: PO_aP for what it does inside the pair, and QO_aP for how it leaves. Quoting one T_2 throws both pieces away.

30.6. A ledger of named baths

Keep “source,” “channel,” and “cluster consequence” on separate lines. The names below are physical objects, not synonyms for “decoherence.”

Microscopic source	Typical microscopic coupling	Main channel after projection	Clustering can suppress when...	Clustering can worsen when...
Host nuclear spins, including ^{13}C in diamond	Hyperfine fields and nuclear flip-flops	Quasistatic dephasing, spectral diffusion; sometimes resonant relaxation	the nuclear field is common across a fixed-magnetization encoding, or isotopic purification removes most bath spins	each constituent samples a different Overhauser field, and the cluster contains more strongly coupled nuclei
Paramagnetic impurities and unintended defects	Magnetic dipolar fields, bath flip-flops, cross-relaxation	Dephasing, relaxation, correlated magnetic bursts	symmetry rejects common field; detuning avoids bath resonances	high defect density supplies both desired coupling and a denser spin bath
Surface spins	Fluctuating magnetic dipoles at interfaces	Dephasing and relaxation, often depth dependent	the cluster is deep and compact compared with the field's correlation length	shallow fabrication exposes every constituent and a nearby fluctuator acts nonuniformly

Microscopic source	Typical microscopic coupling	Main channel after projection	Clustering can suppress when...	Clustering can worsen when...
Charge noise	Electric/Stark shifts; modulation of orbital energies and exchange	Dephasing, bond noise, optical spectral diffusion	a clock point makes the logical splitting first-order insensitive	one trap modulates several bonds or shifts the cluster through an avoided crossing
Strain and electric-field noise	Crystal-field, spin-orbit, Stark, and spin-strain terms	Dephasing, relaxation, coupling disorder	symmetry makes uniform strain an identity operation	long-wavelength strain drives all sites coherently with non-cancelling matrix elements
Phonons	Dynamic strain and orbital/spin-phonon coupling	T_1 , orbital relaxation, leakage, temperature-dependent dephasing	the encoded transition has a vanishing matrix element or no resonant phonon density	collective matrix elements add, or a phonon matches Δ_C
Implantation/processing damage	Vacancies, interstitials, strain fields, traps, paramagnetic complexes	Static disorder plus magnetic, charge, and strain noise	annealing and materials processing remove or passivate damage	adding more implanted constituents multiplies nearby damage and broadens J and Δ_C
Charge-state instability	Ionization/recombination of the active defect	Leakage or erasure; abrupt Hamiltonian change	redundant heralding detects the event	one constituent changing charge destroys the encoded Hamiltonian and perturbs neighbors

Nuclear and paramagnetic baths. In diamond NV systems, coherent evolution coupled to ^{13}C nuclei and nitrogen-related electron-spin baths has been directly observed, and dynamical decoupling can reshape rather than remove the bath spectrum [**Experiment**] [R214]; [R215]. The same lesson carries to a cluster: isotopic purification attacks nuclear abundance, while pulse filtering attacks selected frequencies. Neither guarantees small leakage matrix elements. A deliberately dense array also risks converting spectators, residual implantation products, and off-target charge states into a correlated electron-spin bath.

Surfaces. Measurements with shallow NV centers support a substantial surface-related magnetic-noise contribution [**Experiment**] [R216]. A compact cluster much smaller than

the source distance may see nearly common field noise and reject it. Place one defect much closer to the interface, however, and the common-mode approximation fails. Nanostructuring for optical access can therefore change both noise strength and its spatial covariance. A single-defect T_2 does not determine the encoded T_2 .

Charge, electric field, and optical spectral diffusion. NV experiments have separated regimes dominated by magnetic and electric-field noise [Experiment] [R217], while electric coupling of an individual NV spin has been measured directly [Experiment] [R191]. Charge traps also move optical transition frequencies and can change the defect's charge state. In a cluster, electric noise may shift each logical level, modulate the intra-cluster exchange J_{ij} , or modulate inter-cluster coupling. Those are three different operators and require three sets of $g_{a\mu}$. Charge conversion is more severe than ordinary phase noise: if a constituent ceases to occupy the required charge/spin manifold, it is leakage or erasure, not a small Pauli error. Near-surface NV ensembles exhibit coupled charge dynamics and density-dependent charge-state behavior [Experiment] [R218].

Strain and phonons. Static strain is disorder; time-dependent strain is noise. Spin-strain coupling has been driven and measured in NV–mechanical systems [Experiment] [R219]. Phonons can also relax orbital branches and spins. For inversion-symmetric group-IV vacancy centers, orbital phonon processes are central to temperature-dependent coherence models [Theory/Experiment] [R201]. A long-wavelength acoustic mode can be highly correlated across a nanometre-scale cluster. That helps only if the projected uniform-strain operator is proportional to I ; otherwise the coherent sum can amplify logical drive or leakage.

Implantation damage. Ion implantation provides placement but creates collision cascades, vacancies, interstitials, local strain, and charge traps; annealing converts only part of that landscape into the desired centers [Experiment/Review] [R212]. Damage is therefore not a single named decay channel. It is a generator of static J_{ij} disorder and of magnetic, electric, and strain fluctuators. A credible array model must correlate damage statistics with placement statistics rather than drawing them independently.

Spectral diffusion. The transition frequency wanders as surrounding spins or charges switch. That wandering is a behavior, not an extra microscopic substance. Slow nuclear flip-flops, electron-spin flips, and charge-trap telegraph processes can all produce it [R213]; [R214]; [R217]. Gaussian white noise is often a poor description: one nearby trap yields discrete, non-Gaussian jumps; a broad ensemble of switching rates can yield an approximate $1/f$ spectrum. Echo may refocus slow wander during idle evolution, yet the same drift can detune optical initialization, readout, or cluster-mediated gates.

30.7. Bonds that wander

Let the intended microscopic coupling on bond ij be J_{ij} , in J , and write its error as

$$J_{ij}(t) = \bar{J}_{ij} + \delta J_{ij}^{\text{static}} + \delta J_{ij}^{\text{dyn}}(t).$$

The static term produces a distribution of cluster splittings, leakage gaps, and projected inter-cluster couplings. The dynamic term is *bond noise*. Because a dipolar coupling depends on distance and orientation, positional disorder enters multiplicatively. Exchange is generally even more sensitive to atomic-scale geometry. Projecting $\delta J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$ may create a logical τ_z field, a transverse logical term, or leakage, depending on the encoding.

At array level, retain at least four kinds of correlation:

- **Local independent noise:** separate nuclei or traps near each constituent.
- **Intra-cluster common mode:** long-wavelength magnetic, electric, or strain fluctuations.
- **Inter-cluster correlations:** a surface patch, electrode, laser, or acoustic mode affects several encoded qubits.
- **Control correlations:** pulse-amplitude error, detuning, crosstalk, and global-drive phase noise act coherently by design.

The last item is where clustering can be especially treacherous. A global pulse intended to rotate every constituent can have a matrix element that adds as N , giving error power proportional to N^2 . Fast control can also populate Q . Symmetry that cancels passive common-mode noise may make desired control impossible at the same order, forcing gradient fields or symmetry-breaking pulses that reintroduce sensitivity.

A cluster doublet is an encoded qubit made from physical defect spins. Projecting microscopic noise into Pauli operators does not create topological order. Even if a later many-body Hamiltonian has emergent anyons, errors that change a defect's charge state or leave a cluster doublet occur below the anyon description. They cannot be assumed to become harmless local anyon pairs. A digital simulation that inserts a chosen Pauli channel likewise does not demonstrate the material's native noise correlations.

30.8. What the laboratory actually has

For individual color centers and ensembles, laboratories can measure Ramsey and echo decay, relaxation, noise spectra under pulse sequences, optical spectral diffusion, charge-state switching, strain response, and sometimes spatial dependence near surfaces [R213]; [R214]; [R215]; [R216]; [R217]; [R191]; [R218]; [R219]; [R201]. **[Experiment]** These measurements establish that the named mechanisms are real. They do not yet establish a complete covariance matrix $S_{ab}(\omega)$ for a fabricated, strongly coupled 5–20-defect cluster, much less for a large cluster array.

A responsible characterization sequence would be:

1. Measure each constituent where individual addressability survives: resonance, charge state, T_1 , Ramsey, echo, and optical stability.
2. Spectroscopically identify the cluster doublet and every nearby leakage level, obtaining ω_L and $\Delta_C/\hbar$.
3. Apply calibrated common-mode and gradient magnetic, electric, and strain perturbations to determine the coefficients $g_{a\mu}$.
4. Measure simultaneous time traces or cross-spectra on multiple constituents/clusters rather than inferring independence.
5. Fit static bond disorder separately from dynamic noise.
6. Use the measured spectra and leakage matrix elements in many-body simulations; do not replace them with one average T_2 .

[Proposal] No cited experiment demonstrates that clustering color centers produces a topologically ordered array with a measured passive-protection advantage. The projection framework gives a falsifiable test: measured susceptibilities and spectra must predict logical and leakage rates below the relevant interaction and topological-gap scales. Until that comparison exists, “noise-resilient cluster” is a design hypothesis.

30.9. Ways to get this wrong

1. **Quoting only T_2 .** T_2 folds together a pulse sequence, a noise spectrum, and an operating point. It omits charge erasure, optical instability, correlated errors, and leakage. One number cannot be the noise model.
2. **Calling every fluctuation independent.** Common electrodes, surfaces, laser fields, strain waves, and fabrication damage create correlations. Setting $S_{ab} = 0$ for $a \neq b$ can reverse the predicted benefit of an encoding.
3. **Calling every common fluctuation harmless.** Cancellation requires $\sum_a g_{a\mu} = 0$ for the actual operator. Common exchange or transverse fields need not satisfy it.
4. **Projecting and forgetting Q .** The effective Pauli model is incomplete when noise or control has spectral weight near $\Delta_C/\hbar$. Leakage can grow with the number of constituent levels.
5. **Confusing disorder with decoherence.** Static disorder does not by itself destroy phase in one isolated device, but it changes Hamiltonian parameters, closes local gaps, frustrates calibration, and causes ensemble dephasing. Drift is dynamic disorder.
6. **Assuming echo fixes the architecture.** Echo can reject selected low-frequency terms while also averaging desired interactions. It does not reverse irreversible T_1 , charge conversion, or unobserved leakage.
7. **Optimizing single defects instead of clusters.** The best individual coherence site may yield the wrong interaction graph. The strongest-coupled cluster may sit amid the worst implantation bath. Feasibility is a joint distribution.

8. **Treating topological protection as a solvent.** Topological order can suppress particular local processes under particular energy and temperature conditions. Correlated faults spanning multiple effective sites, leakage below the effective model, and time-dependent Hamiltonian disorder can bypass that argument.

Independent and correlated components contract differently with the cluster susceptibilities. Static bond and onsite disorder, dynamic dephasing and relaxation, charge erasure, and leakage are distinct channels. Clustering suppresses a source only when the relevant symmetry and correlation pattern make its projected logical matrix element small. That, not a borrowed single-center lifetime, is the minimum model needed to test whether an apparent many-body gap survives a realistic defect array.

30.10. Check yourself

1. **What decides whether a microscopic perturbation dephases or relaxes the cluster qubit?**

Its projected coefficients: logical τ_z noise near zero frequency dephases, while τ_x or τ_y noise near ω_L drives transitions.

2. **Show that a perfectly common longitudinal field is invisible inside the three-spin doublet.**

Both logical states have $S_{\text{tot}}^z = +1/2$, so $P(S_1^z + S_2^z + S_3^z)P = \frac{1}{2}I$. The projected operator is a common phase.

3. **What fails if the three site noises are perfectly correlated rather than independent?**

The independent formula $S_L = S_0 \sum_i g_i^2$ is replaced by $S_L = S_0 (\sum_i g_i)^2$. Averaging disappears; a decoherence-free sign pattern still cancels, while same-sign coefficients amplify as N^2 .

4. **What is leakage in this setting?**

A transition from the logical doublet P into another cluster state in Q , driven by microscopic noise or control at the appropriate energy. The matrix elements live in QO_aP .

5. **Is spectral diffusion a separate elementary bath?**

No. It is observed frequency wandering that can arise from nuclear, paramagnetic, or charge dynamics.

6. **What fails if you treat charge-state switching as a small Stark shift?**

A Stark shift may project as dephasing. Changing charge can remove the constituent's required spin and alter the whole cluster Hamiltonian, producing leakage or erasure rather than a Pauli error inside P .

30.11. Sources

- [R212] S. Pezzagna and J. Meijer, “Quantum computer based on color centers in diamond,” *Applied Physics Reviews* **8**, 011308 (2021). DOI: 10.1063/5.0007444.
- [R213] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” *Reviews of Modern Physics* **89**, 035002 (2017). DOI: 10.1103/RevModPhys.89.035002.
- [R214] L. Childress *et al.*, “Coherent dynamics of coupled electron and nuclear spin qubits in diamond,” *Science* **314**, 281–285 (2006). DOI: 10.1126/science.1131871.
- [R215] G. de Lange, Z. H. Wang, D. Ristè, V. V. Dobrovitski, and R. Hanson, “Universal dynamical decoupling of a single solid-state spin from a spin bath,” *Science* **330**, 60–63 (2010). DOI: 10.1126/science.1192739.
- [R216] T. Rosskopf *et al.*, “Investigation of surface magnetic noise by shallow spins in diamond,” *Physical Review Letters* **112**, 147602 (2014). DOI: 10.1103/PhysRevLett.112.147602.
- [R217] P. Jamonneau *et al.*, “Competition between electric field and magnetic field noise in the decoherence of a single spin in diamond,” *Physical Review B* **93**, 024305 (2016). DOI: 10.1103/PhysRevB.93.024305.
- [R191] F. Dolde *et al.*, “Electric-field sensing using single diamond spins,” *Nature Physics* **7**, 459–463 (2011). DOI: 10.1038/nphys1969.
- [R218] S. Dhomkar, H. Jayakumar, P. R. Zangara, and C. A. Meriles, “Charge dynamics in near-surface, variable-density ensembles of nitrogen-vacancy centers in diamond,” *Nano Letters* **18**, 4046–4052 (2018). DOI: 10.1021/acs.nanolett.8b01739.
- [R219] J. Teissier, A. Barfuss, P. Appel, E. Neu, and P. Maletinsky, “Strain coupling of a nitrogen-vacancy center spin to a diamond mechanical oscillator,” *Physical Review Letters* **113**, 020503 (2014). DOI: 10.1103/PhysRevLett.113.020503.
- [R201] K. D. Jahnke *et al.*, “Electron–phonon processes of the silicon-vacancy centre in diamond,” *New Journal of Physics* **17**, 043011 (2015). DOI: 10.1088/1367-2630/17/4/043011.

Chapter 31. Topological protection is not magic

One edge of the lattice flips. The two checks that touch that edge both change their minds. Two anyons sit a lattice spacing apart.

Is this a disaster? Not yet. The pair can walk toward each other and vanish. Every local alarm goes quiet. The secret stored in the winding may still be intact.

Or they walk the long way around the doughnut, meet, and vanish just as quietly. Every local alarm still goes quiet. The secret has flipped.

One local accident is not yet a logical disaster. Topology makes the disaster take a long history. It does not make the history impossible. Protected is not immortal.

Four words get asked to wear the same hat: gap, decoder, threshold, error-free. They are not synonyms. This chapter follows one local fault from its first scream to its possible logical consequence, and refuses to let those four words trade places.

31.1. A footprint is not a lap

Imagine fresh snow on a circular running track. One footprint is local. A continuous trail that goes all the way around the track is global. Inspecting any short segment does not tell you whether the trail eventually closes around the circle.

A topological code works the same way, with less snow and more Pauli matrices. After this paragraph the snow is gone. A local operator normally creates nearby excitations. Further local operators move them. If the excitations meet and disappear, every local alarm can return to normal. Yet their complete path may wind around a hole or handle. That winding acts on the logical state.

This is the central limitation: topology makes a logical fault require an extended *history*, not an impossible history. Locality buys distance. An energy gap buys a Boltzmann penalty for creating excitations. Neither automatically buys a growing energy barrier against moving them.

The spacetime version is even more useful. A data error draws a segment in space. Persistence draws it through time. A faulty syndrome bit draws a time-like segment. Decoding asks which collection of segments most plausibly explains the observed endpoints. A logical failure occurs when the inferred history and the real history differ by a topologically nontrivial loop.

Drop the word “history” and you are left staring at a quiet syndrome, announcing that nothing happened. The quiet syndrome is compatible with both the short walk and the lap.

31.2. Four flips around the doughnut

Use the toric code on a square lattice wrapped periodically in both directions. Put one physical qubit on every edge. Let

$$A_v = \prod_{e \ni v} X_e, \quad B_p = \prod_{e \in \partial p} Z_e,$$

where v labels a vertex, p labels a plaquette, e labels an edge, and X_e and Z_e are Pauli operators on edge e . The ground space has eigenvalue $+1$ for every vertex operator A_v and plaquette operator B_p . A -1 result is a syndrome defect, interpreted in the ideal model as an anyonic excitation [R030].

Take a periodic row containing four horizontal edges, e_1, e_2, e_3, e_4 . Apply Z errors consecutively:

$$E_j = Z_{e_1} Z_{e_2} \cdots Z_{e_j}, \quad j = 1, 2, 3, 4.$$

A single Z_e anticommutes with the two vertex checks at the ends of e . Therefore E_1 creates two violated vertex checks. Multiplying by Z_{e_2} restores the shared vertex and violates the next one: one endpoint has moved. The same happens at e_3 . After e_4 , the moving endpoint meets the first endpoint and both disappear.

Step	Physical action	Visible syndrome	Topological result
0	identity	none	identity
1	Z_{e_1}	two endpoints	pair created
2	$Z_{e_1} Z_{e_2}$	two endpoints	one endpoint moved
3	$Z_{e_1} Z_{e_2} Z_{e_3}$	two endpoints	one endpoint moved again
4	$\bar{Z} = \prod_{i=1}^4 Z_{e_i}$	none	noncontractible logical loop

Why is the last row not the identity? Choose a noncontractible X loop $\bar{X}$ crossing this row once. At the one crossing, $XZ = -ZX$, while all other factors commute. Hence

$$\bar{X}\bar{Z} = -\bar{Z}\bar{X}.$$

An operator that anticommutes with a logical observable changes the encoded state. The syndrome is empty, but the logical qubit has suffered $\tilde{Z}$. This is the local-error-chain mechanism in its least decorative form [R030]; [R141].

For an $L \times L$ periodic lattice, the shortest such loop has L edges. The code distance is therefore $d = L$, and ideal decoding can guarantee correction of at most $\lfloor (d - 1)/2 \rfloor$ adversarial Pauli errors. That statement is a worst-case combinatorial guarantee. A threshold, later in this chapter, is a different kind of claim: an asymptotic statement about a probability distribution over faults and a specified decoder.

The edge spins above may be physical qubits, cluster-encoded pseudospins, or digitally emulated variables. The algebra of an error chain is the same. The protection is not. A simulator can execute the code Hamiltonian while its underlying hardware remains protected only by ordinary active error correction. An analog material realizes passive topological protection only if its own low-energy spectrum, excitations, and local perturbations implement this structure.

31.3. The gap charges for birth, not for walking

The Hamiltonian can do some of the work without anyone measuring a syndrome and feeding back. That is **passive suppression**: the device itself is doing the suppressing. No decoder, no recovery pulse, just the spectrum.

Two mechanisms matter.

First, a many-body **gap** makes excitation creation energetically costly. Define Δ_a as the energy to create one well-separated anyon, in joules (J), and T as absolute temperature, in kelvin (K). A pair costs approximately $2\Delta_a$ in the ideal dilute model. Its thermal weight contains

$$\exp\left(-\frac{2\Delta_a}{k_B T}\right),$$

where k_B is Boltzmann's constant in J/K. The exponent is dimensionless because J divided by (J/K)K is one. This suppresses *creation* when $k_B T \ll \Delta_a$ [R169]. It is a price of birth, not a fence around motion.

Second, nonlocal encoding makes a weak local perturbation poor at distinguishing logical sectors. In suitable gapped topologically ordered Hamiltonians, sufficiently weak local perturbations preserve the phase, and finite-size logical splitting can scale schematically as

$$\delta E(L) \sim E_0 e^{-L/\xi}.$$

Here δE and E_0 are energies, L is a linear system size, and ξ is a correlation length in the same length unit. **[Theory]** This stability is conditional on locality, a spectral gap, topological-order assumptions, and perturbations small enough not to leave the phase [R142]. It is not a blanket theorem about a fabricated, driven, open array.

Passive suppression does not infer an error chain or apply a recovery. If an anyon already exists, moving it can cost little or no additional energy. The toric-code string above never needs more than two endpoints, so its maximum energy above the ground space is constant as L grows. Two-dimensional local stabilizer Hamiltonians therefore lack a macroscopic energy barrier under the assumptions of the Bravyi–Terhal no-go theorem [R168]. Thermal analyses of the ideal two-dimensional Kitaev model likewise show that its encoded information is not a self-correcting memory merely because the ground state is topological [R143]. **[Theory]** The theorem has a scope — two-dimensional stabilizer-code Hamiltonians with local interactions — and should not be inflated into a proof about every non-Abelian phase. The warning, however, travels well: calculate the barrier; do not infer it from the gap.

31.4. Someone has to read the alarms

The four-flip loop left every check quiet. If you want to catch the walk *before* it closes, you have to look, guess a history, and undo it.

That looking-and-guessing is a **decoder**. The hardware does not magically know which pair of endpoints belong together. Someone, or some circuit, has to pick a history.

Active correction measures local checks, repeats measurements when they are noisy, decodes the spacetime syndrome, and applies or tracks a recovery. Its ingredients are operational:

1. extract enough syndrome information without uncontrolled backaction;
2. attach time stamps to repeated rounds;
3. use a decoder matched well enough to the actual noise;
4. complete recovery before errors accumulate beyond the decoder’s capacity.

Active correction can remove thermally created pairs before their worldlines become logical. It can also distinguish a transient detector fault from a persistent data fault by comparing rounds. None of that is passive. The refrigerators do not run minimum-weight matching in their spare time.

What fails if you skip the decoder is exactly the four-flip loop: the alarms go quiet after the lap, and the logical observable has already changed sign.

31.5. A percentage that is not a promise

A *fault-tolerance threshold* is conditional. The name is worth keeping because it is a crossing, not a blessing: below it, growing the code helps; above it, growing the code can make things worse.

Roughly, for a family of growing codes, a specified fault model, syndrome circuit, decoder, and operation schedule, there may be a critical physical-fault parameter p_{th} such that for $p < p_{\text{th}}$, increasing distance drives the logical error per operation downward [R141]. This does not say that every component's quoted "fidelity" may be compared with one universal percentage.

In the canonical independent stochastic model studied by Dennis and collaborators, the toric/surface-code transition is about 10.9% when syndrome measurements are perfect; a phenomenological model with repeated noisy measurements and equal data/measurement error rates gives about 2.9% [R141]. [**Theory/Numerics**] Those are properties of those models, not acceptable error specifications for a defect device. Change leakage, correlations, geometry, boundaries, decoder, or circuit, and the number changes.

Threshold statements must name:

- what counts as a location and a fault;
- whether faults are independent, local stochastic, biased, leaking, or correlated;
- whether state preparation and measurement are noisy;
- whether qubits can disappear or change charge state;
- the decoder and its latency;
- the logical task and error metric.

A gap budget such as $\Delta_{\text{topo}} > k_{\text{B}}T$ is an energy-scale *margin*, not an error-correction threshold. A braid's adiabatic condition is also not a threshold theorem. Vocabulary is cheaper than dilution; use the right noun.

31.6. Zero is a different word

Error-free means the logical channel is exactly the intended one. It is a mathematical identity, not a compliment you award a small number.

An ideal commuting-projector model at zero temperature, initialized exactly in its ground space and acted on by exact operators, can have zero error by assumption. A finite laboratory run instead has a nonzero logical-error probability, even when that probability is impressively small. Below threshold means suppressible with increasing resources, not zero. Passive protection means reduced sensitivity to selected local faults, not zero.

So the four hats, in one place:

- A **gap** charges for creating excitations. It need not charge for moving them.
- A **decoder** guesses a history from alarms and tries to undo it.
- A **threshold** is a crossing for a named noise model, circuit, and decoder.
- **Error-free** is exactly the intended logical channel.

Protected is not immortal. The rest of the chapter is the ways a local accident still becomes a logical disaster.

31.7. Created is not the same as moved

Thermal errors have two stages. Creation pays an energy cost; diffusion may not.

Let a be the lattice spacing in metres and suppose a dilute equilibrium anyon density per site scales as $n_a \sim e^{-\Delta_a/(k_B T)}$. In two dimensions, the characteristic separation then scales as

$$\ell_a \sim \frac{a}{\sqrt{n_a}} \sim a e^{\Delta_a/(2k_B T)}.$$

Both ℓ_a and a have units of metres; the exponential is dimensionless. **[Theory]** This estimate omits species degeneracy, interactions, boundaries, nonequilibrium baths, and kinetic bottlenecks, but it exposes the relevant competition: system size must not be confused with mean thermal separation [R169].

For a hypothetical 10 GHz anyon energy Δ_a/h at $T = 0.10$ K, use $k_B T/h \approx 2.08$ GHz. Then $\Delta_a/(k_B T) \approx 4.8$, so the ideal single-anyon Boltzmann factor is about 8×10^{-3} , while a pair factor is about 7×10^{-5} . This is an illustrative calculation, not a defect-array forecast: the attempt rate, number of creation sites, bath spectrum, and actual many-body gap are still required. Ten thousand opportunities can make a small per-site weight operationally noticeable.

Once present, an anyon can random-walk, be pushed by disorder gradients, or be moved by control crosstalk. A boundary can absorb one member of a pair, leaving a charge whose history is difficult to infer. For non-Abelian anyons, diffusion is worse than a misplaced classical particle: unrecorded exchanges and fusion events act on the fusion space [R015]. Active syndrome extraction and decoding must beat this kinetic process, not merely the initial Boltzmann factor.

What fails if you stop at “the temperature is below the gap” is the second stage. The Boltzmann factor prices birth. It does not pin the walkers.

31.8. Bigger helps one thing and hurts another

A real patch is finite, so three lengths compete: its linear size L , the correlation length ξ , and the characteristic spacing or travel length of unwanted quasiparticles. Increasing L/ξ can exponentially reduce virtual tunnelling between boundaries or around a handle. The associated coherent phase rate is $\delta E/\hbar$, because energy divided by $\hbar$ has units s^{-1} . But a larger area also supplies more places to create thermal pairs and more hardware that can fail. “Bigger” helps one mechanism and hurts another.

Disorder is not one scalar. Weak, bounded, static local perturbations may deform a stable topological phase without destroying it [R142]. **[Theory]** Fabrication disorder can nevertheless vary local gaps, couplings, cluster projectors, and measurement responses. A rare low-gap region becomes a thermal doorway. A sign error or missing interaction may change the effective Hamiltonian rather than merely perturb it. Sufficiently strong disorder can close the mobility or spectral gap and drive the system out of the target phase.

Localization of unwanted quasiparticles can slow diffusion and therefore help memory in some models [R169]. **[Theory/Numerics]** It is not a free upgrade. The same pinning can obstruct intended braids, strand poison quasiparticles near computational anyons, broaden spectroscopy, and invalidate a decoder that assumes homogeneous motion. A protection claim must specify the disorder distribution — including tails — and simulate both equilibrium and dynamics.

31.9. Leaving the space you promised

Two ways to leave the promised description look similar from far away and are not the same accident.

Quasiparticle poisoning means an uncontrolled topological charge enters, leaves, or changes the computational region. The poison is an extra anyon, not a slightly wrong pulse. A thermally created anyon is one source, but poisoning also includes injection from an edge, a reservoir, a driven hotspot, or a nonequilibrium defect. In a non-Abelian device, a stray anyon can fuse with a computational anyon and change its fusion channel. Subsequent nominal braids then implement the wrong unitary [R015]. The error is not necessarily visible to local charge measurements until information has already escaped.

Leakage means a physical constituent leaves the two-level space used by the code. The site is no longer the qubit the decoder was promised. For a defect cluster, define P as the projector onto its intended low-energy doublet and $Q = I - P$ as the projector onto all other cluster states. An ordinary encoded error has support within P . Leakage produces population in Q . Then the projected Pauli model no longer describes the site, and its couplings can spread an error with amplitudes absent from the decoder’s model.

Topological distance does not suppress a single-site leakage event merely because the intended code is topological. Active leakage-reduction schemes periodically remove, reset, swap, or explicitly detect leaked population. Surface-code studies show that untreated leakage can propagate through syndrome circuits and require dedicated handling [R220]. **[Theory/Numerics]** That result is evidence about an active code architecture, not proof that the same protocol works in an interacting defect phase. For an analog cluster array, resetting one cluster may itself quench local couplings and create anyons.

Charge-state switching is especially treacherous. It may simultaneously be leakage from the cluster doublet, a missing or altered coupling in the many-body Hamiltonian, and an optical-readout failure. Counting it once in a Pauli error budget understates the damage.

31.10. Before, during, and after

Topology protects information *after* the system occupies the intended phase and logical sector. Cooling below the gap does not guarantee a known ground state: a finite sample may retain trapped quasiparticles, freeze into the wrong flux sector, or cross a small gap nonadiabatically. Initialization therefore needs a verified preparation protocol — cooling, pumping, syndrome measurement, boundary charge removal, or some combination — and a test of total topological charge.

Control has a window rather than a blessing. A pulse with spectral weight near a leakage transition can populate Q . A local control field strong enough to close the gap along a path can nucleate excitations. An intended adiabatic braid must be slow relative to relevant inverse-gap scales, but making it slower increases exposure to poisoning, drift, and dephasing. The crude timescale $\hbar/\Delta$ has seconds because $(Js)/J = s$; it is only a necessary scale, not a complete adiabatic bound, which also depends on matrix elements and the path's minimum gap [R015]. Topology can make the ideal braid unitary insensitive to smooth geometric deformation. It does not make arbitrary pulse errors, unintended braids, or dynamical phases harmless.

Readout has two layers. Local syndrome readout asks where excitations appear. Logical readout asks for a nonlocal loop, boundary parity, or fusion outcome. A local detector can have assignment error, backaction, crosstalk, and dead time. Repeating a noisy check turns decoding into a three-dimensional spacetime problem; it does not make the check perfect [R141]. A readout result also needs a confidence model conditioned on leakage and charge state. Destructive optical readout may be adequate at the end of a computation while being useless for repeated error correction.

A useful laboratory report therefore gives separate confusion matrices for local checks and logical readout, leakage-conditioned statistics, backaction measurements, cycle time, and decoder latency. Quoting only the best single-defect readout fidelity does not characterize the many-body protocol.

31.11. When many sites fail together

Threshold estimates are often generous only because the model is local. One formal local-stochastic condition bounds the probability that every location in a set S is faulty by a quantity that falls exponentially with $|S|$. Correlations may exist, but high-weight events must become rapidly rarer.

A global magnetic transient, laser-intensity excursion, strain wave, shared microwave phase slip, or calibration error can act on many clusters at once. If one event writes a length- L string with probability proportional to a single-event rate rather than that rate raised to the L th power, increasing code distance may provide little benefit. Long temporal correlations likewise make repeated syndrome rounds agree for the wrong reason. Work on correlated environments shows that fault-tolerance behavior depends on the correlations' spatial and temporal decay rather than on a single per-qubit error probability [R221]. **[Theory]**

Correlations are not automatically fatal. Short-range, sufficiently weak correlations can remain compatible with threshold results under appropriate assumptions. The audit task is to measure or bound the correlation function, map it through the cluster projector, and use that model in decoding and logical-level simulation. Averaging a covariance matrix down to one "error rate" discards exactly the information topology needs.

31.12. What the laboratory actually has

For a defect-cluster proposal, the protection stack should be demonstrated in order:

Layer	Required evidence	What it does not establish
Cluster encoding	isolated doublet, measured leakage gap, projected controls	topological order
Many-body phase	gap, correlations, topological-sector or anyon evidence	useful logical lifetime
Passive suppression	scaling against local perturbations, temperature, and size	active correction or zero error
Syndrome system	repeated check/fusion measurements with backaction characterized	operation below threshold
Active logical memory	logical-error scaling versus distance and rounds under measured noise	universal protected computation
Braiding/control	correct process on the fusion space with poisoning/leakage bounds	error-free gates

Layer	Required evidence	What it does not establish
-------	-------------------	-----------------------------------

[Proposal] A fabricated defect array will have placement disorder, missing or wrong charge states, cluster leakage, local drives, and imperfect optical or microwave readout. The ideal Hamiltonian contains none of these unless they are explicitly added. Consequently, exact diagonalization of a clean patch can establish a model property; it cannot establish passive laboratory protection. Likewise, successful decoding of synthetically injected independent Pauli errors establishes a decoder benchmark; it cannot establish that the material noise is independent or Pauli.

The decisive experiment is a scaling experiment. Vary a meaningful code distance or quasiparticle separation while holding relevant physical conditions fixed, characterize the microscopic noise, and show how logical error changes. Repeat versus temperature to separate thermal activation from control-limited floors, and versus syndrome cadence to separate passive from active benefit. If larger samples only add more poison sources, the data will say so. That is useful science, even when the line slopes the inconvenient way.

31.13. Ways to get this wrong

1. **“The ground-state splitting is exponentially small, so the memory is self-correcting.”** No. Small virtual splitting addresses coherent sector mixing. Thermal creation and diffusion are different processes.
2. **“The temperature is below the gap, so no anyons exist.”** No. Boltzmann suppression is finite, the sample has many sites, and nonequilibrium injection can dominate.
3. **“The physical error rate is below the surface-code threshold.”** Incomplete. Which rate, code, circuit, leakage model, correlations, decoder, and measurement model?
4. **“The braid is topological, so control errors do not matter.”** Smooth path deformations within the same gapped sector may be harmless. Leaving that sector, braiding the wrong charge, or poisoning the fusion space is not.
5. **“Weak disorder localizes anyons, therefore disorder helps.”** Sometimes one kinetic channel improves. Gaps, controls, readout, and intended motion can worsen.
6. **“No syndrome was observed, so no logical error occurred.”** The worked non-contractible loop has empty syndrome.
7. **“Repeated measurement is passive protection.”** It is active error correction, even if software tracks the recovery rather than physically applying it.
8. **“Below threshold means error-free.”** It means logical error can decrease with increasing resources under stated assumptions. Every finite implementation retains residual risk.

31.14. Check yourself

1. **Show that four local Z errors on a noncontractible row can leave no syndrome and still flip the logical qubit.**
 Their endpoints annihilate after the chain closes. The closed chain winds nontrivially and anticommutes with a crossing logical X loop, so $\bar{X}\bar{Z} = -\bar{Z}\bar{X}$.
2. **What fails if you treat the gap as a fence against motion?**
 The gap prices creation. Once two endpoints exist, sliding one of them around the lattice need not cost more energy. The two-dimensional stabilizer no-go is exactly that missing barrier.
3. **What is the difference between passive protection and active correction?**
 Passive protection comes from the Hamiltonian and encoding without syndrome feedback. Active correction measures, decodes, and recovers or tracks errors.
4. **Why is a published threshold not automatically a hardware specification?**
 It belongs to a particular code, noise model, syndrome circuit, decoder, and operation definition; leakage and correlations can change or invalidate it.
5. **How do poisoning and leakage differ?**
 Poisoning changes uncontrolled topological charge in the computational region. Leakage moves a constituent outside its intended local encoded subspace. One physical event can cause both.
6. **What fails if a finite below-threshold device is called error-free?**
 Below threshold describes favorable scaling under stated assumptions. A finite noisy run still has a nonzero logical-error probability. Error-free means the logical channel is exactly the intended one.

A logical fault is a topologically nontrivial spacetime history. A gap suppresses creation but not automatically diffusion. Finite size trades exponentially small virtual splitting against extensive fault opportunities. Disorder can preserve a phase yet create dangerous weak spots. Poisoning and leakage leave the intended computational description. Initialization, control, and readout require separate error models. Correlations can defeat distance-based suppression. Passive suppression, active correction, threshold operation, and error-free operation are four distinct claims. Protected is not immortal.

31.15. Sources

- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0. arXiv:quant-ph/9707021.

- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* **43**, 4452–4505 (2002). DOI: 10.1063/1.1499754. arXiv:quant-ph/0110143.
- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* **11**, 043029 (2009). DOI: 10.1088/1367-2630/11/4/043029. arXiv:0810.1983.
- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* **42**, 065303 (2009). DOI: 10.1088/1751-8113/42/6/065303. arXiv:0810.4584.
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* **88**, 045005 (2016). DOI: 10.1103/RevModPhys.88.045005. arXiv:1411.6643.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195. arXiv:1001.0344.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083. arXiv:0707.1889.
- [R220] A. G. Fowler, “Coping with qubit leakage in topological codes,” *Physical Review A* **88**, 042308 (2013). DOI: 10.1103/PhysRevA.88.042308. arXiv:1308.6642.
- [R221] E. Novais, E. R. Mucciolo, and H. U. Baranger, “Resilient quantum computation in correlated environments: A quantum phase transition perspective,” *Physical Review Letters* **98**, 040501 (2007). DOI: 10.1103/PhysRevLett.98.040501. arXiv:quant-ph/0607155.

Chapter 32. A peak is not a braid

Two other labs want the same thing we want: a leftover you can walk around another leftover, so the memory they share rotates in a way a local bump cannot fake.

The first lab starts with a thin semiconductor wire, an ordinary superconductor glued along it, and a magnet. They hope leftover pieces of electron sit at the two ends and refuse to be ordinary electrons. Those leftovers, if they are what the theory hopes, braid like Ising anyons.

The second lab starts with a flat sheet of electrons, colder than almost anything in the building, in a magnet strong enough to freeze their motion into Landau levels. They hope the electrons lock into a liquid whose leftover swirls, at one particular filling, braid like Fibonacci anyons.

Neither platform is a crystal full of defects. That is ours. Theirs are older, louder, and further along in some rooms. We are going to steal their scorecard: what the braids can compute on paper, what the laboratory has actually seen, and which nouns they are still not allowed to skip.

A zero-bias peak is not a braid. A resistance plateau is not a Fibonacci anyon. Nature is under no obligation to read the press release.

32.1. Allowed, identified, controlled

Imagine finding footprints, identifying the animal, and training it to carry a package. These are three experiments, not one.

1. **A phase is allowed.** A Hamiltonian has a topological region on an ideal phase diagram.
2. **An excitation is identified.** Several independent measurements rule out ordinary states and determine charge, fusion, and statistics.
3. **A computational operation is controlled.** Initialization, a braid or parity-measurement sequence, and readout implement a reproducible unitary on an encoded space.

After this paragraph the footprints are gone. The three layers are the scorecard. Majorana nanowires have strong work at layer one and parts of two. Fractional quantum Hall fluids have conclusive demonstrations of *Abelian* fractional charge and statistics at some filling factors. That does not identify Fibonacci quasiparticles at filling factor $12/5$. Neither substitution is permitted.

The leftovers themselves are not one object wearing four hats.

A leftover sitting at the end of a one-dimensional superconducting segment is an **extrinsic defect mode**. The ends were put there by the fabricator. A vortex in a suitable two-dimensional topological superconductor can behave as an Ising anyon. A digitally programmed Majorana operator is an emulation. A quasihole of an intrinsic quantum Hall fluid is an emergent excitation of the liquid. These objects can share braid mathematics without sharing physical evidence or passive protection.

32.2. What the Ising braid can compute

We first compare the ideal anyon theories. This is a pencil-and-paper experiment. Laboratory status comes later.

The Ising theory has three charges: 1 (vacuum), ψ (a fermion), and σ . Two σ leftovers fuse according to

$$\sigma \times \sigma = 1 + \psi.$$

Encode one qubit in four σ anyons whose total charge is 1. Define $|0\rangle$ as the state in which the first pair fuses to 1, and $|1\rangle$ as the state in which it fuses to ψ . Let B_1 exchange the first two anyons counterclockwise, and let B_2 exchange the middle two. In a common phase convention, after dropping one common global phase,

$$B_1 = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad B_2 = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Now multiply:

$$B_1 B_2 B_1 = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = e^{i\pi/4} H,$$

where $H = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is the Hadamard gate. Thus Ising braids give the phase gate $S = \text{diag}(1, i)$ and H , up to global phases. These belong to the Clifford gate set [R015].

The missing gate is easy to name:

$$T = \text{diag}(1, e^{i\pi/4}).$$

No braid of Ising anyons approximates T arbitrarily well. Ising braiding generates only a finite projective Clifford image, not a dense subset of all one-qubit rotations. Universal

computation therefore needs an added non-topological resource, commonly a specially prepared magic state, a non-topological phase operation, or an appropriate measurement protocol [R015].

Hence non-Abelian is not the same as universal. Drop that distinction and an Ising leftover is asked to do Fibonacci work.

32.3. What the Fibonacci braid can compute

The Fibonacci theory has charges 1 and τ , with

$$\tau \times \tau = 1 + \tau.$$

Encode one qubit in three τ anyons with total charge τ . Define $|0\rangle$ and $|1\rangle$ by whether the first pair fuses to 1 or τ . Let

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

be the golden ratio. A standard choice of the basis-change and exchange matrices is

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad R = \begin{pmatrix} e^{-4\pi i/5} & 0 \\ 0 & e^{3\pi i/5} \end{pmatrix}.$$

The first exchange is $B_1 = R$. To exchange the second and third anyons, first change fusion basis, exchange, and change back:

$$B_2 = F^{-1}RF = FRF,$$

because this convention has $F^{-1} = F$. The relative phase of the two eigenvalues of B_1 is

$$\frac{e^{3\pi i/5}}{e^{-4\pi i/5}} = e^{7\pi i/5} = e^{-3\pi i/5},$$

which is not a Clifford Z -rotation. More importantly, B_1 and B_2 rotate about different axes. Their braid words form a dense representation on the computational space: longer words can approximate an arbitrary one-qubit unitary, and suitable braids of larger encodings supply entangling gates [R015]. Phase conventions can change signs and common phases in these matrices, but not this gate-set conclusion.

Here is the compact result:

Anyon model	Elementary fusion	Braiding supplies	Braiding alone universal?	Missing engineering
Ising/Majorana	$\sigma \times \sigma = 1 + \psi$	Clifford gates such as H and S	No; no dense gate set	non-Clifford resource, parity control, leakage control
Fibonacci	$\tau \times \tau = 1 + \tau$	dense single-qubit rotations and entangling braids	Yes, ideally	compilation, initialization, fusion readout, thermal and leakage control

Universality is a property of the braid representation, not proof that the anyons exist in a sample.

32.4. A wire that wants two leftover ends

Consider a one-dimensional semiconductor with strong spin–orbit coupling, sitting against an ordinary s -wave superconductor, in a magnetic field. Cooper pairing leaks into the semiconductor through the interface. People say the wire is **proximitized**. The name is worth keeping because the pairing is borrowed, not grown in the semiconductor itself.

A minimal Bogoliubov–de Gennes Hamiltonian is

$$H(p) = \left(\frac{p^2}{2m^*} - \mu \right) \tau_z + \alpha p \sigma_y \tau_z + V_Z \sigma_x + \Delta \tau_x.$$

Here p is momentum in kg m s^{-1} ; m^* is the electron effective mass in kg ; μ is chemical potential in joules or electronvolts; α is the spin–orbit coefficient in energy-times-length; V_Z is Zeeman energy; and Δ is the induced superconducting pairing energy. The Pauli matrices σ_a act on spin, while τ_a act on particle–hole space. Every term has units of energy: for example, αp is energy-times-length times inverse length.

For the ideal, uniform, single-band model, the bulk gap closes and reopens at

$$V_Z^2 = \Delta^2 + \mu^2.$$

The region $V_Z > \sqrt{\Delta^2 + \mu^2}$ is topological and supports a leftover mode near each end of a sufficiently long segment [**Theory**] [R222]. Spin–orbit coupling, Zeeman splitting, and ordinary pairing have together manufactured an effective spinless p -wave channel.

That leftover is a **Majorana** operator γ_j . It is its own partner,

$$\gamma_j^\dagger = \gamma_j, \quad \{\gamma_i, \gamma_j\} = 2\delta_{ij},$$

where $\delta_{ij} = 1$ for $i = j$ and zero otherwise. Two Majoranas define one ordinary fermion,

$$f = \frac{\gamma_1 + i\gamma_2}{2}, \quad i\gamma_1\gamma_2 = 2f^\dagger f - 1.$$

The occupation $f^\dagger f$ is stored nonlocally between the two ends. Four Majoranas at fixed total parity provide a qubit. An ideal counterclockwise exchange is represented, up to orientation convention, by

$$U_{ij} = \exp\left(\frac{\pi}{4}\gamma_i\gamma_j\right).$$

These are the same Ising operations used in the worked example.

If the end modes decay over a localization length ξ , their overlap in a clean wire of length L is roughly proportional to $e^{-L/\xi}$, often with an oscillatory prefactor. Increasing L/ξ can suppress coherent splitting. It does not suppress quasiparticle poisoning, a stray low-energy state near one end, a bad tunnel barrier, or a control pulse that closes the gap. Nonlocal encoding protects against a specified class of local perturbations, not against the laboratory as a whole.

32.5. Other leftover-end platforms

Semiconductor nanowires are not the only proposal. Two-dimensional semiconductor–superconductor heterostructures can form gate-defined networks; magnetic-atom chains on superconductors can produce effective topological bands; and vortices or boundaries in candidate intrinsic topological superconductors can host zero-energy states **[Proposal/Theory]** [R223].

The distinction between *intrinsic* and *engineered* matters. In an intrinsic topological superconductor, the superconducting bulk itself carries the relevant topological invariant. In a hybrid, ordinary components collectively realize an effective topological phase under tuned conditions. A sharp zero-energy spectroscopic feature is compatible with a Majorana in either setting, but ordinary Andreev bound states, disorder, and inhomogeneous potentials can imitate important signatures [R223]. Spectroscopy locates a suspect. It does not conduct the trial.

32.6. Electrons in a flat magnet

A two-dimensional electron gas in a perpendicular magnetic field B forms Landau levels. Define the dimensionless filling factor

$$\nu = \frac{nh}{eB},$$

where n is the areal electron density in m^{-2} , h is Planck's constant in J s, and e is the elementary charge in C. The units cancel because eB/h has units of inverse area.

Interactions can stabilize incompressible fractional quantum Hall liquids at rational ν . The $k = 3$ Read–Rezayi state occurs at partial filling $3/5$ in its simplest spin-polarized form and contains a Fibonacci topological sector [**Theory**] [R139]. With two lower Landau levels filled, that corresponds to total filling $13/5$. Particle–hole conjugation within the active Landau level maps it to partial filling $2/5$, hence total filling $12/5$. The conjugate reverses appropriate chiral data but retains Fibonacci fusion content. “Read–Rezayi at $12/5$ ” is therefore often shorthand for this candidate family, not an experimentally stamped label.

The microscopic Coulomb problem does not promise to choose that family. Landau-level mixing, finite quantum-well thickness, disorder, spin polarization, and charge-density order compete. Exact diagonalization and density-matrix-renormalization studies have found regimes consistent with Read–Rezayi or its particle–hole conjugate [**Numerics**] [R225]; [R227]; [R228]. Finite systems and model-dependent corrections leave room for competing Abelian or symmetry-broken interpretations. Agreement with a trial-state overlap or entanglement spectrum is powerful numerical evidence, not a quasiparticle braid.

32.7. Why a Fibonacci leftover would be special

A quasihole in the Fibonacci sector carries charge and topological charge. Its electromagnetic charge affects transport and interferometer phase; its topological charge determines the fusion space. Exchanging quasiholes acts on that space by the F and R matrices above. A decisive program must therefore connect several observables:

- a robust incompressible plateau and activated longitudinal resistance;
- fractional quasiparticle charge;
- edge or thermal measurements compatible with the candidate topological order;
- fusion-channel-dependent interference or projective fusion outcomes;
- noncommuting, order-dependent braid operations consistent with the predicted matrices.

No single item substitutes for the list. A plateau without fusion, or an interferometer without an identified phase, is still only layer one or a fragment of layer two.

32.8. What the laboratory actually has

The evidence snapshot below follows peer-reviewed literature through 2026 and states the strongest justified conclusion, not the most exciting possible interpretation.

Claim	Status	Platform	What was measured or established	Verdict
The ideal spin-orbit nanowire has a topological phase and endpoint Majoranas above the phase boundary	[Theory] established	semiconductor–superconductor	BdG invariant and gap closing/reopening in model Hamiltonians [R222]	candidate mechanism, not material identification
Hybrid and candidate intrinsic superconductors show zero-energy and related signatures	[Experiment] partial	nanowires, planar hybrids, chains, vortex systems	tunnelling and spatial spectroscopy; trivial states can imitate key features [R223]	compatible evidence, platform-dependent and not a braid
Single-shot interferometric parity readout is feasible in an InAs–Al device	[Experiment] demonstrated operation	gate-defined InAs–Al hybrid	up to 1 fF capacitance shift, signal-to-noise ratio 1 in 3.6 μ s, state dwell time above 1 ms near 2 T, and 1% assignment error [R224]	a strong readout primitive; the paper analyzes both trivial and topological origins
Majorana fusion rules and non-Abelian exchange have been conclusively demonstrated in these solid-state devices	not established	Majorana candidates	no accepted controlled exchange establishing the Ising braid representation [R223]; [R224]	remains a required experiment

Claim	Status	Platform	What was measured or established	Verdict
An incompressible 12/5 fractional quantum Hall state exists in high-quality GaAs	[Experiment] demonstrated	GaAs quantum well	activated transport; one 30-nm-well study reported a zero-tilt activation gap near 30 mK [R226]	establishes the plateau, not its topological order
The 12/5 state is the particle–hole-conjugate $k = 3$ Read–Rezayi phase	[Numerics/Theory] supported candidate	cond Landau level	finite-size spectra, entanglement data, and competing-order studies [R225]; [R227]; [R228]	plausible but not uniquely identified experimentally
Fractional statistics can be measured in an electronic quantum Hall interferometer	[Experiment] demonstrated for Abelian anyons	$\nu = 1/3$ Laughlin state	phase slips consistent with quasiparticles encircling localized fractional charge [R131]	validates an important method, but not Fibonacci statistics
Fibonacci fusion and braiding have been observed at 12/5	not established	fractional quantum Hall	no conclusive quasiparticle-identification-and-braid sequence [R139]; [R225]; [R226]; [R227]; [R228]	theoretical candidate, not demonstrated Fibonacci hardware

The thermal scale illustrates the quantum-Hall difficulty. A reported 12/5 activation gap of 30 mK corresponds to

$$k_B T = (8.617 \times 10^{-5} \text{ eV K}^{-1})(0.030 \text{ K}) \approx 2.6 \text{ } \mu\text{eV},$$

where k_B is Boltzmann’s constant [R226]. Energy-per-kelvin times kelvin gives energy, as required. This is an activation scale extracted from transport, not necessarily the pristine many-body gap. Either way, wiring, filtering, and local electrostatic control must avoid heating a fluid whose useful scale is tiny.

32.9. What each platform cannot skip

32.9.1. The wire

1. **Identity before arrays.** Disorder and smooth confinement can create low-energy Andreev states that resemble end Majoranas. Scaling an ambiguous unit cell produces a larger ambiguity.
2. **A hard, uniform gap.** Gates, magnetic field, semiconductor density, and superconductivity must coexist across every segment and junction. The ideal inequality $V_Z > \sqrt{\Delta^2 + \mu^2}$ is not a factory acceptance test.
3. **Networks and motion.** One-dimensional endpoints cannot literally pass each other. Braiding needs junction networks or measurement-only parity sequences, each with calibrated tunnel couplings and protected paths.
4. **Parity lifetime and readout.** Quasiparticle poisoning changes the encoded sector. Fast, repeatable, quantum-nondemolition parity measurements are therefore part of the qubit, not an optional oscilloscope attachment.
5. **Non-universal braids.** Even perfect Ising braiding still needs a high-quality non-Clifford resource. That resource can dominate overhead and weaken the slogan “all gates are topological.”

32.9.2. The fluid

1. **Phase identification.** A $12/5$ plateau must be distinguished from competing orders before its quasiparticles become computational inventory.
2. **Small operating scale.** The fragile state demands ultralow electron temperature, extremely clean material, and low-noise gates [R226].
3. **Controlled quasiparticles.** Gates must create, position, move, and fuse individual quasiholes without closing the gap or unintentionally reconstructing the edge.
4. **Interferometer stability.** Coulomb charging, changing device area, localized quasiparticles, and edge equilibration can alter phase signals. A scalable processor needs many such elements calibrated simultaneously.
5. **Readout and leakage.** Fusion charge must be measured without accidentally adding quasiparticles. Universal braid mathematics does not provide a cable plan.

32.10. Five gates a defect array still has to pass

A defect array may offer site-by-site fabrication and familiar spin readout that a quantum Hall fluid lacks. It pays elsewhere: the desired topological Hamiltonian and anyons have not yet emerged. The comparison suggests five milestone gates.

1. Derive a many-body topological phase from measured microscopic defect parameters, not only from an intended interaction graph.
2. Measure the bulk gap and topological diagnostics while ruling out a prepared but unprotected state.
3. Create localized excitations and establish their fusion channels.
4. Demonstrate order-dependent braid operations or an equivalent measurement-only protocol.
5. Show that logical error decreases when separation, code distance, or another predicted protection parameter increases.

The right benchmark is not “does the platform show a two-level system?” Majorana devices did that long ago. It is whether the full chain from Hamiltonian to protected operation survives alternative explanations.

32.11. Ways to get this wrong

1. **“A zero-bias peak is a Majorana.”** A peak is a spectroscopic observation. Majorana identification requires correlated spatial, gap, parity, fusion, and ultimately statistics evidence; trivial bound states can contribute similar peaks [R223].
2. **“Parity readout proves what carries the parity.”** The 2025 InAs–Al experiment demonstrated an excellent measurement primitive and explicitly treated topologically trivial and non-trivial interpretations [R224]. Instrument capability is not quasiparticle identity.
3. **“The $12/5$ plateau proves Fibonacci order.”** Quantized Hall response and activated transport establish an incompressible phase. They do not uniquely determine its fusion category [R225]; [R226]; [R227]; [R228].
4. **“Quantum Hall anyons have been braided, therefore Fibonacci anyons have been braided.”** The direct interferometric result in Ref. [R131] concerns Abelian Laughlin anyons at $\nu = 1/3$, not the non-Abelian candidate at $12/5$.
5. **“Non-Abelian means universal.”** Ising anyons are non-Abelian but their braids produce Clifford operations only. Fibonacci anyons are computationally universal by braiding in the ideal theory [R015].
6. **“Universal means scalable.”** Universality says a gate set is dense. It says nothing about gap size, poisoning, control wiring, braid time, readout fidelity, fabrication yield, or heat load.
7. **“A simulated braid settles the material question.”** A processor can enact the same matrices without hosting the corresponding emergent quasiparticles. That validates control and theory, not intrinsic topological order.

Keep four nouns apart as you climb. **Candidate phase:** an ideal or numerically studied Hamiltonian supports the order. **Compatible signature:** data agree with a prediction but also admit alternatives. **Quasiparticle identification:** independent observables establish

charge, fusion, and statistics. **Topological qubit:** an encoded state is initialized, operated, and read out with an error advantage tied to topology. Do not skip a noun while climbing this ladder.

32.12. Check yourself

1. **Why are Majorana zero modes associated with Ising rather than Fibonacci computation?**

Their fusion space and exchange operators realize the Ising braid representation, which generates Clifford gates but not a dense universal gate set.

2. **Show that the Ising product $B_1 B_2 B_1$ is a Hadamard, up to a global phase.**

Direct multiplication gives $\frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = e^{i\pi/4} H$. Braiding alone still cannot supply the non-Clifford T gate.

3. **Why does $12/5$ attract Fibonacci interest?**

Its active Landau-level filling is $2/5$, the particle–hole conjugate of the $k = 3$ Read–Rezayi state at $3/5$, whose topological order contains a Fibonacci sector.

4. **What did the 2025 InAs–Al experiment establish?**

Fast single-shot interferometric fermion-parity measurement with 1% assignment error under reported conditions, not conclusive Majorana identity, fusion, or braiding [R224].

5. **What fails if an activated $12/5$ plateau is treated as a Fibonacci qubit?**

The plateau establishes an incompressible phase, not unique fusion content, controlled quasiparticles, fusion-channel readout, a non-Abelian braid test, or scaling of protected logical operations.

6. **What is the central lesson for defect-engineered proposals?**

Measure each link — microscopic Hamiltonian, phase, excitation, statistics, and logical operation — and preserve alternative explanations until an experiment removes them.

The wire has a simple effective Hamiltonian and a live fight over whether the leftover is the leftover it claims. The fluid has a real incompressible liquid and a live fight over which liquid it is. Neither has shown a controlled non-Abelian braid in the solid-state device they want to sell. A defect array is not ahead of them. It has not yet entered the same fight.

32.13. Sources

- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083. arXiv:0707.1889.
- [R222] R. M. Lutchyn, J. D. Sau, and S. Das Sarma, “Majorana fermions and a topological phase transition in semiconductor-superconductor heterostructures,” *Physical Review Letters* **105**, 077001 (2010). DOI: 10.1103/PhysRevLett.105.077001. arXiv:1002.4033.
- [R223] K. Flensberg, F. von Oppen, and A. Stern, “Engineered platforms for topological superconductivity and Majorana zero modes,” *Nature Reviews Materials* **6**, 944–958 (2021). DOI: 10.1038/s41578-021-00336-6. arXiv:2103.05548.
- [R224] Microsoft Azure Quantum, M. Aghaee, A. Alcaraz Ramirez, Z. Alam, *et al.*, “Interferometric single-shot parity measurement in InAs–Al hybrid devices,” *Nature* **638**, 651–655 (2025). DOI: 10.1038/s41586-024-08445-2. arXiv:2401.09549.
- [R139] N. Read and E. Rezayi, “Beyond paired quantum Hall states: Parafermions and incompressible states in the first excited Landau level,” *Physical Review B* **59**, 8084–8092 (1999). DOI: 10.1103/PhysRevB.59.8084. arXiv:cond-mat/9809384.
- [R225] E. H. Rezayi and N. Read, “Non-Abelian quantized Hall states of electrons at filling factors $12/5$ and $13/5$ in the first excited Landau level,” *Physical Review B* **79**, 075306 (2009). DOI: 10.1103/PhysRevB.79.075306. arXiv:cond-mat/0608346.
- [R226] C. Zhang, C. Huan, J. S. Xia, N. S. Sullivan, W. Pan, K. W. Baldwin, K. W. West, L. N. Pfeiffer, and D. C. Tsui, “Spin polarization of the $\nu = 12/5$ fractional quantum Hall state,” *Physical Review B* **85**, 241302(R) (2012). DOI: 10.1103/PhysRevB.85.241302. arXiv:1204.0903.
- [R227] W. Zhu, S. S. Gong, F. D. M. Haldane, and D. N. Sheng, “Fractional quantum Hall states at $\nu = 13/5$ and $12/5$ and their non-Abelian nature,” *Physical Review Letters* **115**, 126805 (2015). DOI: 10.1103/PhysRevLett.115.126805. arXiv:1505.03050.
- [R228] R. S. K. Mong, M. P. Zaletel, F. Pollmann, and Z. Papić, “Fibonacci anyons and charge density order in the $12/5$ and $13/5$ quantum Hall plateaus,” *Physical Review B* **95**, 115136 (2017). DOI: 10.1103/PhysRevB.95.115136. arXiv:1505.02843.
- [R131] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, “Direct observation of anyonic braiding statistics,” *Nature Physics* **16**, 931–936 (2020). DOI: 10.1038/s41567-020-1019-1. arXiv:2006.14115.

Chapter 33. Programmable competitors

A superconducting chip can already fire a short list of microwave pulses and then measure. A cloud of cold atoms, parked by lasers and puffed into giant orbits, can be rearranged into a new graph overnight. Both platforms have prepared states that look like the models of the last few chapters.

A lattice of implanted spins might hide a Fibonacci anyon, but only after growth, placement, and a many-body gap. Every year spent waiting for that Hamiltonian is a year not spent on a processor that can already run a braid circuit. That unpaid bill is *opportunity cost*: the value of the route you did not take.

We will not pick a winner. We will put the same four-spin square on three platforms and compute what each one pays.

33.1. Two different jobs

There are two different jobs. Mix them up and every table in the chapter will lie.

The first job is to *learn the model*. Prepare states, make excitations, measure a braid or a correlation, watch a small system move. The second is to *build protected hardware*. Make the low-energy sector of the physical system itself topologically ordered, so a local bump barely touches the encoded information.

Suppose a circuit prepares the exact ground-state wavefunction of a topological model and then leaves the hardware governed by an unrelated Hamiltonian. Then the state may display topological entanglement at the measurement time, while possessing no energetic protection against later local errors. Hence a prepared state is not a protected phase.

Superconducting processors and Rydberg arrays already have strong evidence for the first job. **[Experiment]** Superconducting processors have prepared toric-code states and moved their excitations [R125], and have digitally simulated Fibonacci string-net states, fusion, and braiding [R138]. **[Experiment]** A Rydberg array has shown signatures of a toric-code-type spin liquid in a programmable analog simulator [R126]. None of those statements, by itself, says that a stored logical qubit enjoyed passive topological protection.

A defect architecture is interesting for the second job only if its nearly static microscopic Hamiltonian produces the model at useful strength. Flexible imitation and passive protection are different products. Drop the distinction and you will credit a digital braid matrix

with a material anyone. That is the counterexample the rest of the chapter is written to prevent.

33.2. The same four-spin square

Place four two-level systems on the edges of one square and label them 1, 2, 3, 4. For system i , the Pauli operator Z_i has eigenvalue $+1$ on $|0\rangle_i$ and -1 on $|1\rangle_i$. The *plaquette operator* is the product of those four questions,

$$P_{\square} = Z_1 Z_2 Z_3 Z_4,$$

and the target Hamiltonian is

$$H_{\square} = -K P_{\square},$$

where $K > 0$ is an energy, measured in joules or electronvolts.

Because $P_{\square}^2 = I$, the only possible eigenvalues are $p = \pm 1$. The $p = +1$ sector has energy $-K$, the $p = -1$ sector has energy $+K$, and the separation is $2K$. Hence a single square already has a parity constraint and a gap.

It does not have topological order. There is no thermodynamic phase, no noncontractible loop, and no pair of locally indistinguishable ground-state sectors. The square is useful because the same term appears as one ingredient in stabilizer Hamiltonians. Fair comparison means all three platforms receive this same small target, the same accounting of controls and ancillas, and the same test for protection.

Drop the last sentence and a platform that can fire six gates will look like a platform that has grown a phase. The gates are real. The phase is a different claim.

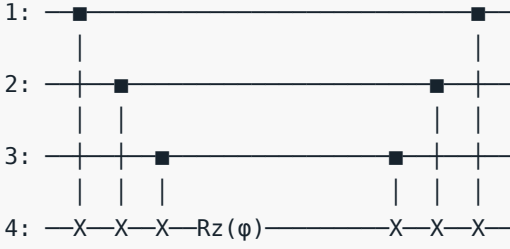
33.2.1. Six gates and a twist

For a short time δt , with units of seconds, the desired evolution is

$$U_{\square}(\delta t) = \exp\left(-\frac{iH_{\square}\delta t}{\hbar}\right) = \exp\left(+i\frac{K\delta t}{\hbar}P_{\square}\right),$$

where $\hbar$ is Planck's reduced constant in joule-seconds. The ratio $K\delta t/\hbar$ is dimensionless.

Let $R_z^{(4)}(\phi) = \exp(-i\phi Z_4/2)$ rotate qubit 4 through angle ϕ , measured in radians. A CNOT with control j and target 4 transforms Z_4 into $Z_j Z_4$ under conjugation. Therefore the sequence



implements $\exp(-i\phi P_{\square}/2)$. Choosing

$$\phi = -\frac{2K\delta t}{\hbar}$$

produces $U_{\square}(\delta t)$ exactly in the ideal gate model. Hence the digital cost of one plaquette step is six two-qubit gates plus one single-qubit rotation.

Sparse connectivity can add routing gates. A parity-measurement construction can instead use an ancilla, but then reset, readout, and feedforward belong in the bill. The circuit is not free just because it is short.

Suppose the target is $H = A + B$, where A and B contain terms that do not commute. A first-order product formula uses

$$e^{-iH\delta t/\hbar} \approx e^{-iA\delta t/\hbar} e^{-iB\delta t/\hbar}.$$

The leading operator error per step is proportional to $\delta t^2[A, B]/\hbar^2$. More, shorter steps reduce this digitization error but require more imperfect gates. Commuting stabilizer terms avoid the product-formula error, not the hardware errors.

33.2.2. Defects have to invent the four-body term

A defect-spin array normally supplies one-spin terms and two-spin interactions — dipolar, exchange, superexchange, strain-mediated, or photon-mediated — not a clean $Z_1 Z_2 Z_3 Z_4$ term. There are two routes.

The *digital route* compiles the CNOT sequence into whatever entangling gates the defects support. It works only if the four spins can be initialized, individually or selectively controlled, and read out, and if the six entangling operations fit inside the error budget. A fixed sparse graph may require a central electron spin, nuclear ancillas, or swaps. **[Experiment]** Pairwise entanglement between individual diamond defect spins has been demonstrated [R080], and a seven-qubit diamond register has implemented fault-tolerant logical operations and flagged stabilizer measurements [R234]. Those results establish valuable primitives. They do not demonstrate a dense, static four-body topological Hamiltonian.

The *analog route* introduces mediator levels or a defect cluster. Virtual excursions can then generate a low-energy four-body interaction. To see the cost without pretending to have a materials design, suppose a particular fourth-order gadget has microscopic coupling energy J , mediator penalty Δ , and a dimensionless coefficient c . Dimensional analysis permits

$$K_{\text{eff}} = c \frac{J^4}{\Delta^3}.$$

Both sides have units of energy. Control of the expansion requires $|J/\Delta| \ll 1$, which also makes $K_{\text{eff}} \ll |J|$. Hence a gadget that is honest about perturbation theory is honest about a small gap.

Lower-order shifts and unwanted couplings must be cancelled or tolerated. **[Proposal]** Defects become competitive for passive Hamiltonian engineering only if the resulting K_{eff} exceeds decoherence, disorder, temperature, leakage, and finite-size scales established in earlier chapters. No pulse diagram can rescue a static gap that is too small.

33.2.3. A chip that already ran the model

A superconducting implementation maps the exact circuit to calibrated microwave pulses and native two-qubit gates. Planar processors commonly provide a fixed or tunable local coupling graph; all-to-one connectivity for this square may be native, mediated by a coupler, or compiled with swaps. Nanofabrication makes repeated layouts possible, while frequency crowding, crosstalk, wiring, leakage beyond the computational doublet, and calibration remain system-level costs [R229].

This is the most direct route to *digital* plaquette evolution. It is also easy to change K , reverse its sign, omit a plaquette, or measure a different string operator in software. **[Experiment]** This control style supported preparation and probing of topologically ordered states on a superconducting processor [R125] and a later digital demonstration of Fibonacci fusion and non-Abelian braid action [R138]. In the Fibonacci experiment, ordinary superconducting qubits simulated the string-net Hilbert space and applied gates representing braids. The hardware did not contain intrinsic material Fibonacci anyons [R138].

A tunable coupler can also engineer driven effective interactions, but “analog superconducting” does not automatically mean passive. If periodic drives, active calibration, or a finite-depth preparation circuit are required, turning them off generally returns a different native Hamiltonian. **[Experiment]** Separately, below-threshold surface-code scaling on a 105-qubit superconducting processor is evidence for active quantum error correction, not evidence for a self-correcting topological material [R230].

Hence the chip that ran the model is a superb instrument for the first job. It is not, by that fact, a casting of the second.

33.2.4. Atoms that refuse to sit next to each other

Park a neutral atom with optical tweezers. A laser can lift its outer electron into a huge orbit. That puffed-up atom is a **Rydberg** atom. The name is worth keeping because the giant orbit is what makes the next atom feel it from far away.

For one commonly used model, the computational state $|1\rangle_i$ is coupled by a laser toward a Rydberg level. Define $n_i = (I - Z_i)/2$ as its occupation projector, Ω_i as a laser Rabi angular frequency in radians per second, δ_i as a detuning in radians per second, and V_{ij} as an interaction angular frequency. A useful rotating-frame Hamiltonian is

$$\frac{H_R}{\hbar} = \sum_i \left(\frac{\Omega_i}{2} X_i - \delta_i n_i \right) + \sum_{i < j} V_{ij} n_i n_j, \quad V_{ij} = \frac{C_6}{r_{ij}^6}.$$

Here X_i flips the two states, r_{ij} is the atom separation in metres, and C_6 has units $\text{rad s}^{-1} \text{m}^6$. The strong r^{-6} interaction creates a *blockade*: nearby atoms cannot all be resonantly excited. Geometry, detuning, and drive strength are therefore programmable Hamiltonian parameters [R231].

The native interaction is pairwise $n_i n_j$, not the product $Z_1 Z_2 Z_3 Z_4$. The plaquette can be produced digitally by compiling the same parity circuit into blockade-mediated entangling gates. A purpose-built multiqubit blockade pulse or an ancilla constraint can reduce depth, but its phase and leakage still require calibration. **[Experiment]** Parallel two-qubit gates with reported 99.5% fidelity on up to 60 atoms, together with low-error three-qubit gates, demonstrate that this route is more than a chalkboard possibility [R232]. They do not establish a native four-body plaquette term.

Rydberg arrays also offer the closest of the three platforms to a rapidly reconfigurable *analog* simulator. The atoms can be rearranged and the physical Hamiltonian above can act continuously rather than as a gate list. **[Experiment]** A constrained 219-atom array reported signatures of a topological spin liquid [R126]. That was a finite, driven atomic simulator with engineered blockade physics. Its evidence is stronger than a classical simulation of the same Hamiltonian. It is not evidence that the apparatus stores an indefinitely passive topological qubit after the lasers are removed.

Drop the lasers and you generally drop the Hamiltonian. That is the remark, not a complaint.

33.3. Count the parts, not the slogans

33.3.1. Who can talk to whom

Connectivity means which pairs can interact strongly enough, selectively enough, and at the same time. It is not the number of lines in a vendor diagram.

Suppose two sites share a long-range tail. Then they are connected in the Hamiltonian, but they are not independently programmable unless the tail can be switched or compiled away. Hence “connected” and “wired on demand” are different predicates.

Defect geometry is largely fixed after fabrication. Dipolar interactions are long-ranged but fall as r^{-3} and carry angular dependence; exchange is shorter-ranged and exponentially sensitive to wavefunction overlap. A beautiful graph on average can still be a poor graph sample by sample.

Superconducting circuits use lithographic nearest-neighbour or coupler graphs that are reproducible, and bus resonators or tunable couplers add options. Long-range logical edges usually consume routing, chip area, frequency allocation, or extra modes.

Rydberg tweezers permit two- and three-dimensional rearrangement, while the blockade radius supplies many-neighbour interactions. Long tails and simultaneous-addressing constraints remain. The graph is programmable. It is not an arbitrary adjacency matrix written in software.

33.3.2. How many useful operations you get

Raw coherence time is a poor scoreboard. A slower qubit can win if its gates are proportionally cleaner, and a long-lived memory can lose if initialization takes most of the experiment. The useful dimensionless quantities are operations per coherence time, error per circuit layer, leakage, and duty cycle.

Superconducting gates are electronically fast and support rapid repetition, but relaxation, dephasing, leakage, and calibration drift operate on the same chip [R229]. Neutral-atom entangling gates exploit strong Rydberg interactions; atom loss, Doppler effects, laser noise, spontaneous emission, rearrangement, and imaging enter the cycle [R231]; [R232]. Defect electron or nuclear spins can be excellent memories, but entangling remote defects, collecting photons, or addressing dense spectra can become the bottleneck.

Comparing one platform’s best memory with another’s full algorithm would be an especially athletic category error. Drop the duty-cycle accounting and the longest-lived spin always wins a race it was not entered in.

33.3.3. Making, resetting, and reading

Superconducting qubits are fabricated as circuits and measured through microwave resonators and amplifiers in a dilution refrigerator. Fabrication is repeatable enough for large processors, but every additional control and readout channel adds cryogenic and calibration load [R229]; [R230].

Rydberg platforms assemble identical atoms from a reservoir rather than fabricating each atom. Rearrangement repairs vacancies before a run; loss can recur during it. State-selective fluorescence gives highly parallel readout, generally destructively for the detected atom, so reloading and sorting matter. **[Experiment]** Reconfigurable arrays have operated an encoded processor with up to 280 physical qubits and 48 logical qubits in sampling circuits [R233]. Those counts describe a logical experiment, not 280 permanently available, error-free atoms.

Defect hosts can hold spins in a compact solid and sometimes permit optical readout, but creating the desired defect, charge state, position, orientation, and spectral line are separate yields. The seven-spin diamond logical experiment [R234] used a local register around one optically active center. It should not be silently redrawn as a wafer-scale lattice of equivalent electronic defect clusters.

33.3.4. Four meanings of scalable

“Scalable” has at least four meanings: more physical sites, more simultaneously high-quality sites, more executable circuit depth, and more logical performance after overhead. Atom count chiefly demonstrates assembly. Chip count chiefly demonstrates integration. Neither alone demonstrates a protected computation.

Superconducting processors require dilution refrigeration and dense classical control. Rydberg processors require ultrahigh vacuum, laser cooling, optical access, stable lasers, and repeated atom handling, though not a millikelvin solid-state stage [R231]. Defects span very different regimes: some spins can operate at room temperature, while high-quality optical interfaces or particular defect species require cryogenic operation.

“Defects need no cryogenics” is therefore not a platform-level fact. Drop the species and the interface, and the slogan becomes a counterexample waiting for a citation.

33.3.5. Changing the energy without compiling

Analog programmability asks which coefficients in H can be changed without decomposing time evolution into gates.

Rydberg arrays offer tunable geometry, local or global detuning, drive amplitude, and strong finite-range constraints. Their native Hamiltonian family is rich but not arbitrary. Superconducting circuits offer tunable frequencies, drives, and couplers, with excellent waveform control but substantial calibration and a tendency toward digital compilation. Defects offer the least post-fabrication reconfiguration; local fields and drives can tune them, but microscopic exchange paths and positions remain fixed.

That inflexibility is a disadvantage for discovery. It is a possible advantage only after the right static Hamiltonian has actually been found. An unchangeable wrong H is not a feature.

33.4. What the laboratory actually has

Evidence in this chapter is assessed through August 2026. The cited experiments are milestones, not purchasing advice; each platform changes faster than a cryostat’s delivery estimate.

The compact comparison is:

Criterion	Crystalline defects	Superconducting processor	Rydberg array
Native interactions	Dipolar, exchange, optical/phonon mediated; strongly materials-dependent	Circuit-mediated two-body gates and tunable couplers	Laser drive plus strong distance-dependent blockade interactions
Graph	Fixed and fabrication-disordered	Lithographic, usually local; routing/couplers extend it	Reconfigurable geometry with interaction tails
Control style	Local microwave/optical control; static interactions possible	Highly developed fast digital pulse control	Strong digital gates and unusually natural analog control
Readout	Species-dependent optical or electrical interface	Multiplexed microwave readout	Parallel fluorescence imaging, with atom-loss/reload cost
Environment	Host- and transition-dependent; room temperature is not universal	Dilution refrigerator and cryogenic wiring	Ultrahigh vacuum, laser cooling, optical infrastructure
Present topological evidence cited here	Defect entanglement and small logical registers, not a material topological phase [R080]; [R234]	Digital toric-code and Fibonacci simulations [R125]; [R138]	Analog spin-liquid signatures in a finite driven array [R126]
Plaquette route	Gate compilation or weak gadget-generated K_{eff}	Short digital circuit; driven analog options	Blockade-compiled gate or constrained analog construction
Passive-protection opportunity	Static host Hamiltonian could qualify if all scale tests pass	Usually active QEC or driven simulation	Emergent analog phase possible during drive; passive memory not automatic

This table does not crown a universal winner because the answer changes with the job.

[Experiment] For programmable exploration today, superconducting processors have directly executed topological-state circuits and digital non-Abelian braid protocols [R125]; [R138], while Rydberg arrays combine flexible geometry with an analog Hamiltonian

that has produced topological-spin-liquid signatures [R126]. **[Experiment]** Defect platforms have compelling local control, entanglement, networking, and small-register results [R080]; [R234], but the cited evidence does not include a defect lattice realizing an emergent non-Abelian phase.

[Proposal] For a static, material Hamiltonian, defects retain a logically distinct opportunity: once fabricated, the interaction network might remain on without a long gate schedule. That statement is an incentive to calculate and measure K_{eff} , disorder, and thermal stability — not evidence that the opportunity has been realized.

33.5. Ways to get this wrong

33.5.1. 1. Calling a prepared state a protected phase

A circuit can prepare the exact ground-state wavefunction of a topological model and then leave it governed by an unrelated hardware Hamiltonian. The state may have topological entanglement signatures at the measurement time, while possessing no energetic protection against later local errors.

33.5.2. 2. Calling active correction passive protection

Surface-code cycles can suppress logical error through repeated syndrome extraction, decoding, and feedback. That is genuine protection and an engineering triumph [R230]. It is *active* protection. A static Hamiltonian gap suppressing local transitions without continual syndrome cycles is *passive* protection. Neither should borrow the other’s adjective.

33.5.3. 3. Calling a Rydberg analog experiment “just digital”

Also wrong. When the laser-driven many-atom Hamiltonian itself realizes a phase over its coherent evolution window, its correlations and excitations are emergent properties of that analog Hamiltonian, not a classical playback of stored answers. But finite lifetime, drive dependence, boundaries, preparation, and readout still limit memory claims.

33.5.4. 4. Treating connectivity as free

A long-range interaction can create unwanted edges as readily as wanted ones. A movable atom consumes transport time. A resonator bus adds modes and crowding. A dipolar tail adds disorder and crosstalk. Every graph must be compiled into simultaneous, calibrated operations or into a quantitatively controlled static Hamiltonian.

33.5.5. 5. Comparing a proposed defect array with demonstrated competitors

A projected defect gap should be labelled [**Proposal**] or [**Speculation**], not placed in the same table cell as a measured gate or correlation. Conversely, present programmable hardware should not be credited with passive Fibonacci protection merely because it displayed a Fibonacci braid matrix.

The superconducting Fibonacci experiment [R138] used physical superconducting qubits to digitally emulate Fibonacci string-net states and braids. The Rydberg spin-liquid experiment [R126] probed correlations emerging from a driven analog Hamiltonian in a finite atomic array. A defect spin is a physical qubit; a register code such as [R234] is an encoded logical qubit. A static defect lattice would possess topological order only if its many-body Hamiltonian, spectrum, correlations, and excitations met the phase criteria. None of the first three observations implies the fourth.

33.6. Check yourself

1. **Show that $P_{\square}^2 = I$ forces a gap of $2K$.**

The only eigenvalues of a square-one Hermitian operator are ± 1 . Then $H_{\square} = -KP_{\square}$ takes the values $-K$ and $+K$, so the sectors are separated by $2K$.

2. **Why is one plaquette not topological order?**

It lacks an extended many-body phase, nonlocal logical sectors, and the scaling properties that make local perturbations ineffective.

3. **What fails if you treat a digitally displayed Fibonacci braid as a material anyon?**

The hardware Hamiltonian need not contain the anyon at all. Ordinary qubits can represent the Hilbert space and apply gates. The braid matrix is then a circuit output, not an excitation of the chip.

4. **Show that the six-CNOT sequence with $\phi = -2K\delta t/\hbar$ produces $U_{\square}(\delta t)$ in the ideal gate model.**

Conjugation by the three CNOTs turns $R_z^{(4)}(\phi)$ into $\exp(-i\phi P_{\square}/2)$. That choice of ϕ is exactly $\exp(+iK\delta t P_{\square}/\hbar)$, which is $U_{\square}(\delta t)$.

5. **What would make a defect implementation scientifically distinct?**

A quantitatively verified static low-energy Hamiltonian with a many-body gap and perturbative stability could provide passive physics without replaying a circuit. Long-lived isolated spins alone do not meet that condition.

6. **What fails if you compare one platform's best memory with another's full algorithm?**

The useful scores are operations per coherence time, error per layer, leakage, and duty cycle. A long-lived spin that cannot be entangled or read on schedule loses the race it was not entered in.

33.7. Sources

- [R229] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, “A quantum engineer’s guide to superconducting qubits,” *Applied Physics Reviews* **6**, 021318 (2019). DOI: 10.1063/1.5089550.
- [R125] K. J. Satzinger *et al.*, “Realizing topologically ordered states on a quantum processor,” *Science* **374**, 1237–1241 (2021). DOI: 10.1126/science.abi8378.
- [R138] S. Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6.
- [R230] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* **638**, 920–926 (2025; published online 2024). DOI: 10.1038/s41586-024-08449-y.
- [R231] A. Browaeys and T. Lahaye, “Many-body physics with individually controlled Rydberg atoms,” *Nature Physics* **16**, 132–142 (2020). DOI: 10.1038/s41567-019-0733-z.
- [R126] G. Semeghini *et al.*, “Probing topological spin liquids on a programmable quantum simulator,” *Science* **374**, 1242–1247 (2021). DOI: 10.1126/science.abi8794.
- [R232] S. J. Evered *et al.*, “High-fidelity parallel entangling gates on a neutral-atom quantum computer,” *Nature* **622**, 268–272 (2023). DOI: 10.1038/s41586-023-06481-y.
- [R233] D. Bluvstein *et al.*, “Logical quantum processor based on reconfigurable atom arrays,” *Nature* **626**, 58–65 (2024). DOI: 10.1038/s41586-023-06927-3.
- [R080] F. Dolde *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R234] M. H. Abobeih *et al.*, “Fault-tolerant operation of a logical qubit in a diamond quantum processor,” *Nature* **606**, 884–889 (2022). DOI: 10.1038/s41586-022-04819-6.

Chapter 34. The coordinate that is not the atom

You type a coordinate. The beam fires. A nitrogen ion punches into diamond and does not stop at the number you typed.

It rattles through the lattice, dumps energy into electrons and nuclei, and comes to rest somewhere nearby. Then you heat the stone. A vacancy wanders. Maybe it finds the nitrogen. Maybe it finds the surface. The glowing center, if one appears, sits where the chemistry finished — not where the file said “here.”

That gap is the whole chapter. A map of commanded sites is not a Hamiltonian.

The Hamiltonian you actually get is drawn by missing centers, extra centers, wrong charges, dark emitters, damaged spins, and bonds that landed a few nanometres off. Chapter 28 already showed why a few nanometres is not a rounding error: electron-spin dipolar coupling scales as $J \propto r^{-3}$, so a small fractional error in bond length is tripled as a fractional error in coupling. A 20 nm graph that wants 20% coupling scatter has almost no room left.

Several processes can pattern emitters, grow a thin dirty layer, or count ions one by one. None cited here has made a dense, uniformly coupled, defect-complete cluster lattice at the sub-nanometre relative tolerance that graph demands.

34.1. Seven inspections, not one aim

Think of building a row of workshops. The truck crosses a property line. The cargo ploughs through the ground. A mobile part wanders during heating. The workshop may or may not assemble, receive the right number of electrons, open its door, or run the machine inside specification.

Those pictures name seven different questions. After this paragraph they die.

Beam accuracy says where the ion crosses the surface. *Straggle* says where it stops after ploughing through the crystal. *Diffusion* says where a mobile vacancy wanders during heating. *Creation yield* says whether the intended complex assembles. *Charge-state yield* says whether it receives the right number of electrons. *Optical usability* says whether initialization and readout work. *Spin usability* says whether the machine inside initializes, stays coherent, and answers within specification.

A bright spot at the right map coordinate can fail the last test. A beautiful spin can fail the map. A complete array needs every indispensable site to pass every inspection, unless the architecture explicitly includes spares and repair.

The picture below follows one nine-site coordinate file through delivery and annealing to the endpoint audit. Dashed rings are commanded positions, green circles are accepted spin-active centers, and offsets or failed sites remain part of the device description.

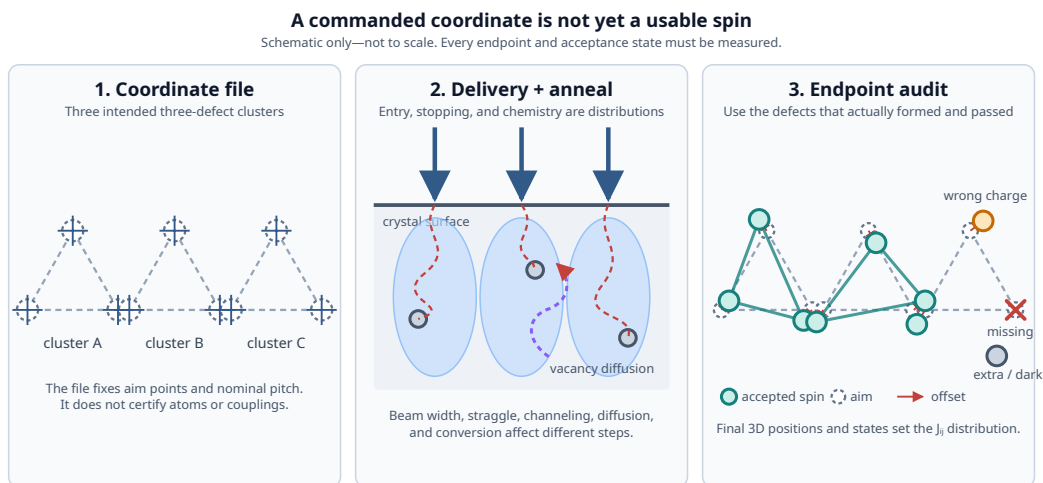


Figure 34.1.: Commanded coordinates do not determine the final defect array. Ion stopping, vacancy diffusion, structural conversion, charge state, and spin acceptance separate the coordinate file from the realized interaction graph.

Every fabrication and acceptance step has a conditional probability. Multiplying a beam-position number by a conversion yield is not enough. One has units of length. The other is a probability. The device needs both.

34.2. Nine spins and a 0.94 nanometre budget

Consider [**Proposal**] three triangular encoded clusters, each containing three electron-spin defects. This nine-site patch is not a topological phase. It is a fabrication coupon: can you place the same intra-cluster interactions more than once?

Let the intended nearest-neighbour separation inside each cluster be $r = 20$ nm. Require root-mean-square radial dipolar-coupling disorder below 20%.

Let σ_{site} be the independent one-dimensional standard deviation of each endpoint along a bond. For $J = C/r^3$, first-order propagation gives

$$\frac{\sigma_J}{J} \approx \frac{3\sqrt{2}\sigma_{\text{site}}}{r}.$$

The ratio is dimensionless because both σ_{site} and r are lengths. Hence the placement allocation is

$$\sigma_{\text{site}} \leq \frac{0.20(20 \text{ nm})}{3\sqrt{2}} = 0.94 \text{ nm}.$$

This is a derived requirement, not a reported capability. It also omits angular-factor disorder, crystallographic site discreteness, exchange variation, and correlated registration error. Drop those omissions and the budget shrinks further.

Now compare it with a measured endpoint distribution. Focused Si implantation followed by annealing produced single SiV centers with a one-axis standard deviation of about 32 nm around target coordinates [R190]. **[Numerics]** If, only for a scale estimate, that distribution were an unbiased Gaussian and the acceptance window were $|x| < 0.94 \text{ nm}$, then

$$P(|x| < 0.94 \text{ nm}) = \text{erf}\left(\frac{0.94}{\sqrt{2}(32)}\right) \approx 0.0235.$$

Requiring the same window independently in two lateral coordinates gives approximately $(0.0235)^2 = 5.5 \times 10^{-4}$ per site before depth is tested. Raising that toy number to nine gives about 4.8×10^{-30} . The calculation is not a forecast — the measured distribution need not be Gaussian into its central nanometre, and the SiV process was not optimized for this coupon. Its purpose is to expose the orders-of-magnitude mismatch between “tens of nanometres” and a sub-nanometre coupling-disorder allocation.

34.3. A ledger, not a single yield

Placement is only half the budget. Define the events

- A : the requested ion or vacancy is delivered;
- C : the desired structural complex forms;
- Q : the desired charge state is occupied;
- O : initialization and optical readout pass;
- S : spin coherence and control pass;
- G : the final three-dimensional position passes.

The *usable-site probability* is the conditional product

$$p_{\text{use}} = P(A)P(C|A)P(Q|A, C)P(O|A, C, Q)P(S|A, C, Q, O)P(G|A, C, Q, O, S).$$

This identity does not assume independence. Each factor is conditional on previous acceptance. Hence a single published “yield” is almost never p_{use} . It is some prefix of the product, taken under the experiment’s own acceptance cuts.

In a deterministic ^{15}N implantation experiment, counted molecular ions were focused with 121(35) nm lateral resolution and produced about 0.6% optically identified $^{15}\text{NV}^-$ centers after annealing [R258]. **[Experiment]** That 0.6% is already a composite of delivery, structural conversion, negative charge, and optical detection — not a separate value for each factor. Even treating 0.006 as an optimistic upper bound on p_{use} would give

$$Y_9 \leq (0.006)^9 \approx 1.0 \times 10^{-20},$$

before imposing the 0.94 nm geometry requirement.

For planning, suppose a future same-process line reaches the following explicitly hypothetical conditional values: 0.995 delivered, 0.50 structural conversion, 0.90 desired charge, 0.85 optical acceptance, 0.90 spin acceptance, and 0.80 geometric acceptance. **[Proposal]** Then

$$p_{\text{use}} = 0.995(0.50)(0.90)(0.85)(0.90)(0.80) = 0.274,$$

and, if sites are independent, $Y_9 = p_{\text{use}}^9 = 8.7 \times 10^{-6}$. Correlated wafer damage or anneal drift could make the array yield still worse. Conversely, measured correlations, spares, repeated implantation, or post-fabrication selection require a different yield model. With no repair, even a 50% nine-site array yield requires

$$p_{\text{use}} \geq 0.5^{1/9} = 0.926.$$

That 92.6% is the useful fabrication target. “One ion on command” is merely its first factor. Drop the later factors and a counted arrival looks like a finished spin.

34.4. Shoot an ingredient, then wait for a workshop

Conventional implantation accelerates an ion species into a crystal, often through a lithographic aperture. The surface-entry distribution is set by aperture geometry, alignment, beam divergence, and vibration.

The ion then loses energy through electronic and nuclear collisions. Its stopping depth and lateral displacement are distributions, called longitudinal and lateral **straggle**. The name is worth keeping because it is leftover scatter, not a typo in the aim. Crystal channeling can add non-Gaussian tails.

The implanted atom is not automatically the desired center. Nitrogen must acquire a neighbouring vacancy to become an NV center. An implanted silicon atom must reach the appropriate split-vacancy configuration for SiV. Annealing supplies mobility and repairs some collision damage, but can also move vacancies to surfaces, aggregates, or unintended impurities.

In one masked CN^- implantation experiment, roughly 50 nitrogen atoms per 80 nm aperture produced a mean 3.5 NV centers — about 7% N-to-NV conversion under an 800 °C, two-hour anneal [R202]. **[Experiment]** It demonstrated patterned ensembles, not exactly one usable NV at each commanded atomic coordinate.

Implant energy presents a recurring trade. Lower energy gives shallower stopping and usually smaller straggle, but near-surface charge noise, vacancy loss, and damage can reduce conversion and coherence. Higher energy can improve vacancy production and conversion while broadening depth and lateral distributions [R260]. **[Experiment]** There is no host-independent “implantation precision”: ion mass, energy, crystal orientation, target composition, and anneal recipe belong beside every number.

34.5. Take the mask off, keep the collisions

A focused ion beam raster-writes coordinates, which helps registration to pillars, waveguides, and cavities. It does not abolish stopping straggle.

For focused silicon implantation into diamond, the beam full width at half maximum was typically below 40 nm, calculated lateral straggle was 19 nm, and the measured created-center distribution had a 32 nm one-axis standard deviation [R190]. **[Experiment]** The measured Si-to-SiV conversion reached about 2.5% at 100 keV; electron irradiation and a second anneal raised a tested bulk-sample value to about 20% [R190]. Those values attach to that SiV process, not to NV, SiC, or sapphire.

Focused writing is attractive for sparse photonic devices because each coordinate can be registered independently. For dense interacting lattices, beam tails, dose calibration, re-deposition, cumulative damage, stage drift, and write time become array variables. A nominal beam spot is a creation *aim*, not a final center covariance matrix.

Suppose the beam were infinitely sharp. The ion would still stop according to a collision distribution, and the vacancy would still wander. Hence a vanishing entry width does not imply a vanishing Hamiltonian error.

34.6. A thin layer with no street address

During growth you can open the impurity tap for only a thin interval, then close it. That thin dirty sheet is *delta doping*: a depth trick, not a street address.

In diamond, nitrogen layers about 1–2 nm thick were grown, then vacancies were supplied by irradiation and annealing; cap thickness selected nominal NV depths from about 5 to 100 nm [R204]. **[Experiment]** This is powerful depth engineering and avoids the implanted nitrogen’s stopping distribution. It does not choose a lateral lattice site: dopants remain random within the plane.

Delta doping is therefore complementary to a localized vacancy source. It narrows one coordinate while another process selects x and y . A 2025 experiment combined a nitrogen delta layer with a localized 200 keV electron beam and reported about 4 nm depth confinement and lateral standard deviation 46(1) nm in 280 nm diamond pillars [R207]. **[Experiment]** Characterized NVs had mean Hahn-echo $T_2 = 98(37) \mu\text{s}$ under those conditions [R207]. This is unusually complete endpoint evidence — position plus a spin metric — but it is not yet a 20 nm-pitch coupled lattice.

Drop the lateral process and you have a well-defined sheet of random sites. That is a useful sample. It is not a graph.

34.7. Write vacancies, not impurity atoms

Energetic electrons transfer momentum to lattice atoms and create vacancies with much less chemical implantation than a heavy ion beam. Uniform irradiation can raise the probability that pre-existing or implanted dopants find vacancies. A focused electron beam can localize vacancy generation laterally, while a delta-doped layer supplies depth selection [R207]. The vacancy-generation volume is still three-dimensional, and mobile vacancies react during annealing.

Electron irradiation therefore changes $P(C|A)$ and the damage landscape. It does not by itself guarantee A for a chosen impurity, Q for the final charge, or S for the spin. In the SiV work, irradiation plus re-annealing improved conversion in a separate tested sample [R190]. Combining its improved yield with another sample’s best placement as though both occurred simultaneously would be a record-value collage.

34.8. Stop when one light appears

A tightly focused femtosecond pulse can create vacancies inside diamond without implanting a foreign atom. Subsequent local heating or furnace annealing lets a vacancy bind to native nitrogen. Optical feedback can stop the write-and-anneal sequence after a single emitter appears, rather than accepting the Poisson limit of one center per blindly dosed site.

Chen and colleagues reported approximately 96% single-NV creation yield with feedback and about 33 nm in-plane positioning deviation [R203]. **[Experiment]** This is a major creation-yield result. It is not 96% sub-nanometre three-dimensional placement, 96% NV[−] charge occupancy under every optical condition, or 96% spin acceptance. The method also requires suitable pre-existing nitrogen near the written vacancy.

Feedback raises $P(C|A)$ for a bright center by watching and stopping. It does not write the covariance matrix of the finished spin. Those remain different columns in the ledger.

34.9. Align through the last hole

A scanning probe can carry a nanometre-scale aperture in its cantilever. The probe images surface markers, aligns locally, and admits ions through the aperture. Persaud and colleagues integrated such a piezoresistive probe with an ion beam and demonstrated single highly charged-ion hits in resist; the terminal aperture limited the spot size [R259]. **[Experiment]** This solves an important registration problem and can bring a stencil close to the surface.

It does not remove aperture convolution, ion straggle, vacancy diffusion, or stochastic conversion. The cited instrument paper did not demonstrate a complete lattice of charge-stable, coherent color centers. “Scanning probe” names the alignment and collimation strategy. It is not a yield certificate.

34.10. Heat until something wanders

Vacancy engineering chooses how many vacancies are generated, where they are generated, and which sinks compete for them. Heavy ions, electrons, neutrons, and lasers create different damage profiles. Annealing temperature and time control vacancy mobility, recombination, aggregation, surface loss, and capture by dopants.

In nitrogen-rich diamond implanted with focused Ar ions, Racke and colleagues measured 0.04–0.79 created NV centers per implanted Ar ion over 12–132 keV [R206]. **[Experiment]** Their diffusion-and-surface-loss model placed an upper bound of roughly 300 nm

on the single-vacancy diffusion length for an 800 °C anneal [R206]. **[Numerics]** That upper bound is not a statement that every final NV is displaced by 300 nm: nitrogen was already distributed in the crystal, and the endpoint is a reaction-and-capture process. It does prove that annealing cannot be represented honestly as “damage erased, position unchanged.”

Annealing can also change charge compensation and optical linewidth by altering residual defects. The correct optimization target is not maximum luminescence alone. It is the joint distribution of final position, desired charge, optical linewidth and stability, spin coherence, and nearby dark paramagnets.

Suppose you treat the furnace as a reset button. Then you will count a bright center as a well-placed spin and miss the capture walk that placed it. The 300 nm bound is the counterexample.

34.11. Count the arrivals, then audit the leftovers

The apparatus can detect or prepare a known number of incident ions and stop at the requested count. That is *deterministic delivery*: it names the count, not the fate.

It removes dose-number uncertainty. It does not make the collision trajectory deterministic, nor does it force an implanted atom into a particular complex or charge state.

Groot-Berning and colleagues extracted single laser-cooled $^{15}\text{N}_2^+$ ions at 5.9 keV, implanted without a mask, and verified $^{15}\text{NV}^-$ centers by ODMR after annealing [R258]. **[Experiment]** The lateral resolution was 121(35) nm and conversion about 0.6% [R258]. This is genuinely deterministic ion-number delivery and genuinely stochastic usable-center creation in the same experiment. The distinction is the central lesson of this chapter.

34.12. What the laboratory actually has

The table keeps unlike metrics in separate columns. A dash means the cited experiment did not establish that quantity as an array-level probability.

Process and conditions	Beam or creation accuracy	Straggle or diffusion	Structural/charge result	Optical/spin usability	Compound-array verdict
CN ⁻ through 80 nm apertures; diamond; 800 °C, 2 h [R202]	Mask-defined 80 nm entry aperture	About 9 nm simulated straggle for the 20 keV N component [R202]	About 7% N-to-NV conversion	Fluorescent NV arrays; no complete spin-acceptance distribution	Patterned sites, not deterministic single centers
Focused Si, diamond, up to 100 keV [R190]	Beam FWHM below 40 nm	19 nm calculated lateral straggle; 32 nm measured one-axis endpoint spread	Up to about 2.5% SiV conversion; separate irradiation/re-anneal test about 20%	Nearly lifetime-limited emitters occurred in nanostructures; not every site qualified	Useful photonic targeting; insufficient evidence for dense complete clusters
N delta layer plus focused electrons, diamond [R207]	20 nm electron spot	About 4 nm depth confinement; 46(1) nm lateral standard deviation	Dose-tunable NV count, not absolute deterministic conversion	Mean Hahn-echo $T_2 = 98(37) \mu\text{s}$ for characterized NVs	Depth and spin evidence coexist; lateral graph remains broad
Femtosecond vacancy writing with feedback, diamond [R203]	Diffraction-scale write focus	Vacancy creation and local annealing set endpoint	About 96% single-NV creation under feedback	Optical detection closes the feedback loop; full charge/spin pass rate separate	High creation yield, about 33 nm in-plane deviation
Counted $^{15}\text{N}_2^+$ at 5.9 keV, diamond [R258]	121(35) nm lateral resolution	Included in measured resolution, not eliminated	About 0.6% $^{15}\text{NV}^-$ conversion	ODMR and coherence characterized for created centers	Deterministic arrivals, very low nine-site compound yield
30 keV C implant through mask, 4H-SiC [R205]	Predetermined array coordinates	Simulated shallow V_{Si} depth about 40 nm [R205]	$19 \pm 4\%$ generation efficiency; $34 \pm 4\%$ probability of a single emitter at optimized dose	Single emitters identified optically	Host-specific patterned array; no demonstrated sub-10-nm pair distribution

The strongest evidence is complementary rather than cumulative. Delta doping gives thin depth. Focused beams and scanning apertures give lateral registration. Electron beams and lasers place vacancy generation. Feedback raises the probability of exactly one bright center. Counted-ion implantation fixes arrival number. Annealing activates complexes. No demonstrated process stack cited here simultaneously supplies sub-nanometre relative placement, near-unity desired-center formation, near-unity desired charge, array-wide optical and spin qualification, and a complete interaction graph.

A counted ion is a fabrication event. A fluorescent color center is a physical emitter. An ODMR-active center is a physical spin qubit candidate. Several coupled spins may form an encoded qubit only after a low-energy subspace is demonstrated. A regular image of bright points is not an interaction graph, and an interaction graph is not an emergent topological phase or a logical topological qubit. Fabrication supplies microscopic degrees of freedom and disorder distributions. It does not supply topological order by vocabulary.

34.13. Ways to get this wrong

1. **Beam width called placement accuracy.** Entry width, simulated stopping straggle, annealed endpoint spread, and device-registration error answer different questions. Report each with axis, statistic, and conditions.
2. **Creation yield called deterministic implantation.** Counted-ion delivery controls A . The 0.6% $^{15}\text{NV}^-$ result [R258] shows directly that C , Q , and detection remain stochastic.
3. **Brightness called charge or spin yield.** A fluorescence threshold can reject dark sites, but it does not by itself establish charge stability during control, T_2 , gate fidelity, or absence of nearby dark spins.
4. **Annealing treated as a reset button.** It mobilizes the vacancies needed for creation and changes damage; it can also broaden capture positions, lose vacancies at a surface, or form unwanted aggregates [R206].
5. **Record values assembled across incompatible samples.** The 96% laser-feedback yield [R203], 1–2 nm delta layer [R204], 32 nm SiV endpoint spread [R190], and a separate long-coherence sample do not describe one fabrication line.
6. **Site yield substituted for array yield.** If every one of N sites is indispensable and independent, $Y_N = p_{\text{use}}^N$. Correlations require a measured joint model, not permission to omit the exponent.
7. **A Gaussian standard deviation treated as the whole disorder model.** Channeling and diffusion create tails; rare close pairs dominate r^{-3} coupling. Supply three-dimensional endpoints, biases, covariance, tail quantiles, missing and extra centers, and dark-defect statistics.

8. **Importing diamond processing into another host.** The 4H-SiC result [R205] has its own ion, energy, vacancy, charge, and anneal physics. Sapphire or hBN needs a new conditioned ledger, not renamed diamond numbers.

A lithography file is a list of commanded coordinates. The Hamiltonian is the joint distribution of the centers that actually formed, charged, glowed, spun, and sat still. Those are different documents.

34.14. Check yourself

1. **What does deterministic implantation determine?**

The number of delivered ions, when individual arrivals are prepared or detected. It does not determine stopping point, complex formation, charge, optical usability, or spin quality.

2. **Show that a 20 nm dipolar graph with 20% radial coupling scatter needs $\sigma_{\text{site}} \lesssim 0.94$ nm.**

First-order propagation gives $\sigma_J/J \approx 3\sqrt{2}\sigma_{\text{site}}/r$. Set the left side to 0.20 and $r = 20$ nm. Then $\sigma_{\text{site}} \leq 0.20 \times 20/(3\sqrt{2}) = 0.94$ nm.

3. **What does delta doping do especially well?**

It confines dopants in depth during growth. By itself it does not assign lateral coordinates.

4. **What fails if you treat annealing as a reset button?**

Vacancies move. Capture positions broaden, surfaces steal vacancies, and unwanted aggregates can form. The 300 nm single-vacancy diffusion bound for an 800 °C anneal is the existence proof that “damage erased, position unchanged” is false.

5. **Show that independent $p_{\text{use}} = 0.90$ on nine indispensable sites yields about 38.7%, and that a 50% array target needs at least 92.6%.**

Under independence, $Y_9 = 0.90^9 \approx 0.387$. The inverse is $p_{\text{use}} \geq 0.5^{1/9} = 0.926$.

6. **What data should enter a Hamiltonian simulation?**

Sampled three-dimensional center coordinates and species, missing/extra-site indicators, charge and spin acceptance, covariance and tails, and the resulting distribution of intended and unintended couplings — not a nominal pitch alone.

34.15. Sources

- [R202] P. Spinicelli *et al.*, “Engineered arrays of NV color centers in diamond based on implantation of CN^- molecules through nanoapertures,” *New Journal of Physics* **13**, 025014 (2011). DOI: 10.1088/1367-2630/13/2/025014; arXiv:1008.1483.
- [R190] T. Schröder *et al.*, “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* **8**, 15376 (2017). DOI: 10.1038/ncomms15376; stable full text.
- [R204] K. Ohno *et al.*, “Engineering shallow spins in diamond with nitrogen delta-doping,” *Applied Physics Letters* **101**, 082413 (2012). DOI: 10.1063/1.4748280; arXiv:1207.2784.
- [R207] S. Kim *et al.*, “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* **16**, 9803 (2025). DOI: 10.1038/s41467-025-64758-4.
- [R203] Y.-C. Chen *et al.*, “Laser writing of individual nitrogen-vacancy defects in diamond with near-unity yield,” *Optica* **6**, 662–667 (2019). DOI: 10.1364/OP-TICA.6.000662.
- [R206] P. Räcke, L. Pietzonka, J. Meijer, D. Spemann, and R. Wunderlich, “Vacancy diffusion and nitrogen-vacancy center formation near the diamond surface,” *Applied Physics Letters* **118**, 204003 (2021). DOI: 10.1063/5.0046031.
- [R205] J. Wang *et al.*, “Efficient generation of an array of single silicon-vacancy defects in silicon carbide,” *Physical Review Applied* **7**, 064021 (2017). DOI: 10.1103/PhysRevApplied.7.064021.
- [R258] K. Groot-Berning, G. Jacob, C. Osterkamp, F. Jelezko, and F. Schmidt-Kaler, “Fabrication of $^{15}\text{NV}^-$ centers in diamond using a deterministic single ion implanter,” *New Journal of Physics* **23**, 063067 (2021). DOI: 10.1088/1367-2630/ac0753; arXiv:2101.01979.
- [R259] A. Persaud, J. A. Liddle, T. Schenkel, J. Bokor, T. Ivanov, and I. W. Rangelow, “Ion implantation with scanning probe alignment,” *Journal of Vacuum Science & Technology B* **23**, 2798–2800 (2005). DOI: 10.1116/1.2062628.
- [R260] S. Pezzagna, B. Naydenov, F. Jelezko, J. Wrachtrup, and J. Meijer, “Creation efficiency of nitrogen-vacancy centres in diamond,” *New Journal of Physics* **12**, 065017 (2010). DOI: 10.1088/1367-2630/12/6/065017.

Chapter 35. One microscope, three neighbors

You park a green laser on one diamond. A single red point answers. In an earlier chapter that was the whole experiment: one missing atom, one spin, one microscope.

Move the defects closer until they can talk. The same microscope still sees a red point. The point is now three neighbors sharing one spot. The microwave that would flip your target also lands on the two next door.

The red flash does not say whose flash it is. A frequency shift that would label your site is, when the pulse is off, just disorder. You need a way to pick one cluster without wrecking the reason you packed them this tight.

35.1. Five jobs, not one flash

An isolated defect that behaves under a microscope is not a controllable array. In a dense patch every laser spot contains neighbors, every microwave line radiates past its intended site, and every frequency shift changes part of the Hamiltonian you were trying to keep uniform.

Five jobs have to be named separately.

1. *Initialize* charge, spin, and encoded-cluster state.
2. *Select* a site or cluster in space or frequency.
3. *Drive* a desired physical or encoded transition.
4. *Measure* it with a stated degree of destructiveness.
5. *Calibrate* the controls without quietly replacing the many-body model you came to test.

The spectral differences that make sites distinguishable are on-site disorder when the control is off. That is the trade, not a footnote.

Previous chapters asked for localized defect states, coherent coupling, a low-energy cluster doublet, the right graph, effective multi-spin terms, and a many-body gap. None of that answers the lab question: which cluster did that pulse hit?

For an analog topological Hamiltonian the knobs are not clerical. A magnetic gradient changes Zeeman energies. Strain and electric fields shift optical and spin transitions and can change the encoded wavefunctions. A focused optical reset can ionize, heat, or project

neighboring defects. Local control perturbs the same system whose symmetry and uniformity are supposed to support a phase.

[Experiment] Optical initialization, coherent microwave control, and optical readout are mature for isolated diamond NV centers, and multi-spin registers around individual centers have been controlled [R074]; [R119]. **[Proposal]** Integrating those primitives with dense defect clusters and an intrinsic topological Hamiltonian remains an architecture, not a demonstrated machine.

Drop the last distinction and a beautiful single-center register will be redrawn as a lattice. It is a register.

35.2. Four ways to pick one

A conductor can cue an orchestra all at once. Singling out one player needs a label: position, pitch, timing, polarization, or a private wire. After this paragraph the orchestra is gone.

In a quantum array that label often *is* a Hamiltonian term. Assign cluster i a distinctive frequency ω_i and you have added a site-dependent energy. That may spoil the equivalence the model requires.

There are four common selectors.

- *Spatial*: focus light or a near field on one location.
- *Spectral*: drive only the transition resonant with one tone.
- *Internal*: use polarization or selection rules to choose an orientation or transition.
- *Temporal*: refocus unwanted evolution with pulse sequences.

Real devices combine them. A modest gradient separates frequencies, a shaped microwave pulse narrows the addressed bandwidth, and an optical channel supplies readout. Each selector has an error budget. The phrase “individually addressable” without that budget is a brochure, not a specification.

Suppose two sites share a line within the driven bandwidth. Then no amount of spatial slogans will keep the spectator idle. Hence a selector is a quantitative window, not a drawing of a beam.

35.3. A neighbor that almost flips

Let target cluster C have an encoded two-level transition at angular frequency ω_C , in radians per second. Let a neighboring spectator S have frequency ω_S . Their detuning is

$$\Delta_S = \omega_S - \omega_C.$$

Drive both with a microwave field resonant with C . Keep only the part of the drive that stays in step with the spin. Let the target *Rabi* angular frequency be Ω , also in radians per second: how fast the resonant pulse rotates the target. A square pulse of duration

$$t_\pi = \frac{\pi}{\Omega}$$

rotates the target by π . The units check: radians are dimensionless, so $(\text{s}^{-1})^{-1} = \text{s}$.

For an ideal two-level spectator that starts in its lower state, the largest possible off-resonant excitation under a constant drive is bounded by

$$P_{S,\max} = \frac{\Omega^2}{\Omega^2 + \Delta_S^2} \approx \left(\frac{\Omega}{\Delta_S} \right)^2 \quad \text{when } |\Delta_S| \gg \Omega.$$

That unwanted leftover on the neighbor is **crosstalk**. The name is worth keeping because it is an operation error on a victim, not a picture of how fat the beam looked.

Suppose the design permits at most 10^{-3} spectator excitation from this mechanism. The ideal bound requires

$$\frac{|\Delta_S|}{\Omega} \gtrsim \sqrt{10^3} \approx 31.6.$$

This is a dimensionless ratio, as it must be. It is not a device promise. Pulse edges, extra levels, inhomogeneous broadening, calibration drift, and interactions add errors. Smooth pulses can reduce spectral sidelobes. They generally take longer.

Now add the cluster's unwanted excited states. Let Δ_{leak} be the energy gap from the encoded doublet to the nearest leakage state, in joules, and define $\omega_{\text{leak}} = \Delta_{\text{leak}}/\hbar$, in radians per second. Selective control needs a window resembling

$$\Gamma_C \ll \Omega \ll \min(|\Delta_S|, \omega_{\text{leak}}),$$

where Γ_C is the target transition linewidth in radians per second. The left inequality makes the pulse faster and clearer than the linewidth. The right inequalities suppress spectator excitation and leakage.

That window is *selectivity*: a bandwidth, not a slogan. If the window is empty, pulse optimization cannot manufacture one.

Faster gates want larger Ω . Selectivity and leakage suppression want smaller Ω . Decoherence objects to waiting while the engineers discuss it. Drop any one of the three inequalities and the remaining two do not define a gate.

35.4. Charge, then spin, then the cluster

A defect cluster can have at least three initialization layers.

Charge and configuration. The intended optical and spin levels exist only in the desired charge state and structural configuration. Optical illumination may both prepare and disturb charge. A bright spot is not proof that every member of a cluster is in the right charge state.

Physical spins. Spin-dependent intersystem crossing allows nonresonant optical pumping of an NV electronic spin. Resonant cryogenic protocols can provide more selective preparation [R074]. These mechanisms are platform-specific. A group-IV vacancy or a defect in SiC has a different level structure and different temperature requirements.

Encoded cluster state. Polarizing each constituent does not automatically prepare a particular eigenstate of an interacting cluster. You may need a sequence of physical-spin rotations, dissipative pumping, measurement and feedback, or an adiabatic ramp from an easily prepared Hamiltonian. The relevant success probability is the overlap with the encoded state, not the brightness of one constituent.

Initialization also has a time at which it is allowed. Resetting a constituent after a correlated many-body phase has been prepared is a local dissipative operation. It can create excitations and erase entanglement. Global reset before preparation is therefore much less invasive than arbitrary mid-run local reset.

Suppose you polarize every physical spin and stop. Then the cluster occupies a product state that may have almost no overlap with the encoded doublet. Brightness is not that overlap.

35.5. Light that cannot pick one

Optical control can select by position, transition frequency, polarization, or a photonic mode. Ordinary far-field focusing is diffraction limited, so several nanometre-spaced defects can occupy one spot. Super-resolution methods use a nonlinear optical response to narrow the effective point-spread function.

[Experiment] Pezzagna and colleagues combined implantation with stimulated-emission-depletion microscopy to optically distinguish closely spaced NV centers [R235]. That is evidence for nanoscale optical selection. It is not evidence for simultaneous low-crosstalk control of a many-body cluster lattice.

Spectral optical addressing uses differing zero-phonon-line frequencies. It can be powerful at cryogenic temperature, where narrow resonant transitions permit spin-selective excitation and single-shot protocols. It is also vulnerable to spectral diffusion, strain variation, electric-field noise, and charge rearrangement. Nanophotonic fabrication can shift and broaden emitters even while improving collection.

[Experiment] Integrated diamond nanophotonics has coupled multiple silicon-vacancy centers to optical modes, and two emitters in a cavity have shown photon-mediated interactions after spectral tuning [R086]; [R238]. Separately, a 2025 platform optically resolved and manipulated more than 100 NV centers in parallel while using shared coherent control [R239]. These results advance routing and multiplexing. They do not demonstrate a uniform, strongly interacting cluster lattice.

Optical frequency differences are useful labels for photon routing. They may be unwelcome disorder if the same orbital states mediate spin interactions. Tuning emitters into mutual optical resonance helps indistinguishable-photon protocols. Tuning them apart helps frequency-selective local addressing. The operating schedule must say which condition is required at each moment.

A resolved optical line is not a resolved cluster. Several cluster transitions may appear, or several defects may share one line. Assignment needs controlled perturbations.

35.6. A field that covers the whole chip

Microwave wavelengths are enormous compared with defect spacing, so a conventional antenna supplies a nearly global magnetic field. Individual selection then comes from frequency differences, local near-field conductors, magnetic gradients, or pulse refocusing.

[Experiment] A microcoil-generated gradient of about 0.1 G nm^{-1} frequency-encoded four NV sites separated by about 100 nm, enabling site-selective Rabi control; each site contained multiple NVs at smaller separations [R236]. Here G is gauss, with $1 \text{ G} = 10^{-4} \text{ T}$, and nm is nanometre. Those values and that geometry belong to a sensing-oriented device, not a demonstrated strongly coupled topological patch.

A local static magnetic gradient changes Zeeman splittings. A local wire can supply a stronger near field but introduces wiring density, dissipation, fabrication variation, and possibly magnetic noise. Frequency multiplexing reduces wire count, yet the minimum separation between channels must exceed linewidths, drive bandwidth, drift, and interaction-induced shifts.

For an encoded cluster, the microwave field couples through a projected physical operator. Let P_C project onto the encoded doublet and let S_j^x be the transverse spin operator of constituent j . The encoded drive is governed by

$$P_C \left(\sum_j g_j S_j^x \right) P_C,$$

where g_j is the local field coupling, with energy units if the drive Hamiltonian is written directly in joules. A highly symmetric cluster under a spatially uniform field may have a vanishing matrix element between its logical states. That can protect against uniform magnetic noise — and make the same uniform microwave field unable to perform the desired gate. Local gradients or deliberately asymmetric couplings can restore control. They also open noise and leakage channels.

Hence symmetry has two faces. It can suppress the noise you fear. It can also suppress the matrix element you need. Drop either face and the remaining slogan is incomplete.

35.7. Looking is also a shove

Room-temperature NV readout usually maps spin-dependent optical dynamics to different fluorescence statistics and averages many repetitions [R074]. Cryogenic resonant excitation can give single-shot, projective electron-spin readout, as demonstrated for an NV center [R199]. Spin-to-charge conversion maps spin information onto a longer-lived charge distinction and can improve the photon budget under suitable conditions [R237].

Every method has a measurement volume and a back-action volume. Excitation light can pump nearby defects, change charge states, and heat a nanostructure. Collected photons from several centers can overlap at one detector. Spectral filters, confocal or super-resolution collection, cavities, and separate waveguides help. None of them makes back-action disappear.

Three measurement claims must be kept separate.

- *Destructive local readout*: the target state may be lost, and neighbor effects are bounded.
- *Repeatable local readout*: the target can be measured again with a stated quantum-nondemolition fidelity.
- *Phase-preserving many-body readout*: the measurement extracts the intended observable without uncontrolled projection or excitation of the surrounding phase.

The first has strong single-defect precedents. The third is an architectural requirement and remains **[Proposal]** for the defect-cluster topological system considered here. Ordinary

fluorescence of one spin is not automatically a stabilizer measurement, and it is not a reading of a logical topological charge.

Suppose you call a bright photon stream a nondemolition loop. Then you have named the detector, not the operator. Repeatability and the measured observable still have to be specified.

35.8. Every knob, written down

A scalable calibration record needs more than one resonance frequency. For each physical defect and encoded cluster it should include:

- charge-state preparation and survival probabilities;
- spin initialization and readout confusion matrices;
- microwave and optical transition frequencies and linewidths;
- Rabi-rate response versus control amplitude;
- pulse phase and timing offsets;
- leakage spectra and encoded-state matrix elements;
- pairwise interaction shifts;
- optical and microwave crosstalk matrices; and
- drift versus time, temperature, and preceding illumination.

An unconstrained N -cluster crosstalk matrix has $N(N - 1)$ off-diagonal entries, so exhaustive measurement scales quadratically. Verified locality and repeated geometry can reduce this burden. Calibration still consumes shots. Practical runs need sacrificial and interleaved references that track both common drift and local outliers.

A crosstalk number is not a beam diameter. It must name the victim subsystem, the unwanted observable or operation, the control sequence, the neighbor state, and the measurement conditions. Drop any of those and the number cannot be audited.

35.9. Everyone at once, or one at a time

Global pulses use fewer channels and can preserve lattice symmetries. They suit polarization, echoes, and repeated bulk sequences. They cannot identify defects or correct local errors unless the Hamiltonian maps those errors to collective signals.

Local controls add routing and calibration freedom. They also add disorder, wiring, heat, and crosstalk. A hybrid with global bulk operations and sparse local controls is [**Proposal**], not proof that sparse controls suffice.

Driving every cluster through the same programmed pulse can digitally or Floquet-engineer an effective evolution. It does not prove that the undriven material has an intrinsic topological phase. Conversely, local optical readout of physical defect spins does not by itself read a logical topological charge. Always name the physical operator, the encoded operator it projects to, and whether the phase exists only under active control.

35.10. Two inequalities that cannot both be true

Let σ_ω be the root-mean-square spread of cluster transition angular frequencies when all addressing controls are off. Let J_{eff} be a characteristic intended intercluster coupling energy and Δ_{mb} the many-body gap, both in joules. A rough uniformity requirement is

$$\hbar\sigma_\omega \ll \min(|J_{\text{eff}}|, \Delta_{\text{mb}}).$$

The left side has units $(\text{J s})(\text{s}^{-1}) = \text{J}$, so the comparison is dimensionally valid. Spectral addressability, however, asks neighboring transition spacings to exceed the driven bandwidth and linewidth:

$$|\omega_i - \omega_j| \gg \max(\Omega, \Gamma_i, \Gamma_j).$$

Permanent random detunings generally cannot satisfy both inequalities when the transition energy participates directly in the target Hamiltonian. Possible escapes are all conditional:

1. apply a *switchable* local shift only during control and return to a uniform idle point;
2. encode the logical transition so its address frequency changes while the relevant static Hamiltonian coefficient does not;
3. compensate known offsets in a rotating frame or with echo pulses, accepting active rather than passive operation; or
4. use spatial selection without large spectral disorder.

Switchability is not free. Turning a local shift on and off can excite leakage states, accumulate unknown phases, or close the local many-body gap. A symmetry-protected encoded state may also be dark to the control operator.

The correct question is not “uniform or addressable?” It is whether a verified control path connects a uniform idle Hamiltonian to a selective operation and back with errors below the phase’s tolerance. Drop the return trip and you have addressed a different Hamiltonian than the one you came to test.

35.11. Nine steps for one cluster

Consider target cluster C inside a dense patch of spectator clusters. The schedule below is for a prepare–probe–read experiment. It does *not* assume that fluorescence can reset or measure C nondestructively in the middle of an established topological state.

Stage	Control action	Required observation or bound	Phase status
0. Sacrificial map	In separate calibration shots, sweep weak microwave/optical probes and fit ω_C , linewidth, leakage transitions, neighbor detunings, and readout confusion matrix.	Stable confidence intervals over the planned run time; crosstalk matrix measured, not inferred from beam size.	Target phase not claimed.
1. Global reset	Prepare charge states and optically polarize all clusters using the platform-specific sequence. Herald or reject runs with wrong charge where possible.	Per-cluster and whole-patch initialization yield reported separately.	No many-body phase yet.
2. Encoded preparation	Apply calibrated global and local pulses, or an adiabatic cluster ramp, to place each cluster in its low-energy encoded state.	Leakage of C and representative spectators bounded by spectroscopy.	Cluster encoding prepared; topology not yet established.
3. Hamiltonian ramp	Turn on the intended intercluster Hamiltonian with a globally specified ramp.	Ramp time justified against the measured finite-patch spectrum and coherence window.	Candidate many-body state prepared.
4. Select C	Apply a reversible local Stark, strain, or Zeeman shift $\delta_{\text{sel}}(t)$, or activate a calibrated gradient. Use a smooth envelope.	During the pulse, $ \delta_{\text{sel}} \gg \Omega, \Gamma$ for spectators; after it, residual shift obeys the uniformity budget.	Local Hamiltonian is perturbed; induced excitations must be counted.
5. Drive C	Apply a shaped encoded microwave pulse with area $\int \Omega(t)dt = \pi$ for a π rotation, or the smaller angle required by the probe.	$\Omega \ll \omega_{\text{leak}}$; measured spectator error and target leakage satisfy the experiment's budget.	Intentional local excitation/rotation; not automatically a logical topological gate.

Stage	Control action	Required observation or bound	Phase status
6. Restore and refocus	Reverse the selection envelope and use a calibrated echo only if it preserves the intended interaction terms. Track the dynamical phase $\phi_C = \int \delta_{\text{sel}}(t)dt$.	Residual frequency and interaction changes return within idle tolerances; ϕ_C is compensated or included in the model.	Candidate bulk Hamiltonian restored.
7. Evolve	Let the patch evolve for the protocol's dwell time, using global refocusing only when its effective Hamiltonian has been derived.	Total schedule shorter than the relevant encoded coherence time; control-induced heating and drift bounded.	Target dynamics under test.
8. Read at endpoint	Stop or reverse the phase-preparation ramp if required, then map the encoded observable onto a readable physical spin and perform optical or spin-to-charge readout.	Report target confusion matrix and neighbor state changes. Repeat over many shots.	Readout may be destructive; no claim of phase survival afterward.

Every invasive step occurs either before phase preparation or is labeled as a perturbation. To claim nondestructive in-phase control, an experiment would need to show that stages 4–6 neither close the local gap nor create uncontrolled quasiparticles, and that stage 8 measures the desired encoded observable through an ancilla or equivalent channel. That evidence does not yet exist for this architecture.

35.12. What the laboratory actually has

The strongest evidence is modular rather than integrated.

Capability	Status through August 2026	What it establishes	What it does not establish
Optical spin preparation and fluorescence readout of single NV centers [R074]	[Experiment]	A single defect can be initialized and observed.	Dense-cluster, phase-preserving reset/readout.
Nanoscale optical distinction of implanted NVs [R235]	[Experiment]	Far-field super-resolution can beat ordinary diffraction for selected centers.	Parallel, low-back-action control at the spacing of a strongly interacting lattice.
Four-site gradient/frequency encoding [R236]	[Experiment]	Microwave site selection and coherent control in a small NV array.	Uniform interacting clusters or topological dynamics.
Entanglement of two separated NV electronic spins [R080]	[Experiment]	Selective control can coexist with a measured two-spin interaction in a small device.	Scaling to a dense, calibrated many-body patch.
Cryogenic single-shot NV readout and spin-to-charge conversion [R199]; [R237]	[Experiment]	Stronger measurement primitives than averaged fluorescence exist.	Neighbor-safe logical topological measurement.
Ten-qubit register around one NV [R119]	[Experiment]	Sophisticated calibration and control of a local electron–nuclear register.	A lattice of equivalent defect clusters with intercluster topology.
Multi-emitter nanophotonics [R086]; [R238]	[Experiment]	Optical routing, tuning, and emitter integration are advancing.	Intrinsic defect-array topological order.
Parallel control and readout of more than 100 resolved NVs [R239]	[Experiment]	Spatially selective optical manipulation and shared coherent control scale beyond a few sites.	Independent local microwave control or a strongly interacting uniform lattice.

The missing integration matters. Isolated-emitter methods may place defects too far apart for strong direct coupling, while dense creation can broaden spectra and confound assignment. Gradients aid selection but split supposedly identical sites. Fabrication, interaction, optical, microwave, thermal, and calibration budgets must therefore be modeled together.

35.13. Ways to get this wrong

1. **A resolved line is not a resolved cluster.** Several cluster transitions may appear, or several defects may share one line. Assignment needs controlled perturbations.
2. **Disorder is not an address bus.** Accidental frequency labels drift and alter the target Hamiltonian.
3. **Spot size is not crosstalk.** Measure unwanted rotations, phases, charge changes, or Hamiltonian shifts on neighbors.
4. **Optical readout is not automatically nondestructive.** Repeatability and the measured observable must be specified.
5. **Physical control is not encoded control.** Project the drive into the logical doublet and bound leakage.
6. **Compensation is active operation.** Continuous echoes or per-site corrections may synthesize useful dynamics, but they do not establish passive protection.
7. **Symmetry breaking can break protection.** A useful gradient may remove degeneracy or noise cancellation. Returning to the symmetric idle point is part of the gate.

35.14. Check yourself

1. **What fails if permanent site-dependent frequencies are used both to address and to keep a uniform Hamiltonian?**
Addressing wants $|\omega_i - \omega_j| \gg \max(\Omega, \Gamma_i, \Gamma_j)$. Uniform many-body physics may require $\hbar\sigma_\omega \ll \min(|J_{\text{eff}}|, \Delta_{\text{mb}})$. Permanent random detunings generally cannot satisfy both when the transition energy sits in the target Hamiltonian.
2. **Show that $P_{S,\text{max}} \leq 10^{-3}$ requires $|\Delta_S|/\Omega \gtrsim 31.6$ for the ideal two-level bound.**
When $|\Delta_S| \gg \Omega$, $P_{S,\text{max}} \approx (\Omega/\Delta_S)^2$. Set the right side to 10^{-3} and invert: $|\Delta_S|/\Omega \gtrsim \sqrt{10^3} \approx 31.6$.
3. **Why is initializing each physical spin not enough?**
An interacting cluster's encoded state is a particular collective eigenstate or subspace. A product of polarized constituents may have poor overlap with it.
4. **What must a crosstalk number specify?**
The victim subsystem, unwanted observable or operation, control sequence, neighbor state, and measurement conditions. Beam geometry alone is insufficient.

5. Show that a uniform field on a highly symmetric encoded cluster can refuse to drive the logical transition.

The encoded drive is $P_C(\sum_j g_j S_j^x)P_C$. If the g_j are equal and the logical states are opposite under a symmetry that flips the sum, the matrix element vanishes. Protection against uniform noise is then also darkness to a uniform microwave.

6. Does driving one encoded cluster implement a topological logical gate?

Not generally. A local operator may create quasiparticles or probe a local degree of freedom. A logical topological action requires the appropriate nonlocal or braided operation and evidence that the phase survives.

Control is not free. Initialization must separately establish charge, physical-spin, and encoded-cluster state. A selective drive needs a nonempty bandwidth window between linewidth and spectator or leakage scales. Crosstalk is an operation error, not a picture of the beam. Optical, microwave, spectral, and local selectors all perturb the device. Global protocols preserve symmetry and reduce wiring but lack arbitrary routing. Local protocols improve routing while adding disorder and calibration overhead. Local spin readout is not logical topological readout.

35.15. Sources

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001; arXiv: 1302.3288.
- [R235] S. Pezzagna, D. Wildanger, P. Mazarov *et al.*, “Nanoscale Engineering and Optical Addressing of Single Spins in Diamond,” *Small* **6**, 2117–2121 (2010). DOI: 10.1002/sml.201000902.
- [R236] H. Zhang, K. Arai, C. Belthangady, J.-C. Jaskula, and R. L. Walsworth, “Selective addressing of solid-state spins at the nanoscale via magnetic resonance frequency encoding,” *npj Quantum Information* **3**, 31 (2017). DOI: 10.1038/s41534-017-0033-3; arXiv: 1701.01154.
- [R080] F. Dolde, I. Jakobi, B. Naydenov *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545; arXiv: 1212.2804.
- [R199] L. Robledo, L. Childress, H. Bernien *et al.*, “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* **477**, 574–578 (2011). DOI: 10.1038/nature10401; arXiv: 1108.1196.
- [R237] B. J. Shields, Q. P. Unterreithmeier, N. P. de Leon, H. Park, and M. D. Lukin, “Efficient Readout of a Single Spin State in Diamond via Spin-to-Charge Conversion,” *Physical Review Letters* **114**, 136402 (2015). DOI: 10.1103/PhysRevLett.114.136402; arXiv: 1410.0370.

- [R119] C. E. Bradley, J. Randall, M. H. Aboeih *et al.*, “A Ten-Qubit Solid-State Spin Register with Quantum Memory up to One Minute,” *Physical Review X* **9**, 031045 (2019). DOI: 10.1103/PhysRevX.9.031045; arXiv: 1905.02094.
- [R086] R. E. Evans, A. Sipahigil, D. D. Sukachev *et al.*, “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* **362**, 662–665 (2018). DOI: 10.1126/science.aau4691; arXiv: 1807.04265.
- [R238] A. Sipahigil, R. E. Evans, D. D. Sukachev *et al.*, “An integrated diamond nanophotonics platform for quantum-optical networks,” *Science* **354**, 847–850 (2016). DOI: 10.1126/science.aah6875; arXiv: 1608.05147.
- [R239] M. Cambria, S. Chand, C. M. Reiter, and S. Kolkowitz, “Scalable Parallel Measurement of Individual Nitrogen-Vacancy Centers,” *Physical Review X* **15**, 031015 (2025). DOI: 10.1103/jdzq-jbfz; arXiv: 2408.11715.

Chapter 36. Reading topological information

You have one spin. A green flash, a handful of red photons, and you call the answer up or down. That is a real question with a real detector. It is also the smallest question in the building.

Someone now claims the crystal is storing a bit that no single spin owns. The information is supposed to live in a *nonlocal sector*: two globally different patterns that look the same under every local check. You cannot walk over and ask the sector how it feels. You have to climb.

Start with the one spin. Then ask whether four neighbors obey a local rule. Then ask what energy a drive can spend. Then drag an open string and see whether charges appear at the ends. Then close a loop that cannot shrink, and ask which sector you are in. Then bring two excitations together and ask what total charge comes back. Each question is a different operator. A claim that does not name one is not yet a claim.

36.1. The questions, in order

A microscope can name the color of one thread. A four-thread check can tell whether a local weaving rule was broken. Neither tells you whether a colored strand winds around the whole tapestry. For that you have to follow an extended path, or compare two globally different patterns. After this paragraph the tapestry is gone.

The quantum version is the same hierarchy, written as operators:

one-site readout	-> physical spin or edge label
local parity	-> stabilizer / nearby excitation
spectral response	-> energy and transition matrix element
open string endpoints	-> created or moved charges
closed loop	-> enclosed charge or logical sector
fusion experiment	-> possible total-charge channels
many-basis data	-> tomography or selected correlators

A quantum measurement is an operator with a probability distribution. Let the data system sit in a state ρ , a density operator. For an observable M with eigenvalues m , the ideal projective odds are

$$P(m) = \text{Tr}(\rho \Pi_m),$$

where Π_m projects onto the eigenspace with eigenvalue m . A readout protocol earns its keep only when the actual detector record can be mapped, with calibrated errors, onto those ideal outcomes.

One distinction will keep coming back. A *local stabilizer* checks whether a small neighborhood satisfies one term of a code Hamiltonian. A *nonlocal logical loop* distinguishes globally different states that can satisfy every local term. Their circuits can look almost identical — both can be parity measurements — while their meanings are not interchangeable.

Attach a short dossier to every claimed signal: the operator, the preparation assumptions, the measurement circuit, the calibration, the error model, and the competing explanations. If any of those is missing, you have a plot, not an identification.

36.2. Four nearby edges

Use the toric-code convention: one physical two-level system on each edge of a square lattice with periodic boundaries. Periodic boundaries turn the sheet into a torus. For edge e , let X_e and Z_e be Pauli operators. Define a plaquette operator

$$B_p = \prod_{e \in \partial p} Z_e,$$

where ∂p is the set of four edges around plaquette p . Its eigenvalues are $b_p = +1$ and -1 . At each vertex s , also define the star operator

$$A_s = \prod_{e \ni s} X_e,$$

where $e \ni s$ means the four edges that meet at s . The ideal ground space has $A_s = +1$ and $B_p = +1$ for every star and plaquette [R030]. **[Theory]**

To measure B_p , bring in an extra two-level system a , an *ancilla*, initialized in $|0\rangle_a$. This spin is not part of the code. It is a helper that is allowed to forget everything except one parity.

For each of the four data edges in ∂p , apply a controlled-NOT with the data edge as control and the ancilla as target. Then measure Z_a . In the computational basis the ancilla records the parity of the four data bits:

$$m_a = z_1 z_2 z_3 z_4 = b_p,$$

where each $z_j = \pm 1$ is the corresponding Z -eigenvalue and m_a is the ancilla outcome. Thus $m_a = -1$ reports a violated plaquette check in the ideal circuit.

For a coherent superposition the same circuit measures the operator B_p , not four pre-existing classical bits. It projects the data into a B_p eigenspace without revealing the individual Z_e values. Repeated ancilla-mediated stabilizer extraction is a working technique in superconducting surface-code experiments [R230]. **[Experiment]** That fact establishes a measurement primitive. It does not establish passive topological order in the processor.

Drop the ancilla and measure the four edges directly, and you still learn the product — but you also learn the factors, and you generally destroy coherence in incompatible bases. The circuit is not a formality.

36.3. A loop that will not shrink

Now choose a closed path C_x that winds once around the torus in the x direction. Define

$$\bar{Z}_x = W_Z(C_x) = \prod_{e \in C_x} Z_e.$$

Use a fresh ancilla and the same style of parity circuit, but couple it to every edge along C_x , then measure the ancilla. The result $\bar{z}_x = \pm 1$ distinguishes two logical sectors. The loop commutes with every stabilizer, so states with different $\bar{z}_x$ can have identical local stabilizer outcomes.

There is a trap. If a closed path C can be shrunk to a point, then in the ideal code its Z loop is a product of the enclosed plaquette stabilizers. It is not an independent logical measurement. A path becomes logical only because it is **noncontractible** on the torus — it cannot be pulled down to a point — or, on a planar surface code, because it connects the appropriate boundaries or encircles a puncture. The name is worth keeping because length is not the point. Topology of the path is the point.

The contrast is now a table:

Measurement	Support	Ideal question	A -1 result means
B_p	four nearby edges	Is this local constraint satisfied?	a local syndrome/excitation is present
$\bar{Z}_x$	a system-spanning noncontractible path	Which logical sector is occupied?	logical eigenvalue -1 , not necessarily a local defect

Measurement	Support	Ideal question	A -1 result means
-------------	---------	----------------	---------------------

The two operators can be measured with similar hardware. One is a Hamiltonian term. The other is a logical observable. Measuring all $B_p = +1$ does not determine $\overline{Z}_x$. Measuring $\overline{Z}_x$ once does not certify that all local constraints hold.

A destructive alternative is to measure every data edge along the chosen path in the Z basis and multiply the outcomes. That estimates the same parity for that shot, but it also reveals individual values and generally destroys coherence in incompatible bases. Destructive measurement is often simpler and can be excellent for a final readout. It is a poor choice if the state must survive for another braid.

Suppose you call every long correlator a logical loop. Then a contractible product of plaquettes will be promoted to a sector label. Length was never the definition.

36.4. Photons are not Pauli Z

A local readout maps a microscopic state onto photons, charge, current, resonator phase, or another classical record. In diamond nitrogen-vacancy centers, confocal fluorescence combined with magnetic resonance enabled detection of individual defect centers [R240], and spin-selective optical protocols later achieved projective readout of a solid-state spin register [R199]. **[Experiment]** Related platforms use spin-dependent tunnelling, spin-to-charge conversion, cavity transmission, or dispersive frequency shifts.

The raw observable is platform-specific. A fluorescence count is not itself Z . Calibration supplies conditional distributions $P(r|z = +1)$ and $P(r|z = -1)$ for detector record r . Preparation errors, state changes during readout, background counts, crosstalk, and threshold choice belong in the inference. Reporting only a thresholded bit discards the information needed to audit that inference.

Local spin readout answers local questions: occupation, polarization, a basis-rotated component, or correlations assembled across repeated shots. It can reconstruct an extended operator by multiplying many local outcomes. It cannot make that operator robust against the growing number of readout errors.

Hence a calibrated transducer is necessary and not sufficient. The operator still has to be named, and the error still grows with the support.

36.5. An extra spin that keeps a secret

For a Pauli product

$$S = P_1 P_2 \cdots P_w,$$

where each P_j is X , Y , or Z on one data system and w is the operator weight, an ancilla can acquire the eigenvalue of S . Basis rotations convert X or Y parity to a Z -parity circuit. Measuring the ancilla then reveals $s = \pm 1$ while ideally preserving superpositions inside the same eigenspace.

The schedule matters. One ancilla fault can propagate through several two-body gates into a correlated data error. Fault-tolerant layouts therefore constrain gate order, add flag ancillas, repeat rounds, and decode a history rather than trusting one bit. Modern surface-code experiments demonstrate repeated stabilizer extraction and below-threshold logical-error scaling in engineered superconducting circuits [R230]. **[Experiment]** These are active error-correction results. They should not be relabelled as evidence for a naturally occurring topological phase.

Stabilizer expectation values also provide an energy estimate for an ideal commuting Hamiltonian such as

$$H = -J \sum_s A_s - J \sum_p B_p,$$

where J is an energy, A_s is a star operator, and B_p is a plaquette operator. Measuring $\langle A_s \rangle$ and $\langle B_p \rangle$ estimates $\langle H \rangle$. It does not, alone, determine the gap, long-range entanglement, or robustness to unmeasured perturbations.

An energy estimate is not a phase identification. Drop the rest of the dossier and a well-satisfied stabilizer looks like a topological certificate.

36.6. A gap is not a name

Spectroscopy drives the system at angular frequency ω and measures absorption, fluorescence, resonator response, or state transfer. Resonance near

$$\hbar\omega = E_n - E_0$$

connects the initial energy E_0 to an excited energy E_n , provided the drive operator has a nonzero transition matrix element. Both sides have units of joules. By varying momentum, position, polarization, and drive operator, an experiment can probe gaps, dispersions, selection rules, bound states, and continuum thresholds.

A spectral gap is necessary for many topological-protection arguments, but a gap is not topological. Ordinary magnets, molecules, and kitchen cupboards can all be gapped; only two of those are routinely diagonalized. Spectroscopy becomes persuasive when the observed states also have the predicted charge, fusion, nonlocal response, and perturbation dependence.

What fails if you stop at the gap? You have shown an energy difference under the probed conditions. You have not shown a nonlocal sector, anyonic statistics, or logical-sector robustness.

36.7. Strings that open and loops that close

An *open string operator* is a product of local operators along a path with endpoints. In the toric code, applying an open string creates, moves, or annihilates excitations at its ends while commuting with stabilizers away from them [R030]. Measuring endpoint syndromes before and after the string checks this rule.

A *closed Wilson loop* $W_a(C)$ represents transporting or testing a charge type a around a closed contour C . The loop is the question, and the charge label a is part of the question. Depending on the model, its expectation value diagnoses confinement, enclosed topological charge, or a logical sector. A contractible loop and a noncontractible logical loop must always be labelled separately.

There are three practical routes:

1. *Ancilla accumulation*: coherently couple one ancilla to every term in the product and read its phase or parity.
2. *Destructive multiplication*: measure all sites in suitable local bases and multiply outcomes shot by shot.
3. *Interferometry*: coherently split a probe path, make one branch encircle a region, recombine the branches, and infer the acquired Abelian phase or non-Abelian action from output probabilities.

Interferometry is conceptually closest to asking what charge lies inside a loop. In an Abelian theory, encircling can add a phase. In a non-Abelian theory, braiding acts as a matrix on a fusion space, so one output intensity is generally insufficient; state preparation and a final fusion-channel measurement are required [R015]. **[Theory]** Visibility loss may arise from dephasing, path distinguishability, leakage, thermal quasiparticles, or a mixture of charges. It is not automatically a non-Abelian signature.

A dim fringe is not a braid matrix. Drop the competing explanations and any lost contrast will be sold as statistics.

36.8. Bringing two charges together

To measure a fusion rule, prepare identified quasiparticles, move them together or implement the equivalent code deformation, and measure the total topological charge of the enclosing region. For Fibonacci charge τ ,

$$\tau \times \tau = 1 + \tau$$

permits vacuum charge 1 or charge τ . A useful experiment must show not merely two histogram peaks but a calibrated mapping from those peaks to charge projectors, repeatability or controlled destructiveness, and the dependence on preparation and braid history.

A region's total charge can be read by a Wilson loop around its boundary, by interferometry, or by converting the charge to local degrees of freedom through a known fusion circuit and then measuring those degrees of freedom. The last option is often called *destructive fusion readout*: bring the anyons together, measure the resulting local syndrome, and accept that the original encoded state is gone.

Digital superconducting-processor experiments have implemented code deformations and non-Abelian graph-vertex braiding [R132], and a 27-transmon experiment prepared a Fibonacci string-net state and measured creation, fusion, and braiding signatures [R138]. **[Experiment: digital simulation]** These are important demonstrations of protocols and controlled Hilbert spaces. They did not observe intrinsic Fibonacci quasiparticles emerging in an unprogrammed defect material; the operations were compiled into gates on ordinary processor qubits [R132]; [R138].

Logical-charge measurement asks the same question at the encoding scale. If logical information is stored in the total charge of a set of anyons or holes, measure a loop enclosing exactly that set, or fuse according to the encoding tree and read the final channel. The loop geometry, orientation, charge label, and chosen fusion basis are part of the observable. “Measure the anyons” is not yet a circuit.

Two peaks alone are spectroscopy with ambitious branding. What fails if you stop there is every test that would distinguish fusion from a pair of ordinary resonances.

36.9. Asking every question

Full state *tomography* measures enough observables to reconstruct ρ . It is a census, not a phase label. For N two-level systems, ρ is a $2^N \times 2^N$ matrix, so unrestricted reconstruction scales exponentially. Small patches can support full tomography. Larger arrays require selected correlators, stabilizer sampling, reduced density matrices, randomized measurements, or classical-shadow protocols [R241]. **[Theory/Protocol]**

Reconstructing a small state close to an ideal code state verifies preparation on that patch. It does not by itself establish thermodynamic ground-state degeneracy or stability as system size grows. Conversely, carefully chosen nonlocal and entanglement observables may test a specific hypothesis more efficiently than full reconstruction.

A complete census of four spins is still a four-spin fact. Drop the scaling tests and a beautiful density matrix will be asked to certify a phase it cannot see.

36.10. What the laboratory already has

The relevant techniques already exist, but not all in one platform. Single defects can be optically located and spin-read [R240]; [R199]. Superconducting processors repeatedly measure high-weight parity checks with ancillas [R230]. A superconducting experiment prepared a toric-code state and probed excitations and topological observables [R125]. **[Experiment: digital state preparation]** A programmable Rydberg array measured signatures consistent with a toric-code-type spin liquid [R126]. **[Experiment: analog quantum simulation]** Non-Abelian and Fibonacci protocols have been digitally implemented on superconducting processors [R132]; [R138].

Those achievements answer different questions. In a gate processor, the operator being measured is usually known because the experimenter compiled it. The challenge is fidelity, scaling, and whether the prepared state retains the predicted structure. In an analog simulator, the Hamiltonian is less directly scripted and phase identification leans more heavily on a dossier of observables. In a defect lattice, additional interfaces are required: local optical or microwave addressability, ancillas or mediator modes for parity, sufficiently nondestructive cycling, and a way to implement loops without the error probability becoming overwhelming.

A credible defect experiment should report at least: raw and corrected outcomes; detector calibration; leakage and loss; control sequence; operator support; error bars; blind or held-out model comparisons where practical; and results versus system size, temperature, preparation time, and controlled perturbations. Readout correction should be shown both applied and unapplied. Otherwise a large “corrected” Wilson loop can become a portrait of the calibration model.

The same measurement words get used in four different rooms. A stabilizer circuit on programmed qubits is active syndrome extraction. A prepared code state is a digital emulation unless it is the equilibrium or dynamically stable phase of the physical Hamiltonian under study. An analog simulator may realize an engineered many-body Hamiltonian, but signatures on one finite patch are not automatically thermodynamic topological order. An emergent anyon is an excitation of such a many-body phase; a logical qubit is information encoded in its nonlocal sector. State the room. The vocabulary will not do it for you.

36.11. Ways to get this wrong

1. **One stabilizer becomes “topology.”** A local constraint can be satisfied by a trivial product state or by a state prepared to pass that one check. Measure the full constraint pattern and independent nonlocal observables.
2. **A long correlator becomes a logical loop.** Length is not enough. The path must have the correct topology and operator content. A contractible loop may be only a product of stabilizers.
3. **Postselection manufactures perfection.** Discarding shots with leakage or unwanted syndromes changes the sampled ensemble. Report acceptance rates and conclusions without postselection.
4. **Readout mitigation assumes the answer.** Independent single-site confusion matrices may fail under correlated crosstalk. Validate the correction on states with known multi-site parity.
5. **A gap becomes a phase identification.** Spectroscopy can show an energy gap. Topological order additionally requires nonlocal structure and robustness.
6. **Fusion-looking populations become fusion rules.** Fusion evidence requires controlled charges, a defined fusion basis, charge-projector calibration, and history-dependent tests. Two peaks alone are spectroscopy with ambitious branding.
7. **Finite prepared-state signatures become passive protection.** The toric-code and Fibonacci processor demonstrations [R125]; [R132]; [R138] measured deliberately prepared or digitally evolved states. **[Experiment]** They do not show that the same hardware passively relaxes into, or protects, that phase.
8. **A Wilson-loop value becomes universal proof.** Loop behavior depends on contour, state, temperature, boundaries, matter content, and noise. Combine it with local syndromes, gap data, entanglement diagnostics, sector structure, and perturbation tests.
9. **Destructive parity is called nondemolition.** Multiplying individually measured spins can estimate a loop but destroys incompatible coherence. A quantum-nondemolition claim requires repeated agreement beyond what state re-preparation or detector memory explains.

A useful evidence hierarchy is cumulative:

Evidence	What it supports	What remains open
Local spin contrast	controllable/readable constituents	interactions and collective phase
Stabilizers near target values	local code constraints	nonlocal sector and long-range entanglement

Evidence	What it supports	What remains open
Gap spectroscopy	energetic isolation	topology of the gapped state
Open-string endpoint rule	candidate quasiparticle motion	statistics and deconfinement
Noncontractible loops	global sector information	robustness and thermodynamic scaling
Fusion and braid matrices	anyonic process within calibrated space	intrinsic emergence versus compiled simulation
Size/perturbation/temperature trends plus entanglement data	coherent phase-identification dossier	ultimate scalability and protection

No row erases the need for the rows beneath it. Strong phase identification combines local checks, independent nonlocal observables, spectral and entanglement information, controlled creation and fusion, scaling, and exclusion of plausible trivial models. Satzinger *et al.* and Semeghini *et al.* illustrate complementary digital and analog dossiers rather than a single decisive meter [R125]; [R126]. **[Experiment]**

36.12. Check yourself

1. What does an ancilla stabilizer measurement reveal?

The eigenvalue of a specified local operator product, ideally without revealing individual data values inside that eigenspace.

2. Show that a contractible Z loop on the torus is not an independent logical observable.

In the ideal code it is a product of the enclosed plaquette operators B_p . Any two ground states that satisfy every $B_p = +1$ therefore agree on that loop. A noncontractible loop cannot be written that way and can distinguish those states.

3. What is the difference between destructive and nondestructive loop readout?

Destructive readout measures constituent sites and multiplies outcomes; nondestructive ancilla readout extracts only the loop eigenvalue while ideally preserving coherence within its eigenspace.

4. What fails if you identify a spectral gap with topological order?

You have shown an energy difference under the probed conditions. Ordinary magnets are gapped too. Nonlocal entanglement, anyonic statistics, and logical-sector robustness remain unshown.

5. What fails if two histogram peaks are called a fusion rule?

Fusion evidence needs identified input charges, a defined fusion basis, calibrated charge projectors, and history-dependent tests. Two peaks are spectroscopy until those are attached.

6. What is the minimum honest label for a gate-compiled Fibonacci experiment?

A digital simulation or emulation of Fibonacci states and operations, unless independent evidence establishes intrinsic emergent Fibonacci order in the hardware Hamiltonian.

Local spins are read through calibrated transducers. Pauli products can be extracted with ancillas or destructive snapshots. Spectroscopy measures energy differences. Open strings test endpoint charges. Closed loops and interferometers probe enclosed or logical charge. Fusion readout projects a defined total-charge channel. Tomography supplies selected state information at exponential full-state cost. None of those tools is a topology meter. A claim needs an observable, and the observable needs a dossier.

36.13. Sources

- [R240] A. Gruber, A. Dräbenstedt, C. Tietz, L. Fleury, J. Wrachtrup, and C. von Borczyskowski, “Scanning Confocal Optical Microscopy and Magnetic Resonance on Single Defect Centers,” *Science* **276**, 2012–2014 (1997). DOI: 10.1126/science.276.5321.2012.
- [R199] L. Robledo, L. Childress, H. Bernien, B. Hensen, P. F. A. Alkemade, and R. Hanson, “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* **477**, 574–578 (2011). DOI: 10.1038/nature10401.
- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0. arXiv: quant-ph/9707021.
- [R230] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* **638**, 920–926 (2025). DOI: 10.1038/s41586-024-08449-y. arXiv: 2408.13687.
- [R125] K. J. Satzinger *et al.*, “Realizing topologically ordered states on a quantum processor,” *Science* **374**, 1237–1241 (2021). DOI: 10.1126/science.abi8378. arXiv: 2104.01180.
- [R126] G. Semeghini *et al.*, “Probing topological spin liquids on a programmable quantum simulator,” *Science* **374**, 1242–1247 (2021). DOI: 10.1126/science.abi8794. arXiv: 2104.04119.
- [R132] Google Quantum AI and Collaborators, “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* **618**, 264–269 (2023). DOI: 10.1038/s41586-023-05954-4. arXiv: 2211.09802.
- [R138] S. Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6. arXiv: 2305.14028.

- [R241] H.-Y. Huang, R. Kueng, and J. Preskill, “Predicting many properties of a quantum system from very few measurements,” *Nature Physics* **16**, 1050–1057 (2020). DOI: 10.1038/s41567-020-0932-7. arXiv: 2002.08953.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian anyons and topological quantum computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083. arXiv: 0707.1889.

Part XI

Verdict

The strongest case, the load-bearing joints, the written nos, the smallest calculation with an exit, and a grade.

Chapter 37. The strongest case, not the loudest

A colleague grants every measurement in the book. Two nearby defect spins can entangle. A crystal plane can make dipoles agree on a sign. A molecule can drop three nitrogens at once. A nucleus next door can remember a state for more than a minute. None of that is a topological phase.

Then she asks what, exactly, is still worth building.

The strongest honest version of the answer is a *steelman*. The name is worth keeping because the opposite — a straw man — is what you build when you want to win a fight. We are not here to win a fight. We are here to see whether grouping defects into clusters gives knobs that a lonely spin does not.

The move that is specific to this architecture is the grouping. A cluster can hide a messy handful of physical spins behind one leftover two-state object. Those leftovers can talk. If the talk is the right shape, and the same from cell to cell, a phase can appear that no single defect ever had.

Every “can” in that paragraph has a different evidence status. Keeping the statuses attached is the whole job.

[Experiment] Several pieces exist separately: coherent coupling of nearby defect spins, small multi-defect motifs, dense two-dimensional dipolar ensembles, local electron–nuclear registers, controlled optical interfaces, and improving three-dimensional placement. **[Theory]** Local two-body Hamiltonians can support emergent topological phases, and cluster projection can generate interactions not obvious in the microscopic model. **[Speculation]** No defect-cluster array has been shown to realize a topologically ordered phase, doubled-Fibonacci order, or intrinsic Fibonacci quasiparticles as of August 2026.

37.1. A leftover, not a nicer spin

Long coherence is useful on many platforms. Bright fluorescence is useful for many experiments. Neither by itself explains why grouping defects should help produce emergent topological order.

Suppose a cluster prices out its expensive motions. What remains is a two-state object sitting behind a gap. The microscopic teeth are physical spins and orbitals. The leftover is a low-energy doublet. The cluster does three jobs: it exposes one useful motion, it makes unwanted motions costly, and it translates available forces into that leftover motion.

For cluster C , write

$$H = \sum_C H_C + \sum_{\langle C,D \rangle} V_{CD}.$$

The internal energy H_C defines a two-state manifold with projector P_C . The jump from that manifold to the nearest leftover state is the cluster gap Δ_C , measured in joules or electronvolts. The talk between clusters, V_{CD} , has a characteristic matrix element v , also an energy. If $v/\Delta_C \ll 1$, the array may be described by leftover two-state objects — *pseudospins* — sitting inside $P = \prod_C P_C$. The object is not a physical spin. It is what the cluster pretends to be after the expensive motions have been priced out.

Hence the structural promise is not that projection makes interactions better by magic. It is that the cluster spectrum, wavefunctions, symmetry, and placement become design variables. A microscopic term that is awkward in the physical-spin basis may project to a useful leftover operator:

$$PV_{CD}P = J_x \tau_C^x \tau_D^x + J_y \tau_C^y \tau_D^y + J_z \tau_C^z \tau_D^z + \text{local terms},$$

where $\tau_C^{x,y,z}$ are Pauli operators acting on cluster C 's doublet. Virtual leakage can add terms proportional to v^2/Δ_C or higher powers. The “instruction set” is the projected Hamiltonian.

Drop reproducibility and the case fails at once. Random spectra and random projected operators give a small box of trouble, not an array. The case succeeds only if cluster-to-cluster spectra and projected operators are reproducible enough that one calibration strategy covers an array.

37.2. Two clusters and one leftover coupling

Take two identical three-spin clusters, A and B . Assume each has a low-energy doublet $\{|\tilde{0}\rangle, |\tilde{1}\rangle\}$ separated by Δ_C from other cluster states. This doublet is the encoded leftover. It is not yet a topological qubit.

Suppose a physical dipolar or exchange interaction joins one spin in A to one spin in B :

$$V_{AB} = v S_{A,3}^z S_{B,1}^z.$$

Here S^z is a dimensionless spin operator and v is an energy. Within the encoded doublets, assume the cluster matrix elements are

$$P_A S_{A,3}^z P_A = a_0 I_A + a_z \tau_A^z, \quad P_B S_{B,1}^z P_B = b_0 I_B + b_z \tau_B^z.$$

The dimensionless coefficients a_0, a_z, b_0, b_z are determined by the cluster eigenstates. First-order projection gives

$$PV_{AB}P = v(a_0I_A + a_z\tau_A^z)(b_0I_B + b_z\tau_B^z).$$

The useful coupling is

$$J_{zz}^{(1)} = va_zb_z.$$

The same expression also produces local fields $va_0b_z\tau_B^z$ and $va_zb_0\tau_A^z$, plus a constant. Hence the microscopic interaction is fixed, but the projected coefficients are controlled by the cluster wavefunctions. Change the internal coupling graph or symmetry, and the effective operator changes without requiring a new fundamental force.

Symmetry may force $a_0 = b_0 = 0$. If it does not, the nuisance terms are real. That is the remark, not a footnote: a designed leftover is useful only if the operators you did not want are either forbidden or pushed to higher order.

At second order, virtual cluster excitations add terms of scale

$$J^{(2)} \sim \alpha \frac{v^2}{\Delta_C},$$

where the dimensionless coefficient α is a signed sum over matrix elements and denominators. **[Theory]** Schrieffer–Wolff perturbation theory makes this statement controlled when $v/\Delta_C \ll 1$ [R025]. **[Proposal]** Repeating such clusters could synthesize an interaction graph unavailable from bare defects. The two-cluster calculation does not establish that a two-dimensional array has topological order.

37.3. Five knobs grouping might turn

The pro case rests on five linked features. Each is a structural knob, not a compliment to diamond.

1. A cluster can convert a messy microscopic space of states into a designed low-energy leftover.
2. Crystal orientation and patterned density can shape the sign and dimensionality of dipolar interactions.
3. Short-range and mediated couplings can, in principle, occupy different levels of the hierarchy.
4. Nearby nuclear spins and optical transitions provide local control and measurement without being the encoded topological degree of freedom.

5. A static solid-state array could realize an always-on analog Hamiltonian, rather than merely compiling its time evolution into gates.

The next five sections take those knobs one at a time. A knob that has been turned in a different sample is not a knob that has been turned here.

37.4. The leftover, the gap, and the jobs they keep

A bare spin provides one local space of states. A cluster can provide an isolated doublet, a leakage gap, selection rules, and projected operators. Those are exactly the objects needed for controlled Hamiltonian reduction.

This could be structurally valuable in two ways.

First, strong couplings inside a cell and weaker couplings between cells can be assigned different jobs. Inside, the interactions create the leftover and its gap. Between cells, they create the many-body model. Geometric separation supplies a natural hierarchy for dipolar coupling because it scales as r^{-3} ; exchange supplies an even sharper short-range dependence.

Second, symmetries can suppress selected local fields or matrix elements. That can move an unwanted operator to higher perturbative order while retaining a useful one. **[Theory]** This is standard encoded-subspace reasoning, not a defect-specific discovery. **[Speculation]** No experimentally validated color-center cluster is currently known to project realistic couplings into the complete doubled-Fibonacci string-net operator set.

Hence the best version of the proposal starts with spectroscopy and exact diagonalization of one repeatable cluster. It does not start by drawing anyons over a lithography image. Drop the isolation of the doublet, and there is nothing left to project.

37.5. Tilting the crystal to pick a sign

Dipolar coupling depends on both separation and angle. That is usually presented as a nuisance. It can also be a design resource.

[Experiment] Hughes and collaborators grew dense, preferentially aligned, two-dimensional NV ensembles in (111)-oriented diamond and showed that this geometry gives the in-plane dipolar interactions a common positive sign in their convention [R242]. The observed interaction-dependent line-shape asymmetry changed with spin polarization, supporting the dipolar interpretation. A 2026 erratum corrected density and sensitivity calculations, so those corrected metrology numbers should not be borrowed

casually; the structural point used here is the orientation-dependent interaction sign [R242].

This result is stronger than “diamond hosts NV centers.” It shows that the host plane, defect axis, and dimensionality can jointly select an interaction pattern. The limitation is equally important: the experiment used an ensemble, not an individually patterned cluster graph, and same-sign dipoles are not a string-net Hamiltonian.

Drop the common orientation and the sign is no longer a knob. Keep the ensemble average and you still have not built a lattice of leftovers.

37.6. Making a cluster in one shot

If a cluster requires three defects within a few nanometres, placing each defect independently compounds yield and registration errors. Correlated implantation offers a different primitive: deliver several dopant atoms from one molecular ion, then let their common straggle define a local motif.

[Experiment] Haruyama and collaborators used a nitrogen-bearing molecular ion to create triple NV centers. Their calculated implanted-nitrogen separation was 9 ± 4 nm; among 7,116 implantation events they identified nine optical triples, one strongly coupled triple and one weakly coupled triple. The total NV creation yield was 7.4%, and the longest reported Hahn-echo T_2 in the measured population was $428 \mu\text{s}$ [R243].

The steelman is not that these yields scale. They plainly do not. The structural lesson is that “cluster creation” need not always mean three independent atomic placement operations. A future process could target the yield and internal spectrum of a motif as a unit. That hypothesis is testable before attempting a lattice.

What fails if the motif is not the fabrication primitive? Then every extra defect multiplies independent placement error, and the cluster spectrum is an accident of three uncorrelated landings.

37.7. Helpers that are not the leftover

Dense analog arrays are hard to interrogate. Defect platforms offer a possible division of labor: electron spins participate in faster interactions; nearby nuclear spins store state or act as helpers; optical transitions initialize and read an electron; photonic structures route selected signals.

[Experiment] A diamond NV register containing one electron spin, one nitrogen nuclear spin, and eight ^{13}C nuclear spins demonstrated pairwise control across ten qubits and protected an arbitrary single-qubit state for more than 75 s at about 3.7 K under dynamical-decoupling control [R119]. This does not demonstrate a cluster of electron-spin defects or passive protection. It demonstrates that a single defect neighborhood can contain addressable helper and memory resources far richer than one two-level system.

[Experiment] Optical photon-mediated interactions have also been induced between two silicon-vacancy centers in a diamond nanocavity [R086]. Such mediation could repair missing graph edges or separate interaction and readout channels. It brings spectral matching, cavity fabrication, drive, and loss requirements; it is not automatically an equilibrium, short-range Hamiltonian.

Hence the structural opportunity is heterogeneity within a repeated cell: data leftover, leakage monitor, readout helper, and photonic port can be physically nearby while serving distinct roles. Call the nuclear memory passive protection and you have renamed a control resource.

37.8. Start lower than Fibonacci

The strongest research program is a ladder, not a leap.

[Theory] Kitaev’s honeycomb model proves that local two-body, bond-dependent spin interactions can yield emergent Majorana fermions and a non-Abelian Ising phase [R017]. Levin–Wen string-net Hamiltonians provide commuting-projector targets for doubled topological orders, including doubled non-Abelian theories [R018]. These are existence proofs for Hamiltonians, not materials demonstrations.

A defect-cluster platform could first test a projected bond-dependent model, then a generated stabilizer, then a small topological patch. Success at an Abelian or Ising-type model would validate the architecture’s Hamiltonian engineering without being renamed “Fibonacci.” The ultimate doubled-Fibonacci claim remains a separate, higher bar.

Drop that order and you test the hardest noun first. A failed Fibonacci headline then cannot say whether the leftover, the graph, or the recoupling data was the missing piece.

37.9. A stack of numbers that have never met

Use frequency units to keep the arithmetic readable. For any energy X , define $\nu_X = X/h$, where Planck’s constant h has units joule-seconds. Consider this deliberately optimistic design point:

Layer	Best-case input	Status	Depends on
Cluster leakage gap	$\nu_{\Delta} = 1 \text{ GHz}$	[Proposal] design target	reproducible cluster spectrum
Intercluster matrix element	$\nu_v = 100 \text{ MHz}$	[Speculation] as an array-wide value	distance, orientation, mediator, disorder
Expansion ratio	$\epsilon = \nu_v / \nu_{\Delta} = 0.10$	derived	two rows above
Desired perturbative order	$q = 2$	[Proposal]	selection rules and projected matrix elements
Path coefficient	$\alpha = 1$	optimistic assumption	interference among virtual paths
Effective interaction	$\nu_{\text{eff}} = \alpha \nu_v \epsilon^{q-1} = 10 \text{ MHz}$	derived	all rows above
Generic next-order scale	$\nu_{\text{corr}} \sim \nu_{\text{eff}} \epsilon = 1 \text{ MHz}$	scaling estimate	no enhanced coefficients or small denominators
Gap fraction	$c = \Delta_{\text{topo}} / J_{\text{eff}} = 0.2$	[Speculation] model assumption	verified many-body phase
Topological gap	$\nu_{\text{topo}} = c \nu_{\text{eff}} = 2 \text{ MHz}$	derived, not measured	correct Hamiltonian and thermodynamic extrapolation
Comparator coherence	$T_2 = 100 \mu\text{s}$, so $1/T_2 = 10 \text{ kHz}$	[Experiment, borrowed] approximately the mean Hahn-echo scale of positioned NVs in one 2025 process [R207]	incompatible sample and control conditions may apply
Disorder target	$\sigma_J / h \lesssim 0.2 \text{ MHz}$	[Proposal] ten-percent-of-gap rule	placement, strain, charge, calibration

The dimensional check is

$$\nu_{\text{eff}} = \frac{\nu_v^2}{\nu_{\Delta}} = \frac{(100 \text{ MHz})^2}{1000 \text{ MHz}} = 10 \text{ MHz}.$$

At this point the optimistic gap exceeds the borrowed decoherence rate by a factor of 200:

$$\nu_{\text{topo}} T_2 = (2 \times 10^6 \text{ s}^{-1})(10^{-4} \text{ s}) = 200.$$

That is the steelman's central numerical opening: a low-order generated interaction could, in principle, outrun a single-defect coherence rate. It is not a device prediction, because the 1 GHz cluster gap, 100 MHz array-wide coupling, coefficient $\alpha = 1$, gap fraction $c = 0.2$, disorder target, and borrowed T_2 have never been co-realized.

Order is decisive. Holding the same $\epsilon = 0.1$, a fourth-order target would have

$$\nu_{\text{eff}}^{(4)} \sim \nu_v \epsilon^3 = 0.1 \text{ MHz}, \quad \nu_{\text{topo}}^{(4)} \sim 20 \text{ kHz},$$

only twice the borrowed $1/T_2$, before disorder or control errors. Hence the best case depends on finding a first- or second-order parent model, or an exact symmetry mechanism, rather than brute-forcing a high-order Fibonacci plaquette.

Temperature is harsher. Since $k_B/h \approx 20.84 \text{ GHz/K}$,

$$\frac{k_B T}{h} \approx 208 \text{ MHz} \quad \text{at } T = 10 \text{ mK}.$$

That exceeds the assumed 2 MHz gap by roughly two orders of magnitude. Equilibrium passive suppression would require a substantially larger gap or an effective temperature well below

$$T_{\text{gap}} = \frac{h \nu_{\text{topo}}}{k_B} \approx 96 \text{ } \mu\text{K}.$$

No cited defect-cluster experiment supplies this full thermal stack. The strongest near-term case is therefore preparation and spectroscopy of an analog phase on coherent timescales, not a passively self-correcting memory. Calling the former the latter would improve the headline and damage the science.

The dependency chain is unforgiving but useful:

```
repeatable motif and placement
    |
    v
cluster spectrum: Delta_C, matrix elements
    |
    v
uniform intercluster coupling v and correct graph
    |
    v
low perturbative order q + bounded corrections
    |
    v
many-body gap Delta_topo and disorder tolerance
    |
    +-----> preparation faster than decoherence
    |
    +-----> k_B T below the relevant gap for passive protection
    |
```

v
local and nonlocal observables identify the phase

This diagram is the strongest case because each box is experimentally or numerically attackable. The architecture can be falsified in modules before a heroic full-array fabrication run. Drop any box and the boxes below it become theatre.

37.10. What the laboratory actually has

The evidence stack is broad but not integrated:

Structurally relevant claim	Status	What was actually shown	What remains missing
Nearby defect electron spins can interact coherently	[Experiment]	Room-temperature entanglement between two diamond defect spins [R080]	repeatable large graph and uniform couplings
A multi-defect motif can be fabricated	[Experiment]	one strongly coupled triple NV motif found in a 7,116-event molecular-implantation data set [R243]	deterministic yield, target spectrum, replication
Crystal orientation can shape dipolar sign	[Experiment]	aligned 2D (111) NV ensembles with same-sign in-plane dipolar geometry [R242]	individually defined cluster lattice and topological Hamiltonian
Defects can be registered to nanostructures	[Experiment]	2025 δ -doping plus electron irradiation gave about 4 nm depth precision, 46(1) nm lateral precision in 280 nm pillars, and mean Hahn-echo $T_2 = 98(37) \mu s$ [R207]	nanometre-scale relative placement required for a chosen direct-coupling graph
One defect neighborhood can host control helpers	[Experiment]	ten-qubit electron–nuclear register and long protected nuclear memory [R119]	parallel operation inside an interacting analog array
A cavity can mediate emitter interactions	[Experiment]	two SiV emitters interacted through one nanocavity [R086]	low-loss, uniform, many-cell Hamiltonian regime

Structurally relevant claim	Status	What was actually shown	What remains missing
Cluster projection can synthesize leftover operators	[Theory]	controlled effective-Hamiltonian formalism [R025]	measured defect-cluster coefficients
Local spin models can host non-Abelian order	[Theory]	solvable honeycomb and string-net models [R017]; [R018]	mapping from available defect terms with a usable gap
Defect clusters realize doubled-Fibonacci order	[Speculation]	no experimental or numerical defect-specific demonstration cited here	essentially the entire integrated chain

A separate proposal showed how NV registers and optical links might be organized into a scalable solid-state processor [R244]. That work supports modular engineering concepts, not the claim that the material Hamiltonian is topologically ordered.

A coupled defect triple is a physical spin cluster. Its isolated doublet may be an encoded leftover. A lattice of such leftovers can digitally emulate or analogously realize a model. Only if the many-body Hamiltonian is in the appropriate phase do emergent anyons exist. Only a demonstrated nonlocal encoding with the relevant error suppression is a topological logical qubit. These nouns describe successive tests, not interchangeable marketing settings.

37.11. Ways the case falls apart

1. **Assembling best values from incompatible samples.** The parameter stack above intentionally does this once and labels it. A final claim may not combine the coherence of a dilute bulk spin, the coupling of a damaged close pair, the placement of a nanostructure process, and the optical linewidth of a cryogenic center as though one device possessed all four.
2. **Treating disorder tolerance as no geometry requirement.** Topological phases can tolerate bounded local perturbations. They do not tolerate an interaction graph that never approximates the target Hamiltonian.
3. **Calling a helper-rich register passive protection.** Nuclear memories and repeated control can improve preparation and readout. They remain active resources unless the static many-body spectrum itself supplies protection.
4. **Counting a generated term but not its competitors.** The desired v^q/Δ^{q-1} term must exceed lower-order local fields, anisotropies, leakage corrections, disorder, and decoherence. A nonzero coefficient is not a victory condition.

5. **Assuming a demonstrated pair interaction scales uniformly.** Dipolar coupling magnifies relative placement error: for $J \propto r^{-3}$, small radial error gives $\delta J/J \approx -3\delta r/r$. A 10% coupling tolerance requires roughly 3% relative distance control before angular errors.
6. **Using “Fibonacci-like” to bridge a missing phase.** Ising non-Abelian order, doubled-Fibonacci string-net order, digital braid emulation, and genuine chiral Fibonacci order have different excitations and computational power.
7. **Ignoring temperature because coherence is long.** T_2 concerns phase memory under specified control. Thermal excitation depends on $k_B T/\Delta_{\text{topo}}$. One inequality cannot substitute for the other.
8. **Overlooking the real structural advantage.** The credible pro case is co-design of motif, spectrum, interaction, helper, and measurement — not an assertion that one host material is universally superior.

37.12. Five things you can test

The steelman is not a finished device. It is five hypotheses, each attackable before anyone fabricates a heroic array.

1. Correlated fabrication can produce a reproducible cluster motif.
2. One motif has an isolated doublet with useful projected operators.
3. Crystal orientation plus local or mediated coupling can realize the required low-order interaction graph.
4. The resulting many-body gap can exceed disorder and decoherence under one set of conditions.
5. Available ancillas can measure phase-defining local and nonlocal observables without replacing analog emergence by digital emulation.

The case succeeds only as a staged, falsifiable research program. The next chapter should attack the dependencies with the greatest leverage — especially uniform coupling, perturbative order, thermal scale, and fabrication yield.

37.13. Check yourself

1. **What is the strongest cluster-specific advantage?**
A cluster’s spectrum and wavefunctions can translate available microscopic couplings into designed leftover operators while placing leakage behind a measurable gap.

2. **Show that first-order projection of $V_{AB} = v S_{A,3}^z S_{B,1}^z$ produces a leftover coupling $J_{zz}^{(1)} = v a_z b_z$.**
 Inside the doublets, $P_A S_{A,3}^z P_A = a_0 I_A + a_z \tau_A^z$ and $P_B S_{B,1}^z P_B = b_0 I_B + b_z \tau_B^z$. Expanding the product, the $\tau^z \tau^z$ coefficient is $v a_z b_z$. Local fields survive unless symmetry forces $a_0 = b_0 = 0$.
3. **What fails if the same $\epsilon = 0.1$ is used at fourth order rather than second?**
 Then $v_{\text{eff}}^{(4)} \sim v \epsilon^3 = 0.1$ MHz and $v_{\text{topo}}^{(4)} \sim 20$ kHz, only twice the borrowed $1/T_2$, before disorder. The optimistic stack outruns coherence only at low perturbative order.
4. **Which number has the greatest leverage in the worked stack?**
 The perturbative order q . At $v/\Delta_C = 0.1$, moving from second to fourth order suppresses the effective term by 10^{-2} .
5. **Show that $v_{\text{topo}} T_2 = 200$ at the optimistic point, and state what that does not establish.**
 $(2 \times 10^6 \text{ s}^{-1})(10^{-4} \text{ s}) = 200$. At most, coherent dynamics might resolve a gap under matched conditions. It does not establish thermal protection, the correct phase, or scalable readout.
6. **What claim remains speculative?**
 That a fabricated defect-cluster array can realize a gapped doubled-Fibonacci Hamiltonian and host its emergent quasiparticles.

Grouping defects is a design move, not a compliment to diamond. The leftover two-state object, the crystal plane, the one-shot motif, the nearby helpers, and the always-on analog Hamiltonian are five knobs that have been turned one at a time. They have never been turned together. The optimistic stack can outrun a borrowed coherence rate only at low perturbative order, and even then it loses to 10 mK by two orders of magnitude. The honest best case is a staged program, not a passively self-correcting Fibonacci crystal.

37.14. Sources

- [R242] L. B. Hughes, S. A. Meynell, W. Wu, S. Parthasarathy, L. Chen, Z. Zhang, Z. Wang, E. J. Davis, K. Mukherjee, N. Y. Yao, and A. C. Bleszynski Jayich, “Strongly Interacting, Two-Dimensional, Dipolar Spin Ensembles in (111)-Oriented Diamond,” *Physical Review X* **15**, 021035 (2025). DOI: 10.1103/PhysRevX.15.021035; arXiv: 2404.10075. See also the 2026 publisher erratum.
- [R243] M. Haruyama et al., “Triple nitrogen-vacancy centre fabrication by C5N4H_n ion implantation,” *Nature Communications* **10**, 2664 (2019). DOI: 10.1038/s41467-019-10529-x.

- [R080] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R207] S. Kim, P. London, D. Yang, L. B. Hughes, J. Ahlers, S. Meynell, W. J. Mitchell, K. Mukherjee, and A. C. Bleszynski Jayich, “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* **16**, 9803 (2025). DOI: 10.1038/s41467-025-64758-4; arXiv: 2502.01198.
- [R119] C. E. Bradley et al., “A Ten-Qubit Solid-State Spin Register with Quantum Memory up to One Minute,” *Physical Review X* **9**, 031045 (2019). DOI: 10.1103/PhysRevX.9.031045; arXiv: 1905.02094.
- [R086] R. E. Evans et al., “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* **362**, 662–665 (2018). DOI: 10.1126/science.aau4691.
- [R025] S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* **326**, 2793–2826 (2011). DOI: 10.1016/j.aop.2011.06.004; arXiv: 1105.0675.
- [R017] A. Kitaev, “Anyons in an exactly solved model and beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv: cond-mat/0506438.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv: cond-mat/0404617.
- [R244] N. Y. Yao, L. Jiang, A. V. Gorshkov, P. C. Maurer, G. Giedke, J. I. Cirac, and M. D. Lukin, “Scalable architecture for a room temperature solid-state quantum information processor,” *Nature Communications* **3**, 800 (2012). DOI: 10.1038/ncomms1788; arXiv: 1012.2864.

Chapter 38. The strongest case against

You do not attack a tower by listing every cracked window. You find the columns.

A color center can be a beautiful sensor, a network node, or a physical qubit and still be a useless brick for an analog topological phase. The proposal in this book is not “put defects in diamond.” It is a chain of arrows, and every arrow has to fire in the same sample, in the same parameter window:

defects → clusters → pseudospins → target graph
→ many-body terms → gapped phase → logical protection.

That chain is *conjunctive*. The name is worth keeping because the logic is ordinary and unforgiving: every arrow must hold at once. Ten separate papers demonstrating ten separate arrows do not demonstrate the chain. The strongest case against is not “diamond is hard.” It is that several individually demanding requirements multiply, while the perturbative steps that create the desired Hamiltonian divide the useful energy.

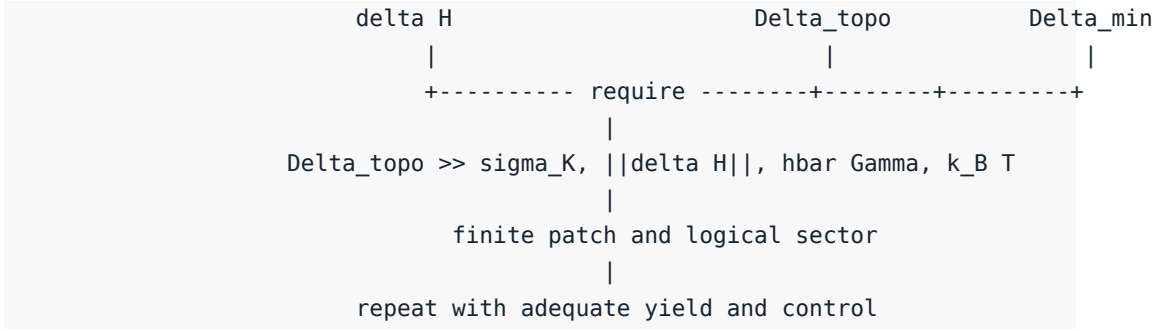
After that picture, call the missing pieces columns only if they sit under the whole claim. Interaction strength, cluster isolation, placement, and graph fidelity are columns. The effective plaquette term is built from them. The topological gap is built from that term. Preparation and thermal stability are judged against that gap. Scaling repeats the entire structure hundreds or thousands of times.

38.1. Two axes, not a complaint list

That picture gives two axes for ranking a crack.

The first axis asks how close the evidence sits to the hardware you actually want. Is the claim measured in a relevant defect array, established only for a component, derived for an ideal model, or merely proposed? Call that closeness *evidence proximity*. A theorem about an ideal Hamiltonian is not a measurement of a fabricated patch.

The second axis asks how many later requirements die if this one fails. A problem with a workaround can be expensive. A failed inequality at the base can be fatal. Call that reach *leverage*.



Now assign **hypothetical**, not measured, values. Let

$$\frac{\Delta_c}{h} = 5 \text{ GHz}, \quad \frac{J}{h} = 100 \text{ MHz}, \quad c = 1,$$

where h is Planck's constant. Then $\epsilon = 0.02$, and

$$\frac{K}{h} = 100 \text{ MHz} \times (0.02)^3 = 800 \text{ Hz}.$$

This passes the projection test comfortably and fails to produce a comfortable many-body scale. If the coherence time is a hypothetical $T_2 = 1 \text{ ms}$, define $\Gamma = 1/T_2 = 10^3 \text{ s}^{-1}$. The associated energy linewidth in frequency units is

$$\frac{\hbar\Gamma}{h} = \frac{\Gamma}{2\pi} = 159 \text{ Hz}.$$

Thus $K/(\hbar\Gamma) \approx 5$, not a robust $\gg 1$ hierarchy. A dimensionless factor smaller than our optimistic $c = 1$, extra perturbative order, or correlated noise can erase the margin.

Placement error is amplified too. For magnetic dipolar coupling, $J \propto r^{-3}$, where r is separation in metres. A small radial error δr gives

$$\frac{\delta J}{J} \approx -3 \frac{\delta r}{r}.$$

Because $K \propto J^4$ in this toy path,

$$\frac{\delta K}{K} \approx 4 \frac{\delta J}{J} \approx -12 \frac{\delta r}{r}.$$

A 5% separation spread therefore produces about a 60% spread in K in this simplified common-coupling estimate. The arithmetic is a sensitivity calculation, not a universal disorder law; independent bonds and angular errors require the actual Hamiltonian.

Preparation inherits the small gap. Let the minimum many-body gap along an interpolation be optimistically $\Delta_{\min} = h \times 800$ Hz, and let the norm of the Hamiltonian change with dimensionless interpolation coordinate s be $\|dH/ds\| = h \times 100$ MHz. A common sufficient adiabatic scale is [R245]

$$t_f \gg \frac{\hbar \|dH/ds\|}{\Delta_{\min}^2} = \frac{1}{2\pi} \frac{10^8}{800^2} \text{ s} \approx 25 \text{ s}.$$

That is roughly 2.5×10^4 hypothetical coherence times. The bound is schedule- and theorem-dependent, but the inverse-gap penalty is not a decorative detail.

Finally, at $T = 10$ mK,

$$\frac{k_B T}{h} = (20.84 \text{ GHz/K})(0.010 \text{ K}) \approx 208 \text{ MHz},$$

which overwhelms an 800 Hz equilibrium gap. Driven preparation can evade thermal equilibrium for a while, but then the claim has changed from a passive thermal phase to a controlled nonequilibrium experiment.

The hierarchy did not collapse because 800 Hz is intrinsically absurd. It collapsed because the same 800 Hz had to beat disorder, decoherence, temperature, and an inverse-gap preparation time simultaneously. Drop any one of those inequalities and the toy still looks promising. Keep all of them and it does not.

38.3. Ten obstacles, ranked

The ranking below is an assessment of the defect-cluster route to an analog, emergent, gapped topological Hamiltonian, not of defect qubits in general. “Direct” means evidence exists for the relevant ingredient; no cited experiment integrates all ingredients. Leverage runs from 1 (localized cost) to 5 (failure blocks the central claim).

Rank	Obstacle	Evidence proximity	Leverage	Why it sits here
1	Generate the required high-order terms	strong theory; architecture-specific implementation is a proposal	5	Without the stabilizer/plaquette algebra, there is no target phase to protect.

Rank	Obstacle	Evidence proximity	Leverage	Why it sits here
2	Retain a usable topological gap	rigorous stability theory starts from an already gapped target; defect value unmeasured	5	Every noise, preparation, and thermal inequality is paid from this one scale.
3	Scale yield and calibration	direct component evidence; no phase-scale array	5	Per-site and per-link imperfections compound with system size.
4	Place the right defects at the right coordinates	direct fabrication evidence	5	Position controls coupling, graph, cluster spectrum, and disorder at once.
5	Make interactions coherent and strong enough	direct pair-level evidence	5	$J \lesssim \hbar\Gamma$ kills coherent Hamiltonian engineering before topology enters.
6	Control disorder	direct evidence for inhomogeneity sources; phase-level tolerance model-dependent	4	Disorder perturbs denominators and effective terms and can close or fill the gap.
7	Realize the right interaction graph	microscopic interactions known; graph conversion is a proposal	4	Extra long-range and angular couplings are Hamiltonian terms, not merely wiring inconvenience.
8	Survive finite temperature	strong theory; thermalization rates platform-dependent	4–5	It is existential for passive memory, but can be deferred for a short proof-of-principle state.

	Rank	Obstacle	Evidence proximity	Leverage	Why it sits here
	9	Prepare the phase before it decoheres	strong adiabatic theory; no integrated demonstration	4	A small minimum gap can make a formally valid Hamiltonian experimentally unreachable.
	10	Initialize, address, drive, and read the array	direct small-register evidence; array extrapolation uncertain	3–4	Severe, but global protocols and improved interfaces offer more workaround space than missing terms do.

Ranks 3–7 are close. A host with mediator-enhanced coupling could move interaction downward while adding control or loss problems. If the goal is an equilibrium passive memory rather than a finite-time phase demonstration, thermal stability rises to rank 3. The table is a decision order, not a law of nature.

38.3.1. The algebra that may never appear

String-net Hamiltonians require local constraints and plaquette recoupling operations with a precise algebra [R018]. Available defect interactions are predominantly two-body. Perturbative gadgets can produce many-body terms [Theory], but higher order suppresses the desired coefficient and also generates lower-order shifts and unwanted operators [R174]. Symmetry can cancel some terms, yet the cancellation itself becomes sensitive to strain, orientation, and coupling imbalance.

The decisive question is not “can a four-body term be written in perturbation theory?” It is

$$\|H_{\text{unwanted}}\|, \sigma_K, \hbar\Gamma \ll K$$

throughout a manufacturable patch. No defect-cluster experiment cited here has measured that hierarchy. [Proposal] Until it does, the central analog-Hamiltonian step is an engineering hypothesis.

What fails if you stop at a nonzero coefficient? You have shown that a term exists in an expansion. You have not shown that it is the Hamiltonian.

38.3.2. A gap that theorems will not invent

Topological-order stability theorems show that certain already-gapped, local ideal Hamiltonians retain separated spectral bands under sufficiently weak local perturbations [R142]. **[Theory]** This is encouraging only after the engineered Hamiltonian lies inside the phase and its perturbations are below the theorem's threshold. It does not say that a weak approximate plaquette term overcomes larger parasitic terms.

Define Δ_{topo} as the many-body energy gap above the relevant ground-state manifold. The useful requirement is

$$\Delta_{\text{topo}} \gg \max(k_B T, \sigma_K, \|H_{\text{unwanted}}\|, \hbar\Gamma, \delta_{\text{fs}}),$$

where σ_K is effective-coupling disorder, T is temperature in kelvin, k_B is Boltzmann's constant, Γ is a decoherence rate in inverse seconds, and δ_{fs} is a finite-size splitting in joules. An experimentally visible spectral gap is still not, by itself, proof of topological order; its excitations, nonlocal operators, and robustness must match the target.

Invoke the stability theorem before the target term dominates, and you have used the conclusion as a hypothesis.

38.3.3. Multiplication as an enemy

Suppose every required site independently works with probability p . An all-good N -site patch then has yield

$$Y_N = p^N.$$

This toy model ignores correlated fabrication and repair. It nevertheless exposes the tax: for $N = 1000$, $p = 0.99$ gives $Y_N \approx 4.3 \times 10^{-5}$, while $p = 0.999$ gives $Y_N \approx 0.368$. Links, orientations, charge states, optical usability, and cluster spectra add separate conditions unless the architecture tolerates vacancies and reroutes around them.

Scaling also means calibrating an extensive number of nonidentical frequencies and couplings. A scheme that needs a separate analog cancellation tone for every parasitic bond may scale in component count while failing in calibration entropy. This is why scaling ranks above several spectacular single-device limitations.

Drop independence and the number can be kinder. Keep it, and $p = 0.99$ is already a closed door at a thousand sites.

38.3.4. The right defect in the right place

[Experiment] A 2025 diamond experiment combined nitrogen delta doping and localized electron irradiation in prefabricated nanopillars, reporting about 4 nm depth precision and 46(1) nm lateral precision in 280 nm-diameter pillars, with mean Hahn-echo $T_2 = 98 \mu\text{s}$ for the created single NV centers [R207]. Those are important fabrication advances under the paper’s conditions. They do not provide arbitrary three-dimensional, few-nanometre, orientation-selected interacting lattices.

The comparison must be made to the architecture’s tolerance, not to the word “nanoscale.” If a dipolar design uses separation r and permits relative coupling spread q , radial placement alone needs roughly $\sigma_r/r \leq q/3$. Exchange-mediated proposals can be more position-sensitive because exchange often changes exponentially with wave-function overlap. Post-fabrication mapping can identify what was built, but topology requires the mapped graph to remain in the desired phase or be repairable.

A 46 nm lateral number does not refute every future process. It does refute treating that 2025 process as already inside a few-nanometre interacting lattice.

38.3.5. A pair is not a lattice

[Experiment] Two engineered NV electron spins separated by approximately 25 nm were entangled at room temperature using their roughly 5 kHz magnetic dipolar coupling [R080]. This is decisive evidence that coherent defect–defect interaction is real, not evidence that the same coupling supplies a clean high-order gap across a large lattice. The experiment used control sequences and a selected pair.

Dipolar interaction offers no independent knobs for strength, angle, and range:

$$J_{ij} \propto \frac{1 - 3 \cos^2 \theta_{ij}}{r_{ij}^3},$$

where θ_{ij} is the angle between the separation vector and quantization axis. Bringing defects closer increases J but generally worsens fabrication damage, spectral crowding, unwanted exchange, and individual addressability. Optical, phononic, or cavity mediators can extend range, but add loss, fabrication, and mode-crowding budgets. They move the bottleneck; they do not delete it.

Hence $J/\hbar = 10\Gamma$ for a pair clears only that pair-level comparison. The topological gap, not bare J , is the later test.

38.3.6. Disorder walks in twice

Microscopic disorder first changes bare frequencies, orientations, and J_{ij} . It then enters perturbative denominators, so it changes effective coefficients nonlinearly. A cluster may cease to have a clean doublet before the array-level topological phase is tested.

Local disorder below a stable phase's threshold need not be fatal [R142]. The hostile case is not weak generic disorder around a healthy gap. It is disorder comparable to a gap already reduced by projection. Echo can refocus certain single-spin frequency offsets but cannot refocus a static error in the Hamiltonian one is trying to realize without also modifying the target dynamics.

Drop the second entrance — the denominators — and disorder looks like a local-field problem. Keep it, and a clean doublet can vanish before anyone asks about anyons.

38.3.7. Extra edges are operators

A string-net or bond-directional model specifies which degrees of freedom couple and which operator acts on each edge or plaquette [R018]. A three-dimensional crystal supplies crystallographic orientations, long-range dipolar tails, and surfaces — not a freely editable graph.

One can encode graph structure into clusters, frequency selection, pulse sequences, or mediators. Each technique adds either smaller effective scales, time-dependent control, or hardware. **[Proposal]** A digitally toggled average Hamiltonian may emulate the graph, but then its protection is control-dependent and should not be advertised as an autonomous material Hamiltonian.

An extra dipolar tail is not wiring inconvenience. It is a term in H .

38.3.8. Heat does not ask permission

At nonzero temperature, excitations appear with rates that depend on the gap and bath dynamics. They can diffuse and enact logical operators. The finite-temperature quantum-memory literature shows that dimensionality, energy barriers, kinetics, and decoding matter, not merely the existence of a zero-temperature gap [R169]. **[Theory]** Two-dimensional local stabilizer Hamiltonians cannot provide a self-correcting quantum memory with a macroscopic energy barrier under the assumptions of the Bravyi–Terhal no-go theorem [R168]. That theorem is not a blanket impossibility result for every non-Abelian, driven, long-range, or actively corrected architecture; it does block casual claims that a 2D local Hamiltonian automatically stores forever.

A short-lived ground-state experiment may operate with active cooling and rapid measurement even when $k_B T > \Delta_{\text{topo}}$. That can demonstrate aspects of a model. It is not passive thermal protection.

The comparison is $k_B T / \Delta_{\text{topo}}$, not the refrigerator temperature alone. A millikelvin number is unimpressive beside a hertz-scale emergent gap.

38.3.9. Getting there before the clock runs out

Even if the final Hamiltonian is in the desired phase, an adiabatic path can pass through a smaller gap Δ_{min} . Adiabatic bounds depend on the full schedule, derivatives of the Hamiltonian, and matrix elements, not only the final gap [R245]. **[Theory]** Increasing system size can shrink Δ_{min} near a phase transition, while decoherence and calibration drift accumulate for longer.

Measurement-assisted or dissipative preparation may be faster. Those are new protocols with ancilla, reset, measurement, and verification requirements. Preparing a wavefunction once also does not prove an equilibrium phase: the subsequent Hamiltonian and response must carry the evidence.

A formally valid Hamiltonian with no experimental window is not a device. It is a matrix.

38.3.10. Control is last because it has more doors

[Experiment] A seven-spin diamond processor has demonstrated fault-tolerant protocol primitives for a five-qubit-code logical qubit, including flagged stabilizer measurements and real-time processing [R234]. The authors explicitly noted that fidelity and qubit number still needed improvement to suppress logical errors below physical errors. This shows sophisticated control in a selected local register, not dense analog control of an extensive defect lattice.

Color-center network reviews likewise identify simultaneous demands on spin coherence, optical interfaces, spectral stability, and integrated fabrication [R246]. Local tuning can correct disorder but may break cluster symmetry; spectral distinguishability helps addressability but creates detuning; dense packing strengthens interaction but increases crosstalk. These are not ten independent knobs.

Control ranks last because workarounds exist. Missing plaquette algebra does not have a workaround that still yields the proposed phase.

38.4. What the laboratory actually has

The evidence through 2026 is asymmetric.

- **[Experiment]** Individual defects, small registers, coherent pair coupling, logical-protocol primitives, and improving registered fabrication are real [R207]; [R080]; [R234].
- **[Theory]** Perturbative generation, stability of ideal topological phases, adiabatic conditions, and finite-temperature limitations are well developed [R174]; [R245]; [R018]; [R142]; [R169]; [R168].
- **[Proposal]** Mapping a particular manufacturable defect-cluster Hamiltonian to a clean non-Abelian string-net model remains a proposal.
- **[Numerics]** Small-model numerics can validate a chosen effective Hamiltonian, but they cannot supply fabrication distributions or decoherence parameters that have not been measured.
- **[Speculation]** Extrapolating current defect components to a scalable, passively protected Fibonacci-like material is scientifically coherent but unsupported as an integrated hardware claim.

The negative verdict is therefore not “impossible.” It is “the central conjunction has no demonstrated margin.” The most impressive experimental results support the bottom edges of the dependency tree, while the claim lives at the top.

38.5. Ways to get this wrong

1. **Counting ingredients instead of checking simultaneous inequalities.** A long T_2 , a strong pair coupling, accurate implantation, and a topological model in four different samples do not form one device. The relevant object is the joint distribution of J , Δ_c , disorder, coherence, yield, and control in the same fabricated patch.
2. **Calling any effective interaction a plaquette term.** An observed four-spin spectral shift is not enough. The operator, sign, competing terms, spatial pattern, and perturbative validity must match the target Hamiltonian. Otherwise a higher-order effect has been demonstrated, not the needed algebra.
3. **Treating a digital workaround as proof of passive emergence.** Pulse synthesis can cancel extra edges and generate multi-body average Hamiltonians. That may be excellent digital or Floquet simulation. If continuous calibration and periodic drive are essential, the protection mechanism is not simply an autonomous equilibrium Hamiltonian.
4. **Using topological stability circularly.** The theorem says a suitable topological phase survives sufficiently weak perturbations. It cannot be invoked before showing that the target term dominates and the system begins inside that phase.
5. **Hiding yield behind postselection.** Selecting one successful pair from a large implanted field demonstrates pair physics. A scalable architecture must count failed

sites, unusable charge states, missing links, calibration time, and the effect of repair on the graph.

6. **Confusing a cold refrigerator with a cold effective Hamiltonian.** The comparison is $k_B T / \Delta_{\text{topo}}$, not the refrigerator temperature alone. A millikelvin number is unimpressive beside a hertz-scale emergent gap.
7. **Collapsing five different claims into one word.** A physical defect qubit can be coherent. A cluster can encode a leftover. A controlled array can digitally emulate a string-net Hamiltonian. An analog material can possess emergent topological order. An encoded state can be actively error-corrected. Failure of the analog defect-cluster route would not negate the other four.

38.6. Check yourself

1. **Why is high-order-term generation ranked above control complexity?**
Because control has possible architectural workarounds, while absence of the target plaquette/stabilizer algebra removes the proposed phase itself. High-order synthesis also sets the small scale inherited by several downstream tests.
2. **Show that the toy numbers give $K/h = 800$ Hz and $K/(\hbar\Gamma) \approx 5$.**
With $J/h = 100$ MHz and $\epsilon = 0.02$, $K/h = 100 \text{ MHz} \times (0.02)^3 = 800$ Hz. For $T_2 = 1$ ms, $\Gamma = 10^3 \text{ s}^{-1}$ and $\hbar\Gamma/h = \Gamma/2\pi = 159$ Hz, so the ratio is about 5, not $\gg 1$.
3. **What fails if cluster isolation Δ_c/J is increased to protect the doublet?**
A fourth-order coefficient scales as $J(J/\Delta_c)^3$. The projection becomes cleaner while the desired term becomes smaller. Better isolation can make the topological gap worse.
4. **Show that a 5% radial spread produces about a 60% spread in K on this fourth-order dipolar path.**
 $\delta J/J \approx -3\delta r/r$ and $K \propto J^4$, so $\delta K/K \approx -12\delta r/r$. Then $12 \times 0.05 = 0.60$. Independent bonds and angles need the actual Hamiltonian.
5. **Does 46 nm lateral positioning precision refute a few-nanometre architecture?**
It refutes neither future improvement nor every fabrication method. It shows that one advanced, high-yield 2025 process did not yet meet that hypothetical tolerance. The architecture must state its tolerance and compare like with like.
6. **What fails if $k_B T > \Delta_{\text{topo}}$? Is every experiment pointless?**
Fast, driven, cooled, measured, or postselected experiments may reveal model dynamics. The result would not establish a passive equilibrium memory at that temperature.

The attack is on the joints, not the windows. Improving a later protocol cannot rescue a missing column.

38.7. Sources

- [R174] S. P. Jordan and E. Farhi, “Perturbative Gadgets at Arbitrary Orders,” *Physical Review A* **77**, 062329 (2008). DOI: 10.1103/PhysRevA.77.062329; arXiv:0802.1874.
- [R245] T. Albash and D. A. Lidar, “Adiabatic quantum computation,” *Reviews of Modern Physics* **90**, 015002 (2018). DOI: 10.1103/RevModPhys.90.015002; arXiv:1611.04471.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv:cond-mat/0404617.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: Stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195; arXiv:1001.0344.
- [R207] S. Kim et al., “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* **16**, 9803 (2025). DOI: 10.1038/s41467-025-64758-4; arXiv:2502.01198.
- [R080] F. Dolde et al., “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545; arXiv:1212.2804.
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* **88**, 045005 (2016). DOI: 10.1103/RevModPhys.88.045005; arXiv:1411.6643.
- [R168] S. Bravyi and B. M. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* **11**, 043029 (2009). DOI: 10.1088/1367-2630/11/4/043029; arXiv:0810.1983.
- [R234] M. H. Abobeih et al., “Fault-tolerant operation of a logical qubit in a diamond quantum processor,” *Nature* **606**, 884–889 (2022). DOI: 10.1038/s41586-022-04819-6; arXiv:2108.01646.
- [R246] M. Ruf, N. H. Wan, H. Choi, D. Englund, and R. Hanson, “Quantum networks based on color centers in diamond,” *Journal of Applied Physics* **130**, 070901 (2021). DOI: 10.1063/5.0056534; arXiv:2105.04341.

Chapter 39. Write the no first

You are about to spend years cooling a crystal so a bit can sit still without anyone watching. Before you book the helium, write the number that would make you stop.

Not the number you hope for. The number that, if the data land on the wrong side of it, ends this particular claim. Write it now, while the cartoon still looks clean.

That written no is a *kill criterion*. The name is worth keeping because it is a door you can actually fall through. A wish list has no door.

A design target says we would like a larger gap. A kill criterion says: if the upper bound you still believe on the activation gap is below this value after measurement X, stop pursuing a memory that sits still at temperature T .

The constraints do not take turns. They hold at the same time. A poor cluster yield reduces patch yield. A weak coupling reduces the gap. A smaller gap makes preparation slower. Slower preparation gives decoherence more time. The product, not the nicest factor, is the project.

This is not theatrical pessimism. It is arithmetic you agree to before you like the answer.

39.1. Five parts of a useful no

A useful trapdoor has five parts:

1. a claim you can still point to after the measurement;
2. a metric with units and a measurement or calculation protocol;
3. a threshold derived from that claim;
4. an uncertainty rule decided before seeing the result;
5. an action: continue, redesign, pivot, or stop.

That fourth part is *preregistration*. The threshold is a promise, not a mood. You do not move the line after you like the plot.

There is no universal magic ratio such as $J/\Gamma > 10$. The required ratio depends on how many interaction times the protocol consumes and how much error is allocated to decoherence. Likewise, $\Delta > k_B T$ is not enough for a thousand-site sample if the allowed probability of even one thermal seed is tiny. The criterion must know the model and the task.

Drop the predeclared action and a missed number becomes a discussion. Keep it, and a missed number is a decision.

39.2. A table that can say no

Consider a deliberately hypothetical target, not a claim about any existing material. [**Proposal**] One thousand cluster leftovers form a two-dimensional analog topological memory with $N_s = 1000$ required sites and $N_b = 1500$ required bonds. The effective interaction frequency is estimated as $\nu_{\text{eff}} = J_{\text{eff}}/h = 20$ MHz, with a credible interval of 10–40 MHz. The measured dephasing time under the intended controls is $T_2 = 100 \mu\text{s}$, interval 50–200 μs . Here J_{eff} is an energy in joules (J), h is Planck’s constant in J·s, and T_2 is in seconds.

The project writes four decisions before the numbers arrive:

- **Continue:** the conservative bound you still believe passes.
- **Resolve:** the intervals cross; obtain the measurement or larger simulation that controls the decision.
- **Redesign:** the present implementation fails, but a named change could alter the controlling equation.
- **Stop/pivot:** even the optimistic bound fails. Stop claiming the stated target; a different target, such as active digital simulation, is a new project.

The numbers below are invented to show the arithmetic. Replace them with the parameter ledger and the model calculations you already have. The table is a rehearsal, not a verdict on a real crystal.

Gate	Hypothetical estimate, including uncertainty	Preregistered threshold	Decision	Required action
Coherent interaction $R_J = J_{\text{eff}}/(\hbar\Gamma)$	1.26×10^4 $[3.1 \times 10^3, 5.0 \times 10^4]$	$R_J \geq 2.0 \times 10^4$	Resolve	Measure T_2 under the complete pulse schedule; do not scale the array yet
Thermal activation $x_T = E_{\text{act}}/(k_B T)$	0.0048 [0.0010, 0.015] at 10 mK	$x_T \geq \ln(N_s/10^{-3}) = 13.8$	Stop/pivot	Abandon passive equilibrium memory for this scale unless the gap or temperature changes by orders of magnitude

Gate	Hypothetical estimate, including uncertainty	Preregistered threshold	Decision	Required action
Coupling- disorder quantile q_I/J	0.08 [0.04, 0.15]	model phase boundary $\eta_c = 0.05$ [0.03, 0.08]	Resolve	Simulate measured disorder tails and missing bonds; RMS alone is insufficient
Placement spread for direct dipolar bonds	$\sigma_r = 1.0$ nm [0.5, 2.0] at $r = 10$ nm	$\sigma_r \leq 0.17$ nm for 5% coupling spread	Redesign	Add tunability or change coupling mechanism; direct fixed dipolar layout fails this tolerance
Unhandled leakage per control location p_u	2×10^{-4} [5×10^{-5} , 8×10^{-4}]	$p_u \leq 10^{-6}$ for 1000 locations and 10^{-3} budget	Redesign	Demonstrate detection/reset before accepting more locations
Preparation window	$t_{\text{ad}} = 100$ μs [30, 500]	$t_{\text{ad}} \leq 0.01T_2 = 0.5-$ 2 μs	Stop/pivot	Change preparation path or use dis- sipative/active preparation; present ramp has no window
Site-only patch yield $Y_s = y_s^{N_s}$	0.0067 [4.3×10^{-5} , 0.368] for $y_s =$ 0.995 [0.990, 0.999]	$Y_s \geq 0.10$, before bond losses	Resolve	Measure correlated yield and test repairable layouts; monolithic scale-up is not justified
Logical scaling, $p_L(d+2)/p_L(d)$	0.80 [0.60, 1.10]	upper credible bound < 1 at fixed physical conditions	Resolve	Add distances and blind decoder analysis; do not extrapolate an improving lifetime yet

Let us check two rows. Define $\Gamma = 1/T_2$, in s^{-1} , and $\hbar = h/(2\pi)$. Then

$$R_J = \frac{J_{\text{eff}}}{\hbar\Gamma} = 2\pi\nu_{\text{eff}}T_2 = 2\pi(20 \times 10^6 \text{ s}^{-1})(100 \times 10^{-6} \text{ s}) \approx 1.26 \times 10^4.$$

The units cancel. If the protocol consumes $\alpha = 20$ inverse-interaction times and receives a decoherence budget $\epsilon_J = 10^{-3}$, the exposure estimate $\Gamma t \approx \alpha/R_J$ requires $R_J \geq \alpha/\epsilon_J = 2.0 \times 10^4$. The central estimate fails, but the interval crosses. The correct action is measurement, not rhetoric.

For the thermal row, suppose a small-patch calculation gives an activation energy $E_{\text{act}}/h = c_{\text{act}}\nu_{\text{eff}}$, with $c_{\text{act}} = 0.05 [0.02, 0.08]$. The central activation frequency is 1 MHz. At $T = 10 \text{ mK}$, $k_B T/h \approx 208 \text{ MHz}$, so $x_T \approx 1/208 = 0.0048$. Even combining the optimistic endpoints gives only $3.2/208 \approx 0.015$. The best case misses 13.8 by nearly three orders of magnitude. More precise fitting cannot rescue the stated passive-memory target.

This table tests an analog memory that is supposed to sit still. A digital simulator may operate with $E_{\text{act}} < k_B T$ because its effective Hamiltonian is enacted by calibrated gates and active error correction rather than thermal equilibrium. Passing a digital circuit benchmark does not pass the analog thermal gate. Conversely, failing passive memory does not prove the hardware useless as physical qubits, sensors, or an emulation platform.

Hence two different claims share the same hardware and not the same door.

39.3. How many ticks before the spin forgets

Define J_{eff} as the coefficient, in J, of the effective term that actually performs the needed low-energy dynamics – not the bare intra-cluster splitting and not the largest microscopic coupling. Define Γ as the measured decay rate, in s^{-1} , under the operating point and control schedule relevant to that term. The dimensionless coherent ratio is

$$R_J = \frac{J_{\text{eff}}}{\hbar\Gamma}.$$

If a protocol requires time

$$t_{\text{req}} = \alpha \frac{\hbar}{J_{\text{eff}}},$$

where α is a dimensionless path or circuit cost obtained from simulation, then first-order Markovian exposure is $\epsilon_{\text{dec}} \approx \Gamma t_{\text{req}} = \alpha/R_J$. Allocating at most ϵ_J to this channel gives the falsifiable criterion

$$R_J \geq \frac{\alpha}{\epsilon_J}.$$

This is why a bare rule $J_{\text{eff}} > \hbar\Gamma$ is weak: it buys only order-one dynamics before decay. It also explains why T_2 measured on an isolated defect cannot be inserted automatically. Clustering, continuous drives, optical cycling, and correlated noise change Γ .

Kill rule. Stop the specified protocol if the upper bound you still believe on R_J remains below the lower bound you still believe on α/ϵ_J after the effective coupling and driven coherence are measured in the same device. If a different protocol reduces α , that is a redesign with a new gate.

What fails if you borrow an isolated-spin echo? You have compared J_{eff} to someone else's Γ . The inequality is then a collage, not a test.

39.4. One seed is already too many

Define E_{act} , in J, as the lowest energy barrier for a locally accessible harmful excitation process. It need not equal the clean spectral gap Δ_{topo} . Boundaries, weak bonds, disorder, and multi-step paths can lower it. For N_c approximately independent creation opportunities, a dilute equilibrium estimate gives

$$P_{\text{seed}} \lesssim N_c g \exp\left(-\frac{E_{\text{act}}}{k_B T}\right),$$

where g is a dimensionless species or channel factor and P_{seed} is the probability of at least one harmful seed. Requiring $P_{\text{seed}} \leq \epsilon_T$ yields

$$\frac{E_{\text{act}}}{k_B T} \geq \ln\left(\frac{N_c g}{\epsilon_T}\right).$$

The exponent is dimensionless: J divided by (J/K)K. Finite-temperature topological memories require kinetic analysis as well as this occupancy estimate; in two-dimensional local stabilizer Hamiltonians, a constant energy barrier prevents passive self-correction under the no-go theorem's assumptions [R168]; [R169]. **[Theory]**

Kill rule. For passive equilibrium memory, stop if the optimistic disorder-renormalized E_{act} , the lowest temperature you still believe, and the smallest useful N_c still fail the logarithmic bound. If active removal is proposed, replace this gate with a measured creation rate, diffusion model, syndrome cadence, and logical-error target. Do not silently keep the word *passive*.

$\Delta > k_B T$ is not the same test. The required ratio grows as $\ln(N_c g / \epsilon_T)$, and diffusion can still dominate after occupancy looks small.

39.5. The weakest bond, not the average

Write the realized Hamiltonian as

$$H = H_{\text{target}} + \delta H, \quad \delta H = \sum_b \delta J_b O_b,$$

where b labels bonds or local terms, δJ_b is a coupling error in J , and O_b is a dimensionless operator. Topological phases can be stable to sufficiently weak local perturbations, but the theorem assumes a gapped target and bounded local perturbations; it does not convert an RMS fabrication number into a universal tolerance [R142]. **[Theory]**

The proper threshold η_c must come from the candidate model's disordered phase diagram or a conservative local-gap bound. A minimal simulation sweeps the *measured distribution*, including missing terms and spatial correlations, and estimates where the gap, topological diagnostic, or logical scaling fails. The fabrication metric should be a high quantile

$$q_J = Q_{1-\alpha/N_b}(|\delta J_b|),$$

not merely the standard deviation. Here Q_p is the p -quantile, N_b is the number of required bonds, and α is the allowed probability that any bond exceeds the quoted value. The $1/N_b$ makes the tail requirement tighten with scale.

Kill rule. Stop monolithic scale-up if even the lower tail you still believe of q_J / J_{eff} lies beyond the upper model tolerance η_c you still believe, or if missing or sign-flipped terms put the realized Hamiltonian in a different phase. A calibratable static offset is not automatically harmless; demonstrate that calibration preserves the many-body gap and does not add control noise.

Report the average and a large patch can still die on one weak bond. The quantile is the object; the RMS is a summary of a different question.

39.6. How far the atoms may wander

Positioning is fatal only through a model-dependent transfer function. Let r be nominal separation and σ_r a small radial uncertainty, both in metres. For a dipolar interaction $J(r) = Cr^{-3}$, differentiation gives

$$\left| \frac{\sigma_J}{J} \right| \approx \left| \frac{d \ln J}{dr} \right| \sigma_r = 3 \frac{\sigma_r}{r}.$$

Thus an allowed fractional coupling spread η_J requires

$$\sigma_r \leq \frac{\eta_J r}{3}.$$

At $r = 10$ nm and $\eta_J = 0.05$, this is 0.17 nm. Length divided by length makes the fractional error dimensionless. Angular uncertainty must be added because dipolar coupling also contains an orientation factor.

For exchange-like coupling $J(r) = J_0 e^{-r/\lambda}$, where λ is a decay length in metres, the corresponding result is $\sigma_J/J \approx \sigma_r/\lambda$. The relevant tolerance may then be much smaller than a nanometre. Defect-platform reviews document that creation, charge-state control, coherence, and placement are separate yields, so a nominal beam diameter is not a complete position distribution [R074]. **[Experiment]**

Kill rule. Stop the *fixed-coupling layout* if the best demonstrated final-position distribution, propagated through the full angular and radial coupling law, exceeds η_c . Tunable couplers, spectroscopy-based calibration, or a less sensitive geometry are redesigns and must include their added noise and wiring costs.

A beam diameter is not σ_r . Straggle, diffusion, conversion, charge selection, and registration enter the realized coupling distribution.

39.7. A leak that stays leaked

Let p_ℓ be leakage probability per relevant control or syndrome location. Let $p_u \leq p_\ell$ be the probability that leakage remains undetected and unremoved long enough to spread. For N_{loc} locations in a task, the dilute approximation gives

$$P_{\text{leak}} \approx N_{\text{loc}} p_u.$$

Allocating ϵ_ℓ to unhandled leakage requires

$$p_u \leq \frac{\epsilon_\ell}{N_{\text{loc}}}.$$

A reset every m cycles changes p_u , but it may also interrupt the analog Hamiltonian. Surface-code studies show that long-lived leakage creates time-correlated, propagating faults and that dedicated leakage-reduction circuitry can restore threshold behavior in

particular active circuits [R220]. **[Theory] [Numerics]** That is a method to test, not permission to assume the same threshold for clusters.

Kill rule. Stop scaling if the optimistic unhandled leakage probability exceeds the task allocation and no compatible detection or reset operation has been demonstrated. Count charge-state switching, departure from the cluster doublet, and loss of a constituent separately before combining them.

Treat leakage as a Pauli error and you have left the modeled local Hilbert space without noticing.

39.8. A window that has to exist

Let the preparation path be $H(s)$, with dimensionless schedule coordinate $s \in [0, 1]$. Define $A = \max_s \|\partial_s H\|$, in J, and $\Delta_{\min}$, in J, as the minimum many-body gap along the actual path. A schematic leading adiabatic estimate is

$$\epsilon_{\text{ad}} \sim \left(\frac{\hbar A}{t_p \Delta_{\min}^2} \right)^2,$$

where t_p is preparation time in seconds. The combination $\hbar A / \Delta_{\min}^2$ has units (J s)/J² = s. Rigorous bounds include endpoint smoothness, higher derivatives, matrix elements, and system-size dependence [R247]. **[Theory]** Solving for an allocated diabatic error ϵ_{ad}^* gives a lower bound

$$t_p \geq t_{\min} = \frac{\hbar A}{\Delta_{\min}^2 \sqrt{\epsilon_{\text{ad}}^*}}.$$

Decoherence supplies an upper bound $t_p \leq t_{\max} \approx \epsilon_{\text{dec}}^* / \Gamma$ in the small-error Markovian estimate. A viable ramp needs

$$\boxed{t_{\min} < t_{\max}}.$$

Kill rule. Stop the adiabatic preparation route if the bounds you still believe leave no overlap, including the size dependence of $\Delta_{\min}$. A dissipative or measurement-assisted route is a new protocol; give it a mixing or convergence time, residual excitation density, and its own decoherence and readout budget.

What fails if the ramp may be arbitrarily slow? Diabatic error falls and open-system error rises. When $t_{\min} > t_{\max}$, slower is not a solution.

39.9. Almost every site is not every site

Let y_s be the probability that a required site has the correct position, species, charge state, local spectrum, and addressability. Let y_b be the probability that a required bond lies within tolerance. If these events were independent, an unrepairable patch yield would be

$$Y_{\text{patch}} = y_s^{N_s} y_b^{N_b}.$$

This is dimensionless. At $y_s = 0.995$ and $N_s = 1000$, the favorable site-only result is $0.995^{1000} \approx 0.0067$. Adding bond failures only lowers it. Independence is not guaranteed: implantation damage, annealing, strain, and optical collection can correlate failures.

Known-loss topological-code thresholds can be high in specifically designed active codes; for example, the idealized surface-code analysis of Stace and collaborators connects tolerable heralded loss to lattice percolation [R248]. **[Theory] [Numerics]** That value must not be borrowed for a string-net Hamiltonian whose interaction graph and terms disappear with a site. A reroutable design passes only after the defective graph is shown to remain in the target phase and the readout identifies losses reliably.

Kill rule. Set an economic or experimental minimum yield $Y_{\min} = 1/N_{\text{attempt,max}}$. Stop monolithic fabrication when the upper Y_{patch} you still believe is below $Y_{\min}$. Continue a repairable architecture only if measured defect maps, an explicit repair algorithm, and phase simulations pass together.

Multiply advertised yields with no denominators and you have multiplied different conditional events as if they were one.

39.10. Bigger has to get better

Small patches can imitate a gap or a topological degeneracy. The extrapolation must be stated. For linear size L , one gapped ansatz is

$$\Delta(L) = \Delta_{\infty} + ae^{-L/\xi},$$

where all Δ and a are in J and L and correlation length ξ share units. A phase claim needs a $\Delta_{\infty} > 0$ you still believe and competing finite-size fits. A logical-memory claim under fixed physical noise might use

$$p_L(d) = A_0 e^{-\alpha d},$$

where code distance d , prefactor A_0 , and exponent α are dimensionless. Improvement requires $\alpha > 0$, or equivalently $p_L(d + \delta d)/p_L(d) < 1$.

Kill rule. Stop claiming scalable protection if the upper improvement ratio you still believe is at least one across preregistered larger sizes under the same noise distribution, or if extrapolation excludes a nonzero thermodynamic gap. Do not let postselection, better control on larger samples, a different decoder, or a lower temperature masquerade as distance scaling. Those may be useful experiments, but they test another curve.

A kill here is not total failure of the hardware. It is failure of the stated scaling claim under the stated noise.

39.11. What the lab has to write down

A kill criterion is only as honest as its uncertainty. Each input should retain a distribution and a condition:

- J_{eff} : inferred from the low-energy spectrum with corrections and fit covariance;
- Γ : measured under the complete operating schedule, not copied from isolated-spin echo;
- E_{act} and Δ_{min} : minimized over boundaries, disorder samples, leakage sectors, and preparation path;
- placement and coupling disorder: joint spatial distributions and tail confidence, not only RMS values;
- leakage: state-resolved rate, dwell-time distribution, propagation, and reset efficacy;
- yield: denominators for creation, correct species, charge, coherence, addressability, and bonds;
- scaling: raw logical counts, decoder frozen before unblinding, and uncertainty on fit and model choice.

[Numerics] Exact diagonalization can establish these quantities only for the Hamiltonian, sizes, boundary conditions, and disorder ensemble actually simulated. **[Experiment]** Spectroscopy can establish a finite-device gap, not by itself thermodynamic topological order. **[Proposal]** Combining the two requires posterior predictive checks: feed measured distributions into the model and ask whether it predicts held-out spectra, dynamics, and logical observables.

Use one-sided intervals for decisions. If passing requires a quantity to be large, compare its lower bound with the threshold. If passing requires it to be small, compare its upper bound. A hard stop is justified when even the optimistic bound fails *after* the measurement has enough power to resolve the decision. An imprecise experiment usually earns “resolve,” not “go” and not “stop.”

39.12. Ways to get this wrong

1. **Universalizing a model threshold.** The roughly 11% perfect-syndrome toric-code threshold belongs to a stochastic Pauli model and decoder assumptions [R141]. It is not a disorder, leakage, or fabrication threshold.
2. **Comparing unlike units.** J_{eff}/h is in hertz; $\Gamma = 1/T_2$ is in s^{-1} ; the energy comparison uses $J_{\text{eff}}/(\hbar\Gamma) = 2\pi(J_{\text{eff}}/h)T_2$. Dropping 2π can move a marginal gate.
3. **Using $\Delta/k_B T > 1$ as a binary test.** The required ratio grows as $\ln(N_c g/\epsilon_T)$, and diffusion can still dominate.
4. **Reporting average disorder for a large patch.** Rare weak bonds, missing sites, and spatially correlated regions govern failure.
5. **Calling nominal implantation precision the final placement error.** Straggle, diffusion, conversion, charge selection, and registration enter the realized coupling distribution.
6. **Treating leakage as a Pauli error.** Leakage leaves the modeled local Hilbert space and can persist or spread.
7. **Optimizing preparation against only one bound.** Adiabaticity imposes a minimum time; decoherence imposes a maximum.
8. **Multiplying advertised yields with no denominators.** Creation yield, correct charge, optical usability, cluster completion, and bond tolerance are different conditional events.
9. **Changing conditions while claiming scaling.** Logical improvement with distance must hold at fixed physical noise and analysis rules.
10. **Interpreting a kill as total failure.** A passive-memory kill may motivate an active simulator, sensor, network node, or smaller proof of principle. That pivot must receive a new claim and new criteria.

39.13. Check yourself

1. **Show that $R_J = 2\pi\nu_{\text{eff}}T_2 \approx 1.26 \times 10^4$ for the hypothetical numbers, and why $R_J > 1$ is usually insufficient.**
 $2\pi(20 \times 10^6 \text{ s}^{-1})(100 \times 10^{-6} \text{ s}) \approx 1.26 \times 10^4$. That permits many interaction cycles, but a protocol costing α interaction times with error budget ϵ_J needs $R_J \geq \alpha/\epsilon_J$. Here $\alpha/\epsilon_J = 2.0 \times 10^4$.
2. **What fails if you use $\Delta > k_B T$ as a binary test?**
 The required ratio grows as $\ln(N_c g/\epsilon_T)$. For $N_s = 1000$ and $\epsilon_T = 10^{-3}$, that logarithm is 13.8, and diffusion can still dominate after occupancy looks small.
3. **Show that a 5% dipolar-coupling tolerance at $r = 10 \text{ nm}$ forces $\sigma_r \leq 0.17 \text{ nm}$.**
 Since $J \propto r^{-3}$, $\sigma_J/J \approx 3\sigma_r/r$, so $\sigma_r \leq \eta_J r/3 = 0.05 \times 10/3 \approx 0.17 \text{ nm}$.

4. **What fails if $t_{\min} > t_{\max}$ even when arbitrarily slow ramps are available?**

The adiabatic lower time bound exceeds the decoherence upper time bound. Slower then reduces diabatic error but increases open-system error. The window does not exist.

5. **Why can 99.5% site yield be poor?**

An unrepairable thousand-site patch has site-only yield $0.995^{1000} \approx 0.0067$, before bond failures.

6. **What makes a result a hard stop rather than “more data needed”?**

A preregistered threshold is missed even by the optimistic bound you still believe after a measurement or calculation capable of resolving the gate.

The eight inequalities are go/no-go gates, not aspirations. A stage advances only when the conservative bounds pass. Intervals that cross trigger the cheapest resolving calculation. Optimistic bounds that fail terminate the stated route or force an explicit redesign.

39.14. Sources

- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* **11**, 043029 (2009). DOI: 10.1088/1367-2630/11/4/043029. arXiv:0810.1983.
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* **88**, 045005 (2016). DOI: 10.1103/RevModPhys.88.045005. arXiv:1411.6643.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195. arXiv:1001.0344.
- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001. arXiv:1302.3288.
- [R220] A. G. Fowler, “Coping with qubit leakage in topological codes,” *Physical Review A* **88**, 042308 (2013). DOI: 10.1103/PhysRevA.88.042308. arXiv:1308.6642.
- [R247] S. Jansen, M.-B. Ruskai, and R. Seiler, “Bounds for the adiabatic approximation with applications to quantum computation,” *Journal of Mathematical Physics* **48**, 102111 (2007). DOI: 10.1063/1.2798382. arXiv:quant-ph/0603175.
- [R248] T. M. Stace, S. D. Barrett, and A. C. Doherty, “Thresholds for topological codes in the presence of loss,” *Physical Review Letters* **102**, 200501 (2009). DOI: 10.1103/PhysRevLett.102.200501. arXiv:0904.3556.
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* **43**, 4452–4505 (2002). DOI: 10.1063/1.1499754. arXiv:quant-ph/0110143.

Chapter 40. Calculations with exits

You sit down with two defects and a four-by-four matrix. You diagonalize it. You look at the two lowest states. If they are not a useful pair, you stop.

That is the whole method. The rest of the chapter is the same door, four more times, each time a little more expensive.

The sequence is cheap at the front on purpose:

two defects $\rightarrow$ one cluster leftover $\rightarrow$ three-to-six-cluster plaquette $\rightarrow$ small patch.

At each arrow you name the inputs, compute the observables, test whether the answer moved when you tightened the calculation, and then either pass a gate you wrote down beforehand or walk out. The aim is not the prettiest picture. It is to kill an unworkable architecture before you rent a supercomputer to decorate it.

A plausible pair does not guarantee an isolated cluster doublet. A plausible doublet does not guarantee the plaquette operator you wanted. A recognizable plaquette term does not guarantee a topological phase. A small patch with a close pair of eigenvalues does not prove topological degeneracy. Every later calculation eats artifacts from the earlier one: calibrated matrices, projectors, uncertainty ranges, effective couplings, leftover error. Enthusiasm is not an artifact.

40.1. Two spins on a page

We begin with a screening calculation, not with a claim about a particular material.

Represent each defect by a spin-1/2. Let X , Y , and Z be the dimensionless Pauli matrices, and let I be the two-dimensional identity. In units where $\hbar = 1$,

$$H_2 = \frac{\omega_1}{2} Z \otimes I + \frac{\omega_2}{2} I \otimes Z + J_x X \otimes X + J_y Y \otimes Y + J_z Z \otimes Z.$$

The coefficients $\omega_1, \omega_2, J_x, J_y, J_z$ must all use the same angular-frequency unit, such as radians per second. Then every eigenvalue of H_2 has that unit. Multiplying by $\hbar$ converts it to energy.

The following NumPy program writes the four-by-four matrix, asks for every eigenvalue, keeps the two lowest eigenvectors as columns of a projector P , calculates the leakage gap, projects three physical control operators, and checks the eigensolver leftover. It prints no invented benchmark values. You have to supply cited, condition-specific parameters.

```
import numpy as np

I = np.eye(2, dtype=complex)
X = np.array([[0, 1], [1, 0]], dtype=complex)
Y = np.array([[0, -1j], [1j, 0]], dtype=complex)
Z = np.diag([1, -1]).astype(complex)
kron = np.kron

def pair_spectrum(w1, w2, Jx, Jy, Jz):
    H = 0.5*w1*kron(Z, I) + 0.5*w2*kron(I, Z)
    H += Jx*kron(X, X) + Jy*kron(Y, Y) + Jz*kron(Z, Z)

    E, U = np.linalg.eigh(H)          # E is ascending
    P = U[:, :2]                     # candidate leftover basis
    Delta_leak = E[2] - E[1]

    physical_ops = {
        "Z1": kron(Z, I),
        "Z2": kron(I, Z),
        "X1": kron(X, I),
    }
    projected_ops = {
        name: P.conj().T @ op @ P
        for name, op in physical_ops.items()
    }

    scale = max(np.linalg.norm(H), 1.0)
    residual = np.linalg.norm(H @ U - U @ np.diag(E)) / scale
    return E, Delta_leak, projected_ops, residual
```

Writing the whole matrix and asking for every eigenpair is *exact diagonalization*. The name is worth keeping because nothing is thrown away except the digits the computer cannot hold. People shorten it to ED.

This program is a pipeline test, not a color-center model. Real defects may need a larger spin, orbital, hyperfine, strain, or charge basis. Replace the toy matrix with the cited microscopic Hamiltonian from Chapters 7–10 and 26.

The output already asks the right questions. Are the two lowest states a useful doublet? Is $\Delta_{\text{leak}} = E_2 - E_1$ large compared with intercluster coupling, drive bandwidth, disorder, and

decoherence expressed in the same units? Do the projected physical operators span useful controls on the encoded pair, or does projection turn them into multiples of the identity? A low pair of eigenvalues with no addressable transition is an elegant paperweight.

40.2. Freeze the numbers first

Before diagonalizing anything, create one machine-readable parameter record. It should contain:

- defect species, charge state, spin and retained orbitals;
- magnetic-field vector, strain tensor, electric field, and temperature;
- pair positions and crystal orientations;
- every Hamiltonian coefficient, unit, sign convention, source, sample conditions, and uncertainty interval;
- which interactions are measured, calculated, fitted, or assumed;
- the architecture’s predeclared kill criteria from Chapter 39.

Use one internal unit. If Hamiltonian coefficients are angular frequencies, convert every linewidth and inverse coherence time to angular frequency before comparison. If they are energies, convert temperature through $k_B T$. A comparison between megahertz and microelectronvolts without conversion is not multidisciplinary research. It is two columns standing near each other.

Stop rule 0. Stop the material-specific calculation if a term essential to the target interaction has neither a cited physical mechanism nor a bounded parameter range. An unconstrained fitting knob can reproduce nearly any tiny spectrum and predict none.

Hence the record is not paperwork. It is the Hamiltonian you are actually studying. Drop a coefficient’s provenance and later “agreement with experiment” has nothing to agree with.

40.3. Two defects

Use dense ED for the smallest basis. If the same small model later needs pulses, collapse operators, or correlation functions, a package built for that job is QuTiP [R249]. For a closed Hermitian matrix, NumPy, SciPy, or Julia’s standard linear algebra is enough.

Sweep the full fabrication uncertainty, not only a best-fit point. For each sample of position, orientation, field, and coupling, store:

1. the low-energy spectrum E_n ;

2. level splittings and avoided crossings;
3. eigenstate composition in the microscopic basis;
4. derivatives $\partial E_n / \partial p$ with respect to uncertain parameter p ;
5. matrix elements of physically available drives and readout operators;
6. leakage under a representative pulse or weak interpair coupling;
7. solver residuals and symmetry quantum numbers.

Convergence test. Enlarge the local basis — for example, add the nearest omitted orbital or hyperfine state — and compare the retained eigenvalues and projected operators. The relevant error is not just an energy shift. A stable spectrum paired with unstable control matrix elements is not converged.

Stop rule 1. Reject the pair design if no connected region of the cited uncertainty range supports the required ordering of states, or if the needed coupling exists only at an isolated fine-tuned point narrower than the fabrication and control tolerance. Also stop if the leakage gap fails the model-specific inequality set in Chapter 39.

What fails if you keep only the best-fit point? You have shown that a Hamiltonian exists in a set of measure zero. The architecture lives in a distribution.

40.4. The whole cluster

Let a cluster contain m defects, each with retained local dimension d . The raw Hilbert-space dimension is d^m . Use all exact symmetries to block-diagonalize it, but check that the perturbations you later add do not break those symmetries.

Dense ED returns every eigenpair and is ideal while the largest symmetry block comfortably fits memory. Sparse Lanczos or Davidson methods return a selected low-energy window once the full matrix is too large. A custom sparse implementation is justified when the Hamiltonian has repeated local terms and exact bit-level structure. In Julia, sparse matrices plus KrylovKit are a direct route; ITensors.jl becomes attractive when a matrix-product representation is natural [R250].

From the two lowest cluster eigenstates, build the rectangular projector

$$P = |\tilde{0}\rangle\langle 0| + |\tilde{1}\rangle\langle 1|,$$

understood as a map from a two-state leftover basis $\{|0\rangle, |1\rangle\}$ into the microscopic cluster space. For each microscopic operator O , compute

$$O_{\text{eff}} = P^\dagger O P.$$

Expand the resulting two-by-two matrix as

$$O_{\text{eff}} = c_0 I + c_x X + c_y Y + c_z Z,$$

with coefficients $c_\mu = \frac{1}{2} \text{Tr}(\sigma_\mu O_{\text{eff}})$, where $\sigma_\mu \in \{I, X, Y, Z\}$. This tells us what local fields, noise, readout, and intercluster couplings become inside the encoded subspace.

Compute the cluster splitting, leakage gap, projected control strengths, and projected noise operators across the uncertainty ensemble. Then couple two complete clusters microscopically and compare their exact low-energy spectrum with the predicted encoded interaction.

[Theory] A Schrieffer–Wolff expansion is controlled only when coupling between retained and discarded sectors is small relative to the relevant energy denominators; direct low-energy diagonalization provides the necessary finite-system benchmark [R025].

Convergence tests. Increase the number of retained cluster states; tighten sparse-eigensolver residuals; compare results with and without each nominally small microscopic term; and verify that phase choices within the doublet do not change basis-invariant conclusions.

Stop rule 2. Stop the cluster architecture if the doublet disappears over realistic disorder, if required controls project nearly to identity while noise projects strongly onto Pauli operators, or if exact two-cluster spectra cannot be reproduced within the predeclared effective-model error budget.

A leftover is certified by isolation, usable projected controls, tolerable projected noise, and agreement between exact coupled-cluster spectra and the effective model. A pretty doublet in an isolated cell is not that list.

40.5. Three to six clusters make a plaquette

This is the decisive gadget test and is still an ED problem. If each cluster is first truncated to a leftover, six clusters span only 2^6 states. Run two versions:

- the **effective calculation**, using one leftover per cluster;
- the **leakage-aware calculation**, retaining additional cluster states and the microscopic intercluster couplings.

Let n be the number of clusters around the candidate plaquette. Project the leakage-aware Hamiltonian into the encoded space and expand it in all Pauli strings,

$$H_{\text{fit}} = \sum_{\alpha} c_{\alpha} \sigma_{\alpha_1} \otimes \cdots \otimes \sigma_{\alpha_n},$$

where each α_j is 0, x , y , or z , and $\sigma_0 = I$. Because Pauli strings are orthogonal under the trace inner product,

$$c_{\alpha} = 2^{-n} \text{Tr}(\sigma_{\alpha}^{\dagger} H_{\text{fit}}).$$

This calculation does not merely ask whether the target plaquette coefficient K is nonzero. It reports every unwanted one-body, two-body, and higher-body coefficient generated at the same time. Compare the full exact low-energy spectrum and eigenvectors with the fitted operator, not just K .

Sweep the intercluster scale λ . The useful window must satisfy two conditions simultaneously: the target term is large enough to beat the relevant noise and temperature criteria, and the effective description remains accurate. If perturbation theory predicts the plaquette at a particular order in λ , fit the scaling only inside a range where exact and effective spectra agree; do not select the fitting interval after seeing the desired exponent [R025].

Calculate spectrum, plaquette gap, target coefficient, norm of unwanted terms, leakage weight, and response to each physically allowed local perturbation. QuTiP remains useful for small driven or open plaquettes [R249]. A sparse ED implementation is usually simpler for closed spectra. Tensor-network methods add approximation without buying useful size here.

Stop rule 3. Abandon this gadget if no parameter interval simultaneously passes the hierarchy, leakage, disorder, and decoherence gates; if cancellation of unwanted terms requires unattainable precision; or if the target operator appears only after projecting away states that strongly mix in the exact calculation.

Hence a nonzero K is not the gate. The gate is K together with every competitor, in a window where leakage-aware ED still agrees.

40.6. A small patch

Start with the least ambitious topological target capable of testing the interaction architecture — typically an Abelian stabilizer or gauge-theory proxy before a doubled-Fibonacci claim. The purpose is to validate emergence from the proposed couplings, not to win the final noun on the first run.

Use ED while the symmetry-reduced Hilbert space and required Krylov vectors fit memory. For N encoded spins, one complex state vector has 2^N entries. If each entry is stored as a 16-byte complex number, its storage is 16×2^N bytes; an iterative solver needs several such vectors plus operator workspace. Do this dimensional estimate before choosing ED.

For long strips and cylinders, write the state as a chain of tensors with a bond that remembers the cut. Growing that bond until the leftover weight stops moving is the *density-matrix renormalization group*, or DMRG, in its matrix-product-state form. ITensor and

TeNPy implement it [R035]; [R250]; [R251]. DMRG is strongest when entanglement across the one-dimensional ordering remains manageable. A two-dimensional lattice mapped onto a cylinder becomes harder as circumference increases; one narrow cylinder is not a two-dimensional thermodynamic limit [R035]; [R252].

For genuinely two-dimensional patches beyond ED, use finite or infinite *projected entangled-pair states* — PEPS — when the target state admits a tractable bond dimension and contractions can be converged [R252]; [R253]. In Julia, TensorKit.jl with MPSKit.jl or PEPSKit.jl provides symmetry-aware tensor infrastructure; ITensors.jl is a mature choice for matrix-product calculations [R250]. Report package versions and contraction algorithms. “Used tensor networks” is not a reproducible method.

Quantum Monte Carlo belongs only where weights are nonnegative or a demonstrated sign-problem treatment applies. Frustrated, complex, or non-Abelian effective interactions commonly remove that advantage. Do not recommend Monte Carlo merely because the lattice has many sites.

Method names do not allocate entanglement capacity. A wide two-dimensional cluster with high entanglement may be worse for DMRG than a smaller ED calculation.

40.7. What to measure, in order

Compute observables in an order that prevents advanced diagnostics from laundering a failed spectrum.

Spectrum, gap, and degeneracy. For each size and boundary condition, obtain several lowest states. Define Δ as the separation between the candidate ground-state manifold and the first excitation, and δ as the splitting within that manifold. Track both with linear size. A small δ at one size may be accidental.

Local indistinguishability and perturbation response. Compare local expectation values and reduced density matrices across candidate ground states. Add each realistic local perturbation separately and in sampled combinations. Track Δ , δ , sector mixing, leakage, and observable drift. The phase claim fails if a generic weak local term distinguishes or mixes the supposed sectors at a scale incompatible with the architecture’s kill criteria.

Entanglement entropy. For a region A , compute

$$S(A) = -\text{Tr}[\rho_A \log \rho_A],$$

where ρ_A is its reduced density matrix. Converge the Schmidt spectrum, not only the reported scalar entropy.

Topological entanglement entropy. Use Kitaev–Preskill or Levin–Wen subtraction geometries to cancel boundary contributions [R254]; [R255]. Regions must be large relative to the correlation length and small relative to the whole system. On a tiny patch, the subtraction is a finite-size diagnostic, not a measured universal constant.

Wilson loops. Define the loop operator from the model’s actual gauge or string algebra. Measure contractible loops of several shapes and noncontractible loops when the boundary permits them. Check scaling with perimeter or enclosed area and compare sectors. One loop expectation value is insufficient.

Fusion. Create separated candidate excitations with open strings, combine them through alternative paths, and project onto charge sectors. Record allowed channels, state counts, and path dependence. On a small patch this tests the intended operator algebra; it does not establish asymptotic quasiparticles.

Modular S and T . Attempt these only after obtaining a resolved quasi-degenerate ground manifold on a torus or equivalent geometry. Construct minimally entangled states, implement or infer the relevant modular transformations, fix arbitrary ground-state phases, and bootstrap uncertainty from numerical truncation. Ground-state entanglement can in principle reveal quasiparticle statistics [R256], but a poorly resolved manifold makes S and T gauge-dependent numerology.

What fails if you extract modular data from accidental degeneracy? Any nearly degenerate finite matrix admits arbitrary basis rotations. The matrices then depend on a phase convention you have not earned.

40.8. Every method has knobs

Every reported point needs more than one knob:

Method	Must vary	Must report	Invalid use
Dense/sparse ED	basis truncation, eigensolver tolerance, symmetry sectors	residuals, number of eigenpairs, omitted levels	claiming a phase from one size
MPS-DMRG	bond dimension, sweeps, initial states, cylinder length/circumference	energy variance, discarded weight, sector, observable drift	treating an unconverged wide cylinder as exact
PEPS/iPEPS	bond dimension, environment dimension, optimizer starts, unit cell	energy/observable convergence and contraction error	reporting one bond dimension
Open-system QuTiP	local basis, time step/solver tolerance, collapse operators	trace preservation, positivity checks, parameter provenance	scaling to a many-body patch by dense matrices

Method	Must vary	Must report	Invalid use
Monte Carlo	size, temperature, sampling length, autocorrelation	sign, effective sample size, finite-size scaling	ignoring a sign problem

For DMRG, use multiple initial states and explicit sector constraints; a variational optimizer may repeatedly find the easiest sector rather than the full ground manifold. For PEPS, vary both state bond dimension and contraction environment dimension. For every method, compare at least one overlapping size against ED. Agreement in the overlap is the handoff certificate.

One PEPS run is not convergence. Bond dimension, environment dimension, unit cell, and initial state all need variation [R252]; [R253]. Low energy alone does not certify the correct sector.

40.9. The run book

The following run book is executable at the level a research repository should implement. It specifies calculations and data products. It asserts no numerical outcome.

Run A — pair audit. Load the parameter record; construct the full two-defect matrix; diagonalize it densely; export sorted eigenvalues, eigenvectors, operator matrix elements, residuals, and parameter derivatives. Repeat over the fabrication-uncertainty ensemble. Cross-check the toy-to-full-model transition by adding omitted local levels.

Decision A. Pass only if a connected robust parameter region survives Stop rule 1. Freeze the retained microscopic basis; do not retune it later to improve the plaquette.

Run B — cluster projection. Assemble the complete intra-cluster Hamiltonian. Compute enough low states to include the candidate doublet and leakage multiplets. Export P , $P^\dagger O_i P$ for every drive, noise, readout, and coupling operator O_i , plus uncertainty samples. Couple two clusters and export the exact-versus-effective spectral error.

Decision B. Pass only if Stop rule 2 is avoided and the error fits the predeclared projection budget.

Run C — plaquette decomposition. Build three-, four-, five-, and six-cluster candidates only when their geometry is physically placeable. Diagonalize both encoded and leakage-aware Hamiltonians. Export all Pauli-string coefficients, low spectra, leakage weights, and perturbation sweeps. Rank designs by a vector of target strength, unwanted-term norm, robustness, and fabrication cost; do not hide tradeoffs in one hand-tuned score.

Decision C. Choose at most one primary and one fallback plaquette. If none passes Stop rule 3, terminate the topological-patch branch and report the best lower-level result.

Run D — ED patch. Tile the chosen plaquette into the smallest open patch, cylinder, and periodic cluster that each answer a distinct question. Export the low-energy manifold, Δ , δ , local reduced states, Wilson loops, excitation and fusion tests, and perturbation response. Repeat for every ED-accessible size and boundary.

Decision D. Continue only if the trends are compatible with a stable gap, decreasing sector splitting, and local indistinguishability. A single favorable size is “unresolved,” not “pass.”

Run E — size extension. If geometry is cylindrical, reproduce the largest ED point with ITensor or TeNPy DMRG, then increase length, circumference, and bond dimension [R035]; [R250]; [R251]. If genuinely two-dimensional scaling is essential, reproduce an ED point with finite PEPS before enlarging the patch [R252]; [R253]. Compute entanglement and Wilson diagnostics first. Attempt modular matrices only after the ground-manifold prerequisites are met [R256].

Decision E. Label the result *positive*, *negative*, or *numerically unresolved*. Failure to converge is not evidence for the phase and not evidence against it. It is evidence that this calculation did not decide.

Preserve inputs, package versions, random seeds, symmetry sectors, solver logs, and immutable outputs. Label direct microscopic ED, projected ED, and tensor-network approximations separately.

Move a stop rule after seeing results and you have renamed calibration as goalpost relocation. Thresholds may be revised when new experimental evidence changes the input contract. They may not be revised merely because the favorite model missed them.

40.10. What the laboratory actually has

The simulation begins from laboratory-conditioned parameters, not symbols selected for convenience. Defect positions are distributed, orientations may be discrete, charge states can change, couplings have sample conditions, and coherence data depend on temperature and pulse sequence.

[Experiment] supplies distributions and bounds: spectra, linewidths, placement statistics, coupling measurements, control matrix elements, and noise correlations.

[Theory] supplies the microscopic Hamiltonian, projection, perturbative expansion, and ideal phase diagnostics.

[Numerics] supplies finite-system spectra, fitted effective operators, wavefunctions, response curves, and controlled extrapolations.

A numerical phase in a Hamiltonian outside experimental uncertainty is a result about that Hamiltonian. It is not evidence that the defect platform realizes the phase. Conversely, a

failed toy Hamiltonian does not rule out every defect architecture; it rules out the declared model and parameter domain. Scope is part of the result.

Before any open-system simulation, first prove that the closed-system Hamiltonian has the desired low-energy structure. Lindblad evolution of a model with no topological sector can calculate decay very accurately. It cannot decay into topology.

40.11. Ways to get this wrong

1. **Starting with the largest patch.** This hides whether failure came from the defect model, cluster truncation, gadget, boundary, or solver. The staged overlaps are not bureaucracy; they locate the broken arrow.
2. **Using DMRG because the system is “large.”** DMRG is a matrix-product variational method, strongest for one-dimensional and quasi-one-dimensional geometries [R035]. A wide two-dimensional cluster with high entanglement may be worse for DMRG than a smaller ED calculation. Method names do not allocate entanglement capacity.
3. **Calling one PEPS run convergence.** PEPS optimization and contraction are approximate. Bond dimension, environment dimension, unit cell, and initial state all need variation [R252]; [R253]. Low energy alone does not certify the correct sector.
4. **Measuring topological entropy too early.** The subtraction formulas assume a scale separation among lattice spacing, correlation length, region size, and system size [R254]; [R255]. A tiny-patch value can be suggestive but not asymptotic evidence.
5. **Extracting modular data from accidental degeneracy.** Any nearly degenerate finite matrix admits arbitrary basis rotations. Modular matrices become meaningful only after the candidate ground manifold, topology, symmetry action, and phase conventions are controlled [R256].
6. **Confusing fitted and emergent interactions.** If a target plaquette term is manually inserted, the simulation tests that term. If it is derived from two-body defect couplings and validated against leakage-aware ED, the simulation tests emergence. Plot these as different models.
7. **Discarding negative uncertainty samples.** A design that works only at nominal couplings but fails across measured placement or strain distributions has failed its engineering objective. Sampling uncertainty is part of the Hamiltonian study, not cosmetic error bars.
8. **Moving stop rules after seeing results.** Thresholds may be revised when new experimental evidence changes the input contract. They may not be revised merely

because the favorite model missed them. That maneuver is called calibration when done before the run and goalpost relocation when done after it.

40.12. Check yourself

1. Why begin with two defects rather than a topological patch?

Because the pair cheaply tests the microscopic interaction, basis, and uncertainty range on which every later projection depends.

2. Show that Pauli-string coefficients on n leftovers are $c_{\alpha} = 2^{-n} \text{Tr}(\sigma_{\alpha}^{\dagger} H_{\text{fit}})$.

Pauli strings are orthogonal under the trace inner product, and $\text{Tr}(I) = 2^n$. The factor 2^{-n} is the inner-product normalization, not a fitting choice. The expansion exists to reveal unwanted terms, not to report only K .

3. What fails if you start with the largest patch?

Failure cannot be located. The defect model, cluster truncation, gadget, boundary, and solver are then one lump. The staged overlaps exist to name the broken arrow.

4. Show that one complex state vector on N encoded spins occupies 16×2^N bytes if each entry is 16 bytes, and say what that decides.

There are 2^N amplitudes. Storage is 16×2^N bytes before the solver's extra vectors. Do this estimate before choosing ED.

5. When is DMRG appropriate?

For one-dimensional or quasi-one-dimensional matrix-product-friendly geometries, with convergence in bond dimension, sweeps, sector, and cylinder dimensions.

6. What fails if modular S and T are extracted from accidental degeneracy?

The matrices become gauge-dependent numerology. A resolved quasi-degenerate ground manifold on suitable geometry, controlled truncation, minimally entangled states, and fixed basis phases must come first.

You now have a versioned parameter record, a pair spectrum, a projector you can defend, a plaquette you can refuse, and three allowed answers at every door. The last gate you actually passed is the result. The most ambitious calculation you attempted is not.

40.13. Sources

- [R249] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP: An open-source Python framework for the dynamics of open quantum systems,” *Computer Physics Communications* **183**, 1760–1772 (2012). DOI: 10.1016/j.cpc.2012.02.021.

- [R035] U. Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics* **326**, 96–192 (2011). DOI: 10.1016/j.aop.2010.09.012; arXiv: 1008.3477.
- [R250] M. Fishman, S. R. White, and E. M. Stoudenmire, “The ITensor Software Library for Tensor Network Calculations,” *SciPost Physics Codebases* **4** (2022). DOI: 10.21468/SciPostPhysCodeb.4; arXiv: 2007.14822.
- [R251] J. Hauschild and F. Pollmann, “Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy),” *SciPost Physics Lecture Notes* **5** (2018). DOI: 10.21468/SciPostPhysLectNotes.5; arXiv: 1805.00055.
- [R252] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete, “Matrix product states and projected entangled pair states: Concepts, symmetries, theorems,” *Reviews of Modern Physics* **93**, 045003 (2021). DOI: 10.1103/RevModPhys.93.045003; arXiv: 2011.12127.
- [R253] J. Jordan, R. Orús, G. Vidal, F. Verstraete, and J. I. Cirac, “Classical simulation of infinite-size quantum lattice systems in two spatial dimensions,” *Physical Review Letters* **101**, 250602 (2008). DOI: 10.1103/PhysRevLett.101.250602; arXiv: cond-mat/0703788.
- [R025] S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* **326**, 2793–2826 (2011). DOI: 10.1016/j.aop.2011.06.004; arXiv: 1105.0675.
- [R254] A. Kitaev and J. Preskill, “Topological entanglement entropy,” *Physical Review Letters* **96**, 110404 (2006). DOI: 10.1103/PhysRevLett.96.110404; arXiv: hep-th/0510092.
- [R255] M. Levin and X.-G. Wen, “Detecting topological order in a ground state wave function,” *Physical Review Letters* **96**, 110405 (2006). DOI: 10.1103/PhysRevLett.96.110405; arXiv: cond-mat/0510613.
- [R256] Y. Zhang, T. Grover, A. Turner, M. Oshikawa, and A. Vishwanath, “Quasi-particle statistics and braiding from ground-state entanglement,” *Physical Review B* **85**, 235151 (2012). DOI: 10.1103/PhysRevB.85.235151; arXiv: 1111.2342.

Chapter 41. The grade

Here is the grade. No ceremony.

The full idea — a crystal packed with defect clusters that become a doubled-Fibonacci phase and quietly protect a logical qubit — is **speculation**. Coherent speculation: nothing in it breaks a known law. It is not **established**. It is not a **straightforward extension** of a device anyone already builds. It is not yet a **credible proposal**, the kind of difficult program you fund because the missing pieces are listed and the first measurement can say no. The interaction that would have to exist and the geometry that would have to be made have not been shown together. It is also not **incompatible**. Incompatible would be cleaner. You could close the notebook.

The nearest honest job is smaller than the dream. Four triangles. Twelve spins. One question: can the pair interactions we already know how to write generate a useful four-body term, or do the ordinary pair terms simply win?

41.1. The first link that is not automatic

The architecture is a chain of arrows:

```
defects -> cluster doublet -> interacting encoded spins
        -> many-body projector -> topological phase -> useful logical qubit
```

Testing the last arrow first is expensive and muddy. A failed large device can blame fabrication, initialization, control, readout, or the phase itself. The better experiment is a tensile test: isolate the first nontrivial link that ordinary pair interactions do not automatically provide.

A three-defect cluster can form a two-state low-energy manifold in a controlled model. That result is interesting. It is not topological. Projecting a physical pair interaction into two cluster doublets normally produces another pair interaction. A toric-code stabilizer needs a product acting on four encoded spins. A doubled-Fibonacci plaquette is more demanding still, because its amplitudes must implement particular recoupling data [R175]; [R018]; [R030].

So the first paper asks a smaller question: is a four-cluster plaquette term generated at a useful scale, with a tolerable list of unwanted operators?

This is deliberately not a Fibonacci paper. A negative answer would prevent years of polishing a Hamiltonian that never existed. A positive answer would earn — not assume — the right to study a patch.

Drop that order and a failed headline cannot say which arrow broke.

41.2. Four triangles, twelve spins

Use negatively charged nitrogen-vacancy centers, NV^- , in diamond, because their spin Hamiltonian and coherent control are unusually well characterized [R074]. Make four nominal clusters, labeled $C = 1, 2, 3, 4$. Each cluster contains three same-orientation NV electron spins arranged as a triangle. Keep the physical levels $|0\rangle = |m_s = 0\rangle$ and $|1\rangle = |m_s = -1\rangle$. The omitted $m_s = +1$ level remains a leakage state.

For a symmetric, resonantly dressed triangle, define Pauli operators τ_i^α on physical defect i , where $\alpha \in \{x, y, z\}$. A useful ideal cluster Hamiltonian is

$$\frac{H_C}{h} = \frac{c}{4} \sum_{i < j \in C} (\tau_i^z \tau_j^z - \tau_i^x \tau_j^x - \tau_i^y \tau_j^y), \quad (41.1)$$

where h is Planck's constant and c is the intracluster dipolar frequency. Its low states are

$$|W_1\rangle = \frac{|100\rangle + |010\rangle + |001\rangle}{\sqrt{3}}, \quad |W_2\rangle = \frac{|011\rangle + |101\rangle + |110\rangle}{\sqrt{3}},$$

separated from the nearest ideal cluster states by $\Delta_C/h = 3c/2$. Call this pair an *encoded leftover*, with encoded Pauli operators X_C, Y_C, Z_C . The two states are a qubit written on three defects, not an anyon.

Place the four triangles on a loop and include every physical dipolar bond, not only the bonds a diagram finds agreeable. Twelve two-level defects give dimension $2^{12} = 4096$, small enough for exact diagonalization. A full spin-1 cross-check has dimension $3^{12} = 531,441$, still reachable with sparse methods for a few low-energy states. These dimensions are exact counts, not performance forecasts.

The target observable is one encoded weight-four coefficient, for example

$$K_4 Z_1 Z_2 Z_3 Z_4,$$

where each Z_C acts on one cluster doublet and K_4 has units of frequency. A single weight-four term does not establish topological order. It establishes only that the proposed microscopic mechanism can cross the pairwise-to-many-body bottleneck.

Hence four encoded spins are the smallest system that can expose a weight-four coefficient and every lower-weight competitor. Three clusters cannot ask the question.

41.3. What the first paper may assume

A defensible title is:

Can dipolar NV triangles generate a useful encoded plaquette interaction? A four-cluster microscopic benchmark.

The paper should report success or failure with equal enthusiasm. It should not contain “Fibonacci phase” in the title, because neither a branching constraint nor a Fibonacci plaquette has yet been derived.

Declare the assumptions before anyone fits a number.

1. **Host and charge state.** Use same-axis NV^- electron spins in isotopically enriched diamond. Include the measured zero-field, Zeeman, strain/electric, hyperfine, drive, and magnetic dipole terms summarized by the established NV Hamiltonian [R074].
2. **Geometry.** Use nominal equilateral three-NV clusters with intracluster side r_C and a four-cluster loop with separation R . Treat both as variables and sample position errors. Do not present $r_C = 5$ nm as fabricated fact.
3. **No borrowed exchange.** Set long-range, lithographically tunable NV–NV exchange to zero unless a separate microscopic calculation or measurement supplies it. Dipolar coupling and microwave control are the supported resources. Entanglement through direct dipolar coupling between two NV electron spins has been demonstrated [**Experiment**] [R080].
4. **Driven status.** Equation (41.1) is a rotating-frame model sustained by a microwave reference. It is an analog driven emulator, not a passive equilibrium ground-state Hamiltonian.
5. **Nuclei and disorder.** Either include the nitrogen and selected ^{13}C nuclear spins explicitly or fix their states and state the preparation assumption. Sample detuning, strain, angular, amplitude, and position disorder separately so their effects remain diagnosable.
6. **No post-selection of the answer.** Fix the cluster basis from isolated-cluster spectroscopy. If local basis rotations are optimized, constrain them to calibrated single-cluster controls and report the unrotated ledger too.
7. **Two protocols, two labels.** Analyze (a) continuous static-in-the-rotating-frame couplings and (b) a predeclared periodic refocusing sequence. Call the second result Floquet engineering, not passive protection.

Fabrication evidence does not yet satisfy these assumptions simultaneously. Nanoaperture work localized implanted NV clusters on roughly the 10-nm scale and reported separations down to about 40 nm [**Experiment**] [R257]. A deterministic single-ion source later demonstrated a 5×5 pattern at $2\text{ }\mu\text{m}$ pitch, with 121(35) nm lateral resolution and about 0.6% NV conversion in that shallow-implantation experiment [**Experiment**] [R258]. These are valuable but different achievements: localization, exact ion count, conversion,

charge stability, orientation, and coherent yield have not been demonstrated together for repeated 5-nm triangles.

What fails if a different encoded basis is chosen for every disorder sample? A large-looking target coefficient that no fixed control realizes. That is optimization of the conclusion.

41.4. The Hamiltonian you actually write

The calculation should begin from physical spin-1 operators $\mathbf{S}_i$, not from the hoped-for stabilizer. Write

$$H(t) = \sum_i \left[D_i(S_i^z)^2 + g_e \mu_B \mathbf{B}_i \cdot \mathbf{S}_i + H_{\text{hf},i} + H_{\text{strain},i} + H_{\text{drive},i}(t) \right] + \sum_{i < j} \frac{\mu_0 (g_e \mu_B)^2}{4\pi r_{ij}^3} [\mathbf{S}_i \cdot \mathbf{S}_j - 3(\mathbf{S}_i \cdot \hat{\mathbf{r}}_{ij})(\mathbf{S}_j \cdot \hat{\mathbf{r}}_{ij})]. \quad (41.2)$$

Here D_i is the zero-field splitting energy, g_e is the electron g factor, μ_B is the Bohr magneton, μ_0 is vacuum permeability, $\mathbf{r}_{ij} = r_{ij} \hat{\mathbf{r}}_{ij}$ joins defects i and j , and the three named H terms describe hyperfine, strain/electric, and microwave interactions. Every term has units of joules. The NV^- ground-state value $D/h \approx 2.87$ GHz and its spin Hamiltonian are experimentally established [R074].

After transforming to the declared rotating frame and checking the rotating-wave approximation against the full model, project onto the four cluster doublets. Let P be that 2^4 -dimensional projector and $Q = I - P$. A Schrieffer–Wolff calculation yields an effective Hamiltonian, while exact low-energy eigenvectors provide a nonperturbative comparison. Expand the result in the complete encoded Pauli basis:

$$\frac{H_{\text{eff}}}{h} = \sum_{P_a \in \{I, X, Y, Z\}^{\otimes 4}} k_a P_a, \quad k_a = \frac{1}{16} \text{Tr} \left(P_a \frac{H_{\text{eff}}}{h} \right). \quad (41.3)$$

The coefficient k_{ZZZZ} is K_4 . All other k_a are competitors, not rounding errors. Perturbative gadgets can generate many-body interactions from two-body resources, but desired terms arrive with energy denominators and with unwanted corrections [**Theory**] [R175].

An illustrative scale check shows why the ledger matters. Electron dipoles have a frequency scale near $52 \text{ MHz}(1 \text{ nm}/r)^3$. At the proposed, not demonstrated, values $r_C = 5 \text{ nm}$ and $R = 10 \text{ nm}$, this gives $c \approx 0.416 \text{ MHz}$, $\Delta_C/h \approx 0.624 \text{ MHz}$, and an unprojected inter-cluster scale $u \lesssim 52 \text{ kHz}$ before geometry factors. Those bare coefficients, about 416 kHz inside a triangle and 52 kHz between triangles, are inputs. They are not the much smaller plaquette scale generated later. If the first surviving connected ring process is fourth order, dimensional analysis gives

$$K_4 \sim \frac{u^4}{(\Delta_C/h)^3} \sim 30 \text{ Hz}. \quad (41.4)$$

The units are $\text{Hz}^4/\text{Hz}^3 = \text{Hz}$. The coefficient and even the leading order depend on symmetry and pulse design, so 30 Hz is **[Proposal]** scale analysis, not a prediction. Its temperature equivalent is $\hbar K_4/k_B \approx 1.4 \text{ nK}$, where k_B is Boltzmann's constant. A passive phase at that scale would be implausible. A driven spectroscopic signal might still be measurable if coherence, calibration, and averaging allow it. The computation decides whether lower-order constructive routes exist or whether larger pair terms simply win.

No demonstrated process repeatedly produces charge-stable, coherent 5-nm NV triangles with acceptable compound yield. Linewidth, disorder, temperature, and perturbative suppression have to be carried together. The geometry and distributions above remain explicit study parameters. They sit inside the book's screening ranges. They are not demonstrated fabrication resources.

Hence $K_4 \neq 0$ is a mechanism check. It is not a phase.

41.5. Six calculations, then stop if you must

The paper can be completed without fabricating twelve precisely placed NV centers.

1. **Calibrate one cluster.** Diagonalize the full 3^3 spin-1 Hamiltonian over magnetic field, drive, strain, hyperfine state, and triangular distortion. Record the doublet splitting, gap Δ_C , leakage matrix elements, and projected single-spin operators.
2. **Audit two clusters.** Diagonalize six spins and compare exact low energies with first- and second-order Schrieffer–Wolff results. This fixes the regime where an encoded Hamiltonian is controlled and reveals the dominant one- and two-cluster Pauli terms.
3. **Fit four clusters.** Use exact diagonalization of the 4096-state physical-qubit model, with selected full-spin-1 cross-checks. Construct H_{eff} by a specified quasi-degenerate projection, expand all 256 encoded Pauli strings using (41.3), and publish the full coefficient table.
4. **Test the drive honestly.** For a periodic sequence of duration T_d , compute the one-period propagator $U(T_d)$ and its quasienergies. Track micromotion, leakage, sensitivity to pulse errors, and the branch choice in $H_F = (i\hbar/T_d) \log U(T_d)$. Do not interpret a quasienergy gap as a thermal gap.
5. **Sample fabrication distributions.** Vary each defect position, orientation, detuning, strain, and drive amplitude. Report medians, tails, failure fractions, and correlations rather than one immaculate geometry.

6. **Attempt a patch only after the gate passes.** If a dominant, robust K_4 exists, place the fitted operators on the smallest periodic stabilizer patch and compute spectrum, low-energy degeneracy, Wilson loops, and response to local perturbations. If the isolated plaquette fails, stop. A topological diagnostic cannot rescue a missing interaction.

Define

$$R_4 = \frac{|K_4|}{\sum_{a \in \mathcal{U}} |k_a|},$$

where $\mathcal{U}$ is the predeclared set of symmetry-allowed, noncommuting unwanted terms after removing calibratable one-cluster fields. Also define ϵ_{spec} as the maximum error of the effective model's sixteen low energies, divided by $|hK_4|$, and p_{leak} as the largest Q -space weight among those states.

A **minimum theoretical success** is: $R_4 > 1$, $\epsilon_{\text{spec}} < 0.1$, $p_{\text{leak}} < 0.01$, and the inequalities remain true across a declared fabrication distribution. A **promising result** is $R_4 > 10$ plus $|K_4|$ larger than independently measured logical decoherence and quasistatic disorder rates. The values 1, 0.1, 0.01, and 10 are project decision thresholds, not universal constants.

Stop or redesign when any of the following occurs:

- no isolated cluster doublet survives realistic hyperfine and geometric disorder;
- eliminating leakage requires u/Δ_C so small that K_4 falls below relevant linewidths;
- lower-weight noncommuting terms remain larger than K_4 after physically allowed calibration;
- a refocusing sequence works only with pulse errors below demonstrated control capability;
- the fitted patch lacks a stable gap or the expected topological-sector structure;
- success depends on exchange, mediators, placement, or cooling not included in the declared platform.

The numerical paper is difficult but tractable. The Hilbert spaces are modest. The real work is gauge-consistent low-energy projection, disorder sampling, and refusing post hoc cancellations. A two-person theory effort could reproduce it from public Hamiltonian parameters without a new fabrication line.

The physical twelve-defect demonstration is substantially harder. It combines nanometre geometry, same-axis and charge-state yield, nuclear-state preparation, individual or patterned microwave phases, spectral discrimination, and long enough coherence. Existing experiments establish important pieces, not their conjunction [R074]; [R080]; [R257]; [R258]. A doubled-Fibonacci material would add correct vertex constraints, conditional recoupling amplitudes, a many-body gap, and phase diagnostics [R018]. That is not the next experiment.

If the four-cluster ledger fails, identify whether geometry, a mediator, or another host changes the scaling before adding system size. If it passes, the physical sequence is one triangle for spectroscopy, then two coupled triangles, and only then the twelve-defect plaquette. A larger topological patch is justified only after the desired term dominates. A Fibonacci claim waits until branching constraints, recoupling amplitudes, a stable gap, and non-Abelian diagnostics are all present.

A Floquet quasienergy gap depends on a clock, drive stability, and continued power. It does not thermalize like a static ground-state gap. Call it passive protection and the claim has changed.

41.6. What the laboratory actually has

Claim	Status	Platform	Evidence	Reference
Single NV spin Hamiltonian, initialization, readout, and microwave control are understood	experimentally demonstrated	Diamond NV ⁻	Extensive spectroscopy and control literature	[R074]
Two separated NV electron spins can be entangled through their interaction	experimentally demonstrated	Diamond NV ⁻	Room-temperature two-spin entanglement experiment	[R080]
Nanometre-localized implanted NV clusters exist	experimentally partial	Diamond NV ⁻	Nanoaperture localization; not repeated deterministic 5-nm triangles	[R257]
Joint exact ion delivery, NV conversion, and nanoscale coherent-array yield	experimentally partial	Diamond NV ⁻	Individual fabrication ingredients exist, but the deterministic-delivery experiment retained coarse focus and low conversion under its stated conditions	[R257]; [R258]

Claim	Status	Platform	Evidence	Reference
Two-body interactions can perturbatively produce encoded many-body terms	theoretically established	Abstract spin gadgets	Controlled expansions with denominators and error bounds	[R175]
The required Fibonacci string-net projectors define doubled-Fibonacci order	theoretically established	Ideal lattice model	Exact commuting-projector construction	[R018]
Digital processors can prepare and braid Fibonacci-model excitations	experimentally demonstrated	Superconducting processors	State preparation, fusion, and digital braiding simulate the target model; they do not establish an autonomous material phase	[R138]; [R165]
A defect lattice could host genuine Fibonacci quasiparticles	speculative	Diamond, sapphire, SiC, others	The cited literature establishes defect qubits, ideal models, and digital simulations, but no defect-material realization	[R074]; [R018]; [R138]; [R165]
Ruby supplies some control ingredients relevant to defect clusters	experimentally partial	$\text{Cr}^{3+}:\text{Al}_2\text{O}_3$	Ensemble spin-cavity coupling has been measured; deterministic single-ion arrays have not	[R089]

41.7. Ways to get this wrong

1. **Optimizing away the conclusion.** Choosing a different encoded basis for every disorder sample can manufacture a large-looking target coefficient that no fixed control realizes.

2. **Calling $K_4 \neq 0$ a phase.** One local term is a mechanism check. Topological order requires a thermodynamic phase or convincing finite-size scaling with the right degeneracy and nonlocal diagnostics.
3. **Calling Floquet quasienergy passive protection.** A driven frame depends on a clock, drive stability, and continued power. It does not thermalize like a static ground-state gap.
4. **Comparing placement records from incompatible processes.** Beam focus, implantation straggle, conversion yield, charge yield, and coherent yield are separate columns.
5. **Using digital Fibonacci demonstrations as material evidence.** Experiments have digitally prepared and braided Fibonacci-model anyons [Experiment] [R138]; [R165]. They establish control protocols and model behavior, not native Fibonacci quasiparticles in the processor substrate.
6. **Hiding the denominator.** A high-order gadget can be mathematically correct and physically useless because its desired coefficient shrinks faster than decoherence and disorder.

41.8. Fifteen questions

1. Can multiple defects act as one effective qubit? Yes, in theory and in controlled few-spin encodings, provided a two-dimensional manifold is spectrally isolated and controls and readout act predictably within it. For the specific three-NV triangle, the ideal driven model supplies a doublet; fabrication and disorder robustness remain unproved. **Verdict: plausible, platform-specific, not yet an array fact.**

2. Can multiple effective defect qubits emulate anyons? Yes. Gates or driven interactions can reproduce string operators, fusion measurements, and braiding statistics. That is an emulator whose underlying errors remain those of the defect controls.

3. Can defect systems digitally simulate Fibonacci anyons? In principle, yes if they provide a universal enough gate set, initialization, and readout. Superconducting processors have already demonstrated digital Fibonacci-model braiding [R138]; [R165], so no known principle excludes a defect processor. The missing evidence is competitive defect-array scale and fidelity, not the algorithm.

4. Could a static or driven defect Hamiltonian genuinely possess Fibonacci topological order? Known physics permits it, but the cited defect interactions do not derive it. A static realization needs the correct low-energy projectors and a stable gap; a driven realization additionally needs Floquet stability and controlled heating. **Current status: speculative.**

5. What is the difference between simulating an anyon and having an emergent anyon? Simulation programs physical qubits to reproduce selected anyon observables. Emergence means the system's own many-body low-energy excitations carry the fusion

and braiding structure robustly throughout a phase. Exact output from a circuit can still lack autonomous Hamiltonian protection.

6. Would clustering defects provide meaningful additional protection? Sometimes. A symmetric encoding can reject a common-mode field or reduce a chosen matrix element, but it can amplify gradients, add leakage, and multiply fabrication channels. Protection must be computed from projected noise operators, not inferred from defect count.

7. Is diamond actually the best substrate? Not categorically. Diamond currently offers the strongest evidence package for optically addressable individual color-center spins, but it is difficult to place and activate dense, identical interacting arrays. “Best” depends on whether the limiting metric is coherence, deterministic fabrication, coupling, optical networking, or passive interaction strength.

8. Is corundum/sapphire competitive? As a low-loss host with mature ensemble spin spectroscopy, potentially; as the specific dense, individually controlled defect-cluster platform studied here, the evidence is weaker. The 2025 ruby result demonstrates Cr^{3+} ensemble–cavity physics, not deterministic single-defect plaquettes [R089]. Sapphire should remain a comparison arm, not be promoted by analogy.

9. What physical interaction is the biggest bottleneck? A strong, controllable, correctly signed **many-body encoded interaction** that dominates ordinary one- and two-body terms. Dipolar coupling exists; the missing step is converting it into the required projector without suppressing the target below noise.

10. What fabrication capability is the biggest bottleneck? Repeated three-dimensional placement and activation of the correct defect species, orientation, charge state, and local environment at the interaction length scale, with high coherent yield. Deterministic ion arrival alone is insufficient [R257]; [R258].

11. What energy scales would be required? The hierarchy must satisfy

$$\frac{\Delta_C}{h} \gg u, \Omega, \sigma, \Gamma_{\text{decoh}}, \quad \frac{\Delta_{\text{topo}}}{h} \gg \frac{k_B T}{h}, \Gamma_{\text{decoh}}, \sigma_{\text{eff}},$$

where the quantities compared are frequencies. Here u is intercluster coupling, Ω a drive scale, σ disorder, Γ_{decoh} a decoherence rate, Δ_C/h the cluster-gap frequency, and Δ_{topo}/h the many-body-gap frequency. The illustrative NV geometry gives sub-megahertz cluster scales and may yield only hertz-scale high-order terms. Those numbers are a warning to calculate, not a device requirement.

12. At what temperature would such a system probably need to operate? There is no honest temperature before there is a measured gap. A passive target needs $k_B T$ comfortably below the energy Δ_{topo} ; choosing a tenfold Boltzmann margin gives $T < \Delta_{\text{topo}}/(10k_B)$. Thus $\Delta_{\text{topo}}/h = 1 \text{ MHz}$ implies $T < 4.8 \mu\text{K}$, while $\Delta_{\text{topo}}/h = 1 \text{ GHz}$ implies $T < 4.8 \text{ mK}$. These are design inequalities, not forecasts, and a driven quasienergy gap does not satisfy them automatically.

13. How many physical defects might be needed per effective degree of freedom?

Three in the scoped triangle model. More elaborate symmetry, leakage suppression, or gadget mediators could raise that count substantially. No evidence presently identifies an optimum, so a universal “5–20” estimate would be decoration rather than calculation.

14. What is the smallest experiment capable of falsifying the idea?

Spectroscopy of one fabricated three-NV triangle can falsify the assumed cluster doublet. To falsify the many-body mechanism, the smallest decisive physical device is four such clusters — twelve defects — because it can extract a weight-four coefficient and its competitors. The four-cluster numerical experiment should come first and can already falsify the Hamiltonian mechanism.

15. What result would make this direction genuinely promising?

A reproducible cluster doublet followed by a measured or microscopically validated many-body coefficient that dominates noncommuting residual terms, exceeds disorder and decoherence, survives realistic position distributions, and produces the expected gap and nonlocal diagnostics on a larger patch. A visually plausible lattice or a digitally scripted braid is not enough.

41.9. How sure we can be

Proposition	Confidence	Explanation
Defect clusters can form encoded leftovers	Medium	Few-spin encoding is standard theory and the ideal NV triangle is solvable, but the specified symmetric cluster array is not demonstrated.
Dense interacting defect arrays are manufacturable	Low	Placement, conversion, charge, orientation, and coherent yield have not been combined at the required repeated nanometre geometry [R257]; [R258].
Effective many-body interactions can be engineered	Low	Gadget theory permits them [R175], but the desired term is suppressed by denominators and competes with larger native terms in this platform.
Toric-code-like phases are plausible	Medium	They require simpler Abelian stabilizer data than Fibonacci order and can be digitally or Floquet emulated; a passive defect realization remains unshown.

Proposition	Confidence	Explanation
Non-Abelian phases are plausible	Low	No derived defect Hamiltonian, gap, or phase diagram supports one here. Digital non-Abelian behavior does not raise the material rating.
Doubled-Fibonacci phase is physically realistic	Low	The ideal target is established [R018], but its branching and recoupling projectors have no supported microscopic defect derivation.
Passive topological protection improves coherence	Low	Conditional theory is sound, but this architecture has neither a demonstrated passive topological gap nor evidence that thermal and leakage errors lie below it.
Diamond is superior to competing platforms	Low	Diamond leads on several single-defect capabilities, not on every system metric; fabrication and coupling may favor another host or a hybrid platform.

The five ranks, used once more so the words stay honest:

- **established** — measured or derived in the relevant system.
- **extension** — a straightforward next device, using existing technology.
- **proposal** — a credible but difficult research program, with a first measurement that can fail.
- **speculation** — physically coherent, but the indispensable pieces are unshown.
- **incompatible** — in conflict with known physics.

For the complete proposal — dense defect clusters producing a genuinely topologically ordered, potentially doubled-Fibonacci material and a passively improved logical qubit — the only justified classification is

highly speculative but physically coherent

It is **not incompatible with known physics**: local Hamiltonians with doubled-Fibonacci order exist, effective interactions can arise perturbatively, and defect spins can be controlled. It is not yet a **credible but difficult research program** at the full architecture level, because the central microscopic interaction and manufacturable geometry have not been demonstrated or even numerically validated together. The scoped four-cluster paper

is a **proposal** in the narrower sense: it is credible precisely because it can return “no” and still succeed scientifically.

41.10. The nearest fallback

The nearest fallback is a **digitally controlled defect-spin network with active error correction**, using single defects or small clusters only where they improve a measured control or noise metric. It would prepare stabilizer or small string-net states with gates, measure syndromes, and treat anyon operations as quantum simulation. Optical links or a cavity may connect sparse modules, relaxing the demand for a perfect dense crystal lattice.

This fallback gives up autonomous Fibonacci topological order and passive self-protection. It retains useful defect coherence, local memories, distributed entanglement, active logical encoding, and a platform for studying anyon protocols. If dense NV fabrication fails, SiC defects, donors, rare-earth ions, trapped ions, neutral atoms, or superconducting circuits should compete on the same measured ledger rather than on substrate loyalty. Digital Fibonacci experiments already show that scientifically meaningful anyon work does not require claiming a native Fibonacci material [R138]; [R165].

Keep digital simulation, driven emulation, and equilibrium emergence in separate columns. Mixing those columns is how a grade inflates.

41.11. Check yourself

1. Why is a four-cluster calculation the first paper?

Because four encoded spins are the smallest system that can expose a weight-four plaquette coefficient and every lower-weight competitor.

2. Show that the ideal triangle (41.1) has gap $\Delta_C/h = 3c/2$, and that a fourth-order K_4 at the proposed 5 nm / 10 nm geometry is about 30 Hz.

The isolated-cluster splitting of (41.1) is $3c/2$. With $c \approx 0.416$ MHz one has $\Delta_C/h \approx 0.624$ MHz and $u \lesssim 52$ kHz, so $K_4 \sim u^4/(\Delta_C/h)^3 \sim 30$ Hz, or $\hbar K_4/k_B \approx 1.4$ nK. That is a warning to compute, not a measured coupling.

3. What fails if $K_4 \neq 0$ is treated as a phase?

One local term is a mechanism check. Dominance, robustness, a gapped phase, and topological diagnostics remain separate tests. A nonzero coefficient does not establish topological order.

4. Why not begin with doubled Fibonacci?

Its vertex and recoupling projectors require more structure than the proposed microscopic Hamiltonian has earned. The simpler stabilizer bottleneck can falsify the route first.

5. What fails if a different encoded basis is chosen for every disorder sample?

A large-looking target coefficient that no fixed control realizes. The cluster basis must be fixed from isolated-cluster spectroscopy, or constrained to calibrated single-cluster controls, with the unrotated ledger reported too.

6. What is the honest fallback?

Use defects as controllable physical or cluster-encoded qubits in an actively corrected digital architecture, and call anyon experiments simulations unless the hardware Hamiltonian itself produces the phase.

41.12. Sources

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001.
- [R080] F. Dolde, I. Jakobi, B. Naydenov, *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545.
- [R257] D. Scarabelli, M. Trusheim, O. Gaathon, D. Englund, and S. J. Wind, “Nanoscale engineering of closely-spaced electronic spins in diamond,” *Nano Letters* **16**, 4982–4990 (2016). DOI: 10.1021/acs.nanolett.6b01692.
- [R258] K. Groot-Berning, T. Kornher, G. Jacob, *et al.*, “Fabrication of $^{15}\text{NV}^-$ centers in diamond using a deterministic single ion implanter,” *New Journal of Physics* **23**, 063067 (2021). DOI: 10.1088/1367-2630/ac0753.
- [R175] S. Bravyi, D. P. DiVincenzo, D. Loss, and B. M. Terhal, “Quantum simulation of many-body Hamiltonians using perturbation theory with bounded-strength interactions,” *Physical Review Letters* **101**, 070503 (2008). DOI: 10.1103/PhysRevLett.101.070503. arXiv: 0803.2686.
- [R018] M. A. Levin and X.-G. Wen, “String-net condensation: A physical mechanism for topological phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110. arXiv: cond-mat/0404617.
- [R030] A. Yu. Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0. arXiv: quant-ph/9707021.
- [R138] S. Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6.

- [R165] Z. K. Mineev *et al.*, “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* **16**, 6225 (2025). DOI: 10.1038/s41467-025-61493-8.
- [R089] Z. Velluire-Pellat, E. Maréchal, C. Feuillet-Palma, *et al.*, “Spin-photon interaction between a ruby crystal and a high-critical-temperature superconducting microwave cavity,” *Communications Physics* **8**, 236 (2025). DOI: 10.1038/s42005-025-02159-1.

Part XII

Mathematical appendices

Tool drawers. Open one when a chapter points at it.

Appendix A. Arrows you can add and stretch

Take a pencil and draw an arrow on a scrap of paper. Draw a second arrow. Lay the second at the tip of the first and call the new path their sum. Stretch either by 2, shrink it by $1/2$, or reverse it by multiplying by -1 .

High-school geometry already knows both operations.

Now let the stretch factor be a complex number. Multiplying by i rotates the arrow a quarter turn. Multiplying by $3 + 4i$ stretches it by 5 and turns it. The arrows still add tip to tail. Nothing physical has been assumed. The set of allowed stretches has simply grown.

A.1. Two operations

A set of objects you can add and complex-scale — with a zero, an opposite for every object, and the rule that 1 times anything is itself — is a *vector space*. We take this as the definition. The objects may be pencil arrows, columns of two complex numbers, or the internal states of a silver atom. The two operations are the whole structure.

Write $\mathbb{C}$ for the complex numbers and $V = \mathbb{C}^2$ for the space of columns of height two. Addition and scaling act entrywise. For vectors $u, v, w \in V$ and scalars $a, b \in \mathbb{C}$, the operations associate and commute where they should, and they distribute over one another. There is no hidden geometry yet.

A.2. Coordinates are not the arrow

The ket $|\psi\rangle$ is the abstract vector. The column you write down depends on which reusable directions you picked.

In the standard pair

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

a normalized state is

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Here $\alpha, \beta \in \mathbb{C}$, and $|\alpha|$ is the complex modulus. Define another pair,

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Substitution — not a new physical assumption — gives

$$|\psi\rangle = \frac{\alpha + \beta}{\sqrt{2}}|+\rangle + \frac{\alpha - \beta}{\sqrt{2}}|-\rangle.$$

The state stayed put. Only its coordinates changed. If U is the matrix whose columns are the new directions written in the old ones, the coordinate columns obey

$$[\psi]_{\text{new}} = U^\dagger [\psi]_{\text{old}}.$$

The dagger $\dagger$ will mean conjugate transpose once we have earned it. This particular U will turn out to preserve lengths, so the state stays normalized.

A finite list $v_1, \dots, v_k$ is linearly independent when

$$a_1 v_1 + \dots + a_k v_k = 0$$

forces $a_1 = \dots = a_k = 0$. Its span is every weighted sum you can form from the list. A *basis* is a linearly independent list whose span is all of V . Every basis of a finite-dimensional space has the same length; that length is the dimension $\dim V$.

A subspace $W \subseteq V$ contains 0 and is closed under addition and complex scaling. The logical doublet of a defect cluster is a two-dimensional subspace of a much larger space. Calling it a subspace says nothing yet about whether the energy operator or the noise keeps a state inside it.

The direct sum $V = W \oplus W'$ means every $v \in V$ has one and only one split $v = w + w'$, with $w \in W$ and $w' \in W'$. Projection methods write the whole space as the desired low-energy piece plus whatever is left.

Remark. Linear independence is a property of a *list*, not of any one arrow. Drop it and the same vector may acquire several coordinate columns in the same spanning set.

A.3. A pairing that makes length

Addition and scaling still do not know length or angle. Suppose we are given a machine that eats two vectors and returns a complex number $\langle u|v\rangle$. We use the physics convention: conjugate-linear in the first argument and linear in the second,

$$\langle au + bw|v\rangle = a^*\langle u|v\rangle + b^*\langle w|v\rangle,$$

$$\langle u|av + bw\rangle = a\langle u|v\rangle + b\langle u|w\rangle.$$

It also satisfies conjugate symmetry, $\langle u|v\rangle = \langle v|u\rangle^*$, and positive definiteness: $\langle v|v\rangle \geq 0$, with equality only for $v = 0$. The star is complex conjugation.

That pairing is an *inner product*. Length, angle, orthogonality, and later probability all sit inside it.

The length is $\|v\| = \sqrt{\langle v|v\rangle}$. Two vectors are orthogonal when their pairing vanishes. A list $\{|e_j\rangle\}$ is orthonormal when $\langle e_j|e_k\rangle = \delta_{jk}$, where the Kronecker delta δ_{jk} equals one for $j = k$ and zero otherwise.

A ket $|v\rangle$ is a vector. Its associated bra $\langle v|$ is the linear functional that maps $|w\rangle$ to $\langle v|w\rangle$. In an orthonormal coordinate basis, taking a ket column to its bra means transpose and complex-conjugate it.

For any orthonormal basis $\{|e_j\rangle\}_{j=1}^d$,

$$I = \sum_{j=1}^d |e_j\rangle\langle e_j|, \quad |v\rangle = \sum_{j=1}^d |e_j\rangle\langle e_j|v\rangle.$$

The first equation is the resolution of the identity. The second says that coordinates are inner products with basis vectors.

Hence, once an orthonormal basis is chosen, the abstract vector and its coordinate column determine each other.

The Cauchy–Schwarz inequality,

$$|\langle u|v\rangle| \leq \|u\| \|v\|,$$

ensures that the overlap of normalized states has modulus at most one. One proof considers the nonnegative length of $|u\rangle - z|v\rangle$ and chooses the complex number z that minimizes it. That is the algebra beneath the statement that Born probabilities lie between zero and one.

In finite dimensions, a complex vector space with a positive-definite inner product is automatically complete in the induced norm, so people also call it a Hilbert space. You do not need that second name to finish a matrix calculation. You need the pairing.

Remark. Drop positive-definiteness and you may still have a Hermitian form. You no longer have a length that vanishes only at zero, and the last step of Cauchy–Schwarz fails.

A.4. Machines that do not break addition

A map $A : V \rightarrow W$ is linear when

$$A(qu + bv) = qA(u) + bA(v).$$

Once bases are chosen, A is a matrix. Its entry in row j , column k is $A_{jk} = \langle e_j | A | e_k \rangle$. Matrix multiplication represents composition: $(AB)|v\rangle = A(B|v\rangle)$, so the rightmost operator acts first.

The kernel $\ker A$ contains vectors mapped to zero. The image $\text{im } A$ contains outputs $A|v\rangle$. The rank–nullity theorem states

$$\dim V = \dim(\ker A) + \dim(\text{im } A)$$

for a linear map whose domain V is finite dimensional. Nothing is lost and nothing is invented. The missing directions go to zero; the rest span the outputs.

A.5. Asking the machine backwards

Given a linear machine A , there is a unique partner $A^\dagger$ defined by

$$\langle u | Av \rangle = \langle A^\dagger u | v \rangle$$

for all vectors u, v . In orthonormal bases its matrix is the conjugate transpose. That partner is the *adjoint*. Lengths, measurement averages, and time evolution all compare A with $A^\dagger$.

Several classes recur.

- A is Hermitian when $A = A^\dagger$. Observables and energy operators are represented by Hermitian operators.

- U is unitary when $U^\dagger U = UU^\dagger = I$. Unitary operators preserve inner products.
- P is an orthogonal projector when $P = P^\dagger = P^2$. Its image is the retained subspace.
- A is positive semidefinite, written $A \geq 0$, when $\langle v|A|v \rangle \geq 0$ for every $|v \rangle$.
- A is normal when $AA^\dagger = A^\dagger A$. Hermitian and unitary operators are normal.

The trace is the basis-independent number

$$\text{Tr } A = \sum_j \langle e_j | A | e_j \rangle.$$

It obeys cyclicity, $\text{Tr}(AB) = \text{Tr}(BA)$, whenever the finite-dimensional products are defined. Cyclicity does not mean arbitrary reordering: generally $\text{Tr}(ABC) = \text{Tr}(BCA)$, not $\text{Tr}(ACB)$.

Remark. Drop the inner product and the adjoint is not defined. You can still transpose a matrix of numbers, but that operation then depends on the basis you used to write the numbers down.

A.6. Directions the machine does not mix

A nonzero vector $|v \rangle$ is an eigenvector of A with eigenvalue $\lambda \in \mathbb{C}$ when

$$A|v \rangle = \lambda|v \rangle.$$

The machine only stretches that arrow. It does not knock it sideways. The eigenvalue is degenerate if its eigenspace has dimension greater than one.

Suppose $A = A^\dagger$ and $A|v \rangle = \lambda|v \rangle$ with $\|v\| = 1$. Then

$$\lambda = \langle v | A | v \rangle = \langle Av | v \rangle = \lambda^*,$$

so λ is real. Hence every eigenvalue of a Hermitian operator is real.

The finite-dimensional *spectral theorem* says that a normal operator has an orthonormal eigenbasis [R001]; [R002]. In particular, a Hermitian operator can be written

$$A = \sum_r \lambda_r P_r,$$

where λ_r are its distinct real eigenvalues and P_r are mutually orthogonal projectors satisfying $P_r P_s = \delta_{rs} P_r$ and $\sum_r P_r = I$. Degeneracy is handled by allowing P_r to project onto a multidimensional eigenspace.

A function of A is then defined spectrally:

$$f(A) = \sum_r f(\lambda_r) P_r.$$

For a Hamiltonian H , measured in joules (J), unitary time evolution for time t , measured in seconds (s), is

$$U(t) = e^{-iHt/\hbar} = \sum_r e^{-i\lambda_r t/\hbar} P_r.$$

The exponent is dimensionless because $(\text{J})(\text{s})/(\text{J s}) = 1$. Here $\hbar$ is the reduced Planck constant.

Hence $U(t)$ is unitary: in the spectral basis every eigenvalue becomes a phase of modulus one, so $U(t)^\dagger U(t) = I$.

Over $\mathbb{C}$, the characteristic polynomial of a finite matrix has a root, so a normal operator has at least one eigenvector. Normalize it. Normality makes the orthogonal complement of that eigenvector invariant under both A and $A^\dagger$. Restrict A to that smaller space and repeat. Induction supplies an orthonormal eigenbasis. This sketch hides no physics, but a complete proof carefully establishes the invariant-complement step [R001]; [R002].

Remark. Drop normality and a finite matrix may lack an orthonormal eigenbasis. It may even be defective: too few independent eigenvectors to span the space. The spectral formula for $f(A)$ is then unavailable in that form.

A.7. The most general two-level energy matrix

Define the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every 2×2 Hermitian Hamiltonian can be written

$$H = cI + h_x \sigma_x + h_y \sigma_y + h_z \sigma_z = cI + \mathbf{h} \cdot \boldsymbol{\sigma},$$

where $c, h_x, h_y, h_z \in \mathbb{R}$ have units of J, $\mathbf{h} = (h_x, h_y, h_z)$, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. Define $h = \sqrt{h_x^2 + h_y^2 + h_z^2}$. For $h \neq 0$, define the unit vector $\mathbf{n} = \mathbf{h}/h$.

The Pauli multiplication rule is

$$\sigma_j \sigma_k = \delta_{jk} I + i \sum_{\ell} \varepsilon_{jkl} \sigma_{\ell},$$

where ε_{jkl} is the antisymmetric Levi-Civita symbol. The symmetric product $n_j n_k$ kills the antisymmetric term, so

$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I.$$

An operator squaring to I has eigenvalues ± 1 . Its projectors are

$$P_{\pm} = \frac{1}{2}(I \pm \mathbf{n} \cdot \boldsymbol{\sigma}),$$

because $P_{\pm}^2 = P_{\pm}$, $P_+ P_- = 0$, and $P_+ + P_- = I$. Therefore the spectral decomposition is

$$H = (c + h)P_+ + (c - h)P_-.$$

The two energies are $E_{\pm} = c \pm h$, and their separation is $2h$. Applying a function spectrally gives

$$\begin{aligned} e^{-iHt/\hbar} &= e^{-i(c+h)t/\hbar} P_+ + e^{-i(c-h)t/\hbar} P_- \\ &= e^{-ict/\hbar} \left[\cos\left(\frac{ht}{\hbar}\right) I - i \sin\left(\frac{ht}{\hbar}\right) \mathbf{n} \cdot \boldsymbol{\sigma} \right]. \end{aligned}$$

This one result covers Zeeman precession, driven qubits in a rotating frame, and effective cluster pseudospins. At $h = 0$, the direction $\mathbf{n}$ is undefined but harmless: $H = cI$, and evolution is only a common phase.

A two-dimensional subspace is mathematically capable of representing a qubit. It is not automatically a controllable physical qubit, an error-corrected logical qubit, or a topological ground space. Those labels add dynamical and operational conditions that linear algebra alone cannot supply.

A.8. Four rewrites that are not the same

Authors may place vectors as columns without kets, use the mathematics convention in which the inner product is linear in its first argument, or set $\hbar = 1$. Check conventions before moving a complex scalar through an inner product. Also distinguish:

- a basis change, which changes coordinates;
- a unitary physical evolution, which changes the state relative to fixed measurement operators;
- a similarity transformation $A' = U^\dagger A U$, which represents the same abstract operator in a new orthonormal basis;
- a projected operator PAP , which generally discards information and is not a basis change.

For infinite-dimensional Hilbert spaces, the spectral theorem becomes a statement about projection-valued measures. Unbounded operators also require domains: the expression AB may not be defined on every vector. The finite-dimensional matrix version is sufficient for the defect clusters and finite lattice models in this book, but continuum papers may need the stronger functional-analytic machinery [R003]; [R005].

A.9. Ways to get this wrong

1. Forgetting complex conjugation. Bras are conjugate-transposed kets; otherwise norms need not be real or positive.
2. Diagonalizing every matrix as if it were Hermitian. A non-normal matrix may lack an orthonormal eigenbasis and may even be defective.
3. Treating degeneracy as a unique basis. The eigenspace is fixed, but any orthonormal basis inside it is allowed.
4. Confusing PAP with A . Projection omits coupling through the complementary subspace; perturbation theory restores some of those virtual effects.
5. Using trace cyclicity as commutativity. Operators remain order sensitive outside the cyclic trace identity.
6. Ignoring units in exponentials. e^{-iHt} silently assumes $\hbar = 1$; restore $tH/\hbar$ before numerical work.

A.10. Check yourself

1. **What is basis independent: a ket or its coordinate column?** The ket is abstract and basis independent; its coordinate column depends on the basis.

2. **Show that every eigenvalue of a Hermitian operator is real.** If $A = A^\dagger$ and $A|v\rangle = \lambda|v\rangle$ with $\|v\| = 1$, then $\lambda = \langle v|A|v\rangle = \langle Av|v\rangle = \lambda^*$.
3. **What extra structure turns a complex vector space into a Hilbert space in finite dimensions?** A positive-definite inner product. Finite-dimensional spaces are automatically complete in the induced norm.
4. **Show that $e^{-iHt/\hbar}$ is unitary when $H = H^\dagger$.** In the spectral basis every eigenvalue of H is real, so every eigenvalue of the exponential is a phase of modulus one.
5. **What fails if you drop normality?** A finite matrix need not have an orthonormal eigenbasis, and the spectral formula for $f(A)$ need not apply.
6. **What is lost in replacing A by PAP ?** Matrix elements connecting the retained subspace to its complement and all action wholly within that complement.

Later chapters may use without rederivation that orthonormal bases resolve the identity; Hermitian operators possess real spectral decompositions; projectors isolate possibly degenerate subspaces; operator functions act eigenvalue by eigenvalue; unitary changes of orthonormal basis preserve inner products; and a projected two-dimensional operator is a linear combination of $I, \sigma_x, \sigma_y, \sigma_z$. Appendix B adds composite systems, incomplete information, and irreversible channels.

A.11. Sources

- [R001] S. Axler, *Linear Algebra Done Right*, 3rd ed., Springer, 2015. DOI: 10.1007/978-3-319-11080-6.
- [R002] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013. DOI: 10.1017/CBO9781139020411.
- [R003] B. C. Hall, *Quantum Theory for Mathematicians*, Springer, 2013. DOI: 10.1007/978-1-4614-7116-5.
- [R004] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press, 2010. DOI: 10.1017/CBO9780511976667.
- [R005] A. Peres, *Quantum Theory: Concepts and Methods*, Kluwer Academic, 1995. DOI: 10.1007/0-306-47120-5.

Appendix B. Two lists that become four

Alice writes two numbers, one for each answer her coin can give:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Bob writes two numbers of his own:

$$\begin{pmatrix} \gamma \\ \delta \end{pmatrix}.$$

If you only care about pairs of answers, you need four slots, not two plus two:

$$\begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}.$$

That bigger list is the whole story when each person has a definite column and you just staple them. Quantum mechanics keeps those four slots and then lets you add two such stapled lists. The extra room is where a pair can refuse to factor.

No crystal yet. No bath. Just two short lists, one longer list, and the bookkeeping that remains after you throw one of the lists away.

B.1. Stapling two columns

Give Alice the answers 0 and 1, written as columns

$$|0\rangle_A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle_A = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Give Bob the same two answers, labeled B . A joint answer is a pair. Write the four pairs as

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

The symbol $\otimes$ is the staple. On a pair of columns it means: take every entry of the first column, multiply the whole second column by that entry, and stack the results. That is why two twos become four, not two plus two.

The space of all such stapled columns, with the usual rule that you may add them and scale them by complex numbers, is their *tensor product*. If Alice's space is called V and Bob's is called W , the staple is linear in each slot separately:

$$(av + bv') \otimes w = a(v \otimes w) + b(v' \otimes w),$$

$$v \otimes (aw + bw') = a(v \otimes w) + b(v \otimes w').$$

Choose a basis $\{|e_j\rangle\}$ for V and a basis $\{|f_k\rangle\}$ for W . Then the stapled pairs $\{|e_j\rangle \otimes |f_k\rangle\}$ are a basis for $V \otimes W$. Consequently the dimensions multiply:

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

For N two-level systems the dimension is 2^N . That exponential is a counting fact. It is not, by itself, a speedup.

Inner products multiply on stapled pairs and then extend by linearity:

$$\langle u \otimes x | v \otimes y \rangle = \langle u | v \rangle \langle x | y \rangle.$$

We write $|a\rangle_A \otimes |b\rangle_B$ as $|ab\rangle$ when the labels are obvious.

If a matrix A acts on V and a matrix B acts on W , the stapled operator is defined by doing each job on its own factor:

$$(A \otimes B)(v \otimes w) = Av \otimes Bw.$$

A local operator on Alice alone is $A \otimes I_W$, often shortened to A after you have said who is who. Operators on distinct factors commute:

$$(A \otimes I_W)(I_V \otimes B) = A \otimes B = (I_V \otimes B)(A \otimes I_W).$$

A vector that really is one staple, $|a\rangle \otimes |b\rangle$, is a *product state*. Alice has a column, Bob has a column, and the pair is just those two columns sitting next to each other.

Remark. Drop linearity in one slot and you no longer have a tensor product. Ordinary multiplication of two numbers of the same length is a different machine, and it does not produce a four-slot list from two two-slot lists.

B.2. A pair you cannot unscrew

Not every four-slot list comes from stapling two two-slot lists. Consider

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Suppose it factored as $(a|0\rangle + b|1\rangle)_A \otimes (c|0\rangle + d|1\rangle)_B$. Expanding would give coefficients ac, ad, bc, bd . Matching $|\Phi^+\rangle$ requires $ac = bd = 1/\sqrt{2}$ and $ad = bc = 0$. The nonzero first pair forces all four scalars to be nonzero, which contradicts the two zero products.

Hence $|\Phi^+\rangle$ cannot be written as one staple. A pure joint state that cannot be written that way is *entangled*. The obstruction is not “the two sides talk.” It is “the four-slot list does not factor.”

There is a tidy test. Every bipartite pure state has orthonormal sets $\{|u_r\rangle_A\}$, $\{|v_r\rangle_B\}$ and nonnegative numbers s_r such that

$$|\psi\rangle_{AB} = \sum_r s_r |u_r\rangle_A |v_r\rangle_B, \quad \sum_r s_r^2 = 1.$$

That expansion is the Schmidt decomposition. The state is a product exactly when one Schmidt coefficient is nonzero. The eigenvalues of either reduced ledger below are the numbers s_r^2 , including the zeros.

B.3. The ledger for a pair

A single ket is enough when you know the exact column. Sometimes you do not. You might have flipped a classical coin and prepared one of two kets. You might have thrown Bob’s list away and kept only Alice’s. Both stories need a ledger that still predicts every local measurement.

Turn the Bell pair into a matrix by writing the ket next to its bra:

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

That matrix is a *density operator*. One object now stores every probability you can extract from the pair, whether the pair is a single ket or a blur of kets.

Any such ledger ρ on a space $\mathcal{H}$ satisfies two rules:

$$\rho \geq 0, \quad \text{Tr } \rho = 1.$$

Positivity already forces Hermiticity in finite dimensions. If a preparation chooses normalized pure states $|\psi_r\rangle$ with classical probabilities $p_r \geq 0$, where $\sum_r p_r = 1$, then

$$\rho = \sum_r p_r |\psi_r\rangle\langle\psi_r|.$$

Different lists $\{p_r, |\psi_r\rangle\}$ can produce the same ρ . Measurement probabilities depend only on ρ , so those lists are indistinguishable unless you kept an extra record.

A state is *pure* exactly when $\rho^2 = \rho$, equivalently $\text{Tr}(\rho^2) = 1$. It is *mixed* when $\text{Tr}(\rho^2) < 1$. The number $\text{Tr}(\rho^2)$ is the purity. The spectral theorem writes

$$\rho = \sum_j q_j |q_j\rangle\langle q_j|,$$

where $q_j \geq 0$ and $\sum_j q_j = 1$. A density operator is a positive matrix with unit trace. That is the whole definition.

For a qubit every density operator has a Bloch representation

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}),$$

where $\mathbf{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, and positivity requires $\|\mathbf{r}\| \leq 1$. Pure states sit on the unit sphere. Mixed states sit inside it.

An observable $M = M^\dagger$ has expectation value

$$\langle M \rangle = \text{Tr}(\rho M).$$

More generally, a positive-operator-valued measure, or *POVM*, is a set of positive operators $\{E_m\}$ satisfying $\sum_m E_m = I$. Outcome m has probability $p(m) = \text{Tr}(\rho E_m)$. A POVM names probabilities. It does not name the leftover state. A quantum instrument supplies the corresponding update maps.

Remark. Drop positivity and $\text{Tr}(\rho M)$ need not be a probability even when $0 \leq M \leq I$. Drop uniqueness of the ensemble and you have not dropped uniqueness of ρ : many lists represent one ledger.

B.4. Throwing one list away

Suppose Bob's half of the Bell pair is inaccessible. Alice still has to predict her own magnet. The right local ledger is obtained by summing Bob's answers out of ρ_{AB} . That sum is the *partial trace*

$$\rho_A = \text{Tr}_B \rho_{AB}.$$

It is a trace on one factor only. The other factor is left standing.

On a rank-one staple the rule is

$$\text{Tr}_B (|i\rangle_A |j\rangle_B \langle k|_A \langle \ell|_B) = \langle \ell|j\rangle_B |i\rangle_A \langle k|_A.$$

The Bell cross terms vanish because $\langle 1|0\rangle = 0$. The diagonal terms survive. Therefore

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I_A.$$

The joint state is pure. Either part alone is as mixed as a fair coin. Hence a local experiment on Alice cannot tell a tossed coin from a missing entangled partner. The ledger does not record which story produced it.

The same map is the unique linear rule that obeys

$$\text{Tr}_B(A \otimes B) = A \text{Tr}(B).$$

For any observable M_A that acts only on Alice, this guarantees

$$\text{Tr}_{AB}[\rho_{AB}(M_A \otimes I_B)] = \text{Tr}_A(\rho_A M_A),$$

where $\rho_A = \text{Tr}_B \rho_{AB}$. That identity is why ρ_A is the correct local state. It reproduces every local average.

A joint ledger is *separable* when it can be written as a classical mixture of product ledgers:

$$\rho_{AB} = \sum_r p_r \rho_A^{(r)} \otimes \rho_B^{(r)}.$$

Otherwise it is entangled. Separable states can still have strong classical correlations. A probability table that refuses to factor, or a nonzero covariance, is not by itself entanglement.

The von Neumann entropy of a ledger is

$$S(\rho) = -\text{Tr}(\rho \log \rho) = -\sum_j q_j \log q_j,$$

with the convention $0 \log 0 = 0$. If the logarithm uses base two, entropy is measured in bits. For a bipartite pure state, $S(\rho_A) = S(\rho_B)$ counts entanglement across that cut. For mixed states the same number also contains ordinary mixture. Local entropy alone is not an entanglement measure.

Remark. Drop purity of the joint state and $S(\rho_A)$ no longer measures entanglement by itself. It still measures mixedness of the reduced ledger.

B.5. What remains after you ignore a partner

Alice's atom can couple to something you never look at and then walk away looking irreversible. The joint motion can still be unitary. The leftover map on Alice is not.

A *quantum channel* $\mathcal{E}$ is a linear map that sends density operators to density operators and does so even if the input is secretly half of a larger pair. Trace preservation means $\text{Tr } \mathcal{E}(X) = \text{Tr } X$. Ordinary positivity means positive inputs stay positive. That is not enough.

Complete positivity is the stronger demand: for an identity map $\mathcal{I}_R$ on an arbitrary reference R , the extended map $\mathcal{I}_R \otimes \mathcal{E}$ must also preserve positivity. The extra word is there because the input may be entangled with an untouched reference [R007]; [R008]. A merely positive map can hand you a nonsense matrix when you apply it to half of a Bell pair.

Every finite-dimensional channel has a Kraus representation

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I.$$

The operators K_k are Kraus operators. The second equation is trace preservation. The list is not unique: a unitary remix of one Kraus list describes the same channel.

A channel can always be realized by adding an environment E in a fixed state, applying a joint unitary U , and tracing E out:

$$\mathcal{E}(\rho) = \text{Tr}_E [U(\rho \otimes |0\rangle_E \langle 0|)U^\dagger].$$

That is a finite-dimensional Stinespring dilation [R007]. It does not claim that a small, memoryless environment is the true bath. It says such a dilation exists on paper.

A useful diagnostic is the Choi operator. If the input dimension is d , define the unnormalized maximally entangled vector $|\Omega\rangle = \sum_{j=1}^d |j\rangle_R |j\rangle_A$. Then

$$J(\mathcal{E}) = (\mathcal{J}_R \otimes \mathcal{E})(|\Omega\rangle\langle\Omega|).$$

Choi's theorem says $\mathcal{E}$ is completely positive exactly when $J(\mathcal{E}) \geq 0$ [R008]. Trace preservation becomes $\text{Tr}_{\text{out}} J(\mathcal{E}) = I_R$, for this unnormalized convention.

Remark. Drop complete positivity and keep ordinary positivity, and the map may still look legal on product states. Apply it to half of $|\Phi^+\rangle$ and the output need not be a density operator.

B.6. A decay that is still unitary somewhere

Amplitude damping is the smallest picture of relaxation: $|1\rangle$ can fall to $|0\rangle$, and an environment keeps the receipt. Let the system be S , the environment be E , and let $p \in [0, 1]$ be the decay probability. Define a unitary on the relevant basis states by

$$U|0\rangle_S|0\rangle_E = |0\rangle_S|0\rangle_E,$$

$$U|1\rangle_S|0\rangle_E = \sqrt{1-p}|1\rangle_S|0\rangle_E + \sqrt{p}|0\rangle_S|1\rangle_E.$$

The two output vectors are normalized and orthogonal, so this isometry can be completed to a unitary on the full four-dimensional space. Prepare E in $|0\rangle_E$. Taking the environment matrix element $K_e = {}_E\langle e|U|0\rangle_E$ for $e = 0, 1$ gives

$$K_0 = |0\rangle\langle 0| + \sqrt{1-p}|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix},$$

$$K_1 = \sqrt{p}|0\rangle\langle 1| = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}.$$

They satisfy

$$K_0^\dagger K_0 + K_1^\dagger K_1 = I,$$

so the channel is trace preserving. The Kraus form makes complete positivity explicit. For an arbitrary input

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix},$$

direct multiplication yields

$$\mathcal{E}_p(\rho) = \begin{pmatrix} \rho_{00} + p\rho_{11} & \sqrt{1-p}\rho_{01} \\ \sqrt{1-p}\rho_{10} & (1-p)\rho_{11} \end{pmatrix}.$$

Population flows from $|1\rangle$ to $|0\rangle$. Coherences shrink by $\sqrt{1-p}$. For a Markovian relaxation model with time $t \geq 0$ and relaxation time $T_1 > 0$, one writes

$$p(t) = 1 - e^{-t/T_1}.$$

Both t and T_1 have units of seconds, so the exponent is dimensionless. The channel then composes as a semigroup: evolving for t_1 and then t_2 equals evolving for $t_1 + t_2$.

The joint $S + E$ motion was unitary. Discarding E made Alice's map irreversible. If the environment keeps a memory and comes back, a one-parameter Markov channel may fail. Relaxation is not “nonunitary fundamental motion” of the combined laboratory. It is unitary motion plus a bookkeeping boundary.

Under time-homogeneous Markov assumptions, the density operator can obey the Gorini–Kossakowski–Sudarshan–Lindblad equation [R009]; [R010],

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\} \right).$$

Here H is Hermitian, L_j are jump operators, $\gamma_j \geq 0$ have units s^{-1} , $[A, B] = AB - BA$, and $\{A, B\} = AB + BA$. This generator preserves trace and complete positivity. It is a model with assumptions, not the definition of all open-system dynamics [R011].

B.7. Clusters, leakage, and a missing partner

For a cluster of N physical defects, the microscopic state lives in a tensor product $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_N$. An encoded doublet is selected by a projector P . The encoded state is not obtained merely by tracing out the other levels. You must say how the system was prepared in P , whether it can leak into $Q = I - P$, and whether a recovery or heralding step maps leakage back.

Correlated noise on two clusters is a joint channel. It need not factor as $\mathcal{E}_A \otimes \mathcal{E}_B$. Replacing it with independent single-cluster channels destroys spatial covariance. A many-body reduced density operator can reveal correlations. Mixedness alone is neither topological order nor its absence; those diagnoses need the nonlocal tests and the scaling that later chapters actually run.

A projection $\rho \mapsto P\rho P$ is trace decreasing when leakage occurred. The missing trace

$$1 - \text{Tr}(P\rho P)$$

is the leakage probability. Renormalizing $P\rho P$ describes the state *conditioned* on no detected leakage. Reporting that conditional state without its success probability makes a failing device look rather athletic.

A tensor product describes composite subsystems. Entanglement is a property of a state relative to that split. An encoded qubit is a chosen subspace. A noisy channel is a map on states. None of those structures, by itself, is an emergent quasiparticle, a topological phase, or a fault-tolerant logical qubit.

B.8. Ways to get this wrong

1. Using $\otimes$ like ordinary multiplication. Dimensions and subsystem order matter. $A \otimes B$ acts on a different space from either factor.
2. Calling every correlated state entangled. Separable mixtures can have nonfactorizing classical statistics.
3. Treating an ensemble decomposition as unique. Many lists $\{p_r, |\psi_r\rangle\}$ represent the same density operator.
4. Using positivity instead of complete positivity. A merely positive map can produce an invalid operator when applied to half of an entangled state.
5. Forgetting trace loss. Postselection and leakage projections are trace-nonincreasing. Renormalization changes what you are allowed to claim.
6. Assuming Kraus operators are physical outcomes. They can label outcomes in a chosen dilation or instrument. Different Kraus lists may describe the same unobserved channel.
7. Equating a Lindblad fit with microscopic truth. Non-Markovian baths, slow spectral diffusion, and correlated fluctuators may violate semigroup assumptions.
8. Tracing out a subsystem and saying it was measured. Ignoring and measuring-with-an-unread-outcome can yield the same reduced state in some setups. Their environmental records and later interventions still differ.

B.9. Check yourself

1. **Show that** $\dim(V \otimes W) = (\dim V)(\dim W)$. If $\{|e_j\rangle\}$ bases V and $\{|f_k\rangle\}$ bases W , the stapled pairs $\{|e_j\rangle \otimes |f_k\rangle\}$ base $V \otimes W$, so the dimensions multiply. For two qubits, $2 \times 2 = 4$.
2. **Show that** $|\Phi^+\rangle$ **is not a product state**. A product would have coefficients ac, ad, bc, bd . Matching $ac = bd = 1/\sqrt{2}$ and $ad = bc = 0$ is impossible.

3. **What two conditions define a density operator?** Positivity and unit trace.
4. **Show that the reduced state of $|\Phi^+\rangle$ is $\frac{1}{2}I$.** Partial-trace the four terms of $|\Phi^+\rangle\langle\Phi^+|$. Cross terms vanish by orthogonality; the diagonals each contribute $\frac{1}{2}|0\rangle\langle 0|$ or $\frac{1}{2}|1\rangle\langle 1|$.
5. **What makes a channel “completely” positive?** It remains positive when tensored with the identity on an arbitrary untouched reference system.
6. **What fails if you report $P\rho P/\text{Tr}(P\rho P)$ without the denominator?** You have conditioned on remaining in P and hidden the leakage probability $1 - \text{Tr}(P\rho P)$.

You now have a staple for joining two lists, a ledger for when a single ket is not enough, and a map for what remains after you throw a list away. Later chapters may use, without rederiving them, that composite dimensions multiply; that pure entangled states can have mixed reductions; that the partial trace is fixed by local expectation values; that density operators are positive and unit trace; that physical deterministic processes are completely positive and trace preserving; that every finite channel admits Kraus and unitary-dilation descriptions; that leakage projections can be trace decreasing; and that independent-channel models fail when the environment correlates two clusters.

B.10. Sources

- [R004] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition*, Cambridge University Press (2010). DOI: 10.1017/CBO9780511976667.
- [R006] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press (2018). DOI: 10.1017/9781316848142.
- [R007] W. F. Stinespring, “Positive functions on C -algebras,” *Proceedings of the American Mathematical Society* **6**, 211–216 (1955). DOI: 10.1090/S0002-9939-1955-0069403-4.
- [R008] M.-D. Choi, “Completely positive linear maps on complex matrices,” *Linear Algebra and its Applications* **10**, 285–290 (1975). DOI: 10.1016/0024-3795(75)90075-0.
- [R009] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Communications in Mathematical Physics* **48**, 119–130 (1976). DOI: 10.1007/BF01608499.
- [R010] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, “Completely positive dynamical semigroups of N -level systems,” *Journal of Mathematical Physics* **17**, 821–825 (1976). DOI: 10.1063/1.522979.
- [R011] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press (2002). DOI: 10.1093/acprof:oso/9780199213900.001.0001.

Appendix C. A knob and a shoelace

A radio knob turns through every angle between here and there. A half turn is a real setting. So is a 37-degree turn. Nothing jumps.

A shoelace is not like that. You either send the left lace over the right or you do not. You can wiggle the lace, stretch it, even yank it, and the knot is still the same knot. The continuous details have been thrown away. What remains is a discrete class: which strand went over which.

This appendix is about those two kinds of reversible motion. One kind can be dialed. The other kind can only be counted. Later we will rotate a spin with the first and exchange particles in a plane with the second. The matrices that come out can look similar. The two motions are not.

C.1. Recipes you can undo

Take a box of moves. “Do this, then that” is a product. There is a do-nothing move. Every move has an undo. If “do A then B then C” never cares where you put the parentheses, you already have the whole structure.

That structure is a *group*. Write G for the set of moves and e for do-nothing. Associativity is $(gh)k = g(hk)$. Every $g \in G$ has an inverse g^{-1} with $gg^{-1} = g^{-1}g = e$. Groups capture reversible composition and nothing else.

Remark. Drop inverses and you have a monoid: you can still concatenate, but you cannot always undo. Drop associativity and the product of three moves is not even well-defined until you parenthesize.

C.2. A twist you can dial

Now give the moves smooth coordinates. Near do-nothing, a tiny twist is just a little arrow. Repeat the tiny twist many times and you build a finite one. The familiar rotations of a plane make a circle: one angle, every value allowed.

A group with that extra smoothness is a *Lie group*. $SO(2)$ is the circle of ordinary plane rotations. You can stop anywhere on it.

The extra structure is a compatible smooth-manifold structure making multiplication and inversion smooth. That is the whole difference between a dial and a click.

C.3. A spin that needs two full turns

Let I be the 2×2 identity matrix and let

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the Pauli Z matrix from Chapter 1. A spin-1/2 rotation through angle θ about the z axis is

$$U_z(\theta) = \exp\left(-\frac{i\theta Z}{2}\right),$$

where $i^2 = -1$ and the matrix exponential is $e^A = \sum_{k=0}^{\infty} A^k/k!$. The matrix Z is the tiny arrow we keep repeating. That arrow is a *generator*. The finite rotation is not a new object. It is the generator, exponentiated.

Since $Z^2 = I$, even and odd powers separate:

$$U_z(\theta) = I \cos \frac{\theta}{2} - iZ \sin \frac{\theta}{2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}.$$

Its determinant is one and $U_z^\dagger U_z = I$, so it belongs to $SU(2)$, the Lie group of 2×2 unitary matrices of determinant one. Acting on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ gives

$$U_z(\theta)|\psi\rangle = e^{-i\theta/2}\alpha|0\rangle + e^{i\theta/2}\beta|1\rangle.$$

The relative phase changes by $e^{i\theta}$. At $\theta = 2\pi$, $U_z = -I$ rather than I . A spinor needs a 4π rotation to return as a vector, although the -1 global phase is invisible for an isolated state. That is the double-cover relation $SU(2) \rightarrow SO(3)$ [R014].

Remark. Drop the factor of $1/2$ in the exponent and a 2π rotation would already be the identity. The extra half is not decoration. It is why a spin-1/2 object sees the double cover.

C.4. Laces that refuse to pass through

Fix n marked points in a plane. Move them without collisions and return to the same set of positions. Two motions count as the same if you can continuously deform one into the other without letting strands pass through one another. Multiplication means performing one motion and then the next.

What remains, after the wiggles have been discarded, is a *braid* [R012]; [R013].

Three strands, time downward

identity	σ_1	σ_2
1 2 3	1 2 3	1 2 3
	\ /	\ /
	x	x
	/ \	/ \

Here σ_i means a chosen overcrossing that exchanges positions i and $i+1$ counterclockwise; σ_i^{-1} is the opposite crossing. Orientation conventions differ, so a source must say which crossing its generator denotes.

Those classes multiply: do this braid, then that one. You can list the allowed crossings and the rules that identify two diagrams. You cannot dial a fractional crossing.

C.5. Three strands and a word that will not commute

The set of those classes is the *braid group*. The product is discrete even when the strands move continuously. On three strands it is

$$B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \rangle.$$

The displayed equality is the braid, or Yang–Baxter, relation. It says that two three-crossing diagrams deform into one another. It does **not** say $\sigma_1 \sigma_2 = \sigma_2 \sigma_1$. Those two words induce different orders of motion and are generally distinct.

For a concrete unitary picture, take three identical particles of a kind called Ising σ , and require the trio to look like one more σ from far away. The leftover quantum state is two-dimensional. On that leftover space define

$$F = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = e^{-i\pi/8} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

The map $\rho : B_3 \rightarrow U(2)$ is

$$\rho(\sigma_1) = D, \quad \rho(\sigma_2) = FDF^{-1}.$$

Because $F^{-1} = F$, direct multiplication gives, after omitting the common phase $e^{-3i\pi/8}$,

$$D_0(FD_0F)D_0 = (FD_0F)D_0(FD_0F) = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

where $D_0 = \text{diag}(1, i)$. Thus $\rho(\sigma_1\sigma_2\sigma_1) = \rho(\sigma_2\sigma_1\sigma_2)$ exactly. The two generator matrices nevertheless fail to commute.

Hence the representation respects the braid relation without forcing the generators to commute.

The images live in a Lie group of unitary matrices. That does not make the braid group itself a continuous group. The photograph is not the city. This dictionary turns braid words into quantum gates, but Ising braids generate only a nonuniversal gate set by themselves [R015].

C.6. The tiny arrow at the identity

A matrix Lie group is a group of invertible matrices that is also a smooth manifold, with smooth multiplication and inversion. Important examples are

$$U(n) = \{U \in \mathbb{C}^{n \times n} : U^\dagger U = I\},$$

$$SU(n) = \{U \in U(n) : \det U = 1\},$$

and

$$SO(n) = \{R \in \mathbb{R}^{n \times n} : R^T R = I, \det R = 1\}.$$

Here $\mathbb{C}^{n \times n}$ denotes complex $n \times n$ matrices, $\mathbb{R}$ denotes real numbers, $\dagger$ is conjugate transpose, and T is transpose.

Stand at the identity and look at the tiny arrows. That tangent space is the *Lie algebra* $\mathfrak{g}$ of G . It is a vector space, not a group of finite moves. It carries the Lie bracket

$$[X, Y] = XY - YX.$$

For $U(n)$, tangent generators X are anti-Hermitian: $X^\dagger = -X$. Physicists often write $X = -iH$ with a Hermitian generator $H = H^\dagger$. Then a one-parameter subgroup is

$$U(t) = e^{tX} = e^{-itH}.$$

If t is time, H has units of angular frequency when $\hbar = 1$, or energy when the exponent is $-itH/\hbar$. The exponent must be dimensionless.

Commutators measure local noncommutativity. For small real parameters s, t ,

$$e^{sX}e^{tY}e^{-sX}e^{-tY} = e^{st[X,Y] + O(s^2t, st^2)}.$$

Thus available Hamiltonian generators and their nested commutators determine which connected unitary transformations are controllable. That is a Lie-algebra question, distinct from asking which discrete braids are available [R014].

Remark. Drop the Lie bracket and you still have a vector space of infinitesimal moves. You no longer know which nested commutators can be synthesized from the generators you can actually apply.

C.7. Far crossings commute; near ones do not

For $n \geq 2$, the Artin braid group B_n has generators

$$\sigma_1, \sigma_2, \dots, \sigma_{n-1}$$

and relations

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{when } |i - j| \geq 2,$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}.$$

The first relation says distant exchanges commute. The second is a local deformation involving three neighboring strands. Inverses cancel: $\sigma_i \sigma_i^{-1} = e$.

For B_2 , there is one generator and no additional relation, so $B_2 \cong \mathbb{Z}$: the integer counts signed crossings. For $n \geq 3$, B_n is non-Abelian. Sending each σ_i to the adjacent transposition $(i \ i + 1)$ defines a surjective homomorphism

$$p : B_n \longrightarrow S_n,$$

where S_n is the permutation group. Its kernel is the *pure braid group* P_n , whose strands return to their individually labeled starting positions. The map forgets winding: σ_i^2 is a nontrivial pure braid even though $\rho(\sigma_i^2)$ is the identity permutation [R012]; [R013].

Hence “everyone went home” is not “nothing happened.”

A braid word is read in an explicitly chosen time order. This book composes operators right to left: in $\rho(\sigma_1\sigma_2)$, $\rho(\sigma_2)$ acts first on a state vector. Diagrammatic sources often stack the left factor first. Neither convention is superior; silently mixing them is.

C.8. Words become matrices

A unitary braid *representation* is a homomorphism

$$\rho : B_n \rightarrow U(V),$$

where V is a finite-dimensional leftover space — a fusion space — and $U(V)$ is its unitary group. It is a dictionary, not an identification. The homomorphism property requires $\rho(b_1b_2) = \rho(b_1)\rho(b_2)$ and guarantees that equivalent braid diagrams act identically. In an anyon theory, F -moves expose the pair being exchanged and R -moves perform the exchange; their consistency supplies the braid relations [R015]; [R016].

[Theory] Braiding is computationally universal on an encoded subspace when braid images can approximate every desired projective unitary there to arbitrary accuracy. “Projective” ignores an overall phase. More precisely, one studies whether the closure of $\rho(B_n)$ is dense in the relevant projective unitary group. The closure is a Lie group even though B_n is discrete. Fibonacci representations can be dense on suitable encodings, whereas Ising representations are not braid-universal [R015].

Deep dive — Walking in the space of point arrangements

Let $M = \mathbb{R}^2$ be the plane and remove all configurations in which two of n points coincide. After quotienting by permutations for indistinguishable particles, the resulting configuration space is $C_n(M)$. The braid group is its fundamental group:

$$B_n \cong \pi_1(C_n(\mathbb{R}^2)).$$

A loop in configuration space is a collision-free exchange history. Homotopy identifies histories that deform into one another. In three spatial dimensions, trajectories can usually pass around one another and only the permutation remains; in two dimensions, winding survives [R013]; [R015].

Remark. Drop the collision ban and the configuration space is no longer punctured. The fundamental group collapses, and there is nothing left for a braid to remember.

C.9. Pulses, paths, and what still breaks

Suppose a laboratory can apply two Hamiltonians, H_x and H_z , to an encoded qubit. Pulse durations are continuous real parameters, so ideal control words have the form

$$U = e^{-it_k H_{j_k}/\hbar} \dots e^{-it_2 H_{j_2}/\hbar} e^{-it_1 H_{j_1}/\hbar}.$$

Here t_r is a time, each H_{j_r} has units of energy, and $\hbar$ has units of energy times time. Every exponent is therefore dimensionless. Whether such pulses generate all of $SU(2)$ is tested from the Lie algebra generated by $-iH_x/\hbar$, $-iH_z/\hbar$, and their commutators. Small calibration errors change the real parameters and usually change the implemented unitary continuously.

A braid protocol instead specifies an integer word

$$b = \sigma_{i_k}^{s_k} \dots \sigma_{i_2}^{s_2} \sigma_{i_1}^{s_1}, \quad s_r \in \{+1, -1\}.$$

Smoothly distorting a strand trajectory without creating a collision leaves b unchanged. That is the source of topological insensitivity in the ideal anyon model. But a real protocol can still fail discretely: an unintended quasiparticle may enter the region, two world lines may be exchanged in the wrong order, or leakage may leave the intended fusion space. Topology removes sensitivity to some path details; it does not turn missing or extra braid generators into the desired word [R015].

Hybrid schemes use both structures. Braids supply a discrete protected gate set, while calibrated continuous operations, measurements, or ancillary states supply gates outside its image. When the braid image is not dense — as for Ising anyons — those non-braid resources are logically essential rather than cosmetic. When the image is dense, compiling a target gate still requires a finite braid approximation and an accuracy metric, commonly operator norm or worst-case projective distance.

C.10. Ways to get this wrong

1. Flipping clockwise versus counterclockwise without saying so. That swap sends σ_i to σ_i^{-1} .

2. Relabeling a fusion-tree basis and treating the new matrices as a new physics. Relabeling conjugates every representation matrix. Eigenvalues of closed operations and measurable probabilities do not move.
3. Treating a global phase as a computational difference. A global phase places gates in $U(d)$ rather than projective $PU(d)$ but does not affect an isolated computational outcome.
4. Programming unitary matrices of a braid representation on ordinary qubits and announcing that the chip now contains anyons. That is a braid *simulation*. It is evidence of correct control, not by itself evidence of emergent anyons.
5. Hearing that gates can be tuned continuously and concluding that their topological protection is continuous. Protection depends on the physical realization and its gap, locality, and error processes.
6. Confusing the road map with the city: a discrete braid group whose image is dense in a Lie group is still discrete.

C.11. Check yourself

1. **What extra structure turns a group into a Lie group?** A compatible smooth-manifold structure, making multiplication and inversion smooth.
2. **Show that** $U_z(2\pi) = -I$. $U_z(\theta) = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$, so $\theta = 2\pi$ yields $\text{diag}(-1, -1) = -I$.
3. **Show that B_3 need not be Abelian.** The defining relation equates three-letter words, not $\sigma_1\sigma_2$ with $\sigma_2\sigma_1$. The Ising matrices D and FDF^{-1} satisfy Yang–Baxter and fail to commute.
4. **Why is a braid more than a permutation?** The permutation forgets winding. A pure braid can return every label to its start while remaining topologically nontrivial.
5. **What fails if you identify B_n with its unitary image?** The image may be dense in a Lie group even though B_n is discrete. The photograph is not the city.
6. **How can a discrete braid group be related to continuous universal gates?** Its unitary image may be dense in a continuous projective unitary group.

C.12. Sources

- [R012] E. Artin, “Theory of Braids,” *Annals of Mathematics* **48**, 101–126 (1947). DOI: 10.2307/1969218.

- [R013] C. Kassel and V. Turaev, *Braid Groups*, Graduate Texts in Mathematics 247, Springer (2008). DOI: 10.1007/978-0-387-68548-9.
- [R014] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Springer (2015). DOI: 10.1007/978-3-319-13467-3.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv:0707.1889.
- [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” *Bulletin of the American Mathematical Society* **55**, 183–238 (2018). DOI: 10.1090/bull/1605; arXiv:1705.06206.

Appendix D. Four charges and a table of crossings

A memory aid, not a definition:

category $\subset$ monoidal category $\subset$ fusion category
 $\subset$ braided/ribbon structures $\subset$ modular tensor category.

The hooks are not set inclusions. Each one adds a picture and a rule. We start with the poorest picture: a labeled wire and an arrow.

D.1. A labeled wire

Chapter 14 already taught you to read a fusion tree. A labeled edge in that tree is a thing you can hold: a charge, or a bag of charges. Draw one of them standing alone.



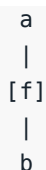
a
|
|
|

That standing wire is an *object*. Later we will have many of them, and we need a word that does not already mean “particle” or “state.” For anyons, an object is a charge type, or a direct sum of charge types.

Finite-dimensional complex Hilbert spaces are objects too. So is a single nameless point, if that is all you need.

D.2. An arrow between wires

Now put a box on the wire. The box is allowed to change the label.



a
|
[f]
|
b

Read top to bottom, as in Chapter 14. The box f is a process that starts at a and ends at b . Write $f : a \rightarrow b$. That process is a *morphism*. It is just “allowed arrow,” and we will spend the rest of the appendix stacking arrows.

For anyons, a morphism is an allowed fusion or splitting: a way to go from one charge (or sum of charges) to another. The vector space of all such arrows from a to b is written $\text{Hom}(a, b)$.

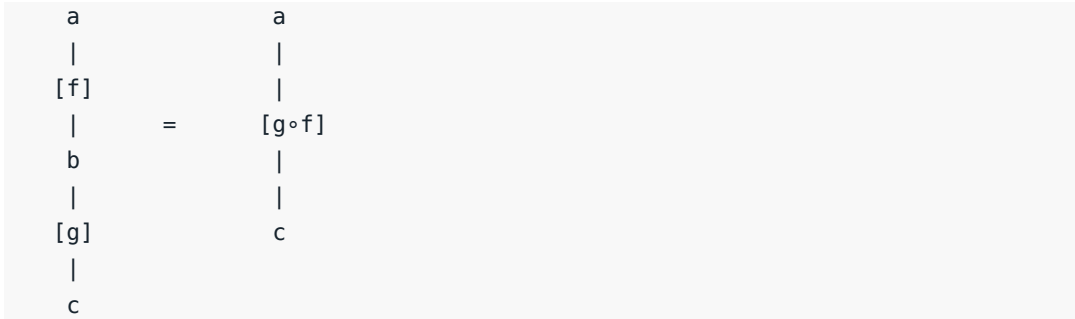
If a and b are distinct simple charges, superselection forbids the arrow:

$$\text{Hom}(a, b) = 0 \quad (a \neq b), \quad \text{Hom}(a, a) \cong \mathbb{C}.$$

The second line says an endomorphism of one simple charge is a complex scalar. It is the categorical version of Schur’s lemma.

D.3. Stacking two boxes

If $f : a \rightarrow b$ and $g : b \rightarrow c$ share the middle label, you may stack them.



The stacked process is the *composite* $g \circ f : a \rightarrow c$. Composition is associative, and a do-nothing box on a wire changes nothing:

$$h \circ (g \circ f) = (h \circ g) \circ f, \quad f \circ \text{id}_a = f = \text{id}_b \circ f.$$

The do-nothing box on a is the *identity* morphism $\text{id}_a : a \rightarrow a$.

Those four ingredients — objects, morphisms, associative composition, identities — are a *category*. The wire and the box are the things you draw. The category is the filing cabinet.

The category $\mathbf{Hilb}_{\text{fd}}$ has finite-dimensional complex Hilbert spaces as objects and linear maps as morphisms. A group is another example: one object, and the invertible morphisms

are the group elements. The abstraction remembers composition and throws away the furniture you do not need.

Remark. Drop identities and you can still stack some arrows, but a wire with no box is no longer a legal process. Drop associativity and three stacked boxes are not a single well-defined arrow.

D.4. Four charges you already know

The toric code has four simple charges

$$\{1, e, m, \varepsilon\},$$

where 1 is vacuum and $\varepsilon = e \otimes m$. We write $\otimes$ for categorical fusion; Chapter 16 often writes $\times$ for the same operation on charge labels. The fusion rules are

$$e \otimes e = m \otimes m = \varepsilon \otimes \varepsilon = 1,$$

$$e \otimes m = \varepsilon, \quad e \otimes \varepsilon = m, \quad m \otimes \varepsilon = e.$$

Every pair has one outcome, so every simple charge has quantum dimension one. The total quantum dimension is

$$\mathcal{D} = \sqrt{d_1^2 + d_e^2 + d_m^2 + d_\varepsilon^2} = 2.$$

Fusion alone does not reveal statistics. The additional braiding says that taking e completely around m contributes -1 , while e around e and m around m contributes $+1$. In the ordered basis $(1, e, m, \varepsilon)$, the normalized modular matrix is

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Our convention defines S_{ab} as $1/\mathcal{D}$ times the quantum trace of a full double braid of a and b . The first row therefore contains $d_a/\mathcal{D} = 1/2$. Direct multiplication gives $S^\dagger S = I$, so S is invertible. No non-vacuum charge braids trivially with every charge: e detects m , m detects e , and ε detects both.

The topological twists, or spins, are

$$\theta_1 = \theta_e = \theta_m = 1, \quad \theta_\varepsilon = -1.$$

Thus e and m have bosonic self-exchange while their composite ε is fermionic. Some authors define a modular T matrix as $T = \text{diag}(\theta_a)$; others multiply it by a framing-dependent common phase involving the chiral central charge. Here we use the unphased convention

$$T = \text{diag}(1, 1, 1, -1).$$

This worked example is a modular tensor category, yet all its anyons are Abelian. “Modular” does not mean “computationally universal.” It means, roughly, that braiding distinguishes all charge types [R017]; [R018].

D.5. Adding numbers to the arrows

A category is $\mathbb{C}$ -linear when every $\text{Hom}(a, b)$ is a complex vector space and composition is linear in each argument. It is *semisimple* when every object is a finite direct sum of simple objects. If a and b are objects, their direct sum $a \oplus b$ represents an alternative sector containing both as summands.

Choose representatives $\{a\}$ of the simple-object classes. In the anyon setting, these are the topological charge labels. “Same class” means isomorphic, not literally identical. An isomorphism $f : a \rightarrow b$ has an inverse $f^{-1} : b \rightarrow a$.

D.6. Putting two wires side by side

A lone wire is not enough. Fusion is two wires standing next to each other.

a	b

A *monoidal category* has a bifunctor $\otimes$ that combines objects and morphisms, a unit object 1 , and natural isomorphisms

$$\alpha_{a,b,c} : (a \otimes b) \otimes c \longrightarrow a \otimes (b \otimes c),$$

$$\lambda_a : 1 \otimes a \rightarrow a, \quad \rho_a : a \otimes 1 \rightarrow a.$$

The map α is the *associator*. Its components become F -matrices after bases are chosen. “Natural” means that the regrouping commutes with all compatible morphisms; it cannot depend on secret details of how an object was presented.

The associator and unit maps obey pentagon and triangle coherence equations. Consequently, all legal ways of removing parentheses agree as maps. One often suppresses parentheses in fusion rules, but the associator remains essential on fusion-space basis states. Appendix E writes the equations explicitly.

Remark. Drop the associator and two parenthesizations of three wires are merely two pictures. They are not yet two bases of one space.

D.7. Finite charges, duals, and a count

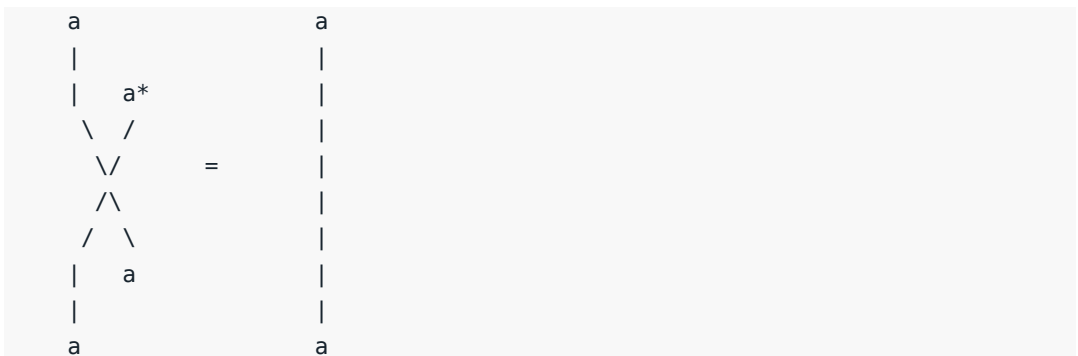
In this book, a *fusion category* means a $\mathbb{C}$ -linear, finite, semisimple, rigid monoidal category with a simple unit object and finite-dimensional morphism spaces [R019]; [R020]. The adjectives do real work:

- **finite**: finitely many simple-object classes;
- **semisimple**: objects split into finite direct sums of simples;
- **rigid**: every object has a dual;
- **simple unit**: $\text{End}(1) \cong \mathbb{C}$.

Rigidity assigns a dual a^* and evaluation/coevaluation morphisms

$$\text{ev}_a : a^* \otimes a \rightarrow 1, \quad \text{coev}_a : 1 \rightarrow a \otimes a^*.$$

They obey “snake” identities: create a dual pair and immediately annihilate the appropriate pair, and the untouched line acts as id_a .



Physically, a^* is the antiparticle $\bar{a}$.

The fusion coefficients are

$$N_{ab}^c = \dim \text{Hom}(c, a \otimes b),$$

which matches the convention that $a \otimes b \cong \bigoplus_c N_{ab}^c c$. Some texts use $\text{Hom}(a \otimes b, c)$ instead; duality relates the conventions, but indices must not be copied blindly.

Define the fusion matrix $(N_a)^c_b = N_{ab}^c$. Its Perron–Frobenius eigenvalue is the *Frobenius–Perron dimension* $\text{FPdim}(a)$. In a unitary fusion category it agrees with the positive categorical, or quantum, dimension d_a used in Chapter 14:

$$d_a d_b = \sum_c N_{ab}^c d_c, \quad d_1 = 1.$$

Outside the unitary setting, categorical dimensions can depend on pivotal structure and need not coincide automatically with Frobenius–Perron dimensions. The book’s physical anyon models assume unitarity unless stated otherwise.

Remark. Drop rigidity and there is no canonical antiparticle. Drop finiteness and the total quantum dimension need not be a finite number.

D.8. Crossing the wires

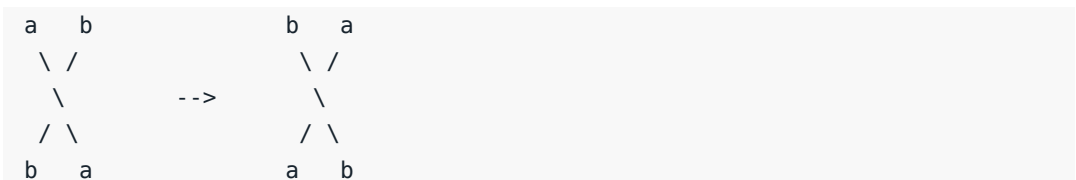
A fusion table still does not say what happens if the particles walk. A *braiding* is a natural isomorphism

$$c_{a,b} : a \otimes b \longrightarrow b \otimes a$$

compatible with the associator through two hexagon equations. In chosen fusion bases, $c_{a,b}$ is encoded by R -symbols. A braided category need not be symmetric: generally

$$c_{b,a} \circ c_{a,b} \neq \text{id}_{a \otimes b}.$$

That double exchange is precisely where anyonic monodromy lives.



A *twist* is a natural automorphism $\theta_a : a \rightarrow a$ representing a 2π twist of a labeled ribbon. On a simple object it is a phase, also called topological spin. Compatibility among duality, braiding, and twist gives a ribbon category. Ribbons remember framing; bare lines do not record self-twist consistently [R017]; [R019].

A *unitary braided fusion category* equips morphism spaces with adjoints and positive inner products so that structural maps can be chosen unitary. This is the setting in which F - and R -moves are quantum-mechanical basis changes and exchanges.

Remark. Drop the hexagons and a collection of R -matrices need not define a representation of the braid group. Drop symmetry and you have not dropped braiding: anyons live in the gap between “swap is involutive” and “swap does not exist.”

D.9. When braiding sees every charge

A charge x is *transparent* if its double braid with every object is the identity:

$$c_{a,x} \circ c_{x,a} = \text{id}_{x \otimes a} \quad \text{for all } a.$$

Transparent objects form the Müger center. A braided fusion category is *nondegenerate* when its only transparent simple object is the vacuum. With compatible ribbon and unitary structure, this is the modular tensor category used for $(2 + 1)$ -dimensional anyon models. Equivalently in this setting, the S matrix is invertible [R019]; [R016].

The adjective “tensor” refers to the monoidal product, not to a particular multidimensional array in software. The adjective “modular” refers to mapping-class-group data and non-degenerate braiding, not to modular computer construction.

What S and T know

The matrices S and T represent standard large diffeomorphisms of a torus on its topological ground-state space, up to convention-dependent phases. S encodes mutual braiding and T topological spins. They obey projective modular-group relations. They are powerful fingerprints but not, in full generality, complete invariants: inequivalent modular categories can share the same modular data. Full F - and R -data retain more information [R019]; [R016].

Look back at the four-charge table. S is invertible, only vacuum is transparent, and $T = \text{diag}(1, 1, 1, -1)$. That is why the toric-code example sits at the top of the memory-aid ladder even though every charge is Abelian.

D.10. When two filing cabinets are the same

A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ maps objects to objects and morphisms to morphisms while preserving identities and composition:

$$F(\mathrm{id}_a) = \mathrm{id}_{F(a)}, \quad F(g \circ f) = F(g) \circ F(f).$$

A monoidal functor also comes with coherent isomorphisms relating $F(a) \otimes F(b)$ to $F(a \otimes b)$ and the two unit objects. A braided monoidal functor must preserve braiding as well. These qualifiers prevent a map from preserving the fusion table while quietly discarding the exchange data.

Two categories are *equivalent* when functors between them preserve their structure up to natural isomorphism; literal equality of object names is unnecessary. For anyon theories, the relevant comparison is normally unitary braided or ribbon equivalence, not merely equality of fusion coefficients. Relabeling e and m in the toric code preserves the full theory because the fusion, twists, and mutual braiding transform consistently. By contrast, a theory and its mirror can share fusion rules while having reversed braiding. They are not generally equivalent as oriented braided theories.

This distinction is practical. A numerical model may identify charge sectors whose labels differ from a reference convention. One should seek a structure-preserving relabeling and compatible basis transformations, then compare gauge-invariant closed operations. Matching names or one fusion table is weaker evidence.

D.11. What the filing cabinet does not know

[Theory] A modular tensor category describes universal long-distance fusion and braiding data of an ideal gapped topological phase. It does not specify the microscopic Hamiltonian, energy gap, quasiparticle mass, coherence time, temperature tolerance, or fabrication route [R017]; [R018]. Those require separate physics.

A physical qubit is a controlled two-level subsystem, not an object of an emergent anyon category by declaration. An encoded qubit is a chosen subspace. It becomes topological only when the physical system supplies the required nonlocal organization and protection. A digital emulator may implement morphisms and modular matrices faithfully without possessing the corresponding equilibrium phase.

Extrinsic symmetry defects can obey generalized, often G -crossed, fusion rules. They are not automatically intrinsic simple objects of the host modular category; gauging the symmetry can change that status [R021]. Doubled phases arising from Drinfeld centers contain a sector and its opposite-chirality partner. “Doubled Fibonacci” is not a casual synonym for chiral Fibonacci order.

D.12. Check yourself

1. **What are the four minimum ingredients of a category?** Objects, morphisms, associative composition, and identity morphisms.
2. **Show that the toric-code S matrix is invertible.** Direct multiplication gives $S^\dagger S = I$. Equivalently, no non-vacuum charge is transparent: e detects m , m detects e , and ε detects both.
3. **What categorical map becomes an F -matrix?** The associator $\alpha_{a,b,c}$ after fusion-space bases are chosen.
4. **Show that a fusion category can lack braiding.** Fusion and associativity supply $\otimes$ and α . They do not by themselves supply an exchange map $c_{a,b}$.
5. **What makes braiding modular?** Nondegeneracy: only vacuum is transparent, equivalently the S matrix is invertible in the unitary ribbon setting.
6. **What fails if you treat modular data as an energy gap?** S and T describe topological data. They do not specify microscopic energy scales.

Later chapters may treat simple objects as charges, $\text{Hom}(c, a \otimes b)$ as fusion spaces, the associator as F , the braiding as R , dual objects as antiparticles, and nondegenerate ribbon braiding as modular structure. Appendix E fixes bases, exposes gauge dependence, and states the pentagon and hexagon equations that make this graphical calculus coherent.

D.13. Sources

- [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv:cond-mat/0506438.
- [R018] M. A. Levin and X.-G. Wen, “String-Net Condensation: A Physical Mechanism for Topological Phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv:cond-mat/0404617.
- [R019] B. Bakalov and A. Kirillov Jr., *Lectures on Tensor Categories and Modular Functors*, University Lecture Series 21, American Mathematical Society (2001). DOI: 10.1090/ulect/021.
- [R020] P. Etingof, S. Gelaki, D. Nikshych, and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society (2015). DOI: 10.1090/surv/205.
- [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” *Bulletin of the American Mathematical Society* **55**, 183–238 (2018). DOI: 10.1090/bull/1605; arXiv:1705.06206.

- [R021] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, “Symmetry Fractionalization, Defects, and Gauging of Topological Phases,” *Physical Review B* **100**, 115147 (2019). DOI: 10.1103/PhysRevB.100.115147; arXiv:1410.4540.

Appendix E. Two trees that must agree

Three leftover particles sit in a line. Call their charges a , b , and c . The bag they make together has a fixed total charge d .

You can pair them two ways. Fuse a with b first, then add c . Or fuse b with c first, then add a . The drawings look different. The particles have not moved. Only the parentheses have.

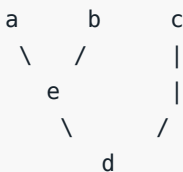
Those two descriptions are two bases of one space. Later braid calculations may use either tree. They are not allowed to depend on which intern parenthesized first.

E.1. Left pairing and right pairing

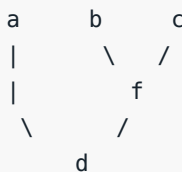
Draw incoming charges at the top and the total at the bottom.

Read top to bottom

$((a\ b)_e\ c)_d$



$(a\ (b\ c)_f)_d$

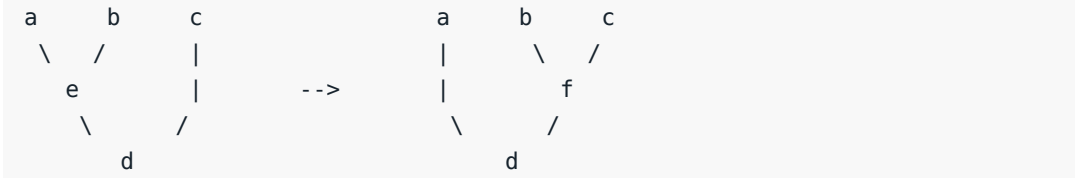


The internal labels e and f are the allowed intermediate charges. If a vertex has more than one independent way to fuse the same three labels, it also carries a basis label. We use orthonormal vertex bases and downward-oriented diagrams.

Each drawing is a *fusion tree*: a choice of parentheses, together with a choice of intermediate charges, for a many-particle state of fixed external labels. We take this as the definition. The tree records a basis, not a claim that one collision happened before another.

E.2. The slide that does not move anyone

The particles stay put. Only the pairing slides:



Write the left-tree state with vertex labels μ on the $a, b \rightarrow e$ fork and ν on the $e, c \rightarrow d$ fork. Write the right-tree state with α on $b, c \rightarrow f$ and β on $a, f \rightarrow d$. The slide is a change of basis:

$$|((ab)_e c)_d; \mu, \nu\rangle = \sum_{f, \alpha, \beta} [F_d^{abc}]_{(e, \mu, \nu)(f, \alpha, \beta)} |(a(bc)_f)_d; \alpha, \beta\rangle.$$

The range of each label is a fusion multiplicity: $\mu = 1, \dots, N_{ab}^e$, $\nu = 1, \dots, N_{ec}^d$, $\alpha = 1, \dots, N_{bc}^f$, and $\beta = 1, \dots, N_{af}^d$. The nonnegative integer N_{ab}^e counts independent ways for a and b to total e .

Those matrix entries are the ***F*-symbols**. They compare two descriptions of the same three-body state. They are not an extra physical event inserted between the particles.

For a unitary anyon theory, F_d^{abc} is a unitary matrix on the allowed fixed- d fusion space. Drop unitarity and the two trees still describe the same abstract space, but probabilities read from one tree need no longer match those read from the other after a naive conjugate transpose.

E.3. The swap that does move them

Now let two charges actually trade places. Keep their total charge c fixed. A chosen counterclockwise exchange is a linear map

$$R_c^{ab} : V_{ab}^c \longrightarrow V_{ba}^c.$$

Here V_{ab}^c is the space of ways for a and b to total c .

The entries of that map are the ***R*-symbols**. This is a real exchange, not another renaming of parentheses.

In a multiplicity-free theory, R_c^{ab} is a phase for each allowed channel c . With multiplicities it is a unitary matrix between the corresponding vertex spaces. A clockwise exchange is

$(R_c^{ba})^{-1}$ with these oriented-domain conventions. It is unsafe to replace every R by a bare complex conjugate without checking labels.

Suppose you drop the orientation. Then a clockwise word and a counterclockwise word become interchangeable on the page, while the operators they name need not be. The later hexagon equations will refuse that sloppiness.

E.4. Three sigmas

The Ising charges $1, \psi$, and σ obey

$$\sigma \times \sigma = 1 + \psi, \quad \psi \times \sigma = \sigma, \quad \psi \times \psi = 1.$$

Take three σ anyons of total charge σ . The left intermediate label is $e \in \{1, \psi\}$. In a standard gauge,

$$F \equiv F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_{\psi}^{\sigma\sigma} = e^{3i\pi/8}.$$

Rows and columns of F are ordered $(1, \psi)$. Direct multiplication gives $F^{\dagger}F = I$ and $F^{-1} = F$. Hence a state with left channel $e = 1$ becomes

$$|1\rangle_L = \frac{|1\rangle_R + |\psi\rangle_R}{\sqrt{2}}.$$

Measuring the right fusion channel therefore gives either result with probability $1/2$. The fusion rule listed the two possibilities. F supplied the amplitudes.

Let

$$B_1 = \begin{pmatrix} R_1^{\sigma\sigma} & 0 \\ 0 & R_{\psi}^{\sigma\sigma} \end{pmatrix} = e^{-i\pi/8} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

exchange the first pair in the left basis. To exchange the second pair, expose its channel with F :

$$B_2 = FB_1F^{-1} = \frac{e^{-i\pi/8}}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Write $D_0 = \text{diag}(1, i)$ and omit the common phase temporarily. Direct multiplication yields

$$D_0(FD_0F)D_0 = (FD_0F)D_0(FD_0F) = \frac{1+i}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Restoring the same phase $e^{-3i\pi/8}$ on both sides proves

$$B_1B_2B_1 = B_2B_1B_2.$$

Hence the local symbols produce a representation of the three-strand braid relation. They also give $B_1B_2 \neq B_2B_1$, so the representation is non-Abelian. Consistency does not imply commutativity [R015]; [R017].

E.5. The same slide, without a basis yet

Before anyone chooses orthonormal forks, the regrouping is a natural isomorphism

$$\alpha_{a,b,c} : (a \otimes b) \otimes c \rightarrow a \otimes (b \otimes c).$$

That map is the *associator*. An F -symbol is a matrix element of α after bases of the relevant fusion spaces have been selected.

For fixed a, b, c, d , unitarity of those matrix elements says

$$\sum_{f,\alpha,\beta} [F_d^{abc}]_{(e,\mu,\nu)(f,\alpha,\beta)} [F_d^{abc}]_{(e',\mu',\nu')(f,\alpha,\beta)}^* = \delta_{ee'} \delta_{\mu\mu'} \delta_{\nu\nu'},$$

where $*$ denotes complex conjugation and δ is the Kronecker delta. This equation presumes the displayed vertex bases are orthonormal. Drop orthonormality and the same abstract α is still an isomorphism; the displayed conjugate-transpose identity is not.

E.6. Four charges, two routes

Four objects can be re-associated from $((a \otimes b) \otimes c) \otimes d$ to $a \otimes (b \otimes (c \otimes d))$ along two routes:

$$\begin{array}{ccc}
 & ((a\ b)\ (c\ d)) & \\
 & / \qquad \qquad \backslash & \\
 ((a\ b)\ c)\ d & & (a\ (b\ (c\ d))) \\
 \backslash & & / \\
 ((a\ (b\ c))\ d) \rightarrow (a\ ((b\ c)\ d))
 \end{array}$$

The upper route uses two associators. The lower route uses three. Equality of those two composite maps is the **pentagon equation**:

$$\alpha_{a,b,c \otimes d} \circ \alpha_{a \otimes b, c, d} = (\text{id}_a \otimes \alpha_{b,c,d}) \circ \alpha_{a,b \otimes c, d} \circ (\alpha_{a,b,c} \otimes \text{id}_d).$$

Both sides have the same source and target. The rightmost map acts first. Expanding this operator equation in fusion-tree bases gives polynomial equations in the entries of the F -matrices, with sums over every allowed intermediate charge and multiplicity label [R022]; [R019].

A quick limiting example is a pointed model in a gauge where every allowed fusion space is one-dimensional and every associator is $F = 1$. The pentagon reads $1 \cdot 1 = 1 \cdot 1 \cdot 1$. Nontrivial pointed categories may instead carry phase-valued associators classified by a group 3-cocycle; then the pentagon is exactly the cocycle condition. For Ising and Fibonacci theories, nontrivial matrix F -symbols solve the same equation.

The pentagon is not optional aesthetic tidying. Without it, a four-anyon amplitude would depend on which sequence of intermediate basis changes a calculator happened to choose.

E.7. Exchange past a pair

Let

$$\beta_{a,b} : a \otimes b \rightarrow b \otimes a$$

be the categorical braiding whose chosen-basis entries are R -symbols. Exchanging a past the composite $b \otimes c$ must agree with exchanging it past b and then past c . Because the parentheses differ along the way, associators surround the exchanges.

With the rightmost map acting first, the positive-crossing equation is

$$\alpha_{b,c,a} \circ \beta_{a,b \otimes c} \circ \alpha_{a,b,c} = (\text{id}_b \otimes \beta_{a,c}) \circ \alpha_{b,a,c} \circ (\beta_{a,b} \otimes \text{id}_c).$$

Both sides map $(a \otimes b) \otimes c$ to $b \otimes (c \otimes a)$. The companion equation moves a composite past c :

$$\alpha_{c,a,b}^{-1} \circ \beta_{a \otimes b, c} \circ \alpha_{a,b,c}^{-1} = (\beta_{a,c} \otimes \text{id}_b) \circ \alpha_{a,c,b}^{-1} \circ (\text{id}_a \otimes \beta_{b,c}).$$

Both sides map $a \otimes (b \otimes c)$ to $(c \otimes a) \otimes b$. These two identities are the **hexagon equations**. Each is a six-sided commute: three associators and three exchanges, drawn as a hexagon, that must give one linear map.

Together with the pentagon, they ensure that F -moves and crossings give an unambiguous value to any equivalent labeled ribbon diagram [R022]; [R023]. Drop either hexagon and exchange ceases to be compatible with fusion on one side of the tensor product. The two equations are not redundant copies of one slogan.

For reference, one component of the first hexagon in a multiplicity-free theory is

$$R_e^{ab} [F_d^{bac}]_{eg} R_g^{ac} = \sum_f [F_d^{abc}]_{ef} R_d^{af} [F_d^{bca}]_{fg}.$$

It follows the convention at the start of this appendix. The sum runs only over allowed intermediate charges f . Sources using upward trees, clockwise R , or the inverse definition of F will display a different-looking but equivalent equation.

E.8. Why the braid relation follows

On three anyons, define B_1 by applying R to the first pair in the chosen left-associated basis. Define B_2 by an F -move, an R -move on the second pair, and the inverse F -move. The hexagon equations make exchange compatible with fusion, while the pentagon makes every re-association route agree. Consequently

$$B_i B_{i+1} B_i = B_{i+1} B_i B_{i+1}, \quad B_i B_j = B_j B_i \quad (|i - j| \geq 2).$$

The Ising calculation above checks the first identity in the smallest matrix example. Satisfying these relations is necessary for a braid representation. It does not imply that its image is dense or computationally universal.

E.9. A full trip around

A single exchange depends on the ordered pair a, b . A full monodromy returns the objects to their original order:

$$M_c^{ab} = R_c^{ab} R_c^{ba}$$

in a multiplicity-free fusion channel c . That product is the *double braid*. One swap still remembers who went first; two swaps is the interferometric trip that brings everyone home.

In a ribbon theory, the balancing relation connects this double braid to the topological twists:

$$M_c^{ab} = \frac{\theta_c}{\theta_a \theta_b},$$

where θ_x is the phase for a 2π twist of simple charge x . This combination is invariant under the vertex rephasings described below. It is therefore closer to an interferometric observable than either single R phase considered in isolation [R015]; [R023].

For the Ising channel $a = b = \sigma$, the standard twists are $\theta_1 = 1$, $\theta_\psi = -1$, and $\theta_\sigma = e^{i\pi/8}$. The vacuum channel gives

$$M_1^{\sigma\sigma} = (e^{-i\pi/8})^2 = e^{-i\pi/4} = \frac{\theta_1}{\theta_\sigma^2},$$

while the ψ channel gives

$$M_\psi^{\sigma\sigma} = (e^{3i\pi/8})^2 = e^{3i\pi/4} = \frac{\theta_\psi}{\theta_\sigma^2}.$$

Hence the relative sign distinguishes the two fusion channels. A common framing convention can shift how individual twist phases are reported, so the convention must accompany the table.

E.10. Relabeling a vertex

Choose a different orthonormal basis at each fusion vertex. In a multiplicity-free theory this is a phase

$$|a, b; c\rangle' = u_c^{ab} |a, b; c\rangle, \quad |u_c^{ab}| = 1.$$

The symbols then transform as

$$[F_d^{abc}]'_{ef} = \frac{u_e^{ab} u_d^{ec}}{u_f^{bc} u_d^{af}} [F_d^{abc}]_{ef},$$

$$[R_c^{ab}]' = \frac{u_c^{ab}}{u_c^{ba}} R_c^{ab}.$$

With fusion multiplicities, each u becomes a unitary matrix and the formulas acquire the corresponding matrix contractions. These rephasings are a **gauge** change: the same physics, written in a different vertex basis. Gauge changes preserve the pentagon and hexagon equations. They also conjugate complete braid operators when the external computational basis is changed.

Therefore an isolated entry such as $[F]_{ef}$ is not generally observable. Gauge-invariant information includes consistently defined fusion probabilities, eigenvalues or traces of closed braid operations, topological spins, and modular data. Even these require a stated framing and overall-phase convention when compared across sources [R015]; [R019].

The **mirror theory** reverses crossings. In a unitary convention its braiding data are related to inverse/complex-conjugate data, but orientation labels and framing phases must be tracked. This is why chiral Fibonacci and its mirror, or a doubled theory containing both, cannot be interchanged by deleting a minus sign.

E.11. What a table does not prove

[Theory] A table of N , F , and R that satisfies unitarity, pentagon, and hexagon is internally consistent topological data. It is not yet proof that a local Hamiltonian realizes the data, nor does it provide a many-body gap [R017]; [R019].

[Numerics] Extracting approximate matrices from a finite system should include basis conventions, finite-size scaling, leakage outside the proposed fusion space, and residual path dependence. Agreement for one braid word cannot test the full coherence structure.

[Experiment] A processor that compiles the Ising matrices above demonstrates the programmed representation to its measured accuracy. Intrinsic anyons require additional evidence that localized excitations, fusion sectors, and braid response emerge from the device's many-body physics. An encoded subspace with cleverly named basis vectors remains an encoded subspace.

E.12. Ways to get this wrong

1. Multiplying braid words in one time convention and drawing them in the opposite convention.
2. Using F where F^{-1} is required when returning to the original tree.

3. Summing over forbidden intermediate labels instead of enforcing the fusion coefficients.
4. Dropping vertex multiplicity indices when $N_{ab}^c > 1$.
5. Comparing raw symbols from different gauges as if disagreement were physical.
6. Treating pentagon and hexagon residuals as harmless numerical noise without comparing them with the claimed precision.
7. Concluding universality merely from noncommuting braid matrices.

E.13. Check yourself

1. **Show that $F_{\sigma}^{\sigma\sigma\sigma}$ equals its own inverse.**
Direct multiplication of the displayed matrix with itself returns I . Combined with $F^{\dagger}F = I$, one also has $F^{-1} = F$.
2. **What does an R -symbol do?**
It represents an oriented exchange of two charges in a specified total fusion channel.
3. **Show that both sides of the pentagon have the same source and target.**
Each composite starts at $((a \otimes b) \otimes c) \otimes d$ and ends at $a \otimes (b \otimes (c \otimes d))$. The two routes are different factorizations of one map.
4. **Why are there two hexagons?**
Braiding must be compatible with tensor products on either side; the equations cover moving one object past a composite and a composite past one object.
5. **Show that $M_1^{\sigma\sigma} = \theta_1/\theta_{\sigma}^2$ from the listed phases.**
 $R_1^{\sigma\sigma} = e^{-i\pi/8}$ squares to $e^{-i\pi/4}$, while $\theta_1/\theta_{\sigma}^2 = 1/e^{i\pi/4} = e^{-i\pi/4}$.
6. **What fails if you drop the pentagon?**
A four-anyon amplitude can depend on which sequence of intermediate basis changes a calculator chose. The symbols no longer name a single linear map.

E.14. Sources

- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics* **80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv:0707.1889.
- [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv:cond-mat/0506438.
- [R022] G. Moore and N. Seiberg, “Classical and Quantum Conformal Field Theory,” *Communications in Mathematical Physics* **123**, 177–254 (1989). DOI: 10.1007/BF01238857.

- [R019] B. Bakalov and A. Kirillov Jr., *Lectures on Tensor Categories and Modular Functors*, University Lecture Series 21, American Mathematical Society (2001). DOI: 10.1090/ulect/021.
- [R023] P. Bonderson, K. Shtengel, and J. K. Slingerland, “Interferometry of Non-Abelian Anyons,” *Annals of Physics* **323**, 2709–2755 (2008). DOI: 10.1016/j.aop.2008.01.012; arXiv:0707.4206.

Appendix F. Rotate the expensive rooms away

You already folded two cheap states and one expensive state down to a leftover 2×2 . The downstairs door was $-g^2/\Delta$. The upstairs room never appeared as a place you would find the system if you measured energy. It appeared as a denominator.

That calculation was a special case. The same folding works for a whole cheap family that may share one energy. We will write the rotation, then check it on two Hubbard sites you can still do by hand.

The move is not amputation. If you simply delete the expensive basis vectors, you throw away every return trip. The honest move is a small unitary rotation of the full space. After the rotation, the cheap block is isolated to a chosen order. You keep that block.

before rotation		after rotation	
P states	<--- V ---> Q states	P block	0
		-----+	-----
		0	Q block

When several cheap states share one unperturbed energy, the one-state textbook formula dies: a denominator hits zero. Treat the whole cheap family as one block. That is **degenerate perturbation theory**. The degeneracy is why you rotate a subspace instead of a single state.

F.1. Split the page, don't tear it

Cut the space with two complementary machines P and Q . They are orthogonal projectors:

$$P^2 = P, \quad Q^2 = Q, \quad PQ = 0, \quad P + Q = I.$$

The identity operator is I . States in P stay on the working page. States in Q leave it. Write the Hamiltonian as a solvable piece plus a kick,

$$H = H_0 + \lambda V,$$

where H_0 is the unperturbed Hamiltonian you can diagonalize, V is a Hermitian perturbation with units of energy, and λ is a dimensionless bookmark. Set λ to one after the expansion, once you have decided that the physical ratio is small.

Assume first that every retained state has the same unperturbed energy E_0 :

$$PH_0P = E_0P,$$

and that no eigenvalue of QH_0Q equals E_0 . The gap to the thrown-away room is

$$\Delta_0 = \min_{\mu \in Q} |E_\mu - E_0| > 0,$$

where E_μ are the eigenvalues of H_0 in Q . Hence the inverse $(E_0 - QH_0Q)^{-1}$ exists on the Q subspace. Drop the gap and the inverse is not an operator on Q ; it is a request for a different splitting.

Split any operator into the part that stays inside a room and the part that is the stair:

$$V_d = PVP + QVQ, \quad V_{od} = PVQ + QVP.$$

The off-diagonal piece is what mixes the two sectors. First-order projection, PHP , simply deletes the stair. We are going to record the return trips instead.

F.2. A small twist that hides the stair

Choose an anti-Hermitian generator $S^\dagger = -S$ that lives only on the stair, $PSP = QSQ = 0$. Then $U = e^S$ is unitary. Define the rotated Hamiltonian

$$\tilde{H} = e^S H e^{-S}.$$

The Baker–Campbell–Hausdorff expansion is

$$\tilde{H} = H + [S, H] + \frac{1}{2!}[S, [S, H]] + \dots,$$

where $[A, B] = AB - BA$ is the commutator. Expand the generator as

$$S = \lambda S_1 + \lambda^2 S_2 + \dots.$$

At first order, the leftover stair is

$$\lambda(V_{\text{od}} + [S_1, H_0]).$$

Cancel it by requiring

$$[H_0, S_1] = V_{\text{od}}.$$

If $|a\rangle$ lies in P and $|\mu\rangle$ lies in Q , this gives

$$\langle a|S_1|\mu\rangle = \frac{\langle a|V|\mu\rangle}{E_0 - E_\mu}.$$

The reverse matrix element is fixed by anti-Hermiticity. Project the rotated Hamiltonian into P and you get, through second order,

$$H_{\text{eff}} = E_0 P + \lambda P V P + \lambda^2 P V Q \frac{1}{E_0 - Q H_0 Q} Q V P + O(\lambda^3).$$

Hence the leftover downstairs operator is not a new physical system. It is the cheap block of a rotated copy of the old one.

The boxed form is Hermitian. A state leaves P through V , acquires an energy denominator in Q , and returns through V . Units check immediately:

$$[V][E^{-1}][V] = E E^{-1} E = E.$$

If $E_\mu > E_0$, the denominator $E_0 - E_\mu$ is negative, so a coupled low-energy state generally shifts downward. The sign is derived. It is not supplied by the word “virtual.” The systematic many-body formulation, including locality and error bounds, is developed by Bravyi, DiVincenzo, and Loss [R025].

This small twist — hide the stair in the cheap block, order by order, without changing the exact spectrum — is the **Schrieffer–Wolff** transformation. The name is worth keeping because it is the standard label for this particular folding, written for a different downstairs problem and now used wherever a cheap block still hears the expensive rooms [R024]; [R025].

When the cheap states are not exactly degenerate

Let $|a\rangle$ and $|b\rangle$ in P have possibly different unperturbed energies E_a and E_b . A Hermitian second-order matrix is

$$(H_{\text{eff}}^{(2)})_{ab} = \frac{1}{2} \sum_{\mu \in Q} V_{a\mu} V_{\mu b} \left(\frac{1}{E_a - E_\mu} + \frac{1}{E_b - E_\mu} \right).$$

Here $V_{a\mu} = \langle a|V|\mu\rangle$. If $E_a = E_b = E_0$, this reduces to the boxed expression. Different leftover-Hamiltonian conventions can differ by a unitary transformation acting only inside P ; spectra and consistently transformed observables agree to the retained order [R025].

F.3. Two electrons that hop and then pay

Take two sites, labeled 1 and 2, with one spin-1/2 fermion per site. Let $c_{i\sigma}^\dagger$ create a fermion of spin $\sigma \in \{\uparrow, \downarrow\}$ at site i , and let $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. The two-site Hubbard Hamiltonian is

$$H_0 = U \sum_{i=1}^2 n_{i\uparrow} n_{i\downarrow},$$

$$V = -t \sum_{\sigma} \left(c_{1\sigma}^\dagger c_{2\sigma} + c_{2\sigma}^\dagger c_{1\sigma} \right).$$

The on-site repulsion $U > 0$ and hopping amplitude t both have units of energy. The retained subspace P contains exactly one fermion on each site, so $E_0 = 0$. The eliminated subspace Q contains a doubly occupied site and an empty site, at energy U . We assume $|t|/U \ll 1$.

First order vanishes: one hop always leaves the one-particle-per-site sector, so $PVP = 0$. At second order,

$$H_{\text{eff}}^{(2)} = -\frac{1}{U} PVQP.$$

Introduce the spin operator on site i ,

$$\mathbf{S}_i = \frac{1}{2} \sum_{\alpha, \beta} c_{i\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} c_{i\beta},$$

where $\boldsymbol{\sigma} = (X, Y, Z)$ is the vector of Pauli matrices. Evaluating the two-hop processes gives [R026]

$$H_{\text{eff}}^{(2)} = \frac{4t^2}{U} \left(\mathbf{S}_1 \cdot \mathbf{S}_2 - \frac{1}{4} n_1 n_2 \right),$$

where $n_i = n_{i\uparrow} + n_{i\downarrow}$. Inside P , $n_1 n_2 = 1$. Define $J_{\text{ex}} = 4t^2/U$. For the spin singlet, $\mathbf{S}_1 \cdot \mathbf{S}_2 = -3/4$, so the shift is $-J_{\text{ex}} = -4t^2/U$. For any triplet, $\mathbf{S}_1 \cdot \mathbf{S}_2 = +1/4$, so the shift is zero. The singlet–triplet separation is therefore

$$\Delta_{ST} = \frac{4t^2}{U}.$$

This is antiferromagnetic *superexchange*. The leftover coupling is not a new downstairs door. It is two hops and one upstairs bill. The generated interaction is $4t(t/U)$, smaller than the bare hopping when $|t|/U \ll 1$.

The Pauli principle supplies the physical distinction. A triplet's symmetric spin state requires an antisymmetric spatial state, blocking the relevant double occupancy. The singlet can virtually double-occupy a site and lower its energy. The algebra and the story agree.

F.4. Six steps before you trust the leftover

For a finite problem, use this sequence.

1. **Choose P physically.** It should contain every state whose energy differences are comparable to the perturbation. Excluding one near-resonant state creates a small denominator rather than a simpler theory.
2. **Diagonalize PVP .** At first order, its eigenvectors are the correct combinations within an exactly degenerate manifold.
3. **Build the resolvent.** Evaluate $(E_0 - QH_0Q)^{-1}$ only on Q . That inverse is the *resolvent* of the thrown-away block: divide by the energy you would have paid upstairs.
4. **Construct H_{eff} .** Keep all operators at the same perturbative order, including unwanted fields and constants.
5. **Transform observables too.** For a microscopic observable O , use

$$O_{\text{eff}} = Pe^S O e^{-S} P.$$

Projecting *POP* alone can miss virtual corrections.

6. **Compare against the full model.** On the smallest tractable cluster, compare low-energy eigenvalues, eigenvectors or subspaces, and observables as $|\lambda|$ varies.

A useful expansion requires matrix elements coupling P and Q to be small relative to the relevant denominators. In a finite system, a rough dimensionless control parameter is $\|V_{\text{od}}\|/\Delta_0$, where $\|\cdot\|$ is the operator norm. In an extended lattice, the norm of the entire perturbation grows with system size; rigorous work instead exploits locality and linked clusters [R025]. Do not reject a local expansion merely because an extensive global norm is extensive, and do not declare it controlled merely because one typical bond is small.

There are two distinct questions. First, does an exact unitary exist that maps the perturbed low-energy spectral subspace back to the reference subspace P ? For a finite-dimensional problem with an isolated spectral band that remains isolated, the answer is yes; this is

the geometric foundation of exact SW theory [R025]; [R027]. Second, does a short Taylor series approximate that unitary accurately? That requires a small parameter and a truncation check. Existence of the exact rotation does not make the second-order formula accurate.

An effective Hamiltonian is also defined only up to an additive scalar and a unitary basis change within P . Additive scalars do not affect dynamics or gaps. Internal basis rotations do change matrix entries, so a term-by-term comparison between two derivations is meaningful only after their conventions are aligned. Compare invariant spectra first, then compare observables transformed in the same convention.

A many-body leftover that looks like a toric-code check can appear this way. That does not, by itself, prove the leftover Hamiltonian is topologically ordered. You still have to study the whole cheap spectrum.

When you later arrange virtual paths so that the first few orders cancel on purpose, put the derived coefficient, the leading remainder, and every symmetry-breaking leftover in the same table. Before a larger calculation uses the expansion, compare the leftover spectrum with the full one on the smallest cluster you can still diagonalize. If an order- m approximation is controlled, the residual should decrease with the next power over a genuine small-parameter window. Fit no exponent from two points. Also compare the retained subspace using projectors, because eigenvectors within a degenerate manifold can rotate without changing the physics.

The original SW transformation was introduced to relate the Anderson and Kondo descriptions [R024]. Modern rigorous treatments show that it is not a license to keep only appealing terms [R025].

F.5. Ways to get this wrong

1. Deleting the Q states and calling PHP the theory. Return trips through Q shift energies and invent couplings.
2. Leaving one near-resonant state out of P . That state is an **intruder**: it walks into the cheap band after the kick is turned on, and the denominator you trusted becomes a land mine.
3. Treating a small typical bond as a controlled expansion on a large lattice, or rejecting a local expansion because an extensive global norm is extensive.
4. Declaring the second-order formula accurate because an exact block-diagonalizing unitary exists. Existence is not a truncation check.
5. Comparing two leftover Hamiltonians term by term before aligning their conventions inside P .
6. Using the leftover Hamiltonian with untransformed microscopic observables.
7. Letting a desired fourth-order term compete, unnoticed, with an unwanted first- or second-order field.

8. Calling the leftover an encoded code, a topological phase, or an anyon. Those are properties of the resulting many-body Hamiltonian and states. The rotation only built an effective local degree of freedom and some leftover interactions.

Watch also for resonance, a Q state approaching the retained energies, and for fabrication that changes both numerators and leakage gaps at once.

F.6. Check yourself

1. **Why not delete the Q states immediately?** Because virtual transitions through Q shift energies and generate interactions in P .
2. **Show that if $E_\mu > E_0$, a coupled low-energy state generally shifts downward at second order.** The resolvent entry is $1/(E_0 - E_\mu) < 0$, and $PVQ(\dots)QVP$ therefore contributes a negative-semidefinite correction on P .
3. **Show that the nondegenerate formula reduces to the boxed one when $E_a = E_b = E_0$.** The two denominators become equal, the average is $1/(E_0 - E_\mu)$, and the sum over μ is the matrix element of $PVQ(E_0 - QH_0Q)^{-1}QVP$.
4. **Why does two-site Hubbard hopping produce $4t^2/U$?** Two hops return to the retained sector; their amplitudes scale as t^2 , the virtual double occupancy costs U , and spin/path multiplicities give the factor four.
5. **What else must be transformed?** Observables and states, not only the Hamiltonian.
6. **What fails if you leave a near-resonant state out of P ?** The denominator you trusted becomes small or zero. The state is an intruder: it walks into the cheap band after the kick is turned on.

F.7. Sources

- [R024] J. R. Schrieffer and P. A. Wolff, “Relation between the Anderson and Kondo Hamiltonians,” *Physical Review* **149**, 491–492 (1966). DOI: 10.1103/PhysRev.149.491.
- [R025] Sergey Bravyi, David P. DiVincenzo, and Daniel Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* **326**, 2793–2826 (2011). DOI: 10.1016/j.aop.2011.06.004; arXiv: 1105.0675.
- [R026] A. H. MacDonald, S. M. Girvin, and D. Yoshioka, “ t/U expansion for the Hubbard model,” *Physical Review B* **37**, 9753–9756 (1988). DOI: 10.1103/PhysRevB.37.9753.

- [R027] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer (1976; corrected printing 1995). DOI: 10.1007/978-3-642-66282-9.

Appendix G. Checks that never read the secret

Three coins sit on a table. They are supposed to show the same face. You may look at whether coin 1 matches coin 2, and whether coin 2 matches coin 3. If both checks pass, all three agree. You still do not know whether they are heads or tails.

That unknown is the secret. The checks never read it.

A general state of n qubits needs 2^n complex amplitudes. Some important states can be named the way those coins were named: by a short list of agreement rules, not by writing every amplitude. “All three bits agree” is shorter than listing eight strings and setting six of them to zero.

G.1. An operator that leaves the good states alone

Suppose a state $|\psi\rangle$ is already one of the allowed ones. There is an operator S that does nothing to it:

$$S|\psi\rangle = |\psi\rangle.$$

The operator leaves the state fixed. That is a **stabilizer**. The good states are the ones this check has already passed: the answer is +1, as if the coins already matched.

A stabilizer code is the common +1 eigenspace of a commuting set of Pauli operators. The compression is what makes error correction, the toric code, and efficient classical simulation of Clifford circuits possible [R028]; [R031]. It does not efficiently describe every quantum state or every gate. Magic remains expensive.

G.2. Four matrices on one qubit

On one qubit the four matrices are

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They obey $X^2 = Y^2 = Z^2 = I$ and, for example, $XZ = -ZX$. The n -qubit **Pauli group** $\mathcal{P}_n$ contains tensor products of these matrices multiplied by phases $\{+1, -1, +i, -i\}$. We abbreviate the tensors: $Z_1 Z_2$ means $Z \otimes Z \otimes I \otimes \dots$.

Two Pauli strings either commute or anticommute. Count the qubit positions at which their nonidentity factors anticommute. The strings commute when that count is even and anticommute when it is odd. Hence commutation is a parity, not a continuum of possibilities.

G.3. A family that must commute, and must not contain $-I$

A **stabilizer group** $\mathcal{S} \subset \mathcal{P}_n$ must be Abelian — its elements commute — and must not contain $-I$. If it contained $-I$, a stabilized state would have to satisfy both $|\psi\rangle = |\psi\rangle$ and $-|\psi\rangle = |\psi\rangle$, leaving only the zero vector.

Suppose $\mathcal{S}$ has r independent generators $S_1, \dots, S_r$. “Independent” means no nonempty product of generators equals I . The code projector is

$$\Pi_{\mathcal{C}} = \prod_{j=1}^r \frac{I + S_j}{2}.$$

Each independent commuting constraint halves the space, so

$$\dim \mathcal{C} = 2^{n-r}.$$

The code therefore encodes

$$k = n - r$$

logical qubits and is denoted $[[n, k, d]]$ once its distance d is known.

The rank of a projector equals its trace. Expanding $\Pi_{\mathcal{C}}$ gives 2^{-r} times a sum of stabilizer elements. Every nonidentity Pauli has trace zero, while $\text{Tr } I = 2^n$. Independence ensures that only the empty product gives I . Therefore

$$\text{Tr } \Pi_{\mathcal{C}} = 2^{-r} 2^n = 2^{n-r}.$$

Drop independence and you overcount the constraints. Redundant checks do not remove another qubit.

G.4. Three qubits that must agree

Take three physical qubits and two generators

$$S_1 = Z_1 Z_2, \quad S_2 = Z_2 Z_3.$$

They commute. The simultaneous +1 condition says qubits 1 and 2 agree and qubits 2 and 3 agree. Thus

$$\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}.$$

There are $n = 3$ physical qubits and $r = 2$ independent generators, so $k = 1$. Choose logical basis states

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle.$$

A physical bit flip X_1 anticommutes with S_1 and commutes with S_2 . Measuring the two stabilizers returns eigenvalues called the **syndrome**. You see only the symptoms — which checks flipped — not the secret those checks were built to ignore. Write +1 as bit 0 and -1 as bit 1:

error	S_1	S_2	syndrome
I	+1	+1	00
X_1	-1	+1	10
X_2	-1	-1	11
X_3	+1	-1	01

The syndrome identifies any single X error without measuring whether the logical state is $|0_L\rangle$ or $|1_L\rangle$. Stabilizer measurement extracts an error relation, not the encoded amplitudes.

Logical Pauli operators must preserve the code space while acting nontrivially inside it. One choice is

$$\overline{X} = X_1 X_2 X_3, \quad \overline{Z} = Z_1.$$

The bar marks a logical operator. Both commute with S_1 and S_2 , and $\overline{XZ} = -\overline{ZX}$. Multiplying a logical operator by a stabilizer changes its physical representative but not its action on code states; for example, Z_1 , Z_2 , and Z_3 are equivalent logical $\overline{Z}$ representatives.

This code corrects one bit flip if the noise is restricted to X errors. As a quantum code against arbitrary single-qubit Pauli errors, however, it has distance one: Z_1 is already a

weight-one undetectable logical operation. Its full label is $[[3, 1, 1]]$ [R028]. Calling it a “single-error-correcting quantum code” without naming the noise model would mix two different jobs.

G.5. Operators that sneak past the checks

The **normalizer** of $\mathcal{S}$ within the Pauli group is

$$N(\mathcal{S}) = \{P \in \mathcal{P}_n : PS = SP \text{ for every } S \in \mathcal{S}\}.$$

These are the Pauli strings that commute with every check. For Pauli operators, commuting with every stabilizer is equivalent to normalizing the group, up to the irrelevant global phases we usually quotient out. A normalizer element can walk past every local check without setting off an alarm.

A Pauli error E has three possibilities:

1. If E anticommutes with at least one generator, it changes a syndrome and is detectable.
2. If $E \in \mathcal{S}$, it acts trivially on every code state.
3. If $E \in N(\mathcal{S}) \setminus \mathcal{S}$, it is an undetectable nontrivial logical Pauli.

The **weight** $\text{wt}(E)$ is the number of qubits on which E is not I . The code distance is

$$d = \min_{E \in N(\mathcal{S}) \setminus \mathcal{S}} \text{wt}(E).$$

A distance- d code detects all Pauli errors of weight less than d and corrects arbitrary errors on at most $\lfloor (d-1)/2 \rfloor$ qubits, assuming ideal syndrome extraction and a suitable decoder [R028]; [R029]. Degeneracy means distinct physical errors can have the same action on the code because they differ by a stabilizer.

Drop the set difference $N(\mathcal{S}) \setminus \mathcal{S}$ and you will count stabilizers themselves as logical operators. They are in the normalizer and they act as the identity on the code.

G.6. Asking the checks, or charging energy for them

The projectors onto the $s_j = \pm 1$ eigenspaces of generator S_j are

$$\Pi_{s_j} = \frac{I + s_j S_j}{2}.$$

In a circuit, an ancilla can acquire the parity associated with S_j and then be measured. In an analog Hamiltonian

$$H_{\text{stab}} = - \sum_j K_j S_j,$$

with energies $K_j > 0$, violating stabilizer S_j changes its term from $-K_j$ to $+K_j$, a cost $2K_j$. Passive energetic enforcement and active repeated syndrome extraction are different mechanisms even though they use the same operators.

A stabilizer is an algebraic constraint. A measured stabilizer code is active quantum error correction. A static stabilizer Hamiltonian energetically favors a code space. Topological order additionally requires locality, system-size-dependent logical operators, and the appropriate many-body phase. One word does not perform all four jobs.

G.7. Two binary lists instead of a matrix

Phases do not affect Pauli commutation, so encode a Pauli string by two binary vectors $\mathbf{x}, \mathbf{z} \in \mathbb{F}_2^n$, where $\mathbb{F}_2 = \{0, 1\}$ with arithmetic modulo two:

$$P \longleftrightarrow (\mathbf{x} \mid \mathbf{z}).$$

At qubit j ,

$$(0, 0) \leftrightarrow I, \quad (1, 0) \leftrightarrow X, \quad (0, 1) \leftrightarrow Z, \quad (1, 1) \leftrightarrow Y$$

up to phase. Multiplication becomes binary vector addition. Two Paulis $u = (\mathbf{x} \mid \mathbf{z})$ and $v = (\mathbf{x}' \mid \mathbf{z}')$ commute exactly when their **symplectic inner product** vanishes:

$$[u, v]_{\text{sp}} = \mathbf{x} \cdot \mathbf{z}' + \mathbf{z} \cdot \mathbf{x}' = 0 \pmod{2}.$$

The word is just a pairing that notices when two strings anticommute. Software will compute this pairing, not stare at matrices.

Place the r generator rows into a binary check matrix

$$H = (H_X \mid H_Z).$$

Commutation of all generator pairs is the matrix condition

$$H_X H_Z^T + H_Z H_X^T = 0 \pmod{2}.$$

For an error vector $e = (\mathbf{x}_e \mid \mathbf{z}_e)$, the syndrome bits are symplectic products with the generator rows. Gaussian elimination over $\mathbb{F}_2$ finds the rank r , redundant checks, logical representatives, and inconsistencies. This is the practical bridge from operator notation to software.

G.8. Stars and plaquettes, written as checks

Place one qubit on each edge of a square lattice with periodic boundaries. For each vertex v , define a star operator

$$A_v = \prod_{e \ni v} X_e,$$

and for each plaquette p , define

$$B_p = \prod_{e \in \partial p} Z_e.$$

A star and plaquette either share zero edges or two edges. Each shared edge contributes one $XZ = -ZX$, so two minus signs cancel. Therefore every A_v commutes with every B_p [R030]. Products of all stars and of all plaquettes equal identity on a periodic connected square lattice, giving two generator dependencies.

For an $L \times L$ torus there are $n = 2L^2$ edge qubits and $r = 2L^2 - 2$ independent stabilizers, hence $k = 2$. **[Theory]** The two logical qubits correspond to noncontractible loop operators [R030]. A shortest logical loop has weight L , so the ideal periodic square-lattice code has distance $d = L$. Boundaries and lattice defects change the counting and must be specified.

An open Z string anticommutes with the two star operators at its endpoints, producing a pair of electric syndromes. Extending the string moves an endpoint without creating a trail of violated checks. A closed contractible string is a stabilizer; a closed noncontractible string is logical. This is the algebraic core of Chapter 16's error-chain picture.

G.9. What a check list cannot prove

Stabilizer algebra proves exact commutation, ground-space dimension, syndromes, and logical-loop structure of the ideal toric code. It does not by itself establish that a microscopic defect Hamiltonian realizes that code.

A gadget may aim to generate $-K_j S_j$ in a low-energy subspace. The Schrieffer–Wolff calculation must also report unwanted terms. If a correction anticommutes with stabilizers, it can create excitations; if it lies in the normalizer but outside the stabilizer, it can split or rotate the logical manifold. Classifying corrections by commutation is more informative than listing coefficients alone.

For a proposed small patch, software should first build the binary check matrix and verify: pairwise commutation, rank, encoded-qubit count, logical representatives, and distance where tractable. Sparse exact diagonalization can then confirm the predicted ground-space dimension and excitation energies. Under non-Pauli perturbations, stabilizer labels need not remain exact; measure expectation values $\langle S_j \rangle$, Wilson loops, and overlap with the ideal code subspace rather than forcing every state into a perfect syndrome box.

The Gottesman–Knill theorem permits efficient classical simulation of stabilizer states under Clifford gates, Pauli measurements, and classical feed-forward [R028]; [R031]. **[Theory]** That efficiency is a property of this restricted computational family, not evidence that the corresponding quantum material is easy to simulate under arbitrary interactions.

G.10. Ways to get this wrong

1. Including $-I$. Then the purported code space is empty.
2. Counting listed generators instead of independent ones. Redundant checks do not remove another qubit.
3. Ignoring phases during state definition. Binary vectors suffice for commutation, but generator signs choose the $+1$ eigenspace.
4. Calling every normalizer element logical. Stabilizers themselves are in the normalizer and act trivially.
5. Quoting distance without boundaries. A planar patch, torus, puncture, and rough/smooth edge have different logical operators.
6. Confusing code distance with an energy gap. Distance counts support size; a Hamiltonian gap has energy units.
7. Assuming a syndrome identifies a unique error. Decoding chooses among equivalence classes and noise-weighted possibilities.

G.11. Check yourself

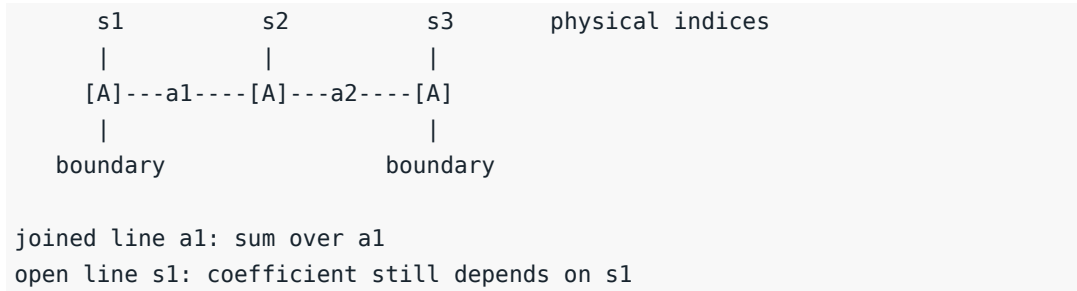
1. **Show that if $-I \in \mathcal{S}$, then the code space is $\{0\}$.**
A stabilized state would have to satisfy both $|\psi\rangle = |\psi\rangle$ and $-|\psi\rangle = |\psi\rangle$.
2. **Show that r independent stabilizers on n qubits encode $k = n - r$ logical qubits.**
The projector $\Pi_{\mathcal{C}}$ has trace 2^{n-r} , which equals $\dim \mathcal{C}$. Hence $k = n - r$.
3. **What makes a Pauli error detectable?**
It anticommutes with at least one stabilizer generator and flips that syndrome bit.
4. **Show that Z_1 , Z_2 , and Z_3 are equivalent logical $\bar{Z}$ representatives for the three-qubit code.**
They differ by products of $S_1 = Z_1 Z_2$ and $S_2 = Z_2 Z_3$, so they agree on every code state.
5. **Why is the repetition code only $[[3, 1, 1]]$ as a full quantum code?**
A weight-one Z operator is an undetectable logical operation.
6. **Show that toric-code stars and plaquettes commute.**
Their supports overlap on an even number of edges, so the minus signs from $XZ = -ZX$ cancel in pairs.

G.12. Sources

- [R028] Daniel Gottesman, *Stabilizer Codes and Quantum Error Correction*, PhD thesis, California Institute of Technology (1997). DOI: 10.7907/rzr7-dt72; arXiv: quant-ph/9705052.
- [R029] A. R. Calderbank, E. M. Rains, P. W. Shor, and N. J. A. Sloane, “Quantum error correction via codes over $\text{GF}(4)$,” *IEEE Transactions on Information Theory* **44**, 1369–1387 (1998). DOI: 10.1109/18.681315; arXiv: quant-ph/9608006.
- [R030] Alexei Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0; arXiv: quant-ph/9707021.
- [R031] Scott Aaronson and Daniel Gottesman, “Improved simulation of stabilizer circuits,” *Physical Review A* **70**, 052328 (2004). DOI: 10.1103/PhysRevA.70.052328; arXiv: quant-ph/0406196.

Appendix H. Boxes and wires

Draw three boxes in a row. Each box has a wire sticking up. Neighboring boxes share a horizontal wire.



That picture is the whole notation. A box is a table of numbers. An open wire is a label you still care about. A wire joining two boxes means: add over that label until it disappears.

The diagram is not decoration. Connectivity specifies the summations.

H.1. A giant table, then smaller ones

A wavefunction for N local degrees of freedom is a table

$$\psi_{s_1 s_2 \dots s_N},$$

with one physical index s_i per site. If each index takes d values, the table has d^N complex entries. That is already too big to store for a modest chain.

We replace the one giant table by smaller multidimensional arrays whose shared indices are summed. Those smaller arrays are **tensors**. The word covers a scalar (no index), a vector v_i (one index), a matrix M_{ij} (two), and a rank- r object $T_{i_1 \dots i_r}$ (r indices) in one breath.

Here “rank” means number of indices, not matrix rank. Mathematics enjoys reusing words. Context has to do some work.

This is like factoring an integer, except the factors have several index slots and may still be difficult to find. The compression works when the state’s correlations fit through moderately sized internal indices. A tensor network is an ansatz family and a notation, not a guarantee of cheap or accurate computation.

H.2. Joined wires are sums

If A_{ij} and B_{jk} share the index j , glue the j wires and add:

$$C_{ik} = \sum_j A_{ij} B_{jk}.$$

That sum is a **contraction**. The shared label is gone. Indices i and k remain free. Every term in an equation must have the same free indices. An index should normally appear at most twice in one monomial: once means free, twice means summed. We will write sums explicitly when ambiguity is possible.

Contraction order changes computational cost but not the exact scalar or tensor obtained. For an approximate contraction, order and truncation do affect the numerical error. Drop exactness and the diagram still names a network; it no longer names a unique number.

H.3. Wires nobody measures

For a quantum state, the open wires that label basis states are physical. The extra wires we invented to factor the table have no local measurement attached.

Those internal wires are **virtual** indices. An auxiliary label is not something a magnet could read.

H.4. How wide is the pipe

If a virtual index takes χ values, χ is its **bond dimension**. That number is the width of the pipe between two boxes. It is not yet an error bar.

H.5. A chain of matrices

Line the boxes up along an open chain of N sites. At site i , write a small matrix $A_i^{s_i}$ of size $\chi_{i-1} \times \chi_i$. Multiply them, and the leftover number is the amplitude of the bit string $s_1 \cdots s_N$.

That product is a **matrix product state**, or MPS:

$$|\psi\rangle = \sum_{s_1, \dots, s_N} \left(A_1^{s_1} A_2^{s_2} \cdots A_N^{s_N} \right) |s_1 s_2 \cdots s_N\rangle.$$

Open boundaries have $\chi_0 = \chi_N = 1$, so the matrix product is a scalar. In fully indexed form,

$$\psi_{s_1 \dots s_N} = \sum_{a_1, \dots, a_{N-1}} (A_1)_{1a_1}^{s_1} (A_2)_{a_1 a_2}^{s_2} \dots (A_N)_{a_{N-1} 1}^{s_N}.$$

The physical index s_i has dimension d_i ; neighboring virtual indices have dimensions χ_{i-1} and χ_i . Site-dependent dimensions are legal and often useful.

H.6. The same state, different numbers

Insert any invertible matrix G and its inverse on a bond:

$$A_i^{s_i} A_{i+1}^{s_{i+1}} = (A_i^{s_i} G)(G^{-1} A_{i+1}^{s_{i+1}}).$$

The state is unchanged. Tensor entries are therefore not unique physical observables. This **gauge freedom** can be used to place an MPS in a canonical form. You may slide an invertible matrix through a wire. Drop invertibility and the insertion need not be undoable; the state can change.

H.7. How much one cut can carry

Across a chosen bond, a state has a Schmidt decomposition

$$|\psi\rangle = \sum_{\alpha=1}^r \lambda_{\alpha} |\alpha_L\rangle |\alpha_R\rangle,$$

where the Schmidt coefficients $\lambda_{\alpha} \geq 0$ satisfy $\sum_{\alpha} \lambda_{\alpha}^2 = 1$, and the left and right Schmidt states are orthonormal. The Schmidt rank r cannot exceed the MPS bond dimension χ on that cut. The bipartite von Neumann entropy is

$$S = - \sum_{\alpha} \lambda_{\alpha}^2 \ln \lambda_{\alpha}^2 \leq \ln \chi.$$

Hence bond dimension is an entanglement capacity. It is not a direct accuracy certificate: a state can require large χ for one cut, and a poorly optimized MPS can waste its available dimension.

H.8. Two matrices that make GHZ

For N qubits, define the Greenberger–Horne–Zeilinger state

$$|\text{GHZ}_N\rangle = \frac{|00 \cdots 0\rangle + |11 \cdots 1\rangle}{\sqrt{2}}.$$

Use bond dimension $\chi = 2$ and the same two matrices at every site:

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Choose a left boundary row and right boundary column

$$\ell = \frac{1}{\sqrt{2}}(1, 1), \quad r = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The amplitude of a bit string $s_1 \cdots s_N$ is

$$\psi_{s_1 \cdots s_N} = \ell A^{s_1} A^{s_2} \cdots A^{s_N} r.$$

If every $s_i = 0$, the product is A^0 and the amplitude is $1/\sqrt{2}$. If every $s_i = 1$, the product is A^1 and the amplitude is $1/\sqrt{2}$. If a string contains both 0 and 1, then at some point the product includes

$$A^0 A^1 = A^1 A^0 = 0,$$

so its amplitude vanishes. Hence the MPS is exactly the GHZ state.

Its norm follows without expanding 2^N amplitudes:

$$\langle \text{GHZ}_N | \text{GHZ}_N \rangle = \frac{1}{2} + \frac{1}{2} = 1.$$

For distinct sites i and j ,

$$\langle Z_i \rangle = 0, \quad \langle Z_i Z_j \rangle = 1.$$

The connected correlation is therefore one. Across any cut, the two Schmidt coefficients are $1/\sqrt{2}$, so $S = \ln 2$, consistent with $\chi = 2$.

This example also gives a warning. The GHZ state has long-range correlations yet tiny bond dimension. Conversely, some short-range-looking problems can have difficult entanglement structures. “More correlation” and “larger bond dimension” are not interchangeable slogans.

H.9. Boxes that eat and spit

An operator has an input wire and an output wire at each site. Stack another layer of boxes on those pairs:

$$O = \sum_{\mathbf{s}, \mathbf{s}'} W_1^{s_1 s'_1} W_2^{s_2 s'_2} \dots W_N^{s_N s'_N} |s_1 \dots s_N\rangle \langle s'_1 \dots s'_N|,$$

where each $W_i^{s_i s'_i}$ is a matrix on virtual indices. The bold symbol $\mathbf{s}$ abbreviates the full sequence $(s_1, \dots, s_N)$. That layered product is a **matrix product operator**, or MPO. It is the MPS idea, now with two physical legs per site.

Expectation values such as $\langle \psi | O | \psi \rangle$ are obtained by stacking the bra MPS, MPO, and ket MPS, then contracting all physical and virtual indices. A one-dimensional slice of the resulting double-layer network takes the virtual data on one cut to the next. That slice is a **transfer matrix**. Its leading eigenvalues govern normalization and, under suitable assumptions, correlation lengths.

MPOs compactly represent sums of local Hamiltonian terms. A compact representation does not make optimization exact; it merely stops the input file from becoming the primary many-body problem.

H.10. The same idea on a grid

An MPS follows a chain. Put the same boxes on a higher-dimensional graph and you have a **projected entangled pair state**, or PEPS [R032]. On a square lattice, a local PEPS tensor can be written

$$A_{lurd}^s,$$

where s is the physical index and l, u, r, d are left, up, right, and down virtual indices. Neighboring virtual indices are contracted. On a general defect graph, use one virtual index per incident edge rather than forcing square-lattice labels.

One construction begins with entangled pairs on graph edges and applies a local map from the virtual spaces meeting at a vertex to its physical space. “Projected entangled pair state” names that construction. Boundary tensors have fewer neighbors or trivial one-dimensional legs.

PEPS naturally match two-dimensional locality, but exact contraction of a generic two-dimensional PEPS is computationally hard; approximation is normally required [R033].

[Theory] This is why Chapter 40 must report both the PEPS bond dimension D and contraction controls such as an environment dimension χ_{env} . Increasing only one can conceal the other's error.

A PEPS is a representation of a state. It may represent an exact fixed-point topological state, a variational approximation to a microscopic ground state, or a completely nontopological state. Running a PEPS algorithm on ordinary hardware is a classical numerical calculation, not physical creation of anyons.

H.11. Hidden symmetries on the wires

Topologically ordered PEPS can possess symmetries acting on virtual indices. A virtual symmetry string may pass through local tensors while leaving the physical state unchanged, and noncontractible virtual strings can label distinct ground states. This framework explains ground-state degeneracy, local indistinguishability, topological entropy, and anyonic sectors for important PEPS classes [R034].

The statement needs restraint. A visually loop-like tensor network is not automatically topological. One must identify the virtual symmetry or algebra, construct the parent Hamiltonian or otherwise establish the phase, and test robustness. Injective PEPS typically describe a unique ground state of an appropriate parent Hamiltonian; topological PEPS require generalized structures such as group or matrix-product-operator symmetries [R034].

H.12. Cutting small pieces, sweeping the rest

For an MPS, sweeping QR or singular-value decompositions can move the orthogonality center and produce left- or right-canonical tensors. At a bond, truncating the Schmidt decomposition to the largest χ coefficients discards weight

$$w_{\text{disc}} = \sum_{\alpha > \chi} \lambda_{\alpha}^2.$$

This quantity is local to that truncation. A small discarded weight is useful evidence but not a global theorem that every observable is accurate.

The density-matrix renormalization group (DMRG) variationally optimizes an MPS, usually one or two sites at a time [R035]. It is especially effective for one-dimensional and quasi-one-dimensional systems whose entanglement across the chosen ordering is manageable. Mapping a wide two-dimensional lattice to a snake-like chain can make the required bond dimension grow rapidly with width.

PEPS algorithms optimize tensors while approximating their environment. Infinite PEPS works directly with a repeating unit cell; finite PEPS retains boundaries. The method must match the question. A translation-invariant infinite ansatz is a poor tool for implantation disorder unless disorder is modeled through a larger cell, sampling, or another representation.

H.13. Seven steps before you trust a number

For a defect-cluster model, proceed in layers:

1. **Declare each physical index.** State whether s_i labels a microscopic spin, a projected cluster pseudospin, or an edge label. Give its dimension d_i .
2. **Declare the graph.** Every virtual bond should correspond to an intended contraction, not necessarily a physical interaction.
3. **Choose an ansatz.** Use exact diagonalization for the smallest full clusters; MPS/DMRG for chains and cylinders; PEPS when two-dimensional connectivity and accessible contraction controls justify it.
4. **State all numerical controls.** Report MPS/PEPS bond dimension, environment dimension, unit-cell size, truncation rule, optimizer tolerance, boundary condition, and initialization strategy.
5. **Converge observables separately.** Energy can converge before a small gap, correlation length, Wilson loop, or entanglement diagnostic. Plot each claimed observable versus the relevant controls.
6. **Use competing initial states.** Near first-order transitions or topological-sector barriers, one initialization can remain metastable. Compare energies and diagnostics from multiple sectors.
7. **Check the effective-model handoff.** Compare a small microscopic calculation with the projected Hamiltonian before applying tensor networks only to the latter.

For Chapter 40's observables:

- obtain the **gap** from excited-state methods or sector-resolved energies, not from ground-state energy convergence alone;
- calculate **entanglement entropy** from Schmidt values for MPS, with geometry and logarithm base stated;
- estimate **topological entanglement entropy** using a subtraction or scaling protocol whose finite-size errors are tested;
- represent **Wilson loops** as MPOs or tensor insertions and examine perimeter/area behavior only over resolved sizes;
- extract **modular data** only after constructing and controlling a basis of ground-state sectors.

No single diagnostic proves topological order in a small noisy patch. Agreement among spectrum, degeneracy pattern, loop observables, entanglement structure, and perturbation response is the stronger case.

H.14. Ways to get this wrong

1. Leaving indices unlabeled. A reader cannot tell physical from virtual or input from output legs.
2. Calling tensor rank matrix rank. State which meaning is intended.
3. Treating gauge-dependent entries as observables. Insertions of GG^{-1} change tensors without changing the state.
4. Reporting bond dimension without convergence. $D = 8$ is an input, not an error bar.
5. Ignoring contraction error. A large PEPS D with a small environment can be less reliable than a smaller, well-converged calculation.
6. Using DMRG outside a sensible geometry. A two-dimensional cylinder may be tractable at one width and misleading at the next.
7. Inferring topology from low variational energy. Competing phases can be close in energy while differing in nonlocal structure.
8. Forgetting the projection error. Accurate solution of an inaccurate effective Hamiltonian is still inaccurate.
9. Inferring topological order from a loop-like drawing. Virtual symmetries and physical diagnostics do the work, not the artwork.

You now have a picture, a few identities, and a checklist. The next time a paper draws boxes and wires, you can read the sums.

H.15. Check yourself

1. **What does a joined tensor-network line mean?**
Sum over the shared index.
2. **Show that inserting G and G^{-1} on a bond leaves the MPS state unchanged.**
 $A_i^{s_i} A_{i+1}^{s_{i+1}} = (A_i^{s_i} G)(G^{-1} A_{i+1}^{s_{i+1}})$ for any invertible G , so every amplitude is the same.
3. **Show that $S \leq \ln \chi$ for an MPS cut of bond dimension χ .**
At most χ Schmidt coefficients can be nonzero, and $-\sum p_\alpha \ln p_\alpha$ on a distribution supported on at most χ outcomes is at most $\ln \chi$.
4. **Show that the GHZ MPS is nonzero only for all-zero and all-one strings.**
Mixed strings contain the zero products $A^0 A^1$ or $A^1 A^0$.

5. **Does a PEPS diagram prove topological order?**

No. Topological structure requires appropriate virtual symmetries and physical diagnostics.

6. **What fails if you report D without an environment dimension?**

A large PEPS bond dimension with a small environment can conceal contraction error. D is an input, not an error bar.

H.16. Sources

- [R032] Frank Verstraete and J. Ignacio Cirac, “Renormalization algorithms for quantum-many body systems in two and higher dimensions,” arXiv: cond-mat/0407066 (2004).
- [R033] Norbert Schuch, Michael M. Wolf, Frank Verstraete, and J. I. Cirac, “Computational complexity of projected entangled pair states,” *Physical Review Letters* **98**, 140506 (2007). DOI: 10.1103/PhysRevLett.98.140506; arXiv: quant-ph/0611050.
- [R034] Norbert Schuch, J. I. Cirac, and David Pérez-García, “PEPS as ground states: Degeneracy and topology,” *Annals of Physics* **325**, 2153–2192 (2010). DOI: 10.1016/j.aop.2010.05.008; arXiv: 1001.3807.
- [R035] Ulrich Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics* **326**, 96–192 (2011). DOI: 10.1016/j.aop.2010.09.012; arXiv: 1008.3477.
- [R036] Román Orús, “A practical introduction to tensor networks: Matrix product states and projected entangled pair states,” *Annals of Physics* **349**, 117–158 (2014). DOI: 10.1016/j.aop.2014.06.013; arXiv: 1306.2164.

Global evidence table

A scoreboard. Each row is one claim, with the kind of support it actually has. A good pair experiment is not a lattice. A circuit that plays anyons is not a crystal that contains them.

Statuses are used as written: **experimentally demonstrated**, **experimentally partial**, **numerically demonstrated**, **theoretically established**, **theoretically proposed**, **speculative**.

Claim	Status	Platform	Evidence	Reference
HOST-01 — Coherent single-NV control	experimentally demonstrated	Diamond NV ⁻	Single-center spectroscopy, optical initialization/readout, and coherent microwave rotations are established. This is a physical-qubit capability, not array or phase evidence.	[R074]
HOST-02 — Diamond is an automatic overall winner	experimentally partial	Diamond color centers	Diamond has the strongest integrated single-node optical-spin evidence in this survey, but demonstrated nanometre-scale uniform interacting arrays are absent; superiority depends on the system metric.	[R074]; [R207]

Claim	Status	Platform	Evidence	Reference
HOST-03 — SiC supplies controllable defect-spin nodes	experimentally demonstrated	4H-SiC divacancies and silicon vacancies	Individual optical isolation/control and millisecond-scale coherence were demonstrated; later work combined a masked 8×8 pattern with a waveguide electron–nuclear register.	[R096]; [R099]
HOST-04 — SiC supplies the required electron-spin interaction graph	experimentally partial	4H-SiC	Patterned implantation and integrated registers establish several fabrication/control ingredients, but not nanometre-registered electron–electron couplings across the pattern.	[R099]; [R205]
HOST-05 — Sapphire matches diamond as an individual-defect platform	experimentally partial	Cr^{3+} :sapphire and sapphire color centers	Ruby experiments demonstrate ensemble ODMR, long low-temperature T_1 , and collective cavity coupling; they do not demonstrate the selected single-ion control, T_2 , readout, and pair-gate stack required here.	[R088]; [R089]

Claim	Status	Platform	Evidence	Reference
COUP-01 — Nearby defects couple coherently	experimentally demonstrated	Diamond NV ⁻ pairs	A selected pair about 25 nm apart showed a roughly 5 kHz dipolar interaction and room- temperature entanglement under refocused optimal control. This establishes a pair primitive only.	[R106]; [R080]
COUP-02 — Cavity photons mediate defect-defect interaction	experimentally demonstrated	Diamond SiV ⁻ emitters in one nanocavity	Two spectrally tuned SiV emitters exhibited photon-mediated interaction through a shared cavity mode; uniform low-loss mediation across a lattice was not shown.	[R086]
COUP-03 — Nanometre- separated defect pairs can have strong, geometry- sensitive direct exchange	numerically demonstrated	Closely spaced diamond spin centers	Atomistic calculations predict exchange can dominate at very short separation and vary strongly with site and orientation; this pair-level result does not establish an extensive calibrated exchange graph.	[R107]

Claim	Status	Platform	Evidence	Reference
COUP-04 — Ordinary coupled defects imply topological order	theoretically established	Any pair or finite defect register	They do not: a measured two-body coupling supplies a Hamiltonian term, whereas topological order additionally requires an extended gapped phase and nonlocal sectors/observables.	[R080]; [R015]
ENC-01 — Encoded defect-associated qubits exist	experimentally demonstrated	NV electron- ¹³ C nuclear-spin registers	Decoherence-protected nuclear-spin subspaces and repeated error correction of a three-spin encoded qubit were demonstrated around NV ancillas. The protection is DFS/active encoding, not emergent topology.	[R116]; [R117]
ENC-02 — A fabricated electronic-defect cluster is an isolated pseudospin	theoretically proposed	Three-color-center electronic-spin cluster	Projection onto an isolated doublet is controlled theory, and one triple-NV motif has been observed, but no cited experiment establishes a reproducible static three-center doublet with measured projected operators.	[R025]; [R243]

Claim	Status	Platform	Evidence	Reference
ENC-03 — Encoding a cluster establishes topological protection	theoretically established	Finite spin clusters	It does not: a locally distinguishable cluster code can suppress specified noise or leakage while lacking a thermodynamic phase, deconfined excitations, and nonlocal logical operators.	[R116]; [R015]
TOPO-01 — Local two-body spin models can host a non-Abelian phase	theoretically established	Kitaev honeycomb model	The exactly solved bond-dependent model has an emergent Ising-type non-Abelian phase. Its anyons are not Fibonacci anyons.	[R017]
TOPO-02 — Engineered analog spin models show topological-phase signatures	experimentally partial	Driven Rydberg-atom arrays	A 219-atom programmable array measured topological-spin-liquid signatures under a driven blockade Hamiltonian; finite size, drive dependence, and preparation prevent an autonomous-memory conclusion.	[R126]

Claim	Status	Platform	Evidence	Reference
TOPO-03 — The Levin–Wen target supports doubled-Fibonacci order	theoretically established	Ideal Fibonacci string-net Hamiltonian	The commuting-projector construction fixes branching and plaquette recoupling data and yields doubled, nonchiral Fibonacci order in the ideal many-body model. It is a target-model theorem, not a defect implementation.	[R018]
SIM-01 — Topologically ordered model states can be digitally prepared	experimentally demonstrated	Superconducting processors	Finite gate circuits prepared toric-code-type states and measured stabilizer, entanglement, and string observables. The native processor Hamiltonian was not thereby converted into a passive topological material.	[R125]

Claim	Status	Platform	Evidence	Reference
SIM-02 — Digital Fibonacci simulation has been demonstrated	experimentally demonstrated	Superconducting transmon processors	Experiments prepared finite doubled-Fibonacci string-net states and executed fusion/braid operations with ordinary transmons, echo, calibration, and/or error mitigation; neither experiment supplied a native Fibonacci gap.	[R138]; [R165]
SIM-03 — A defect processor could digitally simulate Fibonacci anyons	theoretically proposed	Gate-controlled defect-spin processor	Nothing in the algorithm requires the physical qubits themselves to be anyons, but a defect array still needs sufficient gates, connectivity, initialization, readout, and fidelity. Success would remain digital simulation.	[R138]; [R074]

Claim	Status	Platform	Evidence	Reference
EMERG-01 — A correct braid circuit establishes an emergent anyon	theoretically established	Any digital simulator	It does not: emergence requires the hardware many-body Hamiltonian to generate and stabilize localized excitations with the target fusion and braiding data throughout a phase; a compiled unitary establishes only the simulated operation.	[R018]; [R138]
EMERG-02 — Genuine Fibonacci quasiparticles at $12/5$	experimentally partial	Fractional quantum Hall, superconducting, and defect candidates	At $12/5$, transport establishes an incompressible state and numerics support Read–Rezayi-family candidates, but controlled Fibonacci fusion and noncommuting braiding have not been established; digital transmon results are emulations.	[R226]; [R228]; [R138]

Claim	Status	Platform	Evidence	Reference
EMERG-03 — A defect-based Fibonacci phase exists	speculative	Diamond, SiC, sapphire, and other defect lattices	No cited experiment or defect-specific numerical phase calculation demonstrates branching constraints, recoupling terms, a stable many-body gap, topological sectors, and Fibonacci diagnostics in one defect Hamiltonian.	[R018]; [R175]; [R207]; [R080]
GAD-01 — Two-body interactions can generate many-body effective terms	theoretically established	Abstract perturbative gadgets	Controlled low-energy constructions generate n -body terms at scale roughly g^n / Δ^{n-1} with bounded errors when the perturbative hierarchy holds.	[R175]
GAD-02 — Perturbatively generated plaquette interactions exist	theoretically established	Kitaev honeycomb and code-gadget models	Fourth-order virtual processes generate plaquette operators in the anisotropic honeycomb limit, and two-body code gadgets generate toric-code/quantum-double terms. These are model constructions, not defect devices.	[R017]; [R177]

Claim	Status	Platform	Evidence	Reference
GAD-03 — A defect gadget realizes a Fibonacci plaquette	theoretically proposed	Defect-cluster string-net architecture	No cited mapping derives the full Fibonacci plaquette recoupling operator from available defect tensors while bounding lower-order fields, leakage, and cross-gadget paths.	[R175]; [R018]
GAD-04 — The gadget penalty gap is the topological gap	theoretically established	Perturbative Hamiltonian engineering	It is not: the penalty Δ suppresses mediator leakage, while the desired coefficient K and extended model gap are parametrically smaller and must separately beat noise, disorder, and temperature.	[R175]; [R177]
GAP-01 — A weak perturbation preserves an already established gapped topological phase	theoretically established	Local topologically ordered Hamiltonians	Stability theorems protect spectral structure against sufficiently weak bounded local perturbations under their assumptions; they do not manufacture a gap in an approximate target.	[R142]

Claim	Status	Platform	Evidence	Reference
GAP-02 — A defect-cluster topological gap has been established	speculative	Proposed defect-cluster arrays	No connected experiment or validated defect-specific many-body calculation supplies Δ_{topo} ; microscopic splittings, pair couplings, and gadget penalties are not substitutes.	[R142]; [R018]; [R080]
GAP-03 — Passive feasibility requires a gap budget	theoretically established	Static analog topological memory	The relevant phase gap must dominate thermal energy, projected disorder, decoherence broadening, finite-size splitting, and unwanted local remainders with a task-specific margin.	[R169]; [R142]
GAP-04 — A 2D local stabilizer Hamiltonian is automatically a self-correcting memory	theoretically established	Two-dimensional local stabilizer codes	It is not: under the no-go theorem's assumptions the energy barrier is bounded, so a zero-temperature phase gap alone does not give indefinitely improving finite-temperature storage.	[R168]; [R169]

Claim	Status	Platform	Evidence	Reference
FAB-01 — Exact single-ion delivery is possible	experimentally demonstrated	Deterministic nitrogen implantation in diamond	A deterministic implanter detected delivered ions and produced patterned ¹⁵ NV centers, but delivery did not make NV conversion, charge state, coherence, or final relative geometry deterministic.	[R258]
FAB-02 — Focused implantation gives deterministic final defect coordinates	experimentally partial	Focused Si implantation for diamond SiV	A sub-40-nm beam, calculated straggle, and measured roughly 32-nm one-axis center spread were distinct quantities; conversion was process dependent.	[R190]
FAB-03 — Feedback laser writing gives near-unity usable-site yield	experimentally partial	Laser-written diamond NV ⁻	Feedback achieved about 96% single-NV creation under reported conditions with about 33 nm in-plane deviation; it did not establish three-dimensional bond tolerance or full coherent-array yield.	[R203]

Claim	Status	Platform	Evidence	Reference
FAB-04 — Delta doping solves three-dimensional placement	experimentally partial	Delta-doped diamond NV ⁻	Growth confined nitrogen to a roughly 1–2 nm depth layer, while lateral positions remained undetermined; later localized vacancy creation reached about 4 nm depth and 46(1) nm lateral spread.	[R204]; [R207]
FAB-05 — A multi-defect motif can be fabricated reproducibly	experimentally partial	Molecularly implanted diamond NV triples	One strongly coupled triple was identified among 7,116 implantation events, demonstrating a motif can occur but not deterministic repetition, target spectrum, or high yield.	[R243]
FAB-06 — The required dense usable defect lattice is manufacturable	experimentally partial	Diamond and SiC implantation platforms	Beam registration, stopping/straggle, structural conversion, charge state, optical/spin usability, coherence, bond tolerance, and compound array yield have each been demonstrated only in subsets, not jointly at the required repeated geometry.	[R190]; [R203]; [R207]; [R205]

Claim	Status	Platform	Evidence	Reference
CTRL-01 — Dense-array addressability is integrated with strong uniform coupling	experimentally partial	Diamond NV arrays	Nanoscale optical selection, four-site frequency encoding, and parallel measurement of more than 100 resolved NVs are demonstrated separately; low-crosstalk encoded control inside a uniform interacting topological patch is not.	[R235]; [R236]; [R239]
KILL-01 — Cluster-spectrum gate	theoretically proposed	One candidate defect cluster	Stop or redesign if no two-dimensional low-energy manifold remains isolated from leakage across measured hyperfine, strain, and geometric distributions. This is a platform-specific gate, not a universal threshold.	[R025]; [R243]
KILL-02 — Desired-operator dominance gate	theoretically proposed	Three-to-six encoded clusters	Advance only if the target plaquette coefficient exceeds the predeclared sum/norm of noncommuting residual terms and the effective spectrum matches exact low energies with bounded leakage.	[R175]; [R177]

Claim	Status	Platform	Evidence	Reference
KILL-03 — Many-body phase gate	theoretically proposed	Small patches followed by scaling	A nonzero plaquette coefficient is insufficient; require a stable excitation gap, correct low-energy sectors, local indistinguishability, Wilson/fusion data, perturbation robustness, and size trends before a phase claim.	[R018]; [R142]
KILL-04 — Thermal-activation gate	theoretically proposed	Passive analog memory	Use an activation requirement tied to system size and error budget, approximately $E_{\text{act}}/k_B T \gtrsim \ln(N_c g/\epsilon)$ for dilute seeds, then test diffusion kinetics; $\Delta > k_B T$ alone is not a pass.	[R169]; [R168]
KILL-05 — Placement/disorder gate	theoretically proposed	Direct dipolar defect bonds	Because $J \propto r^{-3}$, small radial spread gives $\sigma_J/J \approx 3\sigma_r/r$ before angular errors. Propagate measured tails and missing bonds into the phase model rather than comparing nominal beam spots.	[R106]; [R207]

Claim	Status	Platform	Evidence	Reference
KILL-06 — Preparation-window gate	theoretically proposed	Adiabatic analog preparation	Stop the stated ramp when its adiabatic lower-time bound exceeds the decoherence upper-time bound under the same schedule; making the ramp slower cannot then satisfy both.	[R247]
KILL-07 — Compound-yield gate	theoretically proposed	Large fixed defect arrays	Site, species, charge, coherence, addressability, and bond yields must be conditioned and propagated to patch yield; a high single-step yield does not justify monolithic scale-up without repair.	[R258]; [R190]; [R207]
KILL-08 — Logical-scaling gate	theoretically proposed	Encoded topological memory	Claim protection only when logical error or lifetime improves with size at fixed physical conditions and frozen analysis/decoder rules; a finite prepared-state signature is not scaling evidence.	[R141]; [R125]

Claim	Status	Platform	Evidence	Reference
KILL-09 — Smallest decisive mechanism test	theoretically proposed	Four encoded clusters; candidate twelve-defect plaquette	After one-cluster and two-cluster validation, four clusters are the smallest scoped de- vice/calculation that can expose a weight-four coefficient and every lower-weight competitor; failure there kills the stated mechanism before Fibonacci scaling.	[R175]; [R243]

Glossary

A memory aid, not a prerequisite. If you have not met a word in a chapter yet, skip the entry. “First” means the chapter that taught it here, not the year someone coined it.

Four later distinctions that people mash together:

Ordinary defect qubit. One defect, two levels you can set, flip, and read. Not a crowd, not a particle, not a protected phase. (*Chapter 3, then 7.*)

Encoded defect cluster. Several defects acting as one quieter bit. Still a local gadget. (*Chapter 11.*)

Digital anyon simulation. A computer, or a gate machine, playing the part of an anyon model. Good theatre is not a material. (*Chapter 20.*)

Emergent topological phase. The crystal’s own energy landscape contains the strange particles. A pretty prepared state is not enough. (*Chapters 16 and 21.*)

H.17. A

Abelian anyon. An anyon whose exchange acts on a state by multiplication by a phase. Successive exchanges therefore commute within the relevant sector. Toric-code e and m excitations are standard examples. (*First: Chapter 13, “Anyons and braids.”*)

Active quantum error correction. Protection based on repeated extraction of error information and a recovery action. It differs from passive energetic suppression by a Hamiltonian, although a device may use both. (*First: Chapter 16, “Toric code.”*)

Adjoint ($A^\dagger$). The conjugate transpose of a matrix or linear operator. An operator satisfying $A = A^\dagger$ is Hermitian. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Amplitude. A complex number associated with a possible quantum outcome or component of a state. Probabilities come from squared magnitudes and can depend on interference between amplitudes. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Analog Hamiltonian engineering. Arranging physical couplings so that the device itself evolves under a target Hamiltonian, either statically or in a controlled driven regime. Unlike digital simulation, the target interaction is embodied in the device rather than compiled solely into a gate sequence. Analog engineering does not guarantee that approximations, heating, or unwanted terms are harmless. (*First: Chapter 21, “Analog Hamiltonian engineering.”*)

Anyon. A quasiparticle type possible in two spatial dimensions whose exchanges obey braid statistics more general than the boson/fermion alternatives. The term describes exchange and fusion structure; it does not mean “any unusual particle” or “any state on a quantum processor.” (*First: Chapter 13, “Anyons and braids.”*)

H.18. B

Band. A range of allowed electron energies formed when atomic levels spread through a periodic crystal. Band structure depends on the lattice, chemical composition, and electron interactions. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Band gap. An energy interval with no extended bulk electronic states in an idealized band description. A wide gap can help isolate defect levels, but does not by itself produce long coherence or optical addressability. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Bloch sphere. A geometric representation of the pure states of one two-level system as points on a sphere; mixed states lie inside. It is a qubit-state picture, not the same as a Bloch state in a crystal. (*First: Chapter 3, “Qubits as controllable systems.”*)

Bloch state. A spatially extended single-particle state compatible with lattice periodicity, written as a plane-wave factor times a periodic function. Despite the shared name, it has no direct connection to the qubit Bloch sphere. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Born rule. The rule converting a quantum amplitude into an outcome probability. For a normalized state $|\psi\rangle$ and projector P , the corresponding probability is $\langle\psi|P|\psi\rangle$. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Braid. The history of exchanging particle positions in two dimensions while keeping worldlines from crossing. Braids that cannot be continuously deformed into one another can act differently on an anyonic state space. (*First: Chapter 13, “Anyons and braids.”*)

Braid group. The group generated by neighboring exchanges σ_i , subject to the braid relations. A non-Abelian anyon model assigns matrices to these generators, giving a braid-group representation. (*First: Chapter 13, “Anyons and braids.”*)

H.19. C

Charge state. The net electron-occupation state of a defect relative to a chosen neutral reference, indicated by labels such as NV^- or NV^0 . Different charge states can have entirely different spin and optical properties. (*First: Chapter 7, “Diamond color centers.”*)

Chirality. A handedness of propagation or topological response. A chiral topological phase is not generally equivalent to the nonchiral doubled theory formed by pairing it with its time reverse. (*First: Chapter 19, “Doubled versus chiral Fibonacci.”*)

Coherence. Preservation of phase relationships that allow quantum interference. Coherence is quantified by experiment-specific times and is not synonymous with population lifetime, fidelity, or topological protection. (*First: Chapter 2, “Composite quantum systems”; measured in Chapter 4.*)

Color center. A localized electronic defect complex in a wide-bandgap crystal that produces characteristic optical absorption or emission. Some color centers also carry useful spins; the optical name alone guarantees neither a qubit nor a scalable array. (*First: Chapter 7, “Diamond color centers.”*)

Commuting-projector Hamiltonian. A Hamiltonian written as a sum of mutually commuting local projector-like terms. Such models can be exactly solvable and expose topological structure, but their microscopic realization may require interactions not naturally available in hardware. (*First: Chapter 16, “Toric code.”*)

Correlation length (ξ). A length scale over which local connected correlations appreciably decay in a gapped phase. Finite devices must be large compared with relevant correlation lengths before asymptotic topological arguments become reliable. (*First: Chapter 21, “Analog Hamiltonian engineering.”*)

Crystal field. The electrostatic and covalent environment produced by neighboring atoms, which splits otherwise degenerate electronic orbitals according to local symmetry. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Crystal lattice. The periodic arrangement used to describe an ideal crystal. It provides sites, directions, and symmetries; a real sample also contains boundaries, vibrations, disorder, and defects. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Crosstalk. An unintended response of non-target qubits or couplings during control or readout. In a dense array it can create correlated errors even when each isolated control operation looks accurate. (*First: Chapter 35, “Addressing dense arrays.”*)

H.20. D

Decoherence. Loss of observable phase coherence when a system becomes correlated with uncontrolled degrees of freedom or suffers random evolution. It can turn a pure state into a mixed reduced state without requiring simple energy relaxation. (*First: Chapter 2, “Composite quantum systems.”*)

Decoherence-free subspace (DFS). A subspace on which a specified dominant noise interaction acts identically, so relative quantum information is unaffected by that noise.

The protection is model-dependent and is not topological order. (*First: Chapter 11, “Defect clusters as encoded qubits.”*)

Density operator (density matrix, ρ). A positive, unit-trace operator describing either a pure state or a statistical/marginal mixed state. It is the appropriate state description when classical uncertainty or entanglement with an unobserved environment matters. (*First: Chapter 2, “Composite quantum systems.”*)

Defect. A departure from the periodic structure or composition of a crystal. Point defects include vacancies, substitutions, interstitials, and complexes; extended defects include dislocations and stacking faults. A structural defect is not automatically an optically active spin or a qubit. (*First: Chapter 5, “Crystals, bands, and localized states”; taxonomy in Chapter 6.*)

Defect cluster. A deliberately selected group of nearby, interacting defects treated as one subsystem. “Cluster” says how microscopic constituents are grouped; only a demonstrated isolated low-energy subspace justifies replacing it by an encoded pseudospin. (*First: Chapter 11, “Defect clusters as encoded qubits.”*)

Digital quantum simulation. Representation of a target evolution by gates on programmable qubits, usually after discretization and compilation. Correctly reproducing target observables can validate the simulation without turning the hardware into the simulated material phase. (*First: Chapter 20, “Digital simulation.”*)

Dipolar interaction. Coupling between magnetic dipole moments, scaling as $1/r^3$ and depending on their orientation relative to the displacement vector. Its long reach is useful, while its anisotropy and unwanted couplings complicate lattice design. (*First: Chapter 10, “Defect–defect coupling.”*)

Dislocation. An extended line defect characterized by a mismatch in lattice registry. It can generate strain and electronic states over distances much larger than a point defect. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Disorder. Spatial variation of on-site energies, couplings, fields, positions, or other parameters away from an intended model. Disorder can close a gap, localize excitations, broaden transitions, or occasionally stabilize a regime; its effect must be calculated rather than guessed. (*First: Chapter 21, “Analog Hamiltonian engineering”; budgeted in Chapter 29.*)

Doubled Fibonacci order. A nonchiral topological order that combines Fibonacci topological data with its time-reversed partner, as arises naturally in a corresponding Levin–Wen string-net construction. It is related to, but not synonymous with, a chiral Fibonacci phase. (*First: Chapter 18, “Levin–Wen string nets”; distinction developed in Chapter 19.*)

H.21. E

Effective Hamiltonian. A Hamiltonian for selected low-energy degrees of freedom after higher-energy states have been projected out or treated perturbatively. Its validity depends on scale separation and on the size of omitted corrections. (*First: Chapter 11, “Defect clusters as encoded qubits”; derived systematically in Chapter 22.*)

Eigenstate and eigenvalue. A state $|a\rangle$ satisfying $A|a\rangle = a|a\rangle$. For an observable, a is a possible measurement outcome; for a Hamiltonian, it is an energy. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Emergence. The appearance of collective low-energy degrees of freedom or laws not identifiable with a single microscopic constituent. In this book, an emergent anyon must be an excitation sector of a many-body phase, not merely a relabelled hardware qubit. (*First: Chapter 13, “Anyons and braids”; Hamiltonian example in Chapter 17.*)

Encoded qubit. A two-dimensional information-bearing subspace embedded in a larger Hilbert space, often across several physical constituents. Encoding may suppress a specified noise or leakage process, but is not automatically error correction or topology. (*First: Chapter 3, “Qubits as controllable systems”; cluster construction in Chapter 11.*)

Entanglement. A property of a composite quantum state that cannot be written as a product state, or as an appropriate classical mixture when mixed-state distinctions matter. Correlation alone is not sufficient evidence of entanglement. (*First: Chapter 2, “Composite quantum systems.”*)

Exchange interaction. A short-range spin coupling arising from quantum indistinguishability and overlap of electronic wavefunctions. Unlike magnetic dipolar coupling, its magnitude often changes exponentially with atomic arrangement and is highly sensitive to placement and chemistry. (*First: Chapter 10, “Defect–defect coupling.”*)

H.22. F

F-move (recoupling move). A unitary change of basis between different orders of fusing the same anyons, such as $(a \times b) \times c$ and $a \times (b \times c)$. It does not physically exchange anyons. (*First: Chapter 14, “Fusion categories without the fog.”*)

Fidelity. A quantitative overlap or success measure comparing an actual state, gate, or readout with a target. State, process, average-gate, and readout fidelities are different quantities and require their definitions and conditions. (*First: Chapter 4, “Stability, coherence, and fidelity.”*)

Fibonacci anyon. The nontrivial topological charge τ in the Fibonacci theory, obeying $\tau \times \tau = 1 + \tau$. Its fusion spaces grow with Fibonacci counting, and braiding supports a computationally universal gate set under standard encodings and assumptions. A Fibonacci-like circuit signature is not by itself a genuine Fibonacci quasiparticle. (*First: Chapter 15, “Fibonacci theory.”*)

Finite-size splitting. A small energy difference between states that become exactly degenerate only in an infinite or ideal topological system, often caused by virtual quasiparticle tunnelling across a finite sample. Small splitting is a scale-dependent protection feature, not exact degeneracy in a device. (*First: Chapter 21, “Analog Hamiltonian engineering.”*)

Fusion category. Mathematical data describing particle types, allowed fusion channels, associativity changes, and consistency relations. Additional braiding and nondegeneracy data are needed for the full structures used in topological quantum computation. (*First: Chapter 14, “Fusion categories without the fog.”*)

Fusion channel. A possible total topological charge resulting when specified anyons combine. Multiple channels create a fusion space in which quantum information may be encoded. (*First: Chapter 14, “Fusion categories without the fog.”*)

Fusion rule. A rule $a \times b = \sum_c N_{ab}^c c$ listing which total charges c can result from combining charges a and b , with multiplicities N_{ab}^c . It specifies possibilities, not the complete braid theory. (*First: Chapter 14, “Fusion categories without the fog.”*)

Fusion space. The vector space of consistent fusion histories for a collection of anyons with fixed total charge. Non-Abelian braids act as matrices on this space. A digitally encoded fusion space remains a simulator state space unless it comes from intrinsic quasiparticles. (*First: Chapter 13, “Anyons and braids”; formalized in Chapter 14.*)

H.23. G–H

Gap. An energy separation between designated sectors. This book distinguishes a crystal band gap, a defect-level splitting, a cluster leakage gap, and a many-body topological gap; one cannot substitute for another. (*First: Chapter 5, “Crystals, bands, and localized states”; cluster use in Chapter 11 and topological use in Chapter 16.*)

Genuine Fibonacci anyon. An intrinsic emergent quasiparticle whose fusion and braiding data realize the stated Fibonacci theory, rather than a hardware qubit programmed to reproduce those data. A claim must also say whether the host phase is chiral Fibonacci, doubled Fibonacci, or another explicitly defined theory; “Fibonacci-like” is not a substitute. (*First: Chapter 15, “Fibonacci theory”; phase distinction in Chapter 19.*)

Ground-state degeneracy. The existence of more than one lowest-energy state. In topological order, degeneracy and its dependence on spatial topology are nonlocal features; an accidental local doublet is not equivalent. (*First: Chapter 16, “Toric code.”*)

Hamiltonian (H). The operator specifying a closed system’s energies and generating its unitary time evolution. A proposed hardware Hamiltonian must include both intended terms and corrections available in the real platform. (*First: Chapter 1, “Quantum mechanics foundations”; physical defect form in Chapter 26.*)

Hermitian operator. An operator equal to its adjoint. Its eigenvalues are real, which makes Hermitian operators suitable for representing observables such as energy. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Hilbert space ($\mathcal{H}$). A complex inner-product vector space of quantum states. Its dimension counts independent state amplitudes, not necessarily particles or physical sites. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Homotopy. Classification by continuous deformation without cutting, crossing a forbidden region, or violating stated boundary conditions. It supplies the language behind winding and braids but is not itself quantum topological order. (*First: Chapter 12, “Topology for non-mathematicians.”*)

Hyperfine interaction. Coupling between electronic and nuclear magnetic moments. It can provide useful nuclear memories or resolved control, and can also create dephasing and spectral complexity. (*First: Chapter 6, “The defect zoo and its interactions.”*)

H.24. I–L

Initialization. Preparation of a qubit or many-body system in a known state or sector. High-fidelity single-qubit initialization does not imply successful preparation of a topologically ordered ground state. (*First: Chapter 3, “Qubits as controllable systems.”*)

Interaction graph. A graph whose vertices are degrees of freedom and whose edges mark available couplings, often with type, strength, and direction. Geometric proximity does not automatically create the graph required by a target Hamiltonian. (*First: Chapter 10, “Defect–defect coupling.”*)

Interstitial. An atom occupying a site between the regular lattice sites. It is a point defect and may be mobile or form a complex with other defects. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Ket ($|\psi\rangle$). Notation for a vector in Hilbert space. A bra $\langle\psi|$ is its adjoint, and $\langle\phi|\psi\rangle$ is an inner product. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Leakage. Evolution out of the subspace chosen to represent a qubit or encoded degree of freedom. A cluster’s energy separation from unwanted states suppresses but does not eliminate leakage under strong, noisy, or resonant control. (*First: Chapter 4, “Stability, coherence, and fidelity”; cluster leakage in Chapter 11.*)

Locality. The restriction that Hamiltonian terms or operations act on nearby or few-body degrees of freedom. Topological stability statements are normally claims against sufficiently weak local perturbations, not arbitrary global errors. (*First: Chapter 12, “Topology for non-mathematicians”; many-body use in Chapter 16.*)

Localized state. A state whose spatial weight is concentrated near a defect or region rather than extended across the crystal. Localization can isolate a degree of freedom but may reduce controllable coupling to its neighbors. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Logical qubit. A two-dimensional information-bearing degree of freedom encoded within a larger physical system. It may be an ordinary error-correcting-code qubit, a cluster pseudospin, or a nonlocal topological encoding; the adjective “logical” alone does not specify protection. (*First: Chapter 3, “Qubits as controllable systems.”*)

Low-energy doublet. Two cluster eigenstates selected as an effective qubit and separated from other states by a leakage gap. Their usefulness also requires controllable projected operators, preparation, and readout. A doublet is not an anyon. (*First: Chapter 11, “Defect clusters as encoded qubits.”*)

H.25. M–N

Majorana mode. An emergent degree of freedom represented by an operator equal to its own adjoint; in topological settings, spatially separated Majorana zero modes can encode nonlocal information. Majorana/Ising braiding is not computationally universal by braiding alone and is not Fibonacci order. (*First: Chapter 17, “Kitaev honeycomb model.”*)

Many-body gap. The energy between a many-body ground-state sector and the relevant excitations in a specified size and limit. It controls thermal and perturbative scales but does not alone prove topological order. (*First: Chapter 16, “Toric code”; feasibility conditions in Chapter 21.*)

Measurement (readout). An operation producing a classical outcome with probabilities determined by the quantum state. Readout fidelity, back-action, locality, and which operator is measured all matter. A local spin signal is not automatically a measurement of topological charge. (*First: Chapter 1, “Quantum mechanics foundations”; qubit implementation in Chapter 3.*)

Mixed state. A state described by a density operator with more than one nonzero eigenvalue. It can arise from classical uncertainty or from tracing out part of an entangled state. (*First: Chapter 2, “Composite quantum systems.”*)

Non-Abelian anyon. An anyon for which exchanges act by generally noncommuting matrices on a multidimensional fusion space. “Non-Abelian” does not mean Fibonacci specifically; Ising anyons are a different non-Abelian theory. (*First: Chapter 13, “Anyons and braids.”*)

Noise. Uncontrolled fluctuations or couplings that alter states, gates, measurements, or Hamiltonian parameters. The relevant description includes spectrum, spatial correlation, temporal correlation, and coupling operator—not only a single coherence time. (*First: Chapter 4, “Stability, coherence, and fidelity”; defect-array model in Chapter 30.*)

NV center. A nitrogen-vacancy complex in diamond, commonly discussed in neutral and negatively charged forms. The useful spin, optical, and coherence properties depend on charge state, isotope environment, strain, temperature, and device geometry. (*First: Chapter 7, “Diamond color centers.”*)

H.26. O–P

Operator. A linear map acting on quantum states. Operators can represent observables, transformations, projectors, or parts of a Hamiltonian; their meaning depends on the space on which they act. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Partial trace. The operation that removes an unobserved subsystem from a composite density operator, producing the reduced state of the remaining subsystem. It is how local mixed states arise from globally entangled pure states. (*First: Chapter 2, “Composite quantum systems.”*)

Passive protection. Suppression arising from the system’s spectrum, locality, or Hamiltonian without repeated syndrome measurement and recovery. It is conditional on temperature, system size, noise channels, and timescale, and does not suppress every error. (*First: Chapter 16, “Toric code”; limitations in Chapters 21 and 31.*)

Pauli operators. The matrices X , Y , and Z that generate rotations and describe observables for a two-level system. A physical-spin Pauli and an encoded-pseudospin Pauli act on different Hilbert spaces even if the same symbols are used locally. (*First: Chapter 3, “Qubits as controllable systems.”*)

Perturbative gadget. A construction using auxiliary states and weak couplings so that virtual processes generate a desired effective interaction, often of higher body order. The target term is accompanied by higher-order errors and is usually smaller than the microscopic scales. (*First: Chapter 23, “Perturbative gadgets.”*)

Phonon. A quantized collective vibration of a crystal lattice. Phonons can relax or dephase defect spins, broaden optical transitions, and sometimes mediate useful interactions. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Physical qubit. A directly controlled hardware two-level subsystem. Several physical qubits may encode one logical qubit; no fixed ratio or protection mechanism follows from the term. (*First: Chapter 3, “Qubits as controllable systems.”*)

Projector (P). An operator satisfying $P^2 = P$ that selects a subspace. In cluster engineering, P selects the low-energy encoded sector; projecting an operator gives its action

within that approximation. (*First: Chapter 1, “Quantum mechanics foundations”; cluster use in Chapter 11.*)

Pseudospin. An effective two-level degree of freedom represented with spin-1/2 mathematics even when its states are collective combinations of several microscopic spins or orbitals. It is a modelling object, not necessarily a literal electron spin. (*First: Chapter 11, “Defect clusters as encoded qubits.”*)

H.27. Q–R

Quantum dimension (d_a). A measure of the asymptotic growth of the fusion space when many anyons of type a are added. It need not be an integer; for Fibonacci τ , it is the golden ratio. It is not the ordinary dimension of a single local qubit. (*First: Chapter 14, “Fusion categories without the fog.”*)

Quasiparticle. A collective excitation that behaves as a particle within an effective many-body description. An emergent anyon is a quasiparticle with topological exchange and fusion properties; a bare defect spin is not called an anyon merely because it is localized. (*First: Chapter 13, “Anyons and braids.”*)

Qubit. A controllable quantum degree of freedom with a chosen two-dimensional state space. A usable qubit requires a lifecycle—initialization, coherent operations, and readout—not just two spectral levels. (*First: Chapter 3, “Qubits as controllable systems.”*)

R -move. The unitary associated with exchanging two anyons in a specified fusion channel. Together with fusion data and F -moves, consistent R -moves determine braid actions. (*First: Chapter 14, “Fusion categories without the fog.”*)

Relaxation time (T_1). The characteristic time for population to return toward equilibrium, commonly after an excitation. A long T_1 does not imply a long phase-coherence time T_2 . (*First: Chapter 4, “Stability, coherence, and fidelity.”*)

H.28. S

Schrieffer–Wolff transformation. A perturbative unitary block-diagonalization used to derive a low-energy Hamiltonian by eliminating virtual transitions between selected and high-energy sectors. It is controlled only when couplings are small relative to appropriate energy denominators. (*First: Chapter 22, “Schrieffer–Wolff transformation.”*)

Simulated or emulated anyon. A state, defect, code excitation, or gate action intentionally mapped to an anyon model on hardware that may possess no intrinsic anyonic

phase. Such a simulation can correctly demonstrate model operations while providing active rather than passive protection. (*First: Chapter 13, “Anyons and braids”; experimental treatment in Chapter 20.*)

Spin. An intrinsic quantum angular momentum. In a solid, the effective spin label may include orbital character and crystal-field effects; it must be tied to specified levels and a Hamiltonian. (*First: Chapter 1, “Quantum mechanics foundations”; defect setting in Chapter 6.*)

Spin–orbit coupling. Interaction between spin and orbital motion, strongly shaped by crystal symmetry. It can enable optical selection and large splittings while also opening phonon-mediated relaxation pathways. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Stabilizer. An operator whose specified eigenvalue defines a code or model subspace; a commuting set of stabilizers jointly constrains allowed states. Measuring stabilizers actively is different from realizing the same operators as energetic Hamiltonian terms. (*First: Chapter 16, “Toric code.”*)

Stacking fault. An extended planar defect in which the normal sequence of crystal layers is interrupted. It can alter local electronic structure and strain over a broad region. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Strain. Spatial deformation of a crystal. Through strain coupling it shifts or mixes defect levels and can act as noise, inhomogeneity, a tuning control, or a mediated-interaction channel. (*First: Chapter 6, “The defect zoo and its interactions.”*)

String operator. A product of local operators along a path. In a topologically ordered model, an open string can create excitations at its endpoints and a noncontractible closed string can act logically. (*First: Chapter 16, “Toric code.”*)

String-net. A fluctuating network of labelled strings subject to local branching and re-coupling rules. In a Levin–Wen Hamiltonian, string-net condensation realizes doubled topological orders; a drawn network alone is not evidence of such a phase. (*First: Chapter 18, “Levin–Wen string nets.”*)

Substitutional defect. A lattice site occupied by a different atomic species from the ideal host atom. Its electronic behavior depends on chemistry, local symmetry, and charge compensation. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Superexchange. An effective interaction between localized spins mediated virtually through intermediate orbitals or sites. Its sign and magnitude arise from microscopic hopping and energy costs, not simply distance. (*First: Chapter 10, “Defect–defect coupling”; perturbative derivation in Chapter 22.*)

Superposition. A linear combination of quantum states. It produces new physical predictions because complex amplitudes can interfere; it is not merely ignorance about which classical state is present. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Symmetry-protected subspace. A subspace insulated from selected transitions because a symmetry forbids or suppresses the relevant matrix elements. Breaking the symmetry can remove the protection, and the mechanism is distinct from intrinsic topological order. (*First: Chapter 11, “Defect clusters as encoded qubits.”*)

H.29. T

T_2 and T_2^* . T_2 characterizes homogeneous phase coherence under a specified refocusing convention; T_2^* commonly describes free-induction decay including quasi-static inhomogeneity. Pulse sequence and fitting model must accompany either number. (*First: Chapter 4, “Stability, coherence, and fidelity.”*)

Tensor product ($\otimes$). The mathematical operation combining quantum state spaces: two qubits have $\mathcal{H}_1 \otimes \mathcal{H}_2$, not a simple choice between the two spaces. Tensor products allow entangled states. (*First: Chapter 2, “Composite quantum systems.”*)

Topological charge. A label for an anyon sector specifying how an excitation behaves under fusion and braiding. It is conserved according to fusion rules and need not equal electric charge. (*First: Chapter 14, “Fusion categories without the fog.”*)

Topological error protection. Operational suppression of specified logical errors using nonlocal encoding, a gap, braid structure, active correction, or a combination. It is never a claim of zero error and must name temperature, size, noise, leakage, preparation, and readout assumptions. (*First: Chapter 16, “Toric code”; limits in Chapter 31.*)

Topological invariant. A quantity unchanged under an allowed class of continuous deformations. A winding number is a classical example. Possessing a classical topological invariant does not imply quantum topological order. (*First: Chapter 12, “Topology for non-mathematicians.”*)

Topological order. Intrinsic many-body order characterized by nonlocal structure such as long-range entanglement, topology-dependent ground-state sectors, and anyonic excitations in standard two-dimensional gapped examples. It is not ordinary symmetry breaking, an encoded local cluster, or a digitally prepared wavefunction by itself. (*First: Chapter 12, “Topology for non-mathematicians”; concrete model in Chapter 16.*)

Topological phase. A phase of matter with specified topological organization of its low-energy states and excitations. In this book the unqualified phrase normally means intrinsic topological order; symmetry-protected topological phases are named explicitly when relevant. (*First: Chapter 16, “Toric code.”*)

Topological quantum computation. Storage and processing of information in nonlocal topological degrees of freedom, often using creation, fusion, measurement, and braiding of anyons. Its gate set and protection depend on the anyon theory and physical implementation. (*First: Chapter 13, “Anyons and braids”; Fibonacci case in Chapter 15.*)

Topology. The study of properties unchanged by continuous deformation under stated rules. In this manuscript it provides intuition and mathematics for global sectors, but the word alone does not imply protection or a quantum phase. (*First: Chapter 12, “Topology for non-mathematicians.”*)

Toric code. An exactly solvable spin model with commuting star and plaquette terms, topological ground-state degeneracy, and Abelian e and m anyons. It is a benchmark for topological order, not a Fibonacci model. (*First: Chapter 16, “Toric code.”*)

H.30. U–Z

Unit cell. A repeated building block of a crystal lattice. It contains the basis atoms needed to reconstruct the ideal periodic structure and is distinct from a defect cluster chosen for encoding. (*First: Chapter 5, “Crystals, bands, and localized states.”*)

Unitary evolution. Reversible quantum evolution $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ that preserves inner products and total probability for a closed system. An open subsystem generally requires a channel rather than a unitary acting on that subsystem alone. (*First: Chapter 1, “Quantum mechanics foundations.”*)

Vacancy. A missing atom at a normally occupied lattice site. A vacancy may be mobile, charged, optically dark, or part of a useful complex; the structural label does not determine its qubit properties. (*First: Chapter 6, “The defect zoo and its interactions.”*)

Virtual excitation. An intermediate high-energy state appearing in perturbation theory without being populated as a long-lived real excitation. Virtual paths can generate superexchange or gadget terms, with strengths suppressed by energy denominators. (*First: Chapter 22, “Schrieffer–Wolff transformation.”*)

Wilson loop. A closed nonlocal operator used to diagnose or manipulate gauge and topological sectors. A measured local stabilizer is not the same observable as a noncontractible Wilson loop. (*First: Chapter 16, “Toric code”; measurement in Chapter 36.*)

Winding number. An integer counting how many times a directed loop wraps around a puncture under appropriate conditions. It is robust to allowed deformations but is not by itself a quantum-protected qubit. (*First: Chapter 12, “Topology for non-mathematicians.”*)

Zero-field splitting. A splitting of spin sublevels present without an applied magnetic field, commonly produced by internal spin–spin and crystal-field-related anisotropy. Its value is a spectral parameter, not a coherence time. (*First: Chapter 7, “Diamond color centers.”*)

Annotated bibliography

Each work appears once under its primary subject. Reference labels are global and stable across the manuscript; DOI resolver links are used as the stable URL whenever a DOI exists, with arXiv records included where available.

H.31. Introductory quantum mechanics

- [R001] S. Axler, *Linear Algebra Done Right*, 3rd ed., Springer, 2015. DOI: 10.1007/978-3-319-11080-6.
- [R002] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013. DOI: 10.1017/CBO9781139020411.
- [R003] B. C. Hall, *Quantum Theory for Mathematicians*, Springer, 2013. DOI: 10.1007/978-1-4614-7116-5.
- [R004] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press, 2010. DOI: 10.1017/CBO9780511976667.
- [R005] A. Peres, *Quantum Theory: Concepts and Methods*, Kluwer Academic, 1995. DOI: 10.1007/0-306-47120-5.
- [R006] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. DOI: 10.1017/9781316848142.
- [R007] W. F. Stinespring, “Positive functions on C -algebras,” *Proceedings of the American Mathematical Society** **6**, 211–216 (1955). DOI: 10.1090/S0002-9939-1955-0069403-4.
- [R008] M.-D. Choi, “Completely positive linear maps on complex matrices,” *Linear Algebra and its Applications* **10**, 285–290 (1975). DOI: 10.1016/0024-3795(75)90075-0.
- [R009] G. Lindblad, “On the generators of quantum dynamical semigroups,” *Communications in Mathematical Physics* **48**, 119–130 (1976). DOI: 10.1007/BF01608499.
- [R010] V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, “Completely positive dynamical semigroups of N -level systems,” *Journal of Mathematical Physics* **17**, 821–825 (1976). DOI: 10.1063/1.522979.
- [R011] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, 2002. DOI: 10.1093/acprof:oso/9780199213900.001.0001.

- [R037] M. Born, “Zur Quantenmechanik der Stoßvorgänge,” *Zeitschrift für Physik* **37**, 863–867 (1926). DOI: 10.1007/BF01397477.
- [R038] W. Gerlach and O. Stern, “Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld,” *Zeitschrift für Physik* **9**, 349–352 (1922). DOI: 10.1007/BF01326983.
- [R039] B. Friedrich and D. Herschbach, “Stern and Gerlach: How a Bad Cigar Helped Reorient Atomic Physics,” *Physics Today* **56**(12), 53–59 (2003). DOI: 10.1063/1.1650229.
- [R040] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford University Press (1958; reissued 1981). ISBN: 978-0-19-852011-5. Stable URL: Oxford Academic.
- [R041] E. Schrödinger, “Quantisierung als Eigenwertproblem (Vierte Mitteilung),” *Annalen der Physik* **386**, 109–139 (1926). DOI: 10.1002/andp.19263861802.
- [R042] A. Einstein, B. Podolsky, and N. Rosen, “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?,” *Physical Review* **47**, 777–780 (1935). DOI: 10.1103/PhysRev.47.777.
- [R043] J. S. Bell, “On the Einstein Podolsky Rosen Paradox,” *Physics Physique Fizika* **1**, 195–200 (1964). DOI: 10.1103/PhysicsPhysiqueFizika.1.195.
- [R044] R. F. Werner, “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model,” *Physical Review A* **40**, 4277–4281 (1989). DOI: 10.1103/PhysRevA.40.4277.
- [R045] K. Kraus, “General state changes in quantum theory,” *Annals of Physics* **64**, 311–335 (1971). DOI: 10.1016/0003-4916(71)90108-4.
- [R046] W. H. Zurek, “Decoherence, einselection, and the quantum origins of the classical,” *Reviews of Modern Physics* **75**, 715–775 (2003). DOI: 10.1103/RevModPhys.75.715; arXiv: quant-ph/0105127.
- [R047] M. Schlosshauer, “Decoherence, the measurement problem, and interpretations of quantum mechanics,” *Reviews of Modern Physics* **76**, 1267–1305 (2005). DOI: 10.1103/RevModPhys.76.1267; arXiv: quant-ph/0312059.
- [R048] B. Hensen *et al.*, “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres,” *Nature* **526**, 682–686 (2015). DOI: 10.1038/nature15759; arXiv: 1508.05949.
- [R049] F. Bloch, “Nuclear Induction,” *Physical Review* **70**, 460–474 (1946). DOI: 10.1103/PhysRev.70.460.
- [R050] D. P. DiVincenzo, “The Physical Implementation of Quantum Computation,” *Fortschritte der Physik* **48**, 771–783 (2000). DOI: 10.1002/1521-3978(200009)48:9/11<771::AID-PROP771>3.0.CO;2-E; arXiv: quant-ph/0002077.

- [R051] D. Loss and D. P. DiVincenzo, “Quantum Computation with Quantum Dots,” *Physical Review A* **57**, 120–126 (1998). DOI: 10.1103/PhysRevA.57.120.
- [R052] F. H. L. Koppens, C. Buizert, K. J. Tielrooij, I. T. Vink, K. C. Nowack, T. Meunier, L. P. Kouwenhoven, and L. M. K. Vandersypen, “Driven Coherent Oscillations of a Single Electron Spin in a Quantum Dot,” *Nature* **442**, 766–771 (2006). DOI: 10.1038/nature05065.
- [R053] J. M. Elzerman, R. Hanson, L. H. Willems van Beveren, B. Witkamp, L. M. K. Vandersypen, and L. P. Kouwenhoven, “Single-Shot Read-Out of an Individual Electron Spin in a Quantum Dot,” *Nature* **430**, 431–435 (2004). DOI: 10.1038/nature02693.
- [R054] D. D. Awschalom, R. Hanson, J. Wrachtrup, and B. B. Zhou, “Quantum Technologies with Optically Interfaced Solid-State Spins,” *Nature Photonics* **12**, 516–527 (2018). DOI: 10.1038/s41566-018-0232-2.
- [R055] E. Knill and R. Laflamme, “Theory of Quantum Error-Correcting Codes,” *Physical Review A* **55**, 900–911 (1997). DOI: 10.1103/PhysRevA.55.900; arXiv: quant-ph/9604034.
- [R056] N. F. Ramsey, “A Molecular Beam Resonance Method with Separated Oscillating Fields,” *Physical Review* **78**, 695–699 (1950). DOI: 10.1103/PhysRev.78.695.
- [R057] E. L. Hahn, “Spin Echoes,” *Physical Review* **80**, 580–594 (1950). DOI: 10.1103/PhysRev.80.580.
- [R058] G. Ithier et al., “Decoherence in a superconducting quantum bit circuit,” *Physical Review B* **72**, 134519 (2005). DOI: 10.1103/PhysRevB.72.134519; arXiv: cond-mat/0508588.
- [R059] Ł. Cywiński, R. M. Lutchyn, C. P. Nave, and S. Das Sarma, “How to enhance dephasing time in superconducting qubits,” *Physical Review B* **77**, 174509 (2008). DOI: 10.1103/PhysRevB.77.174509; arXiv: 0712.2225.
- [R060] R. Jozsa, “Fidelity for Mixed Quantum States,” *Journal of Modern Optics* **41**, 2315–2323 (1994). DOI: 10.1080/09500349414552171.
- [R061] M. A. Nielsen, “A simple formula for the average gate fidelity of a quantum dynamical operation,” *Physics Letters A* **303**, 249–252 (2002). DOI: 10.1016/S0375-9601(02)01272-0; arXiv: quant-ph/0205035.
- [R062] E. Magesan et al., “Efficient Measurement of Quantum Gate Error by Interleaved Randomized Benchmarking,” *Physical Review Letters* **109**, 080505 (2012). DOI: 10.1103/PhysRevLett.109.080505; arXiv: 1203.4550.
- [R063] D. Keith et al., “Benchmarking high fidelity single-shot readout of semiconductor qubits,” (2018). arXiv: 1811.03630.

- [R064] C. J. Wood and J. M. Gambetta, “Quantification and characterization of leakage errors,” *Physical Review A* **97**, 032306 (2018). DOI: 10.1103/PhysRevA.97.032306; arXiv: 1704.03081.
- [R065] Google Quantum AI and R. Acharya et al., “Suppressing quantum errors by scaling a surface code logical qubit,” *Nature* **614**, 676–681 (2023). DOI: 10.1038/s41586-022-05434-1; arXiv: 2207.06431.

H.32. Defect centers

- [R066] F. Bloch, “Über die Quantenmechanik der Elektronen in Kristallgittern,” *Zeitschrift für Physik* **52**, 555–600 (1929). DOI: 10.1007/BF01339455.
- [R067] J. C. Slater and G. F. Koster, “Simplified LCAO Method for the Periodic Potential Problem,” *Physical Review* **94**, 1498–1524 (1954). DOI: 10.1103/PhysRev.94.1498.
- [R068] W. Kohn, “Analytic Properties of Bloch Waves and Wannier Functions,” *Physical Review* **115**, 809–821 (1959). DOI: 10.1103/PhysRev.115.809.
- [R069] C. G. Van de Walle and J. Neugebauer, “First-principles calculations for defects and impurities: Applications to III-nitrides,” *Journal of Applied Physics* **95**, 3851–3879 (2004). DOI: 10.1063/1.1682673.
- [R070] C. Freysoldt, B. Grabowski, T. Hickel, J. Neugebauer, G. Kresse, A. Janotti, and C. G. Van de Walle, “First-principles calculations for point defects in solids,” *Reviews of Modern Physics* **86**, 253–305 (2014). DOI: 10.1103/RevModPhys.86.253. This review systematizes first-principles calculations of charged point defects, finite-size corrections, transition levels, and formation energies. It anchors what electronic-structure calculations can and cannot establish about proposed qubit defects.
- [R071] J. R. Weber, W. F. Koehl, J. B. Varley, A. Janotti, B. B. Buckley, C. G. Van de Walle, and D. D. Awschalom, “Quantum computing with defects,” *Proceedings of the National Academy of Sciences* **107**, 8513–8518 (2010). DOI: 10.1073/pnas.1003052107.
- [R072] A. Alkauskas, Q. Yan, and C. G. Van de Walle, “First-principles theory of non-radiative carrier capture via multiphonon emission,” *Physical Review B* **90**, 075202 (2014). DOI: 10.1103/PhysRevB.90.075202.
- [R073] G. Wolfowicz et al., “Quantum guidelines for solid-state spin defects,” *Nature Reviews Materials* **6**, 906–925 (2021). DOI: 10.1038/s41578-021-00306-y. Wolfowicz and collaborators synthesize materials, optical, spin, and coherence criteria for useful solid-state defects. The review is pivotal because it prevents long coherence or bright emission alone from being mistaken for a scalable qubit platform.

- [R078] A. Norambuena, E. Muñoz, H. T. Dinani, A. Jarmola, P. Maletinsky, D. Budker, and J. R. Maze, “Spin-lattice relaxation of individual solid-state spins,” *Physical Review B* **97**, 094304 (2018). DOI: 10.1103/PhysRevB.97.094304; arXiv: 1711.10280.
- [R100] H. L. Stern, Q. Gu, J. Jarman, *et al.*, “Room-temperature optically detected magnetic resonance of single defects in hexagonal boron nitride,” *Nature Communications* **13**, 618 (2022). DOI: 10.1038/s41467-022-28169-z.
- [R101] M. Zhong, M. P. Hedges, R. L. Ahlefeldt, *et al.*, “Optically addressable nuclear spins in a solid with a six-hour coherence time,” *Nature* **517**, 177–180 (2015). DOI: 10.1038/nature14025.
- [R102] J. M. Kindem, A. Ruskuc, J. G. Bartholomew, *et al.*, “Control and single-shot readout of an ion embedded in a nanophotonic cavity,” *Nature* **580**, 201–204 (2020). DOI: 10.1038/s41586-020-2160-9; arXiv: 1907.12161.
- [R103] B. Voisin, J. Bocquel, A. Tankasala, *et al.*, “Valley interference and spin exchange at the atomic scale in silicon,” *Nature Communications* **11**, 6124 (2020). DOI: 10.1038/s41467-020-19835-1.
- [R105] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, “CODATA recommended values of the fundamental physical constants,” NIST/CODATA database (2022 adjustment, accessed 2026), stable URL: <https://physics.nist.gov/cuu/Constants/>.
- [R106] F. Dolde *et al.*, “High-fidelity spin entanglement using optimal control,” *Nature Communications* **5**, 3371 (2014). DOI: 10.1038/ncomms4371; arXiv: 1309.4430. This experiment used optimal control to create high-fidelity entanglement between coupled solid-state spins. It supports the availability of coherent pair interactions while also illustrating the calibration burden hidden by an effective two-spin Hamiltonian.
- [R108] P. W. Anderson, “Antiferromagnetism. Theory of superexchange interaction,” *Physical Review* **79**, 350–356 (1950). DOI: 10.1103/PhysRev.79.350.
- [R112] P. Zanardi and M. Rasetti, “Noiseless quantum codes,” *Physical Review Letters* **79**, 3306–3309 (1997). DOI: 10.1103/PhysRevLett.79.3306.
- [R113] E. Knill, R. Laflamme, and L. Viola, “Theory of quantum error correction for general noise,” *Physical Review Letters* **84**, 2525–2528 (2000). DOI: 10.1103/PhysRevLett.84.2525; arXiv: quant-ph/9908066.
- [R114] D. P. DiVincenzo, D. Bacon, J. Kempe, G. Burkard, and K. B. Whaley, “Universal quantum computation with the exchange interaction,” *Nature* **408**, 339–342 (2000). DOI: 10.1038/35042541; arXiv: quant-ph/0005116.
- [R115] E. A. Laird, J. M. Taylor, D. P. DiVincenzo, C. M. Marcus, M. P. Hanson, and A. C. Gossard, “Coherent spin manipulation in an exchange-only qubit,” *Physical Review B* **82**, 075403 (2010). DOI: 10.1103/PhysRevB.82.075403; arXiv: 1005.0273.

- [R116] A. Reiserer, N. Kalb, M. S. Blok, K. J. M. van Bemmelen, D. J. Twitchen, M. Markham, T. H. Taminiau, and R. Hanson, “Robust quantum-network memory using decoherence-protected subspaces of nuclear spins,” *Physical Review X* **6**, 021040 (2016). DOI: 10.1103/PhysRevX.6.021040; arXiv: 1603.01602. Reiserer and collaborators encoded information in a decoherence-protected subspace of diamond spins. The experiment is direct evidence that small defect clusters can suppress selected noise channels, not that they generate passive topological protection.
- [R117] J. Cramer, N. Kalb, M. A. Rol, B. Hensen, M. S. Blok, M. Markham, D. J. Twitchen, R. Hanson, and T. H. Taminiau, “Repeated quantum error correction on a continuously encoded qubit by real-time feedback,” *Nature Communications* **7**, 11526 (2016). DOI: 10.1038/ncomms11526; arXiv: 1508.01388. Cramer and collaborators repeatedly detected and corrected errors in a diamond nuclear-spin register. The result establishes local encoded-control capability and clarifies the distinction between active quantum error correction and an equilibrium topological phase.
- [R118] H. P. Bartling, M. H. Abobeih, B. Pingault, M. J. Degen, S. J. H. Loenen, C. E. Bradley, J. Randall, M. Markham, D. J. Twitchen, and T. H. Taminiau, “Entanglement of spin-pair qubits with intrinsic dephasing times exceeding a minute,” *Physical Review X* **12**, 011048 (2022). DOI: 10.1103/PhysRevX.12.011048; arXiv: 2103.07961.
- [R119] C. E. Bradley, J. Randall, M. H. Abobeih, R. C. Berrevoets, M. J. Degen, M. A. Bakker, M. Markham, D. J. Twitchen, and T. H. Taminiau, “A ten-qubit solid-state spin register with quantum memory up to one minute,” *Physical Review X* **9**, 031045 (2019). DOI: 10.1103/PhysRevX.9.031045; arXiv: 1905.02094.

H.33. Diamond

- [R074] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg, “The nitrogen-vacancy colour centre in diamond,” *Physics Reports* **528**, 1–45 (2013). DOI: 10.1016/j.physrep.2013.02.001; arXiv: 1302.3288. This comprehensive NV-center review anchors the actual electronic structure, spin Hamiltonian, optical cycle, strain response, and relaxation channels used throughout the manuscript. It prevents the proposed architecture from quietly acquiring interactions or selection rules that diamond does not supply.
- [R075] Á. Gali, “Ab initio theory of the nitrogen-vacancy center in diamond,” *Reviews of Modern Physics* **91**, 015004 (2019). DOI: 10.1103/RevModPhys.91.015004; arXiv: 1906.00047.
- [R076] M. W. Doherty, N. B. Manson, P. Delaney, and L. C. L. Hollenberg, “The negatively charged nitrogen-vacancy centre in diamond: the electronic solution,” *New Journal of Physics* **13**, 025019 (2011). DOI: 10.1088/1367-2630/13/2/025019; arXiv: 1008.5224.

- [R077] P. Udvarhelyi, V. O. Shkolnikov, A. Gali, G. Burkard, and A. Pályi, “Spin-strain interaction in nitrogen-vacancy centers in diamond,” *Physical Review B* **98**, 075201 (2018). DOI: 10.1103/PhysRevB.98.075201; arXiv: 1712.02684.
- [R079] G. Balasubramanian *et al.*, “Ultralong spin coherence time in isotopically engineered diamond,” *Nature Materials* **8**, 383–387 (2009). DOI: 10.1038/nmat2420.
- [R080] F. Dolde *et al.*, “Room-temperature entanglement between single defect spins in diamond,” *Nature Physics* **9**, 139–143 (2013). DOI: 10.1038/nphys2545; arXiv: 1212.2804. Dolde and collaborators demonstrated room-temperature entanglement between two individually addressable NV electronic spins. The result establishes pairwise coherent coupling under favorable geometry; it does not demonstrate a dense, uniform many-body array.
- [R081] M. Iuliano *et al.*, “Unconditionally teleported quantum gates between remote solid-state qubit registers,” *Nature Communications* **17** (2026). DOI: 10.1038/s41467-026-72818-6.
- [R082] C. Bradac, W. Gao, J. Forneris, M. E. Trusheim, and I. Aharonovich, “Quantum nanophotonics with group IV defects in diamond,” *Nature Communications* **10**, 5625 (2019). DOI: 10.1038/s41467-019-13332-w.
- [R083] D. D. Sukachev *et al.*, “Silicon-vacancy spin qubit in diamond: a quantum memory exceeding 10 ms with single-shot state readout,” *Physical Review Letters* **119**, 223602 (2017). DOI: 10.1103/PhysRevLett.119.223602; arXiv: 1708.08852.
- [R084] K. Senkalla, G. Genov, M. H. Metsch, P. Siyushev, and F. Jelezko, “Germanium vacancy in diamond quantum memory exceeding 20 ms,” *Physical Review Letters* **132**, 026901 (2024). DOI: 10.1103/PhysRevLett.132.026901; arXiv: 2308.09666.
- [R085] I. Karapatzakis *et al.*, “Microwave control of the tin-vacancy spin qubit in diamond with a superconducting waveguide,” *Physical Review X* **14**, 031036 (2024). DOI: 10.1103/PhysRevX.14.031036.
- [R086] R. E. Evans *et al.*, “Photon-mediated interactions between quantum emitters in a diamond nanocavity,” *Science* **362**, 662–665 (2018). DOI: 10.1126/science.aau4691; arXiv: 1807.04265.
- [R087] X. Cheng *et al.*, “Laser activation of single group-IV colour centres in diamond,” *Nature Communications* **16**, 5124 (2025). DOI: 10.1038/s41467-025-60373-5.
- [R107] V. R. Kortan, C. Şahin, and M. E. Flatté, “Nanometer-scale exchange interactions between spin centers in diamond,” *Physical Review B* **93**, 220402(R) (2016). DOI: 10.1103/PhysRevB.93.220402; arXiv: 1603.03485.
- [R109] M.-A. Lemonde *et al.*, “Phonon networks with silicon-vacancy centers in diamond waveguides,” *Physical Review Letters* **120**, 213603 (2018). DOI: 10.1103/PhysRevLett.120.213603; arXiv: 1801.01904.

- [R110] A. Bermudez, F. Jelezko, M. B. Plenio, and A. Retzker, “Electron-mediated nuclear-spin interactions between distant nitrogen-vacancy centers,” *Physical Review Letters* **107**, 150503 (2011). DOI: 10.1103/PhysRevLett.107.150503; arXiv: 1107.2617.
- [R111] J. J. Nakane, K. Tahara, K. Kutsuki, and A. Yamakage, “Phonon-mediated spin-spin interaction: A general theory and application to diamond nitrogen vacancy centers,” *Physical Review B* **110**, 064428 (2024). DOI: 10.1103/PhysRevB.110.064428.
- [R120] M.-R. Yun, F.-Q. Guo, L.-L. Yan, E. Liang, Y. Zhang, S.-L. Su, C. X. Shan, and Y. Jia, “Parallel-path implementation of nonadiabatic geometric quantum gates in a decoherence-free subspace with nitrogen-vacancy centers,” *Physical Review A* **105**, 012611 (2022). DOI: 10.1103/PhysRevA.105.012611.
- [R191] F. Dolde, H. Fedder, M. W. Doherty, *et al.*, “Electric-field sensing using single diamond spins,” *Nature Physics* **7**, 459–463 (2011). DOI: 10.1038/nphys1969.
- [R193] E. R. MacQuarrie, T. A. Gosavi, A. M. Moehle, N. R. Jungwirth, S. A. Bhave, and G. D. Fuchs, “Coherent control of a nitrogen-vacancy center spin ensemble with a diamond mechanical resonator,” *Optica* **2**, 233–238 (2015). DOI: 10.1364/OP-TICA.2.000233.
- [R194] A. Jarmola, V. M. Acosta, K. Jensen, S. Chemerisov, and D. Budker, “Temperature- and magnetic-field-dependent longitudinal spin relaxation in nitrogen-vacancy ensembles in diamond,” *Physical Review Letters* **108**, 197601 (2012). DOI: 10.1103/PhysRevLett.108.197601.
- [R197] S. Felton *et al.*, “Hyperfine interaction in the ground state of the negatively charged nitrogen vacancy center in diamond,” *Physical Review B* **79**, 075203 (2009). DOI: 10.1103/PhysRevB.79.075203.
- [R198] L. Robledo *et al.*, “Spin dynamics in the optical cycle of single nitrogen-vacancy centres in diamond,” *New Journal of Physics* **13**, 025013 (2011). DOI: 10.1088/1367-2630/13/2/025013; arXiv: 1010.1192.
- [R200] S. Meesala *et al.*, “Strain engineering of the silicon-vacancy center in diamond,” *Physical Review B* **97**, 205444 (2018). DOI: 10.1103/PhysRevB.97.205444; arXiv: 1801.09833.
- [R201] K. D. Jahnke *et al.*, “Electron–phonon processes of the silicon-vacancy centre in diamond,” *New Journal of Physics* **17**, 043011 (2015). DOI: 10.1088/1367-2630/17/4/043011; arXiv: 1411.2871.
- [R212] S. Pezzagna and J. Meijer, “Quantum computer based on color centers in diamond,” *Applied Physics Reviews* **8**, 011308 (2021). DOI: 10.1063/5.0007444.
- [R214] L. Childress *et al.*, “Coherent dynamics of coupled electron and nuclear spin qubits in diamond,” *Science* **314**, 281–285 (2006). DOI: 10.1126/science.1131871.

- [R216] T. Rosskopf *et al.*, “Investigation of surface magnetic noise by shallow spins in diamond,” *Physical Review Letters* **112**, 147602 (2014). DOI: 10.1103/PhysRevLett.112.147602.
- [R217] P. Jamonneau *et al.*, “Competition between electric field and magnetic field noise in the decoherence of a single spin in diamond,” *Physical Review B* **93**, 024305 (2016). DOI: 10.1103/PhysRevB.93.024305.
- [R218] S. Dhomkar, H. Jayakumar, P. R. Zangara, and C. A. Meriles, “Charge dynamics in near-surface, variable-density ensembles of nitrogen-vacancy centers in diamond,” *Nano Letters* **18**, 4046–4052 (2018). DOI: 10.1021/acs.nanolett.8b01739.
- [R219] J. Teissier, A. Barfuss, P. Appel, E. Neu, and P. Maletinsky, “Strain coupling of a nitrogen-vacancy center spin to a diamond mechanical oscillator,” *Physical Review Letters* **113**, 020503 (2014). DOI: 10.1103/PhysRevLett.113.020503.
- [R234] M. H. Abobeih *et al.*, “Fault-tolerant operation of a logical qubit in a diamond quantum processor,” *Nature* **606**, 884–889 (2022). DOI: 10.1038/s41586-022-04819-6; arXiv: 2108.01646.
- [R235] S. Pezzagna, D. Wildanger, P. Mazarov *et al.*, “Nanoscale Engineering and Optical Addressing of Single Spins in Diamond,” *Small* **6**, 2117–2121 (2010). DOI: 10.1002/smll.201000902.
- [R236] H. Zhang, K. Arai, C. Belthangady, J.-C. Jaskula, and R. L. Walsworth, “Selective addressing of solid-state spins at the nanoscale via magnetic resonance frequency encoding,” *npj Quantum Information* **3**, 31 (2017). DOI: 10.1038/s41534-017-0033-3.
- [R237] B. J. Shields, Q. P. Unterreithmeier, N. P. de Leon, H. Park, and M. D. Lukin, “Efficient Readout of a Single Spin State in Diamond via Spin-to-Charge Conversion,” *Physical Review Letters* **114**, 136402 (2015). DOI: 10.1103/PhysRevLett.114.136402; arXiv: 1410.0370.
- [R238] A. Sipahigil, R. E. Evans, D. D. Sukachev *et al.*, “An integrated diamond nanophotonics platform for quantum-optical networks,” *Science* **354**, 847–850 (2016). DOI: 10.1126/science.aah6875; arXiv: 1608.05147.
- [R239] M. Cambria, S. Chand, C. M. Reiter, and S. Kolkowitz, “Scalable Parallel Measurement of Individual Nitrogen-Vacancy Centers,” *Physical Review X* **15**, 031015 (2025). DOI: 10.1103/jdzq-jbfz; arXiv: 2408.11715. Cambria and collaborators demonstrate scalable parallel optical measurement of individual NV centers. This directly addresses readout throughput, while crosstalk, calibration, and topological-sector measurement remain architecture-level problems.
- [R242] L. B. Hughes, S. A. Meynell, W. Wu, S. Parthasarathy, L. Chen, Z. Zhang, Z. Wang, E. J. Davis, K. Mukherjee, N. Y. Yao, and A. C. Bleszynski Jayich, “Strongly Interacting, Two-Dimensional, Dipolar Spin Ensembles in (111)-Oriented Diamond,”

Physical Review X **15**, 021035 (2025). DOI: 10.1103/PhysRevX.15.021035; arXiv: 2404.10075.

- [R244] N. Y. Yao, L. Jiang, A. V. Gorshkov, P. C. Maurer, G. Giedke, J. I. Cirac, and M. D. Lukin, “Scalable architecture for a room temperature solid-state quantum information processor,” *Nature Communications* **3**, 800 (2012). DOI: 10.1038/ncomms1788; arXiv: 1012.2864.
- [R246] M. Ruf, N. H. Wan, H. Choi, D. Englund, and R. Hanson, “Quantum networks based on color centers in diamond,” *Journal of Applied Physics* **130**, 070901 (2021). DOI: 10.1063/5.0056534; arXiv: 2105.04341.

H.34. Silicon carbide

- [R096] D. J. Christle, A. L. Falk, P. Andrich, *et al.*, “Isolated electron spins in silicon carbide with millisecond coherence times,” *Nature Materials* **14**, 160–163 (2015). DOI: 10.1038/nmat4144; arXiv: 1406.7325. Christle and collaborators established optically addressable isolated defect spins with millisecond-scale coherence in silicon carbide. The work is a foundational demonstration of SiC as a serious solid-state spin host rather than merely a theoretical alternative to diamond.
- [R097] C. P. Anderson, E. O. Glen, C. Zeledon, *et al.*, “Five-second coherence of a single spin with single-shot readout in silicon carbide,” *Science Advances* **8**, eabm5912 (2022). DOI: 10.1126/sciadv.abm5912; arXiv: 2110.01590.
- [R099] H. Hu, Y. Zhou, A. Yi, *et al.*, “Room-temperature waveguide integrated quantum register in a semiconductor photonic platform,” *Nature Communications* **15**, 10256 (2024). DOI: 10.1038/s41467-024-54606-2. This work integrates a room-temperature multiqubit register with a SiC photonic waveguide. It is important system-level evidence for host integration, but it is not a demonstration of autonomous topological order.
- [R104] T. Nishikawa, N. Morioka, H. Abe, *et al.*, “Coherent photoelectrical readout of single spins in silicon carbide at room temperature,” *Nature Communications* **16**, 3405 (2025). DOI: 10.1038/s41467-025-58629-1.
- [R205] J. Wang *et al.*, “Efficient generation of an array of single silicon-vacancy defects in silicon carbide,” *Physical Review Applied* **7**, 064021 (2017). DOI: 10.1103/PhysRevApplied.7.064021. Wang and collaborators generated arrays of individual silicon-vacancy defects in SiC. It is direct fabrication evidence for array creation in that host, not evidence for coherent many-body topological interactions.

H.35. Corundum/sapphire

- [R088] V. K. Sewani, R. J. Stöhr, R. Kolesov, H. H. Vallabhapurapu, T. Simmet, A. Morello, and A. Laucht, “Spin thermometry and spin relaxation of optically detected Cr³⁺ ions in Al₂O₃ (ruby),” *Physical Review B* **102**, 104114 (2020). DOI: 10.1103/PhysRevB.102.104114; arXiv: 2007.07493. This experiment quantifies optical spin thermometry and relaxation for a confocally probed chromium-ion ensemble in ruby. It is direct evidence for optically detected ensemble spin physics in corundum, while leaving single-ion addressability, scalable coupling, and array fabrication open.
- [R089] Z. Velluire-Pellat, E. Maréchal, C. Feuillet-Palma, and N. Bergeal, “Spin-photon interaction between a ruby crystal and a high-critical-temperature superconducting microwave cavity,” *Communications Physics* **8**, 236 (2025). DOI: 10.1038/s42005-025-02159-1.
- [R090] W. G. Farr, D. L. Creedon, M. Goryachev, K. Benmessai, and M. E. Tobar, “Ultrasensitive microwave spectroscopy of paramagnetic impurities of sapphire crystals at millikelvin temperatures,” *Physical Review B* **88**, 224426 (2013). DOI: 10.1103/PhysRevB.88.224426; arXiv: 1311.1049.
- [R091] P. F. Moulton, “Spectroscopic and laser characteristics of Ti:Al₂O₃,” *Journal of the Optical Society of America B* **3**, 125–133 (1986). DOI: 10.1364/JOSAB.3.000125.
- [R092] K. H. Lee and J. H. Crawford Jr., “Luminescence of the F center in sapphire,” *Physical Review B* **19**, 3217–3221 (1979). DOI: 10.1103/PhysRevB.19.3217.
- [R093] C. Wilhelmer, M. E. Turiansky, D. Waldhör, L. Cvitkovich, C. G. Van de Walle, and T. Grasser, “Optical properties of vacancies in aluminum oxide (alpha-Al₂O₃) from first principles,” *Physical Review Materials* **9**, 096202 (2025). DOI: 10.1103/5kvl-hktd.
- [R095] B. D. Evans and M. Stapelbroek, “Optical properties of the F^+ center in crystalline Al₂O₃,” *Physical Review B* **18**, 7089–7098 (1978). DOI: 10.1103/PhysRevB.18.7089.

H.36. Quantum spin systems

- [R147] G. Jackeli and G. Khaliullin, “Mott Insulators in the Strong Spin-Orbit Coupling Limit: From Heisenberg to a Quantum Compass and Kitaev Models,” *Physical Review Letters* **102**, 017205 (2009). DOI: 10.1103/PhysRevLett.102.017205; arXiv: 0809.4658.

- [R148] M. Hermanns, I. Kimchi, and J. Knolle, “Physics of the Kitaev Model: Fractionalization, Dynamic Correlations, and Material Connections,” *Annual Review of Condensed Matter Physics* **9**, 17–33 (2018). DOI: 10.1146/annurev-conmatphys-033117-053934; arXiv: 1705.01740.
- [R149] H. Takagi, T. Takayama, G. Jackeli, G. Khaliullin, and S. E. Nagler, “Concept and realization of Kitaev quantum spin liquids,” *Nature Reviews Physics* **1**, 264–280 (2019). DOI: 10.1038/s42254-019-0038-2; arXiv: 1903.08081.
- [R150] Y. Matsuda, T. Shibauchi, and H.-Y. Kee, “Kitaev quantum spin liquids,” *Reviews of Modern Physics* **97**, 045003 (2025). DOI: 10.1103/3m4m-3v59.
- [R151] A. Banerjee *et al.*, “Neutron scattering in the proximate quantum spin liquid α -RuCl₃,” *Science* **356**, 1055–1059 (2017). DOI: 10.1126/science.aah6015.
- [R152] Y. Kasahara *et al.*, “Majorana quantization and half-integer thermal quantum Hall effect in a Kitaev spin liquid,” *Nature* **559**, 227–231 (2018). DOI: 10.1038/s41586-018-0274-0; arXiv: 1805.05022.
- [R153] É. Lefrançois *et al.*, “Evidence of a Phonon Hall Effect in the Kitaev Spin Liquid Candidate α -RuCl₃,” *Physical Review X* **12**, 021025 (2022). DOI: 10.1103/PhysRevX.12.021025; arXiv: 2111.05493.
- [R154] A. Shragai, E. Horsley, S. Kim, Y.-J. Kim, and B. J. Ramshaw, “Phonon Hall viscosity and the intrinsic thermal Hall effect of α -RuCl₃,” *Nature* **652**, 1166–1172 (2026). DOI: 10.1038/s41586-026-10420-y.
- [R167] M. B. Hastings and T. Koma, “Spectral gap and exponential decay of correlations,” *Communications in Mathematical Physics* **265**, 781–804 (2006). DOI: 10.1007/s00220-006-0030-4; arXiv: math-ph/0507008.
- [R168] S. Bravyi and B. Terhal, “A no-go theorem for a two-dimensional self-correcting quantum memory based on stabilizer codes,” *New Journal of Physics* **11**, 043029 (2009). DOI: 10.1088/1367-2630/11/4/043029; arXiv: 0810.1983. This no-go theorem rules out self-correcting quantum memory for broad classes of two-dimensional stabilizer Hamiltonians. It is a central constraint on claims that two-dimensional topological encoding automatically supplies indefinite passive storage.
- [R169] B. J. Brown, D. Loss, J. K. Pachos, C. N. Self, and J. R. Wootton, “Quantum memories at finite temperature,” *Reviews of Modern Physics* **88**, 045005 (2016). DOI: 10.1103/RevModPhys.88.045005; arXiv: 1411.6643. This review surveys thermal stability and finite-temperature memory across topological codes. It is pivotal for separating zero-temperature phase stability, finite-size lifetime, active decoding, and genuine self-correction.
- [R170] C. Stark, L. Pollet, A. Imamoğlu, and R. Renner, “Localization of toric code defects,” *Physical Review Letters* **107**, 030504 (2011). DOI: 10.1103/PhysRevLett.107.030504; arXiv: 1101.6028.

- [R181] P. Neumann, R. Kolesov, B. Naydenov, *et al.*, “Quantum register based on coupled electron spins in a room-temperature solid,” *Nature Physics* **6**, 249–253 (2010). DOI: 10.1038/nphys1536; arXiv: 1004.5090.
- [R182] E. L. Rosenfeld, L. M. Pham, M. D. Lukin, and R. L. Walsworth, “Sensing coherent dynamics of electronic spin clusters in solids,” *Physical Review Letters* **120**, 243604 (2018). DOI: 10.1103/PhysRevLett.120.243604.
- [R192] A. Barfuss, J. Teissier, E. Neu, A. Nunnenkamp, and P. Maletinsky, “Strong mechanical driving of a single electron spin,” *Nature Physics* **11**, 820–824 (2015). DOI: 10.1038/nphys3411.
- [R195] T.-Y. Hwang, J. Lee, S.-W. Jeon, *et al.*, “Sub-10 nm precision engineering of solid-state defects via nanoscale aperture array mask,” *Nano Letters* **22**, 1672–1679 (2022). DOI: 10.1021/acs.nanolett.1c04699.
- [R196] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th ed., version 3.01, 2024. Stable URL: <https://www.bipm.org/en/publications/si-brochure>. DOI: 10.59161/AUEZ1291.
- [R199] L. Robledo *et al.*, “High-fidelity projective read-out of a solid-state spin quantum register,” *Nature* **477**, 574–578 (2011). DOI: 10.1038/nature10401; arXiv: 1301.0392.
- [R208] P. J. Mohr *et al.*, “CODATA recommended values of the fundamental physical constants: 2022,” *Reviews of Modern Physics* **97**, 025002 (2025). DOI: 10.1103/RevModPhys.97.025002.
- [R213] C. L. Degen, F. Reinhard, and P. Cappellaro, “Quantum sensing,” *Reviews of Modern Physics* **89**, 035002 (2017). DOI: 10.1103/RevModPhys.89.035002.
- [R215] G. de Lange, Z. H. Wang, D. Ristè, V. V. Dobrovitski, and R. Hanson, “Universal dynamical decoupling of a single solid-state spin from a spin bath,” *Science* **330**, 60–63 (2010). DOI: 10.1126/science.1192739.

H.37. Topological order

- [R028] Daniel Gottesman, *Stabilizer Codes and Quantum Error Correction*, PhD thesis, California Institute of Technology (1997). DOI: 10.7907/rzr7-dt72; arXiv: quant-ph/9705052.
- [R029] A. R. Calderbank, E. M. Rains, P. W. Shor, and N. J. A. Sloane, “Quantum error correction via codes over GF(4),” *IEEE Transactions on Information Theory* **44**, 1369–1387 (1998). DOI: 10.1109/18.681315; arXiv: quant-ph/9608006.

- [R030] Alexei Kitaev, “Fault-tolerant quantum computation by anyons,” *Annals of Physics* **303**, 2–30 (2003). DOI: 10.1016/S0003-4916(02)00018-0; arXiv: quant-ph/9707021. Kitaev’s toric-code paper established how nonlocal logical information can be encoded in a gapped two-dimensional many-body system and manipulated through anyonic processes. It remains the canonical baseline for separating topological encoding from self-correction and from active error correction.
- [R031] Scott Aaronson and Daniel Gottesman, “Improved simulation of stabilizer circuits,” *Physical Review A* **70**, 052328 (2004). DOI: 10.1103/PhysRevA.70.052328; arXiv: quant-ph/0406196.
- [R121] N. D. Mermin, “The topological theory of defects in ordered media,” *Reviews of Modern Physics* **51**, 591–648 (1979). DOI: 10.1103/RevModPhys.51.591.
- [R122] X.-G. Wen and Q. Niu, “Ground-state degeneracy of the fractional quantum Hall states in the presence of a random potential and on high-genus Riemann surfaces,” *Physical Review B* **41**, 9377–9396 (1990). DOI: 10.1103/PhysRevB.41.9377.
- [R123] M. B. Hastings and X.-G. Wen, “Quasi-adiabatic continuation of quantum states: The stability of topological ground-state degeneracy and emergent gauge invariance,” *Physical Review B* **72**, 045141 (2005). DOI: 10.1103/PhysRevB.72.045141; arXiv: cond-mat/0503554.
- [R124] X.-G. Wen, “Colloquium: Zoo of quantum-topological phases of matter,” *Reviews of Modern Physics* **89**, 041004 (2017). DOI: 10.1103/RevModPhys.89.041004; arXiv: 1610.03911.
- [R141] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, “Topological quantum memory,” *Journal of Mathematical Physics* **43**, 4452–4505 (2002). DOI: 10.1063/1.1499754; arXiv: quant-ph/0110143. Dennis and collaborators connected toric-code storage to error chains, decoding, and statistical-mechanical thresholds. This paper is foundational for understanding why topological encoding still requires syndrome extraction and decoding in realistic noisy systems.
- [R142] S. Bravyi, M. B. Hastings, and S. Michalakis, “Topological quantum order: stability under local perturbations,” *Journal of Mathematical Physics* **51**, 093512 (2010). DOI: 10.1063/1.3490195; arXiv: 1001.0344. Bravyi, Hastings, and Michalakis proved stability of topological quantum order under sufficiently weak local perturbations for a defined class of Hamiltonians. The theorem protects an already-gapped topological phase; it does not guarantee that a proposed perturbative simulator reaches that phase with a useful gap.
- [R143] R. Alicki, M. Fannes, and M. Horodecki, “On thermalization in Kitaev’s 2D model,” *Journal of Physics A: Mathematical and Theoretical* **42**, 065303 (2009). DOI: 10.1088/1751-8113/42/6/065303; arXiv: 0810.4584.
- [R144] B. M. Terhal, “Quantum error correction for quantum memories,” *Reviews of Modern Physics* **87**, 307–346 (2015). DOI: 10.1103/RevModPhys.87.307; arXiv: 1302.3428.

- [R145] S. Krinner *et al.*, “Realizing repeated quantum error correction in a distance-three surface code,” *Nature* **605**, 669–674 (2022). DOI: 10.1038/s41586-022-04566-8; arXiv: 2112.03708.
- [R146] M. Iqbal *et al.*, “Topological order from measurements and feed-forward on a trapped ion quantum computer,” *Communications Physics* **7**, 205 (2024). DOI: 10.1038/s42005-024-01698-3; arXiv: 2302.01917.
- [R160] M. Müger, “From subfactors to categories and topology II: The quantum double of tensor categories and subfactors,” *Journal of Pure and Applied Algebra* **180**, 159–219 (2003). DOI: 10.1016/S0022-4049(02)00248-7; arXiv: math/0111205.
- [R220] A. G. Fowler, “Coping with qubit leakage in topological codes,” *Physical Review A* **88**, 042308 (2013). DOI: 10.1103/PhysRevA.88.042308; arXiv: 1308.6642.
- [R221] E. Novais, E. R. Mucciolo, and H. U. Baranger, “Resilient quantum computation in correlated environments: A quantum phase transition perspective,” *Physical Review Letters* **98**, 040501 (2007). DOI: 10.1103/PhysRevLett.98.040501; arXiv: quant-ph/0607155.
- [R247] S. Jansen, M.-B. Ruskai, and R. Seiler, “Bounds for the adiabatic approximation with applications to quantum computation,” *Journal of Mathematical Physics* **48**, 102111 (2007). DOI: 10.1063/1.2798382; arXiv: quant-ph/0603175. Jansen, Ruskai, and Seiler give rigorous bounds for the adiabatic approximation with explicit gap dependence. These bounds explain why a small engineered many-body gap can turn state preparation into a prohibitive runtime even when the endpoint Hamiltonian is correct.
- [R248] T. M. Stace, S. D. Barrett, and A. C. Doherty, “Thresholds for topological codes in the presence of loss,” *Physical Review Letters* **102**, 200501 (2009). DOI: 10.1103/PhysRevLett.102.200501; arXiv: 0904.3556.

H.38. Anyons

- [R012] E. Artin, “Theory of Braids,” *Annals of Mathematics* **48**, 101–126 (1947). DOI: 10.2307/1969218.
- [R013] C. Kassel and V. Turaev, *Braid Groups*, Graduate Texts in Mathematics 247, Springer (2008). DOI: 10.1007/978-0-387-68548-9.
- [R014] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Springer (2015). DOI: 10.1007/978-3-319-13467-3.
- [R015] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, “Non-Abelian Anyons and Topological Quantum Computation,” *Reviews of Modern Physics*

- 80**, 1083–1159 (2008). DOI: 10.1103/RevModPhys.80.1083; arXiv: 0707.1889. This review is the standard bridge from non-Abelian statistics to fault-tolerant gate constructions. It also makes the crucial distinction between topological protection in an ideal anyon model and the engineering needed to realize and control that model.
- [R016] E. C. Rowell and Z. Wang, “Mathematics of Topological Quantum Computing,” *Bulletin of the American Mathematical Society* **55**, 183–238 (2018). DOI: 10.1090/bull/1605; arXiv: 1705.06206.
 - [R017] A. Kitaev, “Anyons in an Exactly Solved Model and Beyond,” *Annals of Physics* **321**, 2–111 (2006). DOI: 10.1016/j.aop.2005.10.005; arXiv: cond-mat/0506438. Kitaev’s exact honeycomb solution supplies a rare microscopic spin Hamiltonian with fractionalized excitations and a non-Abelian phase. It is a benchmark for what an engineered spin array must reproduce, not evidence that a defect array already realizes that phase.
 - [R019] B. Bakalov and A. Kirillov Jr., *Lectures on Tensor Categories and Modular Functors*, University Lecture Series 21, American Mathematical Society (2001). DOI: 10.1090/ulect/021.
 - [R020] P. Etingof, S. Gelaki, D. Nikshych, and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society (2015). DOI: 10.1090/surv/205.
 - [R021] M. Barkeshli, P. Bonderson, M. Cheng, and Z. Wang, “Symmetry Fractionalization, Defects, and Gauging of Topological Phases,” *Physical Review B* **100**, 115147 (2019). DOI: 10.1103/PhysRevB.100.115147; arXiv: 1410.4540.
 - [R022] G. Moore and N. Seiberg, “Classical and Quantum Conformal Field Theory,” *Communications in Mathematical Physics* **123**, 177–254 (1989). DOI: 10.1007/BF01238857.
 - [R023] P. Bonderson, K. Shtengel, and J. K. Slingerland, “Interferometry of Non-Abelian Anyons,” *Annals of Physics* **323**, 2709–2755 (2008). DOI: 10.1016/j.aop.2008.01.012; arXiv: 0707.4206.
 - [R127] J. M. Leinaas and J. Myrheim, “On the theory of identical particles,” *Il Nuovo Cimento B* **37**, 1–23 (1977). DOI: 10.1007/BF02727953.
 - [R128] F. Wilczek, “Quantum mechanics of fractional-spin particles,” *Physical Review Letters* **49**, 957–959 (1982). DOI: 10.1103/PhysRevLett.49.957.
 - [R129] Y.-S. Wu, “General theory for quantum statistics in two dimensions,” *Physical Review Letters* **52**, 2103–2106 (1984). DOI: 10.1103/PhysRevLett.52.2103.
 - [R130] H. Bartolomei *et al.*, “Fractional statistics in anyon collisions,” *Science* **368**, 173–177 (2020). DOI: 10.1126/science.aaz5601; arXiv: 2006.13157.

- [R133] A. Stern, “Anyons and the quantum Hall effect—a pedagogical review,” *Annals of Physics* **323**, 204–249 (2008). DOI: 10.1016/j.aop.2007.10.008; arXiv: 0711.4697.
- [R134] D. A. Ivanov, “Non-Abelian statistics of half-quantum vortices in p-wave superconductors,” *Physical Review Letters* **86**, 268–271 (2001). DOI: 10.1103/PhysRevLett.86.268; arXiv: cond-mat/0005069.

H.39. Fibonacci anyons

- [R135] S. Trebst, M. Troyer, Z. Wang, and A. W. W. Ludwig, “A short introduction to Fibonacci anyon models,” *Progress of Theoretical Physics Supplement* **176**, 384–407 (2008). DOI: 10.1143/PTPS.176.384; arXiv: 0902.3275.
- [R136] M. H. Freedman, M. Larsen, and Z. Wang, “A modular functor which is universal for quantum computation,” *Communications in Mathematical Physics* **227**, 605–622 (2002). DOI: 10.1007/s002200200645; arXiv: quant-ph/0001108.
- [R137] L. Hormozi, G. Zikos, N. E. Bonesteel, and S. H. Simon, “Topological quantum compiling,” *Physical Review B* **75**, 165310 (2007). DOI: 10.1103/PhysRevB.75.165310; arXiv: quant-ph/0610111.
- [R139] N. Read and E. Rezayi, “Beyond paired quantum Hall states: Parafermions and incompressible states in the first excited Landau level,” *Physical Review B* **59**, 8084–8092 (1999). DOI: 10.1103/PhysRevB.59.8084; arXiv: cond-mat/9809384. Read and Rezayi identified a sequence of clustered quantum Hall states whose quasiparticles include the Fibonacci case relevant to universal topological computation. The work provides a candidate physical route, but identifying a filling fraction is not by itself conclusive quasiparticle evidence.
- [R225] E. H. Rezayi and N. Read, “Non-Abelian quantized Hall states of electrons at filling factors $12/5$ and $13/5$ in the first excited Landau level,” *Physical Review B* **79**, 075306 (2009). DOI: 10.1103/PhysRevB.79.075306; arXiv: cond-mat/0608346. Rezayi and Read supplied numerical evidence for non-Abelian quantum Hall states at fillings $12/5$ and $13/5$. The result supports a candidate phase assignment but does not directly observe Fibonacci fusion or braiding.
- [R226] C. Zhang, C. Huan, J. S. Xia, N. S. Sullivan, W. Pan, K. W. Baldwin, K. W. West, L. N. Pfeiffer, and D. C. Tsui, “Spin polarization of the $\nu = 12/5$ fractional quantum Hall state,” *Physical Review B* **85**, 241302(R) (2012). DOI: 10.1103/PhysRevB.85.241302. This experiment measured the spin polarization of the $12/5$ fractional quantum Hall state. It constrains candidate orders at that filling without uniquely establishing Fibonacci quasiparticles.

- [R227] W. Zhu, S. S. Gong, F. D. M. Haldane, and D. N. Sheng, “Fractional quantum Hall states at $\nu = 13/5$ and $12/5$ and their non-Abelian nature,” *Physical Review Letters* **115**, 126805 (2015). DOI: 10.1103/PhysRevLett.115.126805; arXiv: 1505.03050.
- [R228] R. S. K. Mong, M. P. Zaletel, F. Pollmann, and Z. Papić, “Fibonacci anyons and charge density order in the $12/5$ and $13/5$ quantum Hall plateaus,” *Physical Review B* **95**, 115136 (2017). DOI: 10.1103/PhysRevB.95.115136; arXiv: 1505.02843. Mong and collaborators found numerical evidence connecting the $12/5$ and $13/5$ plateaus to Fibonacci topological order and charge-density structure. The conclusion remains model- and finite-size-dependent rather than a direct quasiparticle measurement.

H.40. String nets

- [R018] M. A. Levin and X.-G. Wen, “String-Net Condensation: A Physical Mechanism for Topological Phases,” *Physical Review B* **71**, 045110 (2005). DOI: 10.1103/PhysRevB.71.045110; arXiv: cond-mat/0404617. Levin and Wen introduced string-net condensation as a commuting-projector mechanism for broad classes of doubled topological phases. This paper defines the structural target behind the manuscript’s doubled-Fibonacci engineering proposal.
- [R155] Lukasz Fidkowski, Michael Freedman, Chetan Nayak, Kevin Walker, and Zhenghan Wang, “From string nets to nonabelions,” *Communications in Mathematical Physics* **287**, 805–827 (2009). DOI: 10.1007/s00220-009-0757-9; arXiv: cond-mat/0610583.
- [R156] Alexei Kitaev and Liang Kong, “Models for gapped boundaries and domain walls,” *Communications in Mathematical Physics* **313**, 351–373 (2012). DOI: 10.1007/s00220-012-1500-5; arXiv: 1104.5047.
- [R157] Alexander Kirillov Jr., “String-net model of Turaev–Viro invariants,” (2011). DOI: 10.48550/arXiv.1106.6033; arXiv: 1106.6033.
- [R158] Robert König, Greg Kuperberg, and Ben W. Reichardt, “Quantum computation with Turaev–Viro codes,” *Annals of Physics* **325**, 2707–2749 (2010). DOI: 10.1016/j.aop.2010.08.001; arXiv: 1002.2816.
- [R159] Oliver Buerschaper, Miguel Aguado, and Guifré Vidal, “Explicit tensor network representation for the ground states of string-net models,” *Physical Review B* **79**, 085119 (2009). DOI: 10.1103/PhysRevB.79.085119; arXiv: 0809.2393.
- [R161] C.-H. Lin, M. Levin, and F. J. Burnell, “Generalized string-net models: A thorough exposition,” *Physical Review B* **103**, 195155 (2021). DOI: 10.1103/PhysRevB.103.195155; arXiv: 2012.14424.

- [R183] M. D. Schulz, S. Dusuel, K. P. Schmidt, and J. Vidal, “Topological phase transitions in the golden string-net model,” *Physical Review Letters* **110**, 147203 (2013). DOI: 10.1103/PhysRevLett.110.147203; arXiv: 1212.4109.
- [R184] H. Yao and S. A. Kivelson, “An exact chiral spin liquid with non-Abelian anyons,” *Physical Review Letters* **99**, 247203 (2007). DOI: 10.1103/PhysRevLett.99.247203; arXiv: 0708.0040.
- [R185] M. Kargarian, H. Bombin, and M. A. Martin-Delgado, “Topological color codes and two-body quantum lattice Hamiltonians,” *New Journal of Physics* **12**, 025018 (2010). DOI: 10.1088/1367-2630/12/2/025018; arXiv: 0906.4127.
- [R186] G. Kells, J. Kailasvuori, J. K. Slingerland, and J. Vala, “Kaleidoscope of topological phases with multiple Majorana species,” *New Journal of Physics* **13**, 095014 (2011). DOI: 10.1088/1367-2630/13/9/095014; arXiv: 1012.5276.
- [R187] R. Moessner and S. L. Sondhi, “Resonating valence bond phase in the triangular lattice quantum dimer model,” *Physical Review Letters* **86**, 1881–1884 (2001). DOI: 10.1103/PhysRevLett.86.1881; arXiv: cond-mat/0007378.
- [R188] L. Balents, M. P. A. Fisher, and S. M. Girvin, “Fractionalization in an easy-axis kagome antiferromagnet,” *Physical Review B* **65**, 224412 (2002). DOI: 10.1103/PhysRevB.65.224412; arXiv: cond-mat/0110005.
- [R209] Roberto Oliveira and Barbara M. Terhal, “The complexity of quantum spin systems on a two-dimensional square lattice,” *Quantum Information & Computation* **8**, 900–924 (2008). arXiv: quant-ph/0504050.
- [R211] N. Bar-Gill, L. M. Pham, A. Jarmola, D. Budker, and R. L. Walsworth, “Solid-state electronic spin coherence time approaching one second,” *Nature Communications* **4**, 1743 (2013). DOI: 10.1038/ncomms2771.

H.41. Perturbative gadgets

- [R024] J. R. Schrieffer and P. A. Wolff, “Relation between the Anderson and Kondo Hamiltonians,” *Physical Review* **149**, 491–492 (1966). DOI: 10.1103/PhysRev.149.491.
- [R025] Sergey Bravyi, David P. DiVincenzo, and Daniel Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics* **326**, 2793–2826 (2011). DOI: 10.1016/j.aop.2011.06.004; arXiv: 1105.0675. This work puts Schrieffer–Wolff perturbation theory for many-body systems on a controlled footing, including effective Hamiltonians and error bounds. It is the appropriate reference whenever a low-energy interaction is claimed to emerge after eliminating excited cluster states.

- [R026] A. H. MacDonald, S. M. Girvin, and D. Yoshioka, “ t/U expansion for the Hubbard model,” *Physical Review B* **37**, 9753–9756 (1988). DOI: 10.1103/PhysRevB.37.9753.
- [R027] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Springer (1976; corrected printing 1995). DOI: 10.1007/978-3-642-66282-9.
- [R171] H. Feshbach, “Unified theory of nuclear reactions,” *Annals of Physics* **5**, 357–390 (1958). DOI: 10.1016/0003-4916(58)90007-1.
- [R172] J. Hubbard, “Electron correlations in narrow energy bands,” *Proceedings of the Royal Society A* **276**, 238–257 (1963). DOI: 10.1098/rspa.1963.0204.
- [R173] P. W. Anderson, “New Approach to the Theory of Superexchange Interactions,” *Physical Review* **115**, 2–13 (1959). DOI: 10.1103/PhysRev.115.2.
- [R174] S. P. Jordan and E. Farhi, “Perturbative gadgets at arbitrary orders,” *Physical Review A* **77**, 062329 (2008). DOI: 10.1103/PhysRevA.77.062329; arXiv: 0802.1874.
- [R175] S. Bravyi, D. P. DiVincenzo, D. Loss, and B. M. Terhal, “Quantum simulation of many-body Hamiltonians using perturbation theory with bounded-strength interactions,” *Physical Review Letters* **101**, 070503 (2008). DOI: 10.1103/PhysRevLett.101.070503; arXiv: 0803.2686. Bravyi and collaborators make the promise and cost of perturbative gadgets precise: bounded-strength two-body resources can reproduce many-body terms only within a controlled scale hierarchy. The result is therefore both a construction tool and a warning about shrinking effective gaps and amplified fabrication errors.
- [R176] J. Vidal, K. P. Schmidt, and S. Dusuel, “Perturbative approach to an exactly solved problem: Kitaev honeycomb model,” *Physical Review B* **78**, 245121 (2008). DOI: 10.1103/PhysRevB.78.245121; arXiv: 0809.1553.
- [R177] C. G. Brell, S. T. Flammia, S. D. Bartlett, and A. C. Doherty, “Toric codes and quantum doubles from two-body Hamiltonians,” *New Journal of Physics* **13**, 053039 (2011). DOI: 10.1088/1367-2630/13/5/053039; arXiv: 1011.1942. Brell and collaborators explicitly construct toric-code and quantum-double interactions from two-body Hamiltonians using perturbative gadgets. The paper is directly relevant to the proposed defect architecture because it exposes the ancilla and energy-scale overhead needed for such a reduction.
- [R178] R. König, “Simplifying quantum double Hamiltonians using perturbative gadgets,” *Quantum Information and Computation* **10**, 292–324 (2010). DOI: 10.26421/QIC10.3-4-9; arXiv: 0901.1333.
- [R179] C. G. Brell, S. D. Bartlett, and A. C. Doherty, “Perturbative 2-body parent Hamiltonians for projected entangled pair states,” *New Journal of Physics* **16**, 123056 (2014). DOI: 10.1088/1367-2630/16/12/123056; arXiv: 1407.4829.

- [R180] S. A. Ocko and B. Yoshida, “Nonperturbative gadget for topological quantum codes,” *Physical Review Letters* **107**, 250502 (2011). DOI: 10.1103/PhysRevLett.107.250502; arXiv: 1107.2697.
- [R245] T. Albash and D. A. Lidar, “Adiabatic quantum computation,” *Reviews of Modern Physics* **90**, 015002 (2018). DOI: 10.1103/RevModPhys.90.015002; arXiv: 1611.04471.

H.42. Quantum simulation

- [R032] Frank Verstraete and J. Ignacio Cirac, “Renormalization algorithms for quantum-many body systems in two and higher dimensions,” (2004). arXiv: cond-mat/0407066.
- [R033] Norbert Schuch, Michael M. Wolf, Frank Verstraete, and J. Ignacio Cirac, “Computational complexity of projected entangled pair states,” *Physical Review Letters* **98**, 140506 (2007). DOI: 10.1103/PhysRevLett.98.140506; arXiv: quant-ph/0611050.
- [R034] Norbert Schuch, J. Ignacio Cirac, and David Pérez-García, “PEPS as ground states: Degeneracy and topology,” *Annals of Physics* **325**, 2153–2192 (2010). DOI: 10.1016/j.aop.2010.05.008; arXiv: 1001.3807.
- [R035] Ulrich Schollwöck, “The density-matrix renormalization group in the age of matrix product states,” *Annals of Physics* **326**, 96–192 (2011). DOI: 10.1016/j.aop.2010.09.012; arXiv: 1008.3477.
- [R036] Román Orús, “A practical introduction to tensor networks: Matrix product states and projected entangled pair states,” *Annals of Physics* **349**, 117–158 (2014). DOI: 10.1016/j.aop.2014.06.013; arXiv: 1306.2164.
- [R164] Mohsin Iqbal *et al.*, “Non-Abelian topological order and anyons on a trapped-ion processor,” *Nature* **626**, 505–511 (2024). DOI: 10.1038/s41586-023-06934-4; arXiv: 2305.03766.
- [R166] Tomoya Hayata, Yoshimasa Hidaka & Yuta Kikuchi, “Digital quantum simulation of q -deformed SU(2) Yang–Mills theory on a trapped-ion quantum computer,” *Physical Review Research* **8**, 033137 (2026). DOI: 10.1103/vlpv-n8dy; arXiv: 2601.13530.
- [R229] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, “A quantum engineer’s guide to superconducting qubits,” *Applied Physics Reviews* **6**, 021318 (2019). DOI: 10.1063/1.5089550.
- [R231] A. Browaeys and T. Lahaye, “Many-body physics with individually controlled Rydberg atoms,” *Nature Physics* **16**, 132–142 (2020). DOI: 10.1038/s41567-019-0733-z.

- [R232] S. J. Evered *et al.*, “High-fidelity parallel entangling gates on a neutral-atom quantum computer,” *Nature* **622**, 268–272 (2023). DOI: 10.1038/s41586-023-06481-y.
- [R249] J. R. Johansson, P. D. Nation, and F. Nori, “QuTiP: An open-source Python framework for the dynamics of open quantum systems,” *Computer Physics Communications* **183**, 1760–1772 (2012). DOI: 10.1016/j.cpc.2012.02.021.
- [R250] M. Fishman, S. R. White, and E. M. Stoudenmire, “The ITensor Software Library for Tensor Network Calculations,” *SciPost Physics Codebases* **4** (2022). DOI: 10.21468/SciPostPhysCodeb.4; arXiv: 2007.14822.
- [R251] J. Hauschild and F. Pollmann, “Efficient numerical simulations with Tensor Networks: Tensor Network Python (TeNPy),” *SciPost Physics Lecture Notes* **5** (2018). DOI: 10.21468/SciPostPhysLectNotes.5; arXiv: 1805.00055.
- [R252] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete, “Matrix product states and projected entangled pair states: Concepts, symmetries, theorems,” *Reviews of Modern Physics* **93**, 045003 (2021). DOI: 10.1103/RevModPhys.93.045003; arXiv: 2011.12127.
- [R253] J. Jordan, R. Orús, G. Vidal, F. Verstraete, and J. I. Cirac, “Classical simulation of infinite-size quantum lattice systems in two spatial dimensions,” *Physical Review Letters* **101**, 250602 (2008). DOI: 10.1103/PhysRevLett.101.250602; arXiv: cond-mat/0703788.
- [R254] A. Kitaev and J. Preskill, “Topological entanglement entropy,” *Physical Review Letters* **96**, 110404 (2006). DOI: 10.1103/PhysRevLett.96.110404; arXiv: hep-th/0510092.
- [R255] M. Levin and X.-G. Wen, “Detecting topological order in a ground state wave function,” *Physical Review Letters* **96**, 110405 (2006). DOI: 10.1103/PhysRevLett.96.110405; arXiv: cond-mat/0510613.

H.43. Fabrication

- [R094] T. M. Hensen, M. J. A. de Dood, and A. Polman, “Luminescence quantum efficiency and local optical density of states in thin film ruby made by ion implantation,” *Journal of Applied Physics* **88**, 5142–5147 (2000). DOI: 10.1063/1.1314322.
- [R098] C. Babin, R. Stöhr, N. Morioka, *et al.*, “Fabrication and nanophotonic waveguide integration of silicon carbide colour centres with preserved spin-optical coherence,” *Nature Materials* **21**, 67–73 (2022). DOI: 10.1038/s41563-021-01148-3.
- [R189] D. M. Toyli, C. D. Weis, G. D. Fuchs, T. Schenkel, and D. D. Awschalom, “Chip-scale nanofabrication of single spins and spin arrays in diamond,” *Nano Letters* **10**, 3168–3172 (2010). DOI: 10.1021/nl102066q.

- [R190] T. Schröder *et al.*, “Scalable focused ion beam creation of nearly lifetime-limited single quantum emitters in diamond nanostructures,” *Nature Communications* **8**, 15376 (2017). DOI: 10.1038/ncomms15376.
- [R202] P. Spinicelli *et al.*, “Engineered arrays of NV color centers in diamond based on implantation of CN^- molecules through nanoapertures,” *New Journal of Physics* **13**, 025014 (2011). DOI: 10.1088/1367-2630/13/2/025014; arXiv: 1008.1483.
- [R203] Y.-C. Chen *et al.*, “Laser writing of individual nitrogen-vacancy defects in diamond with near-unity yield,” *Optica* **6**, 662–667 (2019). DOI: 10.1364/OP-TICA.6.000662. Chen and collaborators demonstrated laser writing of individual NV defects with near-unity creation yield after optimization. The experiment addresses yield and three-dimensional placement, but optical aberrations, residual damage, and nanometer-scale pair geometry remain separate constraints.
- [R204] K. Ohno *et al.*, “Engineering shallow spins in diamond with nitrogen delta-doping,” *Applied Physics Letters* **101**, 082413 (2012). DOI: 10.1063/1.4748280; arXiv: 1207.2784.
- [R206] P. Räcke, L. Pietzonka, J. Meijer, D. Spemann, and R. Wunderlich, “Vacancy diffusion and nitrogen-vacancy center formation near the diamond surface,” *Applied Physics Letters* **118**, 204003 (2021). DOI: 10.1063/5.0046031.
- [R207] S. Kim *et al.*, “Scalable nanoscale positioning of highly coherent color centers in prefabricated diamond nanostructures,” *Nature Communications* **16**, 9803 (2025). DOI: 10.1038/s41467-025-64758-4; arXiv: 2502.01198. This work demonstrates scalable nanoscale placement of coherent color centers in prefabricated diamond nanostructures. It is among the strongest fabrication results for combining registration and coherence, though the tolerances needed for uniform short-range exchange remain more demanding.
- [R243] M. Haruyama *et al.*, “Triple nitrogen-vacancy centre fabrication by $\text{C}_5\text{N}_4\text{H}_n$ ion implantation,” *Nature Communications* **10**, 2664 (2019). DOI: 10.1038/s41467-019-10529-x. Haruyama and collaborators used molecular implantation to fabricate correlated triples of NV centers. The result is especially relevant to cluster-based proposals, but creation of a triple does not guarantee the geometry, charge state, coherence, and coupling uniformity required by a target Hamiltonian.
- [R257] D. Scarabelli, M. Trusheim, O. Gaathon, D. Englund, and S. J. Wind, “Nanoscale engineering of closely-spaced electronic spins in diamond,” *Nano Letters* **16**, 4982–4990 (2016). DOI: 10.1021/acs.nanolett.6b01692.
- [R258] K. Groot-Berning, G. Jacob, C. Osterkamp, F. Jelezko, and F. Schmidt-Kaler, “Fabrication of $^{15}\text{NV}^-$ centers in diamond using a deterministic single ion implanter,” *New Journal of Physics* **23**, 063067 (2021). DOI: 10.1088/1367-2630/ac0753; arXiv: 2101.01979. Groot-Berning and collaborators demonstrated deterministic single-ion

implantation for creating nitrogen-vacancy centers. The work anchors placement-yield claims at the single-defect level, while conversion yield and post-implant coherence remain distinct metrics.

- [R259] A. Persaud, J. A. Liddle, T. Schenkel, J. Bokor, Tzv. Ivanov, and I. W. Rangelow, “Ion implantation with scanning probe alignment,” *Journal of Vacuum Science & Technology B* **23**, 2798–2800 (2005). DOI: 10.1116/1.2062628. Persaud and collaborators demonstrated ion implantation registered through scanning-probe alignment. The method directly addresses entry-coordinate placement, but stopping straggle and post-implant defect conversion still determine the final three-dimensional site distribution.
- [R260] S. Pezzagna, B. Naydenov, F. Jelezko, J. Wrachtrup, and J. Meijer, “Creation efficiency of nitrogen-vacancy centres in diamond,” *New Journal of Physics* **12**, 065017 (2010). DOI: 10.1088/1367-2630/12/6/065017. Pezzagna and collaborators measured the creation efficiency of implanted nitrogen-vacancy centers. This source is pivotal for separating delivered-ion count from the much smaller probability of obtaining the desired optically active defect.

H.44. Experimental topological quantum computing

- [R125] K. J. Satzinger *et al.*, “Realizing topologically ordered states on a quantum processor,” *Science* **374**, 1237–1241 (2021). DOI: 10.1126/science.abi8378; arXiv: 2104.01180. This superconducting-processor experiment prepared and probed a toric-code-type topologically ordered state. Because the state was digitally assembled and diagnosed on programmable hardware, it demonstrates controllable simulation rather than a naturally gapped topological material.
- [R126] G. Semeghini *et al.*, “Probing topological spin liquids on a programmable quantum simulator,” *Science* **374**, 1242–1247 (2021). DOI: 10.1126/science.abi8794; arXiv: 2104.04119. Semeghini and collaborators probed spin-liquid and topological signatures in a programmable Rydberg-atom array. The work is a major quantum-simulation milestone, but its driven, measured simulator should not be conflated with passive material protection.
- [R131] J. Nakamura, S. Liang, G. C. Gardner, and M. J. Manfra, “Direct observation of anyonic braiding statistics,” *Nature Physics* **16**, 931–936 (2020). DOI: 10.1038/s41567-020-1019-1; arXiv: 2006.14115.
- [R132] M. B. Andersen *et al.* (Google Quantum AI and Collaborators), “Non-Abelian braiding of graph vertices in a superconducting processor,” *Nature* **618**, 264–269 (2023). DOI: 10.1038/s41586-023-05954-4; arXiv: 2210.10255. This experiment implemented non-Abelian braiding of graph defects on a superconducting processor. It

establishes controlled braiding transformations in a digitally engineered code space, not intrinsic non-Abelian quasiparticles in a defect crystal.

- [R138] S. Xu *et al.*, “Non-Abelian braiding of Fibonacci anyons with a superconducting processor,” *Nature Physics* **20**, 1469–1475 (2024). DOI: 10.1038/s41567-024-02529-6; arXiv: 2404.00091. Xu and collaborators digitally prepared, fused, and braided Fibonacci anyons on a superconducting processor. The experiment is pivotal evidence that the target algebra can be enacted, while explicitly not establishing an autonomous Fibonacci material phase.
- [R140] C. F. B. Lo *et al.*, “Universal gates from braiding and fusing anyons on quantum hardware,” *Nature* **655**, 591–597 (2026). DOI: 10.1038/s41586-026-10709-y; arXiv: 2601.20956.
- [R162] Shibo Xu *et al.*, “Digital Simulation of Projective Non-Abelian Anyons with 68 Superconducting Qubits,” *Chinese Physics Letters* **40**, 060301 (2023). DOI: 10.1088/0256-307X/40/6/060301.
- [R163] Yu-ang Fan *et al.*, “Experimental quantum simulation of a topologically protected Hadamard gate via braiding Fibonacci anyons,” *The Innovation* **4**, 100480 (2023). DOI: 10.1016/j.xinn.2023.100480; arXiv: 2210.12145.
- [R165] Zlatko K. Mineev *et al.*, “Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials,” *Nature Communications* **16**, 6225 (2025). DOI: 10.1038/s41467-025-61493-8; arXiv: 2406.12820.
- [R222] R. M. Lutchyn, J. D. Sau, and S. Das Sarma, “Majorana fermions and a topological phase transition in semiconductor-superconductor heterostructures,” *Physical Review Letters* **105**, 077001 (2010). DOI: 10.1103/PhysRevLett.105.077001; arXiv: 1002.4033.
- [R223] K. Flensberg, F. von Oppen, and A. Stern, “Engineered platforms for topological superconductivity and Majorana zero modes,” *Nature Reviews Materials* **6**, 944–958 (2021). DOI: 10.1038/s41578-021-00336-6; arXiv: 2103.05548.
- [R224] Microsoft Azure Quantum, M. Aghaee, A. Alcaraz Ramirez, Z. Alam, *et al.*, “Interferometric single-shot parity measurement in InAs–Al hybrid devices,” *Nature* **638**, 651–655 (2025). DOI: 10.1038/s41586-024-08445-2; arXiv: 2401.09549.
- [R230] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” *Nature* **638**, 920–926 (2025; published online 2024). DOI: 10.1038/s41586-024-08449-y; arXiv: 2408.13687.
- [R233] D. Bluvstein *et al.*, “Logical quantum processor based on reconfigurable atom arrays,” *Nature* **626**, 58–65 (2024). DOI: 10.1038/s41586-023-06927-3.
- [R240] A. Gruber, A. Dräbenstedt, C. Tietz, L. Fleury, J. Wrachtrup, and C. von Borczyskowski, “Scanning Confocal Optical Microscopy and Magnetic Resonance on Single Defect Centers,” *Science* **276**, 2012–2014 (1997). DOI: 10.1126/science.276.5321.2012.

- [R241] H.-Y. Huang, R. Kueng, and J. Preskill, “Predicting many properties of a quantum system from very few measurements,” *Nature Physics* **16**, 1050–1057 (2020). DOI: 10.1038/s41567-020-0932-7; arXiv: 2002.08953.
- [R256] Y. Zhang, T. Grover, A. Turner, M. Oshikawa, and A. Vishwanath, “Quasi-particle statistics and braiding from ground-state entanglement,” *Physical Review B* **85**, 235151 (2012). DOI: 10.1103/PhysRevB.85.235151; arXiv: 1111.2342.